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Filip Rindler

Calculus of Variations

Second Edition

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Filip Rindler

Calculus of Variations

Second Edition

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Preface

The Calculus of Variations has its roots in the first problems of optimality studied in classical antiquity by Archimedes (ca. 287–212 BC) and Zenodorus (ca. 200–140 BC). The beginning of the field as a branch of modern mathematics can be traced back to June 1696, when Johann Bernoulli posed the pivotal *brachistochrone problem*, and Leonhard Euler's eponymous 1766 treatise *Elementa calculi variationum*.

The field has seen a sweeping revolution since the formulation of David Hilbert's 19th, 20th, and 23rd problems in 1900. This is particularly true for the theory of so-called "multiple integrals", that is, integral functionals defined on spaces of vector-valued maps in several variables. Minimization problems for such functionals have been systematically investigated from the 1950s onwards, most notably in the works of Charles B. Morrey Jr., Ennio De Giorgi, and John M. Ball. These developments were further fueled by the adaptation of sophisticated mathematical techniques from measure theory, geometric analysis, and the theory of nonlinear PDEs, and likewise the growing popularity of variational methods in those fields.

On the application side, the discovery of powerful variational principles to investigate questions of material science, in particular in the theories of nonlinear elasticity and microstructure, was (and still is) a rich source of challenging problems, which have shaped the field into its present form.

The intent of this book is to give an introduction to the classical and modern Calculus of Variations with a focus on the theory of integral functionals defined on spaces of vector-valued maps in several variables. It leads the reader from the most fundamental results to topics of current research. I do not claim any originality beyond the design of the exposition, but I have on occasion improved proofs of well-known results using ideas that were not known when the original arguments were conceived.

This is not an encyclopedic work. I aim to show the "big picture", which necessitates the omission of otherwise interesting and important results. Furthermore, more often than not I present only a special case of a general theorem to keep the exposition concise. Naturally, the choice of topics that I treat in detail is biased by my own personal taste.

The presentation of the material is based on a few guiding principles:

- Modern techniques are used whenever this leads to a clearer exposition. Most prominently, Young measures are introduced early since they provide a unified and convenient framework to understand a variety of topics.
- I try to use "reasonable" assumptions, not the most general ones.
- When presented with a choice of how to prove a result, I have usually chosen what is, in my opinion, the most conceptually "insightful" approach, instead of the most "elementary" one.
- Minimization problems over vector-valued maps are considered right from the start and scalar problems are treated as a special case.
- Occasionally, I state recent theorems without giving a proof. The rationale here is that I want the reader to see the frontier of research without compromising the coherence of the text.

This book has three parts: The first of these is the Basic Course and its chapters are intended to be read in order. They can form the basis of a 30-hour or 40-hour lecture course for an advanced undergraduate or graduate audience, with some selection on the part of the lecturer of what to cover in detail and what to omit. In fact, this part is based on lecture notes for a course on the Calculus of Variations that I lectured at the University of Warwick several times.

The second and third part of the book contain further material that is suitable for a topics course, a reading seminar, or self-study. Some results presented in these chapters have so far only been accessible in the research literature and I hope that even seasoned professionals will find something of interest there. The second and third part can be read independently from each other, but it is recommended to follow the chapter order within them, perhaps omitting some particularly challenging proofs during the first reading.

The 140 exercise problems are an integral part of the text and I encourage the reader to attempt as many as possible.

The prerequisites for this book are a good knowledge of functional analysis, measure theory, and some Sobolev space theory. Most of the non-elementary results that are required throughout the book are recalled in the appendix.

The design of this book is strongly influenced by several previous works. I note in particular the lecture notes on microstructure by Müller [248], Dacorogna's treatise [91], Kirchheim's advanced lecture notes on differential inclusions [193], the monograph on Young measures by Pedregal [272], Giusti's text [158], Dolzmann's book on microstructure in materials [123], as well as lecture notes on several related courses by Jan Kristensen and Alexander Mielke.

I am grateful for any comments, corrections, and suggestions. They can be sent via the book's website, where a list of corrections will also be maintained:

www.calculusofvariations.com

The First Edition of this book was published in the Springer *Universitext* series in 2018. Since then, it has been a pleasure to see it being used as the basis for lectures and topics courses on the Calculus of Variations at a number of universities. Many people have sent corrections, suggestions, and remarks, for all of which I am very grateful. This

Second Edition, now in the *Graduate Texts in Mathematics* series, has taken all of these into account, as long as they fit into the overall concept of the book.

The most substantive changes from the First Edition are the following:

- I have moderately rearranged the topics in the Basic Course. Additionally, the introductions to Γ-convergence and to compensated compactness have been moved up from the Advanced Topics part of the First Edition. I have also included a new section on mean coercivity.
- Some sections have been moved to make the sequencing more natural. Moreover, the material has been split into more chapters and some chapters have changed focus slightly.
- The approach to growth conditions on integrands is improved from the First Edition to make the presentation more uniform.
- The quasiconvex envelope is now defined via the sup-formula, like all the other envelopes, whereas the First Edition used the (equivalent) relaxation formula.
- The former Advanced Topics part has been split into two, namely a part on Rigidity & Microstructure and one on Linear-Growth Theory.
- I have extended the treatment of convex integration and moved all the material into its own chapter. In going with the overall theme of the book, I have also added a more recent Young measure approach.
- Many proofs have been improved and a number of arguments that seemed too terse on re-reading have been expanded.
- I have reorganized and extended the exercise problems, of which there are now 140.
- Over 50 new literature references, mostly to research done since the publication of the First Edition, have been added.

I have, however, resisted the temptation to add major new topics. This book is long enough as it is!

I am immensely grateful to my mathematical teachers Jan Kristensen and Alexander Mielke. Through their generosity and enthusiasm in sharing their knowledge, they have provided me with the foundation of my study and research. I also would like to thank the following people for helpful discussions, corrections, and comments: Dimitrios Andreakis, Adolfo Arroyo-Rabasa, John M. Ball, David Bate, Lisa Beck, Paolo Bonicatto, Filippo Cagnetti, Antonín Češík, Guido De Philippis, Giacomo Del Nin, Daniel Faraco, Gilles Francfort, Ed Gallagher, Luca Gennaioli, Francesco Ghiraldin, Richard Gratwick, André Guerra, Martin Jesenko, Kamil Kosiba, Konstantinos Koumatos, Raz Kupferman, Matthias Kurzke, Panu Lahti, Rajnath Laud, Marta Lewicka, Cy Maor, Stefan Müller, David Preiss, Harald Rindler, Angkana Rüland, Bernd Schmidt, Sebastian Schwarzacher, Hanuš Seiner, Giles Shaw, Parth Soneji, Maximilian Stegemeyer, Vladimir Švérak, Florian Theil, Jack Thomas, Ian Tobasco, Peter Topping, Julian Tremmel, Harry Turnbull, Günter von Häfen, Huidong Yang, as well as the participants of various incarnations of the MA4G6 module at the University of Warwick and all the senders of anonymous comments.

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January 2026 *Filip Rindler*

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Part I
Basic Course



Chapter 1
Introduction

In the quest to formulate useful mathematical models of aspects of the world, it turns out on surprisingly many occasions that the model becomes clearer, more compact, or more tractable if one introduces some form of *variational principle*. This means that one can find a quantity, such as an energy or entropy, which obeys a minimization, maximization or saddle-point law.

How much we perceive a variational quantity as "fundamental" or "artificial" depends on the situation at hand. For example, in classical mechanics, one calls forces *conservative* if they are path-independent and hence originate from changing an energy potential. It turns out that many forces in physics are indeed conservative, which seems to imply that the concept of energy should be considered fundamental. On the other hand, the entropy as a measure of *missing information* has a more artificial flavor at first sight – until one considers the *Second Law of Thermodynamics*, which surely must stand as one of the most fundamental laws of science.

But are we really to know for sure what is fundamental and what is artificial? The reader interested in these more philosophical aspects is directed to Wigner's very well-known essay "The Unreasonable Effectiveness of Mathematics in the Natural Sciences" [338] and the book "Mathematics and Optimal Form" by Hildebrandt–Tromba [181] as places to start.

As this is a mathematics textbook, our approach to variational quantities is a pragmatic one: We think of them as providing structure to a problem, which enables one to use powerful *variational methods*. For instance, in elasticity theory it is usually unrealistic to assume that a body will attain a shape that is *globally* energy-minimizing, but this does not mean that a minimum principle cannot be useful in aiding our understanding.

In this book we focus on minimization problems for integral functionals defined on maps from an open and bounded set $\Omega \subset \mathbb{R}^d$ and with values in $\mathbb{R}^m$ ($d, m \in \mathbb{N}$). Thus, we aim to minimize

$$\mathcal{F}[u] := \int_\Omega f(x, u(x), \nabla u(x)) \, \mathrm{d}x, \qquad u \colon \Omega \to \mathbb{R}^m,$$

usually under additional conditions on the boundary values of u and possibly further side constraints.

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From the 1950s onwards, the main research focus has been on variational principles in the *vectorial case* ($d, m > 1$), which exhibit many mathematical difficulties. In particular, it turned out that new forms of (generalized) convexity had to be introduced, most notably Charles B. Morrey Jr.'s *quasiconvexity* [239] and John M. Ball's *polyconvexity* [34]. Another strong driving force of the development in the Calculus of Variations during the second half of the 20th century was the *Italian School* founded by Ennio De Giorgi, which has produced many important discoveries, for instance in regularity theory, geometric problems, and variational convergence. Some further history of the Calculus of Variations can be found in [82, 152, 181].

We start by looking at a parade of examples, which we treat at varying levels of detail. The purpose of these examples is to place the mathematical theory in its applied context and to motivate the themes that have guided the development of the field. We will return to all of these examples once we have developed the necessary mathematical tools.

As some examples treat problems from other scientific disciplines, the reader is asked to take some statements on trust and to simply ignore the sections that are of no interest to them. No knowledge of these examples is required to understand the exposition of the mathematical theory starting in the next chapter.

1.1 The Brachistochrone Problem

In June 1696 Johann Bernoulli published the description of a mathematical problem in the journal *Acta Eruditorum*, see Figure 1.1. Bernoulli also sent a letter containing the problem to Leibniz on 9 June 1696, who returned his solution only a few days later on 16 June, and commented that the problem tempted him "like the apple tempted Eve". Newton also published a solution after the problem had reached him, without giving his identity, but Bernoulli identified him "ex ungue leonem" (from Latin, "by the lion's claw").

Problema novum ad cujus ſolutionem Mathematici invitantur.

Datis in plano verticali duobus punctis A & B (vid. Fig. 5) *aſſignare Mobili M, viam AMB, per quam gravitate ſua deſcendens & moveri incipiens a puncto A, brevissimo tempore perveniat ad alterum punctum B.*

Ut harum rerum amatores inſtigentur & propenſiori animo ferantur ad tentamen hujus problematis, ſciant non conſiſtere in nuda ſpeculatione, ut quidem videtur, ac ſi nullum haberet uſum; habet enim maximum etiam in aliis ſcientiis quam in mechanicis, quod nemo facile crediderit. Interim (ut forte quorundam præcipiti judicio obviam eam) quanquam recta AB ſit breviſſima inter terminos A & B, non tamen illa breviſſimo tempore percurritur; ſed eſt curva AMB Geometris notiſſima, quam ego nominabo, ſi elapſo hoc anno nemo alius eam nominaverit.

Fig. 1.1 The birth certificate of the Calculus of Variations [51] (source: *Hathi Trust Digital Library*)

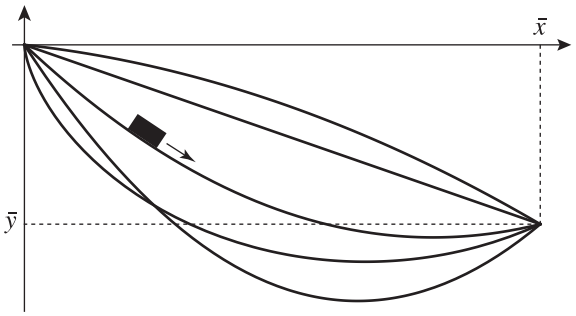


Fig. 1.2 Several slide curves from the origin to $(\bar{x}, \bar{y})$

The problem that the great minds of the time found so irresistible was formulated as follows:

> *Given two points A and B in a vertical* [meaning "not horizontal"] *plane, one shall find a curve AMB for a movable point M, on which it travels from the point A to the other point B in the shortest time, only driven by its own weight.*

The resulting curve is called the *brachistochrone* (from Ancient Greek, "shortest time") curve.

A more precise formulation of the brachistochrone problem is as follows: We look for the curve connecting the origin $(0, 0)$ to the point $(\bar{x}, \bar{y})$, where $\bar{x} > 0$, $\bar{y} < 0$, such that under the gravitational acceleration (in the negative y-direction) a point mass $m > 0$ slides from rest at $(0, 0)$ to $(\bar{x}, \bar{y})$ quickest among all such curves; see Figure 1.2 for an illustration. We parametrize a point (x, y) on the curve by the time $t \geq 0$ that the mass takes to reach it. The kinetic and potential energies of the sliding point mass are, respectively,

$$E_{\text{kin}} = \frac{m}{2}\left[\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2\right] = \frac{m}{2}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2\left[1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2\right],$$
$$E_{\text{pot}} = mgy,$$

where $g \approx 9.81\,\mathrm{m/s^2}$ is the gravitational acceleration on Earth. The total energy $E_{\text{kin}} + E_{\text{pot}}$ is zero at the beginning and conserved along the path. Hence,

$$\frac{m}{2}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2\left[1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2\right] = -mgy.$$

We can solve this for $\mathrm{d}t/\mathrm{d}x$ (denoting by $t = t(x)$ the inverse of the x-parametrization) to get

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \sqrt{\frac{1 + (y')^2}{-2gy}} \qquad \left(\frac{\mathrm{d}t}{\mathrm{d}x} \geq 0\right),$$

where we wrote $y' = \frac{\mathrm{d}y}{\mathrm{d}x}$. Integrating over the whole x-length along the curve from 0 to $\bar{x}$, we obtain for the total slide duration $T[y]$ the expression

$$T[y] = \frac{1}{\sqrt{2g}} \int_0^{\bar{x}} \sqrt{\frac{1 + (y'(x))^2}{-y(x)}}\, \mathrm{d}x.$$

We may drop the constant in front of the integral since it does not affect the minimizer (only the minimum value), and set $\bar{x} = 1$ as the prototypical case, to arrive at the problem

$$\begin{cases} \text{Minimize} & \mathcal{F}[y] := \displaystyle\int_0^1 \sqrt{\frac{1 + (y'(x))^2}{-y(x)}}\, \mathrm{d}x \\ \text{subject to} & y(0) = 0,\ y(1) = \bar{y} < 0. \end{cases}$$

Notice that the integrand is *convex* in $y'(x)$, which will be important for the solution theory. We will come back to this problem in Example 3.26.

1.2 The Isoperimetric Problem

This problem, which dates back to antiquity and is among the oldest questions in the Calculus of Variations, asks to enclose a given area with the shortest possible circumference. We pose it in the following graphical version: Given $\alpha, \beta > 0$, find a map $u\colon [0,1] \to \mathbb{R}$ such that

$$\mathcal{F}[u] := \int_0^1 \sqrt{1 + (u(s)')^2}\, \mathrm{d}s$$

is minimal among all such u with $u(0) = \alpha$, $u(1) = \beta$, and

$$\int_0^1 u(s)\, \mathrm{d}s = A,$$

where $A > 0$ is the prescribed area under the curve, see Figure 1.3. Note that $\mathcal{F}[u]$ is the length of the curve $\gamma(s) := (s, u(s))^\mathsf{T}$. We refer to [56] for more information on the history of this question and the research it inspired.

A special difficulty for this problem arises as follows: The integrand $f(a) := \sqrt{1 + a^2}$ behaves like $|a|$ for large values of $|a|$, so it seems possible that u' concentrates a non-vanishing mass into an arbitrarily small interval as one approaches a minimizer. Hence, a solution could have vertical pieces and it is not clear what kind of candidate functions we should allow in the minimization. We will address this question in Chapter 14, see in particular Example 14.17.

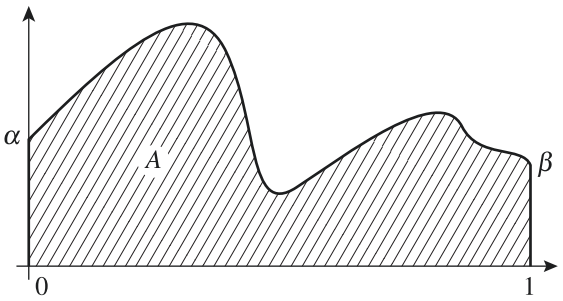


Fig. 1.3 A candidate curve for the isoperimetric problem

1.3 Electrostatics

Consider an electric charge density $\varrho\colon \mathbb{R}^3 \to \mathbb{R}$ (in units of $\mathrm{C/m^3}$) in a three-dimensional vacuum. Let $E\colon \mathbb{R}^3 \to \mathbb{R}^3$ (in $\mathrm{V/m}$) and $B\colon \mathbb{R}^3 \to \mathbb{R}^3$ (in $\mathrm{T} = \mathrm{Vs/m^2}$) be the electric and magnetic fields, respectively, which we assume to be constant in time (hence electro*statics*). The *Gauss law* for electricity reads

$$\nabla \cdot E = \operatorname{div} E = \frac{\varrho}{\varepsilon_0},$$

where $\varepsilon_0 \approx 8.854 \cdot 10^{-12}\,\mathrm{C/(Vm)}$ is the vacuum permittivity (electric constant). Moreover, we have the *Faraday law of induction*

$$\nabla \times E = \operatorname{curl} E = \frac{\mathrm{d}B}{\mathrm{d}t} = 0,$$

where t denotes time. Since E is curl-free, there exists an *electric potential* $\varphi\colon \mathbb{R}^3 \to \mathbb{R}$ (in V) such that

$$E = -\nabla\varphi.$$

Combining this with the Gauss law, we arrive at the *Poisson equation*,

$$\Delta\varphi = \nabla \cdot [\nabla\varphi] = -\frac{\varrho}{\varepsilon_0}. \tag{1.1}$$

We can also look at electrostatics in a variational way: With the norming condition $\varphi(0) = 0$, the electric potential energy $U_E(x; q)$ of a point charge q (in C) at the point $x \in \mathbb{R}^3$ in the electric field E is given by the path integral

$$U_E(x; q) = -\int_0^x qE \cdot \mathrm{d}s = -\int_0^1 qE(hx) \cdot x \,\mathrm{d}h = q\varphi(x),$$

which does not depend on the path chosen since E is a gradient. One can show that the total *electric energy* of our charge distribution ϱ in its own electric field, that is, the energy needed to assemble the charge distribution from infinitesimal point charges brought in from "far away", is given as

$$U_E := \frac{1}{2}\int_{\mathbb{R}^3} \varrho\varphi \,\mathrm{d}x = \frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} (\nabla\cdot E)\varphi \,\mathrm{d}x,$$

which has units of CV = J. Using the identity

$$(\nabla\cdot E)\varphi = \nabla\cdot(E\varphi) - E\cdot(\nabla\varphi),$$

the Gauss–Green theorem, and the natural assumption that φ vanishes at infinity, we get

$$\begin{aligned} U_E &= \frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} \nabla\cdot(E\varphi) - E\cdot(\nabla\varphi)\,\mathrm{d}x \\ &= -\frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} E\cdot(\nabla\varphi)\,\mathrm{d}x \\ &= \frac{\varepsilon_0}{2}\int_{\mathbb{R}^3} |\nabla\varphi|^2\,\mathrm{d}x. \end{aligned}$$

The integral

$$\int_\Omega \frac{1}{2}|\nabla\varphi|^2\,\mathrm{d}x$$

is called the *Dirichlet functional* or the *Dirichlet integral*.

In Example 3.5 we will see that the solutions φ of (1.1) are precisely the minimizers of the variational problem

$$\text{Minimize}\quad \varphi \mapsto U_E - \int_{\mathbb{R}^3} \varrho\varphi\,\mathrm{d}x = \int_{\mathbb{R}^3} \frac{\varepsilon_0}{2}|\nabla\varphi|^2 - \varrho\varphi\,\mathrm{d}x.$$

The second term can be interpreted as the interaction energy between the electric field and the charge density ϱ. The existence and regularity of solutions to this minimization problem will be established in Examples 2.8 and 3.16.

1.4 Stationary States in Quantum Mechanics

The non-relativistic evolution of a quantum mechanical system with N degrees of freedom in an electric field is described completely through its *wave function* $\Psi\colon \mathbb{R}^N\times\mathbb{R}\to\mathbb{C}$, which satisfies the *Schrödinger equation*

$$\mathrm{i}\hbar\frac{\mathrm{d}}{\mathrm{d}t}\Psi(x,t) = \left[\frac{-\hbar^2}{2\mu}\Delta + V(x,t)\right]\Psi(x,t), \qquad (x,t)\in\mathbb{R}^N\times[0,\infty),$$

where $\hbar\approx 1.05\cdot 10^{-34}$ Js is the reduced Planck constant, μ is the reduced mass (in kg), and $V = V(x,t)\in\mathbb{R}$ is the potential energy (in J). The operator $H := -(2\mu)^{-1}\hbar^2\Delta + V$ is called the *Hamiltonian* of the system.

The value of the wave function itself at a given point in spacetime has no obvious physical meaning, but according to the *Copenhagen Interpretation* of quantum mechanics, $x\mapsto|\Psi(x,t)|^2$ is the probability density of finding a particle at the point x in a

measurement at time t. In order for $|\Psi(\bullet, t)|^2$ to be a probability density, we need to impose the side constraint

$$\|\Psi(\bullet, t)\|_{L^2}^2 = \int_{\mathbb{R}^N} |\Psi(x,t)|^2 \, \mathrm{d}x = 1 \qquad \text{for all } t \in [0, \infty).$$

In particular, $|\Psi(x,t)|$ has to decay as $|x| \to \infty$.

Of special interest are the so-called *stationary states*, that is, solutions $\Psi = \Psi(x)$ of the *stationary Schrödinger equation*

$$\left[\frac{-\hbar^2}{2\mu}\Delta + V(x)\right]\Psi(x) = E\Psi(x), \qquad x \in \mathbb{R}^N,$$

where $E > 0$ is an *energy level*. If we are just interested in the lowest-energy state, the so-called *ground state*, we can instead find minimizers of the energy functional

$$\mathcal{E}[\Psi] := \int_{\mathbb{R}^N} \frac{\hbar^2}{4\mu}|\nabla\Psi(x)|^2 + \frac{1}{2}V(x)|\Psi(x)|^2 \, \mathrm{d}x,$$

again under the side constraint

$$\|\Psi\|_{L^2}^2 = 1.$$

The two parts of the integral above correspond to kinetic and potential energy, respectively. We will continue this investigation in Example 3.23.

1.5 Optimal Saving and Consumption

Consider a capitalist worker earning a (constant) wage w per year, which the worker can either spend on consumption or save. Denote by $S(t)$ the accumulated savings at time t, where $t \in [0, T]$ is in years, with $t = 0$ denoting the start of employment and $t = T$ retirement. Let $C(t) \geq 0$ be the consumption rate (consumption per time) at time t. On the saved capital, the worker earns interest, say with gross-continuous rate $\varrho > 0$, meaning that a capital amount $m > 0$ grows as $\mathrm{e}^{\varrho t} m$. If we were given an effective APR $\varrho_1 > 0$ instead of ϱ, then $\varrho = \ln(1 + \varrho_1)$. We further assume that the wages are paid continuously, not in intervals, for simplicity. So, w is really the rate of pay, given in money per time. Then, the worker's savings evolve according to the differential equation

$$\dot{S}(t) = w + \varrho S(t) - C(t). \tag{1.2}$$

We now make the (totally unreasonable!) assumption that the worker's happiness depends only on his consumption rate. Suppose that our worker wants to optimize total life happiness by finding the optimal amount of consumption at any given time. So, if we denote by $U(C)$ the *marginal utility function*, that is, the marginal "happiness" due to the consumption rate C, our worker wants to find $C\colon [0, T] \to \mathbb{R}$ such that

$$\mathcal{H}[C] := \int_0^T U(C(t))\,\mathrm{d}t$$

is maximized. The choice of U depends on our worker's personality, but it is sensible to assume that there is a *law of diminishing returns*, i.e., for twice as much consumption, our worker is happier, but not twice as happy. So, let us assume $U' > 0$ and $U'(C) \to 0$ as $C \to \infty$. Also, we should have $U(0) = 0$ (starvation). Moreover, it is realistic for U to be *concave*, which in particular implies that there are no local maxima. One function that satisfies all of these requirements is

$$U(C) = \ln(1 + C), \qquad C > 0.$$

Let us also assume that the worker starts with no savings, $S(0) = 0$, and wants to retire with savings $S(T) = S_T \geq 0$. Rearranging (1.2) for $C(t)$ and plugging this into the formula for $\mathcal{H}$, we arrive at the *optimal saving problem*

$$\begin{cases} \text{Minimize} & \mathcal{F}[S] := \displaystyle\int_0^T -\ln(1 + w + \varrho S(t) - \dot{S}(t))\,\mathrm{d}t \\ \text{subject to} & S(0) = 0,\ S(T) = S_T \geq 0,\ C(t) := w + \varrho S(t) - \dot{S}(t) \geq 0. \end{cases}$$

This will be solved in Example 3.8.

1.6 Sailing Against the Wind

Every sailor knows how to sail against the wind by "beating to windward": One has to sail at an angle of approximately 45° to the wind (in real boats, the maximum might be at a lower angle, i.e., "closer to the wind"), then *tack*, that is, turn the bow through the wind, and finally, after the sail has caught the wind on the other side, continue again at approximately 45° to the wind. Repeating this procedure makes the boat follow a zig-zag motion, which gives a net movement directly against the wind, see Figure 1.4. A mathematically inclined sailor might ask the question of "how often to tack". In an idealized model we can assume that the wind has the same speed and direction everywhere, tacking costs no time, and the forward sailing speed v_s of the boat depends on the angle α to the wind as follows (at least qualitatively):

$$v_s(\alpha) = v_w \frac{1 - \cos(4\alpha)}{2},$$

where v_w is the maximum speed of the boat at the current wind, which we assume to be constant. Then, $v_s(\alpha)$ is non-negative and has maxima at $\alpha = \pm 45°$.

Assume furthermore that our sailor is sailing along a straight river with the current. The current is fastest in the middle of the river and disappears at the banks. In fact, a good approximation for the flow speed is given by the formula of *Poiseuille (channel) flow*, which can be derived from the flow equations of fluids: At distance r from the center of the river the current's flow speed is approximately

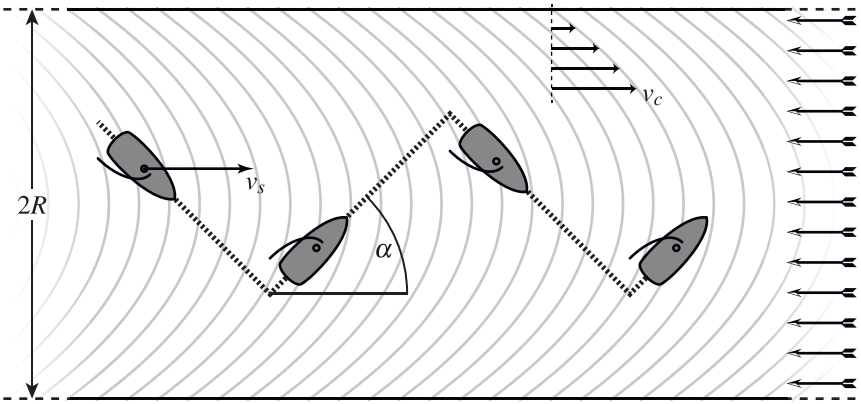


Fig. 1.4 Sailing against the wind in a channel

$$v_c(r) := v_f\left(1 - \frac{r^2}{R^2}\right),$$

where $R > 0$ is half the width of the river.

If we denote by $r(t)$ the distance of the boat from the middle of the channel at time $t \in [0, T]$, then the total speed (called the "velocity made good" in sailing parlance) is

$$\begin{aligned} v_s(t) &:= v_s(\arctan r'(t)) + v_c(r(t)) \\ &= v_w \frac{1 - \cos(4 \arctan r'(t))}{2} + v_f\left(1 - \frac{r(t)^2}{R^2}\right). \end{aligned}$$

The total forward distance traveled over the time interval $[0, T]$ is then

$$\int_0^T v_s(t)\, \mathrm{d}t = \int_0^T v_w \frac{1 - \cos(4 \arctan r'(t))}{2} + v_f\left(1 - \frac{r(t)^2}{R^2}\right) \mathrm{d}t.$$

If we also require the initial and terminal conditions $r(0) = r(T) = 0$, we arrive at the *optimal tacking problem*

$$\begin{cases} \text{Minimize} & \mathcal{F}[r] := \displaystyle\int_0^T v_w \frac{\cos(4 \arctan r'(t)) - 1}{2} + v_f\left(\frac{r(t)^2}{R^2} - 1\right) \mathrm{d}t \\ \text{subject to} & r(0) = r(T) = 0,\ |r(t)| \le R. \end{cases}$$

The key to understanding this problem is the observation that the function given by $a \mapsto (\cos(4 \arctan a) - 1)/2$ has precisely two minima, namely at $a = \pm 1$. We say that this function is a *double-well potential*, see Figure 1.5.

Our intuition tells us that in this idealized model, where tacking costs no time, we should be tacking "infinitely fast" in order to stay in the middle of the river. Later, once we have advanced tools at our disposal, we will make this idea precise, see Example 7.10.

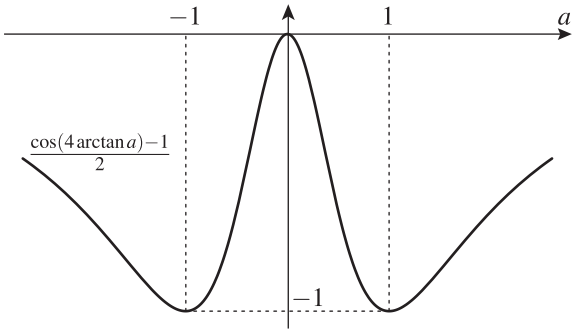


Fig. 1.5 The double-well potential in the sailing example

1.7 Hyperelasticity

Elasticity theory is one of the most important theories of *continuum mechanics*, that is, the study of the mechanics of (idealized) continuous media. We will not go into much detail about elasticity modeling here and refer to [78] for a thorough introduction.

Consider a body of mass occupying a bounded and connected domain $\Omega \subset \mathbb{R}^3$ such that $\partial\Omega$ has Lipschitz regularity (i.e., $\partial\Omega$ looks everywhere locally like a Lipschitz graph). We call Ω the *reference configuration*. If we deform the body, any *material point* $x \in \Omega$ is mapped to a *spatial point* $y(x) \in \mathbb{R}^3$ and we call $y(\Omega)$ the *deformed configuration*, see Figure 1.6. We also require that $y\colon \Omega \to y(\Omega)$ is a differentiable bijection and that it is *orientation-preserving*, i.e.,

$$\det \nabla y(x) > 0, \qquad x \in \Omega.$$

For convenience let us also introduce the *displacement*

$$u(x) := y(x) - x.$$

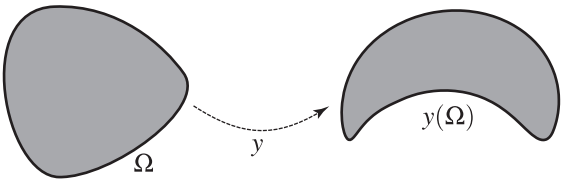


Fig. 1.6 A deformed body

Next, we need a measure of local "stretching", called a *strain tensor*, which should serve as the argument for a local energy density. On physical grounds, *rigid body motions*, that is, deformations of the form $y(x) = Rx + y_0$ with a rotation $R \in \mathbb{R}^{3\times 3}$ ($R^{\mathsf{T}} = R^{-1}$, $\det R = 1$) and $y_0 \in \mathbb{R}^3$, should not cause strain. In this sense, strain measures the deviation of the deformation from a rigid body motion. One common choice

is the *Green–St. Venant strain tensor*

$$G := \frac{1}{2}(\nabla y^{\mathsf{T}} \nabla y - \mathrm{Id}) = \frac{1}{2}\big(\nabla u + \nabla u^{\mathsf{T}} + \nabla u^{\mathsf{T}} \nabla u\big). \tag{1.3}$$

Observe that for a rigid motion $y(x) = Rx + y_0$ as above (with $R^{\mathsf{T}} = R^{-1}$, $\det R = 1$, $y_0 \in \mathbb{R}^3$), it holds that $G = 0$.

We first consider fully nonlinear ("finite strain") elasticity. For our purposes we simply postulate the existence of a stored-energy density $W\colon \mathbb{R}^{3\times 3} \to [0, \infty]$ and an external body force field $b\colon \Omega \to \mathbb{R}^3$ (e.g., gravity) such that

$$\mathcal{F}[y] := \int_\Omega W(\nabla y(x)) - b(x) \cdot y(x)\, \mathrm{d}x$$

represents the total energy stored in the system. If the elastic energy can be written in this way as

$$\int_\Omega W(\nabla y(x))\, \mathrm{d}x,$$

we call the material *hyperelastic*. In applications, W is sometimes given as depending on the Green–St. Venant strain tensor G instead of ∇y, but for the mathematical theory the above form is more convenient. We require several properties of W:

(i) *Norming:* $W(\mathrm{Id}) = 0$ (the undeformed state costs no energy).
(ii) *Frame-indifference:* $W(QA) = W(A)$ for all $Q \in \mathrm{SO}(3)$, $A \in \mathbb{R}^{3\times 3}$.
(iii) *Infinite compression costs infinite energy:* $W(A) \to +\infty$ as $\det A \downarrow 0$.
(iv) *Infinite stretching costs infinite energy:* $W(A) \to +\infty$ as $|A| \to \infty$.

The fundamental task of nonlinear hyperelasticity is to minimize $\mathcal{F}$ as above over all $y\colon \Omega \to \mathbb{R}^3$ with given boundary values. Of course, it is not a priori clear in which space we should look for a solution. Indeed, this depends on the growth properties of W. For example, with the prototypical choice

$$W(A) := \mathrm{dist}(A, \mathrm{SO}(3))^2, \qquad \text{where} \qquad \mathrm{dist}(A, K) := \inf_{B\in K} |A - B|,$$

we would look for square-integrable functions. Note, however, that this W does not satisfy (iii) from our list of requirements.

More realistic in applications are the *Mooney–Rivlin materials*, where W is of the form

$$W(A) := \begin{cases} a|A|^2 + b|\operatorname{cof} A|^2 + \Gamma(\det A) & \text{if } \det A > 0, \\ +\infty & \text{if } \det A \le 0, \end{cases}$$

with $a, b > 0$ and $\Gamma(d) = \alpha d^2 - \beta \log(d)$ for $\alpha, \beta > 0$. If $b = 0$, the material is called *neo-Hookean*. An even larger class is given by the *Ogden materials*, for which

$$W(A) := \sum_{i=1}^{M} a_i \operatorname{tr}\big[(A^{\mathsf{T}}A)^{\gamma_i/2}\big] + \sum_{j=1}^{N} b_j \operatorname{tr} \operatorname{cof}\big[(A^{\mathsf{T}}A)^{\delta_j/2}\big] + \Gamma(\det A),$$

where $M, N \in \mathbb{N}$, $a_i > 0$, $\gamma_i \geq 1$, $b_j > 0$, $\delta_j \geq 1$, and $\Gamma\colon \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ is a convex function with $\Gamma(d) \to +\infty$ as $d \downarrow 0$, $\Gamma(d) = +\infty$ for $d \leq 0$. These materials occur in a wide range of applications. We will consider such problems in Example 6.9.

1.8 Linearized Hyperelasticity

To pass from the setting of the preceding section to *linearized elasticity*, we make the "small strain" assumption that $\nabla u = \nabla y - \mathrm{Id}$ is "small", so that the quadratic term in (1.3) can be neglected. In this case, we work with the *linearized strain tensor*

$$\mathcal{E}u := \mathrm{sym}\nabla u := \frac{1}{2}\left(\nabla u + \nabla u^{\mathsf{T}}\right) \in \mathbb{R}^{3\times 3}_{\mathrm{sym}}.$$

Now, the displacements that do not create strain are precisely the skew-affine maps $u(x) = Zx + u_0$ with $Z^{\mathsf{T}} = -Z$ and $u_0 \in \mathbb{R}$. This becomes more meaningful if we consider a bit more algebra: The Lie group $\mathrm{SO}(3)$ of rotations has as its Lie algebra $\mathrm{Lie}(\mathrm{SO}(3)) = \mathfrak{so}(3)$, the space of all skew-symmetric matrices (see [176]), which then can be seen as "infinitesimal rotations".

For linearized elasticity we consider an energy of the special quadratic form

$$\mathcal{W}[u] := \int_\Omega \frac{1}{2}\mathcal{E}u(x) : \mathbf{C}(x)\,\mathcal{E}u(x)\,\mathrm{d}x,$$

where $\mathbf{C}(x) = \mathbf{C}^{ik}_{jl}(x)$ $(x \in \Omega)$ is a symmetric, positive definite fourth-order tensor, called the *elasticity tensor*. This $\mathbf{C}$ acts on $A \in \mathbb{R}^{d\times d}$ via

$$(\mathbf{C}(x)A)^i_j := \sum_{k,l} \mathbf{C}^{ik}_{jl}(x)A^k_l.$$

More precisely, the symmetry means that

$$A : \mathbf{C}(x)B = (\mathrm{sym}A) : \mathbf{C}(x)(\mathrm{sym}B) = (\mathrm{sym}B) : \mathbf{C}(x)(\mathrm{sym}A)$$

for all $A, B \in \mathbb{R}^{3\times 3}_{\mathrm{sym}}$. The positive-definiteness entails that $A : \mathbf{C}(x)A \geq c|A|^2$ for some $c > 0$ and all $x \in \Omega$, $A \in \mathbb{R}^{3\times 3}_{\mathrm{sym}}$.

For homogeneous, isotropic media, $\mathbf{C}$ does not depend on x or the direction of strain, which translates into the additional condition

$$(AQ) : \mathbf{C}(AQ) = A : \mathbf{C}A \qquad \text{for all } A \in \mathbb{R}^{3\times 3},\ Q \in \mathrm{SO}(3).$$

In this case, it can be shown that $\mathcal{W}$ simplifies to

$$\mathcal{W}[u] = \int_\Omega \mu|\mathcal{E}u(x)|^2 + \frac{1}{2}\left(\kappa - \frac{2}{3}\mu\right)|\operatorname{tr}\mathcal{E}u(x)|^2\,\mathrm{d}x$$

for $\mu > 0$ the *shear modulus* and $\kappa > 0$ the *bulk modulus*, which are material constants. For example, for cold-rolled steel $\mu \approx 75$ GPa and $\kappa \approx 160$ GPa. As in the nonlinear setting, we then consider the minimization problem for the total energy

$$\mathcal{F}[u] := \int_\Omega \mu|\mathcal{E}u(x)|^2 + \frac{1}{2}\Big(\kappa - \frac{2}{3}\mu\Big)|\operatorname{tr}\mathcal{E}u(x)|^2 - b(x)\cdot u(x)\,\mathrm{d}x,$$

where $b\colon \Omega \to \mathbb{R}^3$ is the external body force (now with respect to u). We will consider this functional further in Examples 2.12 and 3.17.

1.9 Microstructure in Crystals

In a single crystal of a metal like iron or an alloy like CuAlNi (Copper–Aluminium–Nickel), the atoms are arranged in a (mostly) regular lattice. Assume that such a material specimen occupies an open, bounded, and connected reference domain $\Omega \subset \mathbb{R}^3$ and that it is subject to external forces. We then want to determine the resulting deformed shape.

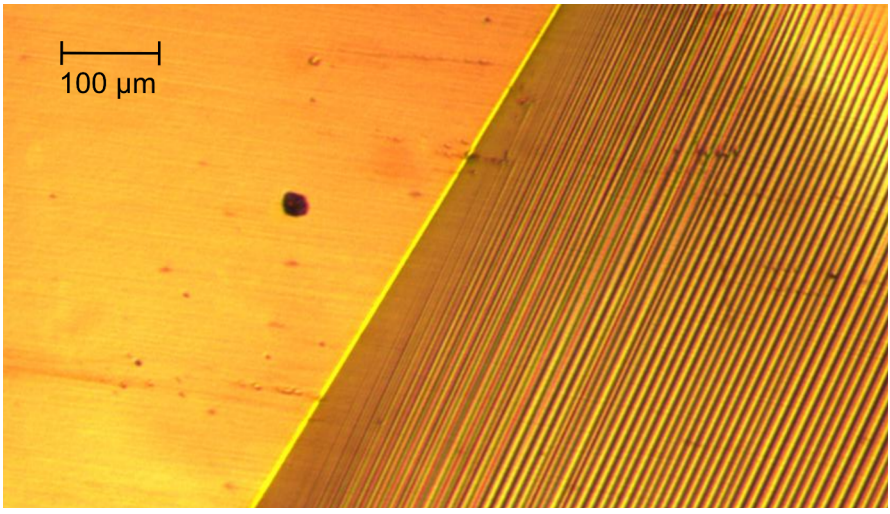


Fig. 1.7 CuAlNi microstructure undergoing a transition from cubic austenite (left) to orthorhombic martensite (right), see [297] for more details on this particular microstructure (source: *original micrograph by Hanuš Seiner, reproduced with kind permission*)

It turns out that on a microscopic scale the deformation of a single crystal sometimes exhibits very fine locally periodic oscillations, that is, the crystal exhibits *microstructure*, see Figure 1.7. This behavior has profound implications for the macroscopic behavior of the material.

The fundamental *Cauchy–Born Hypothesis* postulates that for small shear displacements the lattice atoms will follow this displacement (see [84, 128, 131, 150] for some results on when this assumption is justified). Assuming that the microstructure does not reach down to atomic length scales, we can then model the crystal as a *continuum* and assign the energy density $W(F) \geq 0$ to the linear deformation $x \mapsto Fx$. The crucial point here is that, thanks to the Cauchy–Born Hypothesis, W depends only on F and no other "microscopic structure" of the crystal, at least for small to moderate crystal deformations. In this approach, the total energy of a deformation $y\colon \Omega \to \mathbb{R}^3$ is given as

$$\mathcal{F}[y] := \int_\Omega W(\nabla y(x))\, \mathrm{d}x.$$

Here, on $W\colon \mathbb{R}^{3\times 3} \to [0,\infty)$ we make the following assumptions:

(i) *Norming:* $W(\mathrm{Id}) = 0$ (the undeformed state costs no energy).
(ii) *Frame-indifference:* $W(QA) = W(A)$ for all $Q \in \mathrm{SO}(3)$, $A \in \mathbb{R}^{3\times 3}$.
(iii) *Symmetry-invariance:* $W(AS) = W(S)$ for all $S \in \mathcal{S}$ and all $A \in \mathbb{R}^{3\times 3}$, where $\mathcal{S} \subset \mathrm{SO}(3)$ is the compact *point group* (symmetry group) of the crystal.

The basic variational postulate is that the observed macroscopic deformation is a minimizer of $\mathcal{F}$ under the given boundary conditions. In fact, it is often experimentally observed that the deformation $y\colon \Omega \to \mathbb{R}^3$ is close to a *pointwise* minimizer of the integrand, at least in a very large portion of Ω. Thus, we are led to consider the *differential inclusion*

$$\nabla y(x) \in K, \qquad x \in \Omega,$$

where

$$K := W^{-1}(0) = \left\{ A \in \mathbb{R}^{3\times 3} \,:\, W(A) = \min W \right\}.$$

The set K is compact in the study of crystals, but other applications also lead to differential inclusions with non-compact K.

In concrete applications, one usually has the following (idealized) situation: Above a critical temperature, K is $\mathrm{SO}(3)$, which is the simplest possible set that is compatible with the frame-indifference (ii). This is called the *austenite phase*. Below the critical temperature, however, the material undergoes a solid–solid phase transition to the *martensite phase*, where K is the union of several *wells*, that is,

$$K = \mathrm{SO}(3)U_1 \cup \cdots \cup \mathrm{SO}(3)U_N$$

for distinct matrices $U_1, \ldots, U_N \in \mathbb{R}^{3\times 3}$ with $\det U_i > 0$ ($i = 1, \ldots, N$). By the polar decomposition of matrices with positive determinants into a product of a rotation and a symmetric positive definite matrix, we can assume that all the U_i are symmetric and positive definite.

If $N \geq 2$ and other *compatibility conditions* between the matrices U_i are satisfied, microstructure can indeed be observed. It should be noted that while our model, as formulated above, may imply "infinitely fast" oscillations in the microstructure, in reality other (atomistic) effects limit the length scales that are observed.

As a concrete example, the NiAl (Nickel–Aluminium) alloy undergoes a *cubic-to-tetragonal* phase transition and below the critical temperature we have

$$K = \mathrm{SO}(3)U_1 \cup \mathrm{SO}(3)U_2 \cup \mathrm{SO}(3)U_3$$

with

$$U_1 = \begin{pmatrix} \beta & & \\ & \alpha & \\ & & \alpha \end{pmatrix}, \qquad U_2 = \begin{pmatrix} \alpha & & \\ & \beta & \\ & & \alpha \end{pmatrix}, \qquad U_3 = \begin{pmatrix} \alpha & & \\ & \alpha & \\ & & \beta \end{pmatrix},$$

for $\alpha \approx 0.9392$, $\beta \approx 1.1302$, see [52, 130].

As another example, the CuAlNi alloy undergoes a *cubic-to-orthorhombic* phase transition and below the critical temperature we have

$$K = \mathrm{SO}(3)U_1 \cup \cdots \cup \mathrm{SO}(3)U_6$$

with

$$U_1 = \begin{pmatrix} \xi & 0 & \eta \\ 0 & \beta & 0 \\ \eta & 0 & \xi \end{pmatrix}, \qquad U_2 = \begin{pmatrix} \xi & 0 & -\eta \\ 0 & \beta & 0 \\ -\eta & 0 & \xi \end{pmatrix}, \qquad U_3 = \begin{pmatrix} \xi & \eta & 0 \\ \eta & \xi & 0 \\ 0 & 0 & \beta \end{pmatrix},$$

$$U_4 = \begin{pmatrix} \xi & -\eta & 0 \\ -\eta & \xi & 0 \\ 0 & 0 & \beta \end{pmatrix}, \qquad U_5 = \begin{pmatrix} \beta & 0 & 0 \\ 0 & \xi & \eta \\ 0 & \eta & \xi \end{pmatrix}, \qquad U_6 = \begin{pmatrix} \beta & 0 & 0 \\ 0 & \xi & -\eta \\ 0 & -\eta & \xi \end{pmatrix},$$

where

$$\xi = \frac{\alpha + \gamma}{2}, \qquad \eta = \frac{\alpha - \gamma}{2}$$

for $\alpha \approx 1.0619$, $\beta \approx 0.9178$, $\gamma \approx 1.0230$, see [52, 127].

A first mathematical question that can be asked about such microstructures concerns their effective representation: What are the salient features of the oscillations in the material and how can they be captured mathematically? Moreover, which deformations with linear boundary values $x \mapsto Fx$ have almost zero energy? It turns out that by relying on very high-frequency oscillations, the set of these F can actually be much larger than K and defines a certain "hull" of K. This hull explains the observed microstructure, as we will see in Chapters 11 and 12.

In engineering applications, one striking property of NiAl and CuAlNi is the *shape-memory effect*, where a material specimen "remembers" the shape it had when it was hotter than the critical temperature. After cooling, the specimen can be freely deformed, but when it is again heated above the critical temperature, it "snaps back" into its original shape. This effect is directly related to the formation of microstructure (below the critical temperature), which accommodates the deformations through microstructure changes, but without changing the structure of the crystal lattice bonds. Upon heating the specimen above the critical temperature, all microstructure disappears and the original shape (which is determined by the crystal lattice in the austenite phase) reappears. Note that all the matrices $U_1, U_2, \ldots$ for both NiAl and CuAlNi have determinant very

close to 1, which is a common feature of shape-memory alloys, because it is necessary for the *self-accommodation* effect, where upon cooling through the critical temperature the microstructure arranges itself in a such way that the macroscopic shape does not change. See [52] for a detailed study of the shape-memory effect.

In Chapters 10–12 we will consider the basic principles underlying the phenomenon of microstructure, see in particular Examples 10.10, 12.12. Concrete applications are left to more specialized treatises like [52, 123].

1.10 Composite Elastic Materials

Assume we are given a linearly elastic material specimen (like in Section 1.8) occupying the (bounded, open, connected) reference domain $\Omega \subset \mathbb{R}^3$, whose stored energy for a displacement $u\colon \Omega \to \mathbb{R}^3$ is

$$\mathcal{W}[u] := \int_\Omega \frac{1}{2}\mathcal{E}u(x) : \mathbf{C}(x)\, \mathcal{E}u(x)\, \mathrm{d}x.$$

As in Section 1.8, $\mathbf{C}(x) = \mathbf{C}^{ik}_{jl}(x)$ $(x \in \Omega)$ is a symmetric, positive definite fourth-order elasticity tensor.

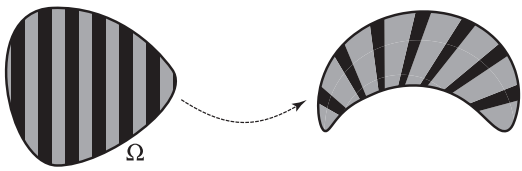


Fig. 1.8 A deformed composite material body

We suppose in addition that $\mathbf{C}$ depends on x in an ε-periodic manner for a small $\varepsilon > 0$. For instance, we could imagine our specimen to be a *composite* consisting of thin alternating material layers of two different types, see Figure 1.8. We denote the elasticity tensors of these layers by $\mathbf{C}_1$ and $\mathbf{C}_2$, respectively, and assume that the layers alternate in the first coordinate direction with thicknesses $\theta\varepsilon$ and $(1-\theta)\varepsilon$, respectively, where $\theta \in (0,1)$. Then, the elastic energy has the form

$$\mathcal{W}_\varepsilon[u] := \int_\Omega \frac{1}{2}\mathcal{E}u(x) : \mathbf{C}\Big(\frac{x}{\varepsilon}\Big)\, \mathcal{E}u(x)\, \mathrm{d}x,$$

where

$$\mathbf{C}(x) = \mathbf{C}_1 + (\mathbf{C}_2 - \mathbf{C}_1)h(x_1), \qquad x \in \mathbb{R}^3,$$

and

$$h(t) := \begin{cases} 0 & \text{if } t - \lfloor t \rfloor \le \theta, \\ 1 & \text{if } t - \lfloor t \rfloor > \theta. \end{cases}$$

Here, $\lfloor t \rfloor$ denotes the largest integer less than or equal to $t \in \mathbb{R}$. The total energy to be minimized is

$$\begin{aligned}\mathcal{F}_\varepsilon[u] &:= \int_\Omega f_\varepsilon\Big(\frac{x}{\varepsilon}, \nabla u(x)\Big) - b(x) \cdot u(x)\, \mathrm{d}x \\ &:= \int_\Omega \frac{1}{2} \mathcal{E}u(x) : \mathbf{C}\Big(\frac{x}{\varepsilon}\Big)\, \mathcal{E}u(x) - b(x) \cdot u(x)\, \mathrm{d}x,\end{aligned}$$

where $b\colon \Omega \to \mathbb{R}^3$ is the external body force.

In many applications, one is predominantly interested in the *homogenized* behavior of the specimen, that is, its large-scale, averaged properties. Mathematically, this corresponds to a form of "variational limit" of the $\mathcal{F}_\varepsilon$, which entails the convergence of minimizers and minimum values. Ideally, we want to compute a homogenized density $f_{\mathrm{hom}}\colon \mathbb{R}^{3\times 3} \to \mathbb{R}$ (not x-dependent), such that $\mathcal{F}_\varepsilon$ "converges" to a limit $\mathcal{F}_0$ of the form

$$\mathcal{F}_0[u] = \int_\Omega f_{\mathrm{hom}}(\nabla u(x)) - b(x) \cdot u(x)\, \mathrm{d}x.$$

The following questions are of importance:

- Does an $\mathcal{F}_0$ as above exist and can it be written as an *integral* functional?
- Does f_{hom} (if it exists) have the same quadratic structure as the f_ε, that is, is there a symmetric, positive definite fourth-order tensor $\mathbf{C}_{\mathrm{hom}}$ such that $f_{\mathrm{hom}}(A) = \frac{1}{2} A^{\mathrm{sym}} : \mathbf{C}_{\mathrm{hom}} A^{\mathrm{sym}}$ (here, A^{sym} is the symmetric part of A)?
- In the special case when $\mathbf{C}_1 = \alpha\, \mathbf{I}$, $\mathbf{C}_2 = \beta\, \mathbf{I}$ for $\alpha, \beta > 0$ (here, $\mathbf{I}$ denotes the tensor such that $A : \mathbf{I}B = A : B$, so $\mathbf{I}^{ik}_{jl} = \delta_{ik}\delta_{jl}$), is $\mathbf{C}_{\mathrm{hom}}$ (if it exists) also of the form $\mathbf{C}_{\mathrm{hom}} = \gamma\, \mathbf{I}$ for some $\gamma > 0$?

We will investigate these questions in Example 9.19.

1.11 Phase Transitions

Consider a (bounded, open, connected) container $\Omega \subset \mathbb{R}^d$ ($d \in \{2, 3\}$ are the physically interesting cases) containing two mixed fluids. We let the *phase field* $\varrho\colon \Omega \to [0, 1]$ model the density of the first fluid and prescribe the relative amounts of the two fluids by requiring that

$$\int_\Omega \varrho(x)\, \mathrm{d}x = \widetilde{\gamma} \in (0, |\Omega|); \tag{1.4}$$

see Figure 1.9. The *Gibbs free energy* of the mixture is given as

$$\mathcal{G}[\varrho] := \int_\Omega W_0(\varrho(x))\, \mathrm{d}x, \qquad \varrho\colon \Omega \to [0, 1],$$

where $W_0\colon \mathbb{R} \to [0, \infty)$. Often, the energy density W_0 has precisely two minima $\alpha, \beta \in [0, 1]$ with $\alpha < \beta$ and thus W_0 is a *double-well potential*. In the simplest case, and we only consider this case in the following, the fluids do not mix well (e.g., water and oil),

and the two local minima of W_0 are located at $\alpha = 0$ (all oil) and $\beta = 1$ (all water). Let us furthermore assume that $W_0(0) = W_0(1) = 0$.

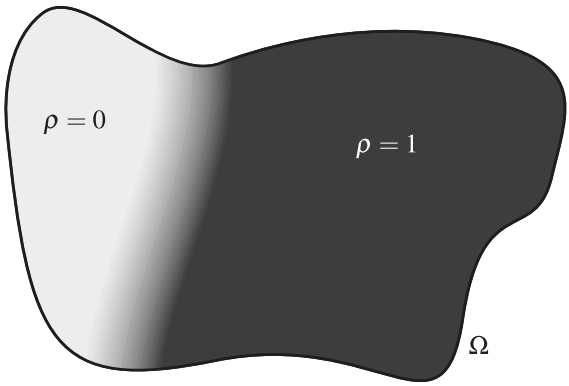


Fig. 1.9 A phase transition

It is a classical problem, first considered by Cahn–Hilliard and Gurtin (see [65, 174, 175]), to determine the equilibrium mixture, i.e., to find a minimizer of $\mathcal{G}$. In order for the problem to be interesting, we assume that both fluids occur, that is,

$$0 < \widetilde{\gamma} < |\Omega|.$$

However, in the above form this problem is not mathematically well-posed: We can just choose any (measurable) function $\varrho\colon \Omega \to [0,1]$ with $\varrho(x) \in \{0,1\}$ for all $x \in \Omega$ such that (1.4) holds. This will be a minimizer of $\mathcal{F}_0$, but this formulation is unsatisfactory: The shape of the two *phases*

$$E_0 := \big\{ x \in \Omega \,:\, \varrho(x) = 0 \big\}, \qquad E_1 := \big\{ x \in \Omega \,:\, \varrho(x) = 1 \big\},$$

is clearly not uniquely determined and *no regularity* can be assumed on the *phase boundary* $\partial E_0 \cap \Omega = \partial E_1 \cap \Omega$. The remedy to this problem comes from physics in the form of the additional assumption that the interface between E_0 and E_1 should have minimal surface area among all competitors. We can incorporate this minimum principle in two different ways.

First, we can penalize changes in the function ϱ by adding a (quadratic) gradient term and set

$$\widetilde{\mathcal{F}}_\varepsilon[\varrho] := \int_\Omega W_0(\varrho(x)) + \varepsilon^2 |\nabla \varrho(x)|^2 \,\mathrm{d}x, \qquad \varrho\colon \Omega \to [0,1],$$

where $\varepsilon > 0$ is a (small) parameter. For purely mathematical reasons it turns out to be beneficial to transform this functional into

$$\mathcal{F}_\varepsilon[u] := \int_\Omega \frac{1}{\varepsilon} W(u(x)) + \varepsilon |\nabla u(x)|^2 \,\mathrm{d}x, \qquad u\colon \Omega \to [-1,1],$$

where $W\colon \mathbb{R} \to [0,\infty)$ is given as (see, for instance, Figure 7.1 for an illustration of such a double-well potential)

$$W(s) := W_0\left(\frac{s+1}{2}\right),$$

so that $-1, +1$ are its global minima and $W(\pm 1) = 0$. This functional is often called the *Allen–Cahn functional*, particularly so when W has the classical form $W(s) = (1-s^2)^2/4$.

From (1.4) it can be verified easily that each ϱ minimizing $\widetilde{\mathcal{F}}_\varepsilon$ corresponds to u minimizing $\mathcal{F}_\varepsilon$ via the transformation

$$\varrho(x) = \frac{u(x)+1}{2}.$$

Indeed, one can compute that for such pairs (u,ϱ),

$$\widetilde{\mathcal{F}}_\varepsilon[\varrho] = \frac{\varepsilon}{2}\mathcal{F}_{\varepsilon/2}[u]$$

and

$$\int_\Omega u(x)\,\mathrm{d}x = \gamma := 2\widetilde{\gamma} - |\Omega| \in (-|\Omega|, |\Omega|).$$

We remark that the balancing of the ε-terms turns out to be necessary if we want to consider the limit as $\varepsilon \downarrow 0$. Notice also that a minimizer of $\mathcal{F}_\varepsilon$ will have a square-integrable gradient and so, besides the *pure phases*

$$E_{\pm 1} := \big\{ x \in \Omega \,:\, u(x) = \pm 1 \big\},$$

there will also be a non-empty *transition region*

$$\Delta := \big\{ x \in \Omega \,:\, u(x) \in (-1,1) \big\}.$$

Intuitively, Δ will shrink to a phase interface surface as $\varepsilon \downarrow 0$ since the regularizing effect of the gradient term in $\mathcal{F}_\varepsilon$ gets weaker as $\varepsilon \downarrow 0$.

An alternative way to model the physical situation is to prescribe that $u\colon \Omega \to \{-1,1\}$ splits the domain into the two pure phases $E_{\pm 1} = E_{\pm 1}(u)$ and the transition region is empty. In this case, we could consider those u minimizing the surface tension between the phases,

$$\mathcal{F}_0[u] := \sigma \mathrm{Per}_\Omega(E_{-1}(u)),$$

to be the physically relevant solutions. Here, the *perimeter* $\mathrm{Per}_\Omega(E_{-1}(u))$ should be understood as the surface area of $\partial E_{-1}(u) \cap \Omega$, at least if $\partial E_{-1}(u) \cap \Omega$ is a smooth manifold (with boundary). In more general situations the definition of this quantity will have to be suitably extended. The constant $\sigma > 0$ takes the role of a *surface tension density*. Notice that for $\mathcal{F}_0$ to be finite, the set $\partial E_{-1}(u) \cap \Omega$ has to have some regularity.

An important question about the above functionals is the following: As $\varepsilon \downarrow 0$, does $\mathcal{F}_\varepsilon$ "converge" to $\mathcal{F}_0$ in a sense that entails the convergence of minimizers and minimum values (for a suitably chosen σ)? We will return to this question in Example 14.24.



Chapter 2
Convexity

In this chapter we start to develop the mathematical theory that will allow us to analyze the prototypical questions presented in the introduction, and many more. The basic minimization problem that we are considering is the following:

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega f(x, u(x), \nabla u(x))\, \mathrm{d}x \\ \text{over all} & u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g. \end{cases}$$

Here, and throughout the text if not stated otherwise, we will make the standard assumption that $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, that is, Ω is open, bounded, connected, and $\partial\Omega$ is locally a Lipschitz graph. The function

$$f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to [0, \infty)$$

is required to be measurable in the first and (jointly) continuous in the second and third arguments, which makes f a so-called *Carathéodory integrand*. Furthermore, in this chapter we (usually) let $p \in (1, \infty)$ and for the prescribed boundary values g we assume

$$g \in \mathrm{W}^{1-1/p,p}(\partial\Omega; \mathbb{R}^m).$$

In this context recall that $\mathrm{W}^{1-1/p,p}(\partial\Omega; \mathbb{R}^m)$ is the space of traces of Sobolev maps in $\mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$; see Appendix A.5 for some background on Sobolev spaces.

The pivotal *Direct Method* of the Calculus of Variations will allow us to prove the existence of minimizers for a large variety of integral functionals. For conceptual clarity, we will first introduce it in an abstract setting, namely for (nonlinear) functionals on Banach spaces. Then we will begin our study of integral functionals, where we will in particular take a close look at the way in which *convexity* properties of f in its last (gradient) argument determine whether $\mathcal{F}$ is *lower semicontinuous* with respect to the weak convergence. Finally, we explain basic aspects of general convex analysis, in particular the Legendre–Fenchel duality.

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F. Rindler, *Calculus of Variations*, Graduate Texts in Mathematics 310,
https://doi.org/10.1007/978-3-032-24483-3_2

2.1 The Direct Method

Fundamental to all of the existence theorems in this book is the conceptually simple, yet powerful, *Direct Method* of the Calculus of Variations. It is called "direct" since we prove the existence of solutions to minimization problems without the detour through a differential equation.

Let X be a complete metric space, for example a Banach space with the norm topology or a closed, convex, and bounded subset of a reflexive Banach space with the weak topology. Denote by

$$\mathcal{F}\colon X \to \mathbb{R}$$

our *objective functional*, which we require to satisfy the following two assumptions:

(H1) **Coercivity:** For all $\Lambda \in \mathbb{R}$, the sublevel set

$$\big\{ u \in X \,:\, \mathcal{F}[u] \le \Lambda \big\} \qquad \text{is sequentially precompact,}$$

that is, if $\mathcal{F}[u_j] \le \Lambda$ for a sequence $(u_j)_j \subset X$ and some $\Lambda \in \mathbb{R}$, then $(u_j)_j$ has a converging subsequence in X.

(H2) **Lower semicontinuity:** For all sequences $(u_j)_j \subset X$ with $u_j \to u$ in X it holds that

$$\mathcal{F}[u] \le \liminf_{j\to\infty} \mathcal{F}[u_j].$$

Note that we use the *sequential* notions of compactness and lower semicontinuity, which are better suited to our needs than the corresponding topological concepts. For more on this point see the notes section at the end of this chapter.

The Direct Method for the abstract problem

$$\text{Minimize } \mathcal{F}[u] \text{ over all } u \in X \tag{2.1}$$

is encapsulated in the following simple result.

Theorem 2.1 *Assume that $\mathcal{F}$ is both coercive and lower semicontinuous. Then, the abstract minimization problem (2.1) has a solution, that is, there exists a $u_* \in X$ with $\mathcal{F}[u_*] = \min\{\, \mathcal{F}[u] : u \in X \,\}$.*

Proof. Let us assume that there exists at least one $u \in X$ such that $\mathcal{F}[u] < +\infty$; otherwise, any $u \in X$ is a "solution" to the (degenerate) minimization problem.

To construct a minimizer we take a **minimizing sequence** $(u_j)_j \subset X$ such that

$$\lim_{j\to\infty} \mathcal{F}[u_j] \to \alpha := \inf\big\{ \mathcal{F}[u] \,:\, u \in X \big\} < +\infty.$$

Then, there exists a $\Lambda \in \mathbb{R}$ such that $\mathcal{F}[u_j] \le \Lambda$ for all $j \in \mathbb{N}$. Hence, by the coercivity, we may select a subsequence, which we do not make explicit in our notation, such that

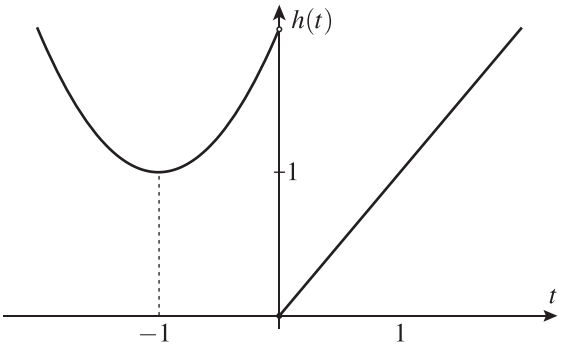
$$u_j \to u_* \in X.$$

By the lower semicontinuity we immediately conclude that

$$\alpha \le \mathcal{F}[u_*] \le \liminf_{j\to\infty} \mathcal{F}[u_j] = \alpha.$$

Thus, $\mathcal{F}[u_*] = \alpha$ and u_* is the sought minimizer. □

Fig. 2.1 The function h



Example 2.2 Using the Direct Method, one can easily see that the lower semicontinuous function

$$h(t) := \begin{cases} t^2 + 2t + 2 & \text{if } t < 0, \\ t & \text{if } t \ge 0, \end{cases}$$

see Figure 2.1, has the minimizer $t_* = 0$. Note that for this function the "Indirect Method", that is, to look for the minimizer in the set of critical points of h, i.e., those $t \in \mathbb{R}$ with $h'(t) = 0$, fails: The only point $t \in \mathbb{R}$ for which $h'(t)$ exist and $h'(t) = 0$ is $t_c = -1$ (where even $h''(t_c) > 0$), but this t_c is not the minimizer.

Despite its nearly trivial proof, the Direct Method is very useful and flexible in applications. Indeed, it splits the difficulty in proving the existence of a minimizer into establishing coercivity and lower semicontinuity. This is a big advantage since we have many tools at our disposal to establish these two hypotheses separately. In particular, for integral functionals, lower semicontinuity is tightly linked to *convexity* properties of the integrand, as we will see shortly.

At this point it is crucial to observe how coercivity and lower semicontinuity interact with the topology on X: If we choose a stronger topology, i.e., one for which there are fewer converging sequences, then it is easier for $\mathcal{F}$ to be lower semicontinuous, but harder for $\mathcal{F}$ to be coercive. The opposite holds if we choose a weaker topology. In the mathematical treatment of a problem from applications, we are most likely in a situation where $\mathcal{F}$ and the set X are given. We then need to find a suitable topology in which we can establish both coercivity and lower semicontinuity.

In this book, X will always be an infinite-dimensional Banach space (or a subset thereof) and we have a real choice between using the strong or the weak convergence. Usually, it turns out that coercivity with respect to the strong convergence is *false* since

strongly compact sets in infinite-dimensional spaces are very restricted, whereas coercivity with respect to the weak convergence is true under reasonable assumptions. On the other hand, while strong lower semicontinuity poses few challenges, lower semicontinuity with respect to weakly converging sequences is a more delicate matter and we will spend considerable time on this topic.

As a result of this discussion, we will almost always use the Direct Method in the following version:

Theorem 2.3 *Let X be a reflexive Banach space or a closed affine subset of a reflexive Banach space and let $\mathcal{F}\colon X \to \mathbb{R}$. Assume the following:*

(WH1) ***Weak coercivity:*** *For all $\Lambda \in \mathbb{R}$ the sublevel set*

$$\big\{ u \in X \,:\, \mathcal{F}[u] \le \Lambda \big\} \qquad \text{is sequentially weakly precompact,}$$

that is, if $\mathcal{F}[u_j] \le \Lambda$ for a sequence $(u_j)_j \subset X$ and some $\Lambda \in \mathbb{R}$, then $(u_j)_j$ has a weakly converging subsequence.

(WH2) ***Weak lower semicontinuity:*** *For all sequences $(u_j)_j \subset X$ with $u_j \rightharpoonup u$ in X (weak convergence) it holds that*

$$\mathcal{F}[u] \le \liminf_{j\to\infty} \mathcal{F}[u_j].$$

Then, the problem

$$\text{Minimize } \mathcal{F}[u] \text{ over all } u \in X$$

has a solution.

The proof of this theorem is analogous to the proof of Theorem 2.1, taking into account the fact that all (strongly) closed affine subsets of a Banach space are weakly closed.

2.2 Functionals with Convex Integrands

As a first instance of the theory of integral functionals to be developed in this book, we now consider the minimization problem for

$$\mathcal{F}[u] := \int_\Omega f(x, \nabla u(x))\, \mathrm{d}x$$

over all $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain and $p \in (1,\infty)$ will be chosen later (depending on growth properties of f). The reader is referred to Appendix A.5 for an overview of Sobolev spaces.

The following lemma gives a criterion for the above integrand to be measurable.

Lemma 2.4 *Let $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$ be a* ***Carathéodory integrand****, that is,*

(i) $x \mapsto f(x,A)$ is Lebesgue-measurable for every fixed $A \in \mathbb{R}^N$;
(ii) $A \mapsto f(x,A)$ is continuous for almost every fixed $x \in \Omega$.

Then, for any Borel-measurable map $V\colon \Omega \to \mathbb{R}^N$ the composition $x \mapsto f(x,V(x))$ is Lebesgue-measurable.

Note that it is common terminology to call both f and $x \mapsto f(x,V(x))$ an "integrand".

Proof. Assume first that V is a simple function,

$$V = \sum_{k=1}^{m} v_k \mathbb{1}_{E_k},$$

where the sets $E_k \subset \Omega$ are Borel-measurable with $\bigcup_{k=1}^m E_k = \Omega$, and $v_k \in \mathbb{R}^N$ ($k = 1,\dots,m$). For $t \in \mathbb{R}$ we have

$$\big\{ x \in \Omega \,:\, f(x,V(x)) > t \big\} = \bigcup_{k=1}^{m} \big\{ x \in E_k \,:\, f(x,v_k) > t \big\},$$

which is a Lebesgue-measurable set by assumption. Hence, the map $x \mapsto f(x,V(x))$ is Lebesgue-measurable.

Turning to the general case, every Borel-measurable function V can be approximated by simple functions V_k with $V_k(x) \to V(x)$ for all $x \in \Omega$ as $k \to \infty$, see Lemma A.6. By the assumed continuity,

$$f(x,V_k(x)) \to f(x,V(x)), \qquad x \in \Omega.$$

We conclude that the right-hand side is Lebesgue-measurable as the pointwise limit of Lebesgue-measurable functions. □

As a consequence, $\mathcal{F}[u]$ is well-defined if f is Carathéodory and if either $f \geq 0$ or if one imposes the **p-growth bound**

$$|f(x,A)| \leq M(1+|A|^p), \qquad (x,A) \in \Omega\times\mathbb{R}^{m\times d},$$

for a $p \in [1,\infty)$ and some $M > 0$. The latter condition also implies the finiteness of $\mathcal{F}[u]$ for all $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.

We next investigate the coercivity of $\mathcal{F}$. If $p \in (1,\infty)$, then the most basic assumption to guarantee coercivity is the **p-coercivity bound**

$$\mu|A|^p - \mu^{-1} \leq f(x,A), \qquad (x,A) \in \Omega\times\mathbb{R}^{m\times d}, \tag{2.2}$$

for some $\mu > 0$. This coercivity bound also determines the exponent p for the Sobolev space where we look for solutions.

Let us remark that in the following we will generically assume that $f \geq 0$ since a uniform lower bound for the integrand is sometimes required. Note that under a p-coercivity bound this positivity is not restrictive since one may always shift f up by μ^{-1} to make it positive, which does not change the minimization problem.

Proposition 2.5 *If the Carathéodory integrand $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$ satisfies the p-coercivity bound (2.2) with $p \in (1,\infty)$, then $\mathcal{F}$ is weakly coercive on the space*

$$\mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m) = \big\{\, u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \;:\; u|_{\partial\Omega} = g \,\big\},$$

where $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

Proof. We need to show that any sequence $(u_j)_j \subset \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ with

$$\sup_{j\in\mathbb{N}} \mathcal{F}[u_j] < \infty$$

is weakly precompact. From (2.2) we get

$$\mu \cdot \sup_{j\in\mathbb{N}} \int_\Omega |\nabla u_j|^p \,\mathrm{d}x - \mu^{-1}|\Omega| \leq \sup_{j\in\mathbb{N}} \mathcal{F}[u_j] < \infty,$$

whereby $\sup_j \|\nabla u_j\|_{\mathrm{L}^p} < \infty$. From the Poincaré inequality, see the version in Theorem A.27 (iii), we therefore get

$$\sup_j \|u_j\|_{\mathrm{W}^{1,p}} \leq C\big(\sup_j \|\nabla u_j\|_{\mathrm{L}^p} + 1\big) < \infty.$$

This finishes the proof since bounded sets in separable and reflexive Banach spaces, like $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ for $p \in (1,\infty)$, are sequentially weakly precompact by Theorem A.1. □

Having settled the question of weak coercivity, we can now investigate the weak lower semicontinuity. The following pivotal result goes back to the work of Leonida Tonelli in the one-dimensional case; the generalization to higher dimensions is due to James Serrin.

Theorem 2.6 (Tonelli 1920, Serrin 1961 [298, 335]) *Let $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand such that*

$$f(x,\cdot) \text{ is convex for almost every } x \in \Omega.$$

Then, $\mathcal{F}$ is weakly lower semicontinuous on $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ for any $p \in (1,\infty)$.

Proof. Step 1. We first establish that $\mathcal{F}$ is *strongly* lower semicontinuous. So let $u_j \to u$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ and $\nabla u_j \to \nabla u$ almost everywhere, which holds after selecting a subsequence (not relabeled), see Appendix A.3. By assumption we have that $f(x,\nabla u_j(x)) \geq 0$. Applying Fatou's Lemma, we immediately conclude that

$$\mathcal{F}[u] = \int_\Omega f(x,\nabla u(x)) \,\mathrm{d}x \leq \liminf_{j\to\infty} \int_\Omega f(x,\nabla u_j(x)) \,\mathrm{d}x = \liminf_{j\to\infty} \mathcal{F}[u_j].$$

Since this holds for all subsequences, it also follows for our original sequence, see Problem 2.1.

Step 2. To prove the claimed weak lower semicontinuity, take $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$ (weakly). We need to show that

$$\mathcal{F}[u] \le \liminf_{j\to\infty} \mathcal{F}[u_j] =: \alpha. \tag{2.3}$$

Taking a subsequence (not relabeled), we can in fact assume that $\mathcal{F}[u_j]$ converges to α.

By the Mazur Lemma A.3 we may find convex combinations

$$v_j = \sum_{n=j}^{N(j)} \theta_n^{(j)} u_n, \qquad \text{where} \qquad \theta_n^{(j)} \in [0,1] \quad \text{and} \quad \sum_{n=j}^{N(j)} \theta_n^{(j)} = 1,$$

such that $v_j \to u$ in $\mathrm{W}^{1,p}$ (strongly). As $f(x,\bullet)$ is convex for almost every x,

$$\mathcal{F}[v_j] = \int_\Omega f\left(x, \sum_{n=j}^{N(j)} \theta_n^{(j)} \nabla u_n(x)\right) \mathrm{d}x \le \sum_{n=j}^{N(j)} \theta_n^{(j)} \mathcal{F}[u_n].$$

Since $\mathcal{F}[u_n] \to \alpha$ as $n \to \infty$ and $\sum_{n=j}^{N(j)} \theta_n^{(j)} = 1$, we arrive at

$$\liminf_{j\to\infty} \mathcal{F}[v_j] \le \alpha.$$

On the other hand, from the first step and since $v_j \to u$ strongly, we have

$$\mathcal{F}[u] \le \liminf_{j\to\infty} \mathcal{F}[v_j] \le \alpha = \liminf_{j\to\infty} \mathcal{F}[u_j].$$

Thus, (2.3) follows and the proof is finished. □

We summarize our findings in the following existence theorem.

Theorem 2.7 *Let $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand such that*

(i) f satisfies the p-coercivity bound (2.2) with $p \in (1,\infty)$;
(ii) $f(x,\bullet)$ is convex for almost every $x \in \Omega$.

Then, the associated functional $\mathcal{F}$ has a minimizer over $\mathrm{W}_g^{1,p}(\Omega;\mathbb{R}^m)$, where $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

Proof. This follows immediately from the Direct Method for the $\mathrm{W}^{1,p}$-weak convergence on $X := \mathrm{W}_g^{1,p}(\Omega;\mathbb{R}^m)$, see Theorem 2.3, together with Proposition 2.5 and the Tonelli–Serrin Theorem 2.6. □

Example 2.8 The **Dirichlet functional** (or **Dirichlet integral**) is

$$\mathcal{F}[u] := \int_\Omega \frac{1}{2}|\nabla u(x)|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m).$$

We already encountered this integral functional when considering electrostatics in Section 1.3. It is easy to see that the Dirichlet functional satisfies all requirements of Theorem 2.7 and so there exists a minimizer for any prescribed boundary values $g \in \mathrm{W}^{1/2,2}(\partial\Omega;\mathbb{R}^m)$.

We next show an illuminating converse to the Tonelli–Serrin Theorem 2.6, namely that the assumed convexity is also necessary.

Proposition 2.9 *Let $\mathcal{F}\colon \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \to [0,\infty)$, $p \in [1,\infty)$, be an integral functional with continuous integrand $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ (not x-dependent). If $\mathcal{F}$ is weakly lower semicontinuous on $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ and if either $d = 1$ or $m = 1$ (the* one-dimensional case *and the* scalar case*, respectively), then f is convex.*

Proof. We only consider the case $m = 1$ and d arbitrary; the other case is proved in a similar manner. Let $a, b \in \mathbb{R}^d$ with $a \neq b$ and $\theta \in (0,1)$. Set $v := \theta a + (1-\theta)b$, $n := b - a$, and

$$u_j(x) := v \cdot x + \frac{1}{j}\varphi_0\big(jx \cdot n - \lfloor jx \cdot n \rfloor\big), \qquad x \in \Omega,$$

where $\lfloor s \rfloor$ denotes the largest integer less than or equal to $s \in \mathbb{R}$, and

$$\varphi_0(t) := \begin{cases} -(1-\theta)t & \text{if } t \in [0,\theta), \\ \theta t - \theta & \text{if } t \in [\theta,1), \end{cases}$$

see Figure 2.2. We have that

$$\nabla u_j(x) = \begin{cases} \theta a + (1-\theta)b - (1-\theta)(b-a) = a & \text{if } jx \cdot n - \lfloor jx \cdot n \rfloor \in [0,\theta), \\ \theta a + (1-\theta)b + \theta(b-a) = b & \text{if } jx \cdot n - \lfloor jx \cdot n \rfloor \in [\theta,1). \end{cases}$$

Hence, $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega)$ and since the second term in the definition of u_j converges to zero uniformly, it holds that $u_j \rightharpoonup v \cdot x$ in $\mathrm{W}^{1,p}$ (here and in the following, "$v \cdot x$" is a shorthand notation for the linear function $x \mapsto v \cdot x$). By the assumed weak lower semicontinuity we conclude that

$$|\Omega| f(\theta a + (1-\theta)b) = \mathcal{F}[v \cdot x] \leq \liminf_{j\to\infty} \mathcal{F}[u_j] = |\Omega|\big(\theta f(a) + (1-\theta) f(b)\big).$$

This proves the claim. □

In the **vectorial case**, i.e., $d \neq 1$ and $m \neq 1$, it turns out that convexity of the integrand (in the gradient variable) is far from being necessary for weak lower semicontinuity. In fact, there is a weaker necessary and sufficient condition for weak lower semicontinuity; we will explore this in Chapter 5.

Finally, we prove the following result concerning the *uniqueness* of the minimizer.

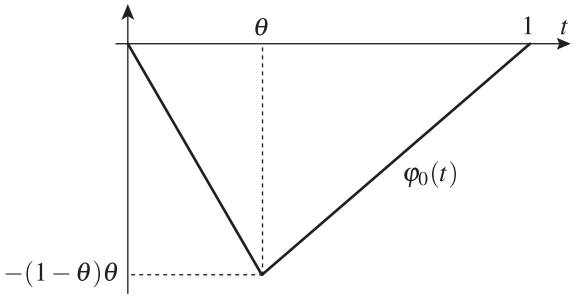


Fig. 2.2 The function φ_0

Proposition 2.10 *Let $\mathcal{F}\colon \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \to [0,\infty)$, $p \in [1,\infty)$, be an integral functional with Carathéodory integrand $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$. If f is **strictly convex**, that is,*

$$f(x,\theta A + (1-\theta)B) < \theta f(x,A) + (1-\theta) f(x,B)$$

for all $x \in \Omega$, $A, B \in \mathbb{R}^{m\times d}$ with $A \neq B$, and $\theta \in (0,1)$, then the minimizer of $\mathcal{F}$ over $\mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ ($g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$), if it exists, is unique.

Proof. Assume there are two different minimizers $u, v \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ of $\mathcal{F}$. Then set

$$w := \frac{1}{2}u + \frac{1}{2}v \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$$

and observe that

$$\mathcal{F}[w] = \int_\Omega f\Big(x, \frac{1}{2}\nabla u(x) + \frac{1}{2}\nabla v(x)\Big) < \frac{1}{2}\mathcal{F}[u] + \frac{1}{2}\mathcal{F}[v] = \min_{\mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)} \mathcal{F},$$

yielding an immediate contradiction. □

2.3 Integrands with u-Dependence

If we try to extend the results in the previous section to more general functionals of the form

$$\mathcal{F}[u] := \int_\Omega f(x, u(x), \nabla u(x))\,\mathrm{d}x,$$

we discover that our proof strategy via the Mazur lemma runs into difficulties: We cannot "pull out" the convex combination inside

$$\int_\Omega f\left(x, \sum_{n=j}^{N(j)} \theta_n^{(j)} u_n(x), \sum_{n=j}^{N(j)} \theta_n^{(j)} \nabla u_n(x)\right)\,\mathrm{d}x$$

any longer. Nevertheless, a lower semicontinuity result analogous to the one for the u-independent case turns out to be true:

Theorem 2.11 *Let $f\colon \Omega\times\mathbb{R}^m\times\mathbb{R}^{m\times d}\to[0,\infty)$ be a Carathéodory integrand, which here means that*

(i) $x\mapsto f(x,v,A)$ is Lebesgue-measurable for every fixed $(v,A)\in\mathbb{R}^m\times\mathbb{R}^{m\times d}$;
(ii) $(v,A)\mapsto f(x,v,A)$ is continuous for almost every fixed $x\in\Omega$.

Assume also that

$$f(x,v,\bullet)\text{ is convex for every }(x,v)\in\Omega\times\mathbb{R}^m.$$

Then, for $p\in(1,\infty)$, the functional

$$\mathcal{F}[u]:=\int_\Omega f(x,u(x),\nabla u(x))\,\mathrm{d}x,\qquad u\in\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

is weakly lower semicontinuous.

While it would be possible to give an elementary proof of this theorem here, we postpone the detailed study of integral functionals with u-dependent integrands until Section 5.6. There, using more advanced techniques, we will establish a much more general lower semicontinuity result, albeit under an additional growth assumption. A proof of the above theorem without this growth assumption can be found in Section 3.2.6 of [91].

Example 2.12 In the prototypical problem of linearized elasticity from Section 1.8 we are tasked to solve

$$\begin{cases}\text{Minimize} & \mathcal{F}[u]:=\displaystyle\int_\Omega \mu|\mathcal{E}u|^2+\frac{1}{2}\Big(\kappa-\frac{2}{3}\mu\Big)|\operatorname{tr}\mathcal{E}u|^2-b\cdot u\,\mathrm{d}x\\ \text{over all} & u\in\mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)\text{ with }u|_{\partial\Omega}=g,\end{cases}$$

where $\mu,\kappa>0$, $b\in\mathrm{L}^2(\Omega;\mathbb{R}^3)$, and $g\in\mathrm{W}^{1/2,2}(\partial\Omega;\mathbb{R}^3)$. It is clear that $\mathcal{F}$ has quadratic growth. We assume that $\kappa-2\mu/3\geq 0$ and $g=0$ for simplicity. Then, we first show the following **Korn inequality**:

$$\|\nabla u\|_{\mathrm{L}^2}\leq\sqrt{2}\|\mathcal{E}u\|_{\mathrm{L}^2}\tag{2.4}$$

for all $u\in\mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$ with $u|_{\partial\Omega}=0$. This can be seen as follows: An elementary computation shows that for $\varphi\in\mathrm{C}_c^\infty(\Omega;\mathbb{R}^3)$ it holds that (with "div" the row-wise divergence)

$$2(\mathcal{E}\varphi:\mathcal{E}\varphi)-\nabla\varphi:\nabla\varphi=\operatorname{div}\big[(\nabla\varphi)\varphi-(\operatorname{div}\varphi)\varphi\big]+(\operatorname{div}\varphi)^2.$$

Thus, by the divergence theorem,

$$\begin{aligned} 2\|\mathcal{E}\varphi\|_{\mathrm{L}^2}^2 - \|\nabla\varphi\|_{\mathrm{L}^2}^2 &= \int_\Omega \operatorname{div}\big[(\nabla\varphi)\varphi - (\operatorname{div}\varphi)\varphi\big]\,\mathrm{d}x + \int_\Omega (\operatorname{div}\varphi)^2\,\mathrm{d}x \\ &= \int_\Omega (\operatorname{div}\varphi)^2\,\mathrm{d}x \\ &\geq 0. \end{aligned}$$

This yields (2.4) for $u = \varphi \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^3)$. The general case follows from the density of $\mathrm{C}_c^\infty(\Omega;\mathbb{R}^3)$ in $\mathrm{W}_0^{1,2}(\Omega;\mathbb{R}^3)$.

Then, using Young's inequality and the Poincaré inequality (see Theorem A.27 (i), denoting the L^2-Poincaré constant by $C_P > 0$), we get for any $\delta > 0$,

$$\begin{aligned} \mathcal{F}[u] &\geq \mu\|\mathcal{E}u\|_{\mathrm{L}^2}^2 - \|b\|_{\mathrm{L}^2}\|u\|_{\mathrm{L}^2} \\ &\geq \mu\|\mathcal{E}u\|_{\mathrm{L}^2}^2 - \frac{1}{2\delta}\|b\|_{\mathrm{L}^2}^2 - \frac{\delta}{2}\|u\|_{\mathrm{L}^2}^2 \\ &\geq \frac{\mu}{2}\|\nabla u\|_{\mathrm{L}^2}^2 - \frac{1}{2\delta}\|b\|_{\mathrm{L}^2}^2 - \frac{C_P^2\delta}{2}\|\nabla u\|_{\mathrm{L}^2}^2. \end{aligned}$$

Choosing $\delta := \mu/(2C_P^2)$, we obtain the coercivity estimate

$$\mathcal{F}[u] \geq \frac{\mu}{4}\|\nabla u\|_{\mathrm{L}^2}^2 - \frac{C_P^2}{\mu}\|b\|_{\mathrm{L}^2}^2.$$

Hence, applying the Poincaré inequality again, $\mathcal{F}[u]$ controls $\|u\|_{\mathrm{W}^{1,2}}$ and our functional is weakly coercive. Moreover, it is clear that the integrand is convex in the $\mathcal{E}u$-argument. Thus, Theorem 2.11 yields the existence of a solution $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$ to our minimization problem of linearized elasticity. In fact, one could also argue using the Tonelli–Serrin Theorem 2.6 and the fact that the lower-order term $\int_\Omega b \cdot u\ \mathrm{d}x$ is weakly continuous on $\mathrm{W}^{1,2}$. More on the topic of linearized elasticity can be found in Sections 6.2 and 6.3 of [78].

2.4 The Lavrentiev Gap Phenomenon

So far, we have chosen the function space in which we look for the solution of a minimization problem from the scale of Sobolev spaces according to a coercivity assumption such as (2.2). However, at first sight, classically differentiable functions may appear to be more appealing. So the question arises whether the infimum value is actually the same when considering different function spaces. More precisely, given two linear or affine spaces $X \subset Y$ such that X is dense in Y, and a functional $\mathcal{F}\colon Y \to \mathbb{R} \cup \{+\infty\}$, we ask whether

$$\inf_X \mathcal{F} = \inf_Y \mathcal{F}.$$

Note that even if the infima agree, it is a priori unlikely that this infimum is *attained* in both spaces unless we have additional regularity of a minimizer (which we will investigate in Section 3.2).

For $X = \mathrm{C}^\infty$ and $Y = \mathrm{W}^{1,p}$ the equality of infima turns out to be true under suitable growth conditions:

Theorem 2.13 *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}$ be a Carathéodory integrand and let $p \in [1,\infty)$ such that there exists an $M > 0$ with*

$$|f(x,v,A)| \le M(1 + |v|^q + |A|^p), \qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d},$$

for some $q \in [1, dp/(d-p)]$ if $p < d$, or $q \in [1,\infty)$ if $p \ge d$. Then, the functional

$$\mathcal{F}[u] := \int_\Omega f(x,u(x),\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

is strongly continuous. Consequently,

$$\inf_{\mathrm{C}^\infty(\Omega;\mathbb{R}^m)} \mathcal{F} = \inf_{\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)} \mathcal{F}.$$

The same equality of infima also holds with fixed boundary values.

Proof. Let $u_j \to u$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, whereby also $u_j \to u$ in $\mathrm{L}^q(\Omega;\mathbb{R}^m)$ by Theorem A.28. Additionally assume that $u_j \to u$, $\nabla u_j \to \nabla u$ almost everywhere, which holds after selecting a subsequence by Lemma A.9. Then, from the growth assumption we get

$$\mathcal{F}[u_j] = \int_\Omega f(x,u_j,\nabla u_j)\,\mathrm{d}x \le \int_\Omega M(1 + |u_j|^q + |\nabla u_j|^p)\,\mathrm{d}x$$

and via Pratt's Theorem A.11 we infer that

$$\mathcal{F}[u_j] \to \mathcal{F}[u].$$

Since this holds for a subsequence of any subsequence of the original sequence $(u_j)_j$, we have established the continuity of $\mathcal{F}$ with respect to the strong convergence in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. The assertion about the equality of infima now follows readily since $\mathrm{C}^\infty(\Omega;\mathbb{R}^m)$ is dense in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.

The equality of the infima under an additional boundary value constraint is a consequence the continuity of the trace operator with respect to the $\mathrm{W}^{1,p}$-convergence, see Theorem A.25, and the fact that any map in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ can be approximated with smooth functions with the same boundary values, see Theorem A.30. □

If we dispense with the above growth assumption, however, the infimum over different spaces may indeed be different – this is called the *Lavrentiev gap phenomenon*, discovered in 1926 by Mikhail Lavrentiev. Here, we give an example between the spaces

$\mathrm{W}^{1,1}$ and $\mathrm{W}^{1,\infty}$ (with boundary conditions):

Example 2.14 (Manià 1934 [221]) Consider the minimization problem

$$\begin{cases} \text{Minimize} \quad \mathcal{F}[u] := \displaystyle\int_0^1 (u(t)^3 - t)^2\, \dot{u}(t)^6 \,\mathrm{d}t \\ \text{subject to} \quad u(0) = 0,\ u(1) = 1 \end{cases}$$

for u from either $\mathrm{W}^{1,1}((0,1))$ or $\mathrm{W}^{1,\infty}((0,1))$. We claim that

$$\inf_{\mathrm{W}^{1,\infty}((0,1))} \mathcal{F} > \inf_{\mathrm{W}^{1,1}((0,1))} \mathcal{F},$$

where here and in the following these infima are to be taken only over functions u with boundary values $u(0) = 0$, $u(1) = 1$.

Clearly, $\mathcal{F} \geq 0$, and for $u_*(t) := t^{1/3} \in (\mathrm{W}^{1,1} \setminus \mathrm{W}^{1,\infty})(0,1)$ we have $\mathcal{F}[u_*] = 0$. Thus,

$$\inf_{\mathrm{W}^{1,1}((0,1))} \mathcal{F} = 0.$$

On the other hand, every $u \in \mathrm{W}^{1,\infty}((0,1))$ is Lipschitz continuous. Thus, also using $u(0) = 0$, $u(1) = 1$, there exists a $\tau \in (0,1)$ with

$$u(t) \leq h(t) := \frac{t^{1/3}}{2} \quad \text{for all } t \in [0,\tau] \qquad \text{and} \qquad u(\tau) = h(\tau).$$

Then, $u(t)^3 - t \leq h(t)^3 - t$ for $t \in [0,\tau]$ and, since both of these terms are negative,

$$(u(t)^3 - t)^2 \geq (h(t)^3 - t)^2 = \frac{7^2}{8^2} t^2 \qquad \text{for all } t \in [0,\tau].$$

We then estimate

$$\mathcal{F}[u] \geq \int_0^\tau (u(t)^3 - t)^2\, \dot{u}(t)^6 \,\mathrm{d}t \geq \frac{7^2}{8^2} \int_0^\tau t^2\, \dot{u}(t)^6 \,\mathrm{d}t.$$

Further, by Hölder's inequality,

$$\begin{aligned} \int_0^\tau \dot{u}(t) \,\mathrm{d}t &= \int_0^\tau t^{-1/3} \cdot t^{1/3}\, \dot{u}(t) \,\mathrm{d}t \\ &\leq \left(\int_0^\tau t^{-2/5} \,\mathrm{d}t \right)^{5/6} \cdot \left(\int_0^\tau t^2\, \dot{u}(t)^6 \,\mathrm{d}t \right)^{1/6} \\ &= \frac{5^{5/6}}{3^{5/6}} \tau^{1/2} \left(\int_0^\tau t^2\, \dot{u}(t)^6 \,\mathrm{d}t \right)^{1/6}. \end{aligned}$$

Because also

$$\int_0^\tau \dot{u}(t) \,\mathrm{d}t = u(\tau) - u(0) = h(\tau) = \frac{\tau^{1/3}}{2},$$

we arrive at

$$\mathcal{F}[u] \geq \frac{7^2 3^5}{8^2 5^5 2^6 \tau} > \frac{7^2 3^5}{8^2 5^5 2^6} > 0.$$

Thus,

$$\inf_{\mathrm{W}^{1,\infty}((0,1))} \mathcal{F} > \inf_{\mathrm{W}^{1,1}((0,1))} \mathcal{F},$$

and $\mathcal{F}$ has been shown to exhibit the Lavrentiev gap phenomenon.

In a more recent example, Ball & Mizel [43] showed that the problem

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_{-1}^{1} (t^4 - u(t)^6)^2 |\dot{u}(t)|^{2m} + \varepsilon \dot{u}(t)^2 \, \mathrm{d}t \\ \text{subject to} & u(-1) = \alpha, \ u(1) = \beta \end{cases}$$

also exhibits the Lavrentiev gap phenomenon between the spaces $\mathrm{W}^{1,2}$ and $\mathrm{W}^{1,\infty}$ if $m \in \mathbb{N}$ satisfies $m > 13$, $\varepsilon > 0$ is sufficiently small, and $-1 \leq \alpha < 0 < \beta \leq 1$. This example is significant because the Ball–Mizel functional is *coercive* on $\mathrm{W}^{1,2}(-1, 1)$ thanks to the second summand in the integrand.

We note that the Lavrentiev gap phenomenon is a major obstacle for the numerical approximation of minimization problems. For instance, standard (piecewise affine) finite element approximations are in $\mathrm{W}^{1,\infty}$ and thus in the presence of the Lavrentiev gap phenomenon (between $\mathrm{W}^{1,p}$ and $\mathrm{W}^{1,\infty}$) we cannot approximate the true solution with such finite elements. Consequently, one is forced to work with non-conforming elements and other advanced schemes. This issue does not only affect "academic" examples such as the ones above, but is also of great concern in applied problems, such as nonlinear elasticity theory.

2.5 Integral Side Constraints

In some minimization problems the class of candidate functions is restricted to include one or more integral side constraints. To establish the existence of a minimizer in these cases, we first need to extend the Direct Method to this scenario.

Theorem 2.15 *Let X be a Banach space or a closed affine subset of a Banach space and let $\mathcal{F}, \mathcal{H} \colon X \to \mathbb{R} \cup \{+\infty\}$. Assume the following:*

(WH1) ***Weak coercivity of $\mathcal{F}$:*** *For all $\Lambda \in \mathbb{R}$ the sublevel set*

$$\big\{ u \in X \ : \ \mathcal{F}[u] \leq \Lambda \big\} \qquad \textit{is sequentially weakly precompact,}$$

that is, if $\mathcal{F}[u_j] \leq \Lambda$ for a sequence $(u_j)_j \subset X$ and some $\Lambda \in \mathbb{R}$, then $(u_j)_j$ has a weakly converging subsequence.

(WH2) ***Weak lower semicontinuity of $\mathcal{F}$:*** *For all sequences $(u_j)_j \subset X$ with $u_j \rightharpoonup u$ in X it holds that*

$$\mathcal{F}[u] \leq \liminf_{j \to \infty} \mathcal{F}[u_j].$$

(WH3) ***Weak continuity of*** $\mathcal{H}$*: For all sequences* $(u_j)_j \subset X$ *with* $u_j \rightharpoonup u$ *in* X *it holds that*

$$\mathcal{H}[u_j] \to \mathcal{H}[u].$$

Assume also that there exists at least one $u_0 \in X$ *with* $\mathcal{H}[u_0] = 0$*. Then, the minimization problem*

$$\text{Minimize } \mathcal{F}[u] \text{ over all } u \in X \text{ with } \mathcal{H}[u] = 0$$

has a solution.

Proof. The proof is almost exactly the same as the one for the standard Direct Method in Theorem 2.3. The only difference is that we need to select the u_j for a minimizing sequence with $\mathcal{H}[u_j] = 0$. By (WH3), this property also holds for any weak limit u_* of a subsequence of the u_j's, which then is the sought minimizer. □

A large class of side constraints can be treated using the following simple result.

Lemma 2.16 *Let* $h\colon \Omega \times \mathbb{R}^m \to \mathbb{R}$ *be a Carathéodory integrand and let* $p \in [1, \infty)$ *such that there exists an* $M > 0$ *with*

$$|h(x, v)| \le M(1 + |v|^q), \qquad (x, v) \in \Omega \times \mathbb{R}^m, \tag{2.5}$$

for some $q \in [1, dp/(d-p))$ *if* $p < d$*, or* $q \in [1, \infty)$ *if* $p \ge d$*. Then, the functional* $\mathcal{H}\colon \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m) \to \mathbb{R}$ *defined through*

$$\mathcal{H}[u] := \int_\Omega h(x, u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m),$$

is weakly continuous.

Proof. We only prove the lemma in the case $p \le d$. The proof for $p > d$ is analogous.

Let $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$, whereby after selecting a subsequence and employing the Rellich–Kondrachov Theorem A.29 and Lemma A.9, $u_j \to u$ in L^q and almost everywhere. By assumption we have

$$\pm h(x, v) + M(1 + |v|^q) \ge 0, \qquad (x, v) \in \Omega \times \mathbb{R}^m.$$

Thus, applying Fatou's lemma separately to these two integrands, we get

$$\liminf_{j\to\infty} \left(\pm\mathcal{H}[u_j] + \int_\Omega M(1 + |u_j|^q)\,\mathrm{d}x \right) \ge \pm\mathcal{H}[u] + \int_\Omega M(1 + |u|^q)\,\mathrm{d}x.$$

Since $\|u_j\|_{\mathrm{L}^q} \to \|u\|_{\mathrm{L}^q}$, we can combine these two assertions to get $\mathcal{H}[u_j] \to \mathcal{H}[u]$. This holds for a subsequence of any subsequence of $(u_j)_j$, hence it also holds for our original sequence. □

Combining this lemma with Theorems 2.7 and 2.15 and also the Rellich–Kondrachov Theorem A.29, we immediately get the following existence result.

Theorem 2.17 *Let $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$ and $h\colon \Omega \times \mathbb{R}^m \to \mathbb{R}$ be Carathéodory integrands such that*

(i) f satisfies the p-coercivity bound (2.2), where $p \in (1,\infty)$;
(ii) $f(x,\bullet)$ is convex for all $x \in \Omega$;
(iii) h satisfies the q-growth condition (2.5) for some $q \in [1, dp/(d-p))$ if $p < d$, or $q \in [1,\infty)$ if $p \geq d$;
(iv) there is at least one $u_0 \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ with $\mathcal{H}[u_0] = 0$.

Then, for all $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$ there exists a minimizer $u_* \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ of the functional

$$\mathcal{F}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m),$$

under the side constraint

$$\mathcal{H}[u] := \int_\Omega h(x,u(x))\,\mathrm{d}x = 0.$$

2.6 Convex Functions and Duality

We finish this chapter by briefly considering the general theory of convex functions.

In all of the following let X be a real reflexive Banach space (finite or infinite-dimensional) with dual space X^*, see Appendix A.2. We denote by $\langle x, x^*\rangle = x^*(x)$ the duality product between $x \in X$ and $x^* \in X^*$. For a set $A \subset X$ we write $\operatorname{co} A$, $\overline{\operatorname{co}}\, A$ for its **convex hull** and **closed convex hull**, respectively. These hulls are defined to be the smallest (closed) convex set containing A, or, equivalently, the intersection of all (closed) convex sets containing A. For $A \subset X$ we furthermore define the **characteristic function** $\chi_A\colon X \to \mathbb{R}\cup\{+\infty\}$ as

$$\chi_A(x) := \frac{1}{\mathbb{1}_A(x)} - 1 = \begin{cases} 0 & \text{if } x \in A,\\ +\infty & \text{if } x \notin A.\end{cases}$$

Let $F\colon X \to \mathbb{R}\cup\{+\infty\}$. The function F is called **proper** if it is not identically $+\infty$. We define the **effective domain** $\operatorname{dom} F \subset X$ and the **epigraph** $\operatorname{epi} F \subset X\times\mathbb{R}$ of F as follows:

$$\operatorname{dom} F := \big\{ x \in X \,:\, F(x) < +\infty \big\},$$
$$\operatorname{epi} F := \big\{ (x,\alpha) \in X\times\mathbb{R} \,:\, F(x) \leq \alpha \big\}.$$

It can be shown (see Problems 2.6, 2.7) that F is convex if and only if $\operatorname{epi} F$ is convex (as a set), and that F is (sequentially) lower semicontinuous if and only if $\operatorname{epi} F$ is (sequentially) closed; this holds with respect to both the strong and the weak convergence.

Lemma 2.18 *If $\dim X < \infty$, then every convex function $F\colon X \to \mathbb{R}\cup\{+\infty\}$ is locally bounded on the interior of its effective domain* $\operatorname{dom} F$.

Proof. Let $n := \dim X < \infty$. If $x \in X$ is in the interior of the effective domain of F, then x lies in the convex hull $\operatorname{co}\{x_1, \dots, x_{n+1}\}$ of $n+1$ affinely independent points x_k (i.e., $\sum_k \alpha_k x_k = 0$ for some $\alpha_k \in \mathbb{R}$ with $\sum_k \alpha_k = 0$ implies $\alpha_1 = \alpha_2 = \cdots = \alpha_{n+1} = 0$) with $F(x_k) < +\infty$. Thus, there exists an open ball around x inside the set $\operatorname{co}\{x_1, \dots, x_{n+1}\}$ on which F is bounded by $\sup\{F(x_1), \dots, F(x_{n+1})\}$. □

Lemma 2.19 *Let $\mathcal{A}$ be a non-empty family of continuous affine functions. Then, the function $F\colon X \to \mathbb{R} \cup \{+\infty\}$ defined through*

$$F(x) := \sup_{a \in \mathcal{A}} a(x)$$

is convex and lower semicontinuous. Conversely, every convex and lower semicontinuous function can be written in this form.

Proof. The convexity of F is clear since all the affine functions $a \in \mathcal{A}$ are in particular convex, and this property is preserved by taking the supremum.

For the lower semicontinuity we just need to realize that the pointwise supremum of continuous functions is always lower semicontinuous. Indeed, for a sequence $x_j \to x$ in X we have for all $\widetilde{a} \in \mathcal{A}$ that

$$\widetilde{a}(x) = \lim_{j\to\infty} \widetilde{a}(x_j) \le \liminf_{j\to\infty} \sup_{a\in\mathcal{A}} a(x_j) = \liminf_{j\to\infty} F(x_j).$$

Taking the supremum over all $\widetilde{a} \in \mathcal{A}$, the lower semicontinuity follows.

For the converse, we may assume that F is proper; otherwise the result is trivial. Let $x \in X$ with $F(x) < +\infty$. The epigraph $\operatorname{epi} F$ of F is closed and convex by assumption. Hence, by the Hahn–Banach Separation Theorem A.5, for every $\gamma < F(x)$ we can find an affine function $a_{x,\gamma}\colon X \to \mathbb{R}$ whose graph separates the point (x, γ) from $\operatorname{epi} F$. This means that $\gamma < a_{x,\gamma}(x) < F(x)$ and $a_{x,\gamma}$ lies everywhere below the graph of F. Letting $\gamma \uparrow F(x)$, we arrive at

$$F(x) = \sup\big\{\, a_{x,\gamma}(x) \;:\; \gamma < F(x) \,\big\}.$$

A similar argument also applies if $F(x) = +\infty$. Collecting all these $a_{x,\gamma}$ for $(x, \gamma) \in X \times \mathbb{R}$ with $\gamma < F(x)$ into the set $\mathcal{A}$, the conclusion follows. □

Proposition 2.20 *If $\dim X < \infty$, every proper convex function $F\colon X \to \mathbb{R} \cup \{+\infty\}$ is continuous on the interior of its effective domain.*

We will prove this in more generality later, see Lemma 5.9 in conjunction with Lemma 2.18.

One important object in the general theory of convex functions is the **(convex) conjugate**, or **Legendre–Fenchel transform**, $F^*\colon X^* \to \mathbb{R} \cup \{+\infty\}$ of a proper (not necessarily convex) function $F\colon X \to \mathbb{R} \cup \{+\infty\}$, which is defined as follows:

$$F^*(x^*) := \sup_{x\in X} \big[\langle x, x^*\rangle - F(x)\big], \qquad x^* \in X^*.$$

Of course, we may restrict to $x \in \operatorname{dom} F$ in the supremum. The intuition here is as follows: For a given $x^* \in X^*$ consider all affine maps on X, given as $x \mapsto \langle x, x^*\rangle - \beta$

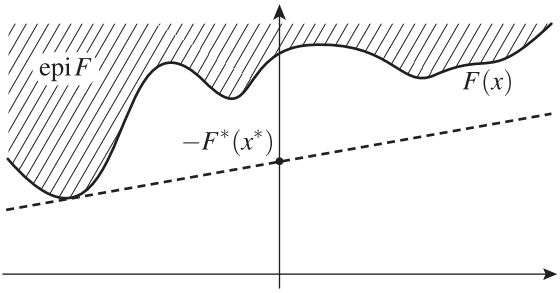


Fig. 2.3 The convex conjugate

with an $x^* \in X^*$ and $\beta \in \mathbb{R}$, that lie below epi F. Then, we claim that $-F^*(x^*)$ is the supremum of the heights at which these affine maps intersect the dependent (vertical) axis, see Figure 2.3. Indeed, fix $x^* \in X^*$ and let $\beta \in \mathbb{R}$ be such that $F(x) \geq \langle x, x^* \rangle - \beta$ for all $x \in X$. Then, $\beta \geq \langle x, x^* \rangle - F(x)$ for all $x \in X$ and hence $\beta \geq F^*(x^*)$. The highest such affine map is $x \mapsto \langle x, x^* \rangle - F^*(x^*)$ (corresponding to the smallest β), which intersects the vertical axis in $-F^*(x^*)$.

The following **Fenchel inequality** is immediate from the definition:

$$\langle x, x^* \rangle \leq F(x) + F^*(x^*), \qquad \text{for all } x \in X, x^* \in X^*. \tag{2.6}$$

We next collect some properties of the conjugate function:

Proposition 2.21 *Let $F, G \colon X \to \mathbb{R} \cup \{+\infty\}$ be proper and $F^*, G^* \colon X^* \to \mathbb{R} \cup \{+\infty\}$ be their conjugates. The following assertions hold for any $x \in X$, $x^* \in X^*$:*

(i) F^* *is convex and lower semicontinuous.*
(ii) $F^*(0) = -\inf F$.
(iii) If $F \leq G$, *then* $G^* \leq F^*$.
(iv) If for $\lambda > 0$ *we denote by* F_λ *the scaled function* $F_\lambda(x) := F(\lambda x)$, *then* $F_\lambda^*(x^*) = F^*(x^*/\lambda)$.
(v) $(\lambda F)^*(x^*) = \lambda F^*(x^*/\lambda)$ *for all* $\lambda > 0$.
(vi) $(F + \gamma)^* = F^* - \gamma$ *for all* $\gamma \in \mathbb{R}$.
(vii) If for $a \in X$ *we denote by* F_a *the translated function* $F_a(x) := F(x - a)$, *then* $F_a^*(x^*) = F^*(x^*) + \langle a, x^* \rangle$.

Proof. The first assertion follows from Lemma 2.19, all the others are straightforward calculations, see Problem 2.8. □

Before progressing, we consider a few canonical examples of convex functions.

Example 2.22 (Support function) Let χ_A be the characteristic function of $A \subset X$. For its conjugate function one gets

$$\sigma_A(x^*) := \chi_A^*(x^*) = \sup_{x \in A} \langle x, x^* \rangle, \qquad x^* \in X^*,$$

which is called the **support function** of A. It is always convex, lower semicontinuous, and **positively** 1**-homogeneous**, i.e., $\sigma_A(\lambda x^*) = \lambda \sigma_A(x^*)$ for all $x^* \in X^*$ and $\lambda \geq 0$, see Problem 2.9.

Example 2.23 Let $p, q \in (1, \infty)$ with $1/p + 1/q = 1$, that is, p, q are **conjugate exponents**. Then,

$$\varphi(t) := \frac{1}{p}|t|^p \qquad \text{and} \qquad \varphi^*(t) := \frac{1}{q}|t|^q, \qquad t \in \mathbb{R},$$

are conjugate. From the Fenchel inequality (2.6) we recover the **Young inequality**

$$xy \le \frac{x^p}{p} + \frac{y^q}{q} \qquad \text{for all } x, y \ge 0.$$

Example 2.24 For the absolute value function $\varphi(t) := |t|$ we get

$$\varphi^*(t) = \chi_{[-1,1]}(t) = \begin{cases} 0 & \text{if } |t| \le 1, \\ +\infty & \text{if } |t| > 1, \end{cases} \qquad t \in \mathbb{R}. \tag{2.7}$$

Example 2.25 The conjugate of the exponential function is

$$\exp^*(t) = \begin{cases} +\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ t\ln(t) - t & \text{if } t > 0, \end{cases} \qquad t \in \mathbb{R}.$$

In this case, (2.6) gives the inequality

$$xy \le \exp(x) + y\ln(y) - y \qquad \text{for all } x, y > 0.$$

Example 2.26 Let $\varphi\colon \mathbb{R} \to [0, +\infty]$ be proper, convex, lower semicontinuous, and satisfy $\varphi(0) = \inf \varphi = 0$. Let $\|\cdot\|, \|\cdot\|_*$ be the norms on X and on X^*, respectively. Then the functions

$$G(x) := \varphi(\|x\|) \qquad \text{and} \qquad G^*(x^*) := \varphi^*(\|x^*\|_*), \qquad x \in X,\ x^* \in X^*,$$

are conjugate. In particular, $\|\cdot\|^p/p$ and $\|\cdot\|^q/q$ for $1/p + 1/q = 1$ are conjugate. The verification of these statements is the task of Problem 2.10.

Example 2.27 Let $X = \mathbb{R}^n$ and let $S \in \mathbb{R}^{n\times n}$ be a symmetric, strictly positive definite matrix. Then,

$$F(x) := \frac{1}{2}x^{\mathsf{T}}Sx \qquad \text{and} \qquad F^*(y) := \frac{1}{2}y^{\mathsf{T}}S^{-1}y, \qquad x, y \in \mathbb{R}^n,$$

are conjugate.

Iterating the construction of the conjugate, we denote by $F^{**}\colon X \to \mathbb{R} \cup \{+\infty\}$ the **biconjugate** of F, that is, the function

$$F^{**}(x) := (F^*)^*(x) = \sup_{x^* \in X^*} \big[\langle x, x^*\rangle - F^*(x^*)\big], \qquad x \in X.$$

Proposition 2.28 *The biconjugate F^{**} is the* ***convex, lower semicontinuous envelope*** *of F, that is, the greatest convex, lower semicontinuous function below F. Moreover, $F^{***} = F^*$.*

Proof. For the moment denote the convex lower semicontinuous envelope of F by F_{clsc}, that is, for $x \in X$,

$$F_{\mathrm{clsc}}(x) := \sup\big\{\, H(x) \;:\; H \le F \text{ convex, lower semicontinuous} \,\big\}.$$

Also define

$$G(x) := \sup\big\{\, a(x) \;:\; a \le F \text{ affine} \,\big\}, \qquad x \in X.$$

Since G is convex and lower semicontinuous by Lemma 2.19, $G \le F_{\mathrm{clsc}}$. On the other hand, for every convex and lower semicontinuous H from the definition of F_{clsc}, we have $H(x) = \sup_{b\in\mathcal{A}} b(x)$ for a collection $\mathcal{A}$ of affine functions $b \le H$ by the same lemma. Clearly, $b \le F$ for all $b \in \mathcal{A}$ and thus b is included in the collection in the definition of G. Hence, $H \le G$, whereby also $F_{\mathrm{clsc}} \le G$. In conclusion, $F_{\mathrm{clsc}} = G$.

In order to see that $G = F^{**}$, we use a similar reasoning to when we identified the intuition behind the conjugate map: Every affine $a \le F$ has the form $a(x) = \langle x, x^*\rangle - \beta$ for some $x^* \in X^*$ and $\beta \in \mathbb{R}$. We can restrict ourselves to such a with β minimal while still preserving the property $a \le F$. We see first that $a \le F$ if and only if $\beta \ge \langle y, x^*\rangle - F(y)$ for all $y \in X$. According to the definition of the conjugate function, this condition is nothing else than $\beta \ge F^*(x^*)$. Thus, β is minimal when $\beta = F^*(x^*)$ and we get

$$F_{\mathrm{clsc}}(x) = G(x) = \sup_{x^*\in X^*} \big[\langle x, x^*\rangle - F^*(x^*)\big] = F^{**}(x), \qquad x \in X.$$

For the second assertion it suffices to observe that F^* is convex and lower semicontinuous by Proposition 2.21 (i) and to apply the first assertion. □

As a particular consequence of the preceding result, we see that conjugation facilitates a bijection between the proper, convex, and lower semicontinuous functions on X and those on X^*, which is self-inverse in the sense above.

Corollary 2.29 $\operatorname{epi} F^{**} = \overline{\operatorname{co}} \operatorname{epi} F$.

Proof. The process of taking the convex lower semicontinuous envelope of F amounts to finding the closed convex hull of the epigraph. □

Example 2.30 For the characteristic function χ_A of $A \subset X$ we get

$$\chi_A^{**} = \sigma_A^* = \chi_{\overline{\operatorname{co}} A}.$$

In particular, A and $\overline{\operatorname{co}} A$ have the same support function.

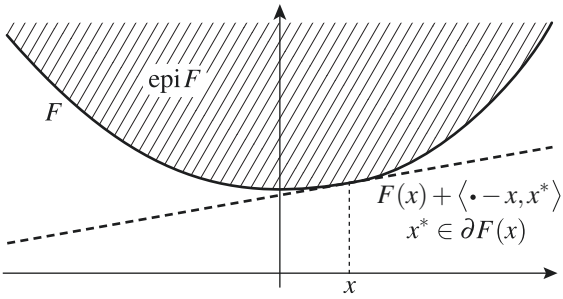


Fig. 2.4 A subgradient

2.7 Subdifferentials

Let X be a reflexive Banach space. The **subdifferential** of a proper function $F\colon X \to \mathbb{R} \cup \{+\infty\}$ is the set-valued map

$$\partial F\colon X \rightrightarrows X^*, \qquad \text{that is,} \qquad \partial F(x) \subset X^* \quad \text{for all } x \in X,$$

defined at $x \in X$ by

$$\partial F(x) := \left\{ x^* \in X^* \ : \ F(x) + \langle y - x, x^* \rangle \le F(y) \text{ for all } y \in X \right\}.$$

Any element x^* of $\partial F(x)$ is called a **subgradient** of F at x. The geometric intuition here is that for all $x^* \in \partial F(x)$ the graph of the affine function $y \mapsto F(x) + \langle y - x, x^* \rangle$ lies everywhere below the interior of epi F and touches the graph of F at $(x, F(x))$, see Figure 2.4.

Example 2.31 Let $F(t) := |t|$, $t \in \mathbb{R}$, which is not differentiable at $t = 0$. Then,

$$\partial F(t) = \operatorname{Sgn}(t) = \begin{cases} \{-1\} & \text{if } t < 0, \\ [-1, 1] & \text{if } t = 0, \\ \{+1\} & \text{if } t > 0, \end{cases} \qquad t \in \mathbb{R}.$$

The function Sgn is called the **(multi-valued) signum function**.

Example 2.32 For the characteristic function χ_K of a compact convex set $K \subset \mathbb{R}^d$ we get

$$\begin{aligned} \partial \chi_K(x) &= N_K(x) \\ &= \begin{cases} \{0\} & \text{if } x \in K \setminus \partial K, \\ \left\{ x^* \in \mathbb{R}^d \ : \ \langle y - x, x^* \rangle \le 0 \text{ for } y \in K \right\} & \text{if } x \in \partial K, \\ \emptyset & \text{if } x \notin K. \end{cases} \end{aligned}$$

The set-valued function N_K is called the **normal cone** to K at x. One can verify its geometric meaning from the definition.

A key property of the subdifferential is that it interacts well with the Legendre–Fenchel conjugate. In particular, equality in the Fenchel inequality (2.6) characterizes subgradients:

Theorem 2.33 *Let $F\colon X \to \mathbb{R} \cup \{+\infty\}$ be proper, lower semicontinuous, and convex, and let $F^*\colon X^* \to \mathbb{R} \cup \{+\infty\}$ be its conjugate. Then, the following are equivalent for $x \in X$, $x^* \in X^*$:*

(i) $x^* \in \partial F(x)$.
(ii) $x \in \partial F^*(x^*)$.
(iii) $\langle x, x^* \rangle = F(x) + F^*(x^*)$.

Proof. (i) ⇔ (iii): We already know "≤" in (iii), this is the Fenchel inequality (2.6). Thus, we only need to show the equivalence of (i) with the direction "≥" of (iii). By definition, $x^* \in \partial F(x)$ is equivalent to

$$\langle x, x^* \rangle - F(x) \geq \langle y, x^* \rangle - F(y) \qquad \text{for all } y \in X.$$

Taking the supremum over all $y \in X$ on the right-hand side, we get that this is further equivalent to

$$\langle x, x^* \rangle - F(x) \geq F^*(x^*).$$

This is the inequality "≥" in (iii).

(ii) ⇔ (iii): This follows in the same way once we recognize that $F = F^{**}$ by Proposition 2.28 since F is proper, convex, and lower semicontinuous. □

We collect several further properties of the subdifferential in the following proposition.

Proposition 2.34 *Let $F\colon X \to \mathbb{R} \cup \{+\infty\}$ be proper and convex and let $x \in X$.*

(i) *$\partial F(x)$ is convex and closed.*
(ii) *$\partial F(x) \neq \emptyset$ if F is finite and continuous at $x \in X$.*
(iii) *If F is **Gâteaux-differentiable** at $x \in X$, that is, there exists an $F'(x) \in X^*$ such that*

$$\lim_{h \downarrow 0} \frac{F(x + hv) - F(x)}{h} = \langle v, F'(x) \rangle \qquad \text{for all } v \in X,$$

then $\partial F(x) = \{F'(x)\}$.

Proof. Ad (i): By Theorem 2.33,

$$\partial F(x) = \left\{ x^* \in X^* \;:\; F^*(x^*) - \langle x, x^* \rangle \leq -F(x) \right\}.$$

The assertion then follows from the fact that the left-hand side of the inequality is convex and lower semicontinuous in x^*, see Proposition 2.21 (i).

Ad (ii): If F is finite and continuous at any point $x \in X$, then the interior of $\operatorname{epi} F$ is non-empty. By the Hahn–Banach Separation Theorem A.5, we can therefore find a supporting hyperplane to the interior of $\operatorname{epi} F$ at x, which is the graph of an affine function

$$a(y) := \left\langle y - x, x^* \right\rangle + F(x), \qquad y \in X,$$

for some $x^* \in X^*$. In particular, $a \le F$ and thus $x^* \in \partial F(x)$.

Ad (iii): We know from (ii) that $\partial F(x)$ is not empty. Moreover, for every $x^* \in \partial F(x)$, that is, $x^* \in X^*$ such that

$$F(x) + \left\langle y - x, x^* \right\rangle \le F(y) \qquad \text{for all } y \in X.$$

Using $y = x + hv$ with $v \in X$, $h > 0$ yields

$$\frac{F(x + hv) - F(x)}{h} \ge \left\langle v, x^* \right\rangle.$$

Letting $h \downarrow 0$ and using the Gâteaux-differentiability, $\langle v, F'(x) \rangle \ge \langle v, x^* \rangle$. Applying this argument with $\pm v$, we arrive at $x^* = F'(x)$. □

The subdifferential behaves analogously to the classical differential in minimization problems:

Theorem 2.35 *Let $F\colon X \to \mathbb{R} \cup \{+\infty\}$ be proper and convex. Then, $x \in X$ is a minimizer for F if and only if $0 \in \partial F(x)$.*

Proof. By the definition of the subdifferential, $0 \in \partial F(x)$ is equivalent to $F(y) \ge F(x)$ for all $y \in X$. □

Notes and Historical Remarks

The basic ideas concerning the Direct Method as well as lower semicontinuity and its connection to convexity are due to Leonida Tonelli and were established in a series of articles in the early 20th century [334–336]. In the 1960s James Serrin generalized the results to higher dimensions [298].

Most of the material in this chapter is very classical and can be found in a variety of books on the Calculus of Variations, we refer in particular to [91, 92, 158] (see, for instance, Theorem 3.23 in [91]).

All of our abstract results on the Direct Method are formulated using sequences and not using general topology tools like nets. This is justified since the weak topology on a separable, reflexive Banach space and the weak*-topology on a dual space with a separable predual are metrizable on norm-bounded sets. Thus, if the functionals under investigation satisfy suitable coerciveness assumptions, one can work with sequences. The only case where one has to be careful is when one uses the weak topology on a non-reflexive Banach space with a non-separable dual space because then the weak topology might not be metrizable. For instance, in the sequence space l^1 (with non-separable dual space l^∞), weak convergence of *sequences* is equivalent to strong convergence, but the weak and strong *topologies* still differ (see Chapter V in [88] for more details). For us more relevant is the observation that norm-bounded sets in $\mathrm{L}^1(\Omega)$ are not weakly pre-compact, either sequentially or topologically (these notions of compactness turn out to

be equivalent by the deep Eberlein–Šmulian theorem). This corresponds to functionals with linear growth, which indeed require a more involved analysis in the space of functions of bounded variation (BV). We will come back to this topic in Chapters 13–15.

The Lavrentiev gap phenomenon was discovered in [218]. Our Example 2.14 is due to Manià; we follow the description in [137]. Tonelli's Regularity Theorem [138, 334] gives regularity and hence the absence of the Lavrentiev gap phenomenon, for some integral functionals with superlinear growth; also see [60, 166–169] for some recent developments in this direction.

Much of the theory of general convex functions was developed by Jean-Jacques Moreau and R. Tyrrell Rockafellar in the 1960s. The books [129, 284] and the more advanced monographs [236, 237, 285] develop these topics in great detail. The theory of subdifferentials has also been extended to non-convex functions; see [285] and [236, 237] for more on this.

Problems

2.1 Let $\mathcal{F}\colon X \to \mathbb{R}$, where X is a complete metric space. Show that if every subsequence of the sequence $(u_j)_j \subset X$ with $u_j \to u$ in X has a further subsequence $(u_{j(k)})_k$ such that

$$\mathcal{F}[u] \le \liminf_{k\to\infty} \mathcal{F}[u_{j(k)}],$$

then also

$$\mathcal{F}[u] \le \liminf_{j\to\infty} \mathcal{F}[u_j].$$

2.2 Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Define

$$V := \left\{ u \in \mathrm{W}^{1,2}(\Omega) \ : \ \int_\Omega u(x)\,\mathrm{d}x = 0 \right\}.$$

Assume furthermore that $f\colon \Omega \times \mathbb{R}^d \to [0,\infty)$ is continuously differentiable with

$$\begin{aligned} \mu|A|^2 \le f(x,A) &\qquad \text{for some } \mu > 0 \text{ and all } (x,A) \in \Omega \times \mathbb{R}^d, \\ |\mathrm{D}_A f(x,A)| \le M(1+|A|^2) &\qquad \text{for some } M > 0 \text{ and all } (x,A) \in \Omega \times \mathbb{R}^d, \end{aligned}$$

and that $A \mapsto f(x,A)$ is convex for all $x \in \Omega$. Finally, let $g \in \mathrm{L}^2(\Omega)$. Consider the following minimization problem:

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega f(x,\nabla u(x)) - g(x)u(x)\,\mathrm{d}x \\ \text{over all} & u \in V. \end{cases}$$

(i) Show that $\mathcal{F}$ is (weakly) coercive on V, that is, there exists a $\mu > 0$ such that

$$\mathcal{F}[u] \ge \mu \|u\|^2_{\mathrm{W}^{1,2}} - \mu^{-1} \qquad \text{for all } u \in V.$$

(ii) Show that $\mathcal{F}$ is also weakly lower semicontinuous on V (weak convergence in $\mathrm{W}^{1,2}$) and hence there exists a minimizer $u_* \in V$ of $\mathcal{F}$ (minimized over V).

This problem is continued in Problem 3.9 in the next chapter.

2.3 Show that the function $f\colon \mathbb{R}^2 \to \mathbb{R}$ given by $f(x,y) = xy$ is **separately convex**, that is, $x \mapsto f(x,y)$ is convex for fixed $y \in \mathbb{R}$ and $y \mapsto f(x,y)$ is convex for fixed $x \in \mathbb{R}$, but f is not convex.

2.4 Let $f\colon \mathbb{R}^d \to [0,\infty)$ be twice continuously differentiable and assume that there are constants $\mu, M > 0$ with

$$\mu|b|^2 \le \mathrm{D}^2 f(a)[b,b] \le M|b|^2 \qquad \text{for all } a, b \in \mathbb{R}^d,$$

where

$$\mathrm{D}^2 f(a)[b,b] := \frac{\mathrm{d}^2}{\mathrm{d}t^2} f(a+tb)\bigg|_{t=0} \qquad \text{for all } a, b \in \mathbb{R}^d.$$

Show that f is convex and that $|f(v)| \le C(1+|v|^2)$ for some $C > 0$ and all $v \in \mathbb{R}^d$.

2.5 Let $f\colon \mathbb{R}^d \to \mathbb{R}$ be convex and fix $x_0 \in \mathbb{R}^d$. Set

$$M := \max_{i=1,\dots,d} \big(|f(x_0 + \mathrm{e}_i) - f(x_0)|, |f(x_0 - \mathrm{e}_i) - f(x_0)|\big).$$

Prove that if $y \in \mathbb{R}^d$ satisfies $|y|_1 := |y_1| + \cdots + |y_d| \le 1$, then $f(x_0 + y) - f(x_0) \le M$.

2.6 Show that $F\colon X \to \mathbb{R} \cup \{+\infty\}$ is convex if and only if epi F is convex (as a set).

2.7 Show that $F\colon X \to \mathbb{R} \cup \{+\infty\}$ is (sequentially) lower semicontinuous if and only if epi F is (sequentially) closed.

2.8 Prove the statements of Proposition 2.21.

2.9 Verify the statements in Example 2.22 about the support function.

2.10 Prove the assertion in Example 2.26.



Chapter 3
Variations

Having seen in the previous chapter how to abstractly prove the existence of solutions to some minimization problems, we now turn to the question of how to compute them (explicitly or numerically). The idea is the following: Let $\mathcal{F}\colon \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m) \to [0,\infty)$ be a functional with minimizer $u_* \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$. Take a path $t \mapsto u_t \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ ($t \in \mathbb{R}$) with $u_0 = u_*$ and consider the behavior of the map

$$t \mapsto \mathcal{F}[u_t]$$

around $t = 0$. If $t \mapsto \mathcal{F}[u_t]$ is differentiable at $t = 0$, then its derivative at $t = 0$ must vanish because of the minimization property. This is analogous to the elementary fact that if $g \in \mathrm{C}^1(\mathbb{R})$ takes its minimum at 0, then $g'(0) = 0$.

The **first variation** $\delta\mathcal{F}[u]$ of $\mathcal{F}$ at $u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ is the linear map

$$\delta\mathcal{F}[u]\colon \mathrm{C}^\infty_c(\Omega;\mathbb{R}^m) \to \mathbb{R}$$

defined as

$$\delta\mathcal{F}[u][\psi] := \lim_{h\downarrow 0} \frac{\mathcal{F}[u+h\psi] - \mathcal{F}[u]}{h}, \qquad \psi \in \mathrm{C}^\infty_c(\Omega;\mathbb{R}^m), \tag{3.1}$$

whenever this is well-defined. By the argument above,

$$\delta\mathcal{F}[u_*] = 0$$

at every minimizer u_* (assuming the variation exists). This yields a partial differential equation, called the *Euler–Lagrange equation*, which minimizers necessarily satisfy in the weak sense. Furthermore, under an additional convexity assumption on $\mathcal{F}$, the Euler–Lagrange equation turns out to be also sufficient for a map to be a minimizer. Thus, at least for well-behaved and convex problems, we can find a minimizer by solving the Euler–Lagrange equation. These "calculations with variations" gave the field its name.

A related topic, which we touch upon in this chapter, is the *regularity theory* of minimizers. It turns out that for so-called *regular variational integrals*, minimizers are always smooth. This is the famous solution to David Hilbert's 19th problem by En-

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F. Rindler, *Calculus of Variations*, Graduate Texts in Mathematics 310,
https://doi.org/10.1007/978-3-032-24483-3_3

nio De Giorgi and John F. Nash, which we briefly outline (without proving the more technical aspects).

If we assume a side constraint as in Section 2.5, the paths $t \mapsto u_t$ above have to take into account this side constraint as well, which leads to the statement that the minimizer satisfies a generalization of the Euler–Lagrange equation, which involves a so-called *Lagrange multiplier*.

Finally, we discuss invariances of the integral functional, that is, nontrivial paths $t \mapsto u_t$ along which $t \mapsto \mathcal{F}[u_t]$ is constant. This leads to a famous theorem by Emmy Noether, which exposes "hidden" conservation laws in minimization problems.

3.1 The Euler–Lagrange Equation

Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand that is additionally assumed to be (totally) differentiable in the last argument. The **directional derivative** $\mathrm{D}_A f(x,v,A) \in \mathbb{R}^{m\times d}$ of $f(x,v,\bullet)$ at $A \in \mathbb{R}^{m\times d}$ in direction $B \in \mathbb{R}^{m\times d}$ is then defined via

$$\mathrm{D}_A f(x,v,A) : B := \lim_{h\downarrow 0} \frac{f(x,v,A+hB) - f(x,v,A)}{h}.$$

If f is differentiable in its second argument, then we likewise define the corresponding directional derivative $\mathrm{D}_v f(x,v,A)\cdot w$ at $v \in \mathbb{R}^m$ in direction $w \in \mathbb{R}^m$. Our notation indicates that the dependence on the direction is linear and the pairing is given by a scalar product, namely the Frobenius product ":" (see Appendix A.1) for $\mathrm{D}_A f$ and the usual Euclidean scalar product "$\cdot$" for $\mathrm{D}_v f$. Indeed, in this sense $\mathrm{D}_A f(x,v,A) \in \mathbb{R}^{m\times d}$ and $\mathrm{D}_v f(x,v,A) \in \mathbb{R}^m$ are given as

$$\mathrm{D}_A f(x,v,A) := \big(\partial_{A_k^j} f(x,v,A)\big)_k^j, \qquad \mathrm{D}_v f(x,v,A) := \big(\partial_{v^j} f(x,v,A)\big)^j,$$

respectively.

We remark that this representation is only true if we a priori assume the (total) differentiability in the respective argument. As is well known, even the linearity is false without this assumption, as can be seen for instance via the classical example

$$f(a,b) := \begin{cases} \dfrac{a^3}{a^2+b^2} & \text{if } (a,b) \neq (0,0), \\ 0 & \text{if } (a,b) = (0,0), \end{cases}$$

whose directional derivative at the origin in direction $(u,v) \neq (0,0)$ exists and equals $u^3/(u^2+v^2)$, which is not linear in (u,v).

The following theorem furnishes the connection between the Calculus of Variations and PDE theory. It is very useful if we want to actually *compute* minimizers, either explicitly or numerically.

Theorem 3.1 *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand that is continuously differentiable in the second and third arguments and that satisfies the growth bounds*

$$|\mathrm{D}_v f(x,v,A)|, |\mathrm{D}_A f(x,v,A)| \le C(1+|v|^{p-1}+|A|^{p-1}) \tag{3.2}$$

for $(x,v,A) \in \Omega\times\mathbb{R}^m\times\mathbb{R}^{m\times d}$, a constant $C>0$, and $p\in[1,\infty)$. If $u_* \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ minimizes the functional

$$\mathcal{F}[u] := \int_\Omega f(x,u(x),\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m),$$

where $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$, then u_* is a weak solution of the ***Euler–Lagrange equation***

$$\begin{cases} -\operatorname{div}\big[\mathrm{D}_A f(x,u,\nabla u)\big] + \mathrm{D}_v f(x,u,\nabla u) = 0 & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{3.3}$$

Here, $u_* \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is called a **weak solution** of (3.3) if

$$\int_\Omega \mathrm{D}_A f(x,u_*,\nabla u_*) : \nabla\psi + \mathrm{D}_v f(x,u_*,\nabla u_*)\cdot\psi \,\mathrm{d}x = 0 \tag{3.4}$$

for all $\psi \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$, where we used the common convention to omit the x-arguments whenever this does not cause any confusion in order to curtail the proliferation of x's, for example in $f(x,u,\nabla u) := f(x,u(x),\nabla u(x))$. The boundary condition $u=g$ on $\partial\Omega$ in (3.3) is to be understood in the sense of trace. Note that the Euler–Lagrange "equation" is actually a *system* of PDEs (or, even more precisely, a *boundary value problem* for this system).

Proof. For all $\psi \in \mathrm{C}^\infty_c(\Omega;\mathbb{R}^m)$ and all $h>0$ we have

$$\mathcal{F}[u_*] \le \mathcal{F}[u_*+h\psi]$$

since $u_*+h\psi \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ is admissible in the minimization. Thus,

$$\begin{aligned} 0 &\le \int_\Omega \frac{f(x,u_*+h\psi,\nabla u_*+h\nabla\psi)-f(x,u_*,\nabla u_*)}{h}\,\mathrm{d}x \\ &= \int_\Omega \frac{1}{h}\int_0^1 \frac{\mathrm{d}}{\mathrm{d}t}\big[f(x,u_*+th\psi,\nabla u_*+th\nabla\psi)\big]\,\mathrm{d}t\,\mathrm{d}x \\ &= \int_\Omega\int_0^1 \mathrm{D}_A f(x,u_*+th\psi,\nabla u_*+th\nabla\psi) : \nabla\psi \\ &\qquad + \mathrm{D}_v f(x,u_*+th\psi,\nabla u_*+th\nabla\psi)\cdot\psi \,\mathrm{d}t\,\mathrm{d}x. \end{aligned}$$

Note that the last integrand is measurable since we assumed continuous differentiability of f in the second and third argument (analogously to Lemma 2.4).

By the growth bounds (3.2) on the derivative, the integrand can be seen to have an h-uniform majorant, namely $C(1 + |u_*| + |\nabla u_*|)^{p-1} \in \mathrm{L}^1(\Omega)$ for some $C > 0$, if we additionally require $h \leq 1$. So, we may apply the Lebesgue dominated convergence theorem to let $h \downarrow 0$ under the double integral. This yields

$$0 \leq \int_\Omega \mathrm{D}_A f(x, u_*, \nabla u_*) : \nabla\psi + \mathrm{D}_v f(x, u_*, \nabla u_*) \cdot \psi \, \mathrm{d}x$$

and we conclude (3.4) for $\psi \in \mathrm{C}_c^\infty(\Omega; \mathbb{R}^m)$ by taking ψ and $-\psi$ in this inequality.

In order to extend the validity of the weak formulation (3.4) to $\psi \in \mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$ we employ a density argument. Indeed, our growth bounds (3.2) imply that the linear functional $T \in \mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)^*$, defined for $\psi \in \mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$ via

$$\langle T, \psi \rangle := \int_\Omega \mathrm{D}_A f(x, u_*, \nabla u_*) : \nabla\psi + \mathrm{D}_v f(x, u_*, \nabla u_*) \cdot \psi \, \mathrm{d}x,$$

satisfies

$$\langle T, \psi \rangle = 0 \qquad \text{for all } \psi \in \mathrm{C}_c^\infty(\Omega; \mathbb{R}^m).$$

and

$$\begin{aligned} \big|\langle T, \psi \rangle\big| &\leq C \int_\Omega \big(1 + |u_*|^{p-1} + |\nabla u_*|^{p-1}\big)\big(|\psi| + |\nabla\psi|\big) \, \mathrm{d}x \\ &\leq C\big(1 + \|u_*\|_{\mathrm{L}^p}^{p-1} + \|\nabla u_*\|_{\mathrm{L}^p}^{p-1}\big)\|\psi\|_{\mathrm{W}^{1,p}}, \end{aligned}$$

where we have employed Hölder's inequality with the conjugate exponents $p/(p-1)$ and p in the last line.

Then, for any $\psi \in \mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$ take a sequence $(\psi_j)_j \subset \mathrm{C}_c^\infty(\Omega; \mathbb{R}^m)$ such that $\psi_j \to \psi$ in $\mathrm{W}^{1,p}$ (such a sequence always exists by the definition of $\mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$). As T is continuous on $\mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$,

$$\langle T, \psi \rangle = \lim_{j \to \infty} \langle T, \psi_j \rangle = 0$$

and (3.4) for $\psi \in \mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$ follows. □

Remark 3.2 The growth bounds (3.2) are fairly universally applicable and included in the hypotheses of several results in this chapter. However, on occasion they can be weakened, for example by utilizing the Sobolev space embedding and compactness results of Theorems A.28, A.29.

Using the notion of first variation, see (3.1), the assertion of Theorem 3.1 can be written as

$$\delta\mathcal{F}[u_*] = 0 \quad \text{if } u_* \text{ minimizes } \mathcal{F} \text{ over } \mathrm{W}_g^{1,p}(\Omega; \mathbb{R}^m).$$

Of course, this condition is only *necessary* for u to be a minimizer. In fact, any (weak) solution of the Euler–Lagrange equation (3.3) is called a **critical point** of $\mathcal{F}$, which

could be a minimizer, a maximizer, or a saddle point. However, under a convexity assumption, the solution to the Euler–Lagrange equation is always a minimizer:

Proposition 3.3 *In the situation of Theorem 3.1, assume furthermore that* $(v, A) \mapsto f(x, v, A)$ *is (jointly) convex for all fixed* $x \in \Omega$. *If* $u_* \in \mathrm{W}_g^{1,p}(\Omega; \mathbb{R}^m)$ *solves (3.3), then* u_* *is a minimizer of* $\mathcal{F}$.

Proof. Let $v \in \mathrm{W}_g^{1,p}(\Omega; \mathbb{R}^m)$ and set $\psi := v - u_* \in \mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$. Consider the function

$$g(t) := \mathcal{F}[u_* + t\psi], \qquad t \in \mathbb{R},$$

which inherits convexity from $\mathcal{F}$. By the same arguments as in the proof of Theorem 3.1, g is differentiable. Since u_* solves the Euler–Lagrange equation, we have

$$\frac{\mathrm{d}}{\mathrm{d}t} g(t)\bigg|_{t=0} = 0.$$

Then, from the convexity, the tangent to g at 0 lies below the graph of g, that is,

$$g(t) \geq g(0) + tg'(0) = g(0), \qquad t \geq 0.$$

Setting $t = 1$, we get $\mathcal{F}[v] \geq \mathcal{F}[u_*]$. As $v \in \mathrm{W}_g^{1,p}(\Omega; \mathbb{R}^m)$ was arbitrary, u_* must be a minimizer. □

Example 3.4 Returning to the Dirichlet functional from Example 2.8, we see that the associated Euler–Lagrange equation is the **Laplace equation**

$$-\Delta u = 0 \quad \text{in } \Omega,$$

where

$$\Delta := \partial_1^2 + \cdots + \partial_d^2$$

is the **Laplace operator**. Solutions u to $-\Delta u = 0$ are called **harmonic maps** (they are always strong solutions by Example 3.16 below, so there is no distinction between weak and strong harmonic maps). Since the Dirichlet functional is convex, by Proposition 3.3 all solutions of the Laplace equation are in fact minimizers of the Dirichlet functional. Furthermore, by Proposition 2.10, it can be seen that the Laplace equation has a unique solution for given boundary values.

Example 3.5 The same assertions as in the previous example also apply to the functional

$$\mathcal{F}[u] := \int_\Omega \frac{1}{2}|\nabla u(x)|^2 - h(x) \cdot u(x)\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega; \mathbb{R}^m),$$

where $h \in \mathrm{L}^2(\Omega; \mathbb{R}^m)$. Here, the Euler–Lagrange equation is the **Poisson equation**

$$-\Delta u = h \quad \text{in } \Omega.$$

Example 3.6 In the linearized elasticity problem from Example 2.12, we may compute the Euler–Lagrange equation to be

$$\begin{cases} -\operatorname{div}\left[2\mu\,\mathcal{E}u + \left(\kappa - \frac{2}{3}\mu\right)(\operatorname{tr}\mathcal{E}u)\mathrm{Id}\right] = b & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases}$$

One crucial consequence of Theorem 3.1 is that we can use all available PDE methods to study minimizers. In this respect we have the following prototypical result.

Proposition 3.7 *In the situation of Theorem 3.1, assume furthermore that f does not depend on v and is quadratic in A, i.e.,*

$$f(x,v,A) = f(x,A) = \frac{1}{2}A : \mathbf{S}(x)A, \qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d},$$

for a fourth-order symmetric tensor field $\mathbf{S}(x) = \mathbf{S}^{ik}_{jl}(x)$ ($x \in \Omega$), which acts on $A \in \mathbb{R}^{m\times d}$ via $(\mathbf{S}(x)A)^i_j := \sum_{k,l} \mathbf{S}^{ik}_{jl}(x)A^k_l$. Then, the Euler–Lagrange equation is the linear PDE

$$\begin{cases} -\operatorname{div}[\mathbf{S}\nabla u] = 0 & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases}$$

Moreover,

$$B : \mathbf{S}(x)B = \mathrm{D}^2_A f(x,A)[B,B] := \frac{\mathrm{d}^2}{\mathrm{d}t^2} f(x, A+tB)\Big|_{t=0}$$

for all $x \in \Omega$ and $A, B \in \mathbb{R}^{m\times d}$. Consequently, $\mathbf{S}(x)$ is positively semidefinite if and only if $f(x,\bullet)$ is convex. If this holds for all $x \in \Omega$, the above PDE is (possibly degenerate) ***elliptic****.*

Note that our requirement to be in the situation of Theorem 3.1 entails further restrictions on the tensor field $\mathbf{S}$, such as uniform boundedness and continuity. The proof is immediate from Theorem 3.1 and the relevant definitions.

We now use the Euler–Lagrange equation to find concrete solutions of variational problems.

Example 3.8 Recall the optimal saving problem from Section 1.5:

$$\begin{cases} \text{Minimize} & \mathcal{F}[S] := \displaystyle\int_0^T -\ln(1 + w + \varrho S(t) - \dot{S}(t))\,\mathrm{d}t \\ \text{subject to} & S(0) = 0,\ S(T) = S_T \geq 0,\ C(t) := w + \varrho S(t) - \dot{S}(t) \geq 0. \end{cases}$$

Since $a \mapsto -\ln(\beta - a)$ is strictly convex for any $\beta > 0$, we know from Proposition 2.10 that if a solution exists then it is unique. The Euler–Lagrange equation may be computed to be

$$-\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{1 + w + \varrho S(t) - \dot{S}(t)}\right] = \frac{\varrho}{1 + w + \varrho S(t) - \dot{S}(t)}. \tag{3.5}$$

Technically, Theorem 3.1 is not applicable since the validity of the growth-bounds is a priori unclear. However, around the solution that will be computed below, they are in fact satisfied and so we can a posteriori justify (3.5). By Proposition 3.3 (again, a posteriori justified), we get that this solution of the Euler–Lagrange equation is also a minimizer of our functional.

In order to find an explicit solution, we rearrange (3.5) into

$$\frac{\varrho\dot{S}(t) - \ddot{S}(t)}{1 + w + \varrho S(t) - \dot{S}(t)} = \varrho.$$

With the modified consumption rate $C_*(t) := 1 + C(t) = 1 + w + \varrho S(t) - \dot{S}(t)$, this is equivalent to

$$\frac{\dot{C}_*(t)}{C_*(t)} = \varrho,$$

and so, if $C(0) = C_0$ (to be determined later),

$$1 + w + \varrho S(t) - \dot{S}(t) = C_*(t) = \mathrm{e}^{\varrho t} C_*(0) = \mathrm{e}^{\varrho t}(1 + C_0).$$

This ordinary differential equation for $S(t)$ can be solved, for example via the Duhamel principle, which yields

$$\begin{aligned} S(t) &= \mathrm{e}^{\varrho t} \cdot 0 + \int_0^t \mathrm{e}^{\varrho(t-s)}(1 + w - \mathrm{e}^{\varrho s} - \mathrm{e}^{\varrho s} C_0)\,\mathrm{d}s \\ &= \frac{\mathrm{e}^{\varrho t} - 1}{\varrho}(1 + w) - t\mathrm{e}^{\varrho t}(1 + C_0), \end{aligned}$$

and C_0 can now be chosen to satisfy the terminal condition $S(T) = S_T$; in fact, $C_0 = (1 - \mathrm{e}^{-\varrho T})(1 + w)/(\varrho T) - \mathrm{e}^{-\varrho T} S_T/T - 1$.

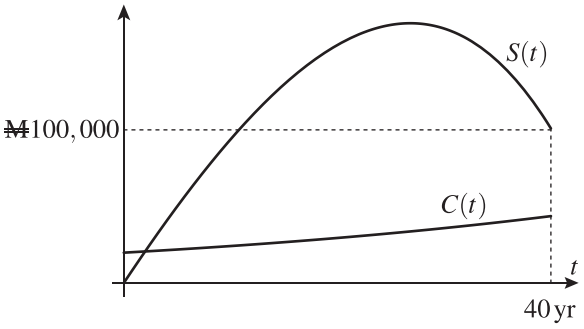


Fig. 3.1 The solution to the optimal saving problem

In Figure 3.1 we see the optimal saving strategy for a worker earning (constant) continuously-paid wages of $w = ₥30,000$ per year and having a savings goal of $S_T = ₥100,000$. The effective APR for savings is set at 2% per year ($\varrho = 0.0198$). The worker has to save for approximately 27 years, reaching savings of just over ₥168,000,

and then starts withdrawing his savings ($\dot{S}(t) < 0$) for the last 13 years. The worker's consumption $C(t) = \mathrm{e}^{\varrho t}(1 + C_0) - 1$ goes up continuously for their whole working life.

Example 3.9 For functions $u = u(t, x) \colon \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}$ consider the functional

$$\mathcal{F}[u] := \int_{\mathbb{R}} \int_{\mathbb{R}^d} \frac{1}{2}\big(-|\partial_t u|^2 + |\nabla_x u|^2\big)\, \mathrm{d}x\, \mathrm{d}t,$$

where $\nabla_x u$ is the gradient of u with respect to $x \in \mathbb{R}^d$. This functional may be interpreted as the usual Dirichlet functional with respect to the Lorentz metric. Then, the Euler–Lagrange equation is the **wave equation**

$$\partial_t^2 u - \Delta u = 0 \quad \text{in } \mathbb{R} \times \mathbb{R}^d.$$

Notice that the integrand of $\mathcal{F}$ is not convex and, indeed, the wave equation is not elliptic, but *hyperbolic*.

It is an important question whether a weak solution of the Euler–Lagrange equation (3.3) is also a **strong solution**, that is, whether $u \in (\mathrm{W}^{1,p} \cap \mathrm{W}^{2,p}_{\mathrm{loc}})(\Omega; \mathbb{R}^m)$ and

$$\begin{cases} -\operatorname{div}\big[\mathrm{D}_A f(x, u(x), \nabla u(x))\big] + \mathrm{D}_v f(x, u(x), \nabla u(x)) = 0 & \text{a.e. in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{3.6}$$

If $u \in \mathrm{C}^2(\Omega; \mathbb{R}^m) \cap \mathrm{C}(\overline{\Omega}; \mathbb{R}^m)$ satisfies this PDE for *every* $x \in \Omega$, then we call u a **classical solution**.

Proposition 3.10 *Let $f \colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m \times d} \to [0, \infty)$ be as in Theorem 3.1. A map $u \in (\mathrm{W}^{1,p} \cap \mathrm{W}^{2,p}_{\mathrm{loc}})(\Omega; \mathbb{R}^m)$ is a weak solution of the Euler–Lagrange equation (3.3) if and only if it is a strong solution.*

Proof. Let first $u \in (\mathrm{W}^{1,p} \cap \mathrm{W}^{2,p}_{\mathrm{loc}})(\Omega; \mathbb{R}^m)$ be a strong solution of the Euler–Lagrange equation, that is, let (3.6) hold. Multiplying (3.6) by a test function $\psi \in \mathrm{W}^{1,p}_0(\Omega; \mathbb{R}^m)$, integrating over Ω, and using the Gauss–Green theorem, we arrive at

$$\int_\Omega \mathrm{D}_A f(x, u, \nabla u) : \nabla\psi + \mathrm{D}_v f(x, u, \nabla u) \cdot \psi \, \mathrm{d}x = 0.$$

So, u also solves (3.3) in the weak sense, i.e., (3.4) holds.

Conversely, if $u \in (\mathrm{W}^{1,p} \cap \mathrm{W}^{2,p}_{\mathrm{loc}})(\Omega; \mathbb{R}^m)$ is a weak solution, then we may integrate by parts in (3.4) for any $\psi \in \mathrm{C}^\infty_c(\Omega; \mathbb{R}^m)$ (or, more precisely, use the definition of weak derivatives) to obtain

$$\int_\Omega \big(-\operatorname{div}\big[\mathrm{D}_A f(x, u, \nabla u)\big] + \mathrm{D}_v f(x, u, \nabla u)\big) \cdot \psi \, \mathrm{d}x = 0.$$

We conclude using the following so-called *Fundamental Lemma* of the Calculus of Variations. □

Lemma 3.11 *Let $\Omega \subset \mathbb{R}^d$ be open. If $g \in \mathrm{L}^1(\Omega)$ satisfies*

$$\int_\Omega g\psi \, \mathrm{d}x = 0 \qquad \text{for all } \psi \in \mathrm{C}_c^\infty(\Omega),$$

then $g \equiv 0$ almost everywhere.

Proof. Let g be extended by zero to all of $\mathbb{R}^d$. Take a family $(\eta_\delta)_{\delta>0}$ of mollifiers such that $\operatorname{supp}\eta_\delta \subset B_\delta$ for all $\delta > 0$, see Appendix A.5, and use the assumption for $\psi(x) := \eta_\delta(y - x)$, where $y \in \Omega$, $\delta > 0$, to get

$$(\eta_\delta \star g)(y) = \int \eta_\delta(y - x)g(x) \, \mathrm{d}x = 0$$

as soon as $\delta < \operatorname{dist}(y, \partial\Omega)$. By Lemma A.32, $\eta_\delta \star g \to g$ as $\delta \downarrow 0$ in $\mathrm{L}^1_{\mathrm{loc}}(\Omega)$. Thus, $g \equiv 0$ almost everywhere. □

3.2 Regularity of Minimizers

We saw at the end of the last section that if a weak solution of the Euler–Lagrange equation has higher regularity (differentiability), then it is also a strong or even a classical solution. Thus, one would like to know how much regularity one can expect from solutions of a variational problem. Such a question was the content of Hilbert's 19th problem [180]:

> *Does every Lagrangian partial differential equation of a regular variational problem have the property of exclusively admitting analytic integrals?*[1]

In modern language, Hilbert asked whether "regular" variational problems (defined below) admit only analytic solutions, i.e., solutions that have a local power series representation.

In this section, we will prove some basic regularity assertions, but we will only sketch the solution of Hilbert's 19th problem as the techniques needed are quite involved. We remark at the outset that many regularity results are very sensitive to the dimensions of the domain and the target space. In particular, the behavior of the scalar case ($m = 1$) and the vector case ($m > 1$) is fundamentally different.

In the spirit of Hilbert's 19th problem, call

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x)) \, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega; \mathbb{R}^m),$$

a **regular variational integral** if $f\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is twice continuously differentiable and there are constants $\mu, M > 0$ with

$$\mu|B|^2 \le \mathrm{D}^2 f(A)[B, B] \le M|B|^2, \qquad A, B \in \mathbb{R}^{m\times d}, \tag{3.7}$$

[1] The German original asks "ob jede Lagrangesche partielle Differentialgleichung eines regulären Variationsproblems die Eigenschaft hat, daß sie nur analytische Integrale zuläßt."

where

$$\mathrm{D}^2 f(A)[B,B] := \frac{\mathrm{d}^2}{\mathrm{d}t^2} f(A+tB)\bigg|_{t=0}.$$

Clearly, regular variational problems are convex. In fact, integrands f that satisfy the lower bound in (3.7) are called **strongly convex**. For example, the Dirichlet functional from Example 2.8 is a regular variational integral.

Since f is twice continuously differentiable,

$$\mathrm{D}^2 f(A)[B_1,B_2] = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\mathrm{d}}{\mathrm{d}s} f(A+sB_1+tB_2)\bigg|_{s,t=0}, \qquad A, B_1, B_2 \in \mathbb{R}^{m\times d},$$

is a symmetric bilinear form in B_1 and B_2. One checks that for $B_1 = B_2 = B$ this agrees with $\mathrm{D}^2 f(A)[B,B]$ as defined above. It also follows from (3.7) using basic linear algebra (e.g., the spectral theorem or the singular value decomposition) that the operator norm of $\mathrm{D}^2 f(A)$ is bounded from above by M. Then,

$$\left|\mathrm{D}^2 f(A)[B_1,B_2]\right| \le M|B_1||B_2| \tag{3.8}$$

and, by the mean value theorem, we further get that $\mathrm{D}f$ is Lipschitz continuous, that is,

$$|\mathrm{D}f(A_1) - \mathrm{D}f(A_2)| \le M|A_1 - A_2|, \qquad A_1, A_2 \in \mathbb{R}^{m\times d}. \tag{3.9}$$

In particular,

$$|\mathrm{D}f(A)| \le C(1+|A|), \qquad A \in \mathbb{R}^{m\times d}, \tag{3.10}$$

for a constant $C > 0$.

The fundamental $\mathrm{W}^{2,2}_{\mathrm{loc}}$-regularity theorem for regular variational integrals is the following.

Theorem 3.12 *Let $\mathcal{F}$ be a regular variational integral. Then, for any minimizer $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ of $\mathcal{F}$ (under any boundary conditions), or, more generally, any weak solution $u_* \in \mathrm{W}^{1,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$ of the corresponding Euler–Lagrange equation, it holds that*

$$u_* \in \mathrm{W}^{2,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m).$$

*Moreover, for any ball $B_{3r}(x_0) \subset \Omega$ the **Caccioppoli inequality***

$$\int_{B_r(x_0)} |\nabla^2 u_*(x)|^2 \,\mathrm{d}x \le \left(\frac{2M}{\mu}\right)^2 \int_{B_{3r}(x_0)} \frac{|\nabla u_*(x) - [\nabla u_*]_{B_{3r}(x_0)}|^2}{r^2} \,\mathrm{d}x \tag{3.11}$$

holds. Consequently, the Euler–Lagrange equation is satisfied strongly,

$$-\operatorname{div} \mathrm{D}f(\nabla u_*) = 0 \quad \text{a.e. in } \Omega.$$

Here,

$$[\nabla u_*]_{B_{3r}(x_0)} := \fint_{B_{3r}(x_0)} \nabla u_* \,\mathrm{d}x = \frac{1}{\omega_d (3r)^d} \int_{B_{3r}(x_0)} \nabla u_* \,\mathrm{d}x,$$

with $\omega_d := |B_1|$ the volume of the d-dimensional unit ball B_1.

Before we come to the proof of the theorem, let us explain the idea by establishing the Caccioppoli inequality (3.11) *assuming* that $u_* \in \mathrm{C}^\infty(\overline{\Omega}; \mathbb{R}^m)$. In this case, for any ball $B_{3r}(x_0) \subset \Omega$ ($x_0 \in \Omega$, $r > 0$) take a cut-off function $\varrho \in \mathrm{C}_c^\infty(\Omega; [0,1])$ such that

$$\mathbb{1}_{B_r(x_0)} \le \varrho \le \mathbb{1}_{B_{2r}(x_0)} \qquad \text{and} \qquad |\nabla \varrho| \le \frac{\alpha}{r}$$

with $\alpha > 1$. Note that it is possible to construct such a ϱ for any $\alpha > 1$. In fact, if we allowed $\varrho \in \mathrm{W}^{1,\infty}(\Omega; [0,1])$, then we could even achieve $\alpha = 1$, but we need the existence of ϱ's second derivatives in the computation below (even though the estimates do not depend on them). Then test the (weak) Euler–Lagrange equation (3.4) with

$$\psi := \partial_k [\varrho^2 \partial_k (u_* - a)]$$

for some to be determined affine map $a \colon \mathbb{R}^d \to \mathbb{R}^m$ and any $k = 1, \dots, d$. Using integration by parts and the interchangeability of derivatives acting on smooth maps,

$$\begin{aligned}
0 &= -\int_\Omega \mathrm{D}f(\nabla u_*) : \nabla\big(\partial_k [\varrho^2 \partial_k (u_* - a)]\big) \,\mathrm{d}x \\
&= \int_\Omega \partial_k (\mathrm{D}f(\nabla u_*)) : \big[\varrho^2 \partial_k \nabla u_* + \partial_k (u_* - a) \otimes \nabla(\varrho^2)\big] \,\mathrm{d}x \\
&= \int_\Omega \varrho^2 \mathrm{D}^2 f(\nabla u_*)[\partial_k \nabla u_*, \partial_k \nabla u_*] \,\mathrm{d}x \\
&\qquad + \int_\Omega \partial_k (\mathrm{D}f(\nabla u_*)) : [\partial_k (u_* - a) \otimes \nabla(\varrho^2)] \,\mathrm{d}x.
\end{aligned}$$

We remark that the tensor product "$\otimes$" in the last line is technically incorrect since we are multiplying a column vector with a row vector, but we include it here and in the following to signify that the result is a *matrix*. Then, using the bounds (3.7), (3.8) on $\mathrm{D}^2 f$ and Young's inequality,

$$\begin{aligned}
\mu \int_\Omega \varrho^2 |\partial_k \nabla u_*|^2 \,\mathrm{d}x &\le \int_\Omega \varrho^2 \mathrm{D}^2 f(\nabla u_*)[\partial_k \nabla u_*, \partial_k \nabla u_*] \,\mathrm{d}x \\
&= -\int_\Omega \partial_k (\mathrm{D}f(\nabla u_*)) : [\partial_k (u_* - a) \otimes \nabla(\varrho^2)] \,\mathrm{d}x \\
&= -\int_\Omega \mathrm{D}^2 f(\nabla u_*) : [\partial_k (u_* - a) \otimes \nabla(\varrho^2), \partial_k \nabla u_*] \,\mathrm{d}x \\
&\le 2M \int_\Omega \varrho |\partial_k \nabla u_*| \cdot |\partial_k (u_* - a)| \cdot |\nabla \varrho| \,\mathrm{d}x
\end{aligned}$$

$$\leq 2M\alpha \int_\Omega |\partial_k \nabla u_*| \cdot \frac{|\partial_k (u_* - a)|}{r} \, \mathrm{d}x$$
$$\leq \frac{\mu\alpha}{2} \int_\Omega |\partial_k \nabla u_*|^2 \, \mathrm{d}x + \frac{2M^2\alpha}{\mu} \int_\Omega \frac{|\partial_k (u_* - a)|^2}{r^2} \, \mathrm{d}x.$$

We then let $\alpha \downarrow 1$ and absorb the first term on the right-hand side into the left-hand side to infer that

$$\frac{\mu}{2} \int_{B_r(x_0)} |\partial_k \nabla u_*|^2 \, \mathrm{d}x \leq \frac{2M^2}{\mu} \int_{B_{3r}(x_0)} \frac{|\partial_k (u_* - a)|^2}{r^2} \, \mathrm{d}x.$$

Multiplying by $2/\mu$, summing over k, and choosing a with $\nabla a = [\nabla u_*]_{B_{3r}(x_0)}$, we arrive at (3.11). This shows that for minimizers that are assumed to be smooth a priori, the first-order derivatives control the second order derivatives. Such a type of assertion, whereby a higher-regularity quantity is controlled by a lower-regularity quantity (albeit on a larger ball), is the key ingredient in a large number of regularity theorems. It just remains to make use of this information without using any a priori assumption of higher regularity.

To this end, we will employ the *difference quotient method*, which is very useful technical tool in regularity theory. For $u \colon \Omega \to \mathbb{R}^m$, define the k'th **difference quotient**, $k = 1, \ldots, d$, of u at $x \in \Omega$ with height $h \in \mathbb{R} \setminus \{0\}$ to be

$$\mathrm{D}_k^h u(x) := \frac{u(x + h\mathrm{e}_k) - u(x)}{h},$$

where $\{\mathrm{e}_1, \ldots, \mathrm{e}_d\}$ is the standard basis of $\mathbb{R}^d$. We also set

$$\mathrm{D}^h u := (\mathrm{D}_1^h u, \ldots, \mathrm{D}_d^h u).$$

The central result for the difference quotient method is the following characterization of Sobolev spaces.

Lemma 3.13 *Let $D \Subset \Omega \subset \mathbb{R}^d$ be open sets, $p \in (1, \infty)$, and $u \in \mathrm{L}^p(\Omega; \mathbb{R}^m)$.*

(i) If $u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$, then

$$\|\mathrm{D}_k^h u\|_{\mathrm{L}^p(D)} \leq \|\partial_k u\|_{\mathrm{L}^p(\Omega)} \qquad \text{for all } k = 1, \ldots, d, \ |h| < \operatorname{dist}(D, \partial\Omega).$$

(ii) If for some $0 < \delta < \operatorname{dist}(D, \partial\Omega)$ and a fixed $C > 0$ it holds that

$$\|\mathrm{D}_k^h u\|_{\mathrm{L}^p(D)} \leq C \qquad \text{for all } k = 1, \ldots, d \text{ and all } |h| < \delta,$$

then $u \in \mathrm{W}^{1,p}(D; \mathbb{R}^m)$ and $\|\partial_k u\|_{\mathrm{L}^p(D)} \leq C$ for all $k = 1, \ldots, d$.

Proof. For (i) assume first that $u \in (\mathrm{W}^{1,p} \cap \mathrm{C}^1)(\Omega; \mathbb{R}^m)$. In this case, by the Fundamental Theorem of Calculus, at $x \in \Omega$ it holds that

$$\mathrm{D}_k^h u(x) = \frac{1}{h} \int_0^1 \frac{\mathrm{d}}{\mathrm{d}t} u(x + th\mathrm{e}_k) \, \mathrm{d}t = \int_0^1 \partial_k u(x + th\mathrm{e}_k) \, \mathrm{d}t.$$

Thus, by Jensen's inequality (see Lemma A.19),

$$\int_D |\mathrm{D}_k^h u(x)|^p \,\mathrm{d}x \le \int_D \int_0^1 |\partial_k u(x + the_k)|^p \,\mathrm{d}t \,\mathrm{d}x \le \int_\Omega |\partial_k u(x)|^p \,\mathrm{d}x,$$

from which the assertion is clear. The general case follows from the density of $(\mathrm{W}^{1,p} \cap \mathrm{C}^1)(\Omega;\mathbb{R}^m)$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.

For (ii), we first observe that for fixed $k = 1,\dots,d$ the family $(\mathrm{D}_k^h u)_{0<h<\delta}$ is uniformly norm-bounded in $\mathrm{L}^p(D;\mathbb{R}^m)$ by assumption. Thus, for an arbitrary fixed sequence of h's tending to zero, there exists a subsequence $h_j \downarrow 0$ with

$$\mathrm{D}_k^{h_j} u \rightharpoonup v_k \quad \text{in } \mathrm{L}^p$$

for some $v_k \in \mathrm{L}^p(D;\mathbb{R}^m)$. Let $\psi \in \mathrm{C}_c^\infty(D;\mathbb{R}^m)$. Using an "integration-by-parts" rule for difference quotients, which is elementary to check, we get

$$\begin{aligned}\int_D v_k \cdot \psi \,\mathrm{d}x &= \lim_{j\to\infty} \int_\Omega \mathrm{D}_k^{h_j} u \cdot \psi \,\mathrm{d}x \\ &= -\lim_{j\to\infty} \int_\Omega u \cdot \mathrm{D}_k^{-h_j} \psi \,\mathrm{d}x \\ &= -\int_D u \cdot \partial_k \psi \,\mathrm{d}x.\end{aligned}$$

Thus, $u \in \mathrm{W}^{1,p}(D;\mathbb{R}^m)$ and $v_k = \partial_k u$. The norm estimate follows from the lower semicontinuity of the norm under weak convergence (which is well-known in functional analysis, but can also be proved using the Tonelli–Serrin Theorem 2.6). □

Proof of Theorem 3.12. The idea is to emulate the a priori non-existent second derivatives using difference quotients and to derive estimates which allow one to conclude that these difference quotients are uniformly (in the height) bounded in L^2. Then we can conclude by the preceding lemma.

Let $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ be a minimizer of $\mathcal{F}$. By Theorem 3.1 (where (3.2) is satisfied by (3.10)),

$$0 = \int_\Omega \mathrm{D}f(\nabla u_*) : \nabla\psi \,\mathrm{d}x \qquad \text{for all } \psi \in \mathrm{W}_0^{1,2}(\Omega;\mathbb{R}^m). \tag{3.12}$$

Fix a ball $B_{3r}(x_0) \subset \Omega$ and take a Lipschitz cut-off function $\varrho \in \mathrm{W}_0^{1,\infty}(\Omega;[0,1])$ such that

$$\mathbb{1}_{B_r(x_0)} \le \varrho \le \mathbb{1}_{B_{2r}(x_0)} \qquad \text{and} \qquad |\nabla\varrho| \le \frac{1}{r}.$$

Then, for any $k = 1,\dots,d$ and $|h| < r$ we let

$$\psi := \mathrm{D}_k^{-h}\left[\varrho^2 \mathrm{D}_k^h (u_* - a)\right] \in \mathrm{W}_0^{1,2}(\Omega;\mathbb{R}^m),$$

where $a\colon \mathbb{R}^d \to \mathbb{R}^m$ is an affine function to be chosen later. We may plug this ψ

into (3.12) to get

$$0 = \int_\Omega \mathrm{D}_k^h(\mathrm{D}f(\nabla u_*)) : \left[\varrho^2 \mathrm{D}_k^h \nabla u_* + \mathrm{D}_k^h(u_* - a) \otimes \nabla(\varrho^2)\right] \mathrm{d}x. \tag{3.13}$$

Here, we again used the "integration-by-parts" formula for difference quotients from the proof of Lemma 3.13.

Next, we estimate, using the assumptions on f,

$$\begin{aligned}
\mu|\mathrm{D}_k^h \nabla u_*|^2 &\le \int_0^1 \mathrm{D}^2 f(\nabla u_* + th\mathrm{D}_k^h \nabla u_*)[\mathrm{D}_k^h \nabla u_*, \mathrm{D}_k^h \nabla u_*] \,\mathrm{d}t \\
&= \frac{1}{h} \mathrm{D}f(\nabla u_* + th\mathrm{D}_k^h \nabla u_*) : \mathrm{D}_k^h \nabla u_* \Big|_{t=0}^1 \\
&= \mathrm{D}_k^h(\mathrm{D}f(\nabla u_*)) : \mathrm{D}_k^h \nabla u_*,
\end{aligned}$$

where for the last line we note that

$$\begin{aligned}
\mathrm{D}_k^h(\mathrm{D}f(\nabla u_*))(x) &= \frac{\mathrm{D}f(\nabla u_*(x + h\mathrm{e}_k)) - \mathrm{D}f(\nabla u_*(x))}{h} \\
&= \frac{\mathrm{D}f(\nabla u_*(x) + h\mathrm{D}_k^h \nabla u_*(x)) - \mathrm{D}f(\nabla u_*(x))}{h}.
\end{aligned}$$

On the other hand, using the Cauchy–Schwarz and Young inequalities,

$$\begin{aligned}
&\left|\mathrm{D}_k^h(\mathrm{D}f(\nabla u_*)) : \left[\mathrm{D}_k^h(u_* - a) \otimes \nabla(\varrho^2)\right]\right| \\
&\quad \le 2\varrho |\mathrm{D}_k^h(\mathrm{D}f(\nabla u_*))| \cdot |\mathrm{D}_k^h(u_* - a)| \cdot |\nabla \varrho| \\
&\quad \le \frac{\mu}{2M^2} \varrho^2 |\mathrm{D}_k^h(\mathrm{D}f(\nabla u_*))|^2 + \frac{2M^2}{\mu} |\mathrm{D}_k^h(u_* - a)|^2 \cdot |\nabla \varrho|^2.
\end{aligned}$$

From the last two estimates and (3.13) we get

$$\begin{aligned}
&\mu \int_\Omega |\mathrm{D}_k^h \nabla u_*|^2 \varrho^2 \,\mathrm{d}x \\
&\quad \le \int_\Omega \mathrm{D}_k^h(\mathrm{D}f(\nabla u_*)) : [\varrho^2 \mathrm{D}_k^h \nabla u_*] \,\mathrm{d}x \\
&\quad = -\int_\Omega \mathrm{D}_k^h(\mathrm{D}f(\nabla u_*)) : \left[\mathrm{D}_k^h(u_* - a) \otimes \nabla(\varrho^2)\right] \mathrm{d}x \\
&\quad \le \int_\Omega \frac{\mu}{2M^2} \varrho^2 |\mathrm{D}_k^h(\mathrm{D}f(\nabla u_*))|^2 + \frac{2M^2}{\mu} |\mathrm{D}_k^h(u_* - a)|^2 \cdot |\nabla \varrho|^2 \,\mathrm{d}x \\
&\quad \le \int_\Omega \frac{\mu}{2} \varrho^2 |\mathrm{D}_k^h \nabla u_*|^2 + \frac{2M^2}{\mu} |\mathrm{D}_k^h(u_* - a)|^2 \cdot |\nabla \varrho|^2 \,\mathrm{d}x,
\end{aligned} \tag{3.14}$$

where in the last line we used the Lipschitz continuity (3.9) of $\mathrm{D}f$ to estimate

$$\begin{aligned}\left|\mathrm{D}_k^h(\mathrm{D}f(\nabla u_*))(x)\right| &= \left|\frac{\mathrm{D}f(\nabla u_*(x+he_k)) - \mathrm{D}f(\nabla u_*(x))}{h}\right| \\ &= \left|\frac{\mathrm{D}f(\nabla u_*(x) + h\mathrm{D}_k^h\nabla u_*(x)) - \mathrm{D}f(\nabla u_*(x))}{h}\right| \\ &\le M|\mathrm{D}_k^h\nabla u_*(x)|.\end{aligned}$$

Absorbing the first term on the right-hand side of (3.14) into the left-hand side and using the properties of ϱ, we arrive at

$$\int_{B_r(x_0)} |\mathrm{D}_k^h\nabla u_*|^2\,\mathrm{d}x \le \left(\frac{2M}{\mu}\right)^2 \int_{B_{2r}(x_0)} \frac{|\mathrm{D}_k^h(u_*-a)|^2}{r^2}\,\mathrm{d}x.$$

Now invoke the difference-quotient lemma, part (i), to deduce that

$$\int_{B_r(x_0)} |\mathrm{D}_k^h\nabla u_*|^2\,\mathrm{d}x \le \left(\frac{2M}{\mu}\right)^2 \int_{B_{3r}(x_0)} \frac{|\partial_k(u_*-a)|^2}{r^2}\,\mathrm{d}x.$$

Applying the difference-quotient lemma again, this time part (ii), we get $u_* \in \mathrm{W}^{2,2}(B_r(x_0);\mathbb{R}^m)$. The Caccioppoli inequality (3.11) follows once we take a with $\nabla a := [\nabla u_*]_{B_{3r}(x_0)}$. □

The $\mathrm{W}^{2,2}_{\mathrm{loc}}$-regularity of solutions can be extended up to and including the boundary if the boundary is smooth enough, yielding a full $\mathrm{W}^{2,2}$-regularity theorem, see Section 6.3.2 in [133] and also [158].

In the special case when f is quadratic, say $\mathrm{D}^2 f(x,v,A) = \mathbf{S}$ for a symmetric and strictly positive-definite fourth-order tensor $\mathbf{S}$, we can iterate, or *bootstrap*, the regularity arguments to conclude the smoothness of a minimizer. Indeed, with the $\mathrm{W}^{2,2}_{\mathrm{loc}}$-regularity result at hand, we may use $\psi = -\partial_k\widetilde{\psi}$ for $\widetilde{\psi} \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m)$, $k = 1,\dots,d$, as test function in the weak formulation of the Euler–Lagrange equation $-\operatorname{div}[\mathrm{D}f(\nabla u_*)] = 0$ to conclude that

$$-\operatorname{div}\big[\mathbf{S}\nabla(\partial_k u_*)\big] = 0 \quad \text{in } \Omega \tag{3.15}$$

holds in the weak sense, i.e.,

$$0 = -\int_\Omega \mathrm{D}f(\nabla u_*) : \nabla(\partial_k\widetilde{\psi})\,\mathrm{d}x = \int_\Omega \big[\mathbf{S}\nabla(\partial_k u_*)\big] : \nabla\widetilde{\psi}\,\mathrm{d}x$$

for all $\widetilde{\psi} \in \mathrm{C}_c^\infty(\Omega;\mathbb{R}^m)$. However, this PDE is itself the Euler–Lagrange equation for the integral functional

$$\int_\Omega \frac{1}{2}\nabla(\partial_k u(x)) : \mathbf{S}\nabla(\partial_k u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{2,2}(\Omega;\mathbb{R}^m).$$

Applying Theorem 3.12 again, we get that $\partial_k u \in \mathrm{W}^{2,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$. Iterating this procedure, we obtain the following higher-regularity result.

Corollary 3.14 *Let $\mathcal{F}$ be a quadratic regular variational integral. Then, for any minimizer $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ of $\mathcal{F}$ it holds that $u_* \in \mathrm{W}^{k,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$ for all $k \in \mathbb{N}$, hence also $u_* \in \mathrm{C}^\infty(\Omega;\mathbb{R}^m)$.*

We also state a regularity result for integral functionals with a lower-order term:

Theorem 3.15 *Let $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ satisfy the same assumptions as in Theorem 3.12 and let $h \in \mathrm{L}^2(\Omega;\mathbb{R}^m)$. Then, minimizers $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ of the functional*

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x)) - h(x)\cdot u(x)\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m),$$

lie in $\mathrm{W}^{2,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$ and satisfy the strong Euler–Lagrange equation

$$-\operatorname{div} \mathrm{D}f(\nabla u_*) = h \quad \text{a.e. in } \Omega.$$

If f is quadratic and $h \in \mathrm{C}^\infty(\Omega;\mathbb{R}^m)$, then $u_* \in \mathrm{C}^\infty(\Omega;\mathbb{R}^m)$.

The proof is the task of Problem 3.4 and an extension is in Problem 3.5.

Example 3.16 For a minimizer $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ of the Dirichlet functional as in Example 2.8, the theory presented so far immediately gives $u_* \in \mathrm{C}^\infty(\Omega;\mathbb{R}^m)$. By Theorem 3.15, the same applies to the functional

$$\mathcal{F}[u] := \int_\Omega \frac{1}{2}|\nabla u(x)|^2 - h(x)\cdot u(x)\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m),$$

where we assume $h \in \mathrm{C}^\infty(\Omega;\mathbb{R}^m)$. Thus, solutions $u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ of the Laplace equation

$$-\Delta u = 0 \quad \text{in } \Omega$$

or the Poisson equation

$$-\Delta u = h \quad \text{in } \Omega$$

with $h \in \mathrm{C}^\infty(\Omega;\mathbb{R}^m)$, are smooth.

Example 3.17 Similarly, for minimizers $u_* \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$ of the problem of linearized elasticity from Section 1.8 and Example 2.12, we also get $u_* \in \mathrm{C}^\infty(\Omega;\mathbb{R}^3)$. The reasoning has to be slightly adjusted, however, to take care of the fact that we are dealing with the symmetric gradient $\mathcal{E}u$ instead of ∇u.

However, for a non-quadratic regular variational integral the Euler–Lagrange equation is *nonlinear* and the bootstrapping procedure fails. Thus, for Hilbert's 19th problem, different methods had to be developed. Full solutions were given, in the scalar case $m = 1$, by Ennio De Giorgi in 1957 [104] and, using different methods, by John F. Nash in 1958 [258]. After the proof was improved by Jürgen Moser in 1960/1961 [241, 242], the results are now collectively referred to as *De Giorgi–Nash–Moser theory*.

The standard solution (following De Giorgi's approach) is based on the *De Giorgi regularity theorem* and the classical *Schauder estimates*, which establish Hölder regularity of weak solutions of a PDE.

Theorem 3.18 (De Giorgi 1957 [104]) *Let $S\colon \Omega \to \mathbb{R}^{d\times d}$ be measurable, symmetric, ($S(x) = S(x)^{\mathsf{T}}$ for $x \in \Omega$), and satisfy the ellipticity and boundedness estimates*

$$\mu|v|^2 \le v^{\mathsf{T}} S(x) v \le M|v|^2, \qquad (x, v) \in \Omega \times \mathbb{R}^d, \tag{3.16}$$

for constants $\mu, M > 0$. If $u \in \mathrm{W}^{1,2}(\Omega)$ is a weak solution of

$$-\operatorname{div}[S\nabla u] = 0, \tag{3.17}$$

then $u \in \mathrm{C}^{0,\alpha_0}_{\mathrm{loc}}(\Omega)$, that is, u is α_0-Hölder continuous, for some $\alpha_0 = \alpha_0(d, M/\mu) \in (0, 1)$.

The surprising fact here is that solutions have additional Hölder regularity even though only *measurability* is assumed on the x-dependence of S!

Theorem 3.19 (Schauder 1934/1937 [293, 294]) *Let $S\colon \Omega \to \mathbb{R}^{d\times d}$ be as above but in addition assumed to be of Hölder class $\mathrm{C}^{n-1,\alpha}$ for some $n \in \mathbb{N}$ and $\alpha \in (0, 1)$. If $u \in \mathrm{W}^{1,2}(\Omega)$ is a weak solution of*

$$-\operatorname{div}[S\nabla u] = 0,$$

then $u \in \mathrm{C}^{n,\alpha}_{\mathrm{loc}}(\Omega)$.

We remark that the Schauder estimates also hold for *systems* of PDEs ($m > 1$), but the De Giorgi regularity theorem does not. The proofs of these results are involved (see the notes section at the end of this chapter for some pointers to the literature), but we at least establish one of their most important consequences, namely the solution of Hilbert's 19th problem in the scalar case (with smoothness instead of analyticity, however):

Theorem 3.20 (De Giorgi 1957, Nash 1958, Moser 1960 [104, 241, 258]) *Let $\mathcal{F}$ be a regular variational integral with an integrand $f\colon \mathbb{R}^d \to [0, \infty)$ that is n times continuously differentiable, where $n \in \{2, 3, \ldots\}$. If $u_* \in \mathrm{W}^{1,2}(\Omega)$ minimizes $\mathcal{F}$, then $u_* \in \mathrm{C}^{n-1,\alpha}_{\mathrm{loc}}(\Omega)$ for some $\alpha \in (0, 1)$. In particular, if f is smooth, then $u_* \in \mathrm{C}^\infty(\Omega)$.*

Proof. We saw in (3.15) that the partial derivatives $\partial_k u_*$, $k = 1, \ldots, d$, of a minimizer $u_* \in \mathrm{W}^{1,2}(\Omega)$ of $\mathcal{F}$ satisfy (3.17) for

$$S(x) := \mathrm{D}^2 f(\nabla u_*(x)), \qquad x \in \Omega.$$

From general properties of the Hessian we conclude that $S(x)$ is symmetric and the upper and lower estimates (3.16) on S follow from the respective properties of $\mathrm{D}^2 f$. However, we cannot conclude (yet) any regularity of S beyond measurability. Nevertheless, we may apply the De Giorgi Regularity Theorem 3.18, whereby $\partial_k u_* \in \mathrm{C}^{0,\alpha_0}_{\mathrm{loc}}(\Omega)$ for some $\alpha_0 \in (0, 1)$ and all $k = 1, \ldots, d$. Hence $u_* \in \mathrm{C}^{1,\alpha_0}_{\mathrm{loc}}(\Omega)$. This is the assertion for $n = 2$.

If $n = 3$, our arguments so far in conjunction with the regularity assumptions on f imply $S(x) = \mathrm{D}^2 f(\nabla u_*(x)) \in \mathrm{C}^{0,\alpha_0}_{\mathrm{loc}}(\Omega)$. Indeed, $\mathrm{D}^2 f$ is locally Lipschitz and ∇u_* is locally α_0-Hölder continuous, hence the composite function is also locally α_0-Hölder continuous. Consequently, the Schauder estimates from Theorem 3.19 apply and yield $\partial_k u_* \in \mathrm{C}^{1,\alpha_0}_{\mathrm{loc}}(\Omega)$ for $k = 1, \dots, d$, whereby $u_* \in \mathrm{C}^{2,\alpha_0}_{\mathrm{loc}}(\Omega) = \mathrm{C}^{n-1,\alpha_0}_{\mathrm{loc}}(\Omega)$.

For higher n, this procedure can be iterated until we run out of f-derivatives and $u_* \in \mathrm{C}^{n-1,\alpha_0}_{\mathrm{loc}}(\Omega)$. □

A slight refinement of the above argument also yields analyticity of u_* if f is analytic. Another refinement shows that in the situation of the De Giorgi–Nash–Moser Theorem, we even have $u_* \in \mathrm{C}^{n-1,\alpha}_{\mathrm{loc}}(\Omega)$ for *all* $\alpha \in (0,1)$. Here one needs the additional Schauder–type result that weak solutions u to $-\operatorname{div}[S\nabla u] = 0$ for $S = S(x)$ that are *continuous* in x, have $\mathrm{C}^{0,\alpha}_{\mathrm{loc}}$-regularity for *all* $\alpha \in (0,1)$. In the above proof we can apply this result at stage $n = 2$ since $S(x) = \mathrm{D}^2 f(\nabla u_*(x))$ is in particular continuous by the De Giorgi Regularity Theorem. Thus, $u_* \in \mathrm{C}^{1,\alpha}_{\mathrm{loc}}(\Omega)$ for *any* $\alpha \in (0,1)$. The other parts of the proof are adapted accordingly. We refer to [300] for details and other improvements.

We close this section by considering the vectorial case $m > 1$. It was shown again by De Giorgi that if $d = m > 2$, then his regularity theorem does not hold:

Example 3.21 (De Giorgi 1968 [105]) Let $d = m > 2$ and define

$$u_*(x) := \frac{x}{|x|^\gamma}, \qquad \gamma := \frac{d}{2}\left(1 - \frac{1}{\sqrt{(2d-2)^2+1}}\right), \qquad x \in B_1.$$

Note that $1 < \gamma < d/2$ and so $u_* \in \mathrm{W}^{1,2}(B_1;\mathbb{R}^d)$ but $u_* \notin \mathrm{L}^\infty_{\mathrm{loc}}(B_1;\mathbb{R}^d)$. By an elementary computation it can be checked that u_* solves the vector-analogue of (3.17), that is,

$$-\operatorname{div}[\mathbf{S}\nabla u_*] = 0$$

for

$$A^{\mathsf{T}}\mathbf{S}(x)A := |A|^2 + \left[\left((d-2)\mathrm{Id} + d \cdot \frac{x \otimes x}{|x|^2}\right) : A\right]^2, \qquad A \in \mathbb{R}^{d\times d}.$$

This quadratic form satisfies all assumptions of a hypothetical vector-analogue of the De Giorgi Regularity Theorem 3.18, which consequently cannot hold.

The explicit solution u_* from De Giorgi's example is clearly also a minimizer of the quadratic variational integral

$$\mathcal{F}_{\mathrm{DG}}[u] := \int_{B_1} \nabla u(x)^{\mathsf{T}} \mathbf{S}(x) \nabla u(x)\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(B_1;\mathbb{R}^2).$$

However, this is not a *regular* variational integral because the integrand depends (non-smoothly) on x.

In principle, despite De Giorgi's example, this still leaves the possibility that the vectorial analogue of Hilbert's 19th problem has a positive solution, just that its proof would have to proceed along a different route. However, Nečas in 1975 [259] gave a

(complicated) example of a regular variational integral for $d \geq 5$ that has a minimizer that is non-C^1 (but still Lipschitz).

For $d = 2$, minimizers to regular variational problems are always as regular as the data allows (as in Theorem 3.20), this is *Morrey's regularity theorem* from 1938, see [238, 240]. If we confine ourselves to Sobolev-regularity, then using the difference quotient technique, one can prove $W^{2,2+\delta}_{\mathrm{loc}}$-regularity for minimizers of regular variational problems for some *dimension-dependent* $\delta > 0$. This result is originally due to Campanato [66]. By the Sobolev embedding theorem this yields $C^{0,\alpha}$-regularity for some $\alpha \in (0,1)$ when $d \leq 4$. In 2008 Kristensen and Melcher [206] established $W^{2,2+\delta}_{\mathrm{loc}}$-regularity for a *dimension-independent* $\delta > 0$ (in fact, $\delta = \mu/(50M)$). We will discuss further regularity results for the vector case in Section 5.7.

On the negative side, the following results are known: Šverák and Yan proved in 2000–2002 [312, 313] that there exist regular variational integrals (with smooth integrands) with the following properties:

- $d \geq 3$, $m \geq 5$ or $d \geq 4$, $m \geq 3$: The minimizer is non-Lipschitz.
- $d \geq 5$, $m \geq 14$: The minimizer is unbounded.

Finally, in 2016 Mooney and Savin [235] were able to give a striking example that in all dimensions $d \geq 3$, $m \geq 2$ there exists a regular variational integral that has a non-Lipschitz minimizer.

3.3 Lagrange Multipliers

We now continue the study of integral side constraints from Section 2.5.

Theorem 3.22 *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to [0,\infty)$ be as in Theorem 3.1, and let $h\colon \Omega \times \mathbb{R}^m \to \mathbb{R}$ be a Carathéodory integrand that is continuously differentiable and satisfies the growth bounds*

$$|h(x,v)| \leq M(1+|v|^p), \qquad |\mathrm{D}_v h(x,v)| \leq M(1+|v|^{p-1}) \tag{3.18}$$

for all $(x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d}$ and some $M > 0$, $p \in [1,\infty)$. Suppose that the map $u_* \in W^{1,p}_g(\Omega;\mathbb{R}^m)$ minimizes the functional

$$\mathcal{F}[u] := \int_\Omega f(x,u(x),\nabla u(x))\,\mathrm{d}x, \qquad u \in W^{1,p}_g(\Omega;\mathbb{R}^m),$$

where $g \in W^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$, under the side constraint

$$\mathcal{H}[u] = \int_\Omega h(x,u(x))\,\mathrm{d}x = 0.$$

Assume furthermore that the consistency condition

$$\delta\mathcal{H}[u_*][w] := \int_\Omega \mathrm{D}_v h(x,u_*(x)) \cdot w(x)\,\mathrm{d}x \neq 0 \tag{3.19}$$

holds for at least one $w \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$. *Then, there exists a* ***Lagrange multiplier*** $\lambda \in \mathbb{R}$ *such that* u_* *is a weak solution of*

$$\begin{cases} -\operatorname{div}\big[\mathrm{D}_A f(x,u,\nabla u)\big] + \mathrm{D}_v f(x,u,\nabla u) = \lambda \mathrm{D}_v h(x,u) & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{3.20}$$

Proof. Let $u_* \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ be a minimizer of $\mathcal{F}$ under the side constraint $\mathcal{H}[u_*] = 0$. From the consistency condition (3.19) we infer that there exists a $w \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$ such that

$$\delta\mathcal{H}[u_*][w] = \int_\Omega \mathrm{D}_v h(x,u_*)\cdot w \,\mathrm{d}x = 1.$$

Now fix any $\psi \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$ and define for $s,t \in \mathbb{R}$,

$$H(s,t) := \mathcal{H}[u_* + s\psi + tw].$$

The function H is continuously differentiable in both s and t and

$$\begin{aligned} \partial_s H(s,t) &= \delta\mathcal{H}[u_* + s\psi + tw][\psi], \\ \partial_t H(s,t) &= \delta\mathcal{H}[u_* + s\psi + tw][w]. \end{aligned}$$

In this context we remark that the variation $\delta\mathcal{H}[u_*]$ exists by similar arguments as in the proof of Theorem 3.1, which in particular rely on the growth bounds (3.18).

Thus,

$$H(0,0) = 0 \qquad \text{and} \qquad \partial_t H(0,0) = 1.$$

By the implicit function theorem there exists a continuously differentiable function $\tau\colon \mathbb{R} \to \mathbb{R}$ such that $\tau(0) = 0$ and

$$H(s,\tau(s)) = 0 \qquad \text{for small } |s|.$$

The chain rule then yields for such s the relation

$$0 = \partial_s[H(s,\tau(s))] = \partial_s H(s,\tau(s)) + \partial_t H(s,\tau(s))\tau'(s),$$

whereby

$$\tau'(0) = -\partial_s H(0,0) = -\int_\Omega \mathrm{D}_v h(x,u_*)\cdot\psi \,\mathrm{d}x. \tag{3.21}$$

Now define for small $|s|$ as above,

$$J(s) := \mathcal{F}[u_* + s\psi + \tau(s)w].$$

Since

$$\mathcal{H}[u_* + s\psi + \tau(s)w] = H(s,\tau(s)) = 0,$$

the continuously differentiable function J has a minimum at $s = 0$ by the minimization property of u_* (this uses the assumed growth bound (3.2) as in the proof of Theorem 3.1). Thus, with the shorthand notations

$$\mathrm{D}_A f := \mathrm{D}_A f(x, u_*, \nabla u_*), \qquad \mathrm{D}_v f := \mathrm{D}_v f(x, u_*, \nabla u_*),$$

we obtain

$$0 = J'(0) = \int_\Omega \mathrm{D}_A f : (\nabla\psi + \tau'(0)\nabla w) + \mathrm{D}_v f \cdot (\psi + \tau'(0)w)\,\mathrm{d}x.$$

Rearranging and using (3.21), we get

$$\begin{aligned}\int_\Omega \mathrm{D}_A f : \nabla\psi + \mathrm{D}_v f \cdot \psi\,\mathrm{d}x &= -\tau'(0)\int_\Omega \mathrm{D}_A f : \nabla w + \mathrm{D}_v f \cdot w\,\mathrm{d}x \\ &= \lambda \int_\Omega \mathrm{D}_v h(x, u_*) \cdot \psi\,\mathrm{d}x, \end{aligned} \tag{3.22}$$

where we have defined

$$\lambda := \int_\Omega \mathrm{D}_A f : \nabla w + \mathrm{D}_v f \cdot w\,\mathrm{d}x.$$

Since $\psi \in \mathrm{W}_0^{1,p}(\Omega;\mathbb{R}^m)$ was arbitrary, (3.22) shows that u_* is a weak solution of (3.20) and the proof is finished. □

Example 3.23 (Stationary Schrödinger equation) When looking for ground states in quantum mechanics as in Section 1.4, we have to minimize

$$\mathcal{E}[\Psi] := \int_{\mathbb{R}^N} \frac{\hbar^2}{4\mu}|\nabla\Psi(x)|^2 + \frac{1}{2}V(x)|\Psi(x)|^2\,\mathrm{d}x$$

over all $\Psi \in \mathrm{W}^{1,2}(\mathbb{R}^N;\mathbb{C})$ under the side constraint

$$\|\Psi\|_{\mathrm{L}^2}^2 = \int_{\mathbb{R}^N} |\Psi(x)|^2\,\mathrm{d}x = 1.$$

From Theorem 2.15 in conjunction with Lemma 2.16 (suitably extended to also apply in the whole space $\Omega = \mathbb{R}^N$) we see that this problem always has at least one solution $\Psi_* \in \mathrm{W}^{1,2}(\mathbb{R}^N;\mathbb{C})$. Theorem 3.22 (likewise extended to the whole space and for side constraints $\mathcal{H}[u] = \alpha \in \mathbb{R}$) yields that this Ψ_* satisfies

$$\left[\frac{-\hbar^2}{2\mu}\Delta + V(x)\right]\Psi_*(x) = E\Psi_*(x), \qquad x \in \Omega,$$

in the weak sense for some $E \in \mathbb{R}$ (the Lagrange multiplier is thus $E/2$). This is precisely the *stationary Schrödinger equation*. One can also show that $E > 0$ is the smallest eigenvalue of the operator $\Psi \mapsto [\frac{-\hbar^2}{2\mu}\Delta + V(x)]\Psi$; the proof of this fact is the task of Problem 3.10.

3.4 Invariances and the Noether Theorem

In physics and other applications of the Calculus of Variations, we are often interested in the *symmetries* of minimizers or, more generally, critical points. These symmetries manifest themselves in other differential or pointwise relations that are automatically satisfied for any minimizer (or critical point). They may be used to identify concrete solutions and are often also interesting in their own right. In this section we only consider $\mathrm{W}^{2,2}_{\mathrm{loc}}$-minimizers (critical points), which is the natural level of regularity by Theorem 3.12.

As a first concrete example of a symmetry, consider the Dirichlet functional

$$\mathcal{F}[u] := \int_\Omega \frac{1}{2}|\nabla u(x)|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega),$$

which was introduced in Example 2.8. We notice that $\mathcal{F}$ is invariant under translations in space: Let $u \in \mathrm{W}^{1,2}(\Omega)$, $\tau \in \mathbb{R}$, $k = 1, \dots, d$, and set

$$u_\tau(x) := u(x + \tau \mathrm{e}_k) \qquad \text{for } x \in D \subset \mathbb{R}^d \text{ with } D_\tau := D + \tau \mathrm{e}_k \subset \Omega.$$

Then, we have the *invariance relation*

$$\int_D \frac{1}{2}|\nabla u_\tau(x)|^2 \,\mathrm{d}x = \int_{D_\tau} \frac{1}{2}|\nabla u(x')|^2 \,\mathrm{d}x'. \tag{3.23}$$

The Dirichlet functional also exhibits an invariance with respect to *scaling*: For $\lambda > 0$ set

$$u_\lambda(x) := \lambda^{(d-2)/2} u(\lambda x), \qquad D_\lambda := \lambda D \subset \Omega. \tag{3.24}$$

It is not hard to see that (3.23) again holds if we set $\lambda = \mathrm{e}^\tau$ (to have $\tau \in \mathbb{R}$ as before).

The main result of this section, Noether's theorem, roughly says that "differentiable invariances of a functional give rise to conservation laws". For instance, the two aforementioned invariances of the Dirichlet functional yield two additional PDEs that any minimizer of the Dirichlet functional must satisfy.

To make this statement precise in the general case, we need a bit of notation: Let $u_* \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega; \mathbb{R}^m)$ be a minimizer (or, more generally, a critical point) of the functional

$$\mathcal{F}[u] := \int_\Omega f(x, u(x), \nabla u(x)) \,\mathrm{d}x,$$

where $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to [0, \infty)$ is assumed to be as in Theorem 3.1. Then, u_* satisfies the strong Euler–Lagrange equation

$$-\operatorname{div}\big[\mathrm{D}_A f(x, u_*, \nabla u_*)\big] + \mathrm{D}_v f(x, u_*, \nabla u_*) = 0 \quad \text{a.e. in } \Omega,$$

see Proposition 3.10. We consider u_* to be extended to all of $\mathbb{R}^d$ (it will not matter below how one extends u_* as long as $u_* \in \mathrm{W}^{2,2}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^m)$).

An invariance is specified through maps $g\colon \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^d$ and $H\colon \mathbb{R}^d \times \mathbb{R} \to \mathbb{R}^m$, which will depend on u_* above, with

$$g(x,0) = x \qquad \text{and} \qquad H(x,0) = u_*(x), \qquad x \in \mathbb{R}^d.$$

We also require that g, H are *continuously differentiable* in their second argument for almost every $x \in \Omega$ and that $H(\bullet,\tau) \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ for every $\tau \in \mathbb{R}$. Then, set for $x \in \mathbb{R}^d$, $\tau \in \mathbb{R}$ and any open set $D \Subset \mathbb{R}^d$,

$$D_\tau := g(D,\tau), \qquad u_\tau(x) := H(x,\tau).$$

One can think of the transformation (g,H) as a form of *homotopy*.

We call $\mathcal{F}$ **invariant** under the transformation defined by (g,H) if

$$\int_D f(x,u_\tau(x),\nabla u_\tau(x))\,\mathrm{d}x = \int_{D_\tau} f(x',u_*(x'),\nabla u_*(x'))\,\mathrm{d}x' \tag{3.25}$$

for all Lipschitz domains $D \subset \mathbb{R}^d$ and for all $\tau \in \mathbb{R}$ sufficiently small such that $D_\tau \Subset \Omega$ (if any such τ exist).

The following result goes back to Emmy Noether and is considered to be one of the most important mathematical theorems ever proved. Its pivotal idea of systematically generating conservation laws from invariances has shown itself to be immensely influential in physics.

Theorem 3.24 (Noether 1918 [263]) *Let $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to [0,\infty)$ be as in Theorem 3.1. Furthermore, let the associated functional $\mathcal{F}$ be invariant under the transformation defined by (g,H) as above, and assume that there exists a majorant $h \in \mathrm{L}^p(\Omega)$ such that*

$$|\partial_\tau H(x,\tau)|, |\nabla \partial_\tau H(x,\tau)|, |\partial_\tau g(x,\tau)| \le h(x) \tag{3.26}$$

for a.e. $x \in \Omega$ and $\tau \in \mathbb{R}$ (near 0). Then, for a minimizer or critical point $u_* \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega;\mathbb{R}^m)$ of the functional $\mathcal{F}$, the conservation law

$$\operatorname{div}\big[\mu^{\mathsf{T}} \mathrm{D}_A f(x,u_*,\nabla u_*) - \nu f(x,u_*,\nabla u_*)\big] = 0 \quad \text{a.e. in } \Omega \tag{3.27}$$

holds, where

$$\mu(x) := \partial_\tau H(x,0) \in \mathbb{R}^m \qquad \text{and} \qquad \nu(x) := \partial_\tau g(x,0) \in \mathbb{R}^d, \qquad x \in \Omega,$$

*are the **Noether multipliers**.*

Corollary 3.25 *If $f = f(v,A)\colon \mathbb{R} \times \mathbb{R}^d \to [0,\infty)$ does not depend on x, then every minimizer (or critical point) $u_* \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega)$ of the corresponding functional $\mathcal{F}$ satisfies*

$$\sum_{i=1}^{d} \partial_i \big[(\partial_k u) \cdot \partial_{A_k} f(u_*,\nabla u_*) - \delta_{ik} f(u_*,\nabla u_*)\big] = 0 \quad \text{a.e. in } \Omega \tag{3.28}$$

for all $k = 1,\dots,d$.

Here,

$$\delta_{ik} := \begin{cases} 1 & \text{if } i = k, \\ 0 & \text{otherwise,} \end{cases}$$

is the **Kronecker delta**.

Proof of Theorem 3.24 and Corollary 3.25. We will differentiate (3.25) at the minimizer (critical point) u_* with respect to τ. To be able to differentiate the left-hand side under the integral, we use the growth assumptions on the derivatives $\mathrm{D}_v f(x, v, A)$, $\mathrm{D}_A f(x, v, A)$ and (3.26) to get a uniform (in τ) majorant, which allows us to move the differentiation under the integral sign. For the right-hand side we need to employ the formula for the differentiation of an integral with respect to a moving domain (this is a special case of the *Reynolds transport theorem*), namely

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \int_{D_\tau} f(x, u_*, \nabla u_*)\, \mathrm{d}x = -\int_{\partial D_\tau} f(x, u_*, \nabla u_*)\, \partial_\tau g(x, \tau) \cdot n \, \mathrm{d}\mathcal{H}^{d-1},$$

where n is the unit inner normal on ∂D_τ and $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional surface (Hausdorff) measure on ∂D_τ; this formula can be checked, at least for D with a smooth boundary (which is all we require below), in an elementary way using the transformation formula for integrals under coordinate changes.

Abbreviating for readability

$$\begin{aligned} f &:= f(x, u_*(x), \nabla u_*(x)), \\ \mathrm{D}_A f &:= \mathrm{D}_A f(x, u_*(x), \nabla u_*(x)), \\ \mathrm{D}_v f &:= \mathrm{D}_v f(x, u_*(x), \nabla u_*(x)), \end{aligned}$$

we get as the result of the differentiation of (3.25) with respect to τ and then setting $\tau = 0$ that

$$\int_D \mathrm{D}_A f : \nabla\mu + \mathrm{D}_v f \cdot \mu \, \mathrm{d}x = -\int_{\partial D} f\nu \cdot n \, \mathrm{d}\mathcal{H}^{d-1},$$

with the Noether multipliers μ, ν defined as in the statement of the theorem. Here we used that μ, $\nabla\mu$, and ν are all p-integrable by (3.26).

Next, we apply the Gauss–Green theorem to obtain

$$\begin{aligned} \int_D \big[-\operatorname{div} \mathrm{D}_A f + \mathrm{D}_v f\big] \cdot \mu \, \mathrm{d}x &= \int_{\partial D} \big[\mu^{\mathsf{T}} \mathrm{D}_A f - \nu f\big] \cdot n \, \mathrm{d}\mathcal{H}^{d-1} \\ &= -\int_D \operatorname{div}\big[\mu^{\mathsf{T}} \mathrm{D}_A f - \nu f\big] \, \mathrm{d}x. \end{aligned}$$

The Euler–Lagrange equation $-\operatorname{div} \mathrm{D}_A f + \mathrm{D}_v f = 0$, which holds strongly for minimizers and critical points of regularity $\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}}$ (see Proposition 3.10), then yields

$$\int_D \operatorname{div}\big[\mu^{\mathsf{T}} \mathrm{D}_A f - \nu f\big] \, \mathrm{d}x = 0,$$

and varying D we conclude (3.27).

The corollary follows by considering the invariance

$$u_\tau(x) := u_*(x + \tau e_k), \qquad D_\tau := D + \tau e_k$$

for $k = 1, \dots, d$, and a computation. □

Example 3.26 In the brachistochrone problem presented in Section 1.1, we were tasked to minimize the functional

$$\mathcal{F}[y] := \int_0^1 \sqrt{\frac{1 + (y')^2}{-y}}\, \mathrm{d}x$$

over all curves $y\colon [0, 1] \to \mathbb{R}$ with $y(0) = 0$, $y(1) = \bar{y} < 0$. The integrand $f(v, a) = \sqrt{-(1 + a^2)/v}$ is independent of x, hence from (3.28) we get

$$y' \cdot \mathrm{D}_a f(y, y') - f(y, y') \equiv \text{const} = -\frac{1}{\sqrt{2r}}$$

for some $r > 0$ (positive constants lead to inadmissible y). So,

$$\frac{(y')^2}{\sqrt{1 + (y')^2} \cdot \sqrt{-y}} - \frac{\sqrt{1 + (y')^2}}{\sqrt{-y}} = -\frac{1}{\sqrt{2r}},$$

which we transform into

$$(y')^2 = -\frac{2r}{y} - 1.$$

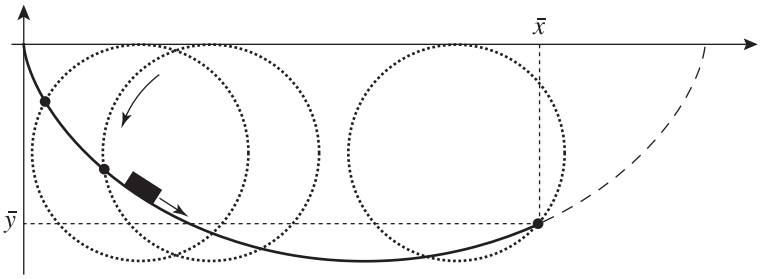


Fig. 3.2 The brachistochrone curve (here, $\bar{x} = 1$)

The solution of this differential equation is called an *inverted cycloid* with radius $r > 0$. It is the curve traced out by a fixed point on a circle of radius r that touches the y-axis at the beginning and then rolls to the right on the bottom of the x-axis, see Figure 3.2. It can be written in parametric form as

$$\begin{aligned} x(t) &= r(t - \sin t), \\ y(t) &= -r(1 - \cos t), \end{aligned} \qquad t \in \mathbb{R}.$$

The radius $r > 0$ has to be chosen to satisfy the two boundary conditions on one cycloid segment.

Of course, we have not properly shown that this is the (unique) solution of the brachistochrone problem for technical reasons (e.g., growth conditions), but this can indeed be proved rigorously, which is left to the reader.

Example 3.27 We return to our canonical example, the Dirichlet functional, see Example 2.8. We know from Example 3.16 that minimizers u_* are smooth. At the beginning of this section we remarked that the Dirichlet functional is invariant under the scaling transformation (3.24) with $\lambda = \mathrm{e}^\tau$. By Noether's theorem, this yields the additional conservation law

$$\operatorname{div}\big[(2\nabla u_*(x)\cdot x + (d-2)u)\nabla u_*(x) - |\nabla u_*(x)|^2 x\big] = 0, \qquad x \in \Omega.$$

Integrating this over $B_r(x_0) \subset \Omega$, $r > 0$, and using the Gauss–Green theorem, we get

$$(d-2)\int_{B_r(x_0)} |\nabla u_*(x)|^2 \,\mathrm{d}x = r\int_{\partial B_r(x_0)} |\nabla u_*(x)|^2 - 2\left(\nabla u_*(x)\cdot\frac{x}{|x|}\right)^2 \mathrm{d}\mathcal{H}^{d-1}$$

and then, after some computations,

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r^{d-2}}\int_{B_r(x_0)} |\nabla u_*(x)|^2 \,\mathrm{d}x\right) = \frac{2}{r^{d-2}}\int_{\partial B_r(x_0)} \left(\nabla u_*(x)\cdot\frac{x}{|x|}\right)^2 \mathrm{d}\mathcal{H}^{d-1} \geq 0.$$

This *monotonicity formula* implies that

$$\frac{1}{r^{d-2}}\int_{B_r(x_0)} |\nabla u_*(x)|^2 \,\mathrm{d}x \quad \text{is increasing in } r > 0.$$

Any harmonic map ($-\Delta u = 0$) that is defined on all of $\mathbb{R}^d$ satisfies this monotonicity formula. For example, this allows us to draw the conclusion that if $d \geq 3$, then u_* cannot be compactly supported. The formula also shows that the growth around a hypothetical singularity has to behave "more smoothly" than $|x|^{-2}$ (in fact, we already know that solutions are smooth). While these are not particularly strong remarks, they serve to illustrate how Noether's theorem restricts the candidates for solutions. Problem 3.8 exploits another invariance of the Dirichlet integral.

We close this section by remarking that the non-obvious conservation laws exhibited in the last examples could in principle have been derived directly, but Noether's theorem gave us a *systematic* way to infer them from (obvious) invariances of the integral functional.

Notes and Historical Remarks

Difference quotients were already considered by Newton. Their application to regularity theory is due to work by Nirenberg in the 1940s and 1950s. Many of the fundamental results of regularity theory (albeit in a non-variational context) can be found in [157]. The books by Giusti [158] and Giaquinta–Martinazzi [156] contain theory relevant for variational questions, whereas [220] treats many questions of "fine regularity" (e.g., pointwise properties of solutions). A recent, very accessible introduction to regularity theory for PDEs is [46], also see the survey [232], which focuses on the Calculus of Variations. A nice framework for Schauder estimates is [300].

The Fundamental Lemma 3.11 of the Calculus of Variations is due to Paul Du Bois-Reymond and is sometimes named after him.

Noether's theorem has many ramifications and can be put into a very general form in Hamiltonian systems and Lie group theory. The idea is to study groups of symmetries and their actions. For an introduction to this diverse field, see [266] and also [337]. Example 3.27 about the monotonicity formula is from [133]. For more on Lagrange multipliers and Noether's theorem, see [154, 155].

Problems

3.1 Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain.

(i) Compute the (weak) Euler–Lagrange equation for a minimizer $u \in \mathrm{W}^{1,2}(\Omega)$ of the functional

$$\mathcal{F}[u] := \int_\Omega \frac{1}{2} \nabla u(x) S(x) \nabla u(x)^\mathsf{T} - g(x) u(x) \, \mathrm{d}x,$$

where $S \colon \Omega \to \mathbb{R}^{d \times d}$, $g \colon \Omega \to \mathbb{R}$ are continuous and $S(x) = S(x)^\mathsf{T}$ for all $x \in \Omega$.

(ii) Assume now that additionally $S(x)$ is continuously differentiable in x and that $u \in \mathrm{W}^{2,2}(\Omega)$ is a minimizer of $\mathcal{F}$ as above. State the strong Euler–Lagrange equation for u and prove that it follows from the weak version.

3.2 Let $\Omega \subset \mathbb{R}^2$ and let $u \in \mathrm{W}^{1,2}(\Omega; \mathbb{R}^2)$. In the two cases

(i) $f(A) = A^i_j$ for some $i, j \in \{1, 2\}$,
(ii) $f(A) = \det A$

show through a direct calculation that

$$-\operatorname{div}[\mathrm{D}_A f(\nabla u)] = 0 \tag{3.29}$$

for *all* $u \in \mathrm{C}^2(\Omega; \mathbb{R}^2)$. This shows that these f are *null-Lagrangians*, i.e., the Euler–Lagrange equation holds for all u.

3.3 Show that also for $d \geq 3$ the above Euler–Lagrange equation (3.29) continues to hold for all $(r \times r)$-minors $f(A) = M(A)$, that is, $M(A)$ is the determinant of a selection of r rows and r columns of A. *Hint:* You can assume that you select the *first* r rows and columns, thus considering only the *principal* minors.

3.4 Prove Theorem 3.15, namely that for functionals of the form

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x)) - h(x) \cdot u(x)\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega; \mathbb{R}^m),$$

where f satisfies the assumptions of Theorem 3.12 and $h \in \mathrm{L}^2(\Omega; \mathbb{R}^m)$, the minimizer u_* has $\mathrm{W}^{2,2}_{\mathrm{loc}}$-regularity. Show furthermore that if f is quadratic and h is smooth, then $u_* \in \mathrm{C}^\infty(\Omega; \mathbb{R}^m)$.

3.5 Prove an analogue of Theorem 3.12 for functionals of the form

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x)) - H(u(x))\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega),$$

where f satisfies the same assumptions as in Theorem 3.12 and $H\colon \mathbb{R}^m \to \mathbb{R}$ is continuously differentiable with $|\mathrm{D}H(v)| \leq C(1 + |v|)$ for some $C > 0$ and all $v \in \mathbb{R}^m$. Is it possible to also allow the weaker growth bound $|\mathrm{D}H(v)| \leq C(1 + |v|^r)$ for some $r > 1$?

3.6 Consider the minimization problem for the problem of linearized elasticity on $\Omega \subset \mathbb{R}^3$,

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega \mu |\mathcal{E}u(x)|^2 + \frac{1}{2}\Big(\kappa - \frac{2}{3}\mu\Big) |\operatorname{tr} \mathcal{E}u(x)|^2 - b(x) \cdot u(x)\, \mathrm{d}x \\ \text{over all} & u \in \mathrm{W}^{1,2}(\Omega; \mathbb{R}^3) \text{ with } u|_{\partial\Omega} = g, \end{cases}$$

where $\mu, \kappa > 0$ are such that $\kappa - \frac{2}{3}\mu \geq 0$, $f \in \mathrm{L}^\infty(\Omega; \mathbb{R}^3)$, $g \in \mathrm{W}^{1/2,2}(\partial\Omega; \mathbb{R}^3)$, and $\mathcal{E}u(x) := (\nabla u(x) + \nabla u(x)^\mathsf{T})/2$. Prove that the Euler–Lagrange equation (satisfied by a minimizer in a weak sense) is

$$\begin{cases} -\operatorname{div}\Big[2\mu\, \mathcal{E}u + \Big(\kappa - \frac{2}{3}\mu\Big)(\operatorname{tr} \mathcal{E}u) I\Big] = b & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega. \end{cases}$$

3.7 In the situation of the previous problem, assume $\kappa - \frac{2}{3}\mu = 0$, $b = 0$, and that $u \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega; \mathbb{R}^3)$ is a minimizer of $\mathcal{F}$ as above. Show, using a suitable Noether symmetry, that for all skew-symmetric $W \in \mathbb{R}^{3\times 3}$ ($W^\mathsf{T} = -W$) it holds that

$$\operatorname{div}[x^\mathsf{T} W^\mathsf{T} \mathcal{E}u(x)] = 0 \qquad \text{for a.e. } x \in \Omega.$$

3.8 Set for a skew-symmetric $W \in \mathbb{R}^{d\times d}$

$$g(x, \tau) := \exp(\tau W)x, \qquad u_\tau = H(x, \tau) := u(\exp(\tau W)x), \qquad \tau \in \mathbb{R}.$$

Show that the Dirichlet functional is invariant under the rotational transformation defined by (g, H) and conclude that any minimizer $u_* \in (\mathrm{W}^{1,2} \cap \mathrm{W}^{2,2}_{\mathrm{loc}})(\Omega; \mathbb{R}^m)$ of the Dirichlet functional (for given boundary values) satisfies the conservation law

$$\operatorname{div}\left[x^{\mathsf{T}} W |\nabla u_*(x)|^2\right] = 0.$$

3.9 In the situation of Exercise 2.2, derive the weak Euler–Lagrange equation. *Hint:* Think about the class of "test variations" ψ that you need to allow.

3.10 In the situation of Example 3.23, show that $E > 0$ is the smallest eigenvalue of the operator $\Psi \mapsto [\frac{-\hbar^2}{2\mu}\Delta + V(x)]\Psi$.

Chapter 4
Young Measures

Before we continue our study of integral functionals, it is convenient to introduce the very versatile theory of *Young measures*, named after their inventor Laurence Chisholm Young. Young measures pervade much of the modern theory of the Calculus of Variations, both as a technical tool and as a means to describe mixing phenomena (e.g., fine phase mixtures in solids exhibiting microstructure, or composites).

Let us motivate this device through the following fundamental question: Assume that we are given a weakly converging sequence $v_j \rightharpoonup v$ in $\mathrm{L}^2(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, and an integral functional

$$\mathcal{F}[w] := \int_\Omega f(x, w(x))\,\mathrm{d}x, \qquad w \in \mathrm{L}^2(\Omega),$$

with $f\colon \Omega \times \mathbb{R} \to \mathbb{R}$ continuous and bounded (for simplicity). Then, $(\mathcal{F}[v_j])_j$ is a bounded sequence of real numbers, which, up to taking a (not relabeled) subsequence, can be assumed to converge to a limit as $j \to \infty$. The starting point of Young measure theory is the question whether there exists some "limit object" for the sequence $(v_j)_j$ itself, from which one can compute the limit of $(\mathcal{F}[v_j])_j$ for *every* integrand f.

Equivalently, we could ask for the weak* limit in $\mathrm{L}^\infty(\Omega)$ of the sequence of compound functions $F_j(x) := f(x, v_j(x))$. It is easy to see that this weak* limit in general is *not* equal to the map $x \mapsto f(x, v(x))$. For example, in $\Omega = (0,1)$, consider the oscillating sequence

$$v_j(x) := \begin{cases} a & \text{if } jx - \lfloor jx \rfloor \in [0, \theta), \\ b & \text{if } jx - \lfloor jx \rfloor \in [\theta, 1), \end{cases} \qquad x \in (0,1),$$

where $a, b \in \mathbb{R}$ with $a \neq b$, $\theta \in (0,1)$. If f is continuous and bounded, say, we see immediately that

$$v_j \overset{*}{\rightharpoonup} \theta a + (1-\theta)b$$

and

$$f(x, v_j) \overset{*}{\rightharpoonup} \theta f(a) + (1-\theta) f(b) =: F(x).$$

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Of course, the right-hand side is not equal to $f(x, \theta a + (1-\theta)b)$ in general. However, we can write

$$F(x) = \big\langle f(x, \bullet), \nu_x \big\rangle = \int f(x, z)\, \mathrm{d}\nu_x(z)$$

with an x-parametrized family of probability measures $\nu_x \in \mathcal{M}^1(\mathbb{R})$ (see Appendices A.3, A.4 for some basic facts of measure theory), namely

$$\nu_x = \theta\delta_a + (1-\theta)\delta_b \qquad \text{for all } x \in (0,1).$$

This family $(\nu_x)_{x\in\Omega}$ reflects the asymptotic distribution of values in the sequence $(v_j)_j$ and will be called the *Young measure* generated by it.

In this chapter we lay the groundwork for the theory of Young measures, in particular regarding their compactness properties. We also introduce the class of *gradient* Young measures, that is, those Young measures where the generating sequence (the $(v_j)_j$ above) consists entirely of gradients. This class will see many uses in subsequent chapters.

4.1 The Fundamental Theorem

We start with a result that nowadays is known as the *Fundamental Theorem* of Young measure theory:

Theorem 4.1 (Young 1937–1942 [339–341]) *Let $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$ be a norm-bounded sequence, where $p \in [1,\infty]$. Then, there exists a subsequence (not relabeled) and a family of probability measures,*

$$(\nu_x)_{x\in\Omega} \subset \mathcal{M}^1(\mathbb{R}^N),$$

*called the **Young measure generated by the (sub)sequence** $(V_j)_j$, such that the following assertions are true:*

*(i) The family $(\nu_x)_x$ is **weakly* measurable**, that is, for all Carathéodory integrands $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$, the compound function*

$$x \mapsto \big\langle f(x, \bullet), \nu_x \big\rangle := \int f(x, A)\, \mathrm{d}\nu_x(A), \qquad x \in \Omega,$$

is Lebesgue-measurable.

(ii) If $p \in [1,\infty)$, it holds that

$$\int_\Omega \int |A|^p \, \mathrm{d}\nu_x(A)\, \mathrm{d}x < \infty,$$

or, if $p = \infty$, there exists a compact set $K \subset \mathbb{R}^N$ such that

$$\operatorname{supp} \nu_x \subset K \qquad \textit{for a.e. } x \in \Omega.$$

(iii) For all Carathéodory integrands $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$ with the property that the family $(f(x,V_j))_j$ is uniformly L^1-bounded and equiintegrable, it holds that

$$f(x,V_j) \rightharpoonup \left(x \mapsto \int f(x,A)\,\mathrm{d}\nu_x(A)\right) \qquad \textit{(weakly) in } \mathrm{L}^1. \tag{4.1}$$

For parametrized measures $\nu = (\nu_x)_{x\in\Omega}$ that satisfy (i) and (ii) above (note that these requirements are independent of the generating sequence $(V_j)_j$), we write

$$\nu = (\nu_x)_x \in \mathbf{Y}^p(\Omega;\mathbb{R}^N).$$

If for the target dimension we have $N = 1$, then we simply write $\mathbf{Y}^p(\Omega)$ instead of $\mathbf{Y}^p(\Omega;\mathbb{R})$.

The **generation** of a Young measure ν by a sequence $(V_j)_j$, i.e., the validity of (iii) for this sequence $(V_j)_j$, will be written symbolically as

$$V_j \overset{\mathbf{Y}}{\to} \nu.$$

The reader is referred to the paragraph after Vitali's Convergence Theorem A.12 for several equivalent ways to express equiintegrability. In particular, recall from the Dunford–Pettis Theorem A.13 that $(f(x,V_j))_j$ is equiintegrable if and only if it is weakly precompact in $\mathrm{L}^1(\Omega)$.

Absorbing a test function for weak convergence into f, the convergence (4.1) can equivalently be written as

$$\int_\Omega f(x,V_j(x))\,\mathrm{d}x \to \int_\Omega\int f(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x = \int_\Omega \big\langle f(x,\bullet),\nu_x\big\rangle\,\mathrm{d}x =: \big\langle\!\big\langle f,\nu\big\rangle\!\big\rangle$$

for all Carathéodory integrands $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$ such that the family $(f(x,V_j))_j$ is uniformly L^1-bounded and equiintegrable. We call $\langle\!\langle f,\nu\rangle\!\rangle$ the **duality pairing** between f and ν.

For $\nu = (\nu_x)_x \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ the **barycenter** $[\nu] \in \mathrm{L}^p(\Omega;\mathbb{R}^N)$ of ν is defined as

$$[\nu](x) := [\nu_x] := \big\langle \mathrm{id},\nu_x\big\rangle = \int A\,\mathrm{d}\nu_x(A), \qquad x\in\Omega.$$

Remark 4.2 The boundedness assumption on the generating sequence $(V_j)_j$ in the Fundamental Theorem can be weakened: For the existence of a Young measure we only need to require the **tightness condition**

$$\lim_{h\uparrow\infty}\sup_{j\in\mathbb{N}} |\{|V_j|\ge h\}| = 0,$$

which, for example, follows from $\sup_j \int_\Omega |V_j|^r\,\mathrm{d}x < \infty$ for any $r>0$. Of course, in this case, statement (ii) needs to be suitably adapted. The proof of these claims is the task of Problem 4.2.

For the proof of the Fundamental Theorem, we first associate with each map V_j an **elementary Young measure**

$$\delta[V_j] = \big(\delta[V_j]_x\big)_{x\in\Omega} \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$$

via

$$\delta[V_j]_x := \delta_{V_j(x)} \qquad \text{for a.e. } x \in \Omega, \tag{4.2}$$

where δ_z denotes the **Dirac mass** at $z \in \mathbb{R}^N$, that is $\delta_z(B) = 1$ if and only if $z \in B$ for any Borel set $B \subset \Omega$. Clearly, $\delta[V_j]_x$ is only defined up to an $\mathcal{L}^d$-negligible set of x's. In fact, we will (implicitly) consider all Young measures to be defined only up to $\mathcal{L}^d$-negligible sets.

We will first show the following *Young measure compactness principle*.

Proposition 4.3 *Let $p \in [1,\infty]$ and let $(\nu^{(j)})_j \subset \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ be a sequence of Young measures. If $p \in [1,\infty)$, assume*

$$\sup_{j\in\mathbb{N}} \big\langle\!\big\langle |\bullet|^p, \nu^{(j)} \big\rangle\!\big\rangle = \sup_{j\in\mathbb{N}} \int_\Omega \int |A|^p \, \mathrm{d}\nu_x^{(j)}(A)\, \mathrm{d}x < \infty \tag{4.3}$$

or, if $p = \infty$, assume that there exists a compact set $K \subset \mathbb{R}^N$ such that

$$\operatorname{supp} \nu_x^{(j)} \subset K \qquad \text{for a.e. } x \in \Omega \text{ and all } j \in \mathbb{N}. \tag{4.4}$$

Then, there exists a subsequence of $(\nu^{(j)})_j$ (not relabeled) and $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ such that the following convergence assertion holds: For all Carathéodory integrands $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$ for which the sequence of functions $x \mapsto \langle f(x,\bullet), \nu_x^{(j)}\rangle$ is uniformly L^1-bounded and the equiintegrability condition

$$\sup_{j\in\mathbb{N}} \big\langle\!\big\langle |f| \mathbb{1}_{\{|f|\geq h\}}, \nu^{(j)} \big\rangle\!\big\rangle \to 0 \qquad \text{as } h \to \infty \tag{4.5}$$

is satisfied, one has

$$\big\langle\!\big\langle f, \nu^{(j)} \big\rangle\!\big\rangle \to \big\langle\!\big\langle f, \nu \big\rangle\!\big\rangle \qquad \text{as } j \to \infty. \tag{4.6}$$

Moreover, if $p < \infty$,

$$\big\langle\!\big\langle |\bullet|^p, \nu \big\rangle\!\big\rangle \leq \liminf_{j\to\infty} \big\langle\!\big\langle |\bullet|^p, \nu^{(j)} \big\rangle\!\big\rangle \tag{4.7}$$

or, if $p = \infty$,

$$\operatorname{supp} \nu_x \subset K \qquad \text{for a.e. } x \in \Omega. \tag{4.8}$$

We say that $\nu^{(j)}$ **converges weakly*** to ν, in symbols

$$\nu^{(j)} \overset{*}{\rightharpoonup} \nu,$$

if

$$\big\langle\!\big\langle f, \nu^{(j)} \big\rangle\!\big\rangle \to \big\langle\!\big\langle f, \nu \big\rangle\!\big\rangle \qquad \text{as } j \to \infty \tag{4.9}$$

for all $f \in C_0(\Omega \times \mathbb{R}^N)$. This is clearly implied by (4.6).

Proof. Step 1. Define the measures

$$\mu^{(j)} := \mathcal{L}^d_x \llcorner \Omega \otimes \nu^{(j)}_x,$$

which is just a shorthand notation for the Radon measures

$$\mu^{(j)} \in \mathcal{M}(\Omega \times \mathbb{R}^N) \cong C_0(\Omega \times \mathbb{R}^N)^*,$$

with "$\cong$" denoting the Riesz isomorphism of Theorem A.22, that are defined through their action

$$\left\langle f, \mu^{(j)} \right\rangle := \int_\Omega \int f(x, A)\, \mathrm{d}\nu^{(j)}_x(A)\, \mathrm{d}x \qquad \text{for all } f \in C_0(\Omega \times \mathbb{R}^N).$$

Note that if $\nu^{(j)} = \delta[V_j]$ with the elementary Young measure $\delta[V_j]$ defined in (4.2) then we recover

$$\left\langle f, \mu^{(j)} \right\rangle = \int_\Omega f(x, V_j(x))\, \mathrm{d}x \qquad \text{for all } f \in C_0(\Omega \times \mathbb{R}^N).$$

Clearly, every $\mu^{(j)}$ is a positive measure and

$$\left|\left\langle f, \mu^{(j)} \right\rangle\right| \le |\Omega| \cdot \|f\|_\infty,$$

whereby the $(\mu^{(j)})_j$ constitute a uniformly bounded sequence in the dual space $C_0(\Omega \times \mathbb{R}^N)^*$. Thus, by the Banach–Alaoglu Theorem A.2, we can select a (not relabeled) subsequence such that there exists a $\mu \in C_0(\Omega \times \mathbb{R}^N)^*$ with

$$\left\langle f, \mu^{(j)} \right\rangle \to \left\langle f, \mu \right\rangle \qquad \text{for all } f \in C_0(\Omega \times \mathbb{R}^N). \tag{4.10}$$

The positivity of the $\mu^{(j)}$ passes to the limit and, using the Riesz Theorem A.22 again, we conclude $\mu \in \mathcal{M}^+(\Omega \times \mathbb{R}^N)$.

Step 2. Next, we will show that μ can again be written in the form $\mu = \mathcal{L}^d_x \llcorner \Omega \otimes \nu_x$ for a weakly* measurable parametrized family $\nu = (\nu_x)_{x \in \Omega} \subset \mathcal{M}^1(\mathbb{R}^N)$ of probability measures. For this we will use the following measure-theoretic *disintegration theorem*, which is proved below.

Theorem 4.4 *Let $\Omega \subset \mathbb{R}^d$ be open and let $\mu \in \mathcal{M}^+(\Omega \times \mathbb{R}^N)$ be a positive Radon measure. Define the measure $\kappa \in \mathcal{M}^+(\Omega)$ via*

$$\kappa(B) := \mu(B \times \mathbb{R}^N) \qquad \textit{for any Borel set } B \subset \Omega. \tag{4.11}$$

Then, there exists a κ-weakly measurable family $(\nu_x)_{x \in \Omega} \subset \mathcal{M}^1(\mathbb{R}^N)$ of probability measures such that*

$$\mu = \kappa(\mathrm{d}x) \otimes \nu_x,$$

that is,

$$\int f \,\mathrm{d}\mu = \int_\Omega \int f(x, A) \,\mathrm{d}\nu_x(A) \,\mathrm{d}\kappa(x) \tag{4.12}$$

for all $f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N)$. *Furthermore, the family* $(\nu_x)_{x\in\Omega} \subset \mathcal{M}^1(\mathbb{R}^N)$ *is* κ*-essentially unique, that is, if another family* $(\nu'_x)_{x\in\Omega} \subset \mathcal{M}^1(\mathbb{R}^N)$ *also has the properties above, then* $\nu'_x = \nu_x$ *for* κ*-almost every* $x \in \Omega$.

With μ disintegrated as in this theorem, we first observe that (4.10) together with Lemma A.20 implies that for every open set $U \subset \Omega$ it holds that

$$\mu(U \times \mathbb{R}^N) \le \liminf_{j\to\infty} \mu^{(j)}(U \times \mathbb{R}^N) = |U|. \tag{4.13}$$

On the other hand, employing the same lemma again, we get for every compact set $K \subset \Omega$ and any $R > 0$ that

$$\begin{aligned}
\mu(K \times \overline{B_R}) &\ge \limsup_{j\to\infty} \mu^{(j)}(K \times \overline{B_R}) \\
&= \limsup_{j\to\infty} \int_K \int_{\{|A|\le R\}} \mathrm{d}\nu_x^{(j)}(A) \,\mathrm{d}x \\
&\ge \limsup_{j\to\infty} \int_K 1 - \int_{\{|A|>R\}} \frac{|A|^p}{R^p} \,\mathrm{d}\nu_x^{(j)}(A) \,\mathrm{d}x \\
&\ge |K| - \frac{1}{R^p} \sup_{j\in\mathbb{N}} \left\langle\!\!\left\langle |\bullet|^p, \nu^{(j)} \right\rangle\!\!\right\rangle.
\end{aligned}$$

Letting $R \to \infty$ and employing (4.3) as well as the inner regularity of Radon measures, see Appendix A.3, we arrive at

$$\mu(K \times \mathbb{R}^N) \ge |K|.$$

Together with (4.13), we obtain for the κ defined in (4.11) that $\kappa = \mathcal{L}^d \llcorner \Omega$. We thus get from the disintegration theorem that

$$\mu = \mathcal{L}^d_x \llcorner \Omega \otimes \nu_x$$

with a weakly* measurable family $(\nu_x)_x$ of probability measures. Thus, for $f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N)$,

$$\lim_{j\to\infty} \left\langle\!\!\left\langle f, \nu^{(j)} \right\rangle\!\!\right\rangle = \lim_{j\to\infty} \left\langle f, \mu^{(j)} \right\rangle = \left\langle f, \mu \right\rangle = \left\langle\!\!\left\langle f, \nu \right\rangle\!\!\right\rangle, \tag{4.14}$$

which is (4.6) for such integrands f.

Step 3. We will now show (4.6) in the case when f is Carathéodory, bounded, $\operatorname{supp} f \subset \Omega \times K$ for a compact set $K \subset \mathbb{R}^N$, and for almost every fixed $x \in \Omega$ the function $f(x, \bullet)$ is uniformly continuous. We will need the following theorem, which is proved below.

Theorem 4.5 (Scorza Dragoni 1948 [296]) *Suppose that $f\colon \Omega \times \mathbb{R}^N \to \mathbb{R}$ is a Carathéodory integrand such that for almost every fixed $x \in \Omega$ the function $f(x,\bullet)$ is uniformly continuous. Then, there exists an increasing sequence of compact sets $S_k \subset \Omega$ $(k \in \mathbb{N})$ with $|\Omega \setminus S_k| \downarrow 0$ such that $f|_{S_k\times\mathbb{R}^N}$ is continuous.*

So, let the compact sets $S_k \Subset \Omega$ $(k \in \mathbb{N})$ with $|\Omega \setminus S_k| \downarrow 0$ be such that $f|_{S_k\times\mathbb{R}^N}$ is continuous for our Carathéodory integrand f. Shrinking S_k a little bit, we can furthermore assume that $\operatorname{dist}(S_k, \partial\Omega) > 1/k$.

We may now extend $f|_{S_k\times\mathbb{R}^N}$ to all of $\Omega\times\mathbb{R}^N$ via the Tietze Extension Theorem A.34, preserving the uniform boundedness with the same maximum norm. Multiplying the so-obtained function with a cut-off function $\varrho_k \in \mathrm{C}_c^\infty(\Omega\times\mathbb{R}^N;[0,1])$ with $\varrho_k \equiv 1$ on $S_k\times K$ (where $K \subset \mathbb{R}^N$ is the compact set with $\operatorname{supp} f \subset \Omega \times K$), we get $f_k \in \mathrm{C}_c(\Omega\times\mathbb{R}^N)$ with $f_k|_{S_k\times\mathbb{R}^N} = f|_{S_k\times\mathbb{R}^N}$.

Since f_k has compact support, the sequence $(\langle f_k(x,\bullet), \nu_x^{(j)}\rangle)_j$ is weakly precompact in $\mathrm{L}^1(\Omega)$ by the Dunford–Pettis Theorem A.13. Thus, (4.14) implies

$$\big\langle f_k(x,\bullet), \nu_x^{(j)}\big\rangle \rightharpoonup \big\langle f_k(x,\bullet), \nu_x\big\rangle \quad \text{in } \mathrm{L}^1(\Omega) \qquad \text{as } j\to\infty.$$

Then also

$$\big\langle f(x,\bullet), \nu_x^{(j)}\big\rangle \rightharpoonup \big\langle f(x,\bullet), \nu_x\big\rangle \quad \text{in } \mathrm{L}^1(S_k) \qquad \text{as } j\to\infty.$$

On the other hand,

$$\begin{aligned}\int_\Omega \Big|\big\langle f(x,\bullet), \nu_x^{(j)}\big\rangle - \mathbb{1}_{S_k}\big\langle f(x,\bullet), \nu_x^{(j)}\big\rangle\Big|\,\mathrm{d}x &= \int_{\Omega\setminus S_k} \Big|\big\langle f(x,\bullet), \nu_x^{(j)}\big\rangle\Big|\,\mathrm{d}x\\ &\le |\Omega\setminus S_k|\cdot\|f\|_\infty\end{aligned}$$

and this converges to zero as $k\to\infty$, uniformly in j, by the boundedness of f. The same estimate holds with ν in place of $\nu^{(j)}$. Therefore, we may conclude that

$$\big\langle f(x,\bullet), \nu_x^{(j)}\big\rangle \rightharpoonup \big\langle f(x,\bullet), \nu_x\big\rangle \quad \text{in } \mathrm{L}^1(\Omega) \qquad \text{as } j\to\infty,$$

which directly yields (4.6) for bounded Carathéodory integrands f with $\operatorname{supp} f \subset \Omega\times K$ for a compact set $K \subset \mathbb{R}^N$.

Step 4. Finally, for a general Carathéodory integrand $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$, choose for any $h\in\mathbb{N}$ a cut-off function $\varrho_h \in \mathrm{C}_c^\infty(\mathbb{R};[0,1])$ with $\varrho_h \equiv 1$ on $(-h,h)$ and $\operatorname{supp}\varrho_h \subset (-2h,2h)$. Set

$$f^h(x,A) := \varrho_h(|A|^{p/2})\varrho_h(f(x,A))f(x,A).$$

Then,

$$\begin{aligned}E_{j,h} &:= \int_\Omega\int |f(x,\bullet) - f^h(x,\bullet)|\,\mathrm{d}\nu_x^{(j)}\,\mathrm{d}x\\ &= \int_\Omega\int \big[1-\varrho_h(|A|^{p/2})\varrho_h(f(x,A))\big]|f(x,A)|\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x\end{aligned}$$

$$
\begin{aligned}
&\le \iint_{\{\,(x,A)\in\Omega\times\mathbb{R}^N\,:\,|A|^{p/2}\ge h \text{ or } |f(x,A)|\ge h\,\}} |f(x,A)|\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x\\
&\le \int_\Omega \int_{\{\,A\in\mathbb{R}^N\,:\,|A|^{p/2}\ge h \text{ and } |f(x,A)|<h\,\}} h\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x\\
&\quad + \iint_{\{\,(x,A)\in\Omega\times\mathbb{R}^N\,:\,|f(x,A)|\ge h\,\}} |f(x,A)|\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x\\
&\le \frac{1}{h}\int_\Omega \int_{\{\,A\in\mathbb{R}^N\,:\,|A|^{p/2}\ge h\,\}} |A|^p\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x\\
&\quad + \iint_{\{\,(x,A)\in\Omega\times\mathbb{R}^N\,:\,|f(x,A)|\ge h\,\}} |f(x,A)|\,\mathrm{d}\nu_x^{(j)}(A)\,\mathrm{d}x\\
&\le \frac{1}{h}\sup_{j\in\mathbb{N}} \left\langle\!\left\langle |\bullet|^p, \nu^{(j)}\right\rangle\!\right\rangle + \sup_{j\in\mathbb{N}} \left\langle\!\left\langle |f|\mathbb{1}_{\{|f|\ge h\}}, \nu^{(j)}\right\rangle\!\right\rangle.
\end{aligned}
$$

Both of these terms converge to zero as $h\to\infty$ by the assumptions from the compactness principle, in particular (4.3) and the equiintegrability condition (4.5). Consequently, for fixed $h\in\mathbb{N}$,

$$
\begin{aligned}
&\limsup_{j\to\infty} \left|\left\langle\!\left\langle f, \nu^{(j)}\right\rangle\!\right\rangle - \left\langle\!\left\langle f, \nu\right\rangle\!\right\rangle\right|\\
&\quad\le \limsup_{j\to\infty} \left(\left|\left\langle\!\left\langle f-f^h, \nu^{(j)}\right\rangle\!\right\rangle\right| + \left|\left\langle\!\left\langle f^h, \nu^{(j)}\right\rangle\!\right\rangle - \left\langle\!\left\langle f^h, \nu\right\rangle\!\right\rangle\right| + \left|\left\langle\!\left\langle f^h-f, \nu\right\rangle\!\right\rangle\right|\right)\\
&\quad\le \sup_{j\in\mathbb{N}} E_{j,h} + \left|\left\langle\!\left\langle f^h-f, \nu\right\rangle\!\right\rangle\right|,
\end{aligned}
$$

where we used the previous step, namely that the Young measure convergence holds for f^h, to see that the middle upper limit vanishes. Now, the first term in the last line vanishes as $h\to\infty$ by the preceding estimate and the second term tends to zero by the pointwise convergence of f^h to f (with majorant $2|f|$). Thus, (4.6) follows also in this case.

Step 5. The last assertion to be shown in the compactness principle is (4.7) if $p\in[1,\infty)$ and (4.8) if $p=\infty$. For (4.7) let $h\in\mathbb{N}$ and define

$$
|A|_h := \min\{|A|, h\}, \qquad A\in\mathbb{R}^N.
$$

Then, by the Young measure representation that we have already established,

$$
\liminf_{j\to\infty} \left\langle\!\left\langle |\bullet|^p, \nu^{(j)}\right\rangle\!\right\rangle \ge \lim_{j\to\infty} \left\langle\!\left\langle |\bullet|_h^p, \nu^{(j)}\right\rangle\!\right\rangle = \left\langle\!\left\langle |\bullet|_h^p, \nu\right\rangle\!\right\rangle \qquad \text{for all } h\in\mathbb{N}.
$$

We conclude by letting $h\to\infty$ and using the monotone convergence theorem.

For (4.8) take any $\varphi\in C_0(\Omega)$, $\psi\in C_0(\mathbb{R}^N)$ with $\operatorname{supp}\psi\cap K=\emptyset$. Then,

$$
\left\langle\!\left\langle \varphi\otimes\psi, \nu\right\rangle\!\right\rangle = \lim_{j\to\infty} \left\langle\!\left\langle \varphi\otimes\psi, \nu^{(j)}\right\rangle\!\right\rangle = 0.
$$

Varying φ and ψ in their respective spaces, the assertion $\operatorname{supp}\nu_x\subset K$ for almost every $x\in\Omega$ follows. □

It remains to show the theorems that were employed in the proof.

Proof of Theorem 4.4. For $\psi \in \mathrm{C}_0(\mathbb{R}^N)$ we define the (signed) measure $\mu_\psi \in \mathcal{M}(\Omega)$ via

$$\mu_\psi(B) := \int_{B\times\mathbb{R}^N} \psi(A)\, \mathrm{d}\mu(x, A)$$

for any Borel set $B \subset \Omega$. We have

$$\mu_\psi(B) \le \|\psi\|_\infty\, \mu(B \times \mathbb{R}^N) = \|\psi\|_\infty\, \kappa(B).$$

By the Besicovitch Differentiation Theorem A.24 there exists a κ-measurable function $h_\psi : \Omega \to \mathbb{R}$ with $|h_\psi| \le \|\psi\|_\infty$ such that

$$\mu_\psi = h_\psi\, \kappa.$$

This construction is evidently linear in ψ, that is, for all $\psi_1, \psi_2 \in \mathrm{C}_0(\mathbb{R}^N)$ and $\alpha, \beta \in \mathbb{R}$ it holds that

$$(h_{\alpha\psi_1+\beta\psi_2})\kappa = \mu_{\alpha\psi_1+\beta\psi_2} = \alpha\mu_{\psi_1} + \beta\mu_{\psi_2} = (\alpha h_{\psi_1} + \beta h_{\psi_2})\kappa.$$

Fix a countable dense family of functions $\mathcal{D} \subset \mathrm{C}_0(\mathbb{R}^N)$. Then, we can find a κ-negligible set $N \subset \Omega$ such that

$$h_{\psi_1}(x) + h_{\psi_2}(x) = h_{\psi_1+\psi_2}(x) \qquad \text{for all } x \in \Omega \setminus N \text{ and all } \psi_1, \psi_2 \in \mathcal{D}.$$

Setting $T_x[\psi] := h_\psi(x)$ for $x \in \Omega \setminus N$ and $\psi \in \mathcal{D}$, we see that $|T_x[\psi]| \le \|\psi\|_\infty$ and thus T_x can be extended to a linear bounded operator on $\mathrm{C}_0(\mathbb{R}^N)$. The Riesz Representation Theorem A.22 then shows that for every $x \in \Omega \setminus N$ there exists a measure $\nu_x \in \mathcal{M}(\mathbb{R}^N)$ with $|\nu_x|(\mathbb{R}^N) \le 1$ such that

$$h_\psi(x) = T_x[\psi] = \int_{\mathbb{R}^N} \psi(A)\, \mathrm{d}\nu_x(A), \qquad \psi \in \mathrm{C}_0(\mathbb{R}^N).$$

Further setting $\nu_x := \delta_0$ at points $x \in N$, we observe that for all $\psi \in \mathcal{D}$ the function

$$x \mapsto \langle \psi, \nu_x \rangle = \begin{cases} T_x[\psi] = h_\psi(x) & \text{if } x \in \Omega \setminus N, \\ \psi(0) & \text{if } x \in N \end{cases}$$

is κ-measurable. By an approximation procedure, the details of which are the content of Problem 4.1, this implies the weak* measurability of $(\nu_x)_{x\in\Omega}$.

For a Borel set $B \subset \Omega$ and $\psi \in \mathcal{D}$, we have

$$\begin{aligned} \int_{\Omega\times\mathbb{R}^N} \mathbb{1}_B(x)\psi(A)\, \mathrm{d}\mu(x, A) &= \mu_\psi(B) \\ &= \int_B h_\psi(x)\, \mathrm{d}\kappa(x) \end{aligned}$$

$$= \int_B \int_{\mathbb{R}^N} \psi(A)\, \mathrm{d}\nu_x(A)\, \mathrm{d}\kappa(x)$$
$$= \int_\Omega \int_{\mathbb{R}^N} \mathbb{1}_B(x)\psi(A)\, \mathrm{d}\nu_x(A)\, \mathrm{d}\kappa(x).$$

This is (4.12) for $f := \mathbb{1}_B \otimes \psi$. By an approximation procedure similar to the one employed in the proof of Theorem 4.1, (4.12) then also holds for all $f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N)$ and also for all $f := \mathbb{1}_{B\times\mathbb{R}^N}$ with $B \subset \Omega$ any Borel set.

To see that the ν_x are indeed probability measures, it suffices to observe

$$\mu(B \times \mathbb{R}^N) = \int_B \nu_x(\mathbb{R}^N)\, \mathrm{d}\kappa(x) \le \int_B 1\, \mathrm{d}\kappa(x) = \mu(B \times \mathbb{R}^N)$$

for all Borel sets $B \subset \Omega$. Hence, $\nu_x(\mathbb{R}^N) = 1$ for κ-almost every $x \in \Omega$, which together with $|\nu_x|(\mathbb{R}^N) \le 1$ implies that ν_x is a probability measure.

The κ-essential uniqueness follows directly by applying (4.12) to all $f := \varphi\otimes\psi$ with $\varphi \in \mathrm{C}_0(\Omega)$, $\psi \in \mathrm{C}_0(\mathbb{R}^N)$. □

Proof of Theorem 4.5. For $j \in \mathbb{N}$ and $x \in \Omega$ let

$$g_j(x) := \sup\left\{ |f(x, A) - f(x, B)| \;:\; A, B \in \mathbb{R}^N,\ |A - B| \le 1/j \right\}.$$

Since f is Carathéodory and $f(x, \bullet)$ is uniformly continuous for almost every $x \in \Omega$, every function g_j is measurable as the pointwise supremum of measurable functions. Furthermore, we have $g_j \to 0$ pointwise almost everywhere as $j \to \infty$.

Let $n \in \mathbb{N}$. By Egorov's Theorem A.14, there exists a compact set $K_0 \subset \Omega$ with $|\Omega \setminus K_0| \le 1/(2n)$ and $g_j \to 0$ uniformly on K_0. Let $(A_i)_i \subset \mathbb{R}^N$ be dense in $\mathbb{R}^N$. By Lusin's Theorem A.17 there are compact sets K_i ($i \in \mathbb{N}$) such that $|\Omega \setminus K_i| \le 1/(2^{i+1}n)$ and $f(\bullet, A_i)$ is uniformly continuous on K_i. For

$$S_n := K_0 \cap \bigcap_{i=1}^{\infty} K_i$$

we estimate

$$|\Omega \setminus S_n| \le \frac{1}{2n}\left(1 + \sum_{i=1}^{\infty} \frac{1}{2^i}\right) = \frac{1}{n}.$$

Consequently, $|\Omega \setminus S_n| \to 0$ as $n \to \infty$.

Let $\varepsilon > 0$ and choose $\delta > 0$ such that $|A - B| \le 2\delta$ for $A, B \in \mathbb{R}^N$ implies $|f(x, A) - f(x, B)| \le \varepsilon$ for all $x \in S_n$. The existence of such a δ follows from the uniform convergence $g_j \to 0$ on $S_n \subset K_0$.

For any $(\bar{x}, \bar{A}) \in S_n \times \mathbb{R}^N$ pick A_i from the dense collection $(A_i)_i$ such that $|\bar{A} - A_i| \le \delta$. For this A_i there exists an $\eta > 0$ such that for all $y \in S_n \subset K_i$ with $|\bar{x} - y| \le \eta$ it holds that $|f(\bar{x}, A_i) - f(y, A_i)| \le \varepsilon$. So, for $(x, A) \in S_n \times \mathbb{R}^N$ with

$$|\bar{x} - x| \le \eta \quad \text{and} \quad |\bar{A} - A| \le \delta,$$

we have $|A_i - A| \le |A_i - \bar{A}| + |\bar{A} - A| \le 2\delta$, and then

$$\begin{aligned}|f(\bar{x},\bar{A}) - f(x,A)| &\le |f(\bar{x},\bar{A}) - f(\bar{x},A_i)| + |f(\bar{x},A_i) - f(x,A_i)| \\ &\quad + |f(x,A_i) - f(x,A)| \\ &\le 3\varepsilon.\end{aligned}$$

Hence, $f|_{S_n\times\mathbb{R}^N}$ is continuous at any $(\bar{x},\bar{A}) \in S_n \times \mathbb{R}^N$. □

The proof of the Fundamental Theorem 4.1 is now a straightforward corollary of the Young measure compactness principle.

Proof of Theorem 4.1. We apply the compactness principle to the sequence $(\delta[V_j])_j$ of elementary Young measures defined in (4.2). The boundedness conditions (4.3), (4.4) are directly implied by the L^p-boundedness assumption on $(V_j)_j$. For $(\delta[V_j])_j$ the assumption (4.5) expresses precisely the equiintegrability of $(f(x,V_j))_j$. Thus, (i)–(iii) follow from the compactness principle in Proposition 4.3. □

Remark 4.6 The Fundamental Theorem 4.1 can also be proved in a more functional analytic way at the expense of relying on a number of abstract results: Let $\mathrm{L}^\infty_{w*}(\Omega;\mathcal{M}(\mathbb{R}^N))$ be the set of essentially bounded weakly* measurable functions defined on Ω with values in the Radon measures $\mathcal{M}(\mathbb{R}^N)$. It turns out (see, for example, [36] or [119, 120]) that $\mathrm{L}^\infty_{w*}(\Omega;\mathcal{M}(\mathbb{R}^N))$ is the dual space to $\mathrm{L}^1(\Omega;\mathrm{C}_0(\mathbb{R}^N))$. One can then show from the assumptions that the maps $\nu^{(j)} = (x \mapsto \nu_x^{(j)})$ form a uniformly bounded set in $\mathrm{L}^\infty_{w*}(\Omega;\mathcal{M}(\mathbb{R}^N))$. By the Banach–Alaoglu Theorem A.2 we can conclude the existence of a weak* limit point ν of the $\nu^{(j)}$'s in $\mathrm{L}^\infty_{w*}(\Omega;\mathcal{M}(\mathbb{R}^N))$. The fact that the limit is a collection of probability measures is then easy to see and the extended representation of limits for Carathéodory integrands f follows as before.

Finally, we show a "lower semicontinuity result" for the duality pairing between a Young measure and a positive integrand, which also holds without the equiintegrability of the compound functions.

Proposition 4.7 *Let $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$, $p \in [1,\infty]$, be a norm-bounded sequence generating the Young measure $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ and let $f\colon \Omega\times\mathbb{R}^N \to [0,\infty)$ be a Carathéodory integrand (not necessarily satisfying the equiintegrability property in (iii) of the Fundamental Theorem 4.1). Then,*

$$\liminf_{j\to\infty} \int_\Omega f(x,V_j(x))\,\mathrm{d}x = \liminf_{j\to\infty} \langle\!\langle f,\delta[V_j]\rangle\!\rangle \ge \langle\!\langle f,\nu\rangle\!\rangle.$$

Proof. For $h\in\mathbb{N}$ define $f_h(x,A) := \min\{f(x,A),h\}$. Then, (iii) from the Fundamental Theorem is applicable for the integrand f_h and we get

$$\int_\Omega f_h(x,V_j(x))\,\mathrm{d}x \to \langle\!\langle f_h,\nu\rangle\!\rangle = \int_\Omega\int f_h(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x.$$

Since $f \ge f_h$, we have

$$\liminf_{j\to\infty} \int_\Omega f(x,V_j(x))\,\mathrm{d}x \ge \langle\!\langle f_h,\nu\rangle\!\rangle.$$

We conclude by letting $h\to\infty$ and using the monotone convergence theorem. □

4.2 Examples

We will now consider a few examples of Young measures $\nu = (\nu_x)_x \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$. These examples will in fact all be **homogeneous** Young measures, that is, the map $x \mapsto \nu_x$ is (almost everywhere) constant with respect to $x \in \Omega$; we then simply write ν in place of ν_x.

In order to identify the Young measure generated by a sequence, the following simple result often turns out to be useful.

Lemma 4.8 *There exists a countable family $(\varphi_k \otimes h_k)_k \subset \mathrm{C}_0(\Omega\times\mathbb{R}^N)$ with the following property: If $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$ is uniformly norm-bounded and $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ is such that*

$$\lim_{j\to\infty} \int_\Omega \varphi_k(x) h_k(V_j(x))\,\mathrm{d}x = \int_\Omega \varphi_k(x)\langle h_k, \nu_x\rangle\,\mathrm{d}x \qquad \text{for all } k \in \mathbb{N},$$

then $V_j \xrightarrow{\mathbf{Y}} \nu$.

Proof. Recall that testing with maps in $\mathrm{C}_0(\Omega\times\mathbb{R}^N)$ already determines Young measure convergence, as we have seen in the proof of Proposition 4.3.

Let $(\varphi_k)_k$ and $(h_l)_l$ be countable dense subsets of $\mathrm{C}_0(\Omega)$ and $\mathrm{C}_0(\mathbb{R}^N)$, respectively. The assertion of the lemma is immediate (with a numbering of $\mathbb{N}\times\mathbb{N}$) once we invoke the basic fact from functional analysis that linear combinations of the functions $f_{k,l} := \varphi_k \otimes h_l$ (that is, $f_{k,l}(x,A) := \varphi_k(x)h_l(A)$) are dense in $\mathrm{C}_0(\Omega\times\mathbb{R}^N)$. □

Example 4.9 In $\Omega := (0,1)$ define $u := \mathbb{1}_{(0,1/2)} - \mathbb{1}_{(1/2,1)}$ and extend this function periodically to all of $\mathbb{R}$. Then, the functions $u_j(x) := u(jx)$ for $j \in \mathbb{N}$ (see Figure 4.1) generate the homogeneous Young measure $\nu \in \mathbf{Y}^\infty((0,1))$ with

$$\nu = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1}.$$

Indeed, for $\varphi \in \mathrm{C}_0((0,1))$, $h \in \mathrm{C}_0(\mathbb{R})$ we have that φ is uniformly continuous, say $|\varphi(x)-\varphi(y)| \le \omega(|x-y|)$ with a **modulus of continuity** $\omega\colon [0,\infty) \to [0,\infty)$, that is, ω is continuous, increasing, and $\omega(0) = 0$. Then,

$$\begin{aligned}
&\lim_{j\to\infty} \int_0^1 \varphi(x) h(u_j(x))\,\mathrm{d}x \\
&\quad= \lim_{j\to\infty} \sum_{k=0}^{j-1} \left[\int_{k/j}^{(k+1)/j} \varphi\left(\frac{k}{j}\right) h(u_j(x))\,\mathrm{d}x + \frac{1}{j}\omega\left(\frac{1}{j}\right)\|h\|_\infty \right] \\
&\quad= \lim_{j\to\infty} \sum_{k=0}^{j-1} \frac{1}{j}\varphi\left(\frac{k}{j}\right) \int_0^1 h(u(y))\,\mathrm{d}y \\
&\quad= \int_0^1 \varphi(x)\,\mathrm{d}x \cdot \left(\frac{1}{2}h(-1) + \frac{1}{2}h(+1)\right).
\end{aligned}$$

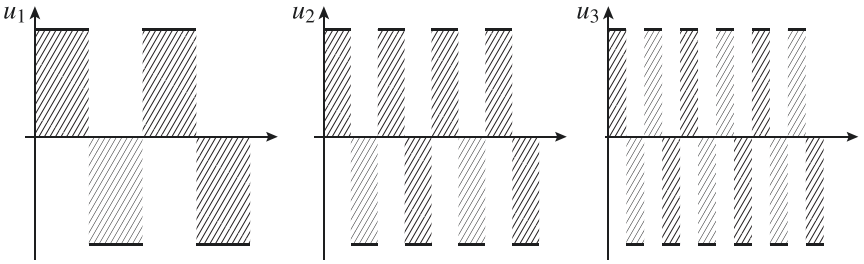


Fig. 4.1 An oscillating sequence

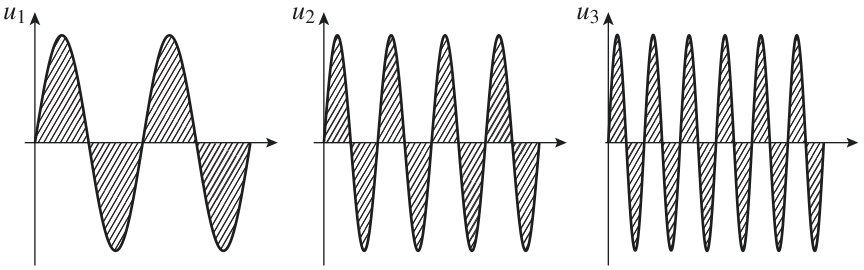


Fig. 4.2 Another oscillating sequence

For the last equality we used that the Riemann sums converge to the integral of φ. By Lemma 4.8, this identifies ν as claimed.

Example 4.10 Take $\Omega := (0,1)$ again, and let $u_j(x) := \sin(2\pi jx)$ for $j \in \mathbb{N}$ (see Figure 4.2). The sequence $(u_j)_j$ generates the homogeneous Young measure $\nu \in \mathbf{Y}^\infty((0,1))$ with

$$\nu = \frac{1}{\pi\sqrt{1-y^2}}\,\mathcal{L}^1_y \llcorner (-1,1).$$

This should be plausible from looking at the oscillating sequence; a formal proof is the task of Problem 4.4.

Example 4.11 Assume that $A, B \in \mathbb{R}^{2\times 2}$ are **rank-one connected**, that is, $B - A = a \otimes n$ for some $a, n \in \mathbb{R}^2$ (this is equivalent to $\operatorname{rank}(A-B) \le 1$ by some basic linear algebra). For $\theta \in (0,1)$ define

$$u(x) := Ax + \left(\int_0^{x\cdot n} \chi(t)\,\mathrm{d}t \right) a, \qquad x \in \mathbb{R}^2,$$

where

$$\chi := \mathbb{1}_L, \qquad L := \bigcup_{z\in\mathbb{Z}} [z, z+1-\theta).$$

If we let $u_j(x) := u(jx)/j$ for $x \in \Omega := (0,1)^2$, then

$$\nabla u_j(x) = A + \chi(jx\cdot n)\, a \otimes n, \qquad x \in (0,1)^2.$$

The gradient sequence $(\nabla u_j)_j$ generates the homogeneous Young measure $\nu \in \mathbf{Y}^\infty((0,1)^2;\mathbb{R}^{2\times 2})$ with

$$\nu = \theta\delta_A + (1-\theta)\delta_B.$$

Note that the last example's construction relies crucially on the rank-one connection between the two matrices A and B.

4.3 Young Measures and Notions of Convergence

Next, we investigate the interaction of Young measure generation with various notions of convergence.

Lemma 4.12 *Let $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$, $p \in (1,\infty]$, be a sequence generating the Young measure $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$. Then, setting*

$$V(x) := [\nu](x) = [\nu_x], \qquad x \in \Omega,$$

it holds that

$$V_j \rightharpoonup V \quad \text{in } \mathrm{L}^p \text{ if } p \in (1,\infty) \qquad \text{or} \qquad V_j \overset{*}{\rightharpoonup} V \quad \text{in } \mathrm{L}^\infty \text{ if } p = \infty.$$

Proof. Norm-bounded sequences in $\mathrm{L}^p(\Omega;\mathbb{R}^N)$ with $p > 1$ are weakly(*) precompact and thus it suffices to identify the limit of any weakly(*) converging subsequence. From the Dunford–Pettis Theorem A.13 it follows that any such (sub)sequence is in fact (1-)equiintegrable. Now apply assertion (iii) of the Fundamental Theorem for the integrand $f(x,A) := \varphi(x)A$ (or, more pedantically, $f_i(A) := \varphi(x)A_i$ for $i = 1,\dots,N$) with any $\varphi \in \mathrm{C}_c^\infty(\Omega)$. This determines the weak(*) limit to be V. □

Lemma 4.13 *Let $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$, $p \in [1,\infty]$, be the Young measure generated by a norm-bounded sequence $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$ and let $\chi\colon \Omega\times\mathbb{R}^N \to [0,1]$ be a Carathéodory integrand. Then,*

$$\chi(\bullet, V_j) \to 0 \text{ in measure} \qquad \Longleftrightarrow \qquad \big\langle \chi(x,\bullet), \nu_x \big\rangle = 0 \text{ for a.e. } x \in \Omega.$$

In particular, for $V \in \mathrm{L}^p(\Omega;\mathbb{R}^N)$,

$$V_j \to V \text{ in measure} \qquad \Longleftrightarrow \qquad \nu_x = \delta_{V(x)} \text{ for a.e. } x \in \Omega,$$

and, for any closed set $F \subset \mathbb{R}^N$,

$$\operatorname{dist}(V_j, F) \to 0 \text{ in measure} \qquad \Longleftrightarrow \qquad \operatorname{supp}\nu_x \subset F \text{ for a.e. } x \in \Omega.$$

Proof. For any "indicator integrand" $\chi\colon \Omega\times\mathbb{R}^N \to [0,1]$ as in the statement of the lemma, and all $\delta \in (0,1)$, the Markov inequality and the Fundamental Theorem 4.1 together imply

$$\limsup_{j\to\infty} \big|\{\, x \in \Omega \,:\, \chi(x, V_j(x)) \geq \delta \,\}\big| \leq \lim_{j\to\infty} \frac{1}{\delta} \int_\Omega \chi(x, V_j(x))\, \mathrm{d}x$$
$$= \frac{1}{\delta} \int_\Omega \int \chi(x, \bullet)\, \mathrm{d}\nu_x\, \mathrm{d}x.$$

On the other hand,

$$\int_\Omega \int \chi(x, \bullet)\, \mathrm{d}\nu_x\, \mathrm{d}x = \lim_{j\to\infty} \int_\Omega \chi(x, V_j(x))\, \mathrm{d}x$$
$$\leq \delta|\Omega| + \limsup_{j\to\infty} \big|\{\, x \in \Omega \,:\, \chi(x, V_j(x)) \geq \delta \,\}\big|.$$

The preceding estimates show that $\chi(x, V_j(x))$ converges to zero in measure if and only if $\langle \chi(x, \bullet), \nu_x \rangle = 0$ for almost every $x \in \Omega$.

For the first additional assertion we choose

$$\chi(x, A) := \frac{|A - V(x)|}{1 + |A - V(x)|}, \qquad (x, A) \in \Omega \times \mathbb{R}^N.$$

Then, the compound maps $x \mapsto \chi(x, V_j(x))$ converge to zero in measure if and only if V_j converges to V in measure, and $\langle \chi(x, \bullet), \nu_x \rangle = 0$ if and only if $\nu_x = \delta_{V(x)}$ almost everywhere.

For the second additional assertion set

$$\chi(x, A) := \frac{\operatorname{dist}(A, F)}{1 + \operatorname{dist}(A, F)}, \qquad (x, A) \in \Omega \times \mathbb{R}^N.$$

In this case, $\chi(\bullet, V_j)$ converge to zero in measure if and only if $\operatorname{dist}(V_j, F) \to 0$ in measure, and $\langle \chi(x, \bullet), \nu_x \rangle = 0$ if and only if $\operatorname{supp} \nu_x \subset F$ almost everywhere. □

4.4 Gradient Young Measures

The most important Young measures for our purposes are those that can be generated by a sequence of *gradients*. Let $\nu \in \mathbf{Y}^p(\Omega; \mathbb{R}^{m\times d})$ (use $\mathbb{R}^N = \mathbb{R}^{m\times d} \cong \mathbb{R}^{md}$ in the theory of the last sections). We say that ν is a $\mathrm{W}^{1,p}$**-gradient Young measure**, where $p \in (1, \infty]$, in symbols

$$\nu \in \mathbf{GY}^p(\Omega; \mathbb{R}^{m\times d}),$$

if there exists a norm-bounded sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$ such that

$$\nabla u_j \xrightarrow{\mathbf{Y}} \nu,$$

i.e., the sequence $(\nabla u_j)_j$ generates ν. Note that it is not required, and in fact never true, that *every* sequence that generates ν is a sequence of gradients. As for examples of gradient Young measures, one observes that all the Young measures considered in Section 4.2 are in fact gradient Young measures.

We deliberately do not extend our definition to $p = 1$ since the theory of Young measures generated by merely L^1-bounded gradients is very special and postponed to Chapter 15; also see the notes at the end of this chapter.

We know from Lemma 4.12 that $\nabla u_j \rightharpoonup \nabla u = [\nu]$ in L^p if $p \in (1,\infty)$, or $\nabla u_j \overset{*}{\rightharpoonup} \nabla u = [\nu]$ in L^∞ if $p = \infty$, for some $u \in W^{1,p}(\Omega;\mathbb{R}^m)$. Any such u is called an **underlying deformation** of ν. Here, the fact that the weak(*) limit of the ∇u_j is a gradient follows by observing that the sequence $(u_j)_j$ may be assumed to be norm-bounded in $W^{1,p}(\Omega;\mathbb{R}^m)$, up to adding constants, by the uniform norm-boundedness of $(\nabla u_j)_j$ in L^p (this is implied by the weak convergence) together with the Poincaré inequality in Theorem A.27 (ii).

We now prove a technical, but immensely useful, result about special generating sequences for gradient Young measures.

Lemma 4.14 *Let $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, $p \in (1,\infty]$, and let $u \in W^{1,p}(\Omega;\mathbb{R}^m)$ be an underlying deformation of ν, that is, $[\nu] = \nabla u$. Then, there exists a norm-bounded sequence $(u_j)_j \subset W^{1,p}(\Omega;\mathbb{R}^m)$ such that*

$$\operatorname{supp}(u_j - u) \Subset \Omega \qquad \text{and} \qquad \nabla u_j \overset{\mathbf{Y}}{\to} \nu.$$

Furthermore, if $p \in (1,\infty)$, we can in addition require that the sequence $(\nabla u_j)_j$ is p-equiintegrable.

For $p = \infty$ instead of the equiintegrability one may obtain a uniform L^∞-bound on $(\nabla u_j)_j$; this will be proved in Zhang's Lemma 8.14. It is the task of Problem 4.9 to show that the preceding lemma cannot hold in the case $p = 1$.

Proof. Step 1. Since Ω has a Lipschitz boundary, we can extend a generating sequence $(\nabla v_j)_j$ for ν to all of $\mathbb{R}^d$ (see Theorem A.26), so from now on we assume $(v_j)_j \subset W^{1,p}(\mathbb{R}^d;\mathbb{R}^m)$ with $\sup_j \|v_j\|_{W^{1,p}} < \infty$.

In the following we will need the maximal function $Mf\colon \mathbb{R}^d \to \mathbb{R}\cup\{+\infty\}$ of $f\colon \mathbb{R}^d \to \mathbb{R}\cup\{+\infty\}$, see Appendix A.6. With this tool at hand, consider the sequence

$$V_j := M\big(|v_j| + |\nabla v_j|\big), \qquad j \in \mathbb{N}.$$

By Theorem A.38 (i), $(V_j)_j$ is uniformly bounded in $L^p(\Omega)$ and we may select a subsequence (not relabeled) such that $(V_j)_j$ generates an (auxiliary) Young measure $\mu \in \mathbf{Y}^p(\Omega)$.

Step 2. For $p \in (1,\infty)$ we first show the claim concerning equiintegrability. Define for $h \in \mathbb{N}$ the *truncation* τ_h via

$$\tau_h s := \begin{cases} s & \text{if } |s| \le h, \\ h\frac{s}{|s|} & \text{if } |s| > h, \end{cases} \qquad s \in \mathbb{R}.$$

For fixed $h \in \mathbb{N}$ the sequence $(\tau_h V_j)_j$ is uniformly bounded in $L^\infty(\Omega)$ and so, by the Young measure representation of limits, we have for every $\varphi \in L^\infty(\Omega)$ that

$$\lim_{h\to\infty}\lim_{j\to\infty}\int_\Omega \varphi(x)\,|\tau_h V_j(x)|^p\,\mathrm{d}x = \lim_{h\to\infty}\int_\Omega \varphi(x)\int |\tau_h s|^p\,\mathrm{d}\mu_x(s)\,\mathrm{d}x$$
$$= \big\langle\!\big\langle \varphi\otimes|\bullet|^p,\mu\big\rangle\!\big\rangle, \tag{4.15}$$

where in the last step we used the monotone convergence theorem.

Now choose for every $k\in\mathbb{N}$ a natural number $j(k) > j(k-1)$, where $j(0) := 0$, such that

$$\left|\lim_{j\to\infty}\int_\Omega |\tau_k V_j(x)|^p\,\mathrm{d}x - \int_\Omega |\tau_k V_n(x)|^p\,\mathrm{d}x\right| \le \frac{1}{k} \tag{4.16}$$

for all $n\ge j(k)$. Let $\psi\in \mathrm{L}^\infty(\Omega)$. We may estimate for $l\le k$ that

$$\int_\Omega \psi(x)\,|\tau_k V_{j(k)}(x)|^p\,\mathrm{d}x$$
$$\le \|\psi\|_{\mathrm{L}^\infty}\cdot\int_\Omega |\tau_k V_{j(k)}(x)|^p\,\mathrm{d}x - \int_\Omega \big(\|\psi\|_{\mathrm{L}^\infty}-\psi(x)\big)|\tau_l V_{j(k)}(x)|^p\,\mathrm{d}x.$$

Thus, using (4.15) for $\varphi := \mathbb{1}_\Omega$, (4.16), and also the Young measure representation of limits for μ,

$$\limsup_{k\to\infty}\int_\Omega \psi(x)\,|\tau_k V_{j(k)}(x)|^p\,\mathrm{d}x$$
$$\le \|\psi\|_{\mathrm{L}^\infty}\cdot\big\langle\!\big\langle \mathbb{1}\otimes|\bullet|^p,\mu\big\rangle\!\big\rangle - \int_\Omega\big(\|\psi\|_{\mathrm{L}^\infty}-\psi(x)\big)\int |\tau_l s|^p\,\mathrm{d}\mu_x(s)\,\mathrm{d}x.$$

Letting $l\to\infty$, we then obtain by the monotone convergence theorem that

$$\limsup_{k\to\infty}\int_\Omega \psi(x)\,|\tau_k V_{j(k)}(x)|^p\,\mathrm{d}x \le \big\langle\!\big\langle \psi\otimes|\bullet|^p,\mu\big\rangle\!\big\rangle.$$

Repeating the same argument for $-\psi$, we conclude that

$$|\tau_k V_{j(k)}|^p \rightharpoonup \big(x\mapsto\big\langle|\bullet|^p,\mu_x\big\rangle\big)\quad \text{in } \mathrm{L}^1.$$

Thus, by the Dunford–Pettis Theorem A.13 the functions $W_k := \tau_k V_{j(k)}$ ($k\in\mathbb{N}$) are uniformly L^p-bounded and p-equiintegrable.

Theorem A.38 (iii) implies that $v_{j(k)}$ is Lipschitz continuous with Lipschitz constant at most Ck on the set

$$S_k := \big\{\, x\in\Omega \,:\, V_{j(k)}(x)\le k\,\big\}.$$

By the Kirszbraun Theorem A.35 we may extend each $v_{j(k)}$ to a function $w_k\colon\mathbb{R}^d\to\mathbb{R}^m$ that is globally Lipschitz continuous with Lipschitz constant at most Ck. Since $w_k = v_{j(k)}$ in S_k, for the gradients ∇w_k (which exist almost everywhere by Rademacher's Theorem A.31) we have

$$|\nabla w_k| = |\nabla v_{j(k)}| \le V_{j(k)} = W_k \qquad \text{a.e. in } S_k,$$
$$|\nabla w_k| \le Ck = CW_k \qquad \text{a.e. in } \Omega\setminus S_k.$$

Consequently, $|\nabla w_k| \leq CW_k$ almost everywhere in Ω and thus $(\nabla w_k)_k$ inherits the p-equiintegrability from $(W_k)_k$. Moreover, by the Markov inequality,

$$|\Omega \setminus S_k| \leq \frac{\|V_{j(k)}\|_{\mathrm{L}^p}^p}{k^p} \to 0 \qquad \text{as } k \to \infty.$$

Therefore, for all $\varphi \in \mathrm{C}_0(\Omega)$ and all $h \in \mathrm{C}_0(\mathbb{R}^{m\times d})$,

$$\int_\Omega |\varphi(x)h(\nabla w_k(x)) - \varphi(x)h(\nabla v_{j(k)}(x))| \,\mathrm{d}x \leq \|\varphi\|_\infty \cdot \|h\|_\infty \cdot |\Omega \setminus S_k| \to 0.$$

Since all such φ, h determine the Young measure (see the proof of the Fundamental Theorem 4.1), we have shown that the p-equiintegrable sequence $(\nabla w_k)_k$ generates the same Young measure ν as $(\nabla v_j)_j$.

Step 3. It remains to perform the boundary adjustment. Since $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ embeds compactly into the space $\mathrm{L}^p(\Omega;\mathbb{R}^m)$ by the Rellich–Kondrachov Theorem A.29, we have $w_k \to u$ in L^p. Let $(\varrho_j)_j \subset \mathrm{C}_c^\infty(\Omega;[0,1])$ be a sequence of cut-off functions such that for the sets $G_j := \{\, x \in \Omega : \varrho_j(x) = 1 \,\}$ it holds that $|\Omega \setminus G_j| \to 0$ as $j \to \infty$. For

$$u_{j,k} := \varrho_j w_k + (1 - \varrho_j)u \in \mathrm{W}_u^{1,p}(\Omega;\mathbb{R}^m)$$

we observe

$$\nabla u_{j,k} = \varrho_j \nabla w_k + (1 - \varrho_j)\nabla u + (w_k - u) \otimes \nabla \varrho_j.$$

For all $\varphi \in \mathrm{C}_0(\Omega)$ and $h \in \mathrm{C}_0(\mathbb{R}^{m\times d})$, we have

$$\int_\Omega |\varphi(x)h(\nabla w_k(x)) - \varphi(x)h(\nabla u_{j,k}(x))| \,\mathrm{d}x \leq \|\varphi\|_\infty \cdot \|h\|_\infty \cdot |\Omega \setminus G_j| \to 0$$

as $j \to \infty$, uniformly in k. As we have remarked before, these φ, h determine the Young measure, so we can now select a diagonal sequence $u_j = u_{j,k(j)}$ such that $(\nabla u_j)_j$ generates ν and

$$\|u_j\|_{\mathrm{W}^{1,p}} \leq \|w_j\|_{\mathrm{W}^{1,p}} + \|u\|_{\mathrm{W}^{1,p}} + 1 \leq C(\|v_j\|_{\mathrm{W}^{1,p}} + \|u\|_{\mathrm{W}^{1,p}} + 1).$$

It is also straightforward to check that the equiintegrability of the gradients is not affected by the cut-off procedure. This so-constructed sequence $(u_j)_j$ satisfies all the requirements from the statement of the lemma. □

The preceding proof is also a nice example of how Young measures can be used as a technical tool (note the way in which the auxiliary Young measure μ is employed).

4.5 Homogeneous Gradient Young Measures

We next discuss some properties of homogeneous gradient Young measures $\nu \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$, for which the map $x \mapsto \nu_x$ is (almost everywhere) constant with respect to x.

It is a particular consequence of the following *averaging principle* (which also applies to non-homogeneous Young measures) that the domain of definition of a homogeneous gradient Young measure can be chosen to be any bounded Lipschitz domain $D \subset \mathbb{R}^d$.

Lemma 4.15 *Let $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, where $p \in (1,\infty]$, such that $[\nu] = \nabla u$ for some $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with linear (or affine) boundary values. Then, for any bounded Lipschitz domain $D \subset \mathbb{R}^d$ there exists a homogeneous gradient Young measure $\overline{\nu} \in \mathbf{GY}^p(D;\mathbb{R}^{m\times d})$ such that*

$$\int h \,\mathrm{d}\overline{\nu} = \fint_\Omega \int h \,\mathrm{d}\nu_x \,\mathrm{d}x \tag{4.17}$$

for all continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth if $p < \infty$, or all continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ (no growth condition) if $p = \infty$. This result remains valid if Ω is a d-dimensional unit cube and u has periodic boundary values.

Proof. We only treat the case $p \in (1,\infty)$, the case $p = \infty$ is in fact easier.

Let $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_j|_{\partial\Omega} = Fx$ in the sense of trace for a fixed matrix $F \in \mathbb{R}^{m\times d}$ (here we denote by "Fx" the linear map $x \mapsto Fx$) and such that $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$. This sequence exists by Lemma 4.14 and the fact that $u|_{\partial\Omega} = Fx$ for some $F \in \mathbb{R}^{m\times d}$. In particular, $\sup_j \|\nabla u_j\|_{\mathrm{L}^p} < \infty$. For every $j \in \mathbb{N}$ choose a Vitali cover of D consisting of rescaled disjoint copies of Ω, see Theorem A.16, i.e.,

$$D = Z^{(j)} \cup \bigcup_{k=1}^{\infty} \Omega(a_k^{(j)}, r_k^{(j)}), \qquad |Z^{(j)}| = 0,$$

with $a_k^{(j)} \in D$, $0 < r_k^{(j)} \le 1/j$ $(k \in \mathbb{N})$, and $\Omega(a,r) := a + r\Omega$. Then define, for $y \in D$,

$$v_j(y) := r_k^{(j)} u_j\left(\frac{y - a_k^{(j)}}{r_k^{(j)}}\right) + Fa_k^{(j)} \qquad \text{if } y \in \Omega(a_k^{(j)}, r_k^{(j)})\ (k \in \mathbb{N}).$$

It is easy to check that there are no jumps over the gluing boundaries and

$$\nabla v_j(y) = \nabla u_j\left(\frac{y - a_k^{(j)}}{r_k^{(j)}}\right) \qquad \text{if } y \in \Omega(a_k^{(j)}, r_k^{(j)})\ (k \in \mathbb{N}).$$

We can then employ a change of variables to compute for all $\varphi \in \mathrm{C}_0(D)$ and all continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth that

$$\begin{aligned}\int_D \varphi(y)\, h(\nabla v_j(y))\,\mathrm{d}y &= \sum_{k=1}^{\infty} \int_{\Omega(a_k^{(j)}, r_k^{(j)})} \varphi(y)\, h\left(\nabla u_j\left(\frac{y-a_k^{(j)}}{r_k^{(j)}}\right)\right)\mathrm{d}y \\ &= \sum_{k=1}^{\infty} (r_k^{(j)})^d\, \varphi(a_k^{(j)}) \int_\Omega h(\nabla u_j(x))\,\mathrm{d}x \\ &\quad + \mathrm{O}\left(\frac{1}{j}\right) \sum_{k=1}^{\infty} (r_k^{(j)})^d\, \|\nabla u_j\|_{\mathrm{L}^p}^p,\end{aligned}$$

where we also used that φ is uniformly continuous. Since $\sum_{k=1}^{\infty} (r_k^{(j)})^d = |D|/|\Omega|$, the error term vanishes as $j \to \infty$. So, using also that the Riemann sums converge to the integral, i.e.,

$$\lim_{j\to\infty} \sum_{k=1}^{\infty} (r_k^{(j)})^d\, \varphi(a_k^{(j)}) = \frac{1}{|\Omega|} \int_D \varphi(x)\,\mathrm{d}x,$$

as well as the Young measure representation of limits, we arrive at

$$\lim_{j\to\infty} \int_D \varphi(y)\, h(\nabla v_j(y))\,\mathrm{d}y = \int_D \varphi(x)\,\mathrm{d}x \cdot \fint_\Omega \int h(A)\,\mathrm{d}\nu_x\,\mathrm{d}x. \tag{4.18}$$

For $\varphi = \mathbb{1}$ and $h(A) := |A|^p$, this gives

$$\limsup_{j\to\infty} \|\nabla v_j\|_{\mathrm{L}^p}^p = \limsup_{j\to\infty} \|\nabla u_j\|_{\mathrm{L}^p}^p < \infty.$$

Thus, there exists a $\overline{\nu} = (\overline{\nu}_y)_{y\in D} \in \mathbf{GY}^p(D;\mathbb{R}^m)$ such that $\nabla v_j \overset{\mathbf{Y}}{\to} \overline{\nu}$, up to selecting a subsequence. Using Lemma 4.14 we may moreover assume that $(\nabla u_j)_j$, and hence also $(\nabla v_j)_j$, is p-equiintegrable. Then, (4.18) implies

$$\int_D \int \varphi(y)\, h(A)\,\mathrm{d}\overline{\nu}_y(A)\,\mathrm{d}y = \int_D \varphi(x)\,\mathrm{d}x \cdot \fint_\Omega \int h(A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x$$

for all φ, h as above. Varying φ, this implies in particular that $(\overline{\nu}_y)_{y\in D} = \overline{\nu}$ is homogeneous. For $\varphi = \mathbb{1}$, we get (4.17).

The additional claim about underlying deformations defined on a unit cube and with periodic boundary values follows analogously since in this case we can "glue" generating functions via a staircase construction; this is the task of Problem 4.10. □

Applying the preceding averaging principle to an elementary gradient Young measure, we get the following result, which is often called the *Riemann–Lebesgue lemma*.

Lemma 4.16 *Let $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, $p \in (1,\infty]$, have linear (or affine) boundary values. Then, there is a homogeneous gradient Young measure $\overline{\delta[\nabla u]} \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ such that*

$$\int h \,\mathrm{d}\overline{\delta[\nabla u]} = \fint_\Omega h(\nabla u(x)) \,\mathrm{d}x \tag{4.19}$$

for all continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth if $p < \infty$, or all continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ (no growth condition) if $p = \infty$. This result remains valid if Ω is a unit cube and u has periodic boundary values.

Notes and Historical Remarks

Laurence Chisholm Young originally introduced the objects that are now called Young measures as "generalized curves/surfaces" in the late 1930s and early 1940s, see [339–341], to treat problems in the Calculus of Variations and optimal control theory that could not be solved using classical methods. His book [342] explains these objects and their applications in great detail (in particular, the "sailing against the wind" example from Section 1.6 is adapted from there). The theory of Young measures is now very mature and there are several monographs [70,272,287] that give overviews of the theory from different points of view. In the spirit of Young's original work, in Chapter 8 we will consider relaxation problems formulated using Young measures.

A second avenue of development – somewhat different from Young's original intention – is to use Young measures as a *technical tool*. This approach is in fact quite old and was probably first adopted in a series of articles by McShane from the 1940s [228–230]. There, the author first finds a Young measure solution to a variational problem, then proves additional properties of the obtained minimizing Young measure, and finally concludes that these properties entail that the Young measure equals a Dirac mass almost everywhere. Hence, the generalized solution is in fact given by an elementary Young measure corresponding to a classical minimizing map.

Many people contributed to Young measure theory from the 1970s onward, including Berliocchi–Lasry [50], Balder [33], Ball [36] and Kristensen [204]. An important breakthrough in this respect was the characterization of the class of gradient Young measures in the early 1990s by Kinderlehrer and Pedregal [190, 191], which we will present in Theorem 8.11. Their result places gradient Young measures in duality with quasiconvex functions (to be defined in the next chapter) via Jensen-type inequalities; another work in this direction is Sychev's article [316]. Young measures can also be used to show regularity, see the work by Dolzmann–Kristensen [125]. Carstensen–Roubíček [69] considered numerical approximations.

Young measure theory was opened up to many new applications in the late 1970s and early 1980s, when Tartar [323, 324, 326] and Murat [254–256] developed the theory of *compensated compactness* and were able to settle many open problems in the theory of hyperbolic conservation laws, see Chapter XVI in [97] for an overview and many references to the vast literature. Another important contributor here was DiPerna, see, for example, [121]. A key point of this strategy (aspects of which go back to McShane)

is that the weak convergence of suitably constrained sequences improves to convergence in measure (or even in norm), hence the name "compensated *compactness*". We discuss compensated compactness theory in Section 8.2.

The disintegration result from Theorem 4.4 is essentially contained in the result from probability theory that *regular conditional probabilities* exist, see, for instance, Theorem 89.1 in [286]. The result as stated also holds for vector-measures, see Theorem 2.28 of [18]. A stronger version of the Scorza Dragoni Theorem 4.5 can be found in Theorem 6.35 of [142]. Lemma 4.14 is a version of the *decomposition lemma* from [146]; another version is in [203].

In the case $p = 1$ the theory of (classical) Young measures is not very satisfactory and some important results such as Lemma 4.12 and Lemma 4.14 do not hold (see Problem 4.9 for a counterexample to Lemma 4.14 in the case $p = 1$). The fundamental reason for this deficiency is that in L^1 norm-bounded sequences are not weakly precompact. A partial remedy can be found by weakening the notion of convergence to be employed in L^1. Then, one can use Chacon's *biting lemma*:

Lemma 4.17 (Chacon 1980 [64]) *Let $(V_j)_j \subset \mathrm{L}^1(\Omega;\mathbb{R}^N)$ be a norm-bounded sequence generating the Young measure $\nu \in \mathbf{Y}^1(\Omega;\mathbb{R}^N)$. Define $V(x) := [\nu_x]$. Then, there exists an increasing sequence $\Omega_k \subset \Omega$ with $|\Omega_k| \uparrow |\Omega|$ such that $V_j \rightharpoonup V$ in $\mathrm{L}^1(\Omega_k;\mathbb{R}^N)$ for all k; this is called* ***biting convergence****.*

More information on this topic can be found in [44] and Chapter 6 of [272]. We will return to the topic of Young measures generated by merely L^1-bounded sequences and develop a much more satisfying theory in Chapter 15.

Problems

4.1 Show that the family $(\nu_x)_{x\in\Omega}$ constructed in the proof of Theorem 4.4 is weakly* measurable. *Hint:* Use the Scorza Dragoni Theorem 4.5.

4.2 Prove that a sequence of measurable maps $V_j : \Omega \to \mathbb{R}^N$ satisfying only the tightness condition

$$\lim_{h\uparrow\infty} \sup_{j\in\mathbb{N}} |\{|V_j| \geq h\}| = 0$$

also generates a Young measure (in a suitable sense).

4.3 Show that for every $p \in [1,\infty)$ any weakly* measurable parametrized measure $(\nu_x)_{x\in\Omega} \subset \mathcal{M}^1(\mathbb{R}^N)$ with $\langle|\bullet|^p, \nu_x\rangle \in \mathrm{L}^p(\Omega)$ (as a function of x) can be generated by a sequence $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$. *Hint:* Approximate a general measure by linear combinations of Dirac masses and use a gluing argument.

4.4 Take $\Omega := (0,1)$ and let $u_j(x) = \sin(2\pi jx)$ for $j \in \mathbb{N}$. Show that the sequence $(u_j)_j$ generates the homogeneous Young measure $\nu \in \mathbf{Y}^\infty((0,1))$ with

$$\nu_x = \frac{1}{\pi\sqrt{1-y^2}}\,\mathcal{L}^1_y \llcorner (-1,1) \qquad \text{for a.e. } x \in (0,1).$$

4.5 Let $a, b \in \mathbb{R}^m$ with $a \neq b$ and let $\theta \in (0,1)$.

(i) Set $\Omega := (0,1)$. Let $\nu = (\nu_x)_{x\in\Omega} \subset \mathcal{M}^1(\mathbb{R}^m)$ be the Young measure with

$$\nu_x = \theta\delta_a + (1-\theta)\delta_b \qquad \text{for a.e. } x \in \Omega.$$

Construct a generating sequence $(V_j)_j \subset \mathrm{L}^\infty(\Omega;\mathbb{R}^m)$ of ν.

(ii) Let $Q := (0,1)^d$ and define

$$A := a \otimes \mathrm{e}_1, \qquad B := b \otimes \mathrm{e}_1 \in \mathbb{R}^{m\times d}.$$

Construct $(u_j)_j \subset \mathrm{W}^{1,\infty}(Q;\mathbb{R}^m)$, based on V_j from the previous problem, such that the sequence $(\nabla u_j)_j$ generates the gradient Young measure $\mu = (\mu_y)_{y\in Q} \in \mathbf{Y}^\infty(Q;\mathbb{R}^{m\times d})$ given as

$$\mu_y = \theta\delta_A + (1-\theta)\delta_B \qquad \text{for a.e. } y \in Q.$$

4.6 Assume for the sequence from the previous problem, part (ii), that $u_j \overset{*}{\rightharpoonup} a$ in $\mathrm{W}^{1,\infty}$ for $a(y) := Fy$ with $F := \theta A + (1-\theta)B$. Based on this, construct a sequence $(V_j)_j \subset \mathrm{W}^{1,\infty}(Q;\mathbb{R}^m)$ such that $v_j \in \mathrm{C}(\overline{Q})$ for all $j \in \mathbb{N}$, $(\nabla v_j)_j$ generates μ, and $v_j|_{\partial Q} = Fx$.

4.7 Let $A, B, C \in \mathbb{R}^{m\times d}$ such that for some $b, c \in \mathbb{R}^m$,

$$B - A = b \otimes \mathrm{e}_1 \qquad \text{and} \qquad C - A = c \otimes \mathrm{e}_1.$$

Let $\theta_A, \theta_B, \theta_C \in (0,1)$ be such that $\theta_A + \theta_B + \theta_C = 1$. Show that $\nu = (\nu_x)_{x\in\Omega}$ ($\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain that you can choose as you like) with

$$\nu_x := \theta_A\delta_A + \theta_B\delta_B + \theta_C\delta_C \qquad \text{for a.e. } x \in \Omega$$

is a (homogenous) $\mathrm{W}^{1,\infty}$-gradient Young measure with barycenter $[\nu] = \theta_A A + \theta_B B + \theta_C C$.

4.8 Let $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ be a Young measure with generating sequence $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$. Show that for every closed set $E \subset \mathbb{R}^N$ with $\nu_x(\partial E) = 0$ it holds that $\nu_x(E)$ is the asymptotic fraction of values of v_j in E, that is,

$$\nu_x(E) = \lim_{r\downarrow 0} \lim_{j\to\infty} \frac{|\{\, x \in B_r(x) : Vj_j(x) \in E \,\}|}{\omega_d r^d},$$

where $\omega_d = |B_1|$. Also show that this formula fails in general when E is not closed.

4.9 Show that in $\Omega := (0,2)^2$ for the map $V\colon \Omega \to \mathbb{R}^2$ given as

$$V(x) := \begin{cases} 0 & \text{if } x \in (0,1)^2, \\ \mathrm{e}_1 & \text{if } x \in (0,2)^2 \setminus (0,1)^2, \end{cases}$$

there cannot exist a $u \in \mathrm{W}^{1,1}((0,2)^2)$ such that $\nabla u = V$. However, prove that the Young measure $\nu \in \mathbf{Y}^1((0,2)^2;\mathbb{R}^2)$ defined by

$$\nu_x := \mathbb{1}_{(0,1)^2}(x)\delta_0 + \mathbb{1}_{(0,2)^2\setminus(0,1)^2}(x)\delta_{\mathrm{e}_1}, \qquad x \in (0,2)^2,$$

is in $\mathbf{GY}^1((0,2)^2;\mathbb{R}^2)$ (which is defined in the same way as at the beginning of the chapter) by exhibiting a norm-bounded sequence $(u_j)_j \subset \mathrm{W}^{1,1}((0,2)^2)$ with $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$. Conclude that Lemma 4.14 cannot be extended to cover the case $p = 1$.

4.10 Prove Lemma 4.15 in the case $\Omega = (-1/2, 1/2)^d$ and underlying deformation u with periodic boundary values.

Chapter 5
Quasiconvexity

We saw in the Tonelli–Serrin Theorem 2.6 that convexity of the integrand (in the gradient variable) implies the weak lower semicontinuity of the corresponding integral functional. Moreover, we proved in Proposition 2.9 that if $d = 1$ or $m = 1$, then convexity of the integrand (in the gradient variable) is also necessary for weak lower semicontinuity.

In the vectorial case ($d, m > 1$), however, such a convexity assumption has serious shortcomings. In particular, it is incompatible with one of the most fundamental principles of continuum mechanics: Assume that the integrand $f = f(A)$ is *frame-indifferent*, that is,

$$f(QA) = f(A) \qquad \text{for all } Q \in \mathrm{SO}(d),\ A \in \mathbb{R}^{d\times d},$$

where $\mathrm{SO}(d)$ is the group of $(d \times d)$-orthogonal matrices with determinant 1 (that is, rotations if $d = 2$ or $d = 3$). Furthermore, suppose that every purely compressive or purely expansive deformation costs energy, i.e.,

$$f(\alpha\,\mathrm{Id}) > f(\mathrm{Id}) \qquad \text{for all } \alpha \in (0,\infty) \setminus \{1\}, \tag{5.1}$$

which is very reasonable in applications. Then, f cannot be convex: Let us for simplicity assume $d = 2$. Set, for a fixed $\gamma \in (0, \pi/2)$,

$$Q := \begin{pmatrix} \cos\gamma & -\sin\gamma \\ \sin\gamma & \cos\gamma \end{pmatrix} \in \mathrm{SO}(2).$$

If f was convex, we would get

$$f((\cos\gamma)\,\mathrm{Id}) \le \frac{1}{2} f(Q) + \frac{1}{2} f(Q^{\mathsf{T}}) = f(\mathrm{Id}),$$

contradicting (5.1). So, convexity is not suitable for many variational problems originating from continuum mechanics.

On the other hand, it turns out that (in the vectorial case) one can find weakly lower semicontinuous integral functionals whose integrands are non-convex. As an instructive example, consider a bounded Lipschitz domain $\Omega \subset \mathbb{R}^2$ and define

$$\mathcal{F}[u] := \int_\Omega |\det \nabla u(x)|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,4}(\Omega;\mathbb{R}^2).$$

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Clearly, the integrand $f(A) := |\det A|^2$ is far from being convex in $A \in \mathbb{R}^{2\times 2}$. However, one may calculate for $u = (u^1, u^2)^\mathsf{T} \in \mathrm{W}^{1,4}(\Omega;\mathbb{R}^2)$ and $\psi \in \mathrm{C}_c^\infty(\Omega)$ as follows:

$$\begin{aligned}\int_\Omega \det \nabla u \cdot \psi \,\mathrm{d}x &= \int_\Omega \big(\partial_1 u^1 \cdot \partial_2 u^2 - \partial_2 u^1 \cdot \partial_1 u^2\big)\,\psi \,\mathrm{d}x \\ &= \int_\Omega \big(\partial_1(u^1 \cdot \partial_2 u^2) - \partial_2(u^1 \cdot \partial_1 u^2)\big)\,\psi \,\mathrm{d}x \\ &= -\int_\Omega (u^1 \cdot \partial_2 u^2)\,\partial_1\psi - (u^1 \cdot \partial_1 u^2)\,\partial_2\psi \,\mathrm{d}x,\end{aligned}$$

where we have used that weak derivatives commute. Let $u_j \rightharpoonup u$ in $\mathrm{W}^{1,4}(\Omega;\mathbb{R}^2)$, for which also $u_j \to u$ in L^4 (strongly) by the Rellich–Kondrachov Theorem A.29. Via the strong $\times$ weak continuity of integrals (which follows from Hölder's inequality) we conclude from the above calculation that

$$\int_\Omega \det \nabla u_j \cdot \psi \,\mathrm{d}x \to \int_\Omega \det \nabla u \cdot \psi \,\mathrm{d}x$$

for all $\psi \in \mathrm{C}_c^\infty(\Omega)$. Additionally, since the (2×2)-determinant is a second-degree polynomial in the matrix entries, $(\det \nabla u_j)_j$ is uniformly bounded in $\mathrm{L}^2(\Omega)$, hence weakly precompact in that space. Thus,

$$\det \nabla u_j \rightharpoonup \det \nabla u \quad \text{in } \mathrm{L}^2.$$

Now, $|\bullet|^2$ is convex, so we may argue as in the proof of the Tonelli–Serrin Theorem 2.6, but with Mazur's lemma applied to the sequence $(\det \nabla u_j)_j$, to conclude that our $\mathcal{F}$ is weakly lower semicontinuous on $\mathrm{W}^{1,4}(\Omega;\mathbb{R}^2)$.

Integrands like the one above fall under Charles B. Morrey Jr.'s concept of *quasiconvexity*, which remedies the shortcomings of convexity for vector-valued variational problems. After exploring some basic properties of quasiconvex functions, we will see how this concept neatly combines with the Young measure theory from the previous chapter to yield an essentially optimal lower semicontinuity theorem (in the case of standard growth bounds).

5.1 Quasiconvexity

The following definition was given in Charles B. Morrey Jr.'s seminar 1952 paper [239] and has proven pivotal in the development of the field.

A locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is called **quasiconvex** if

$$h(A) \le \fint_{B_1} h(A + \nabla\psi(z)) \,\mathrm{d}z \tag{5.2}$$

for all $A \in \mathbb{R}^{m\times d}$ and all $\psi \in \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$. Here note that the integral on the right-hand side is well-defined due to the Borel-measurability and local boundedness of h.

Before we come to the mathematical analysis, let us mention a physical interpretation of quasiconvexity: For $d = m = 3$ suppose that

$$\mathcal{F}[y] := \int_{B_1} h(\nabla y(x))\,\mathrm{d}x, \qquad y \in \mathrm{W}^{1,\infty}(B_1;\mathbb{R}^3),$$

models the physical energy of an elastic body, whose deformation from the reference configuration $\Omega := B_1$ is given through $y\colon B_1 \to \mathbb{R}^3$ (see Section 1.7 for more details on the modelling). A special class of deformations are the *affine* ones, $a(x) = y_0 + Ax$ for some $y_0 \in \mathbb{R}^3$, $A \in \mathbb{R}^{3\times 3}$. Then, quasiconvexity of f entails that

$$\mathcal{F}[a] = \int_{B_1} h(A)\,\mathrm{d}x \le \int_{B_1} h(A + \nabla\psi(x))\,\mathrm{d}x = \mathcal{F}[a+\psi]$$

for all $\psi \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^3)$. This means that the affine deformation a is always energetically favorable over the *internally distorted* deformation $a + \psi$, which is often a reasonable assumption for real materials. This interpretation also holds for any other bounded Lipschitz domain Ω by Lemma 5.2 below.

To justify the name, we also need to convince ourselves that quasiconvexity is indeed a notion of *convexity*: For $A \in \mathbb{R}^{m\times d}$ and $V \in \mathrm{L}^1(B_1;\mathbb{R}^{m\times d})$ with $\int_{B_1} V(x)\,\mathrm{d}x = 0$ define a probability measure $\mu \in \mathcal{M}^1(\mathbb{R}^{m\times d})$ via its action as follows (recall that $\mathcal{M}(\mathbb{R}^{m\times d}) \cong \mathrm{C}_0(\mathbb{R}^{m\times d})^*$ by the Riesz Representation Theorem A.22):

$$\langle h, \mu\rangle := \fint_{B_1} h(A + V(x))\,\mathrm{d}x \qquad \text{for } h \in \mathrm{C}_0(\mathbb{R}^{m\times d}).$$

Since $|\langle h,\mu\rangle| \le \|h\|_\infty$, this μ is easily seen to be an element of the dual space to $\mathrm{C}_0(\mathbb{R}^{m\times d})$. Moreover, μ is a positive measure because $\langle h,\mu\rangle \ge 0$ for $h \ge 0$. Consequently, we may naturally extend the duality pairing $\langle h,\mu\rangle$ to any continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with $x \mapsto h(A + V(x))$ integrable on B_1, or h merely measurable and bounded from below. Evidently $\langle \mathbb{1}, \mu\rangle = 1$, so μ is a probability measure.

The barycenter $[\mu]$ of μ can be computed as follows:

$$[\mu] = \langle \mathrm{id}, \mu\rangle = A + \fint_{B_1} V(x)\,\mathrm{d}x = A.$$

Therefore, if $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is convex, we get from Jensen's inequality (see Lemma A.19),

$$h(A) = h([\mu]) \le \langle h,\mu\rangle = \fint_{B_1} h(A + V(x))\,\mathrm{d}x.$$

In particular, (5.2) holds if we set $V(x) := \nabla\psi(x)$ for any $\psi \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$. Thus, we have shown:

Proposition 5.1 *All convex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ are quasiconvex.*

Two basic properties of quasiconvexity are collected in the following lemma.

Lemma 5.2 *Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be locally bounded and Borel-measurable.*

(i) In the definition of quasiconvexity we can replace the domain B_1 by any bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$.

(ii) If h has p-growth, i.e.,

$$|h(A)| \le M(1 + |A|^p), \qquad A \in \mathbb{R}^{m\times d},$$

for some $p \in [1,\infty)$, $M > 0$, then in the definition (5.2) of quasiconvexity we can replace testing with all $\psi \in \mathrm{W}^{1,\infty}_0(\Omega;\mathbb{R}^m)$ by testing with all $\psi \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$.

Proof. Ad (i). Let $\Omega, \widetilde{\Omega}$ be bounded Lipschitz domains. We will show that if (5.2) holds with the generic domain Ω in place of B_1, that is,

$$h(A) \le \fint_\Omega h(A + \nabla\psi(x))\,\mathrm{d}x, \tag{5.3}$$

then (5.3) also holds for $\widetilde{\Omega}$.

Choose $a \in \widetilde{\Omega}$ and $r > 0$ so small that $a + r\Omega \subset \widetilde{\Omega}$. For a fixed $\psi \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$, considered to be extended by zero to all of $\mathbb{R}^d$, we define the map $\widetilde{\psi} \in \mathrm{W}^{1,p}_0(\widetilde{\Omega};\mathbb{R}^m)$ as follows:

$$\widetilde{\psi}(y) := r\psi\left(\frac{y-a}{r}\right) \qquad y \in \widetilde{\Omega}.$$

Then,

$$\begin{aligned}\int_{\widetilde{\Omega}} h(A + \nabla\widetilde{\psi}(y))\,\mathrm{d}y &= \int_{a+r\Omega} h\left(A + \nabla\psi\left(\frac{y-a}{r}\right)\right)\mathrm{d}y + h(A)\big(|\widetilde{\Omega}| - r^d|\Omega|\big)\\ &= r^d \int_\Omega h(A + \nabla\psi(x)) - h(A)\,\mathrm{d}x + h(A)|\widetilde{\Omega}|\\ &\ge h(A)|\widetilde{\Omega}|,\end{aligned}$$

where we used (5.3) in the last line. This shows (5.3) with $\widetilde{\Omega}$ in place of Ω.

Ad (ii). The second assertion follows since $\mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$ is dense in the space $\mathrm{W}^{1,p}_0(B_1;\mathbb{R}^m)$ and, under a p-growth assumption, for all $A \in \mathbb{R}^{m\times d}$ the integral functional

$$\psi \mapsto \fint_\Omega h(A + \nabla\psi(x))\,\mathrm{d}x, \qquad \psi \in \mathrm{W}^{1,p}_0(B_1;\mathbb{R}^m),$$

is well-defined and $\mathrm{W}^{1,p}$-continuous by Pratt's Theorem A.11 (see the proof of Theorem 2.13 for a similar argument). □

The determinant is the canonical example of a quasiconvex, but not convex, function, as will be proved below.

5.2 Null-Lagrangians

In this section, we will investigate the properties of minors (subdeterminants) as integrands. In particular, we will show that they are quasiconvex, but never convex (except for (1×1)-minors).

Let for $r \in \{1, \dots, \min\{d, m\}\}$,

$$I \in P(m, r) := \left\{ (i_1, i_2, \dots, i_r) \in \{1, \dots, m\}^r \ : \ i_1 < i_2 < \cdots < i_r \right\}$$

and $J \in P(d, r)$, which is defined analogously. We call I and J **ordered multi-indices**. The corresponding $(r \times r)$**-minor** M_J^I is the function given by

$$M_J^I(A) := \det\big(A_J^I\big), \qquad A \in \mathbb{R}^{m \times d},$$

where A_J^I is the $(r \times r)$-matrix consisting of the I-rows and J-columns of A; the number r is called the **rank** of the minor M_J^I.

The first result of this section shows that all minors are **null-Lagrangians**, which by definition is the class of locally bounded Borel-measurable integrands $h \colon \mathbb{R}^{m \times d} \to \mathbb{R}$ such that the value of $\int_\Omega h(\nabla u) \ \mathrm{d}x$ only depends on the boundary values of $u \in \mathrm{W}^{1,\infty}(\Omega; \mathbb{R}^m)$.

Lemma 5.3 *Let $M \colon \mathbb{R}^{m \times d} \to \mathbb{R}$ be an $(r \times r)$-minor, $r \in \{1, \dots, \min\{d, m\}\}$. If $u, v \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$, $p \in [r, \infty]$, with $u - v \in \mathrm{W}_0^{1,p}(\Omega; \mathbb{R}^m)$, then*

$$\int_\Omega M(\nabla u(x)) \ \mathrm{d}x = \int_\Omega M(\nabla v(x)) \ \mathrm{d}x.$$

Proof. In all of the following we will assume that u, v are smooth and $\operatorname{supp}(u - v) \Subset \Omega$, which can be achieved by approximation and a cut-off procedure, see Theorem A.30. We also need the fact that taking the minor M of the gradient commutes with strong convergence, i.e., the strong continuity of $u \mapsto \int_\Omega M(\nabla u) \ \mathrm{d}x$ in $\mathrm{W}^{1,p}$ for $p \geq r$; this follows by Hadamard's inequality $|M(A)| \leq |A|^r$ and Pratt's Theorem A.11 (see the proof of Lemma 2.16 for a similar argument).

All minors of rank one are just the entries of the matrix and the result follows from the Gauss–Green theorem,

$$\int_\Omega \nabla u \ \mathrm{d}x = -\int_{\partial\Omega} u \otimes n \ \mathrm{d}\mathcal{H}^{d-1} = -\int_{\partial\Omega} v \otimes n \ \mathrm{d}\mathcal{H}^{d-1} = \int_\Omega \nabla v \ \mathrm{d}x$$

since $\operatorname{supp}(u - v) \Subset \Omega$. Here, $\mathcal{H}^{d-1}$ is the $(d-1)$-dimensional surface (Hausdorff) measure on $\partial\Omega$ and n is the unit inner normal on $\partial\Omega$.

For higher-rank minors, the crucial observation is that minors of gradients can be written as (row-wise) *divergences*, which we will establish below. So, if $M(\nabla u) = \operatorname{div} G(u, \nabla u)$, then, since $\operatorname{supp}(u - v) \Subset \Omega$,

$$\begin{aligned}\int_\Omega M(\nabla u)\,\mathrm{d}x &= -\int_{\partial\Omega} G(u,\nabla u)\cdot n\,\mathrm{d}\mathcal{H}^{d-1}\\ &= -\int_{\partial\Omega} G(v,\nabla v)\cdot n\,\mathrm{d}\mathcal{H}^{d-1}\\ &= \int_\Omega M(\nabla v)\,\mathrm{d}x\end{aligned}$$

and the result follows.

We first consider the physically most relevant cases $d = m \in \{2, 3\}$.

For $d = m = 2$ and $u = (u^1, u^2)^\mathsf{T}$, the only second-order minor is the Jacobian determinant, for which

$$\begin{aligned}\det\nabla u &= \partial_1 u^1\cdot\partial_2 u^2 - \partial_2 u^1\cdot\partial_1 u^2\\ &= \partial_1(u^1\cdot\partial_2 u^2) - \partial_2(u^1\cdot\partial_1 u^2)\\ &= \operatorname{div}\big(u^1\cdot\partial_2 u^2, -u^1\cdot\partial_1 u^2\big)\end{aligned}$$

since second derivatives of smooth maps commute.

For $d = m = 3$, consider first a second-order minor $M^{\neg k}_{\neg l}(A)$, i.e., the determinant of A after deleting the k'th row and l'th column. Then, analogously to the situation in two dimensions, we get, using cyclic indices $k, l \in \{1, 2, 3\}$ (meaning that for $k = 3$ we formally set $k + 1 := 1$),

$$\begin{aligned}(-1)^{k+l} M^{\neg k}_{\neg l}(\nabla u) &= \partial_{l+1}u^{k+1}\cdot\partial_{l+2}u^{k+2} - \partial_{l+2}u^{k+1}\cdot\partial_{l+1}u^{k+2}\\ &= \partial_{l+1}(u^{k+1}\cdot\partial_{l+2}u^{k+2}) - \partial_{l+2}(u^{k+1}\cdot\partial_{l+1}u^{k+2}).\end{aligned}\tag{5.4}$$

Hence, $M^{\neg k}_{\neg l}(\nabla u)$ can indeed be written as the divergence of a $G(u, \nabla u)$, as claimed.

For the three-dimensional Jacobian determinant, we will show

$$\det\nabla u = \sum_{l=1}^{3}\partial_l u^1\cdot(\operatorname{cof}\nabla u)^1_l = \sum_{l=1}^{3}\partial_l\big(u^1\cdot(\operatorname{cof}\nabla u)^1_l\big),\tag{5.5}$$

where we recall that $(\operatorname{cof} A)^k_l = (-1)^{k+l} M^{\neg k}_{\neg l}(A)$. To see this, use the Cramer formula $(\det A)I = A(\operatorname{cof} A)^\mathsf{T}$, which holds for any square matrix A, to get

$$\det\nabla u = \sum_{l=1}^{3}\partial_l u^1\cdot(\operatorname{cof}\nabla u)^1_l.$$

Then, (5.5) follows from the **Piola identity**

$$\operatorname{div}\operatorname{cof}\nabla u = 0,\tag{5.6}$$

which can be verified directly by a computation.

For general dimensions d, m we use the notation of differential forms to tame the multilinear algebra involved in the proof. One could also argue in an elementary way by induction, but this is more cumbersome.

So, let M be an $(r\times r)$-minor. Reordering $x^1,\dots,x^d$ and $u^1,\dots,u^m$, we can assume without loss of generality that M is a *principal* minor, i.e., M is the determinant of the top-left $(r\times r)$-submatrix. Then,

$$\begin{aligned} M(\nabla u)\,\mathrm{d}x^1\wedge\cdots\wedge\mathrm{d}x^d &= \mathrm{d}u^1\wedge\cdots\wedge\mathrm{d}u^r\wedge\mathrm{d}x^{r+1}\wedge\cdots\wedge\mathrm{d}x^d \\ &= \mathrm{d}(u^1\wedge\mathrm{d}u^2\wedge\cdots\wedge\mathrm{d}u^r\wedge\mathrm{d}x^{r+1}\wedge\cdots\wedge\mathrm{d}x^d). \end{aligned}$$

Thus, the general Stokes theorem gives

$$\begin{aligned} \int_\Omega M(\nabla u)\,\mathrm{d}x^1\wedge\cdots\wedge\mathrm{d}x^d &= \int_\Omega \mathrm{d}(u^1\wedge\mathrm{d}u^2\wedge\cdots\wedge\mathrm{d}u^r\wedge\mathrm{d}x^{r+1}\wedge\cdots\wedge\mathrm{d}x^d) \\ &= \int_{\partial\Omega} u^1\wedge\mathrm{d}u^2\wedge\cdots\wedge\mathrm{d}u^r\wedge\mathrm{d}x^{r+1}\wedge\cdots\wedge\mathrm{d}x^d. \end{aligned}$$

Therefore, $\int_\Omega M(\nabla u)\,\mathrm{d}x^1\wedge\cdots\wedge\mathrm{d}x^d$ only depends on the values of u near $\partial\Omega$. □

As an immediate consequence, we have:

Corollary 5.4 *All $(r\times r)$-minors $M\colon\mathbb{R}^{m\times d}\to\mathbb{R}$ are **quasiaffine**, that is, both M and $-M$ are quasiconvex.*

Proof. Let $A\in\mathbb{R}^{m\times d}$ and $\psi\in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$. Then, by the preceding lemma,

$$M(A)=\fint_{B_1} M(A+\nabla\psi(z))\,\mathrm{d}z.$$

This already implies the claim. □

Minors also enjoy a surprising weak continuity property:

Lemma 5.5 *Let $M\colon\mathbb{R}^{m\times d}\to\mathbb{R}$ be an $(r\times r)$-minor, $r\in\{1,\dots,\min\{d,m\}\}$, and let $(u_j)_j\subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, where $p\in(r,\infty]$. If*

$$u_j\rightharpoonup u \quad \text{in } \mathrm{W}^{1,p}\ (\overset{*}{\rightharpoonup} \text{ in } \mathrm{W}^{1,\infty} \text{ if } p=\infty),$$

then

$$M(\nabla u_j)\rightharpoonup M(\nabla u) \quad \text{in } \mathrm{L}^{p/r}\ (\overset{*}{\rightharpoonup} \text{ if } p=\infty).$$

Proof. We will only prove this lemma in the case $p<\infty$, $d=m\in\{2,3\}$, where we employ the special structure of minors as divergences, as exhibited in the proof of Lemma 5.3. In the general case, one proceeds by induction, see Problem 5.8.

Let $M^{\neg k}_{\neg l}$ be a (2×2)-minor in three dimensions; in two dimensions there is only one (2×2)-minor, the determinant, but we still use the same notation. We rely on (5.4) to observe that with cyclic indices $k,l\in\{1,2,3\}$,

$$\begin{aligned} &(-1)^{k+l}\int_\Omega M^{\neg k}_{\neg l}(\nabla u_j)\cdot\psi\,\mathrm{d}x \\ &\quad= -\int_\Omega (u_j^{k+1}\cdot\partial_{l+2}u_j^{k+2})\cdot\partial_{l+1}\psi-(u_j^{k+1}\cdot\partial_{l+1}u_j^{k+2})\cdot\partial_{l+2}\psi\,\mathrm{d}x \end{aligned}$$

for all $\psi \in C_c^\infty(\Omega)$. Since $u_j \rightharpoonup u$ in $W^{1,p}$, we have $u_j \to u$ in L^p by the Rellich–Kondrachov Theorem A.29. The above expressions under the integral consist of products of one L^p-strongly and one L^p-weakly converging factor, as well as a fixed $L^{p/(p-2)}$-map. Hence, by the strong $\times$ weak continuity of integrals (which follows from Hölder's inequality), the above integral converges as $j \to \infty$ to

$$(-1)^{k+l} \int_\Omega M_{\neg l}^{\neg k}(\nabla u) \cdot \psi \, \mathrm{d}x.$$

Since the sequence $(M(\nabla u_j))_j$ is uniformly norm-bounded in $L^{p/2}$ and $p > r = 2$, it is weakly precompact in this space (see Theorem A.1). From the density of $C_c^\infty(\Omega)$ in $L^{p/2}(\Omega)^* \cong L^{p/(p-2)}(\Omega)$, we thus conclude the sought convergence.

For $d = m = 3$, we additionally need to consider the determinant. However, as a consequence of the above argument in two dimensions, $\operatorname{cof} \nabla u_j \rightharpoonup \operatorname{cof} \nabla u$ in $L^{p/2}$. Then, (5.5) implies

$$\int_\Omega \det \nabla u_j \cdot \psi \, \mathrm{d}x = -\sum_{l=1}^3 \int_\Omega u_j^1 \cdot (\operatorname{cof} \nabla u_j)_l^1 \cdot \partial_l \psi \, \mathrm{d}x$$

for all $\psi \in C_c^\infty(\Omega)$. By a similar reasoning as before, this expression converges to

$$-\sum_{l=1}^3 \int_\Omega u^1 \cdot (\operatorname{cof} \nabla u)_l^1 \cdot \partial_l \psi \, \mathrm{d}x = \int_\Omega \det \nabla u \cdot \psi \, \mathrm{d}x$$

if $u_j \rightharpoonup u$ in $W^{1,p}$. Via a compactness and density argument as above, this convergence then also holds for $\psi \in L^{p/3}(\Omega)^* \cong L^{p/(p-3)}(\Omega)$. □

In fact, it can be shown that *any* quasiaffine function can be written as an affine function of all the minors. This characterization of quasiaffine functions is due to Ball [34]; a different proof (also including further characterizing statements) can be found in Theorem 5.20 of [91]. We will see further non-trivial examples of quasiconvex functions later.

5.3 Rank-One Convexity

An even weaker notion of convexity than quasiconvexity is the following: A locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is called **rank-one convex** if it is convex along any rank-one line, that is,

$$h(\theta A + (1-\theta)B) \le \theta h(A) + (1-\theta)h(B) \tag{5.7}$$

for all $A, B \in \mathbb{R}^{m\times d}$ with $\operatorname{rank}(A-B) \le 1$ and all $\theta \in (0,1)$. In this context, recall that a matrix $F \in \mathbb{R}^{m\times d}$ has rank one if and only if $F = a \otimes b = ab^\mathsf{T}$ for some $a \in \mathbb{R}^m \setminus \{0\}$,

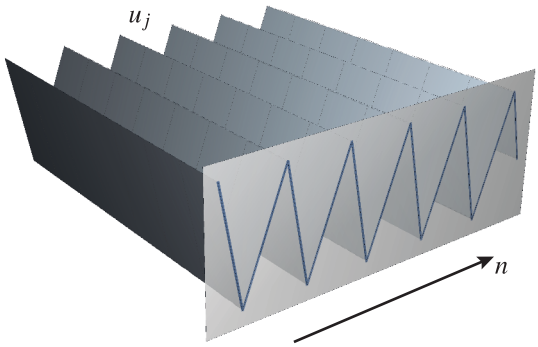


Fig. 5.1 The laminate u_j

$b \in \mathbb{R}^d \setminus \{0\}$. We remark that the local boundedness of h is in fact automatic if (5.7) holds; for this fact see, for instance, the proof of Lemma 2.3 in [195].

Proposition 5.6 *If $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is quasiconvex, then it is rank-one convex.*

Proof. Let $A, B \in \mathbb{R}^{m\times d}$ with $B - A = a \otimes n$ for $a \in \mathbb{R}^m \setminus \{0\}$ and $n \in \mathbb{S}^{d-1}$, the unit sphere in $\mathbb{R}^d$. Denote by $Q_n \subset \mathbb{R}^d$ a unit-volume cube ($|Q_n| = 1$) centered at the origin and with two faces orthogonal to n. We also let $\theta \in (0,1)$.

Step 1. Set $F := \theta A + (1-\theta)B$ and define the sequence of maps $u_j \in \mathrm{W}^{1,\infty}(Q_n;\mathbb{R}^m)$ as follows:

$$u_j(x) := Fx + \frac{1}{j}\varphi_0\big(jx\cdot n - \lfloor jx\cdot n\rfloor\big)a, \qquad x \in Q_n,$$

where

$$\varphi_0(t) := \begin{cases} -(1-\theta)t & \text{if } t \in [0,\theta], \\ \theta t - \theta & \text{if } t \in (\theta, 1], \end{cases}$$

see Figure 2.2 for φ_0 and Figure 5.1 for u_j. Here recall that $\lfloor s\rfloor$ denotes the largest integer less than or equal to $s \in \mathbb{R}$. The sequence $(u_j)_j$ is called a **laminate**.

We calculate

$$\nabla u_j(x) = \begin{cases} F - (1-\theta)a\otimes n = A & \text{if } jx\cdot n - \lfloor jx\cdot n\rfloor \in (0,\theta), \\ F + \theta a \otimes n = B & \text{if } jx\cdot n - \lfloor jx\cdot n\rfloor \in (\theta, 1). \end{cases}$$

Thus,

$$\begin{aligned} &\lim_{j\to\infty} \fint_{Q_n} h(\nabla u_j(x))\,\mathrm{d}x \\ &\quad = \frac{|\{x \in Q_n : jx\cdot n - \lfloor jx\cdot n\rfloor \in (0,\theta)\}|}{|\Omega|}\, h(A) \\ &\qquad + \frac{|\{x \in Q_n : jx\cdot n - \lfloor jx\cdot n\rfloor \in (\theta,1)\}|}{|\Omega|}\, h(B) \\ &\quad = \theta h(A) + (1-\theta)h(B). \end{aligned}$$

Also notice that $u_j \overset{*}{\rightharpoonup} Fx$ in $\mathrm{W}^{1,\infty}$ since φ_0 is uniformly bounded, which implies that $u_j \to Fx$ uniformly and that $\|\nabla u_j\|_{\mathrm{L}^\infty}$ remains uniformly bounded as $j \to \infty$. We will show below that we may replace the sequence $(u_j)_j$ with a sequence $(v_j)_j \subset \mathrm{W}^{1,\infty}_{Fx}(Q_n;\mathbb{R}^m)$ (i.e., with the additional property that $v_j|_{\partial Q} = Fx$) such that still

$$\lim_{j\to\infty} \fint_{Q_n} h(\nabla v_j(x))\,\mathrm{d}x = \theta h(A) + (1-\theta)h(B).$$

By the assumed quasiconvexity of h (also see Lemma 5.2), for all $j \in \mathbb{N}$ it then holds that

$$h(F) \le \fint_{Q_n} h(\nabla v_j(z))\,\mathrm{d}z.$$

Thus, we conclude that

$$h(\theta A + (1-\theta)B) \le \theta h(A) + (1-\theta)h(B)$$

and h is indeed rank-one convex.

Step 2. It remains to construct the sequence $(v_j)_j$, for which we employ a standard cut-off construction: Take a sequence $(\varrho_j)_j \subset \mathrm{C}^\infty_c(Q_n;[0,1])$ of cut-off functions such that with $G_j := \{\, x \in \Omega : \varrho_j(x) = 1 \,\}$ it holds that $|Q_n \setminus G_j| \to 0$ as $j \to \infty$. Set

$$v_{j,k}(x) := \varrho_j(x)u_k(x) + (1-\varrho_j(x))Fx, \qquad x \in \Omega,$$

which lies in $\mathrm{W}^{1,\infty}(Q_n;\mathbb{R}^m)$ and satisfies $v_{j,k}(x) = Fx$ near ∂Q_n. Also,

$$\nabla v_{j,k}(x) = \varrho_j(x)\nabla u_k(x) + (1-\varrho_j(x))F + (u_k(x) - Fx) \otimes \nabla \varrho_j(x).$$

As we have seen above, $u_k \to Fx$ uniformly. Thus, for fixed j,

$$\limsup_{k\to\infty} \|\nabla v_{j,k}\|_{\mathrm{L}^\infty} \le \limsup_{k\to\infty} \|\nabla u_k\|_{\mathrm{L}^\infty} + |F| < \infty$$

because the ∇u_k are uniformly L^∞-bounded. Therefore, we can for every $j \in \mathbb{N}$ choose $k(j) \in \mathbb{N}$ such that $\|\nabla v_{j,k(j)}\|_{\mathrm{L}^\infty}$ is bounded by a constant that is independent of j. As h is assumed to be locally bounded, this implies that there exists a constant $C > 0$ (again independent of j) with

$$\big\|h(\nabla u_{k(j)})\big\|_{\mathrm{L}^\infty} + \big\|h(\nabla v_{j,k(j)})\big\|_{\mathrm{L}^\infty} \le C.$$

Hence, for $v_j := v_{j,k(j)}$, we may estimate

$$\begin{aligned}\lim_{j\to\infty} \int_{Q_n} |h(\nabla v_j) - h(\nabla u_{k(j)})|\,\mathrm{d}x \\ \le \lim_{j\to\infty} \int_{Q_n\setminus G_j} |h(\nabla u_{k(j)})| + |h(\nabla v_{j,k(j)})|\,\mathrm{d}x\end{aligned}$$

$$\leq \lim_{j\to\infty} C|Q_n \setminus G_j|$$
$$= 0.$$

This shows that in Step 1 we may indeed replace $(u_j)_j$ by $(v_j)_j$. □

The following is a standard example for quasiconvexity and rank-one convexity.

Example 5.7 (Alibert–Dacorogna–Marcellini 1988 [7, 93]) For $d = m = 2$ and $\gamma \in \mathbb{R}$ define

$$h_\gamma(A) := |A|^2\big(|A|^2 - 2\gamma \det A\big), \qquad A \in \mathbb{R}^{2\times 2}.$$

For this function it is known that

- h_γ is convex if and only if $|\gamma| \leq \dfrac{2\sqrt{2}}{3} \approx 0.94$,
- h_γ is rank-one convex if and only if $|\gamma| \leq \dfrac{2}{\sqrt{3}} \approx 1.15$,
- h_γ is quasiconvex if and only if $|\gamma| \leq \gamma_{\mathrm{QC}}$ for some $\gamma_{\mathrm{QC}} \in \left(1, \dfrac{2}{\sqrt{3}}\right]$.

It is currently unknown whether $\gamma_{\mathrm{QC}} = 2/\sqrt{3}$. We do not prove these statements here, see Section 5.3.8 in [91] for the details.

Later, we will see in Švérak's Theorem 8.4 that rank-one convexity in general does not imply quasiconvexity.

We end this section with the following observations concerning the growth and continuity properties of rank-one convex (or quasiconvex) functions.

Lemma 5.8 *If $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is rank-one convex and there are $M > 0$, $p \in [1,\infty)$ such that*

$$h(A) \leq M(1 + |A|^p), \qquad A \in \mathbb{R}^{m\times d},$$

then h has p-growth (also from below).

Proof. Let $R > 0$ and choose $F_1 \in \mathbb{R}^{m\times d}$ such that $h(F_1) = \inf_{|A|\leq R} h(A)$. Let $F_1, \dots, F_{2^{md}}$ be the matrices that are obtained from F_1 by flipping the sign of any number of entries. The two matrices of the collection that only differ in the flipped sign at position (i,j) lie on the rank-one line $\mathbb{R}(\mathrm{e}_i \otimes \mathrm{e}_j)$ and average to the zero matrix. Thus, applying the rank-one convexity (at most) md times, we have

$$h(0) \leq \frac{1}{2^{md}} \sum_{k=1}^{2^{md}} h(F_k).$$

Then,

$$2^{md} h(0) \leq (2^{md} - 1) \sup_{|A|\leq R} h(A) + \inf_{|A|\leq R} h(A),$$

from which we conclude that

$$-h(A) \leq M(2^{md} - 1)(1 + R^p) - 2^{md} h(0) \qquad \text{if } |A| \leq R.$$

Hence,

$$-h(A) \le \widetilde{M}(1 + |A|^p), \qquad A \in \mathbb{R}^{m\times d},$$

for some $\widetilde{M} > 0$, and h has been shown to have p-growth. □

Lemma 5.9 *If $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is rank-one convex, then it is locally Lipschitz continuous. If additionally h has p-growth with growth constant $M > 0$, then*

$$|h(A) - h(B)| \le CM(1 + |A|^{p-1} + |B|^{p-1})|A - B|, \qquad A, B \in \mathbb{R}^{m\times d}, \tag{5.8}$$

where $C = C(d, m) > 0$ is a dimensional constant. In particular, a rank-one convex h with linear growth ($p = 1$) is (globally) Lipschitz continuous.

Proof. For any $F \in \mathbb{R}^{m\times d}$ and $r > 0$, we will prove the quantitative bound

$$\operatorname{lip}(h; B_r(F)) \le \sqrt{\min\{d, m\}} \cdot \frac{\operatorname{osc}(h; \overline{B_{6r}(F)})}{3r}, \tag{5.9}$$

where

$$\operatorname{lip}(h; B_r(F)) := \sup_{\substack{A,B\in B_r(F)\\ A\neq B}} \frac{|h(A) - h(B)|}{|A - B|}$$

is the Lipschitz constant of h on the ball $B_r(F) \subset \mathbb{R}^{m\times d}$, and

$$\operatorname{osc}(h; \overline{B_r(F)}) := \sup_{\substack{A,B\in \overline{B_r(F)}\\ A\neq B}} |h(A) - h(B)|$$

is the **oscillation** of h on $\overline{B_r(F)}$. By the local boundedness of h, which is part of our definition of rank-one convexity, the oscillation is bounded on every ball. Thus the Lipschitz constant is locally finite.

To show (5.9), let $A, B \in B_r(F)$ and assume first that $\operatorname{rank}(A - B) \le 1$. Define $X \in \mathbb{R}^{m\times d}$ as the intersection of $\partial B_{2r}(F)$ with the ray starting at B and going through A. Then, because h is convex along this ray,

$$\frac{|h(A) - h(B)|}{|A - B|} \le \frac{|h(X) - h(B)|}{|X - B|} \le \frac{\operatorname{osc}(h; \overline{B_{2r}(F)})}{r} =: \alpha(2r). \tag{5.10}$$

For general $A, B \in B_r(F)$, use the (real) singular value decomposition (see Appendix A.1) to write

$$B - A = \sum_{i=1}^{\min\{d,m\}} \sigma_i P(\mathrm{e}_i \otimes \mathrm{e}_i) Q^{\mathsf{T}},$$

where $\sigma_i \ge 0$ is the i'th singular value, and $P \in \mathbb{R}^{m\times m}$, $Q \in \mathbb{R}^{d\times d}$ are orthogonal matrices. Set

$$A_k := A + \sum_{i=1}^{k-1} \sigma_i P(\mathrm{e}_i \otimes \mathrm{e}_i) Q^{\mathsf{T}}, \qquad k = 1, \dots, \min\{d, m\} + 1,$$

for which we have $A_1 = A$ and $A_{\min\{d,m\}+1} = B$. We obtain (recall that we are employing the Frobenius norm $|X| = \sqrt{\sum_i \sigma_i(X)^2}$)

$$|A_k - F| \le |A - F| + \sqrt{\sum_{i=1}^{k-1} \sigma_i^2} \le |A - F| + |B - A| < 3r$$

and

$$\sum_{k=1}^{\min\{d,m\}} |A_k - A_{k+1}|^2 = \sum_{k=1}^{\min\{d,m\}} \sigma_k^2 = |A - B|^2.$$

Applying (5.10) to $A_k, A_{k+1} \in B_{3r}(F)$, $k = 1, \dots, \min\{d, m\}$, we get

$$\begin{aligned} |h(A) - h(B)| &\le \sum_{k=1}^{\min\{d,m\}} |h(A_k) - h(A_{k+1})| \\ &\le \alpha(6r) \sum_{k=1}^{\min\{d,m\}} |A_k - A_{k+1}| \\ &\le \alpha(6r)\sqrt{\min\{d,m\}} \cdot \left[\sum_{k=1}^{\min\{d,m\}} |A_k - A_{k+1}|^2 \right]^{1/2} \\ &= \alpha(6r)\sqrt{\min\{d,m\}} \cdot |A - B|. \end{aligned}$$

This is (5.9).

If we additionally assume that h has p-growth, then

$$\operatorname{osc}(h; \overline{B_r}) \le 2M(1 + r^p), \qquad r > 0,$$

and so, with $F := 0$, $r := \max\{|A|, |B|\}$ the estimate (5.8) follows from the already established bound (5.9). □

Remark 5.10 An improved argument (see Lemma 2.2 in [42]), where one orders the singular values in a favorable way, allows one to show the better estimate

$$\operatorname{lip}(h; B_r(F)) \le \sqrt{\min\{d,m\}} \cdot \frac{\operatorname{osc}(h; \overline{B_{2r}(F)})}{r}.$$

instead of (5.9).

5.4 A Jensen-Type Inequality for Gradient Young Measures

The connection between Young measure theory and quasiconvexity is furnished by the following fundamental *Jensen-type inequality*:

Lemma 5.11 *Let $\nu \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$, where $p \in (1,\infty]$, be a homogeneous gradient Young measure. Then, for all quasiconvex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth (no growth condition if $p = \infty$) it holds that*

$$h([\nu]) \le \int h \,\mathrm{d}\nu. \tag{5.11}$$

Notice that if h is *convex* the conclusion of this lemma is trivially true by the classical Jensen inequality (and holds for all Young measures, not just for *gradient* Young measures).

Proof. Set $F := [\nu]$ and let $(u_j)_j \subset \mathrm{W}^{1,p}_{Fx}(B_1;\mathbb{R}^m)$ with $\nabla u_j \overset{\mathbf{Y}}{\to} \nu$ and $(\nabla u_j)_j$ p-equiintegrable (if $p < \infty$), which is constructed via Lemma 4.14. Then, from the definition of quasiconvexity, we get

$$h(F) \le \fint_{B_1} h(\nabla u_j(x)) \,\mathrm{d}x$$

for every $j \in \mathbb{N}$. Passing to the Young measure limit as $j \to \infty$ on the right-hand side, for which we note that the family $(h(\nabla u_j))_j$ is equiintegrable by the growth assumption on h, we arrive at

$$h(F) \le \fint_{B_1} \int h \,\mathrm{d}\nu \,\mathrm{d}x = \int h \,\mathrm{d}\nu,$$

which is the sought inequality. □

This result will be of crucial importance in proving weak lower semicontinuity in the next section. It is remarkable that the converse also holds, i.e., the validity of (5.11) for all quasiconvex h with p-growth (no growth condition if $p = \infty$) characterizes the class of homogeneous gradient Young measures that are generated by $\mathrm{W}^{1,p}$-gradients in the class of all homogeneous Young measures. This assertion is the content of the Kinderlehrer–Pedregal Theorem 8.11.

Corollary 5.12 *Let $p \in (1,\infty]$ and let $\nu \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$ be a homogeneous gradient Young measure. Then, for all quasiaffine functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth (no growth condition if $p = \infty$) it holds that*

$$h([\nu]) = \int h \,\mathrm{d}\nu.$$

In particular, the preceding corollary applies to the determinant and, more generally, minors, see Corollary 5.4.

5.5 Lower Semicontinuity

We now return to our central goal, namely to prove existence for minimization problems of the form

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega f(x, \nabla u(x))\, \mathrm{d}x \\ \text{over all} & u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g, \end{cases}$$

where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $p \in (1, \infty)$, the Carathéodory integrand $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0, \infty)$ has p-growth, i.e.

$$f(x, A) \le M(1 + |A|^p), \qquad (x, A) \in \Omega \times \mathbb{R}^{m\times d},$$

for some $M > 0$, and $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega; \mathbb{R}^m)$ specifies the boundary values.

In Chapter 2 we solved this problem in the convex case via the Direct Method, a coercivity result, and, crucially, the Tonelli–Serrin Theorem 2.6 that established the weak lower semicontinuity of the integral functional. In this section, we recycle the Direct Method and the coercivity result, but extend the weak lower semicontinuity to *quasi*convex integrands. Some motivation for this was given at the beginning of the chapter.

Let us first consider how we could approach the proof of weak lower semicontinuity. Suppose that we have a sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$. The (norm-bounded) sequence $(\nabla u_j)_j$ generates the gradient Young measure $\nu \in \mathbf{GY}^p(\Omega; \mathbb{R}^{m\times d})$, up to selecting a subsequence, by the Fundamental Theorem of Young measure theory, Theorem 4.1. If for the moment we additionally assume that the sequence of integrands $(f(x, \nabla u_j(x)))_j$ is equiintegrable, then

$$\mathcal{F}[u_j] \to \int_\Omega \int f(x, A)\, \mathrm{d}\nu_x(A)\, \mathrm{d}x \qquad \text{as } j \to \infty.$$

This is useful, because in order to conclude the weak lower semicontinuity of $\mathcal{F}$, it now suffices to establish the Jensen-type inequality

$$\int f(x, A)\, \mathrm{d}\nu_x(A) \ge f(x, \nabla u(x))$$

for almost every $x \in \Omega$, which we have already seen for *homogeneous* gradient Young measures in Lemma 5.11.

Motivated by this idea, we now show via the fundamental *blow-up technique* (also called the *localization technique*) that ν_{x_0} is a homogeneous gradient Young measure in its own right for almost every $x_0 \in \Omega$:

Proposition 5.13 *Let $\nu = (\nu_x)_x \in \mathbf{GY}^p(\Omega; \mathbb{R}^{m\times d})$, where $p \in (1, \infty)$, be a gradient Young measure. Then, for almost every $x_0 \in \Omega$ the probability measure ν_{x_0} is a homogeneous gradient Young measure in its own right,*

$$\nu_{x_0} = (\nu_{x_0})_{y\in B_1} \in \mathbf{GY}^p(B_1; \mathbb{R}^{m\times d}).$$

Proof. Take a countable collection $(\varphi_k \otimes h_k)_k$ as in Lemma 4.8. Let $x_0 \in \Omega$ be a Lebesgue point of all the functions $x \mapsto \langle h_k, \nu_x\rangle$, $k \in \mathbb{N}$, that is,

$$\lim_{r\downarrow 0} \int_{B_1} \left| \langle h_k, \nu_{x_0+ry}\rangle - \langle h_k, \nu_{x_0}\rangle \right| \mathrm{d}y = 0.$$

By Theorem A.21, almost every point in Ω has this property. Then, at such a point x_0, set

$$v_j^{(r)}(y) := \frac{u_j(x_0 + ry) - [u_j]_{B_r(x_0)}}{r}, \qquad y \in B_1,$$

where $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is a generating sequence for ν, and $[u_j]_{B_r(x_0)} := \fint_{B_r(x_0)} u_j \,\mathrm{d}x$. We compute

$$\begin{aligned}\int_{B_1} \varphi_k(y) h_k(\nabla v_j^{(r)}(y)) \,\mathrm{d}y &= \int_{B_1} \varphi_k(y) h_k(\nabla u_j(x_0 + ry)) \,\mathrm{d}y \\ &= \frac{1}{r^d} \int_{B_r(x_0)} \varphi_k\Big(\frac{x - x_0}{r}\Big) h_k(\nabla u_j(x)) \,\mathrm{d}x\end{aligned}$$

via a change of variables. Letting first $j \to \infty$ and then $r \downarrow 0$, we obtain

$$\begin{aligned}&\lim_{r\downarrow 0} \lim_{j\to\infty} \int_{B_1} \varphi_k(y) h_k(\nabla v_j^{(r)}(y)) \,\mathrm{d}y \\ &\quad= \lim_{r\downarrow 0} \frac{1}{r^d} \int_{B_r(x_0)} \varphi_k\Big(\frac{x - x_0}{r}\Big) \big\langle h_k, \nu_x \big\rangle \,\mathrm{d}x \\ &\quad= \lim_{r\downarrow 0} \int_{B_1} \varphi_k(y) \big\langle h_k, \nu_{x_0+ry} \big\rangle \,\mathrm{d}y \\ &\quad= \int_{B_1} \varphi_k(y) \big\langle h_k, \nu_{x_0} \big\rangle \,\mathrm{d}y,\end{aligned}$$

where the last convergence follows from the Lebesgue point property of x_0.

Similarly,

$$\int_{B_1} |\nabla v_j^{(r)}|^p \,\mathrm{d}y = \int_{B_1} |\nabla u_j(x_0 + ry)|^p \,\mathrm{d}y = \frac{1}{r^d} \int_{B_r(x_0)} |\nabla u_j(x)|^p \,\mathrm{d}x,$$

and the last integral is uniformly bounded in j (for fixed r). Denote by $\lambda \in \mathcal{M}^+(\overline{\Omega})$ the weak* limit of the measures $|\nabla u_j|^p \,\mathcal{L}^d \llcorner \Omega$, which exists after taking a subsequence (not relabeled). If we additionally require of x_0 that

$$\limsup_{r\downarrow 0} \frac{\lambda(\overline{B_r(x_0)})}{r^d} < \infty,$$

which holds at $\mathcal{L}^d$-almost every $x_0 \in \Omega$ (see the Besicovitch Differentiation Theorem A.24), then

$$\limsup_{r\downarrow 0} \lim_{j\to\infty} \int_{B_1} |\nabla v_j^{(r)}|^p \,\mathrm{d}y < \infty.$$

Since also $[v_j^{(r)}]_{B_1} = 0$, the Poincaré inequality from Theorem A.27 (ii) yields that there exists a diagonal sequence $w_n := v_{j(n)}^{r(n)}$ ($n \in \mathbb{N}$) that is uniformly bounded in $\mathrm{W}^{1,p}(B_1;\mathbb{R}^m)$ and such that for all $k \in \mathbb{N}$,

$$\lim_{n\to\infty} \int_{B_1} \varphi_k(y) h_k(\nabla w_n(y))\,\mathrm{d}y = \int_{B_1} \varphi_k(y) \langle h_k, \nu_{x_0}\rangle\,\mathrm{d}y.$$

Therefore, $\nabla w_n \overset{\mathbf{Y}}{\to} \nu_{x_0}$ by Lemma 4.8, where we understand ν_{x_0} as a homogeneous (gradient) Young measure on B_1. □

Remark 5.14 The preceding result also remains true for $p = \infty$, but this needs Zhang's Lemma 8.14, which we will prove in Chapter 8. The proof of this fact is the task of Problem 8.10.

The main weak lower semicontinuity theorem for quasiconvex integrands, called Morrey's theorem, is then a straightforward synthesis of the ideas developed so far. Its first version was proved by Charles B. Morrey, Jr. in 1952 (under additional technical assumptions). The Young measure approach in fact allows us to directly prove a more general statement, which in this form was first established by Acerbi–Fusco [1] (using different methods).

Theorem 5.15 (Morrey 1952 [1,239]) *Let $p \in (1,\infty)$ and let $f\colon \Omega\times\mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand with p-growth and such that*

$$f(x,\cdot) \text{ is quasiconvex for almost every } x \in \Omega.$$

Then, the functional $\mathcal{F}$ is weakly lower semicontinuous on $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$.

Proof. Let $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$. We can select a subsequence $(u_{j(n)})_n$ so that $\mathcal{F}[u_{j(n)}] \to \liminf_{j\to\infty} \mathcal{F}[u_j]$. Assume also that $(\nabla u_{n(j)})_n$ generates a gradient Young measure $\nu = (\nu_x)_x \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ with $[\nu] = \nabla u$, which is true up to taking a further subsequence (not relabeled) by the Fundamental Theorem 4.1.

From Proposition 4.7 we get

$$\liminf_{j\to\infty} \mathcal{F}[u_j] = \lim_{n\to\infty} \mathcal{F}[u_{j(n)}] \geq \langle\!\langle f, \nu\rangle\!\rangle = \int_\Omega \int f(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x.$$

Now, for almost every $x \in \Omega$ we can consider ν_x as a homogeneous Young measure in $\mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$ by the blow-up technique from Proposition 5.13. For this ν_x, the Jensen-type inequality from Lemma 5.11 reads as

$$\int f(x,A)\,\mathrm{d}\nu_x(A) \geq f(x,\nabla u(x)) \qquad \text{for a.e. } x \in \Omega.$$

Combining, we arrive at

$$\liminf_{j\to\infty} \mathcal{F}[u_j] \geq \mathcal{F}[u],$$

which is what we wanted to show. □

At this point it is worthwhile to reflect on the role of Young measures in the proof of the preceding result, namely that they allowed us to split the argument into two parts: First, we passed to the lower limit in the functional via the Young measure. Second, we established a Jensen-type inequality, which then yielded the estimate from below by

the limit functional. It is remarkable that the Young measure preserves exactly the right amount of information to serve as an intermediate object.

Let us further observe that for *convex* integrands the above argument (using only the classical Jensen inequality) gives a different proof of the Tonelli–Serrin Theorem 2.6. While it is vastly more technical than the previously-employed proof strategy, the above reasoning could be considered more conceptual because it clearly separates the limit passage (via Young measures) from the lower bound (via convexity).

We can now sum up and prove the existence of a solution for our minimization problem:

Theorem 5.16 *Let $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand such that*

(i) f has p-growth, where $p \in (1,\infty)$;
(ii) f satisfies the p-coercivity estimate $\mu|A|^p - \mu^{-1} \le f(x,A)$ for some $\mu > 0$ and all $(x,A) \in \Omega \times \mathbb{R}^{m\times d}$;
(iii) $f(x,\bullet)$ is quasiconvex for almost every $x \in \Omega$.

Then, the associated functional $\mathcal{F}$ has a minimizer over $\mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$, where $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

Proof. This follows directly by combining the Direct Method from Theorem 2.3 with the coercivity result in Proposition 2.5 and Morrey's Theorem 5.15. □

Remark 5.17 The coercivity condition (ii) in the preceding theorem can be weakened significantly, see Section 7.5.

The following result shows that quasiconvexity is also necessary for weak lower semicontinuity.

Proposition 5.18 *Let $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ be continuous. If the associated functional*

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m),$$

is weakly lower semicontinuous on $\mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$, then f is quasiconvex. Moreover, it suffices that the weak* lower semicontinuity holds in $\mathrm{W}^{1,\infty}_{Fx}(\Omega;\mathbb{R}^m)$ for a fixed matrix $F \in \mathbb{R}^{m\times d}$.*

Proof. We may assume that $B_1 \Subset \Omega$; otherwise we can translate and rescale the domain. Let $A \in \mathbb{R}^{m\times d}$ and $\psi \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$. We need to show

$$f(A) \le \fint_{B_1} f(A + \nabla\psi(z))\,\mathrm{d}z.$$

Take for every $j \in \mathbb{N}$ a Vitali cover of B_1 consisting of disjoint balls, see Theorem A.16, that is,

$$B_1 = Z^{(j)} \cup \bigcup_{k=1}^{\infty} B_k^{(j)}, \qquad |Z^{(j)}| = 0,$$

with $a_k^{(j)} \in B_1$, $0 < r_k^{(j)} \le 1/j$, and $B_k^{(j)} := B_{r_k^{(j)}}(a_k^{(j)})$ $(k \in \mathbb{N})$. Also fix a smooth and bounded function $w\colon \Omega \setminus B_1 \to \mathbb{R}^m$ with $w(x) = Ax$ for $x \in \partial B_1$ and $w|_{\partial\Omega}$ equal to the prescribed boundary values Fx. Define $u_j \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ via

$$u_j(x) := \begin{cases} Ax + r_k^{(j)} \psi\left(\dfrac{x - a_k^{(j)}}{r_k^{(j)}}\right) & \text{if } x \in B_k^{(j)} \ (k \in \mathbb{N}), \\ w(x) & \text{if } x \in \Omega \setminus B_1, \end{cases} \qquad x \in \Omega.$$

Then, since ψ is uniformly bounded, it is not hard to see that $u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{W}^{1,\infty}$ for

$$u(x) = \begin{cases} Ax & \text{if } x \in B_1, \\ w(x) & \text{if } x \in \Omega \setminus B_1. \end{cases}$$

The weak* lower semicontinuity thus yields, after cancelling the constant part of the functional on $\Omega \setminus B_1$,

$$\begin{aligned} \int_{B_1} f(A)\,\mathrm{d}x &\le \liminf_{j\to\infty} \int_{B_1} f(\nabla u_j(x))\,\mathrm{d}x \\ &= \liminf_{j\to\infty} \sum_{k=1}^{\infty} \int_{B_k^{(j)}} f\left(A + \nabla\psi\left(\frac{x - a_k^{(j)}}{r_k^{(j)}}\right)\right) \mathrm{d}x \\ &= \liminf_{j\to\infty} \sum_{k=1}^{\infty} (r_k^{(j)})^d \int_{B_1} f(A + \nabla\psi(y))\,\mathrm{d}y \\ &= \int_{B_1} f(A + \nabla\psi(y))\,\mathrm{d}y \end{aligned}$$

since $\sum_k (r_k^{(j)})^d = 1$. This is nothing else than quasiconvexity at the arbitrarily chosen matrix A. □

5.6 Integrands with u-Dependence

One very useful feature of our Young measure approach is that it also allows us to derive a lower semicontinuity result for u-dependent integrands with minimal additional effort. So consider the following problem:

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega f(x, u(x), \nabla u(x))\,\mathrm{d}x \\ \text{over all} & u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g, \end{cases}$$

where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $p \in (1,\infty)$, and the Carathéodory integrand $f\colon \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d} \to [0,\infty)$ satisfies the p-growth bound

$$f(x,v,A) \le M(1+|v|^p+|A|^p), \qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d}, \tag{5.12}$$

for some $M > 0$, and $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

The idea is to consider Young measures generated by the pairs $(u_j, \nabla u_j) \in \mathbb{R}^{m+md}$:

Lemma 5.19 *Let $(u_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^M)$ and $(V_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^N)$ be norm-bounded sequences such that for some $u \in \mathrm{L}^p(\Omega;\mathbb{R}^M)$, $\nu \in \mathbf{Y}^p(\Omega;\mathbb{R}^N)$ it holds that*

$$u_j \to u \quad \text{pointwise a.e.} \qquad \text{and} \qquad V_j \overset{\mathbf{Y}}{\to} \nu.$$

Then, $(u_j, V_j) \overset{\mathbf{Y}}{\to} \mu = (\mu_x)_x \in \mathbf{Y}^p(\Omega;\mathbb{R}^{M+N})$ with

$$\mu_x = \delta_{u(x)} \otimes \nu_x \qquad \text{for a.e. } x \in \Omega. \tag{5.13}$$

Moreover, for Carathéodory integrands $f\colon \Omega \times \mathbb{R}^M \times \mathbb{R}^N \to [0,\infty)$ satisfying the p-growth bound (5.12), it holds that

$$\liminf_{j\to\infty} \int_\Omega f(x,u_j(x),V_j(x))\,\mathrm{d}x \ge \int_\Omega \big\langle f(x,u(x),\bullet),\nu_x\big\rangle\,\mathrm{d}x.$$

Proof. By a similar density argument as in the proof of Proposition 5.13, it suffices to test the Young measure convergence with integrands $f(x,v,A) = \varphi(x)\psi(v)h(A)$, where $\varphi \in \mathrm{C}_0(\Omega)$, $\psi \in \mathrm{C}_0(\mathbb{R}^M)$, $h \in \mathrm{C}_0(\mathbb{R}^N)$. We already know from the assumptions that

$$h(V_j) \overset{*}{\rightharpoonup} \big(x \mapsto \big\langle h,\nu_x\big\rangle\big) \quad \text{in } \mathrm{L}^\infty.$$

Furthermore, $\psi(u_j) \to \psi(u)$ almost everywhere and thus (strongly) in L^1 since ψ is bounded. Since the product of an L^∞-weakly* converging sequence and an L^1-strongly converging sequence converges weakly* in the sense of measures (which is elementary to check), we deduce that

$$\int_\Omega \varphi(x)\psi(u_j(x))h(V_j(x))\,\mathrm{d}x \to \int_\Omega \varphi(x)\psi(u(x))\big\langle h,\nu_x\big\rangle\,\mathrm{d}x.$$

This then implies (5.13) by density.

The additional claim follows directly from Proposition 4.7. □

The gist of the preceding lemma is that Young measures allow one to "freeze" $u(x)$ in the integrand. So, we can apply the Jensen-type inequality from Lemma 5.11, just like we did for Morrey's Theorem 5.15, to obtain the following result:

Theorem 5.20 (Acerbi–Fusco 1984 [1]) *Let $p \in (1,\infty)$ and let $f\colon \Omega\times\mathbb{R}^m\times\mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand with p-growth, i.e. (5.12) holds. Assume furthermore that*

$$f(x,v,\bullet) \text{ is quasiconvex for every fixed } (x,v) \in \Omega\times\mathbb{R}^m.$$

Then, the functional $\mathcal{F}$ corresponding to f is weakly lower semicontinuous on $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. If additionally f satisfies the p-coercivity estimate

$$\mu|A|^p - \mu^{-1} \le f(x,v,A), \qquad (x,v,A) \in \Omega\times\mathbb{R}^m\times\mathbb{R}^{m\times d},$$

for some $\mu > 0$, then there exists a minimizer of $\mathcal{F}$ over $\mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$, where $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

Remark 5.21 It is not difficult to extend the previous theorem to integrands f satisfying the more general upper growth condition

$$f(x,v,A) \le M(1+|v|^q+|A|^p), \qquad (x,v,A) \in \Omega \times \mathbb{R}^m \times \mathbb{R}^{m\times d},$$

for some $M > 0$ and $q \in [1, dp/(d-p))$. By the Rellich–Kondrachov Theorem A.29, $u_j \to u$ in L^q for all such q and thus Lemma 5.19 can be suitably generalized. Also, we may only require quasiconvexity of $f(x,v,\bullet)$ for $(x,v) \in (\Omega \setminus Z)\times\mathbb{R}^m$, where $|Z| \subset \Omega$ is a negligible set.

5.7 Regularity of Minimizers

At the end of Section 3.2 we discussed the failure of regularity for minimizers of vector-valued problems. Upon closer inspection, it turns out that in all counterexamples the points where regularity fails form a relatively closed "small" set. This is not a coincidence, as we will see momentarily.

We also note that all the regularity theorems discussed so far (see Section 3.2) required (strong) *convexity* of the integrand. Our discussion in this chapter has, however, shown that convexity is not a good fit for vector-valued problems. To remedy this, one introduces the following notion: A locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is called **strongly (2-)quasiconvex** if there exists a $\gamma > 0$ such that

$$A \mapsto h(A) - \gamma|A|^2 \quad \text{is quasiconvex.}$$

Equivalently, we may require that

$$\gamma \fint_{B_1} |A+\nabla\psi(z)|^2 - |A|^2 \,\mathrm{d}z \le \fint_{B_1} h(A+\nabla\psi(z)) - h(A)\,\mathrm{d}z$$

for all $A \in \mathbb{R}^{m\times d}$ and all $\psi \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$.

The most well-known regularity result in this context is due to Evans (but there is significant overlap with work by Acerbi–Fusco [2]):

Theorem 5.22 (Evans 1986 [132]) *Let $f\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be twice continuously differentiable, strongly (2-)quasiconvex, and assume that there exists an $M > 0$ such that*

$$\mathrm{D}^2 f(A)[B,B] \le M|B|^2, \qquad A, B \in \mathbb{R}^{m\times d}.$$

*Let $u \in \mathrm{W}^{1,2}_g(\Omega;\mathbb{R}^m)$ be a minimizer of $\mathcal{F}$ over the set $\mathrm{W}^{1,2}_g(\Omega;\mathbb{R}^m)$, where $g \in \mathrm{W}^{1/2,2}(\partial\Omega;\mathbb{R}^m)$. Then, there exists a relatively closed **singular set** $\Sigma_u \subset \Omega$ with*

$|\Sigma_u| = 0$ *such that*

$$u \in \mathrm{C}^{1,\alpha}_{\mathrm{loc}}(\Omega \setminus \Sigma_u)$$

for all $\alpha \in (0,1)$.

This theorem is a *partial regularity* result because the regularity does not hold everywhere. It should be noted that while the scalar regularity theory was essentially a theory for PDEs, and hence applies to *all* solutions of the Euler–Lagrange equations, this is not the case for the present result: There is *no* regularity theory for critical points of the Euler–Lagrange equation for a quasiconvex or even polyconvex (see the next chapter) integral functional. This was shown by Müller–Švérak in 2003 [252] (for the quasiconvex case) and Székelyhidi Jr. in 2004 [320] (for the polyconvex case); we quote the first result later in Theorem 12.10.

We finally remark that sometimes better estimates on the "smallness" of Σ_u than merely $|\Sigma_u| = 0$ are available. In fact, for strongly convex integrands it can be shown that the (Hausdorff-)dimension of the singular set is at most $d-2$, see Chapter 2 of [158]. In the strongly quasiconvex case much less is known, but at least for minimizers that happen to be $\mathrm{W}^{1,\infty}$ it was established in 2007 by Kristensen–Mingione that the dimension of the singular set is strictly less than d, see [207]. Many other questions remain open.

Notes and Historical Remarks

The reason why quasiconvexity implies rank-one convexity, as shown in Proposition 5.6, is closely related to **Hadamard's jump condition**: For a matrix-valued function $V\colon \mathbb{R}^d \to \mathbb{R}^{m\times d}$ of the form

$$V(x) = \begin{cases} A & \text{if } x\cdot n \le 0, \\ B & \text{if } x\cdot n > 0, \end{cases}$$

where $A, B \in \mathbb{R}^{m\times d}$ and $n \in \mathbb{S}^{d-1}$, to be the gradient of a function $u\colon \mathbb{R}^d \to \mathbb{R}^m$, it is necessary and sufficient that

$$A - B = a \otimes n \qquad \text{for some } a \in \mathbb{R}^m.$$

The results about null-Lagrangians, in particular Lemmas 5.3 and 5.5, go back to Morrey [240] and Ball [34]. The pivotal proof idea that certain combinations of derivatives might have good convergence properties is also the starting point for the theory of *compensated compactness*; see Section 8.2 for more on this. A general result on why ordinary convexity is inadmissible for realistic problems in nonlinear elasticity can be found in Section 4.8 of [78].

The result that rank-one convex functions are locally Lipschitz continuous, Lemma 5.9, is originally due to Morrey [240], but was well-known for convex functions before, see, for example, Corollary 2.4 in [129]. Our proof with a quantitative bound is from Lemma 2.2 in [42]. A more general version can be found in Lemma 2.3 of [195].

It is possible to prove Morrey's Theorem 5.15 without the use of Young measures, see, for instance, Chapter 8 in [91]. However, many of the ideas are essentially the same, they are just carried out directly without the Young measure machinery (which obscures them somewhat). More on lower semicontinuity and Young measures can be found in the book [272].

For reasons of simplicity, all results in this chapter are formulated for Carathéodory integrands, but many continue to hold for $f\colon \Omega \times \mathbb{R}^N \to [0,\infty)$ that are Borel-measurable and lower semicontinuous in the second argument, so-called **normal integrands**, see [50] and [142].

An argument by Kružík [211], which was refined by Müller, shows the curious fact that for a quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with $m \geq 3$, $d \geq 2$ the function $A \mapsto h(A^{\mathsf{T}})$ may not be quasiconvex. The proof can be found in Section 4.7 of [248]; it is based on Šverák's example of a rank-one convex function that is not quasiconvex (for the same dimensions as above), which we will present as Theorem 8.4.

For minimization problems where the integrand can take negative values, one needs to look carefully at the negative part of the integrand, see Problem 5.6. If the integrand has critical negative growth, then lower semicontinuity only holds if the boundary values are fixed along a sequence or if one imposes *quasiconvexity at the boundary*; see [48] for a survey discussing this topic.

Problems

5.1 Let $\Gamma\colon \mathbb{R} \to \mathbb{R}$ be continuous and convex. Show that the integrand

$$h(A) := \Gamma(\det A), \qquad A \in \mathbb{R}^{d\times d},$$

is quasiconvex.

5.2 Define, with $D := (0,1)^d \subset \mathbb{R}^d$,

$$\mathrm{W}^{1,\infty}_{\mathrm{per}}(D;\mathbb{R}^m) := \left\{ u \in \mathrm{W}^{1,\infty}(\mathbb{R}^d;\mathbb{R}^m) \,:\, u(\bullet + \mathrm{e}_i) = u,\, i = 1,\dots,d \right\}.$$

A locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is called **periodic quasiconvex** if

$$h(A) \leq \int_D h(A + \nabla\psi(z))\,\mathrm{d}z$$

for all $A \in \mathbb{R}^{m\times d}$ and all $\psi \in \mathrm{W}^{1,\infty}_{\mathrm{per}}(D;\mathbb{R}^m)$. Show that periodic quasiconvexity and the usual quasiconvexity are equivalent. *Hint:* Let $\psi \in \mathrm{W}^{1,\infty}_{\mathrm{per}}(D;\mathbb{R}^m)$ and define for $k \in \mathbb{N}$ the function $\psi_k\colon \mathbb{R}^d \to \mathbb{R}^m$ as

$$\psi_k(x) := \frac{1}{k}\psi(kx), \qquad x \in \mathbb{R}^d.$$

Prove that

$$\int_D h(A + \nabla\psi(z))\,\mathrm{d}z = \int_D h(A + \nabla\psi_k(z))\,\mathrm{d}z \qquad \text{for all } A \in \mathbb{R}^{m\times d},\ k \in \mathbb{N},$$

and that $\psi_k \in \mathrm{W}^{1,\infty}_{\mathrm{per}}(D;\mathbb{R}^m)$. You will also need a cut-off argument close to the boundary ∂D.

5.3 Denote by $B := \overline{B_1}$ the closed unit ball in $\mathbb{R}^d$.

(i) Let $w\colon B \to \partial B$ be smooth (a "retraction"). Use $|w(x)|^2 = 1$ for every $x \in B$ to show that $\det \nabla w = 0$ in B.
(ii) Use the fact that the determinant is a null-Lagrangian to conclude that there exists at least one $x \in \partial\Omega$ with $w(x) \neq x$. *Hint:* Use an argument by contradiction.
(iii) Derive a smooth version of the *Brouwer fixed point theorem*: Let $u\colon B \to B$ be smooth. Then u has a fixed point $x_* \in B$, that is, $u(x_*) = x_*$. *Hint:* Argue by contradiction and consider the ray emanating from $u(x)$ and passing through x for all $x \in \Omega$ and reduce to (ii).
(iv) Extend the proof to also apply to merely continuous u.

5.4 Let $f\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be Borel-measurable and strongly 2-quasiconvex, that is, there exists a $\gamma > 0$ such that $A \mapsto f(A) - \gamma|A|^2$ is quasiconvex. Assume furthermore that $|f(A)| \le M(1 + |A|^2)$ for some $M > 0$ and all $A \in \mathbb{R}^{m\times d}$. Show that the functional

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x$$

attains its minimum on $\mathrm{W}^{1,2}_{Fx}(\Omega;\mathbb{R}^m)$ for any $F \in \mathbb{R}^{m\times d}$.

5.5 Show that for $\Omega := (-1,1)^2 \subset \mathbb{R}^2$ the functional

$$\mathcal{F}[u] := \int_\Omega \det(\nabla u_j(x))\,\mathrm{d}x$$

is not weakly lower semicontinuous on $\mathrm{W}^{1,2}(\Omega;\mathbb{R}^2)$ by considering the sequence

$$u_j(x,y) := \frac{(1-|x_2|)^j}{\sqrt{j}}\big(\sin(jx),\cos(jx)\big).$$

5.6 Show that we may extend Morrey's Theorem 5.15 to Carathéodory integrands $f\colon \Omega\times\mathbb{R}^{m\times d} \to \mathbb{R}$ that take negative values as long as

$$-M^{-1}|A|^q - M \le f(A) \le M(1 + |A|^p), \qquad A \in \mathbb{R}^{m\times d},$$

where $q \in (0,p)$ and $M > 0$. *Hint:* Observe that the family of negative parts $(f(\nabla u_j))_j$ is equiintegrable.

5.7 Prove that every quadratic form $q\colon \mathbb{R}^{m\times d} \to \mathbb{R}$, i.e., $q(A) = b(A,A)$ for a bilinear form $b\colon \mathbb{R}^{m\times d}\times\mathbb{R}^{m\times d} \to \mathbb{R}$, is quasiconvex if and only if it is rank-one convex. *Hint:* Use Plancherel's identity (A.6).

5.8 Complete the proof of Lemma 5.5 for higher dimensions. *Hint:* Use the multilinear algebra formulation with differential forms and an induction over the dimension.

5.9 Show that a weaker version of Lemma 5.5 is true if $r = p$, where we only have the convergence of the minors in the sense of distributions.

5.10 A locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is called **$\mathrm{W}^{1,p}$-closed-quasiconvex** if

$$h(F) \leq \fint_{B_1} h(A)\,\mathrm{d}\nu(A)$$

for all $F \in \mathbb{R}^{m\times d}$ and for all homogeneous $\mathrm{W}^{1,p}$-gradient Young measures $\nu \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$ with $[\nu] = F$. Show that for all continuous h that satisfy the p-growth condition $|h(A)| \leq M(1 + |A|^p)$, $\mathrm{W}^{1,p}$-closed-quasiconvexity is equivalent to $\mathrm{W}^{1,p}$-quasiconvexity.



Chapter 6
Polyconvexity

At the beginning of the previous chapter we saw that convexity cannot hold for frame-indifferent integrands (under a mild non-degeneracy condition). Thus, we were led to consider quasiconvex integrands. However, while quasiconvexity is of tremendous importance in the theory of the Calculus of Variations, Morrey's Theorem 5.15 has one major drawback in that we need to require the p-growth bound

$$f(x, A) \leq M(1 + |A|^p), \qquad (x, A) \in \Omega \times \mathbb{R}^{m\times d},$$

for some $M > 0$ and $p \in (1, \infty)$.

Unfortunately, this is not a feasible assumption for nonlinear elasticity theory because it ignores the requirement that infinite compressions should cost infinite energy, as we saw in Section 1.7. Indeed, realistic integrands f for (nonlinear) hyperelastic energy functionals have the property that

$$f(A) \to +\infty \quad \text{as} \quad \det A \downarrow 0 \tag{6.1}$$

and

$$f(A) = +\infty \quad \text{if} \quad \det A \leq 0. \tag{6.2}$$

For instance, the family of matrices

$$A_\alpha := \begin{pmatrix} 1 & 0 \\ 0 & \alpha \end{pmatrix}, \qquad \alpha > 0,$$

satisfies $\det A_\alpha \downarrow 0$ as $\alpha \downarrow 0$, but $|A_\alpha|$ remains uniformly bounded. Thus, the p-growth bound cannot hold at the same time as (6.1).

The question of whether Morrey's Theorem 5.15 for quasiconvex integrands can be extended, in some form, to integrands satisfying (6.1) is a major unsolved problem, see [37]. For the time being, we have to confine ourselves to a more restrictive notion of convexity if we want to allow for (6.1). This type of convexity was introduced by John M. Ball in [34] and is called *polyconvexity*. Ball's accompanying theorem then shows the existence of minimizers for a realistic class of stored-energy functionals in nonlinear elasticity theory, including the so-called Mooney–Rivlin and Ogden materials.

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We will focus on the three-dimensional theory because it is by far the most physically relevant. This restriction eases the notational burden considerably. However, any number of dimensions can be treated in a similar way.

After proving Ball's existence theorem, we also briefly discuss the question of injectivity, which is very relevant for applications.

6.1 Polyconvexity

A function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}\cup\{+\infty\}$ (now we allow the value $+\infty$) is called **polyconvex** if it can be written as a convex function of all the minors of the matrix argument. Concretely, for $d = m = 3$, which is the most interesting case for us, a function $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R}\cup\{+\infty\}$ is called polyconvex if it can be written in the form

$$h(A) = H(A, \operatorname{cof} A, \det A), \qquad A \in \mathbb{R}^{3\times 3},$$

where $H\colon \mathbb{R}^{3\times 3}\times\mathbb{R}^{3\times 3}\times\mathbb{R} \to \mathbb{R}\cup\{+\infty\}$ is convex (as a function on $\mathbb{R}^{3\times 3}\times\mathbb{R}^{3\times 3}\times\mathbb{R} \cong \mathbb{R}^{19}$). Here, cof A denotes the cofactor matrix as defined in Appendix A.1.

While convexity obviously implies polyconvexity, the converse is clearly false, as the determinant function shows. Polyconvex functions in fact constitute the largest practically-relevant class of quasiconvex functions:

Proposition 6.1 *A locally bounded polyconvex function $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R}$ (not taking the value $+\infty$) is quasiconvex.*

Proof. From the definition of polyconvexity we have for all $A \in \mathbb{R}^{3\times 3}$ and all $\psi \in \mathrm{W}^{1,\infty}_{Ax}(B_1;\mathbb{R}^3)$ that

$$\begin{aligned}
&\fint_{B_1} h(\nabla\psi(x))\,\mathrm{d}x \\
&\quad= \fint_{B_1} H(\nabla\psi(x), \operatorname{cof}\nabla\psi(x), \det\nabla\psi(x))\,\mathrm{d}x \\
&\quad\geq H\left(\fint_{B_1} \nabla\psi(x)\,\mathrm{d}x, \fint_{B_1} \operatorname{cof}\nabla\psi(x)\,\mathrm{d}x, \fint_{B_1} \det\nabla\psi(x)\,\mathrm{d}x\right) \\
&\quad= H(A, \operatorname{cof} A, \det A) \\
&\quad= h(A),
\end{aligned}$$

where the inequality follows from Jensen's inequality and for the penultimate equality we used the fact that minors are null-Lagrangians, as proved in Lemma 5.3. □

We will later see in Example 8.2 that the converse of the preceding proposition is not true.

Example 6.2 (Compressible neo-Hookean materials) Functions $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R}$ of the form

$$h(A) := a|A|^2 + \Gamma(\det A)$$

with $a > 0$ and $\Gamma\colon \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ convex, are clearly polyconvex.

Example 6.3 (Compressible Mooney–Rivlin materials) Functions $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R}$ of the form

$$h(A) := a|A|^2 + b|\operatorname{cof} A|^2 + \Gamma(\det A)$$

with $a, b > 0$ and

$$\Gamma(d) = \begin{cases} \alpha d^2 - \beta \log(d) & \text{if } d > 0, \\ +\infty. & \text{if } d \le 0, \end{cases}$$

for some $\alpha, \beta > 0$, are polyconvex. This is obvious once we realize that Γ is convex. See [78, 79] for details.

Example 6.4 (Ogden materials) Let $M, N \in \mathbb{N}$, $a_i > 0$, $\gamma_i \ge 1$, $b_j > 0$, $\delta_j \ge 1$. Also let $\Gamma\colon \mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ be a convex function with $\Gamma(d) \to +\infty$ as $d \downarrow 0$ and $\Gamma(d) = +\infty$ for $d \le 0$. Then, functions $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R} \cup \{+\infty\}$ of the form

$$h(A) := \sum_{i=1}^{M} a_i \operatorname{tr}\left[(A^{\mathsf{T}}A)^{\gamma_i/2}\right] + \sum_{j=1}^{N} b_j \operatorname{tr}\operatorname{cof}\left[(A^{\mathsf{T}}A)^{\delta_j/2}\right] + \Gamma(\det A),$$

where $A \in \mathbb{R}^{3\times 3}$, can be shown to be polyconvex, see Problem 6.6. The corresponding functionals model the elastic energy of so-called *Ogden materials*, which occur frequently in nonlinear elasticity theory; see [78] for the details.

It can also be proved that in three dimensions convex functions of certain combinations of the singular values of a matrix are polyconvex, see Problem 6.10.

Remark 6.5 In general, polyconvex functions are not locally bounded, which was required in our definition of quasiconvexity, see Section 5.1. Indeed, in the most interesting case of a polyconvex h satisfying the conditions (6.1), (6.2), the boundedness will fail to hold on any compact subset of $\mathbb{R}^{3\times 3}$ that contains a singular matrix. On the other hand, by Lemmas 2.18, 2.19 we know that polyconvex h are locally bounded from above on the their effective domain (where they do not take the value $+\infty$) and that they are everywhere locally bounded from below. Thus, the integrals in

$$\int_\Omega h(\nabla u(x))\,\mathrm{d}x = \int_\Omega H(\nabla u(x), \operatorname{cof}\nabla u(x), \det \nabla u(x))\,\mathrm{d}x$$

are well-defined for any $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^3)$, with values in $(-\infty, +\infty]$. However, to avoid issues with well-definedness when only requiring $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ for $p \in [1,\infty)$, and also to circumvent technicalities for x-dependent integrands, we will in the following sections require that the integrands are positive.

6.2 Existence of Minimizers

Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain. In this section we will prove the existence of a solution to the variational problem

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega f(x, \nabla u(x)) - b(x) \cdot u(x) \, \mathrm{d}x \\ \text{over all} & u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^3) \text{ with } \det \nabla u > 0 \text{ a.e. and } u|_{\partial\Omega} = g, \end{cases} \tag{6.3}$$

where $f \colon \Omega \times \mathbb{R}^{3\times 3} \to [0, +\infty]$ is a Carathéodory integrand (with extended-real values, but the definition is analogous) and

$$f(x, \bullet) \text{ is polyconvex for almost every } x \in \Omega.$$

This means that

$$f(x, A) = F(x, A, \operatorname{cof} A, \det A), \qquad (x, A) \in \Omega \times \mathbb{R}^{3\times 3},$$

for an $F \colon \Omega \times \mathbb{R}^{3\times 3} \times \mathbb{R}^{3\times 3} \times \mathbb{R} \to [0, +\infty]$ with

$$(A, B, d) \mapsto F(x, A, B, d)$$

convex (jointly in all arguments) for almost every $x \in \Omega$.

As the only upper growth assumptions on f we impose

$$\begin{cases} f(x, A) \to +\infty & \text{as} \quad \det A \downarrow 0, \\ f(x, A) = +\infty & \text{if} \quad \det A \leq 0. \end{cases}$$

Furthermore, as usual we suppose that $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega; \mathbb{R}^3)$ and $b \in \mathrm{L}^q(\Omega; \mathbb{R}^3)$, where $1/p + 1/q = 1$. The exponent $p \in (1, \infty)$ will remain unspecified for now. Later, when we impose additional conditions on the coercivity of f, we will also specify p.

The first existence result is relatively straightforward:

Theorem 6.6 *If in addition to the above assumptions, the coercivity bound*

$$\mu |A|^p - \mu^{-1} \leq f(x, A), \qquad (x, A) \in \Omega \times \mathbb{R}^{3\times 3}, \tag{6.4}$$

holds for a constant $\mu > 0$ and a $p \in (3, \infty)$, then the problem

$$\textit{Minimize} \quad \mathcal{F}[u] := \int_\Omega f(x, \nabla u(x)) - b(x) \cdot u(x) \, \mathrm{d}x$$

over

$$\mathcal{A} := \left\{ u \in \mathrm{W}^{1,p}_g(\Omega; \mathbb{R}^3) \ : \ \det \nabla u > 0 \ \textit{a.e.} \right\}$$

has a solution whenever the set $\mathcal{A}$ is non-empty.

Proof. We employ the usual Direct Method. For a minimizing sequence $(u_j)_j \subset \mathcal{A}$ for $\mathcal{F}$, that is,

$$\mathcal{F}[u_j] \to \inf_{\mathcal{A}} \mathcal{F},$$

the coercivity estimate (6.4), Young's inequality, and the Poincaré inequality from Theorem A.27 (iii) imply that there is a constant $C > 0$ such that for any $\delta > 0$ it holds that

$$\begin{aligned}\int_\Omega f(x, \nabla u_j(x)) - b(x) \cdot u_j(x)\, \mathrm{d}x \\ &\geq \mu \|\nabla u_j\|_{\mathrm{L}^p}^p - \mu^{-1}|\Omega| - \|b\|_{\mathrm{L}^q} \cdot \|u_j\|_{\mathrm{L}^p} \\ &\geq \frac{\mu}{C}\|u_j\|_{\mathrm{W}^{1,p}}^p - C - \frac{1}{\delta^q q}\|b\|_{\mathrm{L}^q}^q - \frac{\delta^p}{p}\|u_j\|_{\mathrm{W}^{1,p}}^p.\end{aligned}$$

Choosing $\delta = (p\mu/(2C))^{1/p}$, one derives (for a different constant $C > 0$)

$$\sup_{j\in\mathbb{N}} \|u_j\|_{\mathrm{W}^{1,p}} \leq C\left(\sup_{j\in\mathbb{N}} \mathcal{F}[u_j] + 1\right).$$

Thus, we may select a subsequence (not relabeled) such that $u_j \rightharpoonup u_*$ in $\mathrm{W}^{1,p}$.

By Lemma 5.5 and since $p > 3$,

$$\begin{aligned}\det \nabla u_j &\rightharpoonup \det \nabla u_* \quad \text{in } \mathrm{L}^{p/3}, \\ \operatorname{cof} \nabla u_j &\rightharpoonup \operatorname{cof} \nabla u_* \quad \text{in } \mathrm{L}^{p/2}.\end{aligned}$$

Set

$$\begin{aligned}V_j &:= (\nabla u_j, \operatorname{cof} \nabla u_j, \det \nabla u_j), \\ V &:= (\nabla u_*, \operatorname{cof} \nabla u_*, \det \nabla u_*),\end{aligned}$$

for which it holds that

$$V_j \rightharpoonup V \quad \text{in } \mathrm{L}^p \times \mathrm{L}^{p/2} \times \mathrm{L}^{p/3}.$$

Then we may argue exactly as in the proof of the Tonelli–Serrin Theorem 2.6 via Mazur's Lemma A.3 to see that

$$\int_\Omega F(x, V(x))\, \mathrm{d}x \leq \liminf_{j\to\infty} \int_\Omega F(x, V_j(x))\, \mathrm{d}x.$$

In this context we also note that F is continuous with values in $[0, \infty]$ by assumption. Thus, also using that the second (lower-order) part of $\mathcal{F}$ is weakly continuous by Lemma 2.16,

$$\mathcal{F}[u_*] \leq \lim_{j\to\infty} \mathcal{F}[u_j] = \inf_{\mathcal{A}} \mathcal{F} < \infty.$$

In particular, the finiteness of $\mathcal{F}[u_*]$ yields that $\det \nabla u_* > 0$ almost everywhere and $u_*|_{\partial\Omega} = g$ by the weak continuity of the trace. Hence, $u_* \in \mathcal{A}$ and the proof is finished. □

A major drawback of the preceding theorem is that we had to assume $p > 3$. In applications to elasticity theory, however, a more realistic form of f is

$$f(A) = \frac{\lambda}{2}(\operatorname{tr} E)^2 + \mu \operatorname{tr} E^2 + \mathrm{O}(|E|^3), \qquad E = \frac{1}{2}(A^{\mathsf{T}} A - I), \tag{6.5}$$

where $\lambda, \mu > 0$ are the *Lamé constants*. Except for the last term $\mathrm{O}(|E|^3)$, which vanishes as $|E| \downarrow 0$, this energy corresponds to a so-called *St. Venant–Kirchhoff material*. It was shown by Ciarlet–Geymonat [79] (also see Theorem 4.10-2 in [78]) that for any such Lamé constants, there exists a polyconvex function f, corresponding to a compressible *Mooney–Rivlin material*, such that (6.5) holds, that is,

$$f(A) = a|A|^2 + b|\operatorname{cof} A|^2 + \Gamma(\det A) + c,$$

with

$$\Gamma(d) = \begin{cases} \alpha d^2 - \beta \log(d) & \text{if } d > 0, \\ +\infty & \text{if } d \le 0, \end{cases}$$

where $a, b, \alpha, \beta > 0$ and $c \in \mathbb{R}$. Clearly, such an f has only 2-coercivity in $|A|$. Thus, we need an existence theorem for functions with these growth properties.

The core of a refined lower semicontinuity argument will be furnished by the following improvement of Lemma 5.5:

Lemma 6.7 *Let $p, q, r \in [1, \infty)$ with*

$$p \ge 2, \qquad \frac{1}{p} + \frac{1}{q} \le 1, \qquad r \ge 1$$

and assume that the sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega; \mathbb{R}^3)$ satisfies

$$\begin{cases} u_j \rightharpoonup u & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof} \nabla u_j \rightharpoonup H & \text{in } \mathrm{L}^q, \\ \det \nabla u_j \rightharpoonup d & \text{in } \mathrm{L}^r \end{cases}$$

for some $H \in \mathrm{L}^q(\Omega; \mathbb{R}^{3\times 3})$, $d \in \mathrm{L}^r(\Omega)$. Then,

$$H = \operatorname{cof} \nabla u \qquad \text{and} \qquad d = \det \nabla u.$$

Proof. The idea is to use *distributional* versions of the cofactors and determinant of a gradient and to show that they agree with the usual definitions for sufficiently regular functions. To this end we will use the representation of minors as divergences that was already employed in Lemmas 5.3, 5.5.

Step 1. From (5.4) we get that for all $\varphi = (\varphi^1, \varphi^2, \varphi^3)^{\mathsf{T}} \in \mathrm{C}^2(\Omega; \mathbb{R}^3)$,

$$(\operatorname{cof} \nabla \varphi)^k_l = \partial_{l+1}(\varphi^{k+1} \cdot \partial_{l+2} \varphi^{k+2}) - \partial_{l+2}(\varphi^{k+1} \cdot \partial_{l+1} \varphi^{k+2}),$$

where $k, l \in \{1, 2, 3\}$ are cyclic indices. Thus, for all $\psi \in \mathrm{C}^\infty_c(\Omega)$,

$$\begin{aligned}\int_\Omega (\operatorname{cof}\nabla\varphi)^k_l \cdot \psi \,\mathrm{d}x \\ &= -\int_\Omega (\varphi^{k+1}\cdot\partial_{l+2}\varphi^{k+2})\cdot\partial_{l+1}\psi - (\varphi^{k+1}\cdot\partial_{l+1}\varphi^{k+2})\cdot\partial_{l+2}\psi \,\mathrm{d}x \\ &=: \big\langle (\operatorname{Cof}\nabla\varphi)^k_l, \psi\big\rangle. \end{aligned} \tag{6.6}$$

We call the ($\mathbb{R}^{3\times3}$-valued) functional $\operatorname{Cof}\nabla\varphi$ the **distributional cofactors**.

To investigate the continuity properties of $\operatorname{Cof}\nabla\varphi$, we consider

$$\mathcal{G}[\varphi] := \int_\Omega (\varphi^k\cdot\partial_l\varphi^m)\cdot\partial_n\psi \,\mathrm{d}x, \qquad \varphi\in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3),$$

for some $k,l,m,n\in\{1,2,3\}$ and a fixed $\psi\in \mathrm{C}^\infty_c(\Omega)$. We can estimate this integral using the Hölder inequality as follows:

$$|\mathcal{G}[\varphi]| \le \|\varphi\|_{\mathrm{L}^s}\|\nabla\varphi\|_{\mathrm{L}^p}\|\nabla\psi\|_\infty$$

whenever

$$\frac{1}{s}+\frac{1}{p}\le 1. \tag{6.7}$$

In particular, this is true for $s=p\ge 2$. Since $\mathrm{C}^2(\Omega;\mathbb{R}^3)$ is dense in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$, we have "$\operatorname{cof}\nabla\varphi = \operatorname{Cof}\nabla\varphi$", that is, (6.6), also for $\varphi\in\mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$.

Moreover, the above bound yields for any sequence $(\varphi_j)_j\subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ that

$$\mathcal{G}[\varphi_j]\to\mathcal{G}[\varphi] \qquad \text{if} \qquad \begin{cases} \varphi_j\to\varphi & \text{in } \mathrm{L}^s,\\ \nabla\varphi_j\rightharpoonup\nabla\varphi & \text{in } \mathrm{L}^p.\end{cases}$$

If $\varphi_j\rightharpoonup\varphi$ in $\mathrm{W}^{1,p}$, then the first convergence on the right-hand side follows from the Rellich–Kondrachov Theorem A.29 whenever

$$s<\begin{cases}\frac{3p}{3-p} & \text{if } p<3,\\ \infty & \text{if } p\ge 3.\end{cases} \tag{6.8}$$

We can always choose s such that it simultaneously satisfies (6.7) and (6.8), as a quick calculation shows. Thus,

$$\big\langle \operatorname{Cof}\nabla\varphi_j,\psi\big\rangle \to \big\langle\operatorname{Cof}\nabla\varphi,\psi\big\rangle \qquad \text{if} \qquad \varphi_j\rightharpoonup\varphi \quad \text{in } \mathrm{W}^{1,p}. \tag{6.9}$$

Step 2. If $\varphi\in\mathrm{C}^\infty(\Omega;\mathbb{R}^3)$, then we know from (5.5) that

$$\det\nabla\varphi = \sum_{l=1}^3 \partial_l\varphi^1\cdot(\operatorname{cof}\nabla\varphi)^1_l = \sum_{l=1}^3 \partial_l\big(\varphi^1\cdot(\operatorname{cof}\nabla\varphi)^1_l\big).$$

Thus, we have for all $\psi \in C_c^\infty(\Omega)$ that

$$\begin{aligned}\int_\Omega (\det \nabla\varphi)\cdot\psi \,\mathrm{d}x &= \sum_{l=1}^3 \int_\Omega \partial_l \varphi^1 \cdot (\operatorname{cof}\nabla\varphi)_l^1 \cdot \psi \,\mathrm{d}x \\ &= -\sum_{l=1}^3 \int_\Omega \big(\varphi^1 \cdot (\operatorname{cof}\nabla\varphi)_l^1\big)\cdot \partial_l\psi \,\mathrm{d}x. \end{aligned} \tag{6.10}$$

The key idea now is that by Hölder's inequality the last integral is well-defined and finite if only

$$\varphi \in L^p(\Omega;\mathbb{R}^3) \quad \text{with} \quad \operatorname{cof}\nabla\varphi \in L^q(\Omega;\mathbb{R}^3), \quad \text{where} \quad \frac{1}{p}+\frac{1}{q}\le 1.$$

Analogously to the argument for the cofactor matrix, this motivates us to define the **distributional determinant** $\operatorname{Det}\nabla\varphi$ as the linear functional on $C_c^\infty(\Omega)$ given as

$$\big\langle \operatorname{Det}\nabla\varphi, \psi\big\rangle := -\sum_{l=1}^3 \int_\Omega \big(\varphi^1 \cdot (\operatorname{cof}\nabla\varphi)_l^1\big)\cdot \partial_l\psi \,\mathrm{d}x, \qquad \psi \in C_c^\infty(\Omega).$$

From (6.10) we see that if $\varphi \in C^\infty(\Omega;\mathbb{R}^3)$, then "$\det\nabla\varphi = \operatorname{Det}\nabla\varphi$", i.e.,

$$\int_\Omega (\det\nabla\varphi)\cdot\psi \,\mathrm{d}x = \sum_{l=1}^3 \int_\Omega \partial_l\varphi^1 \cdot (\operatorname{cof}\nabla\varphi)_l^1 \cdot \psi \,\mathrm{d}x = \big\langle \operatorname{Det}\nabla\varphi, \psi\big\rangle \tag{6.11}$$

for all $\psi \in C_c^\infty(\Omega)$. We want to show that this equality remains valid if merely $\varphi \in W^{1,p}(\Omega;\mathbb{R}^3)$ with $\operatorname{cof}\nabla\varphi \in L^q(\Omega;\mathbb{R}^{3\times 3})$, where $1/p + 1/q \le 1$.

For the moment fix $\psi \in C_c^\infty(\Omega)$. Define for $\varphi \in C^\infty(\Omega;\mathbb{R}^3)$ and $W \in C^\infty(\Omega;\mathbb{R}^{3\times 3})$,

$$\mathcal{Z}[\varphi, W] := \sum_{l=1}^3 \int_\Omega \partial_l\varphi^1 \cdot W_l^1 \cdot \psi + \varphi^1 \cdot W_l^1 \cdot \partial_l\psi \,\mathrm{d}x.$$

Observe that

$$\mathcal{Z}[\varphi, W] = \sum_{l=1}^3 \int_\Omega W_l^1 \cdot \partial_l(\varphi^1\psi)\,\mathrm{d}x. \tag{6.12}$$

Hölder's inequality furthermore implies

$$|\mathcal{Z}[\varphi, W]| \le C\|\varphi\|_{W^{1,p}}\|W\|_{L^q}\|\psi\|_{W^{1,\infty}} \qquad \text{whenever} \qquad \frac{1}{p}+\frac{1}{q}\le 1.$$

To establish (6.11) for $\varphi \in W^{1,p}(\Omega;\mathbb{R}^3)$ with $\operatorname{cof}\nabla\varphi \in L^q(\Omega;\mathbb{R}^{3\times 3})$, where $1/p+1/q \le 1$, we need to show $\mathcal{Z}[\varphi, \operatorname{cof}\nabla\varphi] = 0$.

If $v \in C^\infty(\Omega;\mathbb{R}^3)$, then we get from the Piola identity (5.6) that

$$\operatorname{div} \operatorname{cof} \nabla v = 0.$$

Thus, using (6.12), we have $\mathcal{Z}[\varphi, \operatorname{cof} \nabla v] = 0$ for $\varphi, v \in C^\infty(\Omega;\mathbb{R}^3)$. Moreover, the map $v \mapsto \operatorname{cof} \nabla v$ is continuous from $W^{1,p}(\Omega;\mathbb{R}^3)$ to $L^1(\Omega;\mathbb{R}^{3\times 3})$ since $|\operatorname{cof} A| \le C|A|^2$ and $p \ge 2$ (one can, for instance, argue using pointwise almost everywhere convergence and Pratt's Theorem A.11). By the density of $C^\infty(\Omega;\mathbb{R}^3)$ in $W^{1,p}(\Omega;\mathbb{R}^3)$ this continuity then yields

$$\mathcal{Z}[\varphi, \operatorname{cof} \nabla v] = 0 \qquad \text{for } \varphi \in C^\infty(\Omega;\mathbb{R}^3) \text{ and } v \in W^{1,p}(\Omega;\mathbb{R}^3).$$

On the other hand, $\mathcal{Z}[\bullet, W]$ is continuous in the first argument with respect to strong convergence in $W^{1,p}$ if $W \in L^q(\Omega;\mathbb{R}^{3\times 3})$ and $1/p + 1/q \le 1$. Hence, another approximation yields that $\mathcal{Z}[\varphi, \operatorname{cof} \nabla v] = 0$ for all $\varphi \in W^{1,p}(\Omega;\mathbb{R}^3)$ and $v \in W^{1,p}(\Omega;\mathbb{R}^3)$ with $\operatorname{cof} \nabla v \in L^q(\Omega;\mathbb{R}^{3\times 3})$. In particular,

$$\mathcal{Z}[\varphi, \operatorname{cof} \nabla \varphi] = 0 \qquad \text{for } \varphi \in W^{1,p}(\Omega;\mathbb{R}^3) \text{ with } \operatorname{cof} \nabla \varphi \in L^q(\Omega;\mathbb{R}^{3\times 3}).$$

Therefore, (6.11) (i.e., "$\det \nabla \varphi = \operatorname{Det} \nabla \varphi$") holds for all such $\varphi \in W^{1,p}(\Omega;\mathbb{R}^3)$.

We see from the definition of the distributional determinant that for a sequence $(\varphi_j)_j \subset W^{1,p}(\Omega;\mathbb{R}^3)$ we have

$$\big\langle \operatorname{Det} \nabla \varphi_j, \psi \big\rangle \to \big\langle \operatorname{Det} \nabla \varphi, \psi \big\rangle \qquad \text{if} \qquad \begin{cases} \varphi_j \to \varphi & \text{in } L^s, \\ \operatorname{cof} \nabla \varphi_j \rightharpoonup \operatorname{cof} \nabla \varphi & \text{in } L^q \end{cases}$$

whenever

$$\frac{1}{s} + \frac{1}{q} \le 1. \tag{6.13}$$

For sequences $\varphi_j \rightharpoonup \varphi$ in $W^{1,p}$ the first convergence on the right-hand side follows, as before, from the Rellich–Kondrachov Theorem A.29 if

$$s < \begin{cases} \frac{3p}{3-p} & \text{if } p < 3, \\ \infty & \text{if } p \ge 3. \end{cases} \tag{6.14}$$

However, since we assumed

$$\frac{1}{p} + \frac{1}{q} \le 1,$$

we can always choose s such that it satisfies (6.13) and (6.14) simultaneously, as can be seen by some elementary algebra. Thus,

$$\big\langle \operatorname{Det} \nabla \varphi_j, \psi \big\rangle \to \big\langle \operatorname{Det} \nabla \varphi, \psi \big\rangle \qquad \text{if} \qquad \begin{cases} \varphi_j \rightharpoonup \varphi & \text{in } W^{1,p}, \\ \operatorname{cof} \nabla \varphi_j \rightharpoonup \operatorname{cof} \nabla \varphi & \text{in } L^q, \end{cases} \tag{6.15}$$

where p, q satisfy the assumptions of the lemma.

Step 3. Assume that, as in the statement of the lemma, we are given a sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ such that for $H \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3})$, $d \in \mathrm{L}^r(\Omega)$ it holds that

$$\begin{cases} u_j \rightharpoonup u & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof} \nabla u_j \rightharpoonup H & \text{in } \mathrm{L}^q, \\ \det \nabla u_j \rightharpoonup d & \text{in } \mathrm{L}^r. \end{cases}$$

Then, (6.6), (6.11) imply that for all $\psi \in \mathrm{C}_c^\infty(\Omega)$,

$$\big\langle \operatorname{Cof} \nabla u_j, \psi \big\rangle \to \big\langle H, \psi \big\rangle \qquad \text{and} \qquad \big\langle \operatorname{Det} \nabla u_j, \psi \big\rangle \to \big\langle d, \psi \big\rangle.$$

On the other hand, from (6.9) and (6.6),

$$\big\langle \operatorname{Cof} \nabla u_j, \psi \big\rangle \to \big\langle \operatorname{Cof} \nabla u, \psi \big\rangle = \int_\Omega (\operatorname{cof} \nabla u)\psi \, \mathrm{d}x,$$

whereby

$$\int_\Omega (\operatorname{cof} \nabla u - H)\psi \, \mathrm{d}x = 0$$

for all $\psi \in \mathrm{C}_c^\infty(\Omega)$. The Fundamental Lemma 3.11 then gives

$$\operatorname{cof} \nabla u = H \in \mathrm{L}^q(\Omega;\mathbb{R}^{3\times 3}).$$

Since we have just shown that $\operatorname{cof} \nabla u_j \rightharpoonup \operatorname{cof} \nabla u$ in L^q, (6.11) and (6.15) imply

$$\big\langle \operatorname{Det} \nabla u_j, \psi \big\rangle \to \big\langle \operatorname{Det} \nabla u, \psi \big\rangle = \int_\Omega (\det \nabla u)\psi \, \mathrm{d}x.$$

Thus, by a similar argument as above,

$$\det \nabla u = d \in \mathrm{L}^r(\Omega).$$

This finishes the proof. □

With this tool at hand, we can now prove Ball's theorem, which is the main existence result for integral functionals with polyconvex integrands:

Theorem 6.8 (Ball 1977 [34, 38]) *Let $p, q, r \in [1, \infty)$ with*

$$p \geq 2, \qquad \frac{1}{p} + \frac{1}{q} \leq 1, \qquad r > 1$$

such that in addition to the assumptions at the beginning of this section it holds that

$$f(x, A) \geq \mu\big(|A|^p + |\operatorname{cof} A|^q + |\det A|^r\big) - \mu^{-1}, \qquad (x, A) \in \Omega \times \mathbb{R}^{3\times 3}, \tag{6.16}$$

for some $\mu > 0$. Then, the problem

$$\textit{Minimize} \quad \mathcal{F}[u] := \int_\Omega f(x, \nabla u(x)) - b(x) \cdot u(x)\, \mathrm{d}x$$

over

$$\mathcal{A} := \left\{ u \in \mathrm{W}^{1,p}_g(\Omega; \mathbb{R}^3) \ : \ \operatorname{cof} \nabla u \in \mathrm{L}^q(\Omega; \mathbb{R}^{3\times 3}),\, \det \nabla u \in \mathrm{L}^r(\Omega; (0, \infty)) \right\}$$

has a solution whenever the set $\mathcal{A}$ is non-empty.

Proof. This follows in an analogous way to the proof of Theorem 6.6, but now we select a subsequence of a minimizing sequence $(u_j)_j \subset \mathcal{A}$ such that for some $u_* \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^3)$, $H \in \mathrm{L}^q(\Omega; \mathbb{R}^{3\times 3})$, $d \in \mathrm{L}^r(\Omega)$ we have

$$\begin{cases} u_j \rightharpoonup u_* & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof} \nabla u_j \rightharpoonup H & \text{in } \mathrm{L}^q, \\ \det \nabla u_j \rightharpoonup d & \text{in } \mathrm{L}^r. \end{cases}$$

This is possible by the usual weak compactness results in conjunction with the coercivity assumption (6.16). Lemma 6.7 then yields $H = \operatorname{cof} \nabla u_*$ and $d = \det \nabla u_*$. Hence,

$$\begin{cases} u_j \rightharpoonup u_* & \text{in } \mathrm{W}^{1,p}, \\ \operatorname{cof} \nabla u_j \rightharpoonup \operatorname{cof} \nabla u_* & \text{in } \mathrm{L}^q, \\ \det \nabla u_j \rightharpoonup \det \nabla u_* & \text{in } \mathrm{L}^r, \end{cases}$$

and we may argue as in Theorem 6.6 to conclude that u_* is a minimizer of our problem over $\mathcal{A}$. □

Example 6.9 Concerning the example from Section 1.7, we can now show that a minimizer exists for the following problem:

$$\begin{cases} \text{Minimize} & \mathcal{F}[y] := \displaystyle\int_\Omega W(\nabla y(x)) - b(x) \cdot y(x)\, \mathrm{d}x \\ \text{over all} & y \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^3) \text{ with } \operatorname{cof} \nabla y \in \mathrm{L}^q(\Omega; \mathbb{R}^{3\times 3}),\, \det \nabla y \in \mathrm{L}^r(\Omega), \\ & \det \nabla y > 0 \text{ a.e., and } y|_{\partial\Omega} = g, \end{cases}$$

if W is any one of the polyconvex integrands exhibited in Examples 6.2, 6.3, or 6.4, $b \in \mathrm{L}^s(\Omega; \mathbb{R}^3)$, and $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega; \mathbb{R}^3)$ with $p, q, r, s \in [1, \infty)$ such that

$$p \geq 2, \qquad \frac{1}{p} + \frac{1}{q} \leq 1, \qquad \frac{3-p}{3p} + \frac{1}{s} < 1, \qquad r > 1.$$

This follows from Theorem 6.8 in conjunction with Lemma 2.16 (for the strong continuity of the second part of $\mathcal{F}$).

6.3 Global Injectivity

For reasons of physical admissibility, one often wants to restrict the set of admissible deformations in a minimization problem, such as the one in Example 6.9, to maps that are **injective almost everywhere**, that is, those $u\colon \Omega \to \mathbb{R}^3$ with

$$\mathcal{H}^0(u^{-1}(x')) = 1 \qquad \text{for almost every } x' \in u(\Omega),$$

where $\mathcal{H}^0$ is the counting measure. Note that if the deformed configuration exhibits self-contact, then we cannot expect complete injectivity.

In the following we will present a classical argument by Ciarlet–Nečas [81] for $\mathrm{W}^{1,p}$-coercive integral functionals with $p > 3$.

Theorem 6.10 *In the situation of Theorem 6.6, in particular $p > 3$, the problem*

$$\textit{Minimize} \quad \mathcal{F}[u] := \int_\Omega f(x, \nabla u(x)) - b(x) \cdot u(x)\, \mathrm{d}x$$

over

$$\mathcal{A} := \left\{ u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^3) \;:\; \det \nabla u > 0 \textit{ a.e., } u \textit{ injective a.e.} \right\}$$

has a solution whenever the set $\mathcal{A}$ is non-empty.

Proof. Let $(u_j)_j \subset \mathcal{A}$ be a minimizing sequence with $u_j \rightharpoonup u_*$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$. The existence proof is analogous to that of Theorem 6.6; we only need to show in addition that u_* is injective almost everywhere.

In the following we will use the fact that for all $v \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ it holds that

$$\int_\Omega |\det \nabla v|\, \mathrm{d}x = \int_{v(\Omega)} \mathcal{H}^0(v^{-1}(x'))\, \mathrm{d}x',$$

where we denote by $\mathcal{H}^0$ the counting measure; see, for example, [224] or [57] for a proof.

For our choice $p > 3$, the space $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^3)$ embeds continuously into $\mathrm{C}(\Omega;\mathbb{R}^3)$, so we have that $u_j \to u_*$ uniformly. Let $U \subset \mathbb{R}^3$ be any precompact open set with $u_*(\Omega) \Subset U$ (note that $u_*(\Omega)$ is bounded). Then, also $u_j(\Omega) \subset U$ for j sufficiently large by the uniform convergence. Hence, for such j,

$$|u_j(\Omega)| \le |U|.$$

Since $\det \nabla u_j$ converges weakly to $\det \nabla u_*$ in $\mathrm{L}^{p/3} \hookrightarrow \mathrm{L}^1$ by Lemma 5.5, and all the u_j are injective almost everywhere,

$$\begin{aligned}\int_\Omega \det \nabla u_*\, \mathrm{d}x &= \lim_{j\to\infty} \int_\Omega \det \nabla u_j\, \mathrm{d}x \\ &= \lim_{j\to\infty} \int_{u_j(\Omega)} \mathcal{H}^0(u_j^{-1}(x'))\, \mathrm{d}x'\end{aligned}$$

$$= \lim_{j\to\infty} |u_j(\Omega)|$$
$$\le |U|.$$

Letting $|U| \downarrow |\overline{u_*(\Omega)}| = |u_*(\Omega)|$ (the last equality follows since $|\partial u_*(\Omega)| = 0$, which again is proved in [224]), we obtain the so-called **Ciarlet–Nečas non-interpenetration condition**

$$\int_\Omega \det \nabla u_* \, \mathrm{d}x \le |u_*(\Omega)|.$$

This in turn gives the estimate

$$|u_*(\Omega)| \le \int_{u_*(\Omega)} \mathcal{H}^0(u_*^{-1}(x')) \, \mathrm{d}x' = \int_\Omega \det \nabla u_* \, \mathrm{d}x \le |u_*(\Omega)|,$$

where we used that $\det \nabla u_* > 0$ almost everywhere (which follows as in the proof of Theorem 6.6). Thus,

$$\mathcal{H}^0(u_*^{-1}(x')) = 1 \qquad \text{for a.e. } x' \in u_*(\Omega),$$

and we have shown the almost everywhere injectivity of u_*. □

Remark 6.11 For coercivity exponents $p < 3$, complex effects such as *cavitation* and *(microscopic) fracture* have to be considered. This is already indicated by the fact that Sobolev functions in $\mathrm{W}^{1,p}_{\mathrm{loc}}(\mathbb{R}^3)$ then are not necessarily continuous. Moreover, the (pointwise) Jacobian determinant loses its physical meaning for such low-regularity maps, as we will see in Example 12.20.

Notes and Historical Remarks

Theorem 6.8 is a refined version of Ball's original result [34] due to Ball–Currie–Olver [38]; also see [39, 40] for further reading. This theorem and its applications to elasticity theory are described in great detail in [78].

Polyconvexity of quadratic forms is equivalent to rank-one convexity if $d = 2$ or $m = 2$, but this does not hold for $d, m \ge 3$. This result, together with pointers to the literature, can be found in Section 5.3.2 of [91].

Many questions about polyconvex integral functionals remain open. In particular, the regularity of solutions and the validity of the Euler–Lagrange equations are largely unknown in the general case. Note that the regularity theory from the previous chapter is not applicable if we do not assume the upper p-growth. These questions are even open for the more restricted situation of the integrands occurring in nonlinear elasticity theory. See [37] for a survey and a collection of challenging open problems. We note that since the publication of [37] counterexamples to uniqueness have been found in [302].

As for the almost injectivity (for $p > 3$), this is in fact sometimes automatic, as shown by Ball [35], but the arguments do not apply to all situations. Tang [322] extended this to $p > 2$, where one has to deal with non-continuous functions, which is quite intricate.

Unfortunately, injectivity almost everywhere does not exclude all unphysical examples. For example, a countable dense set may be mapped into one point. Injectivity *everywhere* is a harder problem. Well-known results in this direction can be found in [35] and [305]. The questions of injectivity, invertibility, and regularity are intimately connected with *cavitation* and *(microscopic) fracture* phenomena, see, for instance, [249] and [177, 178] as places to start.

Finally, we mention in passing the alternative so-called intrinsic approach to elasticity, as pioneered by Ciarlet, see [80].

Problems

6.1 Define $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R}$ via

$$h(A) := \left(|A|^6 + |\operatorname{cof} A|^6\right)^{1/2} + g(\det A), \qquad A \in \mathbb{R}^{3\times 3},$$

where $g\colon \mathbb{R} \to [0,\infty]$ is convex and continuous. Prove that h is polyconvex.

6.2 Show that the function

$$h(A) := \begin{cases} |A| \ln(1+|A|) + \dfrac{1}{\det A} & \text{if } \det A > 0, \\ +\infty & \text{if } \det A \le 0, \end{cases} \qquad A \in \mathbb{R}^{3\times 3},$$

is polyconvex.

6.3 Show that for $u \in (\mathrm{W}^{1,3} \cap \mathrm{C}^2)(\Omega;\mathbb{R}^3)$ it holds that

$$\det \nabla u(x) = \sum_{l=1}^{3} \partial_l (u^1 (\operatorname{cof} \nabla u(x))_l^1), \qquad x \in \Omega.$$

Hint: Use the Piola identity.

6.4 Let $u_j, u \in (\mathrm{W}^{1,3} \cap \mathrm{C}^2)(\Omega;\mathbb{R}^3)$, $j \in \mathbb{N}$, with

$$\operatorname{cof} \nabla u_j,\ \operatorname{cof} \nabla u \in \mathrm{L}^3(\Omega;\mathbb{R}^{3\times 3}).$$

Prove that if

$$u_j \rightharpoonup u \quad \text{in } \mathrm{W}^{1,3} \qquad \text{and} \qquad \operatorname{cof} \nabla u_j \rightharpoonup \operatorname{cof} \nabla u \quad \text{in } \mathrm{L}^3,$$

then $\det \nabla u_j, \det \nabla u \in \mathrm{L}^{3/2}(\Omega)$ and $\det \nabla u_j \rightharpoonup \det \nabla u$ in $\mathrm{L}^{3/2}$.

6.5 In this problem we will construct a rank-one convex function that is not polyconvex.

(i) Find $A_1, A_2, A_3 \in \mathbb{R}^{2\times 2}$ and $\theta_1, \theta_2, \theta_3 \in (0,1)$ such that simultaneously

(a) $\theta_1 + \theta_2 + \theta_3 = 1$;

(b) $\sum_{k=1}^{3} \theta_i \det A_i = \det \left[\sum_{k=1}^{3} \theta_i A_i \right]$;
(c) $\det(A_1 - A_2) \neq 0$, $\det(A_1 - A_3) \neq 0$, $\det(A_2 - A_3) \neq 0$;
(d) $\sum_{k=1}^{3} \theta_i A_i \notin \{A_1, A_2, A_3\}$.

(ii) Define with the A_1, A_2, A_3 from (i) the function $f\colon \mathbb{R}^{2\times 2} \to [0, \infty]$ as

$$f(A) := \begin{cases} 0 & \text{if } A \in \{A_1, A_2, A_3\}, \\ +\infty & \text{otherwise.} \end{cases}$$

Show that f is rank-one convex (extending the definition of rank-one convexity in a suitable way to $(\mathbb{R} \cup \{+\infty\})$-valued functions).
(iii) Show that f is not polyconvex, i.e., there exists no convex function $F\colon \mathbb{R}^{2\times 2}\times\mathbb{R} \to \mathbb{R} \cup \{+\infty\}$ such that

$$f(A) = F(A, \det A), \qquad A \in \mathbb{R}^{2\times 2}.$$

6.6 Show that the function f from Example 6.4 (Ogden materials) is polyconvex. *Hint:* Use Cramer's rule to see that $\operatorname{cof}(AB) = \operatorname{cof}(A)\operatorname{cof}(B)$.

6.7 Set

$$\mathrm{GL}(d) := \left\{ A \in \mathbb{R}^{d\times d} \ : \ \det A \neq 0 \right\} \qquad \text{and}$$
$$\mathrm{GL}^+(d) := \left\{ A \in \mathbb{R}^{d\times d} \ : \ \det A > 0 \right\}.$$

Prove that the convex hull $\mathrm{GL}^+(3)^{**}$ of $\mathrm{GL}^+(3)$ is equal to $\mathrm{GL}(3)$.

6.8 With the notation from the previous problem, define

$$U := \left\{ (A, \operatorname{cof} A, \det A) \in \mathrm{GL}(3) \times \mathrm{GL}(3) \times \mathbb{R} \ : \ A \in \mathrm{GL}^+(3) \right\}$$

and show that

$$U^{**} = \mathrm{GL}(3) \times \mathrm{GL}(3) \times (0, \infty).$$

Hint: Show first:

(i) $(A, H, \delta) \in U$ and $G \in \mathrm{GL}^+(3)$ implies that $(GA, (\operatorname{cof} G)H, (\det G)\delta) \in U^{**}$;
(ii) $(\pm\mathrm{Id}, 0, \delta) \in U$ and $(0, \pm\mathrm{Id}, \delta) \in U^{**}$ for all $\delta > 0$;
(iii) $(A, 0, \delta), (0, H, \delta) \in U^{**}$ for all $A, H \in \mathrm{GL}(3)$ and all $\delta > 0$ (see the previous problem).

6.9 Let $\Phi\colon [0, \infty)^d \to \mathbb{R}$ be symmetric, jointly convex, and increasing (in every variable). Denote by $\sigma_1(A), \ldots, \sigma_d(A) \geq 0$ the singular values of a matrix $A \in \mathbb{R}^{d\times d}$. Then, show that

$$g(A) := \Phi(\sigma_1(A), \ldots, \sigma_d(A)), \qquad A \in \mathbb{R}^{d\times d},$$

is convex.

6.10 Let $h\colon \mathbb{R}^{3\times 3} \to \mathbb{R} \cup \{+\infty\}$ be of the form

$$h(A) := \begin{cases} \Theta\big(\sigma_1,\ \sigma_2,\ \sigma_3,\ \sigma_1\sigma_2,\ \sigma_2\sigma_3,\ \sigma_3\sigma_1,\ \sigma_1\sigma_2\sigma_3\big) & \text{if } \det A > 0, \\ +\infty & \text{if } \det A \le 0, \end{cases}$$

where $\sigma_1, \sigma_2, \sigma_3$ are the three singular values of A. Here, the function $\Theta\colon [0,\infty)^7 \times (0,\infty) \to \mathbb{R}$ is jointly convex, increasing in the first six variables, and

$$\Theta(x_1, x_2, x_3, y_1, y_2, y_3, z) = \Theta(x_{\gamma(1)}, x_{\gamma(2)}, x_{\gamma(3)}, y_{\eta(1)}, y_{\eta(2)}, y_{\eta(3)}, z)$$

for all permutations $\gamma, \eta\colon \{1,2,3\} \to \{1,2,3\}$ and all $x_1, x_2, x_3, y_1, y_2, y_3 \in [0,\infty)$, $z \in (0,\infty)$. Show that h is polyconvex. *Hint:* Use the previous problem.

Chapter 7
Relaxation

Consider the functional

$$\mathcal{F}[u] := \int_0^1 \big(|u'(x)|^2 - 1\big)^2 + |u(x)|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}_0^{1,4}(0,1).$$

The gradient part $a \mapsto (a^2-1)^2$ of the integrand has precisely two distinct minima, see Figure 7.1, which makes it a *double-well potential*. Approximate minimizers u of $\mathcal{F}$ try to satisfy $u' \in \{-1, 1\}$ as closely as possible, while at the same time staying close to zero because of the second term of the integrand. These contradicting requirements lead to minimizing sequences that develop faster and faster oscillations, similar to the ones shown in Figure 5.1. It should be intuitively clear, and is not difficult to show rigorously, that no classical function can be a minimizer of $\mathcal{F}$.

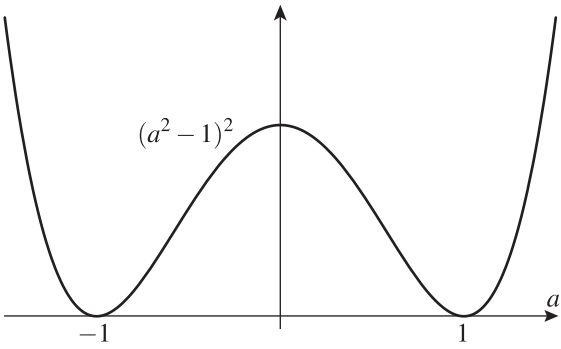


Fig. 7.1 A double-well potential

In this situation, we have two options: First, if we only care about the infimal *value* of a functional $\mathcal{F}\colon X \to \mathbb{R}$ (X a complete metric space), we can compute its **relaxation**, that is, the largest lower semicontinuous functional below $\mathcal{F}$:

$$\mathcal{F}_*[u] := \sup\big\{\, \mathcal{H}[u] \;:\; \mathcal{H}\colon X \to \mathbb{R} \text{ is lower semicontinuous and } \mathcal{H} \le \mathcal{F} \,\big\},$$

where $u \in X$. Here we use the convention $\sup \emptyset := -\infty$.

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If $\mathcal{F}$ is an integral functional as above, then under reasonable assumptions it turns out that $\mathcal{F}_*$ is also an integral functional and its integrand is the *quasiconvex envelope* of the integrand of $\mathcal{F}$. However, a minimizer of $\mathcal{F}_*$ may not say much about the properties of minimizing sequences of our original $\mathcal{F}$. For instance, the volume fractions of the gradient values are "averaged out" in the limit, leading to a loss of information.

Second, we can focus on the minimizing sequences themselves and try to find a generalized limit object that encapsulates all the "interesting" information. The natural candidates for such limit objects are (gradient) Young measures. In fact, applications to the relaxation of integral functionals were the original motivation for introducing them.

Young measure theory allows one to replace a minimization problem over a Sobolev space with a generalized minimization problem over (gradient) Young measures. This generalized minimization problem *always* has a solution, which, however, may not be representable as a function. In this chapter we will consider both approaches in turn.

7.1 Quasiconvex Envelopes

We have seen in Chapter 5 that integral functionals with quasiconvex integrands are weakly lower semicontinuous (assuming suitable growth conditions). Because of the close connection between weak lower semicontinuity and quasiconvexity, we can expect that the relaxation of an integral functional should also be an integral functional with a quasiconvex integrand that is related to the integrand of the original functional.

In this spirit, we define the **quasiconvex envelope** $Qh\colon \mathbb{R}^{m\times d} \to \mathbb{R} \cup \{-\infty\}$ of a locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ as

$$Qh(A) := \sup\big\{\, g(A) \;:\; g\colon \mathbb{R}^{m\times d} \to \mathbb{R} \text{ quasiconvex and } g \le h \,\big\}, \tag{7.1}$$

where $A \in \mathbb{R}^{m\times d}$. The following representation of Qh is in fact much more useful:

Proposition 7.1 *Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be a locally bounded Borel-measurable function with*

$$h \ge h_0$$

for a quasiconvex function $h_0\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ (for instance $h_0 \equiv \alpha \in \mathbb{R}$). Then, the quasiconvex envelope Qh is quasiconvex and continuous. Moreover, Qh can be represented as

$$Qh(A) = \inf\left\{ \fint_{B_1} h(A + \nabla\psi(z))\,\mathrm{d}z \;:\; \psi \in \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m) \right\}, \tag{7.2}$$

where $A \in \mathbb{R}^{m\times d}$. If both h and h_0 have absolute p-growth, $p \in [1,\infty)$, that is,

$$|h_0(A)|, |h(A)| \le M(1 + |A|^p), \qquad A \in \mathbb{R}^{m\times d},$$

for a constant $M > 0$, then also Qh has absolute p-growth.

Remark 7.2 By a similar argument as the one employed in Lemma 5.2 one can see that in (7.2) one may replace the unit ball B_1 by any bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$. Also, if h has p-growth, one may substitute the space $\mathrm{W}_0^{1,p}(B_1;\mathbb{R}^m)$ for $\mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$ via a density argument. Furthermore, employing a covering as in the proof of Proposition 5.18 (with sufficiently small radii), we may restrict the class of ψ in the infimum to those satisfying $\|\psi\|_{\mathrm{L}^\infty} \leq \varepsilon$ for any $\varepsilon > 0$.

Proof. In the following, we denote the function defined by the right-hand side of (7.2) as $Q'h$. Since for $A \in \mathbb{R}^{m\times d}$ we have $Q'h(A) \leq h(A)$ and also

$$Q'h(A) \geq \inf \left\{ \fint_{B_1} h_0(A + \nabla\psi(z))\,\mathrm{d}z \,:\, \psi \in \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m) \right\} \geq h_0(A)$$

by the quasiconvexity of h_0, we know that $Q'h$ is finite everywhere.

Step 1. We first consider the case where h is additionally assumed to be continuous. In this case, $Q'h$ is upper semicontinuous as the infimum over continuous functions (cf. Lemma 2.19), hence Borel-measurable.

Our first goal is to establish that $Q'h$ is quasiconvex, i.e., for any $\psi \in \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$ and any $F \in \mathbb{R}^{m\times d}$ we need to show

$$\fint_{B_1} Q'h(F + \nabla\psi(z))\,\mathrm{d}z \geq Q'h(F). \tag{7.3}$$

Let first $\psi \in \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$ be countably piecewise affine. Suppose that $\psi(x) = v_k + A_k x$ ($v_k \in \mathbb{R}^m$, $A_k \in \mathbb{R}^{m\times d}$) for $x \in D_k$ from a disjoint collection of Lipschitz subdomains $D_k \subset B_1$ ($k \in \mathbb{N}$) with $|B_1 \setminus \bigcup_k D_k| = 0$.

Fix $\varepsilon > 0$. By the definition of $Q'h$ and Remark 7.2 above, for every k we can find $\varphi_k \in \mathrm{W}_0^{1,\infty}(D_k;\mathbb{R}^m)$ such that

$$Q'h(F + A_k) \geq \fint_{D_k} h(F + A_k + \nabla\varphi_k(z))\,\mathrm{d}z - \varepsilon.$$

For $N \in \mathbb{N}$ let $\psi^{(N)} \in \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$ be defined as

$$\psi^{(N)}(x) := \begin{cases} v_k + A_k x + \varphi_k(x) & \text{if } x \in D_k \text{ and } k \leq N, \\ v_k + A_k x & \text{otherwise.} \end{cases}$$

Then, setting $R^{(N)} := \bigcup_{k=N+1}^{\infty} D_k$ we have, with an error term $E^{(N)}$ that changes from line to line,

$$\begin{aligned} \int_{B_1} Q'h(F + \nabla\psi(z))\,\mathrm{d}z &= \sum_{k=1}^{\infty} |D_k|\, Q'h(F + A_k) \\ &\geq \sum_{k=1}^{N} \int_{D_k} h(F + A_k + \nabla\varphi_k(z)) - \varepsilon\,\mathrm{d}z + E^{(N)} \end{aligned}$$

$$
\begin{aligned}
&= \int_{B_1 \setminus R^{(N)}} h(F + \nabla\psi^{(N)}(z)) - \varepsilon \, \mathrm{d}z + E^{(N)} \\
&\geq \int_{B_1} h(F + \nabla\psi^{(N)}(z)) - \varepsilon \, \mathrm{d}z + E^{(N)} \\
&\geq |B_1| \big(Q'h(F) - \varepsilon \big) + E^{(N)},
\end{aligned}
$$

where for the last estimate we used the definition of $Q'h$. The error term $E^{(N)}$ satisfies, using the $\mathrm{W}^{1,\infty}$-boundedness of ψ, the local boundedness of h, and the quasiconvexity of h_0, that

$$
\begin{aligned}
|E^{(N)}| &\leq \max\left\{ - \int_{R^{(N)}} h_0(F + \nabla\psi(z)) \, \mathrm{d}z, 0 \right\} + C|R^{(N)}| \\
&\leq \big(\max\{-h_0(F), 0\} + C \big) |R^{(N)}|.
\end{aligned}
$$

Now let $N \to \infty$ and then $\varepsilon \downarrow 0$ to conclude that (7.3) holds for countably piecewise affine ψ.

Step 2. Next, still assuming the continuity of h, we show that $Q'h$ is continuous, for which we first establish its rank-one convexity. Let $A, B \in \mathbb{R}^{m\times d}$ with $B - A = a \otimes n$ for $a \in \mathbb{R}^m \setminus \{0\}$, $n \in \mathbb{S}^{d-1}$, and let $\theta \in (0,1)$. Then, using a lamination construction as in the proof of Proposition 5.6, but now on B_1 (or a sufficiently large cube containing B_1), and also furnishing the cut-off in a piecewise affine way via Theorem A.30, we can find a sequence of countably piecewise affine maps $\psi_j \in \mathrm{W}^{1,\infty}_0(B_1; \mathbb{R}^m)$ with

$$
\lim_{j\to\infty} \fint_{B_1} Q'h(F + \nabla\psi_j(x)) \, \mathrm{d}x = \theta \, Q'h(A) + (1-\theta) \, Q'h(B).
$$

Combining this with (7.3), which has already been established for our countably affine ψ_j, we obtain

$$
\theta \, Q'h(A) + (1-\theta) \, Q'h(B) \geq Q'h(F) = Q'h(\theta A + (1-\theta)B),
$$

that is, the rank-one convexity of $Q'h$. Hence, Lemma 5.9 yields that $Q'h$ is continuous.

Step 3. With the continuity of $Q'h$ established (in the case where h is assumed continuous), an approximation argument using Lebesgue's dominated convergence theorem in conjunction with the density of countably piecewise affine functions in $\mathrm{W}^{1,\infty}_0(B_1; \mathbb{R}^m)$ with respect to pointwise convergence (see Theorem A.30 and take a subsequence) now allows us to upgrade the inequality (7.3) from countably piecewise affine test maps ψ to arbitrary test maps $\psi \in \mathrm{W}^{1,\infty}_0(B_1; \mathbb{R}^m)$. This shows the quasiconvexity of $Q'h$.

Step 4. It remains to prove that $Qh = Q'h$. By the preceding step we have $Q'h \leq Qh$ since $Q'h$ itself is quasiconvex. On the other hand, for every quasiconvex $g \colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with $g \leq h$ it must hold for all $A \in \mathbb{R}^{m\times d}$ that

$$\begin{aligned} g(A) &\leq \inf \left\{ \fint_{B_1} g(A + \nabla\psi(z)) \, \mathrm{d}z \, : \, \psi \in \mathrm{W}^{1,\infty}_0(B_1; \mathbb{R}^m) \right\} \\ &\leq \inf \left\{ \fint_{B_1} h(A + \nabla\psi(z)) \, \mathrm{d}z \, : \, \psi \in \mathrm{W}^{1,\infty}_0(B_1; \mathbb{R}^m) \right\} \\ &= Q'h(A). \end{aligned}$$

Thus, also $Qh \leq Q'h$.

Step 5. In the case where we do not make a continuity assumption on h, one cannot immediately infer that $Q'h$, as an infimum over *uncountably* many, merely measurable functions, is itself measurable. Thus, the integral in (7.3) is not even well-defined a priori, and so we cannot use the above arguments to directly prove the quasiconvexity of $Q'h$.

To remedy this, one must take a more circuitous route and introduce first $Q''h$, which is defined like $Q'h$, but with the ψ additionally constrained to lie in the set of countably piecewise affine maps with gradients taking values in a fixed countable and dense subset of $\mathbb{R}^{m\times d}$. The set of all such ψ is clearly countable and hence the resulting $Q''h$ is Borel-measurable as a countable infimum over measurable functions. Then, Steps 1 to 3, with $Q''h$ in place of $Q'h$, apply as before. In this context note that the lamination construction in the proof of Proposition 5.6 can be modified to only employ piecewise affine maps with gradients taking values in our countable and dense subset of $\mathbb{R}^{m\times d}$, as long as we restrict the points for which we show rank-one convexity likewise to our countable set. This is, however, enough to apply Lemma 5.9, yielding the local Lipschitz continuity of $Q''h$, which in turn allows us to extend $Q''h$ to all of $\mathbb{R}^{m\times d}$. Working out the details of this step is the task of Problem 7.1.

Analogously to Step 4 we then obtain the chain of inequalities

$$Q''h \leq Qh \leq Q'h.$$

Since trivially $Q'h \leq Q''h$, we conclude that $Qh = Q'h = Q''h$, as required.

Finally note that if both h and h_0 have absolute p-growth, then

$$-M(1 + |\bullet|^p) \leq h_0 \leq Qh \leq h \leq M(1 + |\bullet|^p),$$

whereby also Qh has absolute p-growth. □

7.2 Rigidity for Gradients

As a direct consequence of the representation (7.2), the form of the quasiconvex envelope is intimately tied to questions of *rigidity*, that is, the allowed "shapes" of maps, and sequences of maps, under differential and pointwise constraints. A prototypical rigidity

result is the following:

Theorem 7.3 (Ball–James 1987 [39]) *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $A, B \in \mathbb{R}^{m\times d}$.*

*(i) Suppose that $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ satisfies the **exact two-gradient inclusion***

$$\nabla u \in \{A, B\} \quad \text{a.e. in } \Omega.$$

(a) If $\operatorname{rank}(A-B) \geq 2$, then $\nabla u \equiv A$ or $\nabla u \equiv B$ almost everywhere.
(b) If $B - A = a \otimes n$ for $a \in \mathbb{R}^m$, $n \in \mathbb{S}^{d-1}$ and Ω additionally is assumed to be convex, then there exists a Lipschitz function $h\colon \mathbb{R} \to \mathbb{R}$ with $h' \in \{0,1\}$ almost everywhere and a constant vector $v_0 \in \mathbb{R}^m$ such that

$$u(x) = v_0 + Ax + h(x \cdot n)a.$$

*(ii) Assume $\operatorname{rank}(A-B) \geq 2$ and suppose that a sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ satisfies the **approximate two-gradient inclusion**,*

$$\operatorname{dist}(\nabla u_j, \{A, B\}) \to 0 \quad \text{in measure,}$$

that is, for every $\varepsilon > 0$,

$$\left|\left\{ x \in \Omega \,:\, \operatorname{dist}(\nabla u_j(x), \{A, B\}) > \varepsilon \right\}\right| \to 0 \qquad \text{as } j \to \infty,$$

and that $(u_j)_j$ converges weakly to a limit $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$. Then,*

$$\nabla u_j \to \nabla u \equiv A \quad \text{in measure} \qquad \text{or} \qquad \nabla u_j \to \nabla u \equiv B \quad \text{in measure.}$$

Proof. Ad (i) (a). Assume after a translation that $B = 0$ and thus $\operatorname{rank} A \geq 2$. Then, $\nabla u = Ag$ for a scalar function $g\colon \Omega \to \mathbb{R}$. Mollifying u, we may assume that $g \in \mathrm{C}^\infty(\Omega)$.

The idea of the proof is that the curl of ∇u vanishes, expressed as follows: for all $i, j = 1, \dots, d$ and $k = 1, \dots, m$, it holds that

$$\partial_i [\nabla u]^k_j = \partial_i \partial_j u^k = \partial_j \partial_i u^k = \partial_j [\nabla u]^k_i.$$

For our special $\nabla u = Ag$ this reads as

$$A^k_j \, \partial_i g = A^k_i \, \partial_j g. \tag{7.4}$$

Under the assumption of (i) (a), we claim that $\nabla g \equiv 0$. If otherwise $\xi(x) := \nabla g(x) \neq 0$ for some $x \in \Omega$, then set $a_k(x) := A^k_j / \xi_j(x)$ $(k = 1, \dots, m)$ for any j such that $\xi_j(x) \neq 0$, which is well-defined by the relation (7.4). We obtain

$$A^k_j = a_k(x)\xi_j(x), \qquad \text{i.e.,} \qquad A = a(x) \otimes \xi(x).$$

This, however, is impossible if $\operatorname{rank} A \geq 2$. Hence, $\nabla g \equiv 0$ and u is an affine function since Ω is connected; this property is also stable under mollification.

Ad (i) (b). As in (i) (a) we assume $\nabla u = Ag$, where now $A = a \otimes n$. Pick any $v \in \mathbb{R}^d$ that is orthogonal to n. Then,

$$\frac{\mathrm{d}}{\mathrm{d}t} u(x + tv)\Big|_{t=0} = \nabla u(x) v = a n^{\mathsf{T}} v \, g(x) = 0.$$

This implies that u is constant in direction v. As v was an arbitrary vector orthogonal to n and Ω is assumed convex, $u(x)$ can only depend on $x \cdot n$. This implies the claim.

Ad (ii). Assume once more that $B = 0$ and that there exists a (2×2)-minor M with $M(A) \neq 0$. By assumption, for the sets

$$D_j := \big\{ x \in \Omega \,:\, |\nabla u_j(x) - A| < |\nabla u_j(x)| \big\},$$

we have

$$\nabla u_j - A \mathbb{1}_{D_j} \to 0 \quad \text{in measure}$$

since $|\nabla u_j - A\mathbb{1}_{D_j}| = \operatorname{dist}(\nabla u_j, \{A, 0\})$ by the definition of D_j. Let us assume that, up to selecting a subsequence,

$$\mathbb{1}_{D_j} \overset{*}{\rightharpoonup} \chi \quad \text{in } \mathrm{L}^\infty.$$

In the following we use that for uniformly L^∞-bounded sequences convergence in measure implies weak* convergence in L^∞. Indeed, for any $w \in \mathrm{L}^1(\Omega)$ and any $\varepsilon > 0$ we have

$$\begin{aligned} &\int_\Omega (\nabla u_j - A\mathbb{1}_{D_j}) w \, \mathrm{d}x \\ &\quad \leq \|\nabla u_j - A\mathbb{1}_{D_j}\|_{\mathrm{L}^\infty} \int_{\{|\nabla u_j - A\mathbb{1}_{D_j}| > \varepsilon\}} |w| \, \mathrm{d}x + \varepsilon \|w\|_{\mathrm{L}^1} \\ &\quad \to 0 + \varepsilon \|w\|_{\mathrm{L}^1} \end{aligned}$$

as $j \to \infty$. Since $\varepsilon > 0$ was arbitrary, we obtain the convergence $\nabla u_j - A\mathbb{1}_{D_j} \overset{*}{\rightharpoonup} 0$ in L^∞. Thus,

$$\nabla u_j \overset{*}{\rightharpoonup} A\chi = \nabla u \quad \text{in } \mathrm{L}^\infty.$$

Then, by the weak* continuity of minors proved in Lemma 5.5,

$$\mathrm{w}^*\text{-}\lim_{j\to\infty} M(\nabla u_j) = M(A\chi) = M(A)\chi^2.$$

On the other hand, $M(\nabla u_j) - M(A)\mathbb{1}_{D_j} \to 0$ in measure and then also weakly* in L^∞. Thus,

$$\mathrm{w}^*\text{-}\lim_{j\to\infty} M(\nabla u_j) = M(A) \cdot \mathrm{w}^*\text{-}\lim_{j\to\infty} \mathbb{1}_{D_j} = M(A)\chi.$$

From our earlier assumption $M(A) \neq 0$ we conclude that $\chi = \chi^2$ and hence that there exists a set $D \subset \Omega$ such that

$$\chi = \mathbb{1}_D \qquad \text{and} \qquad \nabla u = A\mathbb{1}_D.$$

Since $\|\mathbb{1}_{D_j}\|_{L^2} \to \|\mathbb{1}_D\|_{L^2}$ (this follows from $\mathbb{1}_{D_j} \overset{*}{\rightharpoonup} \mathbb{1}_D$ in L^∞), the Radon–Riesz Theorem A.15 implies that

$$\mathbb{1}_{D_j} \to \mathbb{1}_D \quad \text{in } L^2 \text{ and in measure.}$$

Thus, combining the above convergence assertions, we arrive at

$$\nabla u_j \to A\mathbb{1}_D = \nabla u \quad \text{in measure.}$$

Part (i) (a) of the present theorem now implies that $\nabla u \equiv A$ or $\nabla u \equiv 0$ $(= B)$ almost everywhere in Ω.

Finally, as we assumed weak* convergence of our original sequence $(u_j)_j$, the limit of the selected subsequence is uniquely determined and so the result holds. □

Remark 7.4 Statement (ii) can be extended to weakly convergent sequences $(u_j)_j \subset W^{1,p}(\Omega;\mathbb{R}^m)$, where $p \in (1,\infty)$. In fact, for $p \in (2,\infty)$ the above proof can be directly adapted, but for $p \in (1,2]$ additional tools are required; see Problem 8.9. A version for $p = 1$ (albeit necessarily under stronger assumptions) is in Theorem 13.14.

As an application of the preceding Ball–James rigidity theorem, we can use the notion of the quasiconvex envelope to introduce an important class of non-trivial quasiconvex functions:

Lemma 7.5 *Let $F \in \mathbb{R}^{m\times d}$ with* $\operatorname{rank} F \geq 2$ *and let $p \in (1,\infty)$. Define*

$$h(A) := \operatorname{dist}(A, \{-F, F\})^p, \qquad A \in \mathbb{R}^{m\times d}.$$

Then, the quasiconvex envelope Qh of h is not convex (at the zero matrix). Moreover, Qh has p-growth.

Remark 7.6 The result remains true for $p = 1$, see Problem 14.2.

Proof. We will show that $Qh(0) > 0$. Then, if Qh was convex at zero, we would have

$$Qh(0) \leq \frac{1}{2}\big(Qh(-F) + Qh(F)\big) \leq \frac{1}{2}\big(h(-F) + h(F)\big) = 0,$$

a contradiction.

Assume to the contrary that $Qh(0) = 0$. Then, by (7.2) in Proposition 7.1 there would exist a sequence $(\psi_j)_j \subset W^{1,\infty}_0(B_1;\mathbb{R}^m)$ with

$$\fint_{B_1} h(\nabla\psi_j(z))\,\mathrm{d}z \to 0,$$

that is, $\operatorname{dist}(\nabla\psi_j, \{-F, F\}) \to 0$ in L^p and then also in measure (see Lemma A.9).

Since $\operatorname{dist}(A, \{-F, F\})^p \geq C^{-1}|A|^p - C$ for some constant $C > 0$, the ψ_j are uniformly $W^{1,p}$-bounded via the Poincaré inequality of Theorem A.27 (i). Taking a subsequence (not relabeled), we may assume that $\psi_j \rightharpoonup \psi$ in $W^{1,p}$. So, the assumptions of the L^p-version of the Ball–James rigidity theorem (see Remark 7.4) are satisfied and

hence $\nabla\psi_j \to \nabla\psi \equiv \pm F$ in measure. The sequence $(\nabla\psi_j)_j$ is also q-equiintegrable for any $q \in (1, p)$ (since it is L^p-bounded, see Appendix A.3). Consequently, Vitali's Convergence Theorem A.12 implies that $\nabla\psi_j \to \nabla\psi \equiv \pm F$ (strongly) in L^q. This is, however, in contradiction to the $\mathrm{W}^{1,q}$-continuity of the trace operator, which implies that the limit ψ must have zero boundary values.

An alternative proof of the lemma, which does not rely on Remark 7.4, may proceed via a more general rigidity result to be established later. Indeed, the convergence $\nabla\psi_j \to \nabla\psi \equiv \pm F$ in measure also follows from Theorem 10.3 together with the first part of the Ball–James rigidity theorem (or one may use directly Theorem 10.5). The other parts of the proof remain the same. □

7.3 Relaxation of Integral Functionals

We first consider the abstract principles of relaxation before moving on to integral functionals. Consider a functional $\mathcal{F}\colon X \to \mathbb{R}$, where X is a reflexive Banach space. Its **(weak) relaxation** $\mathcal{F}_*\colon X \to \mathbb{R} \cup \{-\infty\}$ is defined to be

$$\mathcal{F}_*[u] := \sup\big\{\, \mathcal{H}[u] \;:\; \mathcal{H}\colon X \to \mathbb{R} \text{ weakly lower semicontinuous, } \mathcal{H} \le \mathcal{F} \,\big\},$$

where $u \in X$; see Problem 7.6 for an alternative definition (also cf. Proposition 2.28 in the convex case).

Theorem 7.7 *Let X be a reflexive Banach space and let $\mathcal{F}\colon X \to \mathbb{R}$ be a functional. Assume furthermore:*

(WH1) ***Weak coercivity:*** *For all $\Lambda > 0$ the sublevel set*

$$\big\{\, u \in X \;:\; \mathcal{F}[u] \le \Lambda \,\big\} \qquad \textit{is sequentially weakly precompact.}$$

Then, the relaxation $\mathcal{F}_*$ of $\mathcal{F}$ is weakly lower semicontinuous and

$$\min_X \mathcal{F}_* = \inf_X \mathcal{F}.$$

Proof. The functional $\mathcal{F}_*$ is weakly lower semicontinuous as the supremum of weakly lower semicontinuous functionals. Indeed, if $u_j \rightharpoonup u$ in X, then for all weakly lower semicontinuous $\mathcal{H}\colon X \to \mathbb{R}$ with $\mathcal{H} \le \mathcal{F}$,

$$\mathcal{H}[u] \le \liminf_{j\to\infty} \mathcal{H}[u_j] \le \liminf_{j\to\infty} \mathcal{F}_*[u_j].$$

Taking the supremum over all such $\mathcal{H}$, we see that $\mathcal{F}_*[u] \le \liminf_{j\to\infty} \mathcal{F}_*[u_j]$.

Let $(u_j)_j \subset X$ be a minimizing sequence for $\mathcal{F}$. By the weak coercivity, we may assume that $u_j \rightharpoonup u_*$ in X. Then,

$$\inf_X \mathcal{F} \le \mathcal{F}[u_*] \le \liminf_{j\to\infty} \mathcal{F}_*[u_j] \le \liminf_{j\to\infty} \mathcal{F}[u_j] = \inf_X \mathcal{F}.$$

Hence, $\mathcal{F}_*$ attains its minimum, which is equal to the infimum of $\mathcal{F}$. □

Let now

$$\mathcal{F}[u] := \int_\Omega f(x, \nabla u(x))\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m),$$

where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $p \in (1, \infty)$, and $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0, \infty)$ is a Carathéodory integrand satisfying the p-growth and coercivity assumption

$$\mu|A|^p - \mu^{-1} \le f(x, A) \le M(1 + |A|^p), \qquad (x, A) \in \Omega \times \mathbb{R}^{m\times d}, \tag{7.5}$$

for some $\mu, M > 0$. The main result of this section is the following relaxation theorem:

Theorem 7.8 *Let $\mathcal{F}$ be as above and assume furthermore that there exists a modulus of continuity ω (i.e., $\omega\colon [0, \infty) \to [0, \infty)$ continuous, increasing, and $\omega(0) = 0$) such that*

$$|f(x, A) - f(y, A)| \le \omega(|x - y|)(1 + |A|^p), \qquad x, y \in \Omega,\ A \in \mathbb{R}^{m\times d}. \tag{7.6}$$

Then, the relaxation $\mathcal{F}_*$ of $\mathcal{F}$ is given by

$$\mathcal{F}_*[u] = \int_\Omega Qf(x, \nabla u(x))\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m),$$

where $Qf(x, \bullet)$ denotes the quasiconvex envelope of $f(x, \bullet)$ for $x \in \Omega$. The same conclusion holds if we additionally prescribe fixed boundary values.

Proof. We define

$$\mathcal{G}[u] := \int_\Omega Qf(x, \nabla u(x))\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m).$$

By Proposition 7.1 the integrand $Qf(x, \bullet)$ is quasiconvex and continuous for all $x \in \Omega$. We will moreover see below that $Qf(\bullet, A)$ is continuous for all fixed $A \in \mathbb{R}^{m\times d}$, whereby Qf is in particular Carathéodory, and hence $\mathcal{G}$ is well-defined.

We will show in the following that (a) $\mathcal{G} \le \mathcal{F}_*$ and (b) $\mathcal{G} \ge \mathcal{F}_*$.

To see (a), it suffices to observe that $\mathcal{G}$ is weakly lower semicontinuous by Morrey's Theorem 5.15 and that $\mathcal{G} \le \mathcal{F}$. Thus, from the definition of $\mathcal{F}_*$ we immediately get $\mathcal{G} \le \mathcal{F}_*$.

We will prove (b) in several steps.

Step 1. Let $\varepsilon > 0$ and fix $A \in \mathbb{R}^{m\times d}$. For $x \in \Omega$ let $\psi_x \in \mathrm{W}^{1,p}_0(B_1; \mathbb{R}^m)$ be such that

$$\fint_{B_1} f(x, A + \nabla\psi_x(z))\, \mathrm{d}z \le Qf(x, A) + \varepsilon.$$

Then we use (7.5) to observe

$$\mu \fint_{B_1} |A + \nabla\psi_x(z)|^p\, \mathrm{d}z - \mu^{-1} \le Qf(x, A) + \varepsilon \le M(1 + |A|^p) + \varepsilon.$$

Let now $x, y \in \Omega$ and estimate using (7.6) and the definition of $Qf(x, \bullet)$,

$$
\begin{aligned}
&Qf(y,A) - Qf(x,A)\\
&\quad \le \fint_{B_1} \left| f(y, A + \nabla\psi_x(z)) - f(x, A + \nabla\psi_x(z)) \right| \mathrm{d}z + \varepsilon\\
&\quad \le \omega(|x-y|) \fint_{B_1} 1 + |A + \nabla\psi_x(z)|^p \, \mathrm{d}z + \varepsilon\\
&\quad \le C\omega(|x-y|)(1 + |A|^p) + C\varepsilon,
\end{aligned}
$$

where $C = C(\mu, M)$ is a constant. Letting $\varepsilon \downarrow 0$ and also exchanging the roles of x and y, we see that

$$
\left| Qf(x,A) - Qf(y,A) \right| \le C\omega(|x-y|)(1 + |A|^p) \tag{7.7}
$$

for all $x, y \in \Omega$ and $A \in \mathbb{R}^{m\times d}$. In particular, $Qf(\bullet, A)$ is continuous.

Step 2. Assume first that $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is countably piecewise affine, say $u(x) = v_k + A_k x$ (where $v_k \in \mathbb{R}^m$, $A_k \in \mathbb{R}^{m\times d}$) on the set D_k from a disjoint collection of Lipschitz domains $D_k \subset \Omega$ ($k \in \mathbb{N}$) such that $|\Omega \setminus \bigcup_k D_k| = 0$.

Fix $\varepsilon > 0$. For any $k \in \mathbb{N}$ we may cover D_k up to a negligible set with countably many disjoint balls $B_l^{(k)} := B_{r_l^{(k)}}(x_l^{(k)}) \subset D_k$, where $x_l^{(k)} \in D_k$, $0 < r_l^{(k)} < \varepsilon$ ($l \in \mathbb{N}$), such that

$$
\left| \fint_{B_l^{(k)}} Qf(x, A_k) \, \mathrm{d}x - Qf(x_l^{(k)}, A_k) \right| \le C\omega(\varepsilon)(1 + |A_k|^p). \tag{7.8}
$$

This covering exists by the Vitali Covering Theorem A.16 in conjunction with (7.7).

From Proposition 7.1 and Remark 7.2, in each ball $B_l^{(k)}$ we can find a map $\psi_l^{(k)} \in \mathrm{W}_0^{1,\infty}(B_l^{(k)};\mathbb{R}^m)$ with $\|\psi_l^{(k)}\|_{\mathrm{L}^\infty} \le \varepsilon$ and

$$
\left| Qf(x_l^{(k)}, A_k) - \fint_{B_l^{(k)}} f\big(x_l^{(k)}, A_k + \nabla\psi_l^{(k)}(z)\big) \, \mathrm{d}z \right| \le \varepsilon. \tag{7.9}
$$

Define $v_\varepsilon \colon \Omega \to \mathbb{R}^m$ by setting

$$
v_\varepsilon(x) := u(x) + \psi_l^{(k)}(x) \qquad \text{if } x \in B_l^{(k)} \ (k, l \in \mathbb{N}).
$$

In a similar way to Step 1 one derives that

$$
\mu \fint_{B_l^{(k)}} |A_k + \nabla\psi_l^{(k)}(z)|^p \, \mathrm{d}z - \mu^{-1} \le M(1 + |A_k|^p) + \varepsilon.
$$

Thus,

$$
\begin{aligned}
\int_\Omega |\nabla v_\varepsilon|^p \, \mathrm{d}x &= \sum_{k,l} \int_{B_l^{(k)}} |A_k + \nabla\psi_l^{(k)}(x)|^p \, \mathrm{d}x\\
&\le C \int_\Omega 1 + |\nabla u(x)|^p \, \mathrm{d}x + C\varepsilon\\
&\le C
\end{aligned}
$$

with an ε-independent constant $C > 0$. So, by the Poincaré inequality, $(v_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is uniformly norm-bounded.

From the continuity assumption (7.6) we infer that

$$\left| \fint_{B_l^{(k)}} f(x_l^{(k)}, \nabla v_\varepsilon(x))\,\mathrm{d}x - \fint_{B_l^{(k)}} f(x, \nabla v_\varepsilon(x))\,\mathrm{d}x \right| \leq \omega(\varepsilon) \fint_{B_l^{(k)}} 1 + |\nabla v_\varepsilon(x)|^p\,\mathrm{d}x.$$

We combine this with (7.8), (7.9) to obtain

$$\left| \fint_{B_l^{(k)}} Qf(x, \nabla u(x))\,\mathrm{d}x - \fint_{B_l^{(k)}} f(x, \nabla v_\varepsilon(x))\,\mathrm{d}x \right| \leq C\omega(\varepsilon) \fint_{B_l^{(k)}} 2 + |A_k|^p + |\nabla v_\varepsilon(x)|^p\,\mathrm{d}x + \varepsilon.$$

Multiplying both sides by $|B_l^{(k)}|$ and summing over all k, l, we arrive at

$$\left|\mathcal{G}[u] - \mathcal{F}[v_\varepsilon]\right| \leq C\omega(\varepsilon) \int_\Omega 2 + |\nabla u(x)|^p + |\nabla v_\varepsilon(x)|^p\,\mathrm{d}x + \varepsilon|\Omega|,$$

and the right-hand side vanishes as $\varepsilon \downarrow 0$. For $u_j := v_{1/j}$ we have $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$ since $(u_j)_j$ is weakly precompact in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ and $\|u_j - u\|_{\mathrm{L}^\infty} \leq 1/j \to 0$. Hence, by the weak lower semicontinuity of $\mathcal{F}_*$,

$$\mathcal{G}[u] = \lim_{j\to\infty} \mathcal{F}[u_j] \geq \liminf_{j\to\infty} \mathcal{F}_*[u_j] \geq \mathcal{F}_*[u],$$

which is (b) for countably piecewise affine u.

Step 2. For arbitrary $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ there exists a sequence $(v_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ of countably piecewise affine functions such that $v_j \to u$ in $\mathrm{W}^{1,p}$ and we may also require that $v|_{\partial\Omega} = u|_{\partial\Omega}$ (see Theorem A.30). Since Qf is Carathéodory and has p-growth (as $0 \leq Qf \leq f$), Theorem 2.13 shows that $\mathcal{G}[v_j] \to \mathcal{G}[u]$. Thus, we get using $\mathcal{G}[v_j] \geq \mathcal{F}_*[v_j]$, which was shown in Step 2, as well as the lower semicontinuity of $\mathcal{F}_*$ that

$$\mathcal{G}[u] = \lim_{j\to\infty} \mathcal{G}[v_j] \geq \liminf_{j\to\infty} \mathcal{F}_*[v_j] \geq \mathcal{F}_*[u].$$

This proves (b) on all of $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. □

7.4 Relaxation via Young Measures

As discussed at the beginning of the chapter, the relaxation strategy of Section 7.3 has one serious drawback: While it allows us to find the infimal *value*, the relaxed functional potentially says only very little about the "shape" of minimizing sequences

(e.g., their oscillations). Often, however, this information is decisive. For example, in material science, oscillations in minimizing sequences correspond to crystalline microstructure, see Section 1.9 and also Chapter 11. Therefore, we now implement the second approach outlined in the introduction to the chapter, that is, we extend the minimization problem to a larger space and look for (generalized) solutions there. It turns out that a suitable class of candidate objects for the extended minimization problem is the class of gradient Young measures, which were first mentioned in Section 4.4.

Let us first, in an abstract fashion, collect a few properties that such an extension–relaxation should satisfy. Assume we are given a metric space X and a functional $\mathcal{F}\colon X \to \mathbb{R}$. Then, we extend X to a complete metric space $\overline{X}$ with convergence "$\rightsquigarrow$". For this, we assume that there exists a (usually not continuous) map

$$\iota\colon X \to \overline{X}.$$

We then seek to extend $\mathcal{F}$ to a functional $\overline{\mathcal{F}}\colon \overline{X} \to \mathbb{R} \cup \{+\infty\}$, which we call the **extension–relaxation** of $\mathcal{F}$, such that the following conditions are satisfied:

(i) **Extension property:** $\overline{\mathcal{F}} \circ \iota = \mathcal{F}$.

(ii) **Lower bound:** If $(u_j)_j \subset X$ is (sequentially) precompact, then, up to selecting a subsequence, there exists a $v \in \overline{X}$ such that $\iota(u_j) \rightsquigarrow v$ in $\overline{X}$ and

$$\overline{\mathcal{F}}[v] \le \liminf_{j\to\infty} \mathcal{F}[u_j].$$

(iii) **Recovery sequence:** For all $v \in \overline{X}$ there exists a **recovery sequence** $(u_j)_j \subset X$ with $\iota(u_j) \rightsquigarrow v$ in $\overline{X}$ and such that

$$\lim_{j\to\infty} \mathcal{F}[u_j] = \overline{\mathcal{F}}[v].$$

Intuitively, these conditions entail that we can solve our minimization problem for $\mathcal{F}$ by passing to the extended space $\overline{X}$. In particular, if $(u_j)_j$ is a minimizing and (sequentially) precompact sequence in X, then any limit v in $\overline{X}$ should be considered a *generalized minimizer*. It is easy to see that the above conditions entail the equality of the infima of $\mathcal{F}$ and $\overline{\mathcal{F}}$. If we additionally assume some coercivity, then $\overline{\mathcal{F}}$ attains its minimum, so that

$$\min_{\overline{X}} \overline{\mathcal{F}} = \inf_X \mathcal{F}.$$

Next, we specialize this abstract approach to our prototypical integral functional

$$\mathcal{F}[u] := \int_\Omega f(x, \nabla u(x))\, \mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m),$$

with $\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain, $p \in (1,\infty)$, and $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$ a Carathéodory integrand satisfying the p-growth and coercivity assumption

$$\mu|A|^p - \mu^{-1} \le f(x,A) \le M(1+|A|^p), \qquad (x,A) \in \Omega \times \mathbb{R}^{m\times d}, \tag{7.10}$$

for some $\mu, M > 0$. For X we take a norm-bounded subset of $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with the weak topology and we will use a set of gradient Young measures for $\overline{X}$. For $\nu = (\nu_x)_x \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, we define the **extension–relaxation**

$$\overline{\mathcal{F}}\colon \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d}) \to \mathbb{R}$$

as

$$\overline{\mathcal{F}}[\nu] := \langle\!\langle f, \nu \rangle\!\rangle = \int_\Omega \int f(x,A)\, \mathrm{d}\nu_x(A)\, \mathrm{d}x.$$

Then, our relaxation result takes the following form.

Theorem 7.9 *Let $\mathcal{F}, \overline{\mathcal{F}}$ be as above.*

(i) ***Extension property:*** *For every $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ the elementary Young measure $\delta[\nabla u] = (\delta_{\nabla u(x)})_x \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ satisfies*

$$\mathcal{F}[u] = \overline{\mathcal{F}}[\delta[\nabla u]].$$

(ii) ***Lower bound:*** *If $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is weakly sequentially precompact, then, up to selecting a subsequence, there exists $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ such that $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$ with $[\nu] = \nabla u$ and*

$$\overline{\mathcal{F}}[\nu] \le \liminf_{j\to\infty} \mathcal{F}[u_j]. \tag{7.11}$$

(iii) ***Recovery sequence:*** *For all $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ there exists a recovery sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$ and*

$$\lim_{j\to\infty} \mathcal{F}[u_j] = \overline{\mathcal{F}}[\nu]. \tag{7.12}$$

(iv) ***Equality of minima:*** *$\overline{\mathcal{F}}$ attains its minimum and*

$$\min_{\mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})} \overline{\mathcal{F}} = \inf_{\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)} \mathcal{F}.$$

Furthermore, all these statements remain true if we prescribe boundary values.

Here, for a gradient Young measure the boundary values refer to an underlying deformation (which is only determined up to a constant, of course).

Proof. Ad (i). This follows directly from the definition of $\overline{\mathcal{F}}$.

Ad (ii). The existence of ν is a consequence of the Fundamental Theorem of Young measure theory, Theorem 4.1. The lower bound (7.11) follows from Proposition 4.7.

Ad (iii). Via Lemma 4.14 it is possible to construct a sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with

(a) $u_j|_{\partial\Omega} = u|_{\partial\Omega}$, where $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is an underlying deformation of ν;
(b) the family $(\nabla u_j)_j$ is p-equiintegrable;
(c) $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$.

For this special generating sequence we can now use the statement about representation of limits of integral functionals from the Fundamental Theorem 4.1 (here we need the p-equiintegrability) to get

$$\mathcal{F}[u_j] \to \langle\!\langle f, \nu \rangle\!\rangle = \overline{\mathcal{F}}[\nu],$$

which is nothing else than (7.12).

Ad (iv). This is not hard to see using (ii), (iii), and the coercivity assumption, that is, the lower bound in (7.10), applied to a minimizing sequence for $\mathcal{F}$. □

As a consequence of the preceding theorem, the question then naturally arises whether one can characterize gradient Young measures within the class of general Young measures. We will come back to this in the next chapter.

Example 7.10 In our sailing example from Section 1.6, we were tasked with solving the *optimal tacking problem*

$$\begin{cases} \text{Minimize} & \mathcal{F}[r] := \displaystyle\int_0^T v_w \frac{\cos(4\arctan r'(t)) - 1}{2} + v_f \left(\frac{r(t)^2}{R^2} - 1 \right) \mathrm{d}t \\ \text{subject to} & r(0) = r(T) = 0,\ |r(t)| \le R. \end{cases}$$

Here, because of the additional constraints we can work in any L^p-space, even L^∞. Clearly, the integrand

$$f(r, a) := v_w \frac{\cos(4\arctan a) - 1}{2} + v_f \left(\frac{r^2}{R^2} - 1 \right),$$

where $(r, a) \in [-R, R] \times \mathbb{R}$, is not convex in a, see Figure 1.5. Technically, the preceding theorem is not applicable since f depends on r and a (and $p = \infty$), but it is obvious that we can simply extend it to consider Young measures

$$(\delta_{r(t)} \otimes \nu_t)_{t \in (0,T)} \in \mathbf{Y}^\infty((0,T); \mathbb{R} \times \mathbb{R}) \quad \text{with} \quad [\nu_t] = r'(t)$$

in a similar way to the strategy in Section 5.6, noting that $\mathbf{GY}^\infty((0,T)) = \mathbf{Y}^\infty((0,T))$. We collect all such product Young measures in the set $\overline{X}$, which we equip with the weak* convergence for Young measures, see (4.9).

Then, the extended–relaxed variational problem is

$$\begin{cases} \text{Minimize} & \overline{\mathcal{F}}[\delta_r \otimes \nu] := \langle\!\langle f, \delta_r \otimes \nu \rangle\!\rangle = \displaystyle\int_0^T \int f(r(t), a)\, \mathrm{d}\nu_t(a)\, \mathrm{d}t, \\ \text{over all} & \delta_r \otimes \nu = (\delta_{r(t)} \otimes \nu_t)_t \in \overline{X} \text{ with } r(0) = r(T) = 0,\ |r(t)| \le R. \end{cases}$$

Let us also construct a sequence of approximate solutions that generates the optimal Young measure solution. The first part of the integrand f, namely $a \mapsto v_w(\cos(4\arctan a) - 1)/2$, has two minima with value $-v_w$ at $a = \pm 1$. The second part $r \mapsto v_f(r^2/R^2 - 1)$ attains its minimum $-v_f$ for $r = 0$. Thus,

$$f \ge -(v_w + v_f) =: f_{\min}.$$

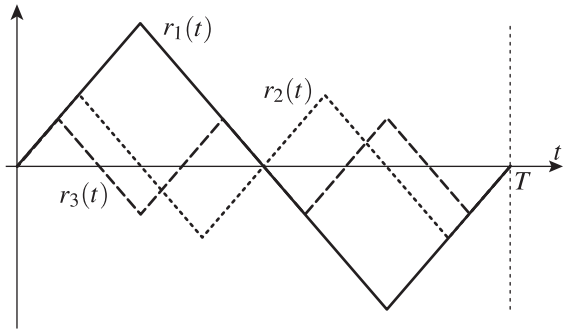


Fig. 7.2 The first few maps in the minimizing sequence for $\mathcal{F}$

We let

$$h(s) := \begin{cases} s & \text{if } s \in [0,1], \\ 2-s & \text{if } s \in (1,3], \\ s-4 & \text{if } s \in (3,4], \end{cases}$$

and consider h to be extended to all $s \in \mathbb{R}$ by periodicity. Then set

$$r_j(t) := \frac{T}{4j} h\left(\frac{4j}{T}t\right), \qquad t \in [0,T],$$

see Figure 7.2. It is easy to see that $r_j' \in \{-1,1\}$ and $r_j \to 0$ uniformly. Thus,

$$\mathcal{F}[r_j] \to T \cdot f_{\min} = \inf \mathcal{F}.$$

By the Fundamental Theorem 4.1 on Young measures, we deduce that we may select a subsequence of j's (not relabeled) such that (see Lemma 5.19)

$$(r_j, r_j') \xrightarrow{\mathbf{Y}} \delta_r \otimes \nu = (\delta_{r(t)} \otimes \nu_t)_t \in \overline{X}.$$

From the construction of r_j we see that

$$\delta_r \otimes \nu = \delta_0 \otimes \left(\frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_{+1}\right).$$

Clearly, this $\delta_r \otimes \nu$ is minimizing for $\overline{\mathcal{F}}$. Since above we have constructed a $\mathrm{W}^{1,\infty}$-bounded generating sequence for this optimal $\delta_r \otimes \nu$, we have also a posteriori justified our choice to work with L^∞-Young measures.

We close this section by showing how the two relaxation approaches are related.

Proposition 7.11 *Let $p \in (1,\infty)$ and let $h\colon \mathbb{R}^{m\times d} \to [0,\infty)$ be continuous and satisfy the p-growth and coercivity assumption*

$$\mu|A|^p - \mu^{-1} \le h(A) \le M(1+|A|^p), \qquad A \in \mathbb{R}^{m\times d},$$

for some $\mu, M > 0$. Then, for all $F \in \mathbb{R}^{m\times d}$ there exists a homogeneous gradient Young measure $\nu^F \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$ with $[\nu^F] = F$ and

$$Qh(F) = \int h \,\mathrm{d}\nu^F. \tag{7.13}$$

Proof. According to (7.1) in Proposition 7.1 and Remark 7.2,

$$Qh(F) = \inf\left\{ \fint_{B_1} h(F + \nabla\psi(z)) \,\mathrm{d}z \,:\, \psi \in \mathrm{W}^{1,p}_0(B_1;\mathbb{R}^m) \right\}.$$

Let $(\psi_j)_j \subset \mathrm{W}^{1,p}_0(B_1;\mathbb{R}^m)$ be a minimizing sequence in the above formula. This minimization problem is admissible in Theorem 7.9, which yields a Young measure minimizer $\nu \in \mathbf{GY}^p(B_1;\mathbb{R}^m)$ such that

$$Qh(F) = \fint_{B_1} \int h \,\mathrm{d}\nu_x \,\mathrm{d}x.$$

It only remains to show that we can replace $(\nu_x)_x$ by a *homogeneous* Young measure ν^F. This, however, follows directly from the averaging principle for Young measures, Lemma 4.15. □

Remark 7.12 Remarkably, the above representation extends to the derivative $\mathrm{D}Qh$ of Qh if h is continuously differentiable, namely

$$\mathrm{D}Qh(F) = \int \mathrm{D}h \,\mathrm{d}\nu^F,$$

as is shown in Proposition 3.6 of [42].

7.5 Mean Coercivity

As we have discussed at length in Chapter 5, quasiconvexity is the natural condition on an integrand to make the corresponding integral functional weakly lower semicontinuous. Using the Young measure relaxation techniques from this chapter, we will now see that *strong* quasiconvexity also allows us to deduce the other ingredient for the Direct Method, namely the weak compactness of minimizing sequences (at least in the situation of x-independent integrands).

Recall that so far we have required the classical *pointwise* p-coercivity bound (2.2) for a continuous integrand $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$, that is,

$$\mu|A|^p - \mu^{-1} \le h(A), \qquad A \in \mathbb{R}^{m\times d}. \tag{7.14}$$

This allowed us to deduce that

$$\mathcal{F}[u] := \int_\Omega h(\nabla u(x)) \,\mathrm{d}x \ge \mu\|\nabla u\|^p_{\mathrm{L}^p} - \mu^{-1}|\Omega| \ge C(\|u\|^p_{\mathrm{W}^{1,p}} - 1) \tag{7.15}$$

for all $u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ (with $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$) and a constant $C > 0$, where we employed the Poincaré inequality for the second inequality.

If (7.15) holds directly, that is,

$$\int_\Omega h(\nabla u(x))\,\mathrm{d}x \geq \bar{\mu}\|u\|^p_{\mathrm{W}^{1,p}} - \bar{\mu}^{-1}|\Omega|, \qquad u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m), \tag{7.16}$$

for a constant $\bar{\mu} > 0$, we call h **$\mathrm{W}^{1,p}$-mean coercive (on Ω and with boundary values g)**.

To avoid technical issues regarding the well-definedness of the above integral, we always understand (7.16) and similar expressions to contain the implicit requirement that the negative part of the integral is finite (which is implied, for instance by the bound $h(A) \geq -C(1 + |A|^p)$).

In the following lemma we observe that the definition of mean coercivity does not, in fact, depend on the domain or the boundary values. Hence we may just speak of "$\mathrm{W}^{1,p}$-mean coercivity" without specifying the domain or the boundary values.

Lemma 7.13 *If a continuous integrand $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ is $\mathrm{W}^{1,p}$-mean coercive on a bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ and with boundary values $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$, then h is also $\mathrm{W}^{1,p}$-mean coercive on any other bounded Lipschitz domain $\widetilde{\Omega} \subset \mathbb{R}^d$ and with any other boundary values $\widetilde{g} \in \mathrm{W}^{1-1/p,p}(\partial\widetilde{\Omega};\mathbb{R}^m)$.*

Proof. Let $w_g \in \mathrm{W}^{1,p}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^m)$ be such that $w_g|_{\partial\Omega} = g$ (in the sense of both inner and outer trace), which always exists by Theorems A.25 and A.26. For a bounded Lipschitz domain $\widetilde{\Omega} \subset \mathbb{R}^d$ and boundary values $\widetilde{g} \in \mathrm{W}^{1-1/p,p}(\partial\widetilde{\Omega};\mathbb{R}^m)$ we assume likewise the existence of $w_{\widetilde{g}} \in \mathrm{W}^{1,p}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^m)$ with $w_{\widetilde{g}}|_{\partial\widetilde{\Omega}} = g$.

Choose $a \in \Omega$ and $r > 0$ so small that $a + r\widetilde{\Omega} \Subset \Omega$, and then fix a cut-off function $\varrho \in \mathrm{C}^\infty_c(\Omega;[0,1])$ with $\varrho \equiv 1$ on a neighborhood of $a + r\widetilde{\Omega}$. Let an arbitrary map $u \in \mathrm{W}^{1,p}_{\widetilde{g}}(\widetilde{\Omega};\mathbb{R}^m)$ be extended to a map $\bar{u} \in \mathrm{W}^{1,p}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^m)$ by setting

$$\bar{u}(x) := \begin{cases} u(x) & \text{if } x \in \widetilde{\Omega}, \\ w_{\widetilde{g}}(x) & \text{if } x \in \mathbb{R}^d \setminus \widetilde{\Omega}, \end{cases}$$

and define $v \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m)$ via

$$v(y) := (1 - \varrho(y))w_g(y) + r\varrho(y)\bar{u}\left(\frac{y-a}{r}\right), \qquad y \in \Omega.$$

We observe that in $\Omega \setminus (a + r\widetilde{\Omega})$,

$$v(y) = (1 - \varrho(y))w_g(y) + r\varrho(y)w_{\widetilde{g}}\left(\frac{y-a}{r}\right) =: z(y), \qquad y \in \Omega \setminus (a + r\widetilde{\Omega}),$$

and the z so defined does not depend on u. Then compute

$$\|\nabla v\|_{\mathrm{L}^p}^p = \int_{a+r\widetilde{\Omega}} \left|\nabla u\left(\frac{y-a}{r}\right)\right|^p \,\mathrm{d}y + \alpha_1 = r^d \|\nabla u\|_{\mathrm{L}^p}^p + \alpha_1$$

with the constant $\alpha_1 := \int_{\Omega\setminus(a+r\widetilde{\Omega})} |z(y)|^p \,\mathrm{d}y \geq 0$, which does not depend on u. In the same way we also obtain

$$\int_\Omega h(\nabla v(y)) \,\mathrm{d}y = r^d \int_{\widetilde{\Omega}} h(\nabla u(x)) \,\mathrm{d}x + \alpha_2$$

with $\alpha_2 \in \mathbb{R}$.

Since h by assumption is $\mathrm{W}^{1,p}$-mean coercive on Ω and with boundary values g, we then estimate as follows:

$$\begin{aligned}
\|\nabla u\|_{\mathrm{L}^p}^p &\leq r^{-d} \|\nabla v\|_{\mathrm{L}^p}^p \\
&\leq r^{-d} \bar{\mu}^{-1} \int_\Omega h(\nabla v(y)) \,\mathrm{d}y + C \\
&\leq \bar{\mu}^{-1} \int_{\widetilde{\Omega}} h(\nabla u(x)) \,\mathrm{d}x + C,
\end{aligned}$$

where the constant $C > 0$ may change from line to line. Via the Poincaré inequality in Theorem A.27 (iii) this implies the $\mathrm{W}^{1,p}$-mean coercivity of h on $\widetilde{\Omega}$ for the boundary values $\widetilde{g}$ (with a modified constant $\bar{\mu}$). □

The $\mathrm{W}^{1,p}$-mean coercivity condition is clearly sufficient to show the weak compactness of minimizing sequences in the Direct Method, so in Theorem 5.16, applied to an x-independent integrand h as above, we may replace condition (ii) by $\mathrm{W}^{1,p}$-mean coercivity. This is a substantive generalization, as can be seen through the following example.

Example 7.14 Consider the integrand

$$h(A) = \left|\frac{A + A^{\mathsf{T}}}{2}\right|^2 - \frac{1}{4} \det A, \qquad A \in \mathbb{R}^{d\times d},\ d \text{ even}.$$

Using Corollary 5.4, we see that h is quasiconvex everywhere. Moreover, h is not (pointwise) 2-coercive since it is even unbounded from below, as can be observed by evaluating $h(\lambda A)$ for $\lambda > 0$ and $A \in \mathbb{R}^{d\times d} \setminus \{0\}$ skew-symmetric with $\det A > 0$ (which requires d to be even). By Hadamard's inequality (see Appendix A.1),

$$h(A) \geq \left|\frac{A + A^{\mathsf{T}}}{2}\right|^2 - \frac{1}{4}|A|^2$$

and then the basic Korn inequality in (2.4) implies that

$$\int_\Omega h(\nabla u(x)) \,\mathrm{d}x \geq \|\mathcal{E}u\|_{\mathrm{L}^2}^2 - \frac{1}{4}\|\nabla u\|_{\mathrm{L}^2}^2 \geq \frac{1}{4}\|\nabla u\|_{\mathrm{L}^2}^2$$

for all $u \in \mathrm{W}^{1,2}_0(\Omega;\mathbb{R}^d)$ on any bounded and connected Lipschitz domain $\Omega \subset \mathbb{R}^d$. Thus, employing also Poincaré's inequality as before, we see that h is $\mathrm{W}^{1,2}$-mean coercive, first for zero boundary values and then also for all boundary values $g \in \mathrm{W}^{1/2,2}(\partial\Omega;\mathbb{R}^d)$ by the preceding Lemma 7.13.

We will now explore the interaction of $\mathrm{W}^{1,p}$-mean coercivity with strong p-quasiconvexity, which we first encountered in Section 5.7 in the special case $p = 2$. For this, call a locally bounded Borel-measurable function $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ **strongly p-quasiconvex at** $A \in \mathbb{R}^{m\times d}$ ($p \in (1,\infty)$) if there exists a $\gamma > 0$ such that

$$A \mapsto h(A) - \gamma V_p(A) \quad \text{is quasiconvex at } A,$$

where

$$V_p(A) := (1 + |A|^2)^{p/2} - 1, \qquad A \in \mathbb{R}^{m\times d}.$$

This means that

$$\gamma \fint_{B_1} V_p(A + \nabla\psi(z)) - V_p(A)\,\mathrm{d}z \le \fint_{B_1} h(A + \nabla\psi(z)) - h(A)\,\mathrm{d}z$$

for all $\psi \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$. If the above condition holds for all $A \in \mathbb{R}^{m\times d}$ with a uniform $\gamma > 0$, then we say that h is **(everywhere) strongly p-quasiconvex**. The statements in Lemma 5.2 also hold for the notion of strong p-quasiconvexity by essentially the same proofs (see Problem 7.7).

Theorem 7.15 (Chen–Kristensen 2017 [75]) *Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be continuous and have p-growth, where $p \in (1,\infty)$. Then, the following conditions are equivalent:*

(i) There exists at least one matrix $F \in \mathbb{R}^{m\times d}$ such that h is strongly p-quasiconvex at F.

(ii) h is $\mathrm{W}^{1,p}$-mean coercive on one, and then any, bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ and with one, and then any, boundary values $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$.

Proof. (i) $\Rightarrow$ (ii): Assume that h is strongly p-quasiconvex at $F \in \mathbb{R}^{m\times d}$. We first show the $\mathrm{W}^{1,p}$-mean coercivity for the unit ball and the special linear boundary values $x \mapsto Fx$. By a density argument the quasiconvexity of $h - \gamma V_p$ for a constant $\gamma > 0$ is equivalent to the validity of

$$\int_{B_1} \gamma V_p(\nabla u(x)) + h(F) - \gamma V_p(F)\,\mathrm{d}x \le \int_{B_1} h(\nabla u(x))\,\mathrm{d}x$$

for all $u \in \mathrm{W}^{1,p}_{Fx}(B_1;\mathbb{R}^m)$. Since

$$|A|^p - 1 \le V_p(A), \qquad A \in \mathbb{R}^{m\times d}, \tag{7.17}$$

this clearly implies the $\mathrm{W}^{1,p}$-mean coercivity of h on B_1 and with the boundary values $x \mapsto Fx$. The $\mathrm{W}^{1,p}$-mean coercivity on an arbitrary bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ and with boundary values $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$ then follows from Lemma 7.13.

(ii) ⇒ (i): Assume that h is $\mathrm{W}^{1,p}$-mean coercive on the unit ball B_1 and with zero boundary values. Note that by Lemma 7.13 this is equivalent to any other variant of $\mathrm{W}^{1,p}$-mean coercivity.

In the proof of Proposition 7.11, here applied to $F = 0$, the pointwise coercivity (and positivity) of h is only used to obtain weak compactness of a minimizing sequence for $Qh(0)$, expressed as an infimum via Proposition 7.1. As discussed above, this argument in fact only requires $\mathrm{W}^{1,p}$-mean coercivity, which is satisfied for the integrand

$$H(A) := h(A) - \gamma V_p(A), \qquad A \in \mathbb{R}^{m\times d},$$

where $\gamma \in (0, \bar{\mu})$ is sufficiently small (as determined below). Here, $\bar{\mu} > 0$ is the constant from the mean coercivity for h. Indeed, we have for $\psi \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$ the estimates

$$\begin{aligned}\int_\Omega h(\nabla\psi(x)) - \gamma V_p(\nabla\psi(x))\,\mathrm{d}x &\geq \int_\Omega \bar{\mu}|\nabla\psi(x)|^p - \bar{\mu} - \gamma V_p(\nabla\psi(x))\,\mathrm{d}x\\ &\geq \int_\Omega \bar{\mu}(1 - C\gamma\bar{\mu}^{-1})|\nabla\psi(x)|^p - C_\gamma\,\mathrm{d}x\end{aligned}$$

for constants $C, C_\gamma > 0$, where we have used the elementary inequality (see Problem 7.8)

$$|A|^p - \theta V_p(A) \geq (1 - C\theta)|A|^p - C_\theta, \qquad A \in \mathbb{R}^{m\times d}, \tag{7.18}$$

for any $\theta > 0$ and constants $C, C_\theta > 0$, only the latter of which depends on θ. Choosing $\gamma > 0$ sufficiently small, we conclude the $\mathrm{W}^{1,p}$-mean coercivity of H. Furthermore, this argument shows that $H \geq G := C^{-1}|\cdot|^p - C$ and G is obviously quasiconvex. This means that the use of Proposition 7.1 in the proof of Proposition 7.11 is justified according to Remark 7.2 (see also Problem 7.2).

Following now the argument of Proposition 7.11 with the aforementioned modifications, we infer that there exists a homogeneous gradient Young measure $\nu \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$ with $[\nu^0] = 0$ and

$$QH(0) = \int H\,\mathrm{d}\nu^0.$$

Via the Jensen inequality for gradient Young measures in Lemma 5.11 and $QH \leq H$ we then get

$$QH(0) \leq \int QH\,\mathrm{d}\nu^0 \leq \int H\,\mathrm{d}\nu^0 = QH(0).$$

Thus,

$$\int H - QH\,\mathrm{d}\nu^0 = 0.$$

Since $H - QH \geq 0$, this in particular implies that for any point $F \in \operatorname{supp}\nu^0$ (of which there is at least one) we have that $H(F) = QH(F)$. By virtue of Proposition 7.1 this is nothing else than the quasiconvexity of $H(A) = h(A) - \gamma V_p(A)$ at F, that is, (i). □

Notes and Historical Remarks

The representation of the quasiconvex envelope in (7.1) is often called *Dacorogna's formula*. One can also equivalently define the quasiconvex envelope through this expression (thanks to Proposition 7.1 and its extensions). This was the case in the First Edition of this book. However, for discontinuous functions (for which the quasiconvex envelope is still continuous!), the present approach is preferable.

The Ball–James Rigidity Theorem 7.3 is from [39]. We will see much more general rigidity results in Chapter 10.

The construction of Lemma 7.5 is from [307]. A different proof of Lemma 7.5 can be found in Section 5.3.9 of [91].

It can be shown that the quasiconvex envelope Qh of an integrand $h\colon \mathbb{R}^{m\times d} \to [0,\infty)$ with p-growth, $p \in [1,\infty]$, and C^1-regularity also has C^1-regularity, see [42] (in fact, more general conditions also imply C^1-regularity for Qh).

The conditions (i)–(iii) at the beginning of Section 7.4 are modeled on the concept of Γ-convergence (introduced by De Giorgi), which we will explore in Chapter 9.

Theorem 7.15 is from [75], also see Section 3 of [159]. We remark that an attempt to define strong p-quasiconvexity as the quasiconvexity of $h - \gamma|\bullet|^p$ for a $\gamma > 0$ (this condition is employed in [75] without being given a name) turns out to result in a notion that is too restrictive for $p < 2$ due to the behavior of $|\bullet|^p$ at the origin (see [159] for details). Let us also mention that for $d = 2$, Example 7.14 is in a sense generic for quasiconvex quadratic forms, see [223] (which goes back to [332]).

Further relaxation formulas can be found in Chapter 11 of the textbook [26]; historically, Dacorogna's lecture notes [90] were also influential.

Problems

7.1 Fill in the details of Step 5 in Proposition 7.1.

7.2 Prove the first claim of Remark 7.2, namely that Proposition 7.1 remains true if the positivity of the integrand h is replaced by the lower bound $h \geq g$ for a quasiconvex $g\colon \mathbb{R}^{m\times d} \to \mathbb{R}$.

7.3 For non-convex domains, statement (i) (b) of the Ball–James Rigidity Theorem 7.3 is false. Construct a counterexample.

7.4 Let L be a linear subspace of $\mathbb{R}^{m\times d}$ with $\operatorname{rank}(A-B) \geq 2$ for all $A, B \in L$, let $K \subset L$ be compact, and let $p \in (1,\infty)$. Show that for

$$h(A) := \operatorname{dist}(A,K)^p, \qquad A \in \mathbb{R}^{m\times d},$$

it holds that Qh is not convex (at zero). Conclude that for $p < 2$ and K not convex, Qh cannot be polyconvex.

7.5 Let $A, B \in \mathbb{R}^{m\times d}$ with $A - B = a \otimes n$ ($a \in \mathbb{R}^m \setminus \{0\}$, $n \in \mathbb{S}^{d-1}$) and let $\theta \in (0,1)$. Set $F := \theta A + (1-\theta)B$ and prove that there exists a uniformly $\mathrm{W}^{1,\infty}$-bounded sequence

$(u_j)_j \subset \mathrm{W}^{1,\infty}_{Fx}(\Omega;\mathbb{R}^m)$ that approximately solves the inclusion $\nabla u \in \{A, B\}$ and $\nabla u_j \overset{*}{\rightharpoonup} F$ in L^∞ but not $\nabla u_j \to F$ in measure.

7.6 Show that under the assumptions of Theorem 7.7 and assuming additionally that X is separable, it holds that

$$\mathcal{F}_*[u] = \inf\left\{ \liminf_{j\to\infty} \mathcal{F}[u_j] \ : \ u_j \rightharpoonup u \text{ in } X \right\}, \qquad u \in X.$$

7.7 Reprove the statements of Lemma 5.2 for strong p-quasiconvexity.

7.8 Prove (7.18) and (7.17).

7.9 Show that in Theorem 7.15, the statements (i), (ii) are furthermore equivalent to

(iii) There is an increasing function $\Theta\colon [0,\infty) \to \mathbb{R}$ with $\Theta(s) \to +\infty$ as $s \to \infty$ such that for one, and then any, bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ and one, and then any, boundary values $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega;\mathbb{R}^m)$, it holds that

$$\int_\Omega h(\nabla u(x))\,\mathrm{d}x \geq \Theta(\|\nabla u\|^p_{\mathrm{L}^p}), \qquad u \in \mathrm{W}^{1,p}_g(\Omega;\mathbb{R}^m).$$

7.10 Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be continuous, have p-growth, and satisfy the pointwise p-coercivity bound (7.14), where $p \in (1,\infty)$. Show that there is at least one matrix $F \in \mathbb{R}^{m\times d}$ such that h is strongly p-quasiconvex at F.



Chapter 8
Quasiconvexity vs. Rank-One Convexity

The four major convexity conditions that play a role in the modern Calculus of Variations satisfy the following implications:

$$\text{convexity} \implies \text{polyconvexity} \implies \text{quasiconvexity} \implies \text{rank-one convexity},$$

where the second and third implications were established in Propositions 6.1 and 5.6, respectively.

While in the scalar case ($d = 1$ or $m = 1$) all four notions are equivalent, this no longer holds in higher dimensions. Proving the non-equivalence between convexity, polyconvexity, and quasiconvexity is relatively straightforward, but the relationship between quasiconvexity and rank-one convexity is a much more subtle affair. Already when he introduced quasiconvexity, Morrey conjectured that, in general, rank-one convexity was a strictly weaker notion:

Conjecture 8.1 (Morrey 1952 [239]) Rank-one convexity does not imply quasiconvexity for $d \geq 2$, $m \geq 2$.

However, a verification of this conjecture proved elusive until Vladimír Šverák's 1992 counterexample, which verified the conjecture, at least for $d \geq 2$, $m \geq 3$. We will consider his ingenious construction as the first theorem of this chapter. The case $d = m = 2$ is still a major unsolved problem; see, for instance, [24, 25, 163] for some recent activity.

On the other hand, in the case of quadratic forms, quasiconvexity and rank-one convexity in fact turn out to be equivalent. This is a consequence of the *compensated compactness* principle embodied in Tartar's Theorem 8.7, which occupies a central role in the field.

One may also switch to a dual point of view and consider the question of how to characterize gradient Young measures. This duality is expressed via the Jensen-type inequalities of Lemma 5.11 that hold for any homogeneous gradient Young measure $\nu \in \mathbf{GY}^p(B_1; \mathbb{R}^{m\times d})$, namely that for all quasiconvex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth,

$$h([\nu]) \leq \int h \, \mathrm{d}\nu. \tag{8.1}$$

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One should understand such a Jensen-type inequality as the reformulation of quasiconvexity in the Young measure context.

It is a surprising fact that the validity of (8.1) for all quasiconvex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth also *characterizes* those Young measures that are generated by some sequence of gradients, as expressed in the Kinderlehrer–Pedregal Theorem 8.11. This hints at the principle that the study of quasiconvexity is really (dual to) the study of the "shape" of sequences of gradients, imparting a much more fundamental flavor to this notion than seeing it as a mere tool to prove the existence of minimizers for integral functionals.

8.1 Comparison Between Generalized Convexity Notions

Define for a locally bounded $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ the **polyconvex envelope** $Ph\colon \mathbb{R}^{m\times d} \to \mathbb{R}\cup\{-\infty\}$ and the **rank-one convex envelope** $Rh\colon \mathbb{R}^{m\times d} \to \mathbb{R}\cup\{-\infty\}$ of h via

$$\begin{aligned} Ph(A) &:= \sup\big\{\, g(A) \;:\; g\colon \mathbb{R}^{m\times d} \to \mathbb{R} \text{ polyconvex and } g \le h \,\big\},\\ Rh(A) &:= \sup\big\{\, g(A) \;:\; g\colon \mathbb{R}^{m\times d} \to \mathbb{R} \text{ rank-one convex and } g \le h \,\big\}, \end{aligned}$$

where $A \in \mathbb{R}^{m\times d}$. As a consequence of the above implications between the convexity notions, we have that

$$h^{**} \le Ph \le Qh \le Rh \le h,$$

where we recall that h^{**} denotes the convex envelope of h. In general, Qh is difficult to compute, so Ph and Rh can give useful lower and upper bounds on Qh.

Clearly, the determinant function is polyconvex but not convex. We next exhibit an example of a quasiconvex, but not polyconvex function (also see Problem 6.5):

Example 8.2 Let $F \in \mathbb{R}^{3\times 3}$ with $\operatorname{rank} F \ge 2$ and let $p \in (1,2)$. Define

$$h(A) := \operatorname{dist}(A, \{-F, F\})^p, \qquad A \in \mathbb{R}^{3\times 3}.$$

Then, the quasiconvex envelope $Qh\colon \mathbb{R}^{3\times 3} \to [0,\infty)$ is quasiconvex by Proposition 7.1, and not convex (at zero) by Lemma 7.5. Since Qh has p-growth and we chose $p < 2$, it can be shown without too much effort that Qh cannot be polyconvex (using the fact that non-constant convex functions have at least linear growth in at least one direction and non-trivial minors have at least quadratic growth).

Example 8.3 (Alibert–Dacorogna–Marcellini 1988 [7, 93]) From Example 5.7 we recall the Alibert–Dacorogna–Marcellini function

$$h_\gamma(A) := |A|^2\big(|A|^2 - 2\gamma \det A\big), \qquad A \in \mathbb{R}^{2\times 2}.$$

It can be shown that h_γ is polyconvex if and only if $|\gamma| \le 1$. Since it is quasiconvex if and only if $|\gamma| \le \gamma_{\mathrm{QC}}$, where $\gamma_{\mathrm{QC}} > 1$, this provides a further example of a quasiconvex function that is not polyconvex. See Section 5.3.8 in [91] for the details.

We now present *Švérak's counterexample* of a rank-one convex, but not quasiconvex, function, which verified Morrey's Conjecture 8.1 for $d \geq 2$, $m \geq 3$.

Theorem 8.4 (Švérak 1992 [309]) *Rank-one convexity does not imply quasiconvexity for $d \geq 2$, $m \geq 3$.*

Proof. We show the non-equivalence of quasiconvexity and rank-one convexity for $d = 2$, $m = 3$ only. Higher dimensions can be treated using an embedding of $\mathbb{R}^{3\times 2}$ into $\mathbb{R}^{m\times d}$. The proof proceeds by exhibiting an explicit function $h\colon \mathbb{R}^{3\times 2} \to \mathbb{R}$ that is rank-one convex but not quasiconvex.

Define the linear subspace L of $\mathbb{R}^{3\times 2}$ as

$$L := \left\{ \begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} \,:\, x, y, z \in \mathbb{R} \right\}$$

and denote by $\mathbf{P}\colon \mathbb{R}^{3\times 2} \to L$ the linear projection onto L given by

$$\mathbf{P}(A) := \begin{pmatrix} a & 0 \\ 0 & d \\ (e+f)/2 & (e+f)/2 \end{pmatrix} \qquad \text{for} \qquad A = \begin{pmatrix} a & b \\ c & d \\ e & f \end{pmatrix} \in \mathbb{R}^{3\times 2}.$$

Also let $g\colon L \to \mathbb{R}$ be defined by

$$g\begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} := -xyz.$$

For $\alpha, \beta > 0$ we set $h_{\alpha,\beta}\colon \mathbb{R}^{3\times 2} \to \mathbb{R}$ to be

$$h_{\alpha,\beta}(A) := g(\mathbf{P}(A)) + \alpha\big(|A|^2 + |A|^4\big) + \beta|A - \mathbf{P}(A)|^2.$$

Below we will prove the following two properties of $h_{\alpha,\beta}$:

(a) For every $\alpha > 0$ sufficiently small and all $\beta > 0$ the function $h_{\alpha,\beta}$ is not quasiconvex.
(b) For every $\alpha > 0$ there exists a $\beta = \beta(\alpha) > 0$ such that $h_{\alpha,\beta}$ is rank-one convex.

This implies the claim for suitable parameters α, β.

Ad (a): For the periodic map $\varphi \in \mathrm{W}^{1,\infty}_{\mathrm{per}}((0,1)^2; \mathbb{R}^3)$ given by

$$\varphi(x_1, x_2) := \frac{1}{2\pi} \begin{pmatrix} \sin(2\pi x_1) \\ \sin(2\pi x_2) \\ \sin(2\pi(x_1 + x_2)) \end{pmatrix}, \qquad x = (x_1, x_2) \in (0,1)^2,$$

we have $\nabla\varphi \in L$ and hence $\mathbf{P}(\nabla\varphi) = \nabla\varphi$. Thus, we may compute in an elementary way

$$\int_{(0,1)^2} g(\nabla\varphi)\, \mathrm{d}x = -\int_0^1 \int_0^1 (\cos 2\pi x_1)^2 (\cos 2\pi x_2)^2 \, \mathrm{d}x_1 \, \mathrm{d}x_2 = -\frac{1}{4} < 0.$$

Then, for $\alpha > 0$ sufficiently small and all $\beta > 0$,

$$\int_{(0,1)^2} h_{\alpha,\beta}(\nabla\varphi)\,\mathrm{d}x < 0 = h_{\alpha,\beta}(0). \tag{8.2}$$

It turns out that in the definition of quasiconvexity we may alternatively test with functions that have periodic boundary values, see Problem 5.2. Hence, (a) follows.

Ad (b): By Problem 8.1, the rank-one convexity of the integrand $h_{\alpha,\beta}$ is equivalent to the **Legendre–Hadamard condition**

$$\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B] := \frac{\mathrm{d}^2}{\mathrm{d}t^2}\, h_{\alpha,\beta}(A+tB)\bigg|_{t=0} \geq 0 \tag{8.3}$$

for all $A, B \in \mathbb{R}^{3\times 2}$ with rank $B \leq 1$.

The function g is a homogeneous polynomial of degree 3, whereby we can find $c > 0$ such that for all $A, B \in \mathbb{R}^{3\times 2}$ with rank $B \leq 1$,

$$G(A,B) := \mathrm{D}^2(g\circ\mathbf{P})(A)[B,B] := \frac{\mathrm{d}^2}{\mathrm{d}t^2}\, g(\mathbf{P}(A+tB))\bigg|_{t=0} \geq -c|A||B|^2.$$

A computation then shows that

$$\begin{aligned}\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B] &= G(A,B) + 2\alpha|B|^2 + 4\alpha|A|^2|B|^2 + 8\alpha(A:B)^2 \\ &\quad + 2\beta|B-\mathbf{P}(B)|^2 \\ &\geq (-c + 4\alpha|A|)|A||B|^2\end{aligned}$$

for some $c > 0$. Thus, for $|A| \geq c/(4\alpha)$, the Legendre–Hadamard condition (and hence the rank-one convexity) holds.

We still need to prove the Legendre–Hadamard condition for $|A| < c/(4\alpha)$. Since $\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B]$ is homogeneous of degree 2 in B, we only need to consider A, B from the compact set

$$K := \left\{ (A,B) \in \mathbb{R}^{3\times 2}\times\mathbb{R}^{3\times 2} \;:\; |A| \leq \frac{c}{4\alpha},\ |B| = 1,\ \operatorname{rank} B = 1 \right\}.$$

By the estimate

$$\mathrm{D}^2 h_{\alpha,\beta}(A)[B,B] \geq G(A,B) + 2\alpha|B|^2 + 2\beta|B-\mathbf{P}(B)|^2 =: \kappa(A,B,\beta)$$

it suffices to show that there exists a $\beta = \beta(\alpha) > 0$ such that $\kappa(A,B,\beta) \geq 0$ for all $(A,B) \in K$. Suppose that this is not the case. Then there is a sequence $\beta_j \to \infty$ and $(A_j,B_j) \in K$ with

$$0 > \kappa(A_j,B_j,\beta_j) = G(A_j,B_j) + 2\alpha + 2\beta_j|B_j - \mathbf{P}(B_j)|^2.$$

As K is compact, we may assume $(A_j, B_j) \to (A, B) \in K$, for which

$$G(A, B) + 2\alpha \le 0, \qquad \mathbf{P}(B) = B, \qquad \text{and} \qquad \operatorname{rank} B = 1.$$

However, since $\operatorname{rank} B = 1$, we have that $g(\mathbf{P}(A + tB)) = 0$ is affine in $t \in \mathbb{R}$. This implies in particular

$$G(A, B) = \mathrm{D}^2(g \circ \mathbf{P})(A)[B, B] = 0,$$

yielding $2\alpha \le 0$, which contradicts our assumption $\alpha > 0$. This completes the proof of (b). □

Based on Švérak's counterexample, it has been shown that quasiconvexity is not a local condition, meaning that there is no pointwise condition involving only the function and a finite number of its derivatives that is both necessary and sufficient for the function to be quasiconvex (verifying a further conjecture by Morrey [239]). More precisely, let $\mathcal{Q}\colon \mathrm{C}^\infty(\mathbb{R}^{m\times d}) \to \mathbf{X}^{m\times d}$ be a nonlinear operator, where we denote by $\mathbf{X}^{m\times d}$ the space of functions from $\mathbb{R}^{m\times d}$ to $[-\infty, +\infty]$. Call $\mathcal{Q}$ **local** if $h = g$ in a neighborhood of $A \in \mathbb{R}^{m\times d}$ implies that also $\mathcal{Q}[h] = \mathcal{Q}[g]$ in a neighborhood of A.

Theorem 8.5 (Kristensen 1999 [205]) *Let $d \ge 2$ and $m \ge 3$. Then, there exists no local (nonlinear) operator $\mathcal{Q}\colon \mathrm{C}^\infty(\mathbb{R}^{m\times d}) \to \mathbf{X}^{m\times d}$ such that*

$$\mathcal{Q}[h] = 0 \quad \Longleftrightarrow \quad h \text{ is quasiconvex}$$

for all $h \in \mathrm{C}^\infty(\mathbb{R}^{m\times d})$.

This is in contrast to rank-one convexity, which is characterized through the local operator

$$\mathcal{R}[h](A) := \inf\left\{ \mathrm{D}^2 h(A)[a \otimes b, a \otimes b] \ : \ a \in \mathbb{R}^m,\ b \in \mathbb{R}^d \right\},$$

where $h \in \mathrm{C}^\infty(\mathbb{R}^{m\times d})$ and $A \in \mathbb{R}^{m\times d}$.

8.2 Compensated Compactness

The theory of *compensated compactness* concerns the surprising convergence properties of some *nonlinear* functions of weakly converging maps in the presence of PDE-constraints (such as curl-freeness, i.e., weak convergence of gradients). We have already seen instances of such results in Section 5.2 when we observed that minors have better than expected convergence properties with respect to the weak(*) convergence of gradients. Here we will consider similar statements concerning certain quadratic forms.

To set the stage, we let $\mathcal{A}$ be a homogeneous first-order linear PDE operator with constant coefficients,

$$\mathcal{A} := \sum_{l=1}^{d} A_l \partial_l, \tag{8.4}$$

where $A_l \in \mathbb{R}^{M\times N}$, $l = 1,\dots,d$ ($M, N \in \mathbb{N}$). In the following, the system of PDEs

$$\mathcal{A}V = 0, \qquad V \in \mathrm{L}^2(\Omega;\mathbb{R}^N),$$

is to be interpreted in the $\mathrm{W}^{-1,2}$-sense, that is, we require

$$\left\langle V, \mathcal{A}^\mathsf{T} w\right\rangle = \left\langle V, \sum_{l=1}^d A_l^\mathsf{T}\partial_l w\right\rangle = 0 \qquad \text{for all } w \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^M).$$

Define the **wave cone** of $\mathcal{A}$ as

$$\Lambda_{\mathcal{A}} := \bigcup_{\xi\in\mathbb{S}^{d-1}} \ker \mathbb{A}(\xi), \qquad \text{where} \qquad \mathbb{A}(\xi) := (2\pi\mathrm{i})\sum_{l=1}^d A_l\xi_l.$$

Here, $\mathbb{A}(\xi)$ is called the **symbol** of $\mathcal{A}$.

Example 8.6 The most important operator $\mathcal{A}$ for our purposes is

$$\mathcal{A}V := \operatorname{curl} V := \left(\partial_j V_i^k - \partial_i V_j^k\right)_{i,j=1,\dots,d}^{k=1,\dots,m}.$$

To explain our particular choice of "curl", we note that for any $u \in \mathrm{C}^2(\mathbb{R}^d;\mathbb{R}^m)$ it holds that

$$\partial_i[\nabla u]_j^k = \partial_i\partial_j u^k = \partial_j\partial_i u^k = \partial_j[\nabla u]_i^k$$

for all $i, j = 1,\dots,d$; $k = 1,\dots,m$. It is a classical result that these *integrability conditions* in fact characterize gradients on simply connected domains. We can now easily compute that $A \in \ker\mathbb{A}(\xi)$ for $\xi \in \mathbb{S}^{d-1}$ if and only if

$$\xi_j A_i^k = \xi_i A_j^k \qquad \text{for all } i, j = 1,\dots,d;\ k = 1,\dots,m.$$

Indeed, these relations are analogous to the integrability conditions, but with "∂_j" replaced by "ξ_j" (this can be justified using the Fourier transform). Thus, for $A \in \ker\mathbb{A}(\xi)$ it must hold that $A_i^k = \xi_i A_j^k/\xi_j$ for all j such that $\xi_j \neq 0$. Clearly, this is only possible if $A = a\otimes\xi$ for some $a \in \mathbb{R}^m$. Indeed, just set $a_k := A_j^k/\xi_j$ for a j with $\xi_j \neq 0$ and notice that this definition does not depend on the choice of j by the above relation. Consequently,

$$\Lambda_{\mathrm{curl}} = \bigcup_{\xi\in\mathbb{S}^{d-1}} \ker\mathbb{A}(\xi) = \left\{ a\otimes\xi \in \mathbb{R}^{m\times d} \,:\, a \in \mathbb{R}^m, \xi \in \mathbb{S}^{d-1}\right\}.$$

One should think of the wave cone $\Lambda_{\mathcal{A}}$ as containing all amplitudes for which $\mathcal{A}$ is "not elliptic". As such, it plays a fundamental role in the study of oscillations in sequences of functions $(V_j)_j \subset \mathrm{L}^2(\Omega;\mathbb{R}^N)$ with $V_j \rightharpoonup V$ in L^2 and $\mathcal{A}V_j \to 0$ in $\mathrm{W}^{-1,2}$. Let us illustrate this observation with a purely formal, yet instructive, argument: For the limit V we must have

$$\mathcal{A}V = 0.$$

Fourier-transforming this (see Appendix A.6), we obtain $\mathbb{A}(\xi)\widehat{V}(\xi) = 0$ for all $\xi \in \mathbb{R}^d$. Hence,

$$\widehat{V}(\xi) \in \ker \mathbb{A}(\xi) \subset \Lambda_{\mathcal{A}} \qquad \text{for all } \xi \in \mathbb{R}^d \setminus \{0\}.$$

Thus, intuitively, all mass of $\widehat{V}_j(\xi)$ outside of $\Lambda_{\mathcal{A}}$ has to disappear in the limit and cannot contribute to oscillations in the sequence $(V_j)_j$.

This intuition is rigorously expressed in the following fundamental result.

Theorem 8.7 (Tartar 1979 [323]) *Suppose that $q\colon \mathbb{R}^N \to \mathbb{R}$ is a quadratic form, i.e., $q(A) = b(A, A)$ for a bilinear form $b\colon \mathbb{R}^{m\times d}\times\mathbb{R}^{m\times d} \to \mathbb{R}$, that satisfies the* ***generalized Legendre–Hadamard condition***

$$q(A) \geq 0 \qquad \text{for all } A \in \Lambda_{\mathcal{A}}. \tag{8.5}$$

Assume furthermore that $(V_j)_j \subset \mathrm{L}^2(\Omega;\mathbb{R}^N)$ with

$$\begin{cases} V_j \rightharpoonup V & \text{in } \mathrm{L}^2, \\ q(V_j) \overset{*}{\rightharpoonup} \mu & \text{in } \mathcal{M}(\Omega), \end{cases}$$

and that

$$(\mathcal{A}V_j)_j \text{ is (strongly) precompact in } \mathrm{W}^{-1,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^M).$$

Then,

$$\mu \geq q(V) \qquad \text{as measures on } \Omega.$$

In particular, if $q(A) = 0$ for all $A \in \Lambda_{\mathcal{A}}$, then $q(V_j) \overset{*}{\rightharpoonup} q(V)$ in $\mathcal{M}(\Omega)$.

Here and in the following we identify $q(V_j)$ with the measure $q(V_j)\,\mathcal{L}^d \llcorner \Omega$. We also remark that condition (8.5) is equivalent to the convexity of q in the directions specified by $\Lambda_{\mathcal{A}}$.

Proof. Step 1. We first show that we may assume that $V \equiv 0$. Indeed, set

$$\widetilde{V}_j := V_j - V.$$

Then, $\widetilde{V}_j \rightharpoonup 0$ in L^2 and $(\mathcal{A}\widetilde{V}_j)_j$ is (strongly) precompact in $\mathrm{W}^{-1,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^m)$. If $q(A) = A^{\mathsf{T}}\mathbf{S}A$ ($A \in \mathbb{R}^N$) with a symmetric matrix $\mathbf{S} \in \mathbb{R}^{N\times N}_{\mathrm{sym}}$, we compute

$$q(\widetilde{V}_j) = q(V_j) - 2V_j^{\mathsf{T}}\mathbf{S}V + q(V) \overset{*}{\rightharpoonup} \mu - q(V) =: \widetilde{\mu} \quad \text{in } \mathcal{M}(\Omega).$$

Thus, it remains to show that $\widetilde{\mu} \geq 0$. In the following we drop the tildes.

Step 2. Let $\varphi \in \mathrm{C}^\infty_c(\Omega)$ and set

$$W_j := \varphi V_j.$$

Then,

$$\mathcal{A}W_j = \varphi\mathcal{A}V_j + \sum_{l=1}^{d} A_l V_j \partial_l \varphi$$

and this sequence is (strongly) precompact in the space $\mathrm{W}^{-1,2}(\Omega;\mathbb{R}^M)$ since the second term is uniformly bounded in $\mathrm{L}^2(\Omega;\mathbb{R}^M)$, which is compactly embedded into $\mathrm{W}^{-1,2}(\Omega;\mathbb{R}^M)$, as can be seen by dualizing the Rellich–Kondrachov Embedding Theorem A.29. Thus, we may assume that

$$\mathcal{A}W_j \to 0 \quad \text{in } \mathrm{W}^{-1,2}. \tag{8.6}$$

Here, the convergence to zero follows by the (strong) precompactness and since $W_j \rightharpoonup 0$ in L^2 implies that $\langle \mathcal{A}W_j, w\rangle = \langle W_j, \mathcal{A}^{\mathsf{T}} w\rangle \to 0$ for all $w \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^M)$. Moreover,

$$q(W_j) = \varphi^2 q(V_j) \overset{*}{\rightharpoonup} \varphi^2 \mu \quad \text{in } \mathcal{M}(\Omega).$$

Step 3. In the following we will show that

$$\int_\Omega \varphi^2 \,\mathrm{d}\mu = \lim_{j\to\infty} \int_\Omega q(W_j)\,\mathrm{d}x \geq 0 \qquad \text{for all } \varphi \in \mathrm{C}_c^\infty(\Omega), \tag{8.7}$$

which will prove the claim. For this, assume that q is extended to $\mathbb{C}^N$ via

$$q(Z) := Z^* \mathbf{S} Z = \overline{Z}^{\mathsf{T}} \mathbf{S} Z, \qquad Z \in \mathbb{C}^N.$$

If $Z = A + \mathrm{i}B \in \Lambda_{\mathcal{A}} + \mathrm{i}\Lambda_{\mathcal{A}}$, a simple calculation together with the condition (8.5) shows

$$q(Z) = q(A) + q(B) \geq 0. \tag{8.8}$$

By the (vector-valued) Parseval relation (A.7),

$$\int_\Omega q(W_j)\,\mathrm{d}x = \int_\Omega W_j^* \mathbf{S} W_j \,\mathrm{d}x = \int \widehat{W}_j^* \mathbf{S} \widehat{W}_j \,\mathrm{d}\xi = \int q(\widehat{W}_j)\,\mathrm{d}\xi.$$

Hence, to prove our claim (8.7), we will in the following establish

$$\lim_{j\to\infty} \int q(\widehat{W}_j)\,\mathrm{d}\xi \geq 0. \tag{8.9}$$

In order to see this, we split the integral's domain into B_1 and $\mathbb{R}^d \setminus B_1$ and prove (8.9) separately for these two parts.

Step 4. On B_1 our claim is straightforward: Since $W_j \rightharpoonup 0$ in L^2 and the W_j's have uniformly bounded support, it holds that

$$\widehat{W}_j(\xi) = \int W_j(x) \mathrm{e}^{-2\pi \mathrm{i} x\cdot\xi} \,\mathrm{d}x \to 0 \qquad \text{for all } \xi \in \mathbb{R}^d.$$

Moreover, $|\widehat{W}_j(\xi)| \leq \|W_j\|_{\mathrm{L}^1} \leq C$ for all $\xi \in \mathbb{R}^d$ and a uniform constant $C > 0$. Thus,

$$\widehat{W}_j \to 0 \quad \text{in } \mathrm{L}^2_{\mathrm{loc}}.$$

Consequently,

$$\lim_{j\to\infty}\int_{B_1} q(\widehat{W}_j)\,\mathrm{d}\xi = 0. \tag{8.10}$$

Step 5. We next show that for all $\delta > 0$ there exists a constant $C_\delta > 0$ such that

$$q(Z) \geq -\delta|Z|^2 - C_\delta|\mathbb{A}(\eta)Z|^2 \tag{8.11}$$

for all $Z \in \mathbb{C}^N$ and all $\eta \in \mathbb{S}^{d-1}$. Indeed, if this was not the case there would exist $\delta > 0$ and sequences $(Z_m)_m \subset \mathbb{C}^N$ with $|Z_m| = 1$ and $(\eta_m)_m \subset \mathbb{S}^{d-1}$ with the property

$$q(Z_m) < -\delta|Z_m|^2 - m|\mathbb{A}(\eta_m)Z_m|^2.$$

Without loss of generality we may assume that $\eta_m \to \eta$ and $Z_m \to Z$ with $|Z| = 1$. Then, there is a constant $C > 0$ such that

$$|\mathbb{A}(\eta_m)Z_m|^2 \leq \frac{C}{m},$$

and thus $Z \in \ker_{\mathbb{C}} \mathbb{A}(\eta) \subset \Lambda_{\mathcal{A}} + \mathrm{i}\Lambda_{\mathcal{A}}$. By (8.8), $q(Z) \geq 0$. On the other hand,

$$q(Z) = \lim_{m\to\infty} q(Z_m) \leq -\delta,$$

a contradiction. Hence, (8.11) must hold.

Step 6. From (8.6) we get via Lemma A.37 that

$$(\mathrm{id} - \Delta)^{-1/2}[\mathcal{A}W_j] \to 0 \quad \text{in } \mathrm{L}^2,$$

and via Plancherel's identity (A.6) this gives

$$\frac{1}{(1 + 4\pi^2|\xi|^2)^{1/2}}\mathbb{A}(\xi)\widehat{W}_j(\xi) \to 0 \quad \text{in } \mathrm{L}^2.$$

Since $(1 + 4\pi^2|\xi|^2)^{1/2} \sim |\xi|$ for $|\xi| \geq 1$, we have

$$\mathbb{A}\left(\frac{\xi}{|\xi|}\right)\widehat{W}_j(\xi) = \frac{1}{|\xi|}\mathbb{A}(\xi)\widehat{W}_j(\xi) \to 0 \quad \text{in } \mathrm{L}^2(\mathbb{R}^d \setminus B_1). \tag{8.12}$$

Applying (8.11) for $Z := \widehat{W}_j(\xi)$ and $\eta := \xi/|\xi|$, we get

$$\int_{\mathbb{R}^d\setminus B_1} q(\widehat{W}_j(\xi))\,\mathrm{d}\xi \geq -\int_{\mathbb{R}^d\setminus B_1} \delta|\widehat{W}_j(\xi)|^2\,\mathrm{d}\xi - C_\delta\int_{\mathbb{R}^d\setminus B_1}\left|\mathbb{A}\left(\frac{\xi}{|\xi|}\right)\widehat{W}_j(\xi)\right|^2\mathrm{d}\xi.$$

Combine this with (8.12) to deduce that

$$\lim_{j\to\infty}\int_{\mathbb{R}^d\setminus B_1} q(\widehat{W}_j(\xi))\,\mathrm{d}\xi \geq -C\delta$$

for a constant $C > 0$, where we also used that the $\widehat{W}_j$ are uniformly norm-bounded in $\mathrm{L}^2(\mathbb{R}^d)$. Thus, as $\delta > 0$ was arbitrary,

$$\lim_{j\to\infty} \int_{\mathbb{R}^d\setminus B_1} q(\widehat{W}_j(\xi))\,\mathrm{d}\xi \geq 0. \tag{8.13}$$

From (8.10) and (8.13) we conclude (8.9), finishing the proof in the case $q \geq 0$ on $\Lambda_{\mathcal{A}}$.

If $q \equiv 0$ on $\Lambda_{\mathcal{A}}$, simply apply the first part of the theorem separately to q and to $-q$. □

As the most well-known compensated compactness result we have the following *div-curl lemma*:

Lemma 8.8 (Murat–Tartar 1974 [254]) *Let $(u_j)_j, (v_j)_j \subset \mathrm{L}^2(\Omega;\mathbb{R}^d)$ be such that*

$$u_j \rightharpoonup u, \quad v_j \rightharpoonup v \qquad \text{in } \mathrm{L}^2$$

and

$$(\operatorname{div} u_j)_j, (\operatorname{curl} v_j)_j \text{ are precompact in } \mathrm{W}^{-1,2}_{\mathrm{loc}}.$$

Then,

$$u_j \cdot v_j \overset{*}{\rightharpoonup} u \cdot v \quad \text{in } \mathcal{M}(\Omega).$$

Proof. We set $V_j := (u_j, v_j)$ and $\mathcal{A} := (\operatorname{div}, \operatorname{curl})$. One may compute (see Problem 8.2) that

$$\Lambda_{\mathcal{A}} = \left\{ (a,b) \in \mathbb{R}^d \times \mathbb{R}^d \;:\; a \perp b \right\}.$$

Thus, for the quadratic function $q(a,b) := a \cdot b$ we have that q vanishes on $\Lambda_{\mathcal{A}}$ and the conclusion follows from Tartar's theorem. □

One direct application of Tartar's Theorem 8.7 is the following result on the equivalence of quasiconvexity and rank-one convexity for quadratic forms:

Proposition 8.9 *Let $q\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be a quadratic form. Then, q is quasiconvex if and only if it is rank-one convex, which is furthermore equivalent to the* Legendre–Hadamard condition

$$q(A) \geq 0 \qquad \text{for all } A \in \mathbb{R}^{m\times d} \text{ with } \operatorname{rank} A \leq 1. \tag{8.14}$$

Proof. The fact that (8.14) is equivalent to rank-one convexity for a quadratic form q follows from the fact that for all $F, A \in \mathbb{R}^{m\times d}$ with $\operatorname{rank} A \leq 1$ it holds that

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2} q(F + tA)\bigg|_{t=0} = 2q(A).$$

Then, by Problem 8.1, q is convex along rank-one lines if and only if (8.14) holds.

The quasiconvexity of a quadratic form q with $q \geq 0$ on the rank-one cone is a consequence of Tartar's Theorem 8.7 for $\mathcal{A} := \operatorname{curl}$, as defined in Example 8.6. Indeed, assume that we are given a sequence $(u_j)_j \subset \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ with $u_j|_{\partial\Omega} = Fx$ and $u_j \rightharpoonup Fx$ in $\mathrm{W}^{1,2}$. Extending all u_j by Fx to a larger domain $\Omega' \ni \Omega$ and applying

Tartar's theorem there, we obtain that

$$q(F) \le \liminf_{j\to\infty} \int_\Omega q(\nabla u_j(x))\,\mathrm{d}x.$$

Invoking Proposition 5.18, we observe that this (restricted) weak lower semicontinuity assertion is enough to conclude the quasiconvexity of q. □

Example 8.10 The quadratic form

$$q(A) := |A|^2 - (\operatorname{tr} A)^2, \qquad A \in \mathbb{R}^{2\times 2},$$

is quasiconvex: Since

$$|a\otimes b|^2 - (\operatorname{tr} a\otimes b)^2 = |a|^2\cdot|b|^2 - |a\cdot b|^2 \ge 0$$

for all $a, b \in \mathbb{R}^2$, this follows directly from Proposition 8.9.

8.3 Characterization of Gradient Young Measures

Is every Young measure also a gradient Young measure? In the previous chapter we replaced a minimization problem over a Sobolev space by its extension–relaxation, defined on the space of *gradient* Young measures, so this is a natural question.

For *inhomogeneous* Young measures the answer is clearly negative since the barycenter of a gradient Young measure must be a gradient (i.e., curl-free), so the elementary Young measure $\delta[V]$ for V with $\operatorname{curl} V \not\equiv 0$ provides an immediate counterexample.

The question of whether all *homogeneous* Young measures are gradient Young measures is more intricate since then the barycenter is constant and hence trivially a gradient. Still, there are homogeneous Young measures that are not gradient Young measures, but proving that no generating sequence of gradients can be found is often not straightforward. One possibility is to show that there is a quasiconvex function such that the Jensen-type inequality of Lemma 5.11 fails. This strategy is used to good effect in Chapters 10, 11.

Here we consider a more elementary argument: Let $A, B \in \mathbb{R}^{m\times d}$ with $A \ne B$ and $\theta \in (0,1)$. Consider the homogeneous Young measure

$$\nu := \theta\delta_A + (1-\theta)\delta_B \in \mathbf{Y}^\infty(B_1;\mathbb{R}^{m\times d}). \tag{8.15}$$

We know from Example 4.11 that for $\operatorname{rank}(A-B) \le 1$, ν is a gradient Young measure.

The case $\operatorname{rank}(A-B) \ge 2$ can be investigated via the Ball–James rigidity result in Theorem 7.3: Assume that there is a sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}(B_1;\mathbb{R}^m)$ with $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$. By Lemma 4.13 it follows that

$$\operatorname{dist}(\nabla u_j, \{A, B\}) \to 0 \quad \text{in measure.}$$

Then, however, from statement (ii) of the Ball–James Rigidity Theorem we get $\nabla u_j \to A$ or $\nabla u_j \to B$ in measure, either of which yields a contradiction (again by Lemma 4.13).

Up to now the class of gradient Young measures has only been specified extrinsically, i.e., through the existence of a generating sequence of gradients. Intuitively, trying to find an *intrinsic* characterization of gradient Young measures amounts to understanding the (asymptotic) oscillations that can occur in sequences of gradients, and it should be clear by now that this is a useful endeavor.

Recall that in Lemma 5.11 we showed that a homogeneous gradient Young measure $\nu = (\nu)_x \in \mathbf{GY}^p(B_1; \mathbb{R}^{m\times d})$ satisfies the Jensen-type inequality

$$h([\nu]) \le \int h \, \mathrm{d}\nu$$

for all quasiconvex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth. The following important result shows that these Jensen-type inequalities (essentially) characterize gradient Young measures.

Theorem 8.11 (Kinderlehrer–Pedregal 1991/1994 [190, 191]) *Assume that $\nu \in \mathbf{Y}^p(\Omega; \mathbb{R}^{m\times d})$, $p \in (1, \infty]$, is a Young measure with $[\nu] = \nabla u$ for some underlying deformation $u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^m)$. Then, $\nu \in \mathbf{GY}^p(\Omega; \mathbb{R}^{m\times d})$ if and only if for almost every $x \in \Omega$ the Jensen-type inequality*

$$h(\nabla u(x)) \le \int h \, \mathrm{d}\nu_x \tag{8.16}$$

holds for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth if $p \in (1, \infty)$ (no growth condition if $p = \infty$).

Remark 8.12 It will follow from the proof that we only need to verify (8.16) for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ such that

$$\lim_{|A|\to\infty} \frac{h(A)}{1 + |A|^p} \quad \text{exists (in } \mathbb{R}) \tag{8.17}$$

if $p \in (1, \infty)$. Note that by Lemma 5.11, the condition (8.17) is not needed for the Jensen-type inequality (8.16) to hold. For another strengthening of the Kinderlehrer–Pedregal theorem, see Problem 8.4.

The proof starts by reducing to the case of homogeneous Young measures ν. Then, we will show that the set of homogeneous gradient Young measures is convex and closed with respect to a suitable (weak-type) topology. It turns out that the Jensen-type inequalities express that ν cannot be separated from this set by any hyperplane. Thus, the geometric Hahn–Banach theorem implies that ν actually lies in the set of gradient Young measures. Evidently, this is a *non-constructive* argument, which does not produce an explicit generating sequence.

For the functional-analytic setup we let $\mathcal{M}_p(\mathbb{R}^{m\times d})$ for $p \in (1,\infty)$ denote the class of finite signed measures on $\mathbb{R}^{m\times d}$ with bounded p'th-order absolute moment, that is,

$$\mathcal{M}_p(\mathbb{R}^{m\times d}) := \left\{ \mu \in \mathcal{M}(\mathbb{R}^{m\times d};\mathbb{R}) \ : \ \int 1 + |A|^p \, \mathrm{d}\mu(A) < \infty \right\}.$$

We also define the following class of integrands:

$$\mathbf{I}^p(\mathbb{R}^{m\times d}) := \left\{ h \in \mathrm{C}(\mathbb{R}^{m\times d}) \ : \ \lim_{|A|\to\infty} \frac{h(A)}{1+|A|^p} \ \text{ exists (in } \mathbb{R}) \right\}, \tag{8.18}$$

which is a separable Banach space when equipped with the norm

$$\|h\|_{\mathbf{I}^p} := \left\| \frac{h}{1+|\bullet|^p} \right\|_\infty, \qquad h \in \mathbf{I}^p(\mathbb{R}^{m\times d}).$$

For the separability, see Problem 8.3.

The key observation is that an integrand $h \in \mathbf{I}^p(\mathbb{R}^{m\times d})$ can be viewed as a linear functional on $\mathcal{M}_p(\mathbb{R}^{m\times d})$ via the duality pairing

$$\big\langle h, \mu \big\rangle := \int h \, \mathrm{d}\mu, \qquad \mu \in \mathcal{M}_p(\mathbb{R}^{m\times d}).$$

Note that the functionals $\langle h, \bullet\rangle$, $h \in \mathbf{I}^p(\mathbb{R}^{m\times d})$, separate the points of $\mathcal{M}_p(\mathbb{R}^{m\times d})$, that is, for $\mu_1, \mu_2 \in \mathcal{M}_p(\mathbb{R}^{m\times d})$ with $\mu_1 \neq \mu_2$ there is an integrand $h \in \mathbf{I}^p(\mathbb{R}^{m\times d})$ such that $\langle \mu_1, h\rangle \neq \langle \mu_2, h\rangle$.

Let τ_p be the weakest topology on $\mathcal{M}_p(\mathbb{R}^{m\times d})$ that makes all our functionals $\langle h, \bullet\rangle$ continuous. It is a general result of topology (see Lemma A.4) that the topological space $(\mathcal{M}_p(\mathbb{R}^{m\times d}), \tau_p)$ is a locally convex topological vector space and its dual space $(\mathcal{M}_p(\mathbb{R}^{m\times d}), \tau_p)^*$ can be identified with $\mathbf{I}^p(\mathbb{R}^{m\times d})$ via the above duality pairing.

The set of homogeneous $\mathrm{W}^{1,p}$-gradient Young measures with barycenter $F \in \mathbb{R}^{m\times d}$ is defined as

$$\mathbf{GY}^p_{\mathrm{hom}}(F) := \big\{ \mu \in \mathcal{M}^1(\mathbb{R}^{m\times d}) \ : \ \mu \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d}), \ [\mu] = F \big\},$$

which in the following we view as a subset of the space $\mathcal{M}_p(\mathbb{R}^{m\times d})$.

Lemma 8.13 *For any $F \in \mathbb{R}^{m\times d}$ the set $\mathbf{GY}^p_{\mathrm{hom}}(F)$ is convex and τ_p-closed in $\mathcal{M}_p(\mathbb{R}^{m\times d})$.*

Proof. Step 1: Convexity. Recall that homogeneous Young measures have generating sequences on any bounded Lipschitz domain, see Lemma 4.15; we will use this fact several times in the sequel.

Let $\mu_1, \mu_2 \in \mathbf{GY}^p_{\mathrm{hom}}(F)$ and $\theta \in (0,1)$. Choose a Lipschitz subdomain $D_1 \subset B_1$ with

$$|D_1| = \theta\omega_d,$$

where we recall that $\omega_d := |B_1|$. We assume that μ_1 is a Young measure on D_1 and μ_2 is a Young measure on $D_2 := B_1 \backslash D_1$. Let $(u_j)_j \subset \mathrm{W}^{1,p}_{Fx}(D_1;\mathbb{R}^m)$, $(v_j)_j \subset \mathrm{W}^{1,p}_{Fx}(D_2;\mathbb{R}^m)$

with $\nabla u_j \xrightarrow{\mathbf{Y}} \mu_1$, $\nabla v_j \xrightarrow{\mathbf{Y}} \mu_2$ and $(\nabla u_j)_j$, $(\nabla v_j)_j$ p-equiintegrable (if $p < \infty$), which can be constructed using Lemma 4.14. We define $(w_j)_j \subset \mathrm{W}^{1,p}_{Fx}(B_1;\mathbb{R}^m)$ through

$$w_j(x) := \begin{cases} u_j(x) & \text{if } x \in D_1, \\ v_j(x) & \text{if } x \in D_2, \end{cases}$$

which is a $\mathrm{W}^{1,p}$-bounded sequence. Thus, up to selecting a subsequence, we may assume that $\nabla w_j \xrightarrow{\mathbf{Y}} \nu \in \mathbf{GY}^p(B_1;\mathbb{R}^m)$.

For all continuous $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth (no growth condition if $p = \infty$) we find via the Fundamental Theorem in conjunction with the p-equiintegrability of $(\nabla w_j)_j$ (if $p < \infty$) that

$$\begin{aligned} \fint_{B_1} \int h \,\mathrm{d}\nu_x \,\mathrm{d}x &= \frac{1}{\omega_d} \cdot \lim_{j\to\infty} \left[\int_{D_1} h(\nabla u_j(x)) \,\mathrm{d}x + \int_{D_2} h(\nabla v_j(y)) \,\mathrm{d}y \right] \\ &= \theta \int h \,\mathrm{d}\mu_1 + (1-\theta) \int h \,\mathrm{d}\mu_2. \end{aligned}$$

Then use the averaging principle from Lemma 4.15 to get $\overline{\nu} \in \mathbf{GY}^p_{\mathrm{hom}}(F)$ with

$$\int h \,\mathrm{d}\overline{\nu} = \fint_{B_1} \int h \,\mathrm{d}\nu_x \,\mathrm{d}x = \theta \int h \,\mathrm{d}\mu_1 + (1-\theta) \int h \,\mathrm{d}\mu_2$$

and the convexity of $\mathbf{GY}^p_{\mathrm{hom}}(F)$ follows.

Step 2: τ_p-closedness. If $\mu \in \mathcal{M}_p(\mathbb{R}^{m\times d})$ lies in the (topological) τ_p-closure of $\mathbf{GY}^p_{\mathrm{hom}}(F)$, then we observe first that μ must be a probability measure. Indeed, the weak* topology on $\mathrm{C}_0(\mathbb{R}^{m\times d})^*$ is weaker than the τ_p-topology on $\mathcal{M}_p(\mathbb{R}^{m\times d})$, so μ is a positive measure. Moreover, $\mathbb{1} \in \mathbf{I}^p(\mathbb{R}^{m\times d})$, whereby we infer that $\mu \in \mathcal{M}^1(\mathbb{R}^{m\times d})$.

Take a countable collection $(h_k)_k \subset \mathrm{C}_0(\mathbb{R}^{m\times d}) \subset \mathbf{I}^p(\mathbb{R}^{m\times d})$ that determines Young measures (as in Lemma 4.8). Then, by the (topological) definition of closure for the τ_p-topology on $\mathcal{M}_p(\mathbb{R}^{m\times d})$, for all $j \in \mathbb{N}$ there exists a $\mu_j \in \mathbf{GY}^p_{\mathrm{hom}}(F)$ with

$$\left| \int h_k \,\mathrm{d}(\mu_j - \mu) \right| \leq \frac{1}{j} \qquad \text{for all } k \leq j.$$

For every μ_j we find $u_j \in \mathrm{W}^{1,p}(B_1;\mathbb{R}^m)$ with $\int_\Omega u_j \,\mathrm{d}x = 0$ such that

$$\left| \int_{B_1} h_k(\nabla u_j) \,\mathrm{d}x - \int_{B_1} \int h_k \,\mathrm{d}\mu_j \,\mathrm{d}x \right| \leq \frac{1}{j} \qquad \text{for all } k \leq j.$$

Additionally, we may assume, adding $h_0(A) := |A|^p$ to our collection $(h_k)_k$ and using the Poincaré inequality, that the sequence $(u_j)_j$ is uniformly $\mathrm{W}^{1,p}$-bounded (see the proof of Proposition 2.5 for a more precise argument). Hence, up to a subsequence, $\nabla u_j \xrightarrow{\mathbf{Y}} \nu \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$.

By the averaging principle, Lemma 4.15, we may furthermore assume that $\nu = \overline{\nu}$ is homogeneous. Let $(\nabla v_j)_j$ be the averaged generating sequence obtained from $(\nabla u_j)_j$ in the proof of that lemma. Since the integrands h_k are independent of x,

$$\left| \int_{B_1} h_k(\nabla v_j)\,\mathrm{d}x - \int_{B_1} \int h_k\,\mathrm{d}\mu\,\mathrm{d}x \right| \leq \frac{C}{j} \qquad \text{for all } k \leq j,$$

and so, letting $j \to \infty$,

$$\int h_k\,\mathrm{d}\nu = \int h_k\,\mathrm{d}\mu \qquad \text{for all } k \in \mathbb{N}.$$

Thus, we conclude that $\mu = \nu$ is a homogeneous gradient Young measure. □

Proof of Theorem 8.11. By Lemma 5.11 in conjunction with Proposition 5.13, only the sufficiency of (8.16) remains to be established.

Step 1. We first prove the result for homogeneous Young measures and $p \in (1, \infty)$, so let $\mu \in \mathbf{Y}^p(B_1; \mathbb{R}^{m\times d}) \subset \mathcal{M}^1(\mathbb{R}^{m\times d})$ be homogeneous with

$$h(F) \leq \int h\,\mathrm{d}\mu, \qquad F := [\mu], \tag{8.19}$$

for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth. Now, for any $g \in \mathbf{I}^p(\mathbb{R}^{m\times d})$ set $g_\alpha := \max\{g, \alpha\}$, $\alpha \in \mathbb{R}$. By Proposition 7.1 and the remark following it, we know that Qg_α is quasiconvex. Then, (8.19) implies

$$Qg(F) \leq Qg_\alpha(F) \leq \int Qg_\alpha\,\mathrm{d}\mu \leq \int g_\alpha\,\mathrm{d}\mu.$$

Hence, by the monotone convergence theorem,

$$Qg(F) \leq \int g\,\mathrm{d}\mu \tag{8.20}$$

since $g_\alpha \downarrow g$ as $\alpha \downarrow -\infty$ (this also uses the p-growth of g).

Assume that μ is not a gradient Young measure. From the preceding lemma we know that $\mathbf{GY}^p_{\mathrm{hom}}(F)$ is convex and τ_p-closed in $\mathcal{M}_p(\mathbb{R}^{m\times d})$. Then, applying the Hahn–Banach separation theorem in the version of Theorem A.5 in conjunction with the fact that $\mathbf{I}^p(\mathbb{R}^{m\times d})$ is the dual space of the locally convex topological vector space $(\mathcal{M}_p(\mathbb{R}^{m\times d}), \tau_p)$, there is a $g_* \in \mathbf{I}^p(\mathbb{R}^{m\times d})$ such that

$$\int g_*\,\mathrm{d}\mu = \langle g_*, \mu\rangle < \inf_{\nu\in\mathbf{GY}^p_{\mathrm{hom}}(F)} \langle g_*, \nu\rangle = \inf_{\nu\in\mathbf{GY}^p_{\mathrm{hom}}(F)} \int g_*\,\mathrm{d}\nu.$$

In particular, for $\psi \in \mathrm{W}^{1,\infty}_0(B_1; \mathbb{R}^m)$ we may test this with the homogeneous Young measure $\nu := \overline{\delta_{F+\nabla\psi}}$ originating from the Riemann–Lebesgue Lemma 4.16. So, by an

analogous argument to the proof of Proposition 7.1 we see that

$$\inf_{\nu\in\mathbf{GY}^p_{\mathrm{hom}}(F)} \int g_* \,\mathrm{d}\nu \le Qg_*(F).$$

In this context observe that while we do not have a quasiconvex lower bound $g_0 \le g_*$, the above infimum is a priori bounded from below. Moreover, our $g_* \in \mathbf{I}^p(\mathbb{R}^{m\times d})$ is continuous and satisfies an absolute p-growth bound. This suffices to run the arguments of Steps 1–3 in the proof of Proposition 7.1.

Consequently,

$$\int g_* \,\mathrm{d}\mu < Qg_*(F),$$

which contradicts (8.20).

Step 2. We now treat the inhomogeneous case for $p \in (1,\infty)$, but first assuming that $u = 0$ almost everywhere. So let $\nu = (\nu_x)_x \in \mathbf{Y}^p(\Omega;\mathbb{R}^{m\times d})$ satisfy the Jensen-type inequality (8.16) for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth. Take a countable collection $(\varphi_k \otimes h_k)_k$ as in Lemma 4.8. By the first step we have that $\nu_x \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$ for almost every $x \in \Omega$. Moreover, almost every $x \in \Omega$ is a Lebesgue point of the functions

$$x \mapsto \langle h_k, \nu_x\rangle \ \ (l \in \mathbb{N}) \qquad \text{and} \qquad x \mapsto \langle |\bullet|^p, \nu_x\rangle,$$

see Theorem A.21.

Fix $\varepsilon \in (0,1)$ and cover Ω up to a negligible set with a countable Vitali collection of disjoint balls $B_{r_n}(a_n)$, that is,

$$\Omega = Z \cup \bigcup_{n=1}^{\infty} B_{r_n}(a_n), \qquad |Z| = 0,$$

where $a_n \in \Omega$, $r_n > 0$ are such that (8.16) holds at $x = a_n$, and

$$\left| \fint_{B_{r_n}(a_n)} \langle h_k, \nu_y\rangle \,\mathrm{d}y - \langle h_k, \nu_{a_n}\rangle \right| \le \varepsilon \qquad \text{for all } k \in \mathbb{N}, \tag{8.21}$$

as well as

$$\left| \fint_{B_{r_n}(a_n)} \langle |\bullet|^p, \nu_y\rangle \,\mathrm{d}y - \langle |\bullet|^p, \nu_{a_n}\rangle \right| \le \varepsilon. \tag{8.22}$$

These estimates can be achieved by the Lebesgue point property of almost every $x \in \Omega$ and the fact that in the Vitali cover we may choose every radius $r_n > 0$ to be arbitrarily small.

For each $n \in \mathbb{N}$ take a generating sequence $(v_j^{(n)})_j \subset \mathrm{W}^{1,p}_0(B_1;\mathbb{R}^m)$ with $\nabla v_j^{(n)} \xrightarrow{\mathbf{Y}} \nu_{a_n}$, cf. Lemma 4.14. Define for $j \in \mathbb{N}$, the map $w_j^\varepsilon\colon B_1 \to \mathbb{R}^m$ via

$$w_j^\varepsilon(x) := r_n v_j^{(n)}\left(\frac{x-a_n}{r_n}\right) \qquad \text{if } x \in B_{r_n}(a_n)\ (n \in \mathbb{N}).$$

We then estimate for all $j \in \mathbb{N}$,

$$\begin{aligned}\int_\Omega |\nabla w_j^\varepsilon|^p \,\mathrm{d}x &= \sum_{n=1}^\infty \int_{B_{r_n}(a_n)} \left|\nabla v_j^{(n)}\left(\frac{x-a_n}{r_n}\right)\right|^p \,\mathrm{d}x \\ &= \sum_{n=1}^\infty r_n^d \int_{B_1} |\nabla v_j^{(n)}|^p \,\mathrm{d}y \\ &\le \sum_{n=1}^\infty |B_{r_n}(a_n)| \cdot \left(\left\langle |\bullet|^p, \nu_{a_n}\right\rangle + \varepsilon\right) \\ &\le \sum_{n=1}^\infty \int_{B_{r_n}(a_n)} \left\langle |\bullet|^p, \nu_y\right\rangle \,\mathrm{d}y + 2\varepsilon|\Omega| \\ &< \infty,\end{aligned}$$

where we discarded some leading elements of $(v_j^{(n)})_j$ for every $n \in \mathbb{N}$ and also used (8.22). By the Poincaré inequality from Theorem A.27 (i) we thus get that $(w_j^\varepsilon)_j$ is uniformly norm-bounded in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ and so, selecting a subsequence, we may assume that

$$\nabla w_j^\varepsilon \overset{\mathbf{Y}}{\rightharpoonup} \nu^\varepsilon \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d}).$$

By a similar calculation as above, this time also using the uniform continuity of φ_k and (8.21), we get for every $k \in \mathbb{N}$,

$$\begin{aligned}\left\langle\!\left\langle \varphi_k \otimes h_k, \nu^\varepsilon \right\rangle\!\right\rangle &= \lim_{j\to\infty} \int_\Omega \varphi_k(x) h_k(\nabla w_j^\varepsilon(x)) \,\mathrm{d}x \\ &= \lim_{j\to\infty} \sum_{n=1}^\infty r_n^d \int_{B_1} \varphi_k(a_n) h_k(\nabla v_j^{(n)}(y)) \,\mathrm{d}y + E(\varepsilon) \\ &= \sum_{n=1}^\infty |B_{r_n}(a_n)| \cdot \varphi_k(a_n) \cdot \left\langle h_k, \nu_{a_n}\right\rangle + E(\varepsilon) \\ &= \sum_{n=1}^\infty \varphi_k(a_n) \int_{B_{r_n}(a_n)} \left\langle h_k, \nu_y\right\rangle \,\mathrm{d}y + E(\varepsilon) \\ &= \int_\Omega \varphi_k(y) \left\langle h_k, \nu_y\right\rangle \,\mathrm{d}y + E(\varepsilon) \\ &= \left\langle\!\left\langle \varphi_k \otimes h_k, \nu \right\rangle\!\right\rangle + E(\varepsilon).\end{aligned}$$

Here, $E(\varepsilon)$ is an error term that may change from line to line and vanishes as $\varepsilon \downarrow 0$. Thus, as $\varepsilon \downarrow 0$ we get that

$$\nu^\varepsilon \overset{*}{\rightharpoonup} \nu \quad \text{in } \mathbf{Y}^p(\Omega;\mathbb{R}^{m\times d}),$$

that is,

$$\left\langle\!\left\langle f, \nu^\varepsilon \right\rangle\!\right\rangle \to \left\langle\!\left\langle f, \nu \right\rangle\!\right\rangle \qquad \text{for all } f \in \mathrm{C}_0(\Omega \times \mathbb{R}^N),$$

see (4.9). As all the ν^ε are gradient Young measures with

$$\sup_{j\in\mathbb{N}} \left\langle\!\left\langle |\bullet|^p, \nu^\varepsilon \right\rangle\!\right\rangle < \infty,$$

a diagonal argument (similar to Step 2 in the proof of Lemma 8.13) yields that ν is also a gradient Young measure.

Step 3. Let now $p \in (1,\infty)$ and let u not be identically zero but, without loss of generality, finite everywhere. Then define the **shifted Young measure** $\widehat{\nu} = (\widehat{\nu}_x)_x \in \mathbf{Y}^p(\Omega;\mathbb{R}^{m\times d})$ by $\widehat{\nu}_x := \nu_x \star \delta_{-\nabla u(x)}$, that is,

$$\int h \,\mathrm{d}\widehat{\nu}_x = \int h(A - \nabla u(x)) \,\mathrm{d}\nu_x \qquad \text{for all } h \in \mathrm{C}_0(\mathbb{R}^{m\times d}), \text{ a.e. } x \in \Omega.$$

We have $[\widehat{\nu}] = 0$ and the Jensen-type inequalities still hold for $\widehat{\nu}_x$ at almost every $x \in \Omega$: Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be quasiconvex and have p-growth. Then, $\widetilde{h}(A) := h(A - \nabla u(x))$ is also quasiconvex with p-growth, and so,

$$h([\widehat{\nu}_x]) = h([\nu_x] - \nabla u(x)) \le \int h(A - \nabla u(x)) \,\mathrm{d}\nu_x(A) = \int h \,\mathrm{d}\widehat{\nu}_x.$$

Thus, the previous step applies to $\widehat{\nu}$. Consequently, $\widehat{\nu} \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ and there is a sequence $(w_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ such that $\nabla w_j \xrightarrow{\mathbf{Y}} \widehat{\nu}$. Then, for the inversely shifted maps

$$u_j(x) := w_j(x) + u(x)$$

we have $\nabla u_j \xrightarrow{\mathbf{Y}} \widehat{\nu} \star \delta_{\nabla u(x)} = \nu$ and thus $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$. This finishes the proof for $p \in (1,\infty)$.

Step 4. For the sufficiency part of Theorem 8.11 in the case $p = \infty$, we simply apply the previous steps for one exponent $q \in (1,\infty)$. This yields that $\nu \in \mathbf{GY}^q(\Omega;\mathbb{R}^{m\times d})$. On the other hand, since $\nu \in \mathbf{Y}^\infty(\Omega;\mathbb{R}^{m\times d})$, there exists a compact set $K \subset \mathbb{R}^{m\times d}$ with $\operatorname{supp}\nu_x \subset K$ for almost every $x \in \Omega$, see the Fundamental Theorem 4.1. Zhang's lemma below then implies that $\nu \in \mathbf{GY}^\infty(\Omega;\mathbb{R}^{m\times d})$. □

Lemma 8.14 (Zhang 1992 [343]) *Let $\nu \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ for some $p \in (1,\infty)$ and suppose that there exists a compact set $K \subset \mathbb{R}^{m\times d}$ with*

$$\operatorname{supp}\nu_x \subset K \qquad \text{for a.e. } x \in \Omega.$$

Then, $\nu \in \mathbf{GY}^\infty(\Omega;\mathbb{R}^{m\times d})$. Moreover, there is a sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with

$$\operatorname{supp}(u_j - u) \Subset \Omega, \qquad \nabla u_j \xrightarrow{\mathbf{Y}} \nu.$$

and

$$\|\nabla u_j\|_{\mathrm{L}^\infty} \le C|K|_\infty, \qquad \text{where} \qquad |K|_\infty := \sup\big\{\,|A| \,:\, A \in K\,\big\},$$

where $C = C(d,m) > 0$ is a dimensional constant.

Proof. Without loss of generality we may assume that K is convex.

Step 1. We suppose first that $[\nu] = 0$ and thus that there is a sequence $(v_j)_j \subset \mathrm{W}^{1,p}(\mathbb{R}^d;\mathbb{R}^m)$ with $v_j \rightharpoonup 0$ in $\mathrm{W}^{1,p}$ and $\nabla v_j \xrightarrow{\mathbf{Y}} \nu$. Define, using the maximal operator M from Appendix A.6,

$$V_j := \mathrm{M}\big(|v_j| + |\nabla v_j|\big), \qquad j \in \mathbb{N},$$

and

$$G_j := \big\{ x \in \Omega \,:\, V_j(x) \le 8|K|_\infty \big\}.$$

Theorem A.38 (iii) implies that v_j is Lipschitz continuous on G_j with Lipschitz constant $C|K|_\infty$ for some dimensional constant $C = C(d,m) > 0$. Hence, by the Kirszbraun Theorem A.35, we may extend $v_j|_{G_j}$ to a Lipschitz function $u_j \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ without changing the Lipschitz constant. Consequently,

$$\|\nabla u_j\|_{\mathrm{L}^\infty} \le C|K|_\infty$$

for all $j \in \mathbb{N}$.

By the weak-type estimate on the maximal function, see Theorem A.38 (ii), we furthermore get

$$\begin{aligned} |\Omega \setminus G_j| &\le \big|\big\{ x \in \Omega \,:\, |\mathrm{M}v_j(x)| \ge 4|K|_\infty \big\}\big| \\ &\quad + \big|\big\{ x \in \Omega \,:\, |\mathrm{M}(\nabla v_j)(x)| \ge 4|K|_\infty \big\}\big| \\ &\le \frac{C}{|K|_\infty} \int_\Omega |v_j| \,\mathrm{d}x + \frac{C}{|K|_\infty} \int_{\{|\nabla v_j| \ge 2|K|_\infty\}} |\nabla v_j| \,\mathrm{d}x \\ &\le \frac{C}{|K|_\infty} \int_\Omega |v_j| \,\mathrm{d}x + \frac{C}{|K|_\infty} \int_\Omega g(|\nabla v_j|) \,\mathrm{d}x, \end{aligned} \tag{8.23}$$

where $g\colon [0,\infty) \to [0,\infty)$ is given as

$$g(s) := \begin{cases} 0 & \text{if } s \le |K|_\infty, \\ 2(s - |K|_\infty) & \text{if } |K|_\infty < s < 2|K|_\infty, \\ s & \text{if } s \ge 2|K|_\infty. \end{cases}$$

The first term in (8.23) tends to zero since $v_j \to 0$ in L^1 by the compact embedding from $\mathrm{W}^{1,p}$ into L^1. By the Young measure representation (note that $(g(|\nabla v_j|))_j$ is uniformly L^p-bounded, hence (1-)equiintegrable), for the second term we obtain

$$\int_\Omega g(|\nabla v_j|) \,\mathrm{d}x \to \int_\Omega \int g(|A|) \,\mathrm{d}\nu_x(A) \,\mathrm{d}x = 0$$

as $j \to \infty$. The equality on the right-hand side holds since $\operatorname{supp} \nu_x \subset K$ for almost every $x \in \Omega$. Therefore, for all $\varphi \in \mathrm{C}_0(\Omega)$ and $h \in \mathrm{C}_0(\mathbb{R}^m)$,

$$\int_\Omega |\varphi h(\nabla u_j) - \varphi h(\nabla v_j)| \,\mathrm{d}x \le \|\varphi\|_\infty \cdot \|h\|_\infty \cdot |\Omega \setminus G_j| \to 0,$$

and we may conclude that also the uniformly $\mathrm{W}^{1,\infty}$-norm bounded sequence $(\nabla u_j)_j$ generates ν, which therefore has been shown to lie in $\mathbf{GY}^\infty(\Omega;\mathbb{R}^{m\times d})$.

Step 2. If $[\nu] = \nabla u \neq 0$ for some $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$, we consider the shifted Young measure $\widehat{\nu} = (\widehat{\nu}_x)_x \in \mathbf{Y}^p(\Omega;\mathbb{R}^{m\times d})$ defined via $\widehat{\nu}_x := \nu_x \star \delta_{-\nabla u(x)}$, that is,

$$\int h \,\mathrm{d}\widehat{\nu}_x = \int h(A - \nabla u(x)) \,\mathrm{d}\nu_x \qquad \text{for all } h \in \mathrm{C}_0(\mathbb{R}^{m\times d}).$$

Then, since $\nabla u \in K$ almost everywhere (as we assumed that K is convex) we have

$$\operatorname{supp} \widehat{\nu}_x \subset K - K := \left\{ A - B \,:\, A, B \in K \right\} \subset B_{2|K|_\infty}.$$

Thus, the first step applies to $\widehat{\nu}$ and yields a sequence $(\widehat{u}_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with $\nabla \widehat{u}_j \xrightarrow{\mathbf{Y}} \widehat{\nu} \in \mathbf{GY}^\infty(\Omega;\mathbb{R}^{m\times d})$ and $\|\nabla \widehat{u}_j\|_{\mathrm{L}^\infty} \leq 2C|K|_\infty$. Setting

$$u_j := \widehat{u}_j + u,$$

we get $\|\nabla u_j\|_{\mathrm{L}^\infty} \leq (2C+1)|K|_\infty$ and $\nabla u_j \xrightarrow{\mathbf{Y}} \nu \in \mathbf{GY}^\infty(\Omega;\mathbb{R}^{m\times d})$. This concludes the proof.

Step 3. The only part we have not shown so far is that u_j can be adjusted so that $\operatorname{supp}(u_j - u) \Subset \Omega$. We omit the proof since it is the same as the standard localization argument in Step 3 of the proof of Lemma 4.14. One only needs to observe that adjusting a generating sequence near the boundary of Ω can be achieved without increasing the Lipschitz constant by more than a dimensional factor. □

Remark 8.15 Zhang's result as stated here is far from sharp. A refined version shows that the constant C can be chosen arbitrarily close to 1 and for the sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$ one can achieve $\operatorname{dist}(\nabla u_j, K) \to 0$ in L^∞. This is proved in [247].

Notes and Historical Remarks

A more combinatorial presentation of Švérak's counterexample, Theorem 8.4, due to James, can be found in Example 11.11; also see Section 4.7 of [248]. Recently, a very interesting algebraic approach, which yields a new class of rank-one convex but not quasiconvex functions, was developed in [163]. Research on the two-dimensional case is ongoing and only partial results are known, see for instance [246] for the case of diagonal matrices and the recent [172] for a conditional result in the isotropic case. The works [24, 25, 163] contain further recent developments.

The Murat–Tartar Div-Curl Lemma 8.8 was first published in [254], but established four years before in 1974. Another proof of this fundamental result using elliptic regularity theory can be found in Theorem 16.2.1 of [97].

Our setup for the proof of the Kinderlehrer–Pedregal Theorem 8.11 is inspired by the proof in [209] for the characterization of BV-Young measures (stated in Theorem 15.19).

The truncation argument used in Zhang's Lemma 8.14 seems to be originally due to Acerbi–Fusco [1, 3]. The book [220] makes use of this technique in regularity theory and also contains several refinements.

Problems

8.1 Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be twice continuously differentiable. Show that then h is rank-one convex if and only if h satisfies the Legendre–Hadamard condition, that is,

$$\begin{aligned} \mathrm{D}^2 h(A)[a\otimes b, a\otimes b] &= \frac{\mathrm{d}^2}{\mathrm{d}t^2}\, h(A + ta\otimes b)\Big|_{t=0} \\ &= \sum_{i,k=1}^{m}\sum_{j,l=1}^{d} \frac{\partial^2 h(A)}{\partial A^i_j \partial A^k_l} a^i b^j a^k b^l \\ &\geq 0 \end{aligned}$$

for all $A \in \mathbb{R}^{m\times d}$ and all $a \in \mathbb{R}^m$, $b \in \mathbb{R}^d$.

8.2 Let "$\mathcal{A} := (\mathrm{div}, \mathrm{curl})$", which needs to be defined properly in the sense of (8.4). Then compute that

$$\ker \Lambda_{\mathcal{A}} = \big\{ (a,b) \in \mathbb{R}^d \times \mathbb{R}^d \,:\, a \perp b \big\}.$$

Also show that $q(a,b) := a \cdot b$ can be written as a quadratic form.

8.3 Show that $\mathbf{I}^p(\mathbb{R}^{m\times d})$ (defined in (8.18)) is isomorphic to the (separable) space $\mathrm{C}(\alpha\mathbb{R}^{m\times d})$, where $\alpha\mathbb{R}^{m\times d}$ is the Alexandrov (one-point) compactification of $\mathbb{R}^{m\times d}$, or, equivalently, $\mathbf{I}^p(\mathbb{R}^{m\times d})$ is isometrically isomorphic to the set of all $\varphi \in \mathrm{C}(\overline{\mathbb{B}^{m\times d}})$ with $\varphi|_{\partial\mathbb{B}^{m\times d}}$ constant, where $\mathbb{B}^{m\times d}$ denotes the open unit ball in $\mathbb{R}^{m\times d}$ (with respect to the Frobenius norm). Conclude that $\mathbf{I}^p(\mathbb{R}^{m\times d})$ is separable.

8.4 Show that in the Kinderlehrer–Pedregal Theorem 8.11 in the case $p \in (1,\infty)$ it suffices to verify (8.16) for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ that are bounded from below and for which

$$\lim_{|A|\to\infty} \frac{h(A)}{1+|A|} \quad \text{exists (in } \mathbb{R}).$$

8.5 Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and $F \in \mathbb{R}^{m\times d}$ with $\mathrm{rank}\, F = 1$. Consider the functional

$$\mathcal{F}[u] := \int_\Omega \mathrm{dist}(\nabla u(x), \{-F, F\})^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m).$$

Construct a sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with $u_j \overset{*}{\rightharpoonup} 0$ in $\mathrm{W}^{1,\infty}$ and $\mathcal{F}[u_j] = 0$ for all $j \in \mathbb{N}$. Conclude that $\mathcal{F}$ is not lower semicontinuous with respect to weak* convergence in $\mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$.

8.6 In the situation of Problem 8.5 now assume that $\operatorname{rank} F \geq 2$ and that Ω is convex. Is the resulting functional $\mathcal{F}$ weakly lower semicontinuous?

8.7 Show that the integrand in the integral functional from Problem 8.5 is not rank-one convex.

8.8 Take the rank-one convex but not quasiconvex function h from the proof of Šverák's Theorem 8.4 (which has 4-growth). Define

$$\mathcal{F}[u] := \int_{(0,1)^2} h(\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,4}((0,1)^2;\mathbb{R}^3).$$

Show that there exists a sequence of maps $(u_j) \subset \mathrm{W}^{1,4}((0,1)^2;\mathbb{R}^3)$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,4}$ and such that $\mathcal{F}[u] > \liminf_{j\to\infty} \mathcal{F}[u_j]$.

8.9 Let $K \subset \mathbb{R}^{m\times d}$ be compact and non-empty. Also assume there is a norm-bounded sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, $p \in (1,\infty)$ with $\operatorname{dist}(\nabla u_j, K) \to 0$ in measure. Show that then there exists a sequence $(v_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ such that also $\operatorname{dist}(\nabla v_j, K) \to 0$ in measure. Conclude that (ii) in the Ball–James rigidity theorem continues to hold for sequences that converge weakly in $\mathrm{W}^{1,p}$, where $p \in (1,\infty)$. *Hint:* Use an argument like in the proof of Zhang's Lemma 8.14.

8.10 Prove the claim of Remark 5.14.



Chapter 9
Γ-Convergence

Sometimes, a functional of interest depends on the value of a parameter, say a small $\varepsilon > 0$, as we have seen with the functionals $\mathcal{F}_\varepsilon$ from the example on composite elastic materials in Section 1.10. In such problems, the goal usually lies not in minimizing $\mathcal{F}_\varepsilon$ for one particular value of ε, but in determining the asymptotic limit of the minimization problems as $\varepsilon \downarrow 0$. Concretely, we need to identify, if possible, a *limit functional* $\mathcal{F}_0$ such that (approximate) minimizers of the $\mathcal{F}_\varepsilon$ converge to a minimizer of $\mathcal{F}_0$ as $\varepsilon \downarrow 0$. The idea here is that we believe $\mathcal{F}_\varepsilon$ for some small $\varepsilon > 0$ to be the "true" model and $\mathcal{F}_0$ to be the "effective" model in the limit $\varepsilon \downarrow 0$.

A prototypical situation is the following *periodic homogenization problem*: Let the integrand $f\colon \mathbb{R}^d \times \mathbb{R}^{m\times d} \to \mathbb{R}$ be periodic in the first argument, say $f(x + z, A) = f(x, A)$ for all $x \in \mathbb{R}^d$, $z \in \mathbb{Z}^d$, $A \in \mathbb{R}^{m\times d}$, and consider

$$\mathcal{F}_\varepsilon[u] := \int_\Omega f\Big(\frac{x}{\varepsilon}, \nabla u(x)\Big)\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m).$$

Minimizers (or approximate minimizers) of $\mathcal{F}_\varepsilon$ usually display oscillations with length scale ε in order to accommodate the oscillations of the integrand. Naturally, we expect these oscillations to "average out" as $\varepsilon \downarrow 0$. Thus, the limit functional may be conjectured not to depend on x. On the other hand, it is reasonable to expect that the integrand of the limit functional reflects directional properties of the allowed oscillations such as anisotropy and the averaging via the harmonic mean, not the arithmetic mean, in some directions.

The theory of Γ-convergence, which was introduced by Ennio De Giorgi in the 1970s, is a powerful framework in which one can investigate a large variety of such questions. Its basic notion is a convergence of functionals that is compatible with the Direct Method. In particular, under an appropriate uniform coercivity hypothesis, Γ-convergence entails the convergence of the infimal values of the functionals $\mathcal{F}_\varepsilon$ to the minimum value of the limit functional $\mathcal{F}_0$ and the convergence of minimizers (or approximate minimizers) of the $\mathcal{F}_\varepsilon$ to a minimizer of $\mathcal{F}_0$.

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9.1 Abstract Γ-Convergence

Let X be a complete metric space. The functional $\mathcal{F}_\infty : X \to \mathbb{R}$ is called the **(sequential) Γ-limit** of the functionals $\mathcal{F}_k : X \to \mathbb{R}$, $k \in \mathbb{N}$, if the following two conditions are satisfied:

(H1) For all sequences $(u_k)_k \subset X$ with $u_k \to u$ in X, the **lim inf-inequality** holds:

$$\mathcal{F}_\infty[u] \le \liminf_{k\to\infty} \mathcal{F}_k[u_k].$$

(H2) For all $u \in X$ there exists a **recovery sequence** $(u_k)_k \subset X$, that is, $u_k \to u$ in X and

$$\mathcal{F}_\infty[u] = \lim_{k\to\infty} \mathcal{F}_k[u_k].$$

The Γ-limit of the sequence $(\mathcal{F}_k)_k$, if it exists, is uniquely determined (see Problem 9.2) and is denoted by $\Gamma\text{-}\lim_k \mathcal{F}_k$.

One observes that, assuming the first condition, the second condition in the definition of Γ-convergence can be replaced by the following:

(H2') For all $u \in X$ there exists a sequence $u_k \to u$ in X such that the **lim sup-inequality** holds:

$$\mathcal{F}_\infty[u] \ge \limsup_{k\to\infty} \mathcal{F}_k[u_k].$$

We first note that pointwise convergence does not in general imply Γ-convergence:

Example 9.1 On $X = \mathbb{R}$ define for $k \in \mathbb{N}$,

$$\mathcal{F}_k[x] := -\delta_{-1/k}(x) + \delta_{1/k}(x) = \begin{cases} \pm 1 & \text{if } x = \pm 1/k, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$\mathcal{F}_\infty[x] := -\delta_0(x) = \begin{cases} -1 & \text{if } x = 0, \\ 0 & \text{otherwise.} \end{cases}$$

We claim that $\Gamma\text{-}\lim_k \mathcal{F}_k = \mathcal{F}_\infty$. Indeed, on each open set $U \Subset \mathbb{R} \setminus \{0\}$, the functions $\mathcal{F}_k$ converge to the zero function uniformly. Together with the lower semicontinuity of $\mathcal{F}_\infty$, this implies Γ-convergence on $\mathbb{R} \setminus \{0\}$ (see Problem 9.1). For $x = 0$ we have $\Gamma\text{-}\lim \mathcal{F}_k[0] = -1$ by using the recovery sequence $x_k := -1/k$ (the lim inf-inequality is trivial).

This example also shows that the Γ-limit does not necessarily coincide with the pointwise limit (which for our $\mathcal{F}_k$ is the zero function). In fact, even for the constant sequence of functionals $(\mathcal{F}_1)_k$, the Γ-limit is $-\delta_{-1} \ne \mathcal{F}_1$, as can be easily verified. One can also show that $\Gamma\text{-}\lim_k(-\mathcal{F}_k) = \mathcal{F}_\infty$, hence $\Gamma\text{-}\lim_k(-\mathcal{F}_k) \ne -\Gamma\text{-}\lim_k \mathcal{F}_k$ and consequently, taking the Γ-limit is not a linear operation.

Despite these counterintuitive properties, the theory of Γ-convergence is very useful and widely used in the Calculus of Variations. One of the reasons for this is the following fact.

Proposition 9.2 *$\mathcal{F}_\infty = \Gamma\text{-}\lim_k \mathcal{F}_k$, if it exists, is (sequentially) lower semicontinuous.*

Proof. We need to prove $\mathcal{F}_\infty[u] \le \liminf_{j\to\infty} \mathcal{F}_\infty[u_j]$ for all $u_j \to u$ in X. For every u_j choose a recovery sequence $(u_k^{(j)})_k$, i.e., $u_k^{(j)} \to u_j$ as $k \to \infty$ and $\mathcal{F}_\infty[u_j] = \lim_{k\to\infty} \mathcal{F}_k[u_k^{(j)}]$. Next, denoting the metric in X by d, we choose a strictly increasing sequence of indices $(k(j))_j$ such that

$$d\big(u_{k(j)}^{(j)}, u_j\big) \le \frac{1}{j} \qquad \text{and} \qquad \big|\mathcal{F}_{k(j)}(u_{k(j)}^{(j)}) - \mathcal{F}_\infty[u_j]\big| \le \frac{1}{j}.$$

Then set

$$\widetilde{u}_k := \begin{cases} u_{k(j)}^{(j)}, & \text{if } k = k(j) \text{ for some } j \in \mathbb{N}, \\ u, & \text{otherwise.} \end{cases}$$

We have that $\widetilde{u}_k \to u$ and the lim inf-inequality implies

$$\mathcal{F}_\infty[u] \le \liminf_{k\to\infty} \mathcal{F}_k[\widetilde{u}_k] \le \liminf_{j\to\infty} \mathcal{F}_{k(j)}[u_{k(j)}^{(j)}] = \liminf_{j\to\infty} \mathcal{F}_\infty[u_j].$$

Thus we have shown the (sequential) lower semicontinuity of $\mathcal{F}_\infty$. □

As a consequence of the preceding proposition, the notion of Γ-convergence cannot be generated by a topology, because any constant sequence of a non-lower semicontinuous function has a different Γ-limit (cf. Example 9.1). It turns out, however, that in spaces of lower semicontinuous functions, Γ-convergence is in fact generated by a topology, see Chapter 10 of [98].

The most important property of Γ-convergence is that it entails the convergence of minima (or infima) and of the corresponding minimizers (or approximate minimizers) if the following notion of uniform coercivity is satisfied: A family $(\mathcal{F}_k)_k$ of functionals $\mathcal{F}_k : X \to \mathbb{R}$ is called **equicoercive** if there exists a compact set $K \subset X$ with the property that

$$\inf_X \mathcal{F}_k = \inf_K \mathcal{F}_k \qquad \text{for all } k \in \mathbb{N}.$$

Clearly, without equicoercivity no convergence of minima or minimizers can be expected, as the sequence $\mathcal{F}_k := -\delta_k$ ($k \in \mathbb{N}$) on the space $X := \mathbb{R}$ shows.

Theorem 9.3 *Let $\mathcal{F}_k : X \to \mathbb{R}$, $k \in \mathbb{N}$, be equicoercive functionals and assume that $\mathcal{F}_\infty = \Gamma\text{-}\lim_k \mathcal{F}_k$ exists. Then, $\mathcal{F}_\infty$ has a minimizer and*

$$\min_X \mathcal{F}_\infty = \lim_{k\to\infty} \inf_X \mathcal{F}_k.$$

In addition, all accumulation points of any precompact sequence $(u_k)_k \subset X$ with the property that $\lim_{k\to\infty} \mathcal{F}_k[u_k] = \lim_{k\to\infty} \inf_X \mathcal{F}_k$ are minimizers of $\mathcal{F}_\infty$.

Proof. Denote by K the compact set from the equicoercivity of the $\mathcal{F}_k$. For all $k \in \mathbb{N}$ choose $u_k \in K$ such that $|\mathcal{F}_k[u_k] - \inf_X \mathcal{F}_k| \le 1/k$. Then,

$$\liminf_{k\to\infty} \mathcal{F}_k[u_k] = \liminf_{k\to\infty} \inf_X \mathcal{F}_k.$$

Because all the u_k lie in the compact set K, we can select a subsequence $(u_{k(j)})_j$ with $u_{k(j)} \to u_* \in X$ as $j \to \infty$ and

$$\lim_{j\to\infty} \mathcal{F}_{k(j)}[u_{k(j)}] = \liminf_{k\to\infty} \inf_X \mathcal{F}_k.$$

Then define the sequence $(\widetilde{u}_k)_k$ as follows:

$$\widetilde{u}_k := \begin{cases} u_{k(j)} & \text{if } k = k(j) \text{ for some } j \in \mathbb{N}, \\ u_* & \text{otherwise.} \end{cases}$$

We have $\widetilde{u}_k \to u_*$ and, by the lim inf-inequality,

$$\inf_X \mathcal{F}_\infty \le \mathcal{F}_\infty[u_*] \le \liminf_{k\to\infty} \mathcal{F}_k[\widetilde{u}_k] \le \lim_{j\to\infty} \mathcal{F}_{k(j)}[u_{k(j)}] = \liminf_{k\to\infty} \inf_X \mathcal{F}_k. \tag{9.1}$$

Fix $\varepsilon > 0$. Let $u \in X$ be such that $\mathcal{F}_\infty[u] \le \inf_X \mathcal{F}_\infty + \varepsilon$ and take a recovery sequence $u_k \to u$ for u. We estimate

$$\limsup_{k\to\infty} \inf_X \mathcal{F}_k \le \lim_{k\to\infty} \mathcal{F}_k[u_k] = \mathcal{F}_\infty[u] \le \inf_X \mathcal{F}_\infty + \varepsilon.$$

Let then $\varepsilon \downarrow 0$ to get

$$\limsup_{k\to\infty} \inf_X \mathcal{F}_k \le \inf_X \mathcal{F}_\infty. \tag{9.2}$$

Combining (9.1) and (9.2), the first assertion of the theorem follows.

The second claim is clear if we take a sequence converging to a given accumulation point in the argument leading up to (9.1). □

For some situations it is furthermore useful to introduce the following two finer notions: The **(sequential) Γ-lower limit** $\Gamma\text{-}\liminf_k \mathcal{F}_k$ and the **(sequential) Γ-upper limit** $\Gamma\text{-}\limsup_k \mathcal{F}_k$ of the sequence of functionals $\mathcal{F}_k \colon X \to \mathbb{R}$, $k \in \mathbb{N}$, are, respectively,

$$\Gamma\text{-}\liminf_k \mathcal{F}_k[u] := \inf \left\{ \liminf_{k\to\infty} \mathcal{F}_k[u_k] \ : \ u_k \to u \text{ in } X \right\},$$
$$\Gamma\text{-}\limsup_k \mathcal{F}_k[u] := \inf \left\{ \limsup_{k\to\infty} \mathcal{F}_k[u_k] \ : \ u_k \to u \text{ in } X \right\},$$

where $u \in X$.

The following lemma gives us an indirect way to establish that a sequence of functionals Γ-converges.

Lemma 9.4 *For $\mathcal{F}_k, \mathcal{F}_\infty \colon X \to \mathbb{R}$, $k \in \mathbb{N}$, it holds that $\mathcal{F}_\infty = \Gamma\text{-}\lim_k \mathcal{F}_k$ if and only if $\Gamma\text{-}\liminf_k \mathcal{F}_k = \Gamma\text{-}\limsup_k \mathcal{F}_k = \mathcal{F}_\infty$.*

Proof. Assume first that $\Gamma\text{-}\liminf_k \mathcal{F}_k = \Gamma\text{-}\limsup_k \mathcal{F}_k = \mathcal{F}_\infty$. Then, for all $u_k \to u$ in X we have

$$\mathcal{F}_\infty[u] = \Gamma\text{-}\liminf_k \mathcal{F}_k[u] \le \liminf_{k\to\infty} \mathcal{F}_k[u_k],$$

that is, the lim inf-inequality holds. Moreover, for $u \in X$ and $n \in \mathbb{N}$ choose a sequence $(u_k^{(n)})_k$ with $u_k^{(n)} \to u$ as $k \to \infty$ and

$$\mathcal{F}_\infty[u] \geq \limsup_{k\to\infty} \mathcal{F}_k[u_k^{(n)}] - \frac{1}{n}.$$

It is always possible to find such a sequence since $\mathcal{F}_\infty[u] = \Gamma\text{-}\limsup_k \mathcal{F}_k[u]$. Now we inductively combine the sequences $(u_k^{(n)})_k$ into one sequence $\widetilde{u}_k$ as follows: Let $(K(n))_n$ denote a growing sequence of indices such that

$$d(u_k^{(n)}, u) \leq \frac{1}{n} \quad \text{and} \quad \mathcal{F}_\infty[u] \geq \mathcal{F}_k[u_k^{(n)}] - \frac{2}{n} \qquad \text{for all } k \geq K(n),$$

which exists by the definition of the upper limit. Then define

$$(\widetilde{u}_k)_k := \left(u_{K(1)}^{(1)}, u_{K(1)+1}^{(1)}, \dots, u_{K(2)}^{(2)}, u_{K(2)+1}^{(2)}, \dots, u_{K(l)}^{(l)}, u_{K(l)+1}^{(l)}, \dots\right)$$

and observe that $\widetilde{u}_k \to u$ as well as

$$\mathcal{F}_\infty[u] \geq \limsup_{k\to\infty} \mathcal{F}_k[\widetilde{u}_k].$$

This proves the lim sup-inequality and hence the existence of a recovery sequence.

For the other direction assume $\mathcal{F}_\infty = \Gamma\text{-}\lim_k \mathcal{F}_k$. It is easy to see that for all $u \in X$ it holds that

$$\mathcal{F}_\infty[u] \leq \Gamma\text{-}\liminf_k \mathcal{F}_k[u] \leq \Gamma\text{-}\limsup_k \mathcal{F}_k[u] \leq \mathcal{F}_\infty[u],$$

where the last inequality follows since recovery sequences are admissible in the definition of the Γ-upper limit. Thus, $\Gamma\text{-}\liminf_k \mathcal{F}_k = \Gamma\text{-}\limsup_k \mathcal{F}_k = \mathcal{F}_\infty$. □

Recall from Chapter 7 that the relaxation $\mathcal{F}_*\colon X \to \mathbb{R}\cup\{-\infty\}$ of a functional $\mathcal{F}\colon X \to \mathbb{R}$ is defined via

$$\mathcal{F}_*[u] := \sup\left\{ \mathcal{H}[u] \;:\; \mathcal{H}\colon X \to \mathbb{R} \text{ is lower semicontinuous and } \mathcal{H} \leq \mathcal{F} \right\},$$

where $u \in X$; as before we set $\sup\emptyset := -\infty$. The relaxation can also be expressed via Γ-convergence as follows:

Proposition 9.5 *It holds that $\mathcal{F}_* = \Gamma\text{-}\lim \mathcal{F}$, where $\Gamma\text{-}\lim \mathcal{F}$ denotes the Γ-limit of the constant sequence $(\mathcal{F})_k$, with the definition of the Γ-limit extended to also allow the value $-\infty$.*

Proof. First, we show that $\Gamma\text{-}\lim \mathcal{F}$ always exists (with values in $\mathbb{R}\cup\{-\infty\}$). Indeed, by the definition of the Γ-lower limit we can for all $u \in X$ and all $\varepsilon > 0$ find a sequence $u_k \to u$ such that

$$\liminf_{k\to\infty} \mathcal{F}[u_k] \leq \Gamma\text{-}\liminf_k \mathcal{F}[u] + \varepsilon < \infty.$$

Up to taking a subsequence of the u_k's, we may also assume that

$$\lim_{k\to\infty} \mathcal{F}[u_k] = \liminf_{k\to\infty} \mathcal{F}[u_k] \in [-\infty, \infty).$$

Using this sequence in the definition of the Γ-upper limit and letting $\varepsilon \to 0$, we conclude that $\Gamma\text{-}\limsup_k \mathcal{F}[u] \le \Gamma\text{-}\liminf_k \mathcal{F}[u]$. Then, Lemma 9.4 shows that $\Gamma\text{-}\lim_k \mathcal{F}$ exists.

For any lower semicontinuous function $\mathcal{H} \le \mathcal{F}$, we have $\mathcal{H} = \Gamma\text{-}\lim \mathcal{H}$ since for all $u_k \to u$ it holds that

$$\begin{aligned}\mathcal{H}[u] &\le \inf\left\{ \liminf_{k\to\infty} \mathcal{H}[u_k] \;:\; u_k \to u \right\} \\ &= \Gamma\text{-}\liminf_k \mathcal{H}[u] \\ &\le \Gamma\text{-}\limsup_k \mathcal{H}[u] \\ &\le \mathcal{H}[u]\end{aligned}$$

since the constant sequence is admissible in the definition of the Γ-upper limit. Thus $\mathcal{H} = \Gamma\text{-}\lim \mathcal{H}$ by Lemma 9.4. Consequently,

$$\mathcal{H} = \Gamma\text{-}\lim \mathcal{H} \le \Gamma\text{-}\lim \mathcal{F}.$$

Taking the supremum over all such $\mathcal{H}$, we obtain $\mathcal{F}_* \le \Gamma\text{-}\lim \mathcal{F}$.

To prove the converse inequality, note that $\Gamma\text{-}\lim \mathcal{F}$ is lower semicontinuous by Proposition 9.2 and $\Gamma\text{-}\lim \mathcal{F} \le \mathcal{F}$ (see above). Then it follows immediately from the definition of $\mathcal{F}_*$ that also $\Gamma\text{-}\lim \mathcal{F} \le \mathcal{F}_*$. □

9.2 Periodic Homogenization

We now turn to the investigation of highly-oscillatory integrands. Define for all $\varepsilon > 0$ the functional

$$\mathcal{F}_\varepsilon[u] := \int_\Omega f\Big(\frac{x}{\varepsilon}, \nabla u(x)\Big)\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

where $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain and $f\colon \mathbb{R}^d \times \mathbb{R}^{m\times d} \to [0,\infty)$ is a Carathéodory integrand that satisfies the following standard p-growth and p-coercivity assumption

$$\mu|A|^p \le f(x,A) \le M(1+|A|^p), \qquad (x,A) \in \mathbb{R}^d \times \mathbb{R}^{m\times d}, \tag{9.3}$$

for some $p \in (1,\infty)$ and constants $\mu, M > 0$. To enable slightly shorter arguments and without loss of generality, the assumed lower bound here does not contain an additive constant. We also assume the *periodicity condition*

$$f(x+z,A) = f(x,A) \qquad \text{for all } x \in \mathbb{R}^d,\ z \in \mathbb{Z}^d,\ A \in \mathbb{R}^{m\times d},$$

as well as the *local Lipschitz condition*

$$|f(x,A) - f(x,B)| \le C(1+|A|^{p-1}+|B|^{p-1})|A-B| \tag{9.4}$$

for all $x \in \Omega$, $A, B \in \mathbb{R}^{m\times d}$ and some constant $C > 0$.

The main result of this section is the following *homogenization theorem*:

Theorem 9.6 (Braides 1985, Müller 1987 [61, 244]) *The functionals $\mathcal{F}_\varepsilon$ Γ-converge with respect to the weak $\mathrm{W}^{1,p}$-convergence to the functional*

$$\mathcal{F}_0[u] := \int_\Omega f_{\mathrm{hom}}(\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m),$$

*as $\varepsilon \downarrow 0$. Here, $f_{\mathrm{hom}}\colon \mathbb{R}^{m\times d} \to [0,\infty)$ is the **homogenized integrand**, given by the **asymptotic homogenization formula***

$$f_{\mathrm{hom}}(A) := \inf_{k\in\mathbb{N}} \inf_{\psi\in \mathrm{W}^{1,p}_{\mathrm{per}}((0,k)^d;\mathbb{R}^m)} \fint_{(0,k)^d} f(x, A+\nabla\psi(x))\,\mathrm{d}x,$$

where $A \in \mathbb{R}^{m\times d}$. Moreover, in the definition of $\mathcal{F}_0$ the integrand f_{hom} may equivalently be replaced by

$$f_{\mathrm{hom},0}(A) := \inf_{k\in\mathbb{N}} \inf_{\psi\in \mathrm{W}^{1,p}_{0}((0,k)^d;\mathbb{R}^m)} \fint_{(0,k)^d} f(x, A+\nabla\psi(x))\,\mathrm{d}x.$$

Above, $\mathrm{W}^{1,p}_{\mathrm{per}}((0,k)^d;\mathbb{R}^m)$ is the $\mathrm{W}^{1,p}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^m)$-closure of smooth k-periodic maps (meaning that $u(x+kz) = u(x)$ for $x \in \mathbb{R}^d$, $z \in \mathbb{Z}^d$).

Note that the formula for $f_{\mathrm{hom},0}$ does *not* agree with the formula (7.1) for the quasiconvex envelope because the first argument of f is not held fixed. Indeed, it is intuitively evident that the optimal ψ needs to take into account how f varies in its first argument.

Remark 9.7 The local Lipschitz condition (9.4) is in fact not needed if one follows a more involved proof, see [61, 63]. However, we showed in Lemma 5.9 that (9.4) holds if $f(x,\bullet)$ is quasiconvex for all x, which is necessary for weak lower semicontinuity by Proposition 5.18. So, unless we want to consider non-quasiconvex integrands, assuming (9.4) is no restriction.

We will establish this theorem via several lemmas. For technical reasons we first consider $\mathcal{F}_0$ to be defined with $f_{\mathrm{hom},0}$ in place of f_{hom} and then show that $f_{\mathrm{hom},0} = f_{\mathrm{hom}}$.

Lemma 9.8 *Let $F \in \mathbb{R}^{m\times d}$. Then, there exists $(u_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}_{Fx}(\Omega;\mathbb{R}^m)$ with $u_\varepsilon \rightharpoonup Fx$ in $\mathrm{W}^{1,p}$ as $\varepsilon \downarrow 0$ and*

$$\mathcal{F}_0[Fx] = \lim_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon].$$

Proof. Fix $\delta > 0$ and choose $\psi_\delta \in \mathrm{W}^{1,p}_0((0,k)^d;\mathbb{R}^m)$ for some $k \in \mathbb{N}$ (depending on δ) such that

$$f_{\mathrm{hom},0}(F) \le \fint_{(0,k)^d} f(y, F+\nabla\psi_\delta(y))\,\mathrm{d}y \le f_{\mathrm{hom},0}(F) + \delta. \tag{9.5}$$

For $\eta > 0$ we denote by $\Omega^\eta \subset \Omega$ the union of all cubes from the lattice of open cubes $(0,\eta)^d + \eta\mathbb{Z}^d$ that are wholly contained in Ω. Consider ψ_δ to be extended to all of $\mathbb{R}^d$ by periodicity and define $u_{\delta,\varepsilon}\colon \Omega \to \mathbb{R}^m$ via

$$u_{\delta,\varepsilon}(x) := \begin{cases} Fx + \varepsilon\psi_\delta\left(\frac{x}{\varepsilon}\right) & \text{if } x \in \Omega^{\varepsilon k}, \\ Fx & \text{if } x \in \Omega \setminus \Omega^{\varepsilon k}. \end{cases}$$

A straightforward calculation shows that $u_{\delta,\varepsilon} \in \mathrm{W}^{1,p}_{Fx}(\Omega;\mathbb{R}^m)$ and $u_{\delta,\varepsilon} \to Fx$ in L^p as $\varepsilon \downarrow 0$.

For every cube $Q \in (0,\varepsilon k)^d + \varepsilon k\mathbb{Z}^d$ in $\Omega^{\varepsilon k}$ we have

$$\begin{aligned} \fint_Q f\Big(\frac{x}{\varepsilon}, \nabla u_{\delta,\varepsilon}(x)\Big)\,\mathrm{d}x &= \fint_Q f\Big(\frac{x}{\varepsilon}, F + \nabla\psi_\delta\Big(\frac{x}{\varepsilon}\Big)\Big)\,\mathrm{d}x \\ &= \fint_{(0,k)^d} f(y, F + \nabla\psi_\delta(y))\,\mathrm{d}y. \end{aligned}$$

Combining this with (9.5) and summing over all cubes Q contained in $\Omega^{\varepsilon k}$, we get

$$|\Omega^{\varepsilon k}| f_{\mathrm{hom},0}(F) \le \int_{\Omega^{\varepsilon k}} f\Big(\frac{x}{\varepsilon}, \nabla u_{\delta,\varepsilon}(x)\Big)\,\mathrm{d}x \le |\Omega^{\varepsilon k}|\big(f_{\mathrm{hom},0}(F) + \delta\big).$$

We now let $\varepsilon \downarrow 0$ and use the growth bound on f together with $|\Omega \setminus \Omega^{\varepsilon k}| \to 0$ to conclude that

$$\mathcal{F}_0[Fx] \le \liminf_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_{\delta,\varepsilon}] \le \limsup_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_{\delta,\varepsilon}] \le \mathcal{F}_0[Fx] + \delta|\Omega|.$$

Thus, we have

$$\limsup_{\delta\downarrow 0} \limsup_{\varepsilon\downarrow 0} \Big[\big|\mathcal{F}_\varepsilon[u_{\delta,\varepsilon}] - \mathcal{F}_0[Fx]\big| + \|u_{\delta,\varepsilon} - Fx\|_{\mathrm{L}^p}\Big] = 0.$$

Using a diagonal sequence, see Problem 9.5, we can construct a function $\delta\colon (0,\infty) \to (0,\infty)$ such that for $u_\varepsilon := u_{\delta(\varepsilon),\varepsilon}$ it holds that

$$\mathcal{F}_\varepsilon[u_\varepsilon] \to \mathcal{F}_0[Fx] \qquad \text{and} \qquad u_\varepsilon \to Fx \quad \text{in } \mathrm{L}^p.$$

Since by the coercivity bound in (9.3) we have that $(\nabla u_\varepsilon)_\varepsilon$ is uniformly bounded in L^p, we obtain that also $u_\varepsilon \rightharpoonup Fx$ in $\mathrm{W}^{1,p}$ (the weak limit is already determined by $u_\varepsilon \to Fx$ in L^p). □

Lemma 9.9 *Let $(u_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_\varepsilon \rightharpoonup Fx$ in $\mathrm{W}^{1,p}$ as $\varepsilon \downarrow 0$, where $F \in \mathbb{R}^{m\times d}$. Then,*

$$\mathcal{F}_0[Fx] \le \liminf_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon].$$

Proof. Step 1. Assume first that $\Omega = Q$ is an open cube with all edges parallel to the coordinate axes and side length $s > 0$. Assume furthermore that the u_ε all have the same linear boundary values as the limit, i.e., $(u_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}_{Fx}(Q;\mathbb{R}^m)$ with $u_\varepsilon \rightharpoonup Fx$.

For fixed $\varepsilon > 0$ let $k \in \mathbb{N}$ be the smallest natural number such that $\varepsilon k \geq s + \varepsilon$ and denote by $Q^\varepsilon \supset Q$ an open cube with side length εk and such that all vertices of Q^ε lie in $\varepsilon\mathbb{Z}^d$, i.e., $Q^\varepsilon = \varepsilon z_0 + (0, \varepsilon k)^d$ for some $z_0 \in \mathbb{Z}^d$. Then extend u_ε to Q^ε by setting

$$u_\varepsilon(x) := Fx \qquad \text{for } x \in Q^\varepsilon \setminus Q.$$

Hence,

$$\int_Q f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x = \int_{Q^\varepsilon} f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x - \int_{Q^\varepsilon \setminus Q} f\Big(\frac{x}{\varepsilon}, F\Big)\,\mathrm{d}x.$$

Since the second term vanishes as $\varepsilon \downarrow 0$ by the growth bounds on f and $|Q^\varepsilon \setminus Q| \to 0$, we get with the change of variables $x = \varepsilon(z_0 + y)$ and setting

$$\begin{aligned}
&\psi_\varepsilon := u_\varepsilon - Fx \in \mathrm{W}^{1,p}_0(Q^\varepsilon; \mathbb{R}^m),\\
&\widetilde{\psi}_\varepsilon := \varepsilon^{-1}\psi_\varepsilon(\varepsilon(z_0 + \bullet)) \in \mathrm{W}^{1,p}_0((0,k)^d; \mathbb{R}^m)
\end{aligned}$$

that

$$\begin{aligned}
&\liminf_{\varepsilon\downarrow 0} \int_Q f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x\\
&\quad \geq \liminf_{\varepsilon\downarrow 0} \int_{Q^\varepsilon} f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x\\
&\quad = \liminf_{\varepsilon\downarrow 0} (\varepsilon k)^d \fint_{(0,k)^d} f\big(y, F + \nabla\psi_\varepsilon(\varepsilon(z_0 + y))\big)\,\mathrm{d}y\\
&\quad = \liminf_{\varepsilon\downarrow 0} (\varepsilon k)^d \fint_{(0,k)^d} f(y, F + \nabla\widetilde{\psi}_\varepsilon(y))\,\mathrm{d}y\\
&\quad \geq \liminf_{\varepsilon\downarrow 0} (\varepsilon k)^d f_{\mathrm{hom},0}(F)\\
&\quad \geq |Q| f_{\mathrm{hom},0}(F).
\end{aligned}$$

Here, we also used the definition of $f_{\mathrm{hom},0}$ and $\varepsilon k \geq s$. This proves the claim if $\Omega = Q$ and $(u_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}_{Fx}(Q; \mathbb{R}^m)$.

Step 2. We now assume that Ω is an arbitrary bounded Lipschitz domain, but we still require that the u_ε have linear boundary values, that is, $(u_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}_{Fx}(\Omega; \mathbb{R}^m)$ with $u_\varepsilon \rightharpoonup Fx$ in $\mathrm{W}^{1,p}$. Let $Q \ni \Omega$ be a cube with all edges parallel to the coordinate axes. By the previous lemma applied in the domain $Q \setminus \overline{\Omega}$ there exists a sequence $(v_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}_{Fx}(Q \setminus \overline{\Omega}; \mathbb{R}^m)$ with $v_\varepsilon \rightharpoonup Fx$ in $\mathrm{W}^{1,p}$ and

$$\int_{Q\setminus\overline{\Omega}} f\Big(\frac{x}{\varepsilon}, \nabla v_\varepsilon(x)\Big)\,\mathrm{d}x \to |Q \setminus \overline{\Omega}|\, f_{\mathrm{hom},0}(F) \qquad \text{as } \varepsilon \downarrow 0. \tag{9.6}$$

Define for $\varepsilon > 0$,

$$w_\varepsilon(x) := \begin{cases} u_\varepsilon(x) & \text{if } x \in \Omega,\\ v_\varepsilon(x) & \text{if } x \in Q \setminus \overline{\Omega}. \end{cases}$$

These maps lie in $\mathrm{W}^{1,p}(Q;\mathbb{R}^m)$ since the boundary values agree over the gluing boundary. By Step 1 we have that

$$|Q| f_{\mathrm{hom},0}(F) \le \liminf_{\varepsilon\downarrow 0} \int_Q f\Big(\frac{x}{\varepsilon}, \nabla w_\varepsilon(x)\Big)\,\mathrm{d}x.$$

Hence, combining this with (9.6), we arrive at

$$\mathcal{F}_0[Fx] \le \liminf_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon].$$

Step 3. Finally, we remove the restriction on the boundary values via a cut-off procedure. So, suppose that $(u_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_\varepsilon \rightharpoonup Fx$ in $\mathrm{W}^{1,p}$. Then let $n \in \mathbb{N}$ and let $\Omega_0 \Subset \Omega$ be a Lipschitz subdomain. With $R := \operatorname{dist}(\Omega_0, \partial\Omega)$ set

$$\Omega_i := \left\{ x \in \Omega \;:\; \operatorname{dist}(x,\Omega_0) < \frac{iR}{n} \right\} \qquad \text{for } i = 1,\dots,n.$$

Next, choose cut-off functions $\varrho_i \in \mathrm{C}_0^\infty(\Omega;[0,1])$ such that

$$\mathbb{1}_{\Omega_{i-1}} \le \varrho_i \le \mathbb{1}_{\Omega_i}, \qquad |\nabla \varrho_i| \le \frac{2n}{R}$$

and define $u_\varepsilon^{(i)} \in \mathrm{W}^{1,p}_{Fx}(\Omega;\mathbb{R}^m)$ by

$$u_\varepsilon^{(i)}(x) := Fx + \varrho_i(x)(u_\varepsilon(x) - Fx), \qquad x \in \Omega,$$

for which

$$\nabla u_\varepsilon^{(i)}(x) = F + \varrho_i(x)(\nabla u_\varepsilon(x) - F) + (u_\varepsilon(x) - Fx) \otimes \nabla \varrho_i(x).$$

From the growth bound on f we get, with an i,n-independent constant $C > 0$,

$$\begin{aligned}
\mathcal{F}_\varepsilon[u_\varepsilon^{(i)}] &= \int_{\Omega_{i-1}} f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x \\
&\quad + \int_{\Omega_i\setminus\Omega_{i-1}} f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon^{(i)}(x)\Big)\,\mathrm{d}x + \int_{\Omega\setminus\Omega_i} f\Big(\frac{x}{\varepsilon}, F\Big)\,\mathrm{d}x \\
&\le \int_\Omega f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x \\
&\quad + C\int_{\Omega_i\setminus\Omega_{i-1}} 1 + |F|^p + |\nabla u_\varepsilon(x) - F|^p\,\mathrm{d}x \\
&\quad + C\Big(\frac{2n}{R}\Big)^p \|u_\varepsilon - Fx\|_{\mathrm{L}^p}^p + C(1+|F|^p)|\Omega\setminus\Omega_i| \\
&\le \mathcal{F}_\varepsilon[u_\varepsilon] + C\int_{\Omega_i\setminus\Omega_{i-1}} |\nabla u_\varepsilon(x) - F|^p\,\mathrm{d}x \\
&\quad + C\Big(\frac{2n}{R}\Big)^p \|u_\varepsilon - Fx\|_{\mathrm{L}^p}^p + C|\Omega\setminus\Omega_0|.
\end{aligned}$$

On the other hand, $u_\varepsilon^{(i)} \rightharpoonup Fx$ in $\mathrm{W}^{1,p}$ and so, by Step 2,

$$\begin{aligned}\mathcal{F}_0[Fx] &\le \liminf_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon^{(i)}] \\ &\le \liminf_{\varepsilon\downarrow 0} \left[\mathcal{F}_\varepsilon[u_\varepsilon] + C\int_{\Omega_i\setminus\Omega_{i-1}} |\nabla u_\varepsilon - F|^p \,\mathrm{d}x\right] + C|\Omega\setminus\Omega_0|.\end{aligned}$$

Now sum this over $i = 1,\dots,n$, use the superadditivity of the lower limit (that is, $\liminf_{j\to\infty} a_j + \liminf_{j\to\infty} b_j \le \liminf_{j\to\infty}(a_j + b_j)$), and divide by n to see that

$$\mathcal{F}_0[Fx] \le \liminf_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon] + \frac{C}{n}\cdot \limsup_{\varepsilon\downarrow 0}\int_\Omega |\nabla u_\varepsilon - F|^p \,\mathrm{d}x + C|\Omega\setminus\Omega_0|.$$

Then we may let $n\to\infty$ and $\Omega_0\uparrow\Omega$, that is, $|\Omega\setminus\Omega_0|\to 0$, to conclude. □

Lemma 9.10 *The integrand $f_{\mathrm{hom},0}\colon \mathbb{R}^{m\times d}\to[0,\infty)$ from Theorem 9.6 satisfies the growth condition (9.3) and the local Lipschitz condition (9.4).*

Proof. Step 1. Since $\psi\equiv 0$ is admissible in the definition of $f_{\mathrm{hom},0}$, we immediately have $f_{\mathrm{hom},0}(x,A)\le M(1+|A|^p)$. For the lower bound at $A\in\mathbb{R}^{m\times d}$, fix $\delta>0$ and choose $\psi_\delta\in \mathrm{W}^{1,p}_0((0,k)^d;\mathbb{R}^m)$ for some $k\in\mathbb{N}$ (depending on δ) such that

$$f_{\mathrm{hom},0}(A) + \delta \ge \fint_{(0,k)^d} f(y, A+\nabla\psi_\delta(y))\,\mathrm{d}y.$$

Then we can estimate, using (9.3) for f and Jensen's inequality,

$$f_{\mathrm{hom},0}(A) + \delta \ge \mu \fint_{(0,k)^d} |A+\nabla\psi_\delta(y)|^p \,\mathrm{d}y \ge \mu|A|^p,$$

which, letting $\delta\downarrow 0$, establishes the lower bound in (9.3) for $f_{\mathrm{hom},0}$.

Step 2. To show the local Lipschitz condition (9.4) for $f_{\mathrm{hom},0}$, we fix $A,B\in\mathbb{R}^{m\times d}$ and use Lemma 9.8 on the domain $(0,1)^d$ to get a (recovery) sequence $(u_\varepsilon)_\varepsilon \subset \mathrm{W}^{1,p}_{Ax}((0,1)^d;\mathbb{R}^m)$ with $u_\varepsilon\rightharpoonup Ax$ in $\mathrm{W}^{1,p}$ and

$$f_{\mathrm{hom},0}(A) = \int_{(0,1)^d} f_{\mathrm{hom},0}(A)\,\mathrm{d}x = \lim_{\varepsilon\downarrow 0}\int_{(0,1)^d} f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x.$$

From (9.3) for $f_{\mathrm{hom},0}$ (proved in Step 1) we infer that

$$\limsup_{\varepsilon\downarrow 0} \|\nabla u_\varepsilon\|_{\mathrm{L}^p}^p \le \frac{1}{\mu} f_{\mathrm{hom},0}(A) \le \frac{M}{\mu}(1+|A|^p). \tag{9.7}$$

Define now $v_\varepsilon\in \mathrm{W}^{1,p}_{Bx}((0,1)^d;\mathbb{R}^m)$ by

$$v_\varepsilon(x) := (B-A)x + u_\varepsilon(x), \qquad x\in(0,1)^d.$$

It is straightforward to see that

$$\begin{aligned}\limsup_{\varepsilon\downarrow 0} \|\nabla v_\varepsilon\|_{\mathrm{L}^p}^p &\le C \cdot \limsup_{\varepsilon\downarrow 0} \big(\|\nabla u_\varepsilon\|_{\mathrm{L}^p}^p + |A|^p + |B|^p\big) \\ &\le C(1 + |A|^p + |B|^p). \end{aligned} \tag{9.8}$$

Furthermore, since $v_\varepsilon \rightharpoonup Bx$ in $\mathrm{W}^{1,p}$, Lemma 9.9 implies

$$f_{\mathrm{hom},0}(B) = \int_{(0,1)^d} f_{\mathrm{hom},0}(B)\,\mathrm{d}x \le \liminf_{\varepsilon\downarrow 0} \int_{(0,1)^d} f\Big(\frac{x}{\varepsilon}, \nabla v_\varepsilon(x)\Big)\,\mathrm{d}x.$$

Thus, by the local Lipschitz continuity (9.4) for f,

$$\begin{aligned} &f_{\mathrm{hom},0}(B) - f_{\mathrm{hom},0}(A) \\ &\quad\le \liminf_{\varepsilon\downarrow 0} \int_{(0,1)^d} f\Big(\frac{x}{\varepsilon}, \nabla v_\varepsilon(x)\Big) - f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x \\ &\quad\le C \cdot \liminf_{\varepsilon\downarrow 0} \int_{(0,1)^d} (1 + |\nabla u_\varepsilon|^{p-1} + |\nabla v_\varepsilon|^{p-1}) \cdot |\nabla u_\varepsilon - \nabla v_\varepsilon|\,\mathrm{d}x \\ &\quad\le C \cdot \limsup_{\varepsilon\downarrow 0} \big(1 + \|\nabla u_\varepsilon\|_{\mathrm{L}^p}^p + \|\nabla v_\varepsilon\|_{\mathrm{L}^p}^p\big)^{(p-1)/p} \cdot |A - B| \\ &\quad\le C(1 + |A|^{p-1} + |B|^{p-1})|A - B|, \end{aligned}$$

where we also used (9.7) and (9.8). Interchanging the roles of A and B, we have thus shown (9.4) for $f_{\mathrm{hom},0}$. □

Lemma 9.11 *It holds that* $f_{\mathrm{hom}} = f_{\mathrm{hom},0}$.

Proof. Clearly, $f_{\mathrm{hom}} \le f_{\mathrm{hom},0}$ since $\mathrm{W}_0^{1,p}((0,k)^d;\mathbb{R}^m)$ is contained in the space $\mathrm{W}_{\mathrm{per}}^{1,p}((0,k)^d;\mathbb{R}^m)$.

To see the other inequality, let $A \in \mathbb{R}^{m\times d}$, $\psi \in \mathrm{W}_{\mathrm{per}}^{1,p}((0,k)^d;\mathbb{R}^m)$ and define $u_\varepsilon \in \mathrm{W}^{1,p}((0,k)^d;\mathbb{R}^m)$ by

$$u_\varepsilon(x) := Ax + \varepsilon\psi\Big(\frac{x}{\varepsilon}\Big), \qquad x \in (0,k)^d.$$

Since $u_\varepsilon \rightharpoonup Ax$ in $\mathrm{W}^{1,p}$, we can apply Lemma 9.9 on the domain $(0,k)^d$ to deduce that

$$\begin{aligned} f_{\mathrm{hom},0}(A) &\le \liminf_{\varepsilon\downarrow 0} \int_{(0,k)^d} f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big)\,\mathrm{d}x \\ &= \fint_{(0,k)^d} f(y, A + \nabla\psi(y))\,\mathrm{d}y. \end{aligned}$$

Here, for the limit passage we utilized the oscillatory nature of the function $f(\varepsilon^{-1}x, \nabla u_\varepsilon(x)) = h(\varepsilon^{-1}x)$, where $h(y) := f(y, A + \nabla\psi(y))$; see Problem 9.4. Taking the infimum over all $\psi \in \mathrm{W}_{\mathrm{per}}^{1,p}((0,k)^d;\mathbb{R}^m)$, we arrive at

$$f_{\mathrm{hom},0}(A) \le f_{\mathrm{hom}}(A),$$

which implies the claim. □

Proof of Theorem 9.6. Step 1: Lim inf-inequality. Assume that $u_\varepsilon \rightharpoonup u$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. Given $\delta > 0$, it can be seen via an approximation argument that there exists a finite partition of Ω into finitely many disjoint open sets $\Omega_i \subset \Omega$, $i = 1,\dots,N$, and a negligible set $Z \subset \Omega$, that is,

$$\Omega = Z \cup \bigcup_{i=1}^{N} \Omega_i, \qquad |Z| = 0,$$

such that

$$\sum_{i=1}^{N} \int_{\Omega_i} |\nabla u(x) - A_i|^p \,\mathrm{d}x < \delta^p, \tag{9.9}$$

where

$$A_i := [\nabla u]_{\Omega_i} = \fint_{\Omega_i} \nabla u(x)\,\mathrm{d}x.$$

Then define $v_\varepsilon \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ by

$$v_\varepsilon(x) := A_i x + u_\varepsilon(x) - u(x) \qquad \text{if } x \in \Omega_i \ (i = 1,\dots,N).$$

It is obvious that $v_\varepsilon \rightharpoonup A_i x$ in $\mathrm{W}^{1,p}(\Omega_i;\mathbb{R}^m)$.

Below we will show the estimates

$$\left| \mathcal{F}_\varepsilon[u_\varepsilon] - \sum_{i=1}^{N} \int_{\Omega_i} f\Big(\frac{x}{\varepsilon}, \nabla v_\varepsilon(x)\Big)\,\mathrm{d}x \right| \le C\delta, \tag{9.10}$$

$$\left| \mathcal{F}_0[u] - \sum_{i=1}^{N} \int_{\Omega_i} f_{\mathrm{hom}}(A_i)\,\mathrm{d}x \right| \le C\delta \tag{9.11}$$

for some constant $C > 0$ that does not depend on ε or δ. On the other hand, from Lemma 9.9 (extended to affine maps, which is trivial) in conjunction with Lemma 9.11, we get

$$\sum_{i=1}^{N} \int_{\Omega_i} f_{\mathrm{hom}}(A_i)\,\mathrm{d}x \le \liminf_{\varepsilon\downarrow 0} \sum_{i=1}^{N} \int_{\Omega_i} f\Big(\frac{x}{\varepsilon}, \nabla v_\varepsilon(x)\Big)\,\mathrm{d}x.$$

Thus, combining this with (9.10) and (9.11),

$$\mathcal{F}_0[u] \le \liminf_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon] + C\delta.$$

As $\delta > 0$ was arbitrary, we arrive at the lim inf-inequality.

It remains to show (9.10) and (9.11). For the first assertion we estimate, using the local Lipschitz continuity (9.4) of f and Hölder's inequality for sums, that

$$\begin{aligned}
&\left|\mathcal{F}_\varepsilon[u_\varepsilon] - \sum_{i=1}^N \int_{\Omega_i} f\Big(\frac{x}{\varepsilon}, \nabla v_\varepsilon(x)\Big)\,\mathrm{d}x\right| \\
&\quad\le \sum_{i=1}^N \int_{\Omega_i} \left|f\Big(\frac{x}{\varepsilon}, \nabla u_\varepsilon(x)\Big) - f\Big(\frac{x}{\varepsilon}, \nabla v_\varepsilon(x)\Big)\right| \mathrm{d}x \\
&\quad\le C \sum_{i=1}^N \int_{\Omega_i} \big(1 + |\nabla u_\varepsilon|^{p-1} + |\nabla v_\varepsilon|^{p-1}\big)|\nabla u_\varepsilon - \nabla v_\varepsilon|\,\mathrm{d}x \\
&\quad\le C \sum_{i=1}^N \left[\big\|1 + |\nabla u_\varepsilon| + |\nabla v_\varepsilon|\big\|_{\mathrm{L}^p(\Omega_i)}^{p-1} \cdot \left(\int_{\Omega_i} |\nabla u(x) - A_i|^p\,\mathrm{d}x\right)^{1/p}\right] \\
&\quad\le C\big\|1 + |\nabla u_\varepsilon| + |\nabla v_\varepsilon|\big\|_{\mathrm{L}^p(\Omega)}^{p-1} \cdot \left(\sum_{i=1}^N \int_{\Omega_i} |\nabla u(x) - A_i|^p\,\mathrm{d}x\right)^{1/p} \\
&\quad\le C\delta,
\end{aligned}$$

where for the last estimate we also employed (9.9). The second assertion is shown in a similar fashion using Lemma 9.10.

Step 2: Recovery sequence. Let $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. Then, for every $\delta > 0$ there exists a $u_\delta \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ that is countably piecewise affine and such that $\|u - u_\delta\|_{\mathrm{W}^{1,p}} < \delta$ (see Theorem A.30). By the same estimate (based on Lemma 9.10) as above, we get

$$\big|\mathcal{F}_0[u_\delta] - \mathcal{F}_0[u]\big| \le C\big(1 + \|\nabla u_\delta\|_{\mathrm{L}^p}^p + \|\nabla u\|_{\mathrm{L}^p}^p\big)^{(p-1)/p}\|\nabla u_\delta - \nabla u\|_{\mathrm{L}^p} \to 0$$

as $\delta \downarrow 0$.

Suppose that $\Omega = Z \cup \bigcup_i \Omega_i$ is a decomposition of Ω, up to the negligible set Z, into disjoint open patches Ω_i ($i \in \mathbb{N}$), on every one of which u_δ is affine, say $\nabla u_\delta(x) = A_i \in \mathbb{R}^{m\times d}$ for $x \in \Omega_i$. We remark that the Ω_i of course depend on δ, but we suppress this in our notation, which should not cause any confusion. Now apply Lemma 9.8 (extended to affine maps) in conjunction with Lemma 9.11 in every Ω_i separately to get sequences $(u_{\delta,\varepsilon}^{(i)})_\varepsilon \subset \mathrm{W}^{1,p}(\Omega_i;\mathbb{R}^m)$ with

$$u_{\delta,\varepsilon}^{(i)}|_{\partial\Omega_i} = u_\delta|_{\partial\Omega_i}$$

as well as

$$u_{\delta,\varepsilon}^{(i)} \rightharpoonup u_\delta \quad \text{in } \mathrm{W}^{1,p}(\Omega_i;\mathbb{R}^m) \qquad \text{as } \varepsilon \downarrow 0 \ (\delta \text{ held fixed})$$

and

$$\lim_{\varepsilon\downarrow 0} \int_{\Omega_i} f\Big(\frac{x}{\varepsilon}, \nabla u_{\delta,\varepsilon}^{(i)}(x)\Big)\,\mathrm{d}x = |\Omega_i| f_{\mathrm{hom}}(A_i).$$

Then define $u_{\delta,\varepsilon} \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ via

$$u_{\delta,\varepsilon}(x) := u^{(i)}_{\delta,\varepsilon}(x) \qquad \text{if } x \in \Omega_i \ (i \in \mathbb{N}).$$

By construction, it holds that $u_{\delta,\varepsilon} \rightharpoonup u_\delta$ in $\mathrm{W}^{1,p}$ as $\varepsilon \downarrow 0$ and

$$\lim_{\varepsilon\downarrow 0} \int_\Omega f\Big(\frac{x}{\varepsilon}, \nabla u_{\delta,\varepsilon}(x)\Big)\,\mathrm{d}x = \int_\Omega f_{\mathrm{hom}}(\nabla u_\delta(x))\,\mathrm{d}x. \tag{9.12}$$

For

$$v_{\delta,\varepsilon} := u_{\delta,\varepsilon} + u - u_\delta \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$$

we then have $v_{\delta,\varepsilon} \rightharpoonup u$ in $\mathrm{W}^{1,p}$ as $\varepsilon \downarrow 0$ and δ held fixed. From the Lipschitz assumption (9.4) we may furthermore derive

$$\begin{aligned}\big|\mathcal{F}_\varepsilon[u_{\delta,\varepsilon}] - \mathcal{F}_\varepsilon[v_{\delta,\varepsilon}]\big| &\le C\big(1 + \|\nabla u_{\delta,\varepsilon}\|_{\mathrm{L}^p}^p + \|\nabla u_{\delta,\varepsilon}\|_{\mathrm{L}^p}^p\big)^{(p-1)/p}\|\nabla u_{\delta,\varepsilon} - \nabla v_{\delta,\varepsilon}\|_{\mathrm{L}^p}\\ &\le C\|\nabla u - \nabla u_\delta\|_{\mathrm{L}^p}.\end{aligned}$$

Thus, using (9.12),

$$\limsup_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[v_{\delta,\varepsilon}] \le \limsup_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[u_{\delta,\varepsilon}] + C\delta \le \mathcal{F}_0[u_\delta] + C\delta.$$

In a similar fashion (with Lemma 9.10 replacing (9.4)) one also gets

$$\mathcal{F}_0[u_\delta] - C\delta \le \liminf_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[v_{\delta,\varepsilon}].$$

Consequently,

$$\lim_{\delta\downarrow 0}\lim_{\varepsilon\downarrow 0} \mathcal{F}_\varepsilon[v_{\delta,\varepsilon}] = \mathcal{F}_0[u].$$

Thus we can finish the proof by the diagonal argument from Problem 9.5, just like in the proof of Lemma 9.8. □

9.3 Convex Homogenization

In the convex case, the homogenization formula from Theorem 9.6 simplifies and we have the following theorem.

Theorem 9.12 (Marcellini 1978 [222]) *In the situation of Theorem 9.6, if additionally $A \mapsto f(x,A)$ is assumed to be convex for almost every $x \in \Omega$, then the asymptotic homogenization formula simplifies to the* ***cell problem formula***

$$f_{\mathrm{hom}}(A) = \inf_{\varphi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)} \int_{(0,1)^d} f(x, A + \nabla\varphi(x))\,\mathrm{d}x,$$

where $A \in \mathbb{R}^{m\times d}$.

Remark 9.13 This result for the convex case also holds for more general upper growth assumptions on f than the one in (9.3), but then the proof becomes more involved, see [244]. Moreover, for the scalar case $m = 1$, the convexity condition is not necessary, see Problem 9.9.

Proof. Let

$$\widehat{f}_{\mathrm{hom}}(A) := \inf_{\varphi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)} \int_{(0,1)^d} f(x, A + \nabla\varphi(x))\,\mathrm{d}x, \qquad A \in \mathbb{R}^{m\times d}.$$

We will show in the following that

$$f_{\mathrm{hom}}(A) = \inf_{\psi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,k)^d;\mathbb{R}^m)} \fint_{(0,k)^d} f(x, A + \nabla\psi(x))\,\mathrm{d}x \geq \widehat{f}_{\mathrm{hom}}(A) \tag{9.13}$$

for every $A \in \mathbb{R}^{m\times d}$ and every $k \in \mathbb{N}$. From this the conclusion follows immediately since $f_{\mathrm{hom}} \leq \widehat{f}_{\mathrm{hom}}$ is trivially true as well.

By Problem 9.8 we may assume that f is smooth in its second argument. Since the values of both f_{hom} and $\widehat{f}_{\mathrm{hom}}$ are defined via minimization problems with convex, coercive integrands, by Theorem 2.7 there exist minimizers $\varphi_* \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,k)^d;\mathbb{R}^m)$ and $\widehat{\varphi}_* \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)$, respectively. Moreover, by a slight modification of Theorem 3.1 for periodic spaces of candidate functions, we have the Euler–Lagrange equation

$$\int_{(0,1)^d} \mathrm{D}_A f(x, A + \nabla\widehat{\varphi}_*(x)) : \nabla\psi(x)\,\mathrm{d}x = 0$$

for all $\psi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)$, see Problem 9.7. This expression is well-defined since (9.4) implies the bound $|\mathrm{D}_A f(x,A)| \leq C(1 + |A|^{p-1})$ with a constant $C > 0$ (and this bound is unchanged by the procedure of Problem 9.8). From the convexity of f we then get

$$\begin{aligned}\int_{(0,k)^d} & f(x, A + \nabla\varphi_*(x)) - f(x, A + \nabla\widehat{\varphi}_*(x))\,\mathrm{d}x \\ &\geq \int_{(0,k)^d} \mathrm{D}_A f(x, A + \nabla\widehat{\varphi}_*(x)) : [\nabla\varphi_*(x) - \nabla\widehat{\varphi}_*(x)]\,\mathrm{d}x.\end{aligned}$$

Set $v := \varphi_* - \widehat{\varphi}_*$ and write the right-hand side as follows, using the periodicity of $\widehat{\varphi}_*$ and of f with respect to the first argument:

$$\begin{aligned}&\int_{(0,k)^d} \mathrm{D}_A f(x, A + \nabla\widehat{\varphi}_*(x)) : \nabla v(x)\,\mathrm{d}x \\ &\quad= \sum_{z\in\{0,\dots,k-1\}^d} \int_{(0,1)^d} \mathrm{D}_A f(x+z, A + \nabla\widehat{\varphi}_*(x+z)) : \nabla v(x+z)\,\mathrm{d}x \\ &\quad= \int_{(0,1)^d} \mathrm{D}_A f(x, A + \nabla\widehat{\varphi}_*(x)) : \nabla w(x)\,\mathrm{d}x,\end{aligned}$$

where

$$w(x) := \sum_{z\in\{0,\dots,k-1\}^d} v(x+z), \qquad x \in (0,1)^d.$$

Then, $w \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)$, as can be checked easily, and thus, by the Euler–Lagrange equation,

$$\begin{aligned}\int_{(0,k)^d} & f(x, A+\nabla\varphi_*(x)) - f(x, A+\nabla\widehat{\varphi}_*(x))\,\mathrm{d}x \\ &\geq \int_{(0,k)^d} \mathrm{D}_A f(x, A+\nabla\widehat{\varphi}_*(x)) : \nabla w(x)\,\mathrm{d}x \\ &= 0.\end{aligned}$$

This shows (9.13) and completes the proof. □

We remark that in the general non-convex case the reduction to the cell problem on $(0,1)^d$ is not possible. A counterexample can be found in [244] or in Section 14.4 of [63].

9.4 Quadratic Homogenization

Finally, we give a more concrete formula for f_{hom} in the case when f is a symmetric and positive definite quadratic form (see Appendix A.1 for precise definitions of these notions).

Theorem 9.14 *In the situation of Theorem 9.6 (for $p = 2$), assume that f has the form*

$$f(x,A) = A : \mathbf{S}(x)\,A, \qquad (x,A) \in \mathbb{R}^d \times \mathbb{R}^{m\times d},$$

with a symmetric and uniformly positive definite fourth-order tensor field $\mathbf{S}(x) = \mathbf{S}^{ik}_{jl}(x)$ that is measurable and 1-periodic in $x \in \mathbb{R}^d$. Then, there exists a symmetric and positive definite fourth-order tensor $\mathbf{S}_{\mathrm{hom}} = [\mathbf{S}_{\mathrm{hom}}]^{ik}_{jl}$ (which is independent of x) such that

$$f_{\mathrm{hom}}(A) = A : \mathbf{S}_{\mathrm{hom}}\,A, \qquad A \in \mathbb{R}^{m\times d}.$$

Moreover, the entries $[\mathbf{S}_{\mathrm{hom}}]^{ik}_{jl}$ ($i,k = 1,\dots,m$, $j,l \in \{1,\dots,d\}$) of the tensor $\mathbf{S}_{\mathrm{hom}}$ are given as

$$[\mathbf{S}_{\mathrm{hom}}]^{ik}_{jl} = \int_{(0,1)^d} (\mathrm{e}_i \otimes \mathrm{e}_j + \nabla\varphi_{i,j}(x)) : \mathbf{S}(x)\,(\mathrm{e}_k \otimes \mathrm{e}_l + \nabla\varphi_{k,l}(x))\,\mathrm{d}x, \tag{9.14}$$

*where $\varphi_{i,j} \in \mathrm{W}^{1,2}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)$ is the (unique up to constants) weak solution of the **cell problem***

$$\begin{cases} -\operatorname{div}\big[\mathbf{S}(x)(\mathrm{e}_i \otimes \mathrm{e}_j + \nabla\varphi_{i,j}(x))\big] = 0, & x \in (0,1)^d, \\ \varphi_{i,j} \text{ has periodic boundary values.} \end{cases}$$

We also explicitly formulate this theorem for the scalar case:

Corollary 9.15 *In the situation of Theorem 9.6 (for $p = 2$), assume that $m = 1$ and that f has the form*

$$f(x,\xi) = \xi^{\mathsf{T}} S(x)\,\xi, \qquad (x,\xi) \in \mathbb{R}^d \times \mathbb{R}^d,$$

with a symmetric, uniformly positive definite, measurable, and 1-periodic matrix field $S\colon \mathbb{R}^d \to \mathbb{R}^{d\times d}_{\mathrm{sym}}$. Then, there exists a symmetric and positive definite matrix $S_{\mathrm{hom}} \in \mathbb{R}^{d\times d}_{\mathrm{sym}}$ such that

$$f_{\mathrm{hom}}(\xi) = \xi^{\mathsf{T}} S_{\mathrm{hom}}\,\xi, \qquad \xi \in \mathbb{R}^d.$$

Moreover, the entries $[S_{\mathrm{hom}}]_l^k$ $(k,l \in \{1,\dots,d\})$ of the matrix S_{hom} are given as

$$[S_{\mathrm{hom}}]_l^k = \int_{(0,1)^d} (\mathrm{e}_k + \nabla\varphi_k(x))^{\mathsf{T}} S(x)\,(\mathrm{e}_l + \nabla\varphi_l(x))\,\mathrm{d}x, \tag{9.15}$$

where $\varphi_k \in \mathrm{W}^{1,2}_{\mathrm{per}}((0,1)^d)$ is the (unique up to constants) weak solution of the cell problem

$$\begin{cases} -\operatorname{div}\big[S(x)(\mathrm{e}_k + \nabla\varphi_k(x))\big] = 0, & x \in (0,1)^d, \\ \varphi_k \text{ has periodic boundary values.} \end{cases}$$

We will use the following elementary characterization of quadratic forms, whose proof is the task of Problem 9.10.

Lemma 9.16 *Let X be a Banach space and let $F\colon X \to [0,\infty)$. Then, F is a quadratic form, that is, $F(x) = b(x,x)$ with $b\colon X \times X \to \mathbb{R}$ bilinear, if and only if it satisfies the following conditions:*

(i) $F(0) = 0$;
(ii) $F(tx) = t^2 F(x)$ *for all* $x \in X$, $t > 0$;
(iii) $F(x+y) + F(x-y) \le 2F(x) + 2F(y)$ *for every* $x, y \in X$.

From this lemma we may infer an abstract result about Γ-convergence:

Proposition 9.17 *Let X be a Banach space and let $\mathcal{F}_k\colon X \to [0,\infty)$, $k \in \mathbb{N}$, be positive definite quadratic forms that Γ-converge to $\mathcal{F}_\infty\colon X \to \mathbb{R}$. Then, $\mathcal{F}_\infty$ is also a positive definite quadratic form.*

Proof. It is easy to see that $\mathcal{F}_\infty \ge 0$. We will show the conditions (i)–(iii) of Lemma 9.16 for $\mathcal{F}_\infty$.

For (i), we notice by the lim inf-inequality that

$$\mathcal{F}_\infty[0] \le \liminf_{j\to\infty} \mathcal{F}_k[0] = 0.$$

Thus, $\mathcal{F}_\infty[0] = 0$.

For (ii), let $u \in X$ and $t > 0$. Take a recovery sequence $(u_k)_k \subset X$ of u. Clearly, $tu_k \to tu$ in X. Thus, using the lim inf-inequality, the positive 2-homogeneity of the $\mathcal{F}_k$'s, and the recovery property of $(u_k)_k$,

$$\mathcal{F}_\infty[tu] \le \liminf_{k\to\infty} \mathcal{F}_k[tu_k] = t^2 \cdot \lim_{k\to\infty} \mathcal{F}_k[u_k] = t^2\mathcal{F}_\infty[u].$$

Hence,

$$\mathcal{F}_\infty[u] = \mathcal{F}_\infty[t^{-1}tu] \le t^{-2}\mathcal{F}_\infty[tu] \le \mathcal{F}_\infty[u],$$

from which we conclude that $\mathcal{F}_\infty[tu] = t^2\mathcal{F}_\infty[u]$.

Finally, for (iii), we take $u, v \in X$ and let $(u_k)_k, (v_k)_k \subset X$ be recovery sequences for u and v, respectively, Then it suffices to observe by the lim inf-inequality and the recovery property of $(u_k)_k, (v_k)_k$ together with property (iii) for $\mathcal{F}_k$ that

$$\begin{aligned}
&\mathcal{F}_\infty[u+v] + \mathcal{F}_\infty[u-v] - 2\mathcal{F}_\infty[u] - 2\mathcal{F}_\infty[v] \\
&\quad\le \liminf_{j\to\infty} \big(\mathcal{F}_k[u_k+v_k] + \mathcal{F}_k[u_k-v_k] - 2\mathcal{F}_k[u_k] - 2\mathcal{F}_k[v_k]\big) \\
&\quad\le 0.
\end{aligned}$$

This shows (iii) for $\mathcal{F}_\infty$. Hence, by Lemma 9.16, $\mathcal{F}_\infty$ is a quadratic form. □

Proof of Theorem 9.14. Via Proposition 9.17 we immediately have

$$f_{\mathrm{hom}}(A) = A : \mathbf{S}_{\mathrm{hom}}\, A = \sum_{i,k=1}^{m} \sum_{j,l=1}^{d} [\mathbf{S}_{\mathrm{hom}}]_{jl}^{ik} A_j^i A_l^k \tag{9.16}$$

for a positive definite fourth-order tensor $\mathbf{S}_{\mathrm{hom}} = [\mathbf{S}_{\mathrm{hom}}]_{jl}^{ik}$ as in the statement of the theorem. It only remains to show the formula (9.14).

From Marcellini's Theorem 9.12 we know that for $A \in \mathbb{R}^{m\times d}$ it holds that

$$f_{\mathrm{hom}}(A) = \inf_{\varphi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)} \int_{(0,1)^d} (A + \nabla\varphi(x)) : \mathbf{S}(x)\,(A + \nabla\varphi(x))\,\mathrm{d}x.$$

The minimizer $\varphi_A \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m)$ to this minimization problem exists, is unique up to constants (by a suitable adaptation of Proposition 2.10), and satisfies the Euler–Lagrange equation (see Problem 9.7)

$$-\operatorname{div}\big[\mathbf{S}(x)(A + \nabla\varphi_A(x))\big] = 0, \qquad x \in (0,1)^d,$$

weakly. Moreover, φ_A is linear in the matrix $A \in \mathbb{R}^{m\times d}$, as can be seen directly from the Euler–Lagrange equation. Thus, abbreviating

$$\varphi_{i,j} := \varphi_{\mathrm{e}_i \otimes \mathrm{e}_j},$$

we may write

$$\varphi_A = \sum_{i=1}^{m} \sum_{j=1}^{d} A_j^i \varphi_{i,j}.$$

Hence, $f_{\mathrm{hom}}(A)$ is equal to

$$\sum_{i,k=1}^{m}\sum_{j,l=1}^{d}\left(\int_{(0,1)^d}(\mathrm{e}_i\otimes\mathrm{e}_j+\nabla\varphi_{i,j}(x)):\mathbf{S}(x)\,(\mathrm{e}_k\otimes\mathrm{e}_l+\nabla\varphi_{k,l}(x))\,\mathrm{d}x\right)A^i_jA^k_l.$$

Comparing this to (9.16) term by term, we obtain (9.14). □

Example 9.18 We now consider the case $n=1$ as in Corollary 9.15 and assume furthermore that

$$S(x)=h_1(x_1)\cdots h_d(x_d)\,\mathrm{Id},\qquad x\in\mathbb{R}^d,$$

where $h_1,\dots,h_d\colon\mathbb{R}\to[0,\infty)$ are 1-periodic Borel functions such that there exist constants $\alpha,\beta>0$ with $\alpha\le h_i\le\beta$ for $i=1,\dots,d$. In this case, we use formula (9.15) to infer that

$$[S_{\mathrm{hom}}]^k_k=\int_{(0,1)^d}h_1(x_1)\cdots h_d(x_d)\,|\mathrm{e}_k+\nabla\varphi_k(x)|^2\,\mathrm{d}x,\tag{9.17}$$

where $\varphi_k\in\mathrm{W}^{1,2}_{\mathrm{per}}((0,1)^d)$ is the (unique up to additive constants) weak solution of the cell problem

$$-\operatorname{div}\big[h_1(x_1)\cdots h_d(x_d)(\mathrm{e}_k+\nabla\varphi_k(x))\big]=0,\qquad x\in(0,1)^d.\tag{9.18}$$

We can also deduce that, with a slight abuse of notation, $\varphi_k(x)=\varphi_k(x_k)$. Indeed, choosing the ansatz that φ_k only depends on x_k, the cell problem reads

$$[h_k(t)(1+\varphi_k'(t))]'=0,\qquad t\in(0,1),\tag{9.19}$$

which has a unique periodic solution (up to additive constants). Since the solution to (9.18) is unique (up to additive constants), it must be given by this $\varphi_k(x)=\varphi_k(x_k)$.

From the special structure of φ_k, we also deduce via (9.15) that

$$[S_{\mathrm{hom}}]^k_l=0\qquad\text{if } k\neq l.$$

Furthermore, as a consequence of (9.19),

$$h_k(t)(1+\varphi_k'(t))\equiv\text{const}=:\beta,$$

and, since $\int_0^1 1+\varphi_k'\,\mathrm{d}s=1$ (by the periodicity), we may conclude that

$$\beta=\left(\int_0^1\frac{1}{h_k(s)}\,\mathrm{d}s\right)^{-1}.$$

Hence,

$$1+\varphi_k'(t)=\frac{1}{h_k(t)}\left(\int_0^1\frac{1}{h_k(s)}\,\mathrm{d}s\right)^{-1},\qquad t\in[0,1].$$

Plugging this into (9.17), we calculate

$$\begin{aligned}[S_{\mathrm{hom}}]_k^k &= \int_{(0,1)^d} h_1(x_1)\cdots h_d(x_d)\,\frac{1}{h_k(x_k)^2}\left(\int_0^1 \frac{1}{h_k(s)}\,\mathrm{d}s\right)^{-2}\mathrm{d}x\\ &= \overline{h_1}\cdots\overline{h_{k-1}}\cdot\underline{h_k}\cdot\overline{h_{k+1}}\cdots\overline{h_d},\end{aligned}$$

where we denoted the *arithmetic mean* of h_i ($i = 1,\dots,d$) by

$$\overline{h_i} := \int_0^1 h_i(s)\,\mathrm{d}s$$

and the *harmonic mean* of h_i by

$$\underline{h_i} := \left(\int_0^1 \frac{1}{h_i(s)}\,\mathrm{d}s\right)^{-1}.$$

Example 9.19 In Section 1.10 we were led to identify the variational limit as $\varepsilon \downarrow 0$ of the functionals

$$\begin{aligned}\mathcal{F}_\varepsilon[u] &:= \int_\Omega \frac{1}{2}\mathcal{E}u(x) : \mathbf{C}\Big(\frac{x}{\varepsilon}\Big)\,\mathcal{E}u(x) - b(x)\cdot u(x)\,\mathrm{d}x\\ &= \int_\Omega f\Big(\frac{x}{\varepsilon},\nabla u(x)\Big) - b(x)\cdot u(x)\,\mathrm{d}x,\end{aligned}$$

where

$$\mathbf{C}(x) = \mathbf{C}_1 + (\mathbf{C}_2 - \mathbf{C}_1)h(x_1), \qquad x\in\mathbb{R}^3,$$

and

$$h(t) := \begin{cases} 0 & \text{if } t - \lfloor t\rfloor \le \theta,\\ 1 & \text{if } t - \lfloor t\rfloor > \theta,\end{cases}$$

for some $\theta \in (0,1)$, and $b \in \mathrm{L}^2(\Omega;\mathbb{R}^3)$ (say). From Theorem 9.14 in conjunction with the straightforward fact that the addition of a strongly continuous ε-independent functional commutes with Γ-convergence, we infer that indeed the $\mathcal{F}_\varepsilon$ Γ-converge with respect to the weak topology in $\mathrm{W}^{1,2}(\Omega;\mathbb{R}^3)$ to the functional

$$\mathcal{F}_0[u] = \int_\Omega \frac{1}{2}\mathcal{E}u(x) : \mathbf{C}_{\mathrm{hom}}\,\mathcal{E}u(x) - b(x)\cdot u(x)\,\mathrm{d}x$$

for a symmetric and positive definite fourth-order tensor $\mathbf{C}_{\mathrm{hom}} = [\mathbf{C}_{\mathrm{hom}}]^{ik}_{jl}$.

We finally consider the special case when

$$\mathbf{C}_1 = \alpha\,\mathbf{I}, \qquad \mathbf{C}_2 = \beta\,\mathbf{I}$$

where $\alpha,\beta > 0$ and $\mathbf{I}$ denotes the tensor such that $A : \mathbf{I}B = A : B$ (that is, $\mathbf{I}^{ik}_{jl} = \delta_{ik}\delta_{jl}$). In this case,

$$f(x,A) = \big(\alpha + (\beta-\alpha)h(x_1)\big)|A|^2, \qquad (x,A)\in\mathbb{R}^3\times\mathbb{R}^{3\times 3}.$$

We can adapt the previous example to see that

$$[\mathbf{C}_{\text{hom}}]^{ii}_{11} = \left(\frac{\theta}{\alpha} + \frac{1-\theta}{\beta}\right)^{-1},$$
$$[\mathbf{C}_{\text{hom}}]^{ii}_{jj} = \theta\alpha + (1-\theta)\beta \qquad \text{if } j \neq 1,$$

and $[\mathbf{C}_{\text{hom}}]^{ik}_{jl} = 0$ for all other values of i, j, k, l. Equivalently,

$$[\mathbf{C}_{\text{hom}}]^{ik}_{jl} = \delta_{ik}\delta_{jl}\left[\left(\frac{\theta}{\alpha} + \frac{1-\theta}{\beta}\right)^{-1}\delta_{j1} + \big(\theta\alpha + (1-\theta)\beta\big)(1-\delta_{j1})\right].$$

In particular, $\mathbf{C}_{\text{hom}}$ can no longer be written as $\gamma\mathbf{I}$ for some $\gamma \in \mathbb{R}$. This is also not surprising since the anisotropic lamination structure (in the first coordinate direction) is reflected in $\mathbf{C}_{\text{hom}}$.

Notes and Historical Remarks

The theory of Γ-convergence was introduced by Ennio De Giorgi in the 1970s and is widely used in the Calculus of Variations. All results from Section 9.1 are essentially due to De Giorgi, see [106]. The book [62] provides a thorough introduction to Γ-convergence with many applications to homogenization theory, phase transition, and free discontinuity problems. The monograph [98] treats Γ-convergence in general topological spaces and also considers many applications to integral functionals.

Our proof of Theorem 9.6 is the one in [244], which was obtained independently of Braides' proof in [61].

Another powerful technique in the theory of Γ-convergence is the *compactness method*: It can often be shown by a compactness argument that the Γ-limit of a (sub)sequence of integral functionals exists as some abstract functional. The task is then to identify this Γ-limit. For this, one can use a technique, which seems to be due to De Giorgi–Letta, Fusco, and Braides, where one first shows that the Γ-limit parametrized on the domain is a measure, which is then in a second step seen to be given by an integral functional. The book [63] uses this technique to present many homogenization results in a unified framework. In fact, Chapter 14 of [63] proves Theorem 9.6 and several extensions using this technique.

For pointers to the history of the homogenization theory for integral functionals we refer the reader to the notes sections in [62, 63, 98].

In a non-variational context, that is, on the level of PDEs, several techniques have been developed for homogenization, which can also cope with completely non-periodic situations. We mention in particular G-convergence (see [98]), H-convergence (see [8]) and two-scale convergence (see [261] and also [8]).

There are also theories of asymptotic measures (akin to Young measures, but on the Fourier side) that allow one to pass to the limit in some quadratic expressions, which have been developed by Tartar [325] under the name "H-measures" and, independently,

by Gérard [153], who called them "micro-local defect measures"; see the survey articles [148, 328] for more on this topic.

Problems

9.1 Let X be a complete metric space and let $\mathcal{F}_k \colon X \to \mathbb{R}$, $k \in \mathbb{N}$, be a sequence of functionals. Assume furthermore that the $\mathcal{F}_k$ converge locally uniformly to a functional $\mathcal{F}_\infty \colon X \to \mathbb{R}$, i.e., for all $u \in X$ there exists some open neighborhood $U \Subset X$ of u such that $\sup_{u \in U} |\mathcal{F}_k[u] - \mathcal{F}_\infty[u]| \to 0$ as $k \to \infty$, and that $\mathcal{F}_\infty$ is lower semicontinuous. Prove that the $\mathcal{F}_k$ also Γ-converge to $\mathcal{F}_\infty$.

9.2 Show that the Γ-limit, if it exists, is uniquely determined.

9.3 Let $\mathcal{F}_k \colon X \to \mathbb{R}$, $k \in \mathbb{N}$, be equicoercive functionals on a complete metric space. Prove that $\Gamma\text{-}\liminf_k \mathcal{F}_k$ admits a minimizer and that

$$\min_X \Gamma\text{-}\liminf_k \mathcal{F}_k = \liminf_{k\to\infty} \inf_X \mathcal{F}_k .$$

Also show that

$$\limsup_{k\to\infty} \inf_X \mathcal{F}_k \le \inf_X \Gamma\text{-}\limsup_k \mathcal{F}_k .$$

Find a sequence of equicoercive functionals such that the Γ-upper limit does not fulfill the reverse inequality (this is contrary to the situation for the Γ-lower limit).

9.4 Prove that if $g \in \mathrm{L}^p_{\mathrm{loc}}(\mathbb{R}^d)$ is k-periodic for some $k \in \mathbb{N}$, that is, $g(x + kz) = g(x)$ for $x \in \mathbb{R}^d$ and $z \in \mathbb{Z}$, then the maps $h_\varepsilon(x) := g(x/\varepsilon)$ converge weakly in $\mathrm{L}^1_{\mathrm{loc}}$ to the constant map

$$h_0(x) := \fint_{(0,k)^d} g(y)\, \mathrm{d}y, \qquad x \in \mathbb{R}^d .$$

Hint: Inspect the proof of Lemma 4.16.

9.5 Let $f \colon (0,\infty) \times (0,\infty) \to \mathbb{R}$. Show that there exists a $\delta \colon (0,\infty) \to (0,\infty)$ such that $\delta(\varepsilon) \to 0$ as $\varepsilon \downarrow 0$ and

$$\limsup_{\varepsilon\downarrow 0} f(\varepsilon, \delta(\varepsilon)) \le \limsup_{\delta\downarrow 0} \limsup_{\varepsilon\downarrow 0} f(\varepsilon, \delta).$$

9.6 Let f be as in Marcellini's Homogenization Theorem 9.12. Show that the problem

$$\begin{cases} \text{Minimize} & \displaystyle\int_{(0,1)^d} f(x, F + \nabla\varphi(x))\, \mathrm{d}x \\ \text{over all} & \varphi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d; \mathbb{R}^m) \end{cases}$$

has a solution $\varphi_* \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d; \mathbb{R}^m)$.

9.7 In the situation of the preceding exercise, prove that φ_* is a weak solution of the Euler–Lagrange equation

$$-\operatorname{div}\big[\mathrm{D}_A f(x, F + \nabla\varphi_*(x))\big] = 0, \qquad x \in (0,1)^d,$$

that is, $|\mathrm{D}_A f(x,A)| \le CM(1+|A|^p)$ for $C > 0$ and

$$\int_\Omega \mathrm{D}_A f(x, F + \nabla\varphi_*(x)) : \nabla\psi(x)\,\mathrm{d}x = 0 \qquad \text{for all } \psi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,1)^d;\mathbb{R}^m).$$

9.8 Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $f\colon \mathbb{R}^d \times \mathbb{R}^{m\times d} \to [0,\infty)$ be a Carathéodory integrand. Define the **partial regularization** $f_\delta\colon \mathbb{R}^d \times \mathbb{R}^{m\times d} \to [0,\infty)$ of f as

$$f_\delta(x,A) := \int \eta_\delta(A-B) f(x,B)\,\mathrm{d}B, \qquad (x,A) \in \mathbb{R}^d \times \mathbb{R}^{m\times d},$$

where $(\eta_\delta)_{\delta>0}$ is a radially symmetric and positive family of mollifiers on $\mathbb{R}^{m\times d}$. Show that $f_\delta \to f$ pointwise. Show also that if f satisfies any of the following conditions, then so does f_δ:

(i) $\mu|A|^p - \mu^{-1} \le f(x,A) \le M(1+|A|^p)$ for all $(x,A) \in \mathbb{R}^d \times \mathbb{R}^{m\times d}$ and some $p \in (1,\infty)$, $\mu, M > 0$;
(ii) $x \mapsto f(x,A)$ is $(-1/2,1/2)^d$-periodic for all $A \in \mathbb{R}^{m\times d}$;
(iii) $|f(x,A) - f(x,B)| \le C(1+|A|^{p-1}+|B|^{p-1})|A-B|$ for all $x \in \Omega$, $A, B \in \mathbb{R}^{m\times d}$ and some $C > 0$.

9.9 In the situation of Marcellini's Homogenization Theorem 9.12, show that if $m = 1$ then

$$f_{\mathrm{hom}}(A) = \inf_{\varphi \in \mathrm{W}^{1,p}_{\mathrm{per}}((0,k)^d)} \int_{(0,k)^d} f^{**}(x, A + \nabla\varphi(x))\,\mathrm{d}x,$$

where f^{**} denotes the convex envelope of f with respect to the second argument. Conclude that for $m = 1$ Marcellini's Homogenization Theorem 9.12 also holds without assuming the convexity of f in the second argument. *Hint:* Extend a relaxation theorem to periodic integrands.

9.10 Prove Lemma 9.16.

Part II
Rigidity & Microstructure



Chapter 10
Rigidity

Suppose we are trying to minimize the functional

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x,$$

where $f\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ $(d, m \geq 2)$ is continuous, over a (Sobolev) class of functions $u\colon \Omega \to \mathbb{R}^m$ with prescribed boundary values. As usual, we also assume that $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain. Let us associate with this problem the "differential inclusion"

$$\nabla u(x) \in K := \left\{ A \in \mathbb{R}^{m\times d} \,:\, f(A) = \min f \right\}, \qquad x \in \Omega,$$

where $\min f$ denotes the pointwise minimum of f (assumed to exist in $\mathbb{R}$) and so K is the (usually compact) set of pointwise minimizing values.

If there exists a $u\colon \Omega \to \mathbb{R}^m$ with $\nabla u \in K$ almost everywhere that satisfies the prescribed boundary values, then this u obviously is a minimizer of $\mathcal{F}$. Conversely, a generic minimizer u of $\mathcal{F}$ does not necessarily satisfy $\nabla u \in K$ almost everywhere, particularly if K is "small". However, the differential inclusion should hold at least in some approximate sense and we can thus hope to understand the "shape" of minimizers or approximate minimizers by studying it.

Motivated by these considerations, this chapter is devoted to analyzing solutions to the **differential inclusion**

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \\ \nabla u \in K \quad \text{in } \Omega \end{cases} \tag{10.1}$$

for a non-empty compact set $K \subset \mathbb{R}^{m\times d}$. Obviously, if $A \in K$ and $v_0 \in \mathbb{R}^m$, then $u(x) := v_0 + Ax$ solves (10.1) exactly. The question is whether there are other, non-affine solutions. In the following, we mostly focus on the $\mathrm{W}^{1,\infty}$-theory. By Zhang's Lemma 8.14 this is no restriction as long as K is compact, which is the most common case (see Problem 8.9).

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There are two notions of solution for (10.1) that are relevant for our investigation:

- A map $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is an **exact solution** to (10.1) if the differential inclusion holds pointwise almost everywhere, that is,

$$\nabla u(x) \in K \qquad \text{for a.e. } x \in \Omega.$$

- A uniformly norm-bounded sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is an **approximate solution** to (10.1) if

$$\operatorname{dist}(\nabla u_j, K) \to 0 \quad \text{in measure,}$$

that is, for every $\varepsilon > 0$,

$$\left|\left\{ x \in \Omega \,:\, \operatorname{dist}(\nabla u_j(x), K) > \varepsilon \right\}\right| \to 0 \qquad \text{as } j \to \infty.$$

There is no universal theory for (10.1) that is able to handle all possible sets K, but some techniques are widely applicable and we present a selection of them. Whereas this chapter focuses on situations where we have *rigidity*, that is, exact or approximate solutions to the differential inclusion are necessarily affine, the next two chapters treat the complementary case where non-affine solutions do occur.

10.1 Rigidity for Gradient Inclusions

We start our investigation into differential inclusions with the smallest non-trivial set $K \subset \mathbb{R}^{m\times d}$ ($d, m \geq 2$) and consider the **two-gradient inclusion**

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \\ \nabla u \in K := \{A, B\} \quad \text{in } \Omega, \end{cases} \tag{10.2}$$

where $A, B \in \mathbb{R}^{m\times d}$ with $A \neq B$. By now it should come as no surprise that the behavior of (10.2) depends decisively on whether or not A, B are rank-one connected.

Let us first assume that these matrices are not rank-one connected, i.e., $\operatorname{rank}(A-B) \geq 2$. Then, the Ball–James Rigidity Theorem 7.3 (i) (a) implies that any exact solution $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ to (10.2) is in fact affine,

$$u(x) = v_0 + Fx \qquad \text{with } v_0 \in \mathbb{R}^m \text{ and } F = A \text{ or } F = B.$$

Turning to approximate solutions, let us assume that $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is uniformly norm-bounded and

$$\operatorname{dist}(\nabla u_j, \{A, B\}) \to 0 \quad \text{in measure.}$$

Then, assertion (ii) of the Ball–James Rigidity Theorem yields that, up to selecting a subsequence,

$$\nabla u_j \to A \quad \text{in measure} \qquad \text{or} \qquad \nabla u_j \to B \quad \text{in measure.}$$

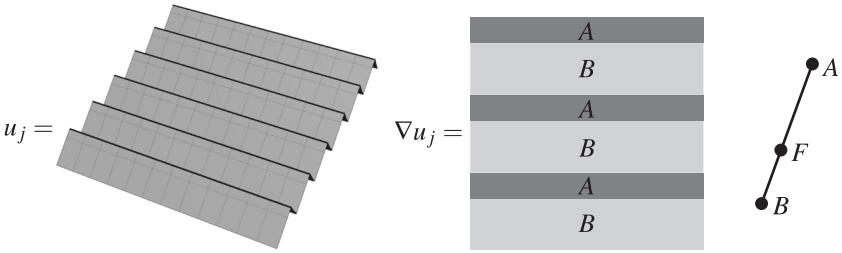


Fig. 10.1 The laminate u_j: map, gradient schematic, and rank-one diagram

We now consider the case $\operatorname{rank}(A - B) \leq 1$, i.e., we assume that there exist $a \in \mathbb{R}^m \setminus \{0\}$, $n \in \mathbb{S}^{d-1}$ such that

$$B - A = a \otimes n.$$

To see what kind of non-trivial solutions to (10.2) are now possible, recall the lamination construction from the proof of Proposition 5.6. There, on an oriented unit-volume cube Q_n with two faces orthogonal to n, we constructed an exact and non-affine solution of (10.2). In fact, in the same way one can construct many solutions: For any $\theta \in (0, 1)$ let $F := \theta A + (1 - \theta)B$ and define $u_j \in \mathrm{W}^{1,\infty}(Q_n; \mathbb{R}^m)$, $j \in \mathbb{N}$, as the map

$$u_j(x) := Fx + \frac{1}{j}\psi(jx \cdot n)a \qquad x \in Q_n,$$

where

$$\psi(t) := \begin{cases} -(1-\theta)t & \text{if } t - \lfloor t \rfloor \in [0, \theta), \\ \theta t - \theta & \text{if } t - \lfloor t \rfloor \in (\theta, 1), \end{cases}$$

see Figure 10.1. For the gradients we compute

$$\nabla u_j(x) = \begin{cases} F - (1-\theta)a \otimes n = A & \text{if } jx \cdot n - \lfloor jx \cdot n \rfloor \in [0, \theta), \\ F - \theta a \otimes n = B & \text{if } jx \cdot n - \lfloor jx \cdot n \rfloor \in (\theta, 1). \end{cases}$$

Clearly, any such u_j solves (10.2) exactly and it holds that $u_j \overset{*}{\rightharpoonup} Fx$ in $\mathrm{W}^{1,\infty}$, but not $u_j \to Fx$ in measure.

We can further modify the sequence $(u_j)_j$, without disturbing the convergence assertions, to agree with Fx on ∂Q_n by a standard cut-off construction near the boundary (as in Step 3 of Lemma 4.14). If we embed a rescaled copy of Q_n in any given Lipschitz domain Ω and extend the (modified) maps u_j by Fx, we observe that non-trivial approximate solutions also exist in other domains.

These observations inspire the following definitions:

- The differential inclusion (10.1) is called **rigid for exact solutions** if all of its exact solutions (without assuming any boundary conditions) are affine.
- The differential inclusion (10.1) is called **rigid for approximate solutions** if for all approximate solutions $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega; \mathbb{R}^m)$ with

$$u_j \overset{*}{\rightharpoonup} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m) \qquad \text{and} \qquad u_j|_{\partial\Omega} = Fx \text{ for some } F \in \mathbb{R}^{m\times d},$$

it holds that

$$\nabla u_j \to \nabla u \quad \text{in measure} \qquad \text{and} \qquad \nabla u \equiv \text{const} = F. \tag{10.3}$$

- The differential inclusion (10.1) is called **strongly rigid** if in the situation of the previous definition, (10.3) holds without any condition on the boundary values $u_j|_{\partial\Omega}$.

Neither of the first two notions of rigidity implies the other, as we will see in Theorem 10.16, Theorem 11.1, and Problem 10.5. Of course, strong rigidity implies the other two rigidity notions.

With these definitions we can rephrase the Ball–James Rigidity Theorem 7.3 as follows:

Theorem 10.1 (Ball–James 1987 [39])

(i) If $\operatorname{rank}(A-B) \geq 2$, *then the two-gradient inclusion* (10.2) *is strongly rigid; in particular, it is rigid for exact and for approximate solutions.*
(ii) If $\operatorname{rank}(A-B) = 1$, *then the two-gradient inclusion* (10.2) *is not rigid for exact and not rigid for approximate solutions.*

This result provides a complete solution to the first non-trivial case of the so-called N**-gradient inclusion**:

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \\ \nabla u \in K := \{A_1,\dots,A_N\} \quad \text{in } \Omega \end{cases} \tag{10.4}$$

for distinct matrices $A_1,\dots,A_N \in \mathbb{R}^{m\times d}$. As discussed above for the two-gradient inclusion, when there is a rank-one connection in K, i.e., if $A_i - A_j = a\otimes n$ for some $i,j \in \{1,\dots,N\}$, $a \in \mathbb{R}^m \setminus \{0\}$, $n \in \mathbb{S}^{d-1}$, then we can construct laminates whose gradients oscillate between A_i and A_j. In this case, the differential inclusion is always not rigid for exact and not rigid for approximate solutions. In this chapter we focus on the situation where there are no rank-one connections in K, that is, the following **incompatibility relation** holds:

$$\operatorname{rank}(A_i - A_j) \geq 2 \qquad \text{for all } i,j \in \{1,\dots,N\} \text{ with } i \neq j. \tag{10.5}$$

We will come back to the N-gradient inclusion in Sections 10.4 and 10.5.

10.2 Linear Inclusions

We next consider **linear differential inclusions**, where the gradient of a $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$-map, $p \in [1,\infty]$, is restricted to lie in a linear subspace $L \subset \mathbb{R}^{m\times d}$, that is,

$$\begin{cases} u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m), \\ \nabla u \in L \quad \text{in } \Omega. \end{cases} \tag{10.6}$$

Since L is not compact, it is natural to look for solutions also in spaces of maps with unbounded gradients.

We first observe that if there is a nontrivial rank-one connection in L, that is, there are $A, B \in L$ with $\operatorname{rank}(A-B) = 1$, then we can construct a laminate whose gradient oscillates between A and B; this is done in the same fashion as in the previous sections. Hence, in this case, the differential inclusion is not rigid for both exact and approximate solutions. We can even construct smooth non-constant solutions:

Example 10.2 Let $P_0 := a \otimes n$ with $a \in \mathbb{R}^m$, $n \in \mathbb{S}^{d-1}$. Then, for all $j \in \mathbb{N}$, the maps

$$u_j(x) := \frac{1}{j}\sin(jx \cdot n)a, \qquad x \in \mathbb{R}^d,$$

satisfy

$$\nabla u_j(x) = \cos(jx \cdot n)(a \otimes n) \in \operatorname{span}\{a \otimes n\} =: L.$$

We now prove a general result about (10.6). While necessarily falling short of rigidity, it still expresses the *ellipticity* of the differential inclusion, which can be interpreted as a weaker version of rigidity.

Theorem 10.3 *Let $L \subset \mathbb{R}^{m\times d}$ be a linear subspace that contains no rank-one connections, that is, for all $A, B \in L$ with $A \neq B$ it holds that $\operatorname{rank}(A-B) \geq 2$.*

(i) If $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, $p \in [1,\infty]$, satisfies (10.6) exactly, then u is smooth.
(ii) If $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ ($u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ if $p = \infty$) and

$$\operatorname{dist}(\nabla u_j, L) \to 0 \quad \text{in measure,}$$

then

$$\nabla u_j \to \nabla u \quad \text{in measure} \qquad \text{and} \qquad \nabla u \in L \quad \text{a.e.}$$

Moreover, u is smooth.

Proof. The idea of the proof is that $\nabla u \in L$ can be written as the PDE

$$\mathbf{P}(\nabla u) = 0, \tag{10.7}$$

where we denote by $\mathbf{P}\colon \mathbb{R}^{m\times d} \to \mathbb{R}^{m\times d}$ the orthogonal projection onto the orthogonal complement $L^\perp$ of L. It turns out that under the assumptions of the theorem this PDE is *elliptic* and as such the usual higher-integrability estimates apply. However, since the argument is based on the Fourier-transform, which only operates on the whole space, we need to cut off suitably and this adds some complexity to the implementation of this strategy.

We will only prove the statement for $p \in (1,\infty)$. The case $p = \infty$ can obviously be reduced to the case $p = 2$ and for $p = 1$ the proof is the task of Problem 10.3.

Ad (i). Step 1. For every smooth cut-off function $\varrho \in \mathrm{C}_c^\infty(\Omega;[0,1])$ with $\varrho \equiv 1$ on an open set $U \Subset \Omega$, the function $w := \varrho u$ satisfies

$$\nabla w = \varrho \nabla u + u \otimes \nabla \varrho.$$

Here, as elsewhere, we write the technically superfluous symbol "$\otimes$" even between a column and a row vector to emphasize that $u \otimes \nabla \varrho$ is a *matrix* field. Combining this equation with (10.7), we get

$$\mathbf{P}(\nabla w) = \mathbf{P}(u \otimes \nabla \varrho) =: R \in \mathrm{L}^p(\mathbb{R}^d;\mathbb{R}^{m\times d}). \tag{10.8}$$

Hence, applying the Fourier transform (see Appendix A.6) to both sides of (10.8) and considering $\mathbf{P}$ to be identified with its complexification (that is, $\mathbf{P}(A + \mathrm{i}B) := \mathbf{P}(A) + \mathrm{i}\mathbf{P}(B)$ for $A, B \in \mathbb{R}^{m\times d}$), we arrive at

$$\mathbf{P}\big(\widehat{\nabla w}(\xi)\big) = \widehat{R}(\xi), \qquad \xi \in \mathbb{R}^d. \tag{10.9}$$

Step 2. We will now rewrite (10.9) as the multiplier equation

$$\widehat{\nabla w}(\xi) = \mathbf{M}(\xi)\widehat{R}(\xi), \qquad \xi \in \mathbb{R}^d \setminus \{0\}, \tag{10.10}$$

where $\mathbf{M}(\xi)\colon \mathbb{R}^{m\times d} \to \mathbb{R}^{m\times d}$ is smooth and positively 0-homogeneous in ξ. For this, observe first that $\mathbf{P}(a \otimes \xi) \neq 0$ for any $a \in \mathbb{C}^m \setminus \{0\}$, $\xi \in \mathbb{R}^d \setminus \{0\}$ by the assumption that L does not contain rank-one connections. Thus, employing also the homogeneity of the projection, for some constant $C > 0$ we have the *ellipticity estimate*

$$|a \otimes \xi| \le C|\mathbf{P}(a \otimes \xi)| \qquad \text{for all } a \in \mathbb{C}^m, \xi \in \mathbb{R}^d.$$

The (complexified) projection $\mathbf{P}\colon \mathbb{C}^{m\times d} \to \mathbb{C}^{m\times d}$ has kernel $L^{\mathbb{C}} := \operatorname{span}_{\mathbb{C}} L$, the complex span of L, which in the following we also denote just by L. Hence, $\mathbf{P}$ descends to the quotient

$$[\mathbf{P}]\colon \mathbb{C}^{m\times d}/L \to \operatorname{ran}\mathbf{P},$$

and $[\mathbf{P}]$ is an invertible linear map.

Assume that

$$L = \operatorname{span}\{F_1, \dots, F_k\}$$

and for $\xi \in \mathbb{R}^d \setminus \{0\}$ extend this to a $\mathbb{C}$-basis

$$\{F_1, \dots, F_k, \mathrm{e}_1 \otimes \xi, \dots, \mathrm{e}_d \otimes \xi, G_{d+1}(\xi), \dots, G_{md-k}(\xi)\}$$

of $\mathbb{C}^{m\times d}$ with the additional property that the matrices $G_{d+1}(\xi), \dots, G_{md-k}(\xi)$ depend smoothly on ξ and are 1-homogeneous in ξ (i.e., $G_{d+1}(\alpha\xi) = \alpha G_{d+1}(\xi)$ for all $\alpha \in \mathbb{R}$). In this context note that in particular $\mathrm{e}_i \otimes \xi \notin L$ by assumption.

Furthermore, for $\xi \in \mathbb{R}^d \setminus \{0\}$ denote by $\mathbf{Q}(\xi)\colon \mathbb{C}^{m\times d} \to \mathbb{C}^{m\times d}$ the (non-orthogonal) projection with

$$\begin{aligned} \operatorname{ran} \mathbf{Q}(\xi) &= \operatorname{span}\big\{\mathrm{e}_1 \otimes \xi, \dots, \mathrm{e}_d \otimes \xi, G_{d+1}(\xi), \dots, G_{md-k}(\xi)\big\}, \\ \ker \mathbf{Q}(\xi) &= L. \end{aligned}$$

If we interpret $\mathrm{e}_1 \otimes \xi, \dots, \mathrm{e}_d \otimes \xi, G_{d+1}(\xi), \dots, G_{md-k}(\xi)$ as vectors in $\mathbb{R}^{md}$ and collect them into the columns of the matrix $X(\xi) \in \mathbb{R}^{md\times(md-k)}$, and if we further let $Y \in \mathbb{R}^{md\times(md-k)}$ be a matrix whose columns comprise an orthonormal basis of $L^\perp$, then, up to a change in sign for one of the G_l's, there exists a constant $c > 0$ such that

$$\det(Y^\mathsf{T} X(\xi)) \geq c > 0 \qquad \text{for all } \xi \in \mathbb{S}^{d-1}.$$

Indeed, if $\det(Y^\mathsf{T} X(\xi))$ was not uniformly bounded away from zero for all $\xi \in \mathbb{S}^{d-1}$, then by compactness there would exist a $\xi_0 \in \mathbb{S}^{d-1}$ with $\det(Y^\mathsf{T} X(\xi_0)) = 0$, a contradiction. We can then write $\mathbf{Q}(\xi)$ explicitly as

$$\mathbf{Q}(\xi) = X(\xi)(Y^\mathsf{T} X(\xi))^{-1} Y^\mathsf{T}.$$

This implies that $\mathbf{Q}(\xi)$ depends positively 0-homogeneously and smoothly on $\xi \in \mathbb{R}^d \setminus \{0\}$.

Also $\mathbf{Q}(\xi)$ for $\xi \neq 0$ descends to the quotient

$$[\mathbf{Q}(\xi)]\colon \mathbb{C}^{m\times d}/L \to \operatorname{ran} \mathbf{Q}(\xi),$$

which is now invertible. It is not difficult to see that $\xi \mapsto [\mathbf{Q}(\xi)]$ is still positively 0-homogeneous and smooth in $\xi \neq 0$ (by utilizing the basis given above). Since $\widehat{w}(\xi) \otimes \xi \in \operatorname{ran} \mathbf{Q}(\xi)$ by construction, we have

$$[\mathbf{Q}(\xi)]^{-1}(\widehat{w}(\xi) \otimes \xi) = [\widehat{w}(\xi) \otimes \xi],$$

where $[\widehat{w}(\xi) \otimes \xi]$ designates the equivalence class of $\widehat{w}(\xi) \otimes \xi$ in $\mathbb{C}^{m\times d}/L$. This fact in conjunction with the formula

$$\widehat{\nabla w}(\xi) = (2\pi \mathrm{i})\, \widehat{w}(\xi) \otimes \xi, \qquad \xi \in \mathbb{R}^d,$$

which follows directly from the definition of the Fourier transform (see Appendix A.6), allows us to deduce from (10.9) that

$$(2\pi \mathrm{i})\, [\mathbf{P}][\mathbf{Q}(\xi)]^{-1}(\widehat{w}(\xi) \otimes \xi) = \widehat{R}(\xi), \qquad \xi \in \mathbb{R}^d \setminus \{0\}.$$

Then,

$$\widehat{\nabla w}(\xi) = (2\pi \mathrm{i})\, \widehat{w}(\xi) \otimes \xi = [\mathbf{Q}(\xi)][\mathbf{P}]^{-1} \widehat{R}(\xi), \qquad \xi \in \mathbb{R}^d \setminus \{0\}.$$

The multiplier $\mathbf{M}(\xi)\colon \mathbb{R}^{m\times d} \to \mathbb{R}^{m\times d}$ for $\xi \in \mathbb{R}^d \setminus \{0\}$ is thus given by

$$\mathbf{M}(\xi) := [\mathbf{Q}(\xi)][\mathbf{P}]^{-1},$$

which is smooth and positively 0-homogeneous in ξ. Consequently, we have shown the multiplier equation (10.10).

Step 3. If $p = 2$ we may use Plancherel's identity (A.6) to estimate

$$\|\nabla w\|_{\mathrm{L}^2} = \|\widehat{\nabla w}\|_{\mathrm{L}^2} \le \|\mathbf{M}\|_\infty \|\widehat{R}\|_{\mathrm{L}^2} = \|\mathbf{M}\|_\infty \|R\|_{\mathrm{L}^2} \le C\|u\|_{\mathrm{L}^2}$$

for some constant $C > 0$ that depends on L though the L^∞-norm of $\mathbf{M}$. If $p \in (1, \infty)$, we likewise infer from the Mihlin Multiplier Theorem A.36 that

$$\|\nabla w\|_{\mathrm{L}^p} \le C\|\mathbf{M}\|_{\mathrm{C}^{\lfloor d/2 \rfloor + 1}} \|R\|_{\mathrm{L}^p} \le C\|u\|_{\mathrm{L}^p}.$$

Also using $\varrho \nabla u = \nabla w - u \otimes \nabla \varrho$, we thus arrive at the estimate

$$\|\nabla u\|_{\mathrm{L}^p(U)} \le \|\nabla w\|_{\mathrm{L}^p(\Omega)} + \|u \otimes \nabla \varrho\|_{\mathrm{L}^p(\Omega)} \le C\|u\|_{\mathrm{L}^p(\Omega)}$$

for some constant $C > 0$. Differentiating (10.7) in a weak sense (or using mollification), applying the above argument to $\partial_j u$ ($j = 1, \dots, d$), and iterating (*bootstrapping* as in the argument for Corollary 3.14), we conclude that u is smooth.

Ad (ii). The assumptions imply

$$u_j \to u, \ \mathbf{P}(\nabla u_j) \to 0 \quad \text{in } \mathrm{L}^q \text{ for all } q \in (1, p).$$

Indeed, the first assertion follows from the Rellich–Kondrachov Theorem A.29. For the second assertion we observe by Vitali's Convergence Theorem A.12 that

$$\int_\Omega |\mathbf{P}(\nabla u_j)|^q \, \mathrm{d}x \le \int_\Omega \operatorname{dist}(\nabla u_j, L)^q \, \mathrm{d}x \to 0$$

since the ∇u_j are L^p-uniformly bounded, and hence q-equiintegrable by Markov's inequality.

Moreover, by the weak continuity of the linear operator $\mathbf{P}$, we get that

$$\mathbf{P}(\nabla u) = 0 \quad \text{a.e. in } \Omega. \tag{10.11}$$

Thus, for every smooth cut-off function $\varrho \in \mathrm{C}_c^\infty(\Omega; [0, 1])$ we have

$$\mathbf{P}\big(\nabla(\varrho(u_j - u))\big) = \varrho \mathbf{P}(\nabla u_j) + \mathbf{P}((u_j - u) \otimes \nabla \varrho) \to 0 \quad \text{in } \mathrm{L}^q.$$

Consequently, by a similar estimate as before, the $(\mathrm{L}^q \to \mathrm{L}^q)$-boundedness of the Fourier multiplier operator corresponding to the multiplier $\mathbf{M}(\xi)$ from (10.10) implies that $\nabla(\varrho(u_j - u)) \to 0$ in L^q, hence also $\nabla u_j \to \nabla u$ in measure by Lemma A.9. Together with (10.11) and (i), this yields the assertions of (ii). □

The preceding theorem has some immediate and interesting applications in the theory of elliptic PDEs such as the following.

Example 10.4 Let

$$L := \mathbb{R}^{d \times d}_{\mathrm{sym,dev}} := \big\{ A \in \mathbb{R}^{d \times d} \ : \ A^\mathsf{T} = A, \ \operatorname{tr} A = 0 \big\}.$$

This linear space does not contain any rank-one connections: Assume that $\operatorname{rank}(A-B) \le 1$ for $A, B \in L$. Then, $A - B = a \otimes b$ for some $a, b \in \mathbb{R}^d$ and since $a \cdot b = \operatorname{tr}(a \otimes b) = \operatorname{tr}(A - B) = 0$, we must have that a is orthogonal to b. On the other hand, $b \otimes a = (A-B)^{\mathsf{T}} = A - B = a \otimes b$, so a is also parallel to b, yielding $A = B$.

Any $v \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^d)$, $p \in [1,\infty]$, that satisfies

$$\nabla v \in L \quad \text{a.e.}$$

is the gradient of a map $u \in \mathrm{W}^{2,p}(\Omega)$ satisfying

$$\Delta u = 0 \quad \text{a.e.},$$

i.e., u is harmonic, and vice versa. Indeed, if $\nabla v \in L$ almost everywhere, the symmetry requirement in the definition of L implies that ∇v is in fact a Hessian, $\nabla v = \nabla^2 u$ for a $u \in \mathrm{W}^{2,p}(\Omega)$. Then, $\Delta u = \operatorname{tr} \nabla^2 u = \operatorname{tr} \nabla v = 0$ almost everywhere. The other direction is similar.

Thus, Theorem 10.3 (i) yields a well-known result, namely that all (weakly) harmonic maps must be smooth; we already proved this in Example 3.16 using classical elliptic regularity theory. The conclusion (ii) of Theorem 10.3 is perhaps less well-known for harmonic maps. Clearly, we can find examples of non-affine harmonic maps in $\mathrm{W}^{1,\infty}(\Omega)$, so Theorem 10.3 cannot in general be strengthened to yield full rigidity for exact solutions.

Another related example of a linear differential inclusion is the topic of Problem 10.1.

A special case of the linear inclusion (10.6), which is often interesting, is the case when the linear space L is one-dimensional, i.e.,

$$L = \operatorname{span}\{P_0\} = \left\{ \lambda P_0 \,:\, \lambda \in \mathbb{R} \right\}, \qquad P_0 \in \mathbb{R}^{m\times d} \text{ a fixed matrix.}$$

The resulting **polar differential inclusion**

$$\begin{cases} u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m), \\ \nabla u \in \operatorname{span}\{P_0\} \quad \text{in } \Omega, \end{cases} \tag{10.12}$$

where as before $p \in [1,\infty]$, could equivalently be stated by requiring that

$$\nabla u = P_0 g \quad \text{a.e.}$$

for some scalar function $g\colon \Omega \to \mathbb{R}$, which then is an additional unknown in the problem. If we suppose (without loss of generality) that $|P_0| = 1$, this formulation explains the name "polar inclusion". An example of (10.12) was already exhibited in Example 10.2.

The reason why we are interested in polar differential inclusions is that they occur naturally in a wide variety of variational problems as soon as we blow-up ("magnify") around a point; see the proof of Proposition 5.13 for an example of this technique.

For the polar inclusion, stronger results than for the general linear inclusion are available, which are summarized in the following theorem.

Theorem 10.5 *Let $\Omega \subset \mathbb{R}^d$ be open, bounded, and connected.*

(i) If $\operatorname{rank} P_0 \geq 2$, *then* (10.12) *is strongly rigid:*

(a) If $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$, $p \in [1,\infty]$, satisfies (10.12) *exactly, then u is affine.*

(b) If $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ ($u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ if $p = \infty$) and

$$\operatorname{dist}(\nabla u_j, \operatorname{span}\{P_0\}) \to 0 \quad \text{in measure,}$$

then

$$\nabla u_j \to \nabla u \quad \text{in measure}$$

and u is affine.

(ii) If $\operatorname{rank} P_0 = 1$, *then* (10.12) *is not rigid for exact solutions and not rigid for approximate solutions.*

The proof of this theorem is the task of Problem 10.4.

10.3 The Young Measure Approach

Before we can investigate more complicated differential inclusions, we need to develop a more sophisticated approach to rigidity, which is built on the theory of Young measures.

To motivate the basic strategy, we associate with our compact set $K \subset \mathbb{R}^{m\times d}$ the functional $\mathcal{F}^K \colon \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m) \to \mathbb{R}$, given by

$$\mathcal{F}^K[u] := \int_\Omega \operatorname{dist}(\nabla u(x), K)^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m).$$

Then, just like in Chapter 7, we can ask about the relaxation $\mathcal{F}^K_*$ of $\mathcal{F}^K$. Theorem 7.8 tells us that

$$\mathcal{F}^K_*[u] = \int_\Omega Q \operatorname{dist}(\nabla u(x), K)^2 \,\mathrm{d}x,$$

where $Q\operatorname{dist}(A,K)^2 := Q[\operatorname{dist}(\bullet,K)^2](A)$ is the quasiconvex envelope of the integrand $\operatorname{dist}(\bullet,K)^2$, which was defined in (7.1), evaluated at $A \in \mathbb{R}^{m\times d}$. Clearly, $Q\operatorname{dist}(A,K)^2 \geq 0$ and $Q\operatorname{dist}(A,K)^2 = 0$ if $\operatorname{dist}(A,K)^2 = 0$, that is, $A \in K$.

If we apply the present chapter's overarching philosophy that the pointwise minimizer set for the integrand determines the admissible microstructure, we can now define a *relaxation hull* $\overline{K}$ of K via

$$\overline{K} := \left\{ F \in \mathbb{R}^{m\times d} \;:\; Q\operatorname{dist}(F,K)^2 = 0 \right\} \supset K.$$

According to Proposition 7.11, for all $F \in \mathbb{R}^{m\times d}$ there exists a homogeneous gradient Young measure $\mu^F \in \mathbf{GY}^2(B_1;\mathbb{R}^{m\times d})$ with $[\mu^F] = F$ and

$$Q\operatorname{dist}(F,K)^2 = \int \operatorname{dist}(\bullet,K)^2 \,\mathrm{d}\mu^F.$$

By Zhang's Lemma 8.14, we may also assume that $\mu^F \in \mathbf{GY}^\infty(B_1;\mathbb{R}^{m\times d})$.

If $F \in \overline{K}$, then μ^F must be supported on K. On the other hand, for any homogeneous gradient Young measure $\mu \in \mathbf{GY}^\infty(B_1;\mathbb{R}^{m\times d})$ with $[\mu] = F \in \mathbb{R}^{m\times d}$ and $\operatorname{supp}\mu \subset K$, there exists a sequence $(\psi_j)_j \subset \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$ with $\|\nabla\psi_j\|_{\mathrm{L}^\infty}$ uniformly bounded and $F + \nabla\psi_j \xrightarrow{\mathbf{Y}} \mu$ by Zhang's Lemma 8.14. Via Proposition 7.1 and the Young measure representation of limits we thus obtain

$$\begin{aligned} Q\operatorname{dist}(F,K)^2 &\le \lim_{j\to\infty} \fint_{B_1} \operatorname{dist}(F+\nabla\psi_j(z),K)^2 \,\mathrm{d}z \\ &= \int \operatorname{dist}(\bullet,K)^2 \,\mathrm{d}\mu \\ &= 0. \end{aligned}$$

Hence, we have shown that $\overline{K}$ consists precisely of all barycenters of homogeneous $\mathrm{W}^{1,\infty}$-gradient Young measures that are supported on K:

Lemma 10.6 *The relaxation hull*

$$\overline{K} := \left\{ F \in \mathbb{R}^{m\times d} \;:\; Q\operatorname{dist}(F,K)^2 = 0 \right\} \supset K$$

of a compact set $K \subset \mathbb{R}^{m\times d}$ equals the ***quasiconvex hull*** K^{qc} *of K, defined via*

$$K^{\mathrm{qc}} := \left\{ [\mu] \;:\; \mu \in \mathcal{M}^{\mathrm{qc}}(K) \right\}, \tag{10.13}$$

where

$$\mathcal{M}^{\mathrm{qc}}(K) := \left\{ \mu \in \mathbf{GY}^\infty(B_1;\mathbb{R}^{m\times d}) \;:\; \mu \text{ homogeneous, } \operatorname{supp}\mu \subset K \right\}.$$

The set $\mathcal{M}^{\mathrm{qc}}(K)$ is a subset of $\mathcal{M}^1(K)$, the set of all probability measures supported on K. By the Kinderlehrer–Pedregal Theorem 8.11 it can be characterized as

$$\mathcal{M}^{\mathrm{qc}}(K) = \left\{ \mu \in \mathcal{M}^1(K) \;:\; h([\mu]) \le \int h \,\mathrm{d}\mu \text{ for all quasiconvex } h \in \mathrm{C}(\mathbb{R}^{m\times d}) \right\},$$

which explains the notation "$\mathcal{M}^{\mathrm{qc}}(K)$". It is the task of Problem 10.6 to show that for a compact set $K \subset \mathbb{R}^{m\times d}$, the quasiconvex hull K^{qc} is also compact.

Example 10.7 Let $K = \{A, B\}$ with $\operatorname{rank}(A-B) \ge 2$. Then, by Theorem 10.1, $K^{\mathrm{qc}} = K$. Thus, in general $K^{\mathrm{qc}} \neq K^{**}$, the convexification of K.

The above discussion leads to the *Young measure approach* to the study of approximate rigidity for the differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \\ \nabla u \in K \quad \text{in } \Omega, \end{cases} \tag{10.14}$$

with $K \subset \mathbb{R}^{m\times d}$ compact and non-empty. The basic ideas are expressed in the following results.

Lemma 10.8 *Assume that every measure in $\mathcal{M}^{\mathrm{qc}}(K)$ is a Dirac mass. Then:*

(i) $K^{\mathrm{qc}} = K$.
(ii) If $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ *satisfies*

$$\nabla u \in K \quad a.e. \qquad and \qquad u|_{\partial\Omega} = Fx$$

for some $F \in \mathbb{R}^{m\times d}$, then $\nabla u = F$ almost everywhere.

It is important to note that (ii) is not the same as rigidity for exact solutions since here we additionally assume linear boundary values.

Proof. Ad (i). For $F \in K^{\mathrm{qc}}$ by definition there exists a probability measure $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ with $[\mu] = F$. By assumption, μ is the Dirac mass δ_F and, since $\operatorname{supp}\mu \subset K$, necessarily $F \in K$. Thus, $K^{\mathrm{qc}} \subset K$; the opposite inclusion is trivial.

Ad (ii). Thanks to the linear boundary values of u, we may apply the Riemann–Lebesgue lemma for gradient Young measures, Lemma 4.16. This yields $\overline{\nu} := \overline{\delta[\nabla u]} \in \mathcal{M}^{\mathrm{qc}}(K)$ with $[\overline{\nu}] = F$. By assumption, $\overline{\nu} = \delta_F$. Hence, ∇u must be constant (see (4.19)). □

Proposition 10.9

(i) The differential inclusion (10.14) is rigid for approximate solutions if and only if every measure in $\mathcal{M}^{\mathrm{qc}}(K)$ is a Dirac mass.
(ii) The differential inclusion (10.14) is strongly rigid if and only if it is rigid for exact solutions and every measure in $\mathcal{M}^{\mathrm{qc}}(K)$ is a Dirac mass.
(iii) The differential inclusion (10.14) is strongly rigid if and only if it is rigid for exact solutions and rigid for approximate solutions.

In assertion (ii), the additional assumption of exact rigidity cannot be dispensed with, see Problem 10.5.

Proof. Ad (i). Assume first that (10.14) is rigid for approximate solutions. To see the assertion concerning Young measures, let $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ with $[\mu] = F \in \mathbb{R}^{m\times d}$. By Zhang's Lemma 8.14 there exists a sequence $(\psi_j)_j \subset \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$ such that the norms $\|\nabla\psi_j\|_{\mathrm{L}^\infty}$ are uniformly bounded and $F + \nabla\psi_j \xrightarrow{\mathbf{Y}} \mu$. Moreover,

$$\psi_j \overset{*}{\rightharpoonup} 0 \quad \text{in } \mathrm{W}^{1,\infty} \qquad \text{and} \qquad \operatorname{dist}(F + \nabla\psi_j, K) \to 0 \quad \text{in measure}$$

by Lemma 4.13. Thus, the rigidity for approximate solutions implies that

$$F + \nabla\psi_j \to F \quad \text{in measure.}$$

Another application of Lemma 4.13 then yields $\mu = \delta_F$.

For the converse implication, assume that every measure in $\mathcal{M}^{\mathrm{qc}}(K)$ is a Dirac mass and let $(u_j)_j \subset \mathrm{W}^{1,\infty}_{Fx}(\Omega;\mathbb{R}^m)$ for some $F \in \mathbb{R}^{m\times d}$ such that

$$u_j \overset{*}{\rightharpoonup} u \in \mathrm{W}^{1,\infty}_{Fx}(\Omega;\mathbb{R}^m) \qquad \text{and} \qquad \operatorname{dist}(\nabla u_j, K) \to 0 \quad \text{in measure.}$$

Up to selecting a subsequence, we may further assume

$$\nabla u_j \overset{\mathbf{Y}}{\to} \nu \in \mathbf{GY}^{\infty}(\Omega;\mathbb{R}^{m\times d})$$

with $[\nu_x] = \nabla u(x)$ and $\operatorname{supp}\nu_x \subset K$ for almost every $x \in \Omega$ by Lemma 4.13.

Almost every ν_x is a homogeneous gradient Young measure in its own right, $\nu_x \in \mathcal{M}^{\mathrm{qc}}(K)$, via the blow-up principle in Proposition 5.13. Then, by assumption, $\nu_x = \delta_{\nabla u(x)}$ and so, once more Lemma 4.13 implies that $\nabla u_j \to \nabla u$ in measure. Moreover, since $u|_{\partial\Omega} = Fx$, part (ii) of the preceding Lemma 10.8 immediately yields that ∇u is constant almost everywhere. Thus, we have established that (10.14) is rigid for approximate solutions.

Ad (ii). The proof of the first implications ("only if") follows from the first implication in (i) since strong rigidity implies rigidity for exact and for approximate solutions.

For the other direction ("if") we trace through the preceding proof for the second implication of (i), but no longer assume the boundary condition $u_j|_{\partial\Omega} = Fx$ for some $F \in \mathbb{R}^{m\times d}$. In fact, the only place where we used this condition was when we proved that ∇u is a constant (via Lemma 10.8) and this now follows directly from the assumed rigidity for exact solutions.

Ad (iii). This is a direct consequence of (i) and (ii). □

It is sometimes possible by "testing" with carefully selected quasiconvex functions to show that any Young measure supported on a set K necessarily must be a Dirac mass. Together with exact rigidity, which must be established separately, this then implies strong rigidity and $K^{\mathrm{qc}} = K$. We will see examples of this strategy in the following sections.

Example 10.10 In the physical context of crystalline microstructure as described in Section 1.9, let us assume that the integrand W looks locally similar to the squared distance from the pointwise minimizer set

$$K := W^{-1}(0) = \big\{ A \in \mathbb{R}^{3\times 3} \;:\; W(A) = \min W \big\},$$

that is,

$$W \approx \operatorname{dist}(\bullet, K)^2 \quad \text{near } K.$$

From the above reasoning we get that precisely for $F \in \overline{K} = K^{\mathrm{qc}}$ there exists a sequence $(\psi_j)_j \subset \mathrm{W}^{1,\infty}_0(\Omega;\mathbb{R}^m)$ with $\|\nabla\psi_j\|_{\mathrm{L}^\infty}$ uniformly bounded, and such that

$$\int_\Omega W(F + \nabla\psi_j(x), K)^2 \,\mathrm{d}x \approx \int_\Omega \operatorname{dist}(F + \nabla\psi_j(x), K)^2 \,\mathrm{d}x \to 0$$

as $j \to \infty$. Consequently, the material can realize the linear boundary values Fx with almost no expenditure of energy. However, the internal microstructure of the material

may be very complicated, as expressed by fine oscillations in $(\psi_j)_j$. The upshot of this discussion is that if we want to understand the macroscopic behavior of a crystal that develops microstructure, it is the quasiconvex hull K^{qc} of the pointwise minimizer set K that determines the energetic behavior. For instance, if K^{qc} contains an open set, then we have very soft, fluid-like behavior.

10.4 Three-Gradient Inclusions

We now return to the N-gradient inclusion (10.4). It turns out that the three-gradient inclusion behaves similarly to the two-gradient inclusion, but the proof of this fact is much more involved.

Theorem 10.11 (Šverák 1991 [306]) *Assume that $K := \{A_1, A_2, A_3\} \subset \mathbb{R}^{m\times d}$ contains no rank-one connection. Then, the inclusion (10.4) is strongly rigid and $K^{\mathrm{qc}} = K$.*

We will show the rigidity for exact solutions and the rigidity for approximate solutions separately, which suffices for strong rigidity by Proposition 10.9 (iii). We start with the rigidity for exact solutions, for which we first prove a dimension-reduction lemma.

Lemma 10.12 *Let $K = \{A_1, \dots, A_N\} \subset \mathbb{R}^{m\times d}$ contain no rank-one connections. If $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is a non-affine map with $\nabla u \in K$ almost everywhere, then there exists a set $\widetilde{K} = \{\widetilde{A}_1, \dots, \widetilde{A}_N\} \subset \mathbb{R}^{2\times 2}$ without rank-one connections and a non-affine map $\widetilde{u} \in \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^2)$ with $\nabla\widetilde{u} \in \widetilde{K}$ almost everywhere.*

Proof. As u is not affine, we may find $y_1, y_2 \in B_\varepsilon(x_0) \subset B_{3\varepsilon}(x_0) \subset \Omega$ for a sufficiently small $\varepsilon > 0$ and $x_0 \in \Omega$ such that

$$u(y_1 + y_2 - x_0) + u(x_0) \neq u(y_1) + u(y_2).$$

Now pick $P_0 \in \mathbb{R}^{d\times 2}$, $Q_0 \in \mathbb{R}^{2\times m}$ with

$$P_0 \mathrm{e}_1 = y_1 - x_0, \qquad P_0 \mathrm{e}_2 = y_2 - x_0,$$

and

$$Q_0\big[u(y_1 + y_2 - x_0) + u(x_0) - u(y_1) - u(y_2)\big] \neq 0.$$

Since rank-two (invertible) matrices are dense in $\mathbb{R}^{2\times 2}$ and $K = \{A_1, \dots, A_N\}$ is a finite set, we may find $P \in \mathbb{R}^{d\times 2}$, $Q \in \mathbb{R}^{2\times m}$ close to P_0, Q_0, respectively, such that, possibly lowering $\varepsilon > 0$, the following conditions are satisfied:

(a) $\operatorname{rank}(Q(A_i - A_j)P) = 2$ for all $i \neq j$;
(b) $Pz + x \in \Omega$ for all $z \in (0,1)^2$, $x \in B_\varepsilon(x_0)$;
(c) the non-affinity condition

$$Q\big[u(P\mathrm{e}_1 + P\mathrm{e}_2 + x) + u(x) - u(P\mathrm{e}_1 + x) - u(P\mathrm{e}_2 + x)\big] \neq 0 \tag{10.15}$$

holds for all $x \in B_\varepsilon(x_0)$.

Note that in (c) we have used the continuity of u, which follows by the Sobolev Embedding Theorem A.28.

For almost every $x_1 \in \mathbb{R}^d$ and almost every $z \in \mathbb{R}^2$ (where the exceptional negligible set may depend on x_1) with $Pz + x_1 \in \Omega$ we have

$$\nabla_z [Qu(Pz + x_1)] \in \widetilde{K} := \{QA_1P, \dots, QA_NP\} \subset \mathbb{R}^{2\times 2}$$

since otherwise $\nabla u \in K$ would be violated on a set of non-zero measure. Pick $x_1 \in B_\varepsilon(x_0)$ with this property and observe that the map $\widetilde{u}\colon (0,1)^2 \to \mathbb{R}^2$ given as

$$\widetilde{u}(z) := Qu(Pz + x_1), \qquad z \in (0,1)^2,$$

is not affine by (10.15). □

Proof of Theorem 10.11: Rigidity for exact solutions. Assume that there is a non-affine $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with $\nabla u \in K$ almost everywhere. By Lemma 10.12 we can assume that $d = m = 2$ and $\Omega = (0,1)^2$.

We may suppose without loss of generality that $A_1 = 0$ (this transforms u into $\widetilde{u}(x) := u(x) - A_1x$), whereby $\det A_2, \det A_3 \neq 0$ by the incompatibility relation (10.5), and that $A_2 = \mathrm{Id}$ (this transforms $\widetilde{u}$ into $\widehat{u}(x) := \widetilde{u}(A_2^{-1}x)$). Finally, by a change of variables and utilizing the Jordan normal form, see Appendix A.1, we may assume that A_3 can be written either as

$$A_3 = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \qquad \text{with} \qquad a \neq 0,\ c \notin \{0,1\},$$

or

$$A_3 = \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \qquad \text{with} \qquad a^2 + b^2 \neq 0.$$

Here, the additional conditions on the coefficients follow from the incompatibility relation (10.5).

In the first case, for $u = (u^1, u^2)^\mathsf{T}$ we have $\partial_1 u^2 = 0$ since all matrices A_1, A_2, A_3 have a zero as their $(2,1)$-element. Hence, with a slight abuse of notation, $u^2(x) = u^2(x_2)$. Clearly, since $c \notin \{0,1\}$, the value of $\partial_2 u^2(x) = \partial_2 u^2(x_2)$ determines which of the matrices A_1, A_2, A_3 our gradient $\nabla u(x)$ takes at x. Thus, $\nabla u(x)$ depends only on x_2 and therefore

$$\partial_1^2 u = \partial_2 \partial_1 u = \partial_1 \partial_2 u = 0$$

in the sense of distributions. It follows that

$$\nabla u(x) = \begin{pmatrix} \gamma & \partial_2 u^1(x_2) \\ 0 & \partial_2 u^2(x_2) \end{pmatrix} = (\gamma, 0) \otimes \mathrm{e}_1 + \partial_2 u(x_2) \otimes \mathrm{e}_2$$

for some $\gamma \in \mathbb{R}$. Hence, $\operatorname{rank}(\nabla u(x) - \nabla u(y)) \leq 1$ for all $x, y \in \Omega$. By our incompatibility assumption, this is only possible if ∇u is constant.

In the second case, we see that $\nabla u \in L$, where

$$L := \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \ : \ a, b \in \mathbb{R} \right\} \subset \mathbb{R}^{2\times 2}$$

once we observe that $A_1, A_2, A_3 \in L$. The space L does not contain rank-one connections since non-zero skew-symmetric matrices have at least rank two (as can be proved by elementary linear algebra). Hence, Theorem 10.3 (i) implies that u is smooth. As K is disconnected, ∇u must be constant. □

The proof of rigidity for approximate solutions is more involved. We will establish this result by a dimension-reduction to (2×2)-matrices and then employ the *separation method*, which entails testing with special quasiconvex functions that vanish on K.

We start by proving a generally useful rigidity lemma.

Lemma 10.13 *Let $K \subset \mathbb{R}^{2\times 2}$ be compact and non-empty such that*

$$\det(A - B) > 0 \qquad \text{for all } A, B \in K \text{ with } A \neq B.$$

Then, K is rigid for approximate solutions.

Proof. By Proposition 10.9 (i) it suffices to check that all $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ are Dirac masses. We will use the elementary formula

$$\det(A - B) = \det A - \operatorname{cof} A : B + \det B, \qquad A, B \in \mathbb{R}^{2\times 2}. \tag{10.16}$$

From the fact that $0 \leq \det(A - B)$ for all $A, B \in \operatorname{supp} \mu$ in conjunction with (10.16) and Corollary 5.12, we see that

$$\begin{aligned} 0 &\leq \int\int \det(A - B)\, \mathrm{d}\mu(A)\, \mathrm{d}\mu(B) \\ &= \int\int \det A - \operatorname{cof} A : B + \det B \ \mathrm{d}\mu(A)\, \mathrm{d}\mu(B) \\ &= \int \det [\mu] - \operatorname{cof} [\mu] : B + \det B \ \mathrm{d}\mu(B) \\ &= \det [\mu] - \operatorname{cof} [\mu] : [\mu] + \det [\mu] \\ &= \det([\mu] - [\mu]) \\ &= 0. \end{aligned}$$

Thus, $\det(A - B) = 0$ for $(\mu \otimes \mu)$-almost every $(A, B) \in K \times K$. On the other hand, by assumption, $\det(A - B) > 0$ for all such A, B with $A \neq B$. Thus, μ must be a Dirac mass. □

Next, we will need a special quasiconvex function on symmetric matrices:

Lemma 10.14 *Define* $\det^{++}\colon \mathbb{R}^{2\times 2}_{\mathrm{sym}} \to [0,\infty)$ *by*

$$\det^{++}(A) := \begin{cases} \det A & \text{if } A \text{ is positive semidefinite,} \\ 0 & \text{otherwise,} \end{cases} \qquad A \in \mathbb{R}^{2\times 2}_{\mathrm{sym}}.$$

Then, $\det^{++}$ *is* ***quasiconvex on symmetric matrices****, that is,*

$$\det^{++}(A) \le \fint_{B_1} \det^{++}(A + \nabla^2\psi(z))\,\mathrm{d}z \tag{10.17}$$

for all $A \in \mathbb{R}^{2\times 2}_{\mathrm{sym}}$ *and all* $\psi \in \mathrm{W}^{2,2}_c(B_1)$.

Here and in the following we set

$$\mathrm{W}^{2,q}_c(\Omega;\mathbb{R}^m) := \left\{ u \in \mathrm{W}^{2,q}(\Omega;\mathbb{R}^m) \;:\; \operatorname{supp} u \Subset \Omega \right\}$$

for $\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain and $q \in [1,\infty]$.

Proof. Let $A \in \mathbb{R}^{2\times 2}_{\mathrm{sym}}$ be positive semidefinite (the assertion is trivial otherwise). We will only show (10.17) for $\psi \in \mathrm{C}^\infty_c(B_1)$; the general case follows by approximation. Since we are dealing with symmetric matrices, which can be orthogonally diagonalized, we may assume that $A = \mathrm{Id}$ via a coordinate transformation. Define

$$u(z) := \frac{1}{2}|z|^2 + \psi(z), \qquad z \in B_1,$$

and set

$$D := \left\{ z \in B_1 \;:\; \nabla^2 u(z) \text{ positive semidefinite} \right\}.$$

We claim that $B_1 \subset \nabla u(D)$, the image of D under ∇u. Indeed, let $x_0 \in B_1$ be arbitrary and take a point $z_0 \in B_1$ such that $z \mapsto u(z) - x_0 \cdot z$ attains its minimum at z_0. Note that such a minimizer z_0 exists in B_1 because $z \mapsto \frac{1}{2}|z|^2 - x_0 \cdot z$ attains its minimum in B_1 and ψ has compact support in B_1. Differentiating, we get $\nabla u(z_0) = x_0$. Moreover, $\nabla^2 u(z_0)$ is positive semidefinite. Thus, $B_1 \subset \nabla u(D)$ and then

$$\omega_d \le |\nabla u(D)| \le \int_D \det \nabla^2 u(z)\,\mathrm{d}z = \int_{B_1} \det^{++}(\mathrm{Id} + \nabla^2\psi(z))\,\mathrm{d}z.$$

Since $\det^{++}(\mathrm{Id}) = 1$, (10.17) has been shown to hold for $A = \mathrm{Id}$. □

We now prove the analogue of the Jensen-type inequality from Lemma 5.11 for functions that are quasiconvex on symmetric matrices.

Lemma 10.15 *Let $K \subset \mathbb{R}^{d\times d}_{\mathrm{sym}}$ and $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$. Then, for all $h\colon \mathbb{R}^{d\times d}_{\mathrm{sym}} \to \mathbb{R}$ that are quasiconvex on symmetric matrices (in the sense defined above) and have p-growth for some $p \in [1,\infty)$, the Jensen-type inequality*

$$h([\mu]) \le \int h \,\mathrm{d}\mu$$

holds.

Proof. We will show below that for every $q \in (1,\infty)$ there exists a norm-bounded sequence $(\psi_j)_j \subset (\mathrm{W}^{1,2}_0 \cap \mathrm{W}^{2,q})(B_1)$ such that $F + \nabla^2\psi_j \overset{\mathbf{Y}}{\to} \mu$, where $F := [\mu]$. Then, if $q > p$,

$$h(F) \le \lim_{j\to\infty} \fint_{B_1} h(F + \nabla^2\psi_j(z))\,\mathrm{d}z = \int h \,\mathrm{d}\mu,$$

where the inequality follows by a cut-off procedure as in Lemma 4.14 (Step 3 of the proof) and the quasiconvexity on symmetric matrices. The limit representation via μ is due to the Fundamental Theorem 4.1 together with the equiintegrability of the integrands.

To show the existence of the sequence $(\psi_j)_j$ as claimed above, assume without loss of generality that $[\mu] = F = 0$ and let the sequence $(v_j)_j \subset \mathrm{C}^\infty_c(B_1;\mathbb{R}^d)$ be such that $\nabla v_j \overset{\mathbf{Y}}{\to} \mu$ (see Lemmas 4.14 and 8.14). Define $\psi_j \in (\mathrm{W}^{1,2}_0 \cap \mathrm{W}^{2,2})(B_1)$ as the (unique) solution of

$$\begin{cases} \Delta\psi_j = \operatorname{div} v_j & \text{in } B_1, \\ \psi_j = 0 & \text{on } \partial B_1. \end{cases}$$

By standard results, such a solution ψ_j exists in $(\mathrm{W}^{1,2}_0 \cap \mathrm{W}^{2,2})(B_1)$ and $\|\psi_j\|_{\mathrm{W}^{2,2}} \le C\|v_j\|_{\mathrm{W}^{1,2}}$ (we proved this in Examples 3.5 and 3.16, also taking into account Section 6.3.2 in [133] for the regularity up to the boundary). It is a well-known fact that even

$$\|\psi_j\|_{\mathrm{W}^{2,q}} \le C_q\|v_j\|_{\mathrm{W}^{1,\infty}}$$

for any $q \in (1,\infty)$ and a constant $C_q > 0$. This can be seen by the bootstrapping procedure explained in Section 3.2 and the fact that $\mathrm{W}^{k,2}(B_1)$ embeds into $\mathrm{W}^{2,q}(B_1)$ for sufficiently large k, see Theorem A.28.

Set $w_j := v_j - \nabla\psi_j$, which satisfies $\operatorname{div} w_j = 0$ (this gives the Helmholtz decomposition of v_j). Thus, with the usual definition $\operatorname{curl} w_j := \nabla w_j - (\nabla w_j)^{\mathsf{T}}$, we may calculate, using integration by parts,

$$\begin{aligned} &\int_{B_1} |\operatorname{curl} w_j|^2 \,\mathrm{d}x \\ &\quad= \int_{B_1} |\nabla w_j - (\nabla w_j)^{\mathsf{T}}|^2 + 2(\operatorname{div} w_j)^2 \,\mathrm{d}x \\ &\quad= \int_{B_1} |\nabla w_j|^2 - 2\nabla w_j : (\nabla w_j)^{\mathsf{T}} + |(\nabla w_j)^{\mathsf{T}}|^2 + 2\nabla w_j : (\nabla w_j)^{\mathsf{T}} \,\mathrm{d}x \\ &\quad= \int_{B_1} 2|\nabla w_j|^2 \,\mathrm{d}x. \end{aligned}$$

On the other hand, $\operatorname{curl} w_j = \operatorname{curl} v_j$, and by Young measure representation,

$$\int_{B_1} |\operatorname{curl} w_j|^2 \,\mathrm{d}x = \int_{B_1} |\nabla v_j - (\nabla v_j)^{\mathsf{T}}|^2 \,\mathrm{d}x \to \int_{B_1} \int |A - A^{\mathsf{T}}|^2 \,\mathrm{d}\mu(A)\,\mathrm{d}x = 0$$

since μ is supported on symmetric matrices. Thus, $\nabla w_j \to 0$ in L^2 and the sequence $\nabla^2 \psi_j = \nabla v_j - \nabla w_j$ generates μ, as desired. □

We can now complete the proof of Theorem 10.11.

Proof of Theorem 10.11: Rigidity for approximate solutions. We will show that every Young measure in $\mathcal{M}^{\mathrm{qc}}(K)$ is a Dirac mass, which implies the claim by Proposition 10.9 (i).

Assume that $A_3 = 0$ (by an affine shift). Let L be the linear subspace of $\mathbb{R}^{m\times d}$ spanned by the matrices A_1, A_2. We distinguish three cases.

Case 1: L contains at most one rank-one direction. If there is a rank-one direction in L, then let $L_0 \subset L$ be the rank-one line through the origin; otherwise set $L_0 := \{0\}$. Denote by $\mathbf{P}\colon \mathbb{R}^{m\times d} \to L$ and $\mathbf{Q}\colon L \to L_0$ the orthogonal projections onto L and L_0, respectively. With

$$g(A) := |A - \mathbf{Q}(A)|^2, \qquad A \in L,$$
$$f_{\varepsilon,k}(A) := -g(\mathbf{P}(A)) + \varepsilon|\mathbf{QP}(A)|^2 + k|A - \mathbf{P}(A)|^2, \qquad A \in \mathbb{R}^{m\times d},$$

we claim that for every $\varepsilon > 0$ there is a $k \in \mathbb{N}$ such that for all $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ with $F := [\mu]$ the inequality

$$f_{\varepsilon,k}(F) \le \int f_{\varepsilon,k} \,\mathrm{d}\mu \tag{10.18}$$

holds. Then, since μ is supported in L, whereby also $F \in L$, we may let $\varepsilon \downarrow 0$ to conclude the "reverse Jensen inequality" for g, namely

$$g(F) \ge \int g \,\mathrm{d}\mu.$$

Since g is strictly convex, this implies that μ is a Dirac mass.

It remains to show (10.18). For this, we first note that for all rank-one matrices $A \in \mathbb{R}^{m\times d}$ with $|A| = 1$ and $-g(\mathbf{P}(A)) + \varepsilon|\mathbf{QP}(A)|^2 \le 0$, it holds that

$$|A - \mathbf{P}(A)|^2 \ge c \tag{10.19}$$

for some constant $c > 0$ that does not depend on A. Indeed, $|\mathbf{P}(A) - \mathbf{QP}(A)|^2 = g(\mathbf{P}(A)) \ge \varepsilon|\mathbf{QP}(A)|^2$ implies that A has positive distance from L_0. On the other hand, the rank-one cone intersects L only in L_0, so that we obtain that A has positive distance from L, hence $|A - \mathbf{P}(A)|^2 > 0$. Then we can minimize over the compact set of the A as above to obtain (10.19). Thus, since every term in the definition of $f_{\varepsilon,k}$ is positively

2-homogeneous, for every $\varepsilon > 0$ we can find $k \in \mathbb{N}$ such that $f_{\varepsilon,k}$ is positive on the rank-one cone.

We now have two ways to conclude. As the first option, we may invoke Tartar's theorem from the theory of compensated compactness in the version of Proposition 8.9, to see that

$$f_{\varepsilon,k}(F) \le \liminf_{j\to\infty} \fint_{B_1} f_{\varepsilon,k}(\nabla u_j)\,\mathrm{d}x$$

for a sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}_{Fx}(B_1;\mathbb{R}^m)$ such that $\nabla u_j \overset{\mathbf{Y}}{\to} \mu$ (also use Lemma 4.14 and Zhang's Lemma 8.14, keeping the boundary values intact). This directly implies (10.18).

As the second option, the positivity of $f_{\varepsilon,k}$ on the rank-one cone implies that $f_{\varepsilon,k}$ is rank-one convex (one can check the Legendre–Hadamard condition (8.3)), which in turn implies that $f_{\varepsilon,k}$ is also quasiconvex via Problem 5.7. Then, (10.18) follows from the Jensen-type inequality for gradient Young measures proved in Lemma 5.11.

Case 2: $d = m = 2$ and L contains the rank-one directions $\mathrm{e}_1 \otimes \mathrm{e}_1$, $\mathrm{e}_2 \otimes \mathrm{e}_2$. In this case, A_1, A_2 are diagonal. We can also assume that $\det A_1, \det A_2 > 0$ and $A_3 = 0$. Indeed, at least two of the three numbers $\det(A_i - A_j)$ for $i, j \in \{1,2,3\}$ and $i < j$ must have the same sign. Multiplying the first row of the matrices by -1 (which corresponds to flipping the sign of the first component of a generating sequence of μ) and exchanging indices, we may thus assume that $\det(A_1 - A_3) > 0$, $\det(A_2 - A_3) > 0$. Then we shift all matrices by $-A_3$ to obtain $\widetilde{A}_1 := A_1 - A_3$, $\widetilde{A}_2 := A_2 - A_3$, $\widetilde{A}_3 := 0$, of which the first two have positive determinant. In the following we drop the tildes.

We can further multiply the first and second components of generating sequences for μ with scalars to reduce to the situation where $K = \{A_1, A_2, A_3\}$ with

$$A_1 = \mathrm{Id}, \qquad A_2 = \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}, \qquad A_3 = 0$$

for

$$\alpha > \beta > 0 \qquad \text{or} \qquad 0 > \alpha > \beta$$

since $\det A_2 > 0$. Note also that $\alpha = 1$ or $\beta = 1$ is impossible since $\operatorname{rank}(A_2 - A_1) = 2$.

If $\alpha < 1$, then also $\beta < 1$ and so all the differences $A_i - A_j$ for $i, j \in \{1,2,3\}$ have positive determinant. Thus, Lemma 10.13 implies that any $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ is a Dirac mass. If $\alpha > 1$ and $\beta > 1$, a similar argument applies, again yielding that any $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ is a Dirac mass.

It remains to investigate the situation where $\alpha > 1$ and $0 < \beta < 1$. First, for any $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$, Corollary 5.12 shows that $\det F = \langle \det, \mu \rangle$, where $F := [\mu]$. Writing

$$\mu = \theta_1 \delta_{A_1} + \theta_2 \delta_{A_2} + \theta_3 \delta_{A_3}$$

for $\theta_1, \theta_2, \theta_3 \in [0,1]$ with $\theta_1 + \theta_2 + \theta_3 = 1$, this implies the relation

$$(\theta_1 + \theta_2\alpha)(\theta_1 + \theta_2\beta) = \theta_1 + \theta_2\alpha\beta. \tag{10.20}$$

Second, from Lemma 10.14 we infer that the function $h\colon \mathbb{R}^{2\times 2}_{\mathrm{sym}} \to \mathbb{R}$ given by

$$h(A) := \det^{++}(A - X), \qquad \text{where} \qquad X := \begin{pmatrix} 1 & \\ & \beta \end{pmatrix},$$

is quasiconvex on symmetric matrices. By the Jensen-type inequality proved in Lemma 10.15, we thus get

$$\det^{++}(F - X) \le \int \det^{++}(A - X)\, \mathrm{d}\mu(A) = 0,$$

where the last equality holds since the three matrices $A_1 - X$, $A_2 - X$, $A_3 - X$ are not positive definite. Hence, also

$$F - X = \begin{pmatrix} \theta_1 + \theta_2\alpha - 1 & \\ & \theta_1 + \theta_2\beta - \beta \end{pmatrix}$$

cannot be positive definite, so at least one entry in this matrix needs to be nonpositive. Combining this fact with (10.20) we get that either

$$\theta_1 + \theta_2\alpha\beta \le \theta_1 + \theta_2\beta \qquad \text{or} \qquad \theta_1 + \theta_2\alpha\beta \le \theta_1\beta + \theta_2\alpha\beta.$$

In the first case, $\alpha > 1$ implies that $\theta_2 = 0$, which, however, by (10.20) immediately gives $\theta_1^2 = \theta_1$, whereby μ must be a Dirac mass. In the second case, from $\beta < 1$ we get $\theta_1 = 0$ and hence (10.20) implies $\theta_2^2 = \theta_2$, which again yields that μ is a Dirac mass.

Case 3: $d \ge 2$, $m \ge 2$ and L contains at least two (different) rank-one directions. We will reduce this case to the previous one via a dimension-reduction argument. Let $a_1 \otimes b_1$ and $a_2 \otimes b_2$ be two rank-one matrices in L with a_1, a_2 and b_1, b_2 linearly independent. By a change of variables we can assume that a_1, b_1 are the first unit vector e_1 (in $\mathbb{R}^m$ and $\mathbb{R}^d$, respectively) and a_2, b_2 are the second unit vector e_2.

Let $(u_j)_j \subset \mathrm{W}^{1,\infty}((0,1)^d;\mathbb{R}^m)$ be such that $\nabla u_j \xrightarrow{\mathbf{Y}} \mu$ with $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$. Then, for fixed $z = (z_3, \dots, z_d) \in (0,1)^{d-2}$, we define $v_j^{(z)} \in \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^2)$ by

$$v_j^{(z)}(x_1, x_2) := \begin{pmatrix} u_j^1(x_1, x_2, z) \\ u_j^2(x_1, x_2, z) \end{pmatrix}, \qquad (x_1, x_2) \in (0,1)^2.$$

For $\mathcal{L}^{d-2}$-almost every $z \in (0,1)^{d-2}$ it holds that $\nabla v_j^{(z)} \xrightarrow{\mathbf{Y}} \nu^{(z)} \in \mathcal{M}^{\mathrm{qc}}(\widehat{K})$ with

$$\widehat{K} := \{A_1, A_2, 0\} = \left\{ \begin{pmatrix} \gamma_1 & \\ & \gamma_2 \end{pmatrix}, \begin{pmatrix} \eta_1 & \\ & \eta_2 \end{pmatrix}, 0 \right\} \subset \mathbb{R}^{2\times 2},$$

with $\gamma_1, \gamma_2, \eta_1, \eta_2 \in \mathbb{R} \setminus \{0\}$. Here, A_1 and A_2 can be taken as diagonal since they both lie in L, which is spanned by $\mathrm{e}_1 \otimes \mathrm{e}_1$ and $\mathrm{e}_2 \otimes \mathrm{e}_2$ by the change of variables above.

Thus, Case 2 is applicable and we obtain that $\nu^{(z)}$ is a Dirac mass for $\mathcal{L}^{d-2}$-almost every $z \in (0,1)^{d-2}$. Since $\mu = \mathcal{L}_z^{d-2} \llcorner (0,1)^{d-2} \otimes \nu^{(z)}$ and we know that μ is a *homogeneous* Young measure, that is, $\nu^{(z)}$ does not actually depend on z, we conclude that also μ is a Dirac mass. □

10.5 Four- and Five-Gradient Inclusions

Rigidity for exact solutions also holds for the four-gradient inclusion:

Theorem 10.16 (Chlebík–Kirchheim 2002 [77]) *Assume that the four-element set $K := \{A_1, A_2, A_3, A_4\} \subset \mathbb{R}^{m\times d}$ contains no rank-one connection. Then,* (10.4) *is rigid for exact solutions.*

The proof is beyond the scope of this book. It uses a reduction to the Monge–Ampère equation (like the original proof for the three-gradient inclusion in [306]), where one has to distinguish the elliptic case $\det \nabla^2 u > 0$ and the hyperbolic case $\det \nabla^2 u < 0$; see [77, 193] for the details.

Despite this result on rigidity for exact solutions, approximate solutions of the four-gradient inclusion are *not* rigid, as we will see in the next chapter by means of the so-called T_4-configuration.

Finally, the five-gradient inclusion loses all rigidity:

Theorem 10.17 (Kirchheim–Preiss 2003 [193]) *There are five matrices $A_1, \dots, A_5 \in \mathbb{R}^{2\times 2}_{\mathrm{sym}}$ without rank-one connections such that* (10.4) *is not rigid for exact solutions and not rigid for approximate solutions. In particular, there exists a non-affine $u \in \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^2)$ with*

$$\nabla u(x) \in K_{\mathrm{KP}} := \{A_1, \dots, A_5\} \qquad \text{for a.e. } x \in (0,1)^2.$$

Moreover, $K_{\mathrm{KP}}^{\mathrm{qc}} \supsetneq K_{\mathrm{KP}}$.

It is remarkable that the matrices $A_1, \dots, A_5$ can be chosen to be *symmetric*. The proof is very geometric, see Section 4.3 in [193] and also [273], where the quasiconvex hull $K_{\mathrm{KP}}^{\mathrm{qc}}$ is identified explicitly. A recent improvement and generalization can be found in [147].

10.6 The One-Well Inclusion

So far we have treated differential inclusions with discrete and with linear constraints. However, despite their importance, none of these situations are applicable to the problems of crystal microstructure described in Example 1.9, where we are interested in differential inclusions of the form

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^d), \\ \nabla u \in K := \mathrm{SO}(d)U_1 \cup \cdots \cup \mathrm{SO}(d)U_N \quad \text{in } \Omega \end{cases} \tag{10.21}$$

for distinct matrices $U_1, \dots, U_N \in \mathbb{R}^{d\times d}$. Every set $\mathrm{SO}(d)U$ for $U \in \mathbb{R}^{d\times d}$ is called a **well**. We will only consider the physically most relevant case where

$$\det U_1, \det U_2, \dots, \det U_N > 0.$$

By the polar decomposition of matrices (see Appendix A.1) we may write $U_i = Q\widetilde{U}_i$ with $Q \in \mathrm{SO}(d)$ and $\widetilde{U}_i$ symmetric and positive definite. As we can absorb Q into the $\mathrm{SO}(d)$-factor in the well, we assume in all of the following that the U_i are symmetric and positive definite to start with.

First, we observe that there are no rank-one connections in the well $\mathrm{SO}(d)$. Indeed, assume that there were $Q, R \in \mathrm{SO}(d)$ with $R = Q + a \otimes b$ for some $a, b \in \mathbb{R}^d \setminus \{0\}$ with $|a| = 1$ (without loss of generality). Then, since $\mathrm{SO}(d)$ is a group,

$$S := RQ^{\mathsf{T}} = \mathrm{Id} + a \otimes (Qb) =: \mathrm{Id} + a \otimes \widehat{b} \in \mathrm{SO}(d).$$

Then, $SS^{\mathsf{T}} = \mathrm{Id} = S^{\mathsf{T}}S$ and thus

$$\mathrm{Id} + a \otimes \widehat{b} + \widehat{b} \otimes a + |\widehat{b}|^2 a \otimes a = \mathrm{Id} + a \otimes \widehat{b} + \widehat{b} \otimes a + |a|^2\, \widehat{b} \otimes \widehat{b}.$$

So, $\widehat{b} = \lambda a$ for some $\lambda \neq 0$. Now take a rotation $P \in \mathrm{SO}(d)$ with $Pa = \mathrm{e}_1$. Then,

$$\det(PSP^{\mathsf{T}}) = \det(\mathrm{Id} + \lambda \mathrm{e}_1 \otimes \mathrm{e}_1) = 1 + \lambda \neq 1,$$

contradicting the group property of $\mathrm{SO}(d)$. Hence, there are no rank-one connections in $\mathrm{SO}(d)$ and one may hope that the one-well inclusion is rigid. This is indeed true and in fact a classical result.

Theorem 10.18 (Reshetnyak 1967 [276]) *For $K := \mathrm{SO}(d)$ the one-well inclusion (10.21) is strongly rigid. In particular, $K^{\mathrm{qc}} = K$.*

Remark 10.19 The same result holds if $u \in \mathrm{W}^{1,\infty}$ is replaced with $u \in \mathrm{W}^{1,d}$ in (10.21), as can be seen from the proof.

Proof. Step 1: Rigidity for exact solutions. Since $\operatorname{cof} Q = Q$ for all $Q \in \mathrm{SO}(d)$ (see Appendix A.1) we have for every $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^d)$ with $\nabla u \in \mathrm{SO}(d)$ almost everywhere that

$$\operatorname{div} \operatorname{cof} \nabla u = \Delta u.$$

On the other hand, by the Piola identity (5.6), $\operatorname{div} \operatorname{cof} \nabla u = 0$. Thus, our u is harmonic and hence smooth (see Example 3.16 or Example 10.4). Recalling that $|\cdot|$ for us always means the Frobenius norm,

$$|\nabla u|^2 = \operatorname{tr}\big((\nabla u)^{\mathsf{T}} \nabla u\big) = \operatorname{tr}(\mathrm{Id}) = d,$$

and so, by means of a computation based on the product rule,

$$2|\nabla^2 u|^2 = 2\nabla u \cdot \nabla \Delta u + 2|\nabla^2 u|^2 = \Delta\big[|\nabla u|^2\big] = 0.$$

Thus, ∇u is constant.

Step 2: Strong rigidity. As a preparation define

$$g(A) := |A|^d - d^{d/2} \det A, \qquad A \in \mathbb{R}^{d\times d},$$

which we claim satisfies $g \geq 0$. To see this, we assume $\det A > 0$ and use the polar decomposition to write $A = QS$ with $Q \in \mathrm{SO}(d)$ and $S \in \mathbb{R}^{d\times d}_{\mathrm{sym}}$ positive definite. Then diagonalize S as $G^{\mathsf{T}} S G = D = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$ with $G \in \mathbb{R}^{d\times d}$ orthogonal and $\lambda_1, \dots, \lambda_d \geq 0$. So,

$$\begin{aligned}
\det A &= \lambda_1 \cdots \lambda_d \\
&\leq \left(\frac{\lambda_1 + \cdots + \lambda_d}{d}\right)^d \\
&\leq d^{-d/2}\big(\lambda_1^2 + \cdots + \lambda_d^2\big)^{d/2} \\
&= d^{-d/2}|D|^d \\
&= d^{-d/2}|A|^d,
\end{aligned}$$

where we used the vector norm inequality $|\cdot|_1 \leq \sqrt{d}|\cdot|_2$ and the orthogonal invariance of the Frobenius norm.

Furthermore, $g(A) = 0$ if and only if $A = \alpha Q$ for some $Q \in \mathrm{SO}(d)$ and $\alpha \geq 0$. Indeed, $g(A) = 0$ if and only if the above inequalities are in fact equalities, which is the case if and only if $\lambda_1 = \cdots = \lambda_d =: \alpha \geq 0$, that is, $A = \alpha Q$.

Now suppose that $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^d)$ is an approximate solution of (10.21), that is, $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^d)$ with $u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{W}^{1,\infty}$ for some map $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^d)$ and $\mathrm{dist}(\nabla u_j, \mathrm{SO}(d)) \to 0$ in measure. Then, using that the L^d-norm is weakly* lower semicontinuous and the determinant is weakly* continuous by Lemma 5.5, as well as the dominated convergence theorem,

$$\begin{aligned}
0 &\leq \int_\Omega g(\nabla u)\,\mathrm{d}x \\
&= \int_\Omega |\nabla u|^d - d^{d/2} \det \nabla u \,\mathrm{d}x \\
&\leq \liminf_{j\to\infty} \int_\Omega |\nabla u_j|^d - d^{d/2} \det \nabla u_j \,\mathrm{d}x \\
&= \liminf_{j\to\infty} \int_\Omega g(\nabla u_j)\,\mathrm{d}x \\
&= 0.
\end{aligned}$$

Hence, all these inequalities are in fact equalities and we conclude that

$$g(\nabla u) = 0 \quad \text{a.e.} \qquad \text{and} \qquad \|\nabla u_j\|_{\mathrm{L}^d} \to \|\nabla u_j\|_{\mathrm{L}^d}.$$

This implies that $\nabla u_j \to \nabla u$ in L^d by the Radon–Riesz Theorem A.15. Consequently,

$$\nabla u_j \to \nabla u \quad \text{in measure and almost everywhere.}$$

Moreover, $\nabla u(x) = \alpha(x)Q(x)$ with $\alpha(x) \geq 0$ and $Q(x) \in \mathrm{SO}(d)$ for almost every $x \in \Omega$.

On the other hand, $\lim_{j\to\infty} |\nabla u_j|^2 = d$ almost everywhere, whereby also $|\nabla u|^2 = d$ almost everywhere. Thus, $\alpha \equiv 1$ and so u is an exact solution to our differential inclusion $\nabla u \in \mathrm{SO}(d)$. Then, we may conclude that ∇u is constant via the exact rigidity proved in Step 1. This establishes the strong rigidity. The fact that $K^{\mathrm{qc}} = K$ then immediately follows from Proposition 10.9 and Lemma 10.8. □

We mention that there is also a *quantitative* one-well rigidity estimate:

Theorem 10.20 (Friesecke–James–Müller 2002 [149])
Let $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^d)$, where $p \in (1,\infty)$. Then,

$$\inf_{Q\in\mathrm{SO}(d)} \int_\Omega |\nabla u - Q|^p \,\mathrm{d}x \leq C \int_\Omega \mathrm{dist}(\nabla u, \mathrm{SO}(d))^p \,\mathrm{d}x,$$

where $C = C(\Omega, p) > 0$ is a constant that only depends on Ω and p.

This result can be seen as a nonlinear analogue of the **Korn inequality**, which says that for any map $u \in \mathrm{L}^p(\Omega;\mathbb{R}^d)$, $p \in (1,\infty)$, such that the (weak) symmetric gradient $\mathcal{E}u := (\nabla u + \nabla u^T)/2$ exists and lies in $\mathrm{L}^p(\Omega;\mathbb{R}^{d\times d}_{\mathrm{sym}})$, it holds that $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^d)$ and

$$\|\nabla u\|_{\mathrm{L}^p} \leq C_K \big(\|u\|^p_{\mathrm{L}^p} + \|\mathcal{E}u\|^p_{\mathrm{L}^p}\big)^{1/p}, \tag{10.22}$$

for a constant $C_K = C_K(\Omega, p) > 0$, see Proposition I.1.1 in [330] and also (2.4) (in a special case). A proof of Theorem 10.20 can be found in the original work [149] for $p = 2$. That proof in fact also works for $p \in (1,\infty)$ with only minor modifications, as was observed in [87].

10.7 Multi-Well Inclusions in 2D

On the topic of multi-well problems, we first consider the situation in two dimensions. The reason that this case is easier than the corresponding problem in higher dimensions is that for $A, B \in \mathbb{R}^{2\times 2}$ we have the very useful equivalence

$$\det(A - B) \neq 0 \qquad \Longleftrightarrow \qquad \mathrm{rank}(A - B) = 2,$$

which is of course false in higher dimensions.

The two-dimensional N-well problem without rank-one connections (and positive determinants) can be completely analyzed as follows.

Theorem 10.21 (Švérak 1993 [310, 311]) *Let*

$$K := \mathrm{SO}(2)U_1 \cup \cdots \cup \mathrm{SO}(2)U_N$$

for $U_1, \dots, U_N \in \mathbb{R}^{2\times 2}$ with $\det U_1, \det U_2, \dots, \det U_N > 0$. If K does not contain any rank-one connections, then the multi-well inclusion (10.21) is strongly rigid.

For the proof we will need an elliptic regularity lemma.

Lemma 10.22 *Let $K \subset \mathbb{R}^{2\times 2}$ be compact and non-empty with the property that*

$$\det(A - B) \geq \beta |A - B|^2 \qquad \text{for all } A, B \in K \text{ and some } \beta > 0. \tag{10.23}$$

Then, for $u \in \mathrm{W}^{1,\infty}(\Omega; \mathbb{R}^2)$ satisfying $\nabla u \in K$ almost everywhere it holds that $u \in \mathrm{W}^{2,2}_{\mathrm{loc}}(\Omega; \mathbb{R}^2)$.

Proof. Recall from Section 3.2 the definition of the k'th difference quotient ($k = 1, 2$) of u,

$$\mathrm{D}^h_k u(x) := \frac{u(x + h\mathrm{e}_k) - u(x)}{h}, \qquad x \in \Omega,$$

where $h \in \mathbb{R} \setminus \{0\}$ with $|h|$ small enough such that $B_h(x) \Subset \Omega$. Applying (10.23) to $\varrho(x)(\nabla u(x + h\mathrm{e}_k) - \nabla u(x))$, where $\varrho \in \mathrm{C}^\infty_c(\Omega; [0, 1])$ is a cut-off function, and integrating, we get

$$\beta \int_\Omega \varrho^2 |\mathrm{D}^h_k \nabla u|^2 \,\mathrm{d}x \leq \int_\Omega \det[\varrho \mathrm{D}^h_k \nabla u] \,\mathrm{d}x. \tag{10.24}$$

By the identity (10.16),

$$\begin{aligned}
&\int_\Omega \det[\varrho \mathrm{D}^h_k \nabla u] \,\mathrm{d}x \\
&\quad = \int_\Omega \det\big[\nabla(\varrho \mathrm{D}^h_k u) - \mathrm{D}^h_k u \otimes \nabla \varrho\big] \,\mathrm{d}x \\
&\quad = \int_\Omega \det[\nabla(\varrho \mathrm{D}^h_k u)] - \mathrm{cof}\,[\nabla(\varrho \mathrm{D}^h_k u)] : (\mathrm{D}^h_k u \otimes \nabla \varrho) + \det[\mathrm{D}^h_k u \otimes \nabla \varrho] \,\mathrm{d}x \\
&\quad \leq 0 + C \int_\Omega |\nabla(\varrho \mathrm{D}^h_k u)| \cdot |\mathrm{D}^h_k u| \cdot |\nabla \varrho| + |\mathrm{D}^h_k u|^2 \cdot |\nabla \varrho|^2 \,\mathrm{d}x \\
&\quad \leq C \int_\Omega \varrho |\mathrm{D}^h_k \nabla u| \cdot |\mathrm{D}^h_k u| \cdot |\nabla \varrho| \,\mathrm{d}x + 2C \int_\Omega |\mathrm{D}^h_k u|^2 \cdot |\nabla \varrho|^2 \,\mathrm{d}x \\
&\quad \leq \frac{\beta}{2} \int_\Omega \varrho^2 |\mathrm{D}^h_k \nabla u|^2 \,\mathrm{d}x + \widetilde{C} \int_\Omega |\mathrm{D}^h_k u|^2 \cdot |\nabla \varrho|^2 \,\mathrm{d}x,
\end{aligned}$$

where we also used that the determinant is a null-Lagrangian, see Lemma 5.3, and Young's inequality (with $\delta := \beta/C$). Combine this with (10.24) to get the estimate

$$\int_\Omega \varrho^2 |\mathrm{D}_k^h \nabla u|^2 \,\mathrm{d}x \le \frac{2\widetilde{C}}{\beta} \int_\Omega |\mathrm{D}_k^h u|^2 \cdot |\nabla \varrho|^2 \,\mathrm{d}x,$$

where the term on the right-hand side is bounded uniformly in h by virtue of Lemma 3.13 (i). By part (ii) of that lemma we then conclude that $u \in \mathrm{W}^{2,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^2)$. □

Proof of Theorem 10.21. The idea is to reduce to the one-well inclusion.

Since there are no rank-one connections in K, we have for all $i, j \in \{1,\dots,N\}, i \neq j$, and all $Q, R \in \mathrm{SO}(2)$ that

$$\det(QU_i - RU_j) \neq 0.$$

This expression must be positive for all $Q, R \in \mathrm{SO}(d)$ or negative for all $Q, R \in \mathrm{SO}(d)$ since $\mathrm{SO}(2)\times\mathrm{SO}(2)$ is connected, see Problem 10.7. In the following we only treat the case

$$\det(A - B) > 0 \qquad \text{for all } A, B \in K \text{ with } A \neq B; \tag{10.25}$$

the other case is analogous.

With a view to applying the previous lemma, we next prove the quantitative estimate

$$\det(A - B) \ge \beta |A - B|^2 \qquad \text{for all } A, B \in K \text{ and some } \beta > 0. \tag{10.26}$$

Set

$$K_i := \mathrm{SO}(2)U_i, \qquad i = 1,\dots,N,$$

which are pairwise disjoint sets. Moreover, if $A \in K_i$ and $B \in K_j$ for $i \neq j$, then $|A - B| > 0$. The continuous and bounded function

$$(A, B) \in K_i \times K_j \mapsto g(A, B) := \frac{\det(A - B)}{|A - B|^2}$$

is strictly positive by (10.25) on the compact and non-empty set $K_i \times K_j$, hence bounded from below. Thus, for A, B from different wells, (10.26) holds.

Fix i and assume without loss of generality that $U_i = \mathrm{Id}$. For $A, B \in K_i$, that is, $A, B \in \mathrm{SO}(2)$, observe that

$$\frac{\det(A - B)}{|A - B|^2} = \frac{1}{2} \cdot \frac{\det(\mathrm{Id} - A^\mathsf{T} B)}{|\mathrm{Id} - A^\mathsf{T} B|^2}.$$

It is well known that the tangent space of $\mathrm{SO}(2)$ at the identity matrix consists of all skew-symmetric matrices (this is really a restatement of the fact that the Lie algebra of the Lie group $\mathrm{SO}(2)$ is $\mathfrak{so}(2)$, the vector space of all skew-symmetric matrices, see [176]). In particular, if $Q_j \to \mathrm{Id}$ for $A_j, B_j \in \mathrm{SO}(2)$, then

$$\frac{\mathrm{Id} - Q_j}{|\mathrm{Id} - Q_j|} \to W, \qquad \text{where } W^\mathsf{T} = -W, |W| = 1.$$

Thus, for $Q_j := A_j^\top B_j \to \mathrm{Id}$ we have

$$\lim_{j\to\infty} \frac{\det(\mathrm{Id} - Q_j)}{|\mathrm{Id} - Q_j|^2} = \cdot \lim_{j\to\infty} \det\left(\frac{\mathrm{Id} - Q_j}{|\mathrm{Id} - Q_j|}\right) = \det W > 0.$$

Here we used that a non-zero skew-symmetric matrix $W \in \mathbb{R}^{2\times 2}$ has two non-zero eigenvalues of the form $\pm i\alpha$, $\alpha \in \mathbb{R}$, and thus $\det W = \alpha^2 > 0$. This implies that the function g is continuous, strictly positive, and bounded on $\mathrm{SO}(2)\times\mathrm{SO}(2)$. Thus, (10.26) also holds for A, B from the same well.

The rigidity for exact solutions $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^2)$ with $\nabla u \in K$ almost everywhere now follows in a straightforward manner: By Lemma 10.22 in conjunction with (10.26) we conclude that $\nabla u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^{2\times 2})$. Then, $\nabla u \in K_i$ almost everywhere for one $i \in \{1,\dots,N\}$; otherwise there would be a jump in ∇u, which is impossible for $\mathrm{W}^{1,2}$-regular maps. Thus, Theorem 10.18 becomes applicable and we conclude that ∇u is almost everywhere constant. This shows the rigidity for exact solutions.

With (10.25) already established, the rigidity for approximate solutions follows directly from Lemma 10.13. We can hence conclude strong rigidity by invoking Proposition 10.9 (iii). □

A special case of the preceding result concerns the two-well problem in two space dimensions:

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^2), \\ \nabla u \in K := \mathrm{SO}(2)U_1 \cup \mathrm{SO}(2)U_2, \end{cases} \tag{10.27}$$

where $U_1, U_2 \in \mathbb{R}^{2\times 2}$, $U_1 \neq U_2$, $\det U_1, \det U_2 > 0$. We can normalize U_1, U_2 to the case

$$U_1 = \mathrm{Id} = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix} \quad \neq \quad U_2 = \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}, \qquad 0 < \alpha \le \beta, \qquad \alpha\beta \ge 1.$$

Indeed, assume that $\det U_2 \ge \det U_1 > 0$. Then, right-multiplying by U_1^{-1} (which transforms u correspondingly), we can reduce to the case when $U_1 = \mathrm{Id}$ and $\det U_2 \ge 1$. Now use the polar decomposition for matrices (see Appendix A.1) and a diagonalization (of the resulting symmetric and positive definite matrices) to further reduce to the case when $U_2 = \mathrm{diag}(\alpha,\beta)$ with $0 < \alpha \le \beta$. Because $\det U_2 \ge 1$, it necessarily also follows that $\alpha\beta \ge 1$.

In this case, we can explicitly expose the rank-one connections in K:

Lemma 10.23 *Let $0 < \alpha \le \beta$ with $\alpha\beta \ge 1$ and set*

$$K := \mathrm{SO}(2) \cup \mathrm{SO}(2) \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}.$$

Then:

(i) If $\alpha > 1$, then there are no rank-one connections in K.
(ii) If $\alpha = 1$, then every matrix in K is rank-one connected to exactly one other matrix in K.

(iii) If $\alpha < 1$, then every matrix in K is rank-one connected to exactly two other matrices in K.

Proof. Recall that for $A, B \in K$ with $A \neq B$ we have that $\operatorname{rank}(A-B) = 1$ if and only if $\det(A-B) = 0$. We showed toward the beginning of Section 10.6 that the only possible rank-one connections in K are *between* the wells. So, for a fixed $Q \in \mathrm{SO}(2)$ we are trying to find all $R \in \mathrm{SO}(2)$ such that

$$0 = \det\left(Q - R\begin{pmatrix}\alpha & \\ & \beta\end{pmatrix}\right) = \det\left(\mathrm{Id} - Q^{\mathsf{T}}R\begin{pmatrix}\alpha & \\ & \beta\end{pmatrix}\right).$$

Via the identification $R \leftrightarrow Q^{\mathsf{T}}R =: S \in \mathrm{SO}(2)$ we are thus trying to find all

$$S = \begin{pmatrix} a & b \\ -b & a\end{pmatrix} \qquad \text{with} \qquad a^2 + b^2 = 1$$

such that

$$0 = \det\left(\mathrm{Id} - S\begin{pmatrix}\alpha & \\ & \beta\end{pmatrix}\right) = \det\begin{pmatrix}1-\alpha a & -\beta b \\ \alpha b & 1-\beta a\end{pmatrix} = 1 - (\alpha+\beta)a + \alpha\beta. \tag{10.28}$$

So, solvability is equivalent to finding $a \leq 1$ (because of the condition $a^2 + b^2 = 1$) with $a = (1+\alpha\beta)/(\alpha+\beta) \in (0, \infty)$.

Ad (i). For $\alpha > 1$ we have (using $\beta > 1$, which always follows from the above conditions) that $0 < (1-\alpha)(1-\beta) = (1+\alpha\beta) - (\alpha+\beta)$, whereby necessarily $a > 1$. Thus, there is no solution $S \in \mathrm{SO}(2)$ to (10.28) and consequently, for all $Q, R \in \mathrm{SO}(2)$ we have that Q and $R\,\mathrm{diag}(\alpha, \beta)$ are not rank-one connected.

Ad (ii), (iii). By a similar argument as for (i), in the case $\alpha \leq 1$ we have instead $0 \geq (1+\alpha\beta) - (\alpha+\beta)$ and so (10.28) is solvable with $a \leq 1$ and $b := \pm\sqrt{1-a^2}$. This yields $S, S^{\mathsf{T}} \in \mathrm{SO}(2)$ such that Q is rank-one connected to $R_1 := QS$ and $R_2 := QS^{\mathsf{T}} \in \mathrm{SO}(2)$.

If $\alpha = 1$, then $a = 1$, $b = 0$, $S = S^{\mathsf{T}}$, and $R_1 = R_2$. Hence, (ii) follows.

Finally, if $\alpha < 1$, we have $a < 1$, whereby $S \neq S^{\mathsf{T}}$ and $R_1 \neq R_2$. Thus, (iii) is also established. □

From this discussion in conjunction with Theorem 10.21 we conclude the following rigidity result:

Corollary 10.24 *Let*

$$K = \mathrm{SO}(2) \cup \mathrm{SO}(2)\begin{pmatrix}\alpha & \\ & \beta\end{pmatrix}, \qquad 1 < \alpha \leq \beta.$$

Then the two-well inclusion (10.27) *is strongly rigid.*

10.8 Two-Well Inclusions in 3D

The study of the two-well problem in three (and higher) dimensions is still incomplete and in particular the following conjecture remains open in full generality.

Conjecture 10.25 (Kinderlehrer 1988, reported in [124, 226]) If

$$K := \mathrm{SO}(3)U_1 \cup \mathrm{SO}(3)U_2,$$

where $U_1, U_2 \in \mathbb{R}^{3\times 3}$ with $\det U_1, \det U_2 > 0$, contains no rank-one connections, then the differential inclusion $\nabla u \in K$ is rigid for approximate solutions.

By similar arguments as in the last section, it can be seen that we may reduce to the case

$$K = \mathrm{SO}(3) \cup \mathrm{SO}(3)U$$

with

$$U = \begin{pmatrix} \alpha_1 & & \\ & \alpha_2 & \\ & & \alpha_3 \end{pmatrix}, \qquad \alpha_1 \geq \alpha_2 \geq \alpha_3 > 0. \tag{10.29}$$

Moreover, a calculation shows that there are no rank-one connections in K if and only if $\alpha_2 \neq 1$, see [186]. However, all currently known rigidity results for the three-dimensional two-well problem additionally need a form of *strong incompatibility* between the wells. We will only survey some available results here.

Theorem 10.26 (Matos 1992 [226]) *Let K be as in (10.29) such that $\alpha_2 \neq 1$. If, with cyclic indices $i \in \{1, 2, 3\}$, it holds that*

$$(1 - \alpha_i)(1 - \alpha_{i-1}\alpha_{i+1}) \geq 0 \qquad \text{for at least one } i \in \{1, 2, 3\},$$

then the two-well inclusion (10.21) is strongly rigid.

Another result is the following:

Theorem 10.27 (Dolzmann–Kirchheim–Müller–Šverák 2000 [124]) *Let K be as in (10.29) with $\alpha_2 \neq 1$. If*

$$\alpha_1 \geq \alpha_2 > 1 > \alpha_3 \geq \frac{1}{3} \qquad \text{or} \qquad 3 > \alpha_1 > 1 > \alpha_2 \geq \alpha_3 > 0,$$

then the two-well inclusion (10.21) is rigid for approximate solutions.

Finally, we remark that for the two-well problem there is a quantitative rigidity result, even in arbitrary dimensions. Call disjoint non-empty compact sets $K_1, K_2 \subset \mathbb{R}^{m\times d}$ **strongly incompatible** if for every gradient Young measure $\nu \in \mathbf{GY}^\infty(\Omega; \mathbb{R}^{m\times d})$ ($\Omega \subset \mathbb{R}^d$ any Lipschitz domain) the following implication holds:

$$\operatorname{supp} \nu_x \subset K_1 \cup K_2 \text{ a.e.} \quad \Longrightarrow \quad \operatorname{supp} \nu_x \subset K_1 \text{ a.e. or } \operatorname{supp} \nu_x \subset K_2 \text{ a.e.}$$

In this case we have a strong quantitative rigidity statement:

Theorem 10.28 (Chaudhuri–Müller 2004 [74]) *Let $K_1, K_2 \subset \mathbb{R}^{m\times d}$ be disjoint, non-empty, compact, and strongly incompatible sets and let $p \in [1,\infty)$. Abbreviate, for any $K \subset \mathbb{R}^{m\times d}$ non-empty and compact,*

$$\mathcal{D}^p(K) := \int_\Omega \operatorname{dist}(\nabla u(x), K)^p \, \mathrm{d}x.$$

Then,

$$\min\{\mathcal{D}^p(K_1), \mathcal{D}^p(K_2)\} \le C \cdot \mathcal{D}^p(K_1 \cup K_2),$$

where $C = C(\Omega, p) > 0$ is a constant.

Note that here, curiously, the case $p = 1$ is allowed in contrast to Theorem 10.20. A result for more than two wells can be found in [76].

Notes and Historical Remarks

The origin of the rigidity theory as presented in this chapter lies in the Murat–Tartar Div-Curl Lemma 8.8 and the Ball–James Rigidity Theorem 10.1 (also see Theorem 7.3). Indeed, one may view the strong rigidity of a differential inclusion $\nabla u \in K$, where $K \subset \mathbb{R}^{m\times d}$ is compact, as the following "weak-to-strong" convergence principle:

$$\left.\begin{aligned} (V_j)_j &\subset \mathrm{L}^\infty(\Omega;\mathbb{R}^{m\times d}) \\ V_j &\overset{*}{\rightharpoonup} V \quad \text{in } \mathrm{L}^\infty \\ \operatorname{curl} V_j &= 0 \quad \text{as distributions} \\ \operatorname{dist}(V_j, K) &\to 0 \quad \text{in measure} \end{aligned}\right\} \quad \Longrightarrow \quad V_j \to V \quad \text{in measure}$$

Here, $\operatorname{curl} V := (\partial_i V^k_j - \partial_j V^k_i)_{i,j,k}$ (in the sense of distributions). This formulation stresses that it is the interplay between the differential constraint $\operatorname{curl} V_j = 0$ and the nonlinear pointwise condition $\operatorname{dist}(V_j, K) \to 0$ that ensures convergence in measure, which is equivalent to the absence of oscillations. Exploring this interaction between pointwise and differential constraints systematically is the core idea of the general theory of *compensated compactness*, of which we have seen some aspects in Section 8.2.

The term "rigidity" itself is unfortunately used in different ways by different authors. In fact, *any* kind of restriction on the shape of a map is sometimes called "rigidity". Our definitions of rigidity for exact and for approximate solutions follow most closely those in Kirchheim's influential lecture notes [193]. In particular, we require linear boundary values along approximate solutions, but no boundary condition for exact solutions. This is explained as follows: Many differential inclusions with a discrete set K are trivially rigid when we impose affine boundary conditions, even if there are rank-one connections in K. On the other hand, rigidity for approximate solutions is most interesting when imposing linear (or affine) boundary conditions, see the discussion in Section 10.3.

Theorem 10.3 is contained in a more general result of Tartar [324]. It is called the "span restriction" in [54]. A special case of Theorem 10.5 (i) for $d = m = 2$ was shown in Lemma 1.4 of [107], also see the proof of Theorem 3.95 in [18].

Theorem 10.11 was first established in full in [306], which was never published in a journal. The rigidity for exact solutions was known before, namely through more general results in [187] and an unpublished manuscript by Zhang. The presented proof of rigidity for exact solutions, however, is due to Kirchheim and reproduced in [248]. Our proof of approximate rigidity follows an idea from [308] and also uses Lemma 10.13, which is from [310]. Another proof is in [23] based on the theory of quasiregular mappings (which also mentions an unpublished similar argument by Ball & James). A recent result in a similar direction, related to the Aviles–Giga functional, is in [215].

The Young measure approach of Section 10.3 is a reimagining of the ideas contained in Tartar's work [323, 324, 326, 327] and the lecture notes [193] of Kirchheim.

The first part of Theorem 10.18 is (a version of) the classical Liouville theorem (also see Problem 10.2); the extension to Sobolev functions as well as part (ii) is the work of Reshetnyak [276]. Our proof is due to Kinderlehrer [189].

Proofs of Korn's inequality (10.22) and generalizations can be found in [118, 160–162, 182, 262, 329, 330].

The rigidity for the two-well problem in two dimensions and its extension to the N-well problem are from [310, 311]. More general N-well problems for $N \geq 3$ in three dimensions were studied by Kirchheim [192, 193]. Some recent investigations into various notions of "incompatibility" between sets $K_1, \dots, K_n$ can be found in [41] and the references cited therein.

Differential inclusions have also been considered for other constraints, e.g., for symmetric gradients [67, 68, 114, 290, 291], divergence-free vector fields [151, 275], as well as general PDE-constraints [110, 145] (and the references contained therein). However, the general study of rigidity for differential inclusions is far from complete and results are usually limited to special situations rather than broad classes of problems. The interested reader can, for instance, consult [22, 173] for some recent progress on quantitative estimates and a number of open problems.

Problems

10.1 Consider a complex function $f : \mathbb{C} \to \mathbb{C}$ as a mapping $f = (u, v)$ from $\mathbb{R}^2$ to itself (identify $z = x + \mathrm{i}y \in \mathbb{C}$ with $(x, y) \in \mathbb{R}^2$). Find a subspace $L \subset \mathbb{R}^{2\times 2}$ such that $\nabla f(x, y) \in L$ for all $(x, y) \in \mathbb{R}^2$ is equivalent to the (weak) *Cauchy–Riemann equations*:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Thus prove that all (weakly) holomorphic functions are smooth.

10.2 Using the idea of the previous problem, prove the following two well-known theorems from complex analysis:

(i) Let $f\colon \mathbb{C} \to \mathbb{C}$ be holomorphic with $|f| \equiv \text{const}$. Then, f is constant.
(ii) Prove *Montel's theorem*: A sequence of uniformly bounded holomorphic functions $f_j\colon \mathbb{D} \to \mathbb{C}$ on the unit disc $\mathbb{D} \subset \mathbb{C}$ (or any other bounded open domain) that converges to $f\colon \mathbb{D} \to \mathbb{C}$ pointwise or in measure converges in fact locally uniformly and the limit f is itself holomorphic.

10.3 Prove the statements in Theorem 10.3 for the case $p = 1$. *Hint:* Use the weak-type estimates for Fourier multipliers from Theorem A.36.

10.4 Prove Theorem 10.5. *Hint:* Inspect the proof of the Ball–James Rigidity Theorem 7.3 and also use Theorem 10.3. Alternatively, prove the rigidity for exact solutions for rank $P_0 \geq 2$ in the following alternative way: For $u \in \mathrm{C}^1(\Omega;\mathbb{R}^m)$ satisfying $\nabla u(x) = P_0 g(x)$ with $g \in \mathrm{C}(\Omega)$ first show the projection relation $\nabla g(x) = (\xi \cdot \nabla g(x)^{\mathsf{T}})\xi^{\mathsf{T}}$ and conclude from there.

10.5 Find a non-empty compact set $K \subset \mathbb{R}^{m\times d}$ such that $\nabla u \in K$ is not rigid for exact solutions, but rigid for approximate solutions.

10.6 Show that if the set $K \subset \mathbb{R}^{m\times d}$ is compact, then K^{qc} is also compact.

10.7 Let $K \subset \mathbb{R}^{2\times 2}$ be a compact, non-empty and *connected* set without rank-one connections. Show that it either holds that $\det(A - B) > 0$ for all $A, B \in K$ with $A \neq B$ or $\det(A - B) < 0$ for all $A, B \in K$ with $A \neq B$. Conclude that K is rigid for approximate solutions and $K^{\mathrm{qc}} = K$. This proves Tartar's conjecture for sets K as above and was established by Šverák in 1993. *Hint:* Adapt the arguments from Section 10.7.

10.8 Let $K \subset \mathbb{R}^{2\times 2}$ be a compact, non-empty and *connected* set without rank-one connections. Show that if K is a closed connected smooth manifold and *elliptic* in the sense that for all $A \in K$ the tangent space of K at A does not contain rank-one directions, then either $\det(A - B) > c|A - B|^2$ for all $A, B \in K$ or $\det(A - B) < -c|A - B|^2$ for all $A, B \in K$. Conclude that every $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^2)$ with $\nabla u \in K$ is in $\mathrm{W}^{2,2}_{\mathrm{loc}}(\Omega;\mathbb{R}^2)$.

10.9 Let

$$K := \mathrm{SO}(2) \cup \mathrm{SO}(2)U, \quad U := \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}, \quad 0 < \alpha \leq 1 \leq \beta, \quad \alpha\beta \geq 1.$$

Show that for every $A \in K^{**}$ we can find unique $y, z \in \mathbb{R}^2$ with

$$|y|^2 = y_1^2 + y_2^2 \leq 1 \qquad \text{and} \qquad |z|^2 = z_1^2 + z_2^2 \leq 1$$

such that

$$A = \begin{pmatrix} y_1 & -y_2 \\ y_2 & y_1 \end{pmatrix} + \begin{pmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{pmatrix} U.$$

Show furthermore that the mapping $(y, z) \mapsto A$ is a diffeomorphism.

10.10 Use the method of proof we presented for Theorem 10.3 to give an alternative proof of Lemma 7.5.



Chapter 11
Microstructure

In this chapter we continue the analysis of approximate solutions to the differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \\ \nabla u \in K \quad \text{in } \Omega, \end{cases} \tag{11.1}$$

where, as before, $K \subset \mathbb{R}^{m\times d}$ $(d, m \geq 2)$ is assumed to be compact and non-empty. Recall that this means that we are looking for uniformly norm-bounded sequences $(u_j)_j \subset \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with

$$\operatorname{dist}(\nabla u_j, K) \to 0 \quad \text{in measure.}$$

Unlike in the previous chapter, however, and motivated by the example on crystal microstructure in Section 1.9, we now investigate the case

$$K^{\mathrm{qc}} \supsetneq K,$$

that is, the occurrence of *microstructure*. This necessarily requires non-Dirac measures in $\mathcal{M}^{\mathrm{qc}}(K)$ and hence by Proposition 10.9 the set K cannot be rigid for approximate solutions. So, we will be interested in solution sequences $(u_j)_j$ to our differential inclusion that do not converge in measure.

As we have seen in the last chapter, the structure of the quasiconvex hull K^{qc} of K is intimately related to the rank-one connections in K. On the other hand, it is an important observation that there are sets K without rank-one connections that are not rigid for approximate solutions. The most striking example of this is the T_4-configuration K_{T4}, to be encountered shortly, which consists of four matrices that are not rank-one connected, but for which

$$K_{\mathrm{T4}}^{\mathrm{qc}} \supsetneq K_{\mathrm{T4}}.$$

To compute the quasiconvex hull of K_{T4} and of other sets, we will introduce lower and upper bounds on K^{qc} in the form of the lamination-convex hull, the rank-one convex hull, and the polyconvex hull of K. If the upper and lower bounds agree, they must be equal to the quasiconvex hull as well.

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F. Rindler, *Calculus of Variations*, Graduate Texts in Mathematics 310,
https://doi.org/10.1007/978-3-032-24483-3_11

11.1 The T_4-Configuration

The following construction shows that a set with four matrices that are not rank-one connected is not necessarily rigid for approximate solutions, even though it must be rigid for exact solutions by the Chlebík–Kirchheim Theorem 10.16. This observation is due to Tartar [327] and anticipates a core tenet of the convex integration method to be introduced in the next chapter.

The T_4**-configuration** is the set

$$K_{\mathrm{T4}} := \{A_1, A_2, A_3, A_4\} \subset \mathbb{R}^{2\times 2}$$

with

$$A_1 := \begin{pmatrix} 3 & \\ & 1 \end{pmatrix}, \qquad A_2 := \begin{pmatrix} -1 & \\ & 3 \end{pmatrix}, \qquad A_3 := -A_1, \qquad A_4 := -A_2.$$

It is elementary to see that these four matrices do not have rank-one connections.

Theorem 11.1 (Tartar 1993 [327]) *The differential inclusion* (11.1) *for* $K := K_{\mathrm{T4}}$ *is not rigid for approximate solutions, that is, there exists a weakly* converging sequence* $(u_j)_j \subset \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^2)$ *such that*

$$\operatorname{dist}(\nabla u_j, K_{\mathrm{T4}}) \to 0 \quad \textit{in measure,}$$

but $(\nabla u_j)_j$ *does not converge in measure. Moreover,* $K_{\mathrm{T4}}^{\mathrm{qc}} \supsetneq K_{\mathrm{T4}}$.

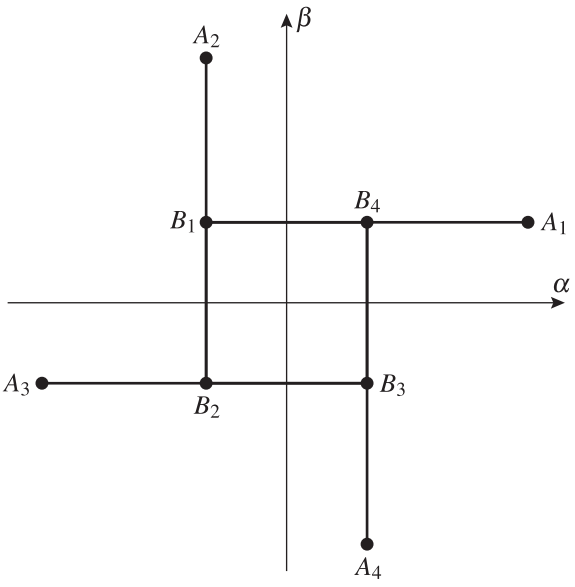


Fig. 11.1 The rank-one diagram of the T_4-configuration in the space of (2×2)-diagonal matrices with entries α, β (solid lines signify rank-one connections); notice the four T-shapes in the diagram

Proof. We define the intermediate matrices

$$B_1 := \begin{pmatrix} -1 & \\ & 1 \end{pmatrix}, \qquad B_2 := \begin{pmatrix} -1 & \\ & -1 \end{pmatrix}, \qquad B_3 := -B_1, \qquad B_4 := -B_2;$$

see Figure 11.1 for a schematic representation. The idea of the proof is to build a laminate of *infinite* order supported on K_{T4} and with barycenter B_4.

First, write B_4 as a rank-one convex combination between A_1 and B_1, which is possible since $B_4 = (A_1 + B_1)/2$ and $\operatorname{rank}(A_1 - B_1) = 1$. We have $A_1 \in K_{\mathrm{T4}}$ and $B_1 = (A_2 + B_2)/2$. Again, $A_2 \in K_{\mathrm{T4}}$. We will iterate this procedure and construct the corresponding *higher-order laminates* (see the next section for a precise definition), which yields a sequence that satisfies the differential inclusion approximately. See Figure 11.2 for an illustration.

To implement this strategy rigorously, we use the construction from the proof of Proposition 5.6. Suppose $A, B \in \mathbb{R}^{2\times 2}$ with $A - B = a \otimes n$, where $a \in \mathbb{R}^2 \setminus \{0\}$, $n \in \mathbb{S}^{d-1}$, and $\theta \in (0,1)$. For any $\varepsilon > 0$ let $D_n \subset \mathbb{R}^2$ be any rectangular parallelepiped with two faces orthogonal to n and set $F := \theta A + (1-\theta)B$. Then that proof constructed a map $v \in \mathrm{W}^{1,\infty}_{Fx}(D_n; \mathbb{R}^2)$ such that the following conditions hold:

(a) $\|v\|_{\mathrm{W}^{1,\infty}} \le C(1 + |A| + |B|)$ for some $C > 0$;

(b) $\displaystyle\sum_{k=1}^{N} |R_i^A| \le \theta |D_n|, \ \sum_{k=1}^{N} |R_i^B| \le (1-\theta)|D_n|, \ \left| D_n \setminus \left(\bigcup_{i=1}^{N} R_i^A \cup \bigcup_{i=1}^{N} R_i^B \right) \right| \le \varepsilon,$
where R_i^A, R_i^B $(i = 1, \dots, N)$ are disjoint rectangular (open) parallelepipeds with two faces orthogonal to n (some of which may be empty);

(c) $\nabla v \equiv A$ a.e. in R_i^A and $\nabla v \equiv B$ a.e. in R_i^B $(i = 1, \dots, N)$.

Fix $\delta > 0$. Apply this construction with $A := A_1$, $B := B_1$, $\theta := 1/2$ (hence $F = B_4$), and $\varepsilon := \delta/2$ in $D_n := (0,1)^2$. This yields $u_1^{(\delta)}$ satisfying the above properties. Then apply the construction again in every $R_i^{B_1}$, $i = 1, \dots, N_1$, with $A := A_2$, $B := B_2$, $\theta := 1/2$ (hence $F = B_1$), and $\varepsilon := \delta/(2^2 N_1)$ to get $u_2^{(\delta)}$, which is defined in $\bigcup_{i=1}^{N} R_i^{B_1}$. Note that we may just replace $u_1^{(\delta)}$ by $u_2^{(\delta)}$ in the $R_i^{B_1}$s since the boundary values agree.

Let $[k] := ((k-1) \bmod 4) + 1$ denote the cyclic index corresponding to $k \in \mathbb{N}$. We then proceed with the above construction for $k = 2, 3, \dots$ in the following way: In the k'th step, on the N_{k-1} parallelepipeds $R_i^{B_{[k-1]}}$, $i = 1, \dots, N_{k-1}$, with the intermediate matrix $B_{[k-1]} \in \{B_1, B_2, B_3, B_4\}$ in it, use the lamination construction with

$$A := A_{[k]}, \qquad B := B_{[k]}, \qquad \theta := \frac{1}{2}, \qquad \varepsilon := \frac{\delta}{2^k N_{k-1}},$$

so that $F = B_{[k-1]}$. In this way we successively eliminate more and more of the intermediate matrices B_1, B_2, B_3, B_4.

Thus, we have constructed a sequence $(u_l^{(\delta)})_l \subset \mathrm{W}^{1,\infty}((0,1)^2; \mathbb{R}^2)$ that is uniformly bounded in $\mathrm{W}^{1,\infty}$, satisfies $u_l^{(\delta)}|_{\partial(0,1)^2} = B_4 x$, and such that

$$\left| \left\{ x \in (0,1)^2 \ : \ \nabla u_l^{(\delta)}(x) \notin K_{\mathrm{T4}} \right\} \right| \le 2^{-l} + \sum_{k=1}^{l} \delta 2^{-k} \le 2^{-l} + \delta.$$

Here, the first summand corresponds to the union of the sets R_i^B, $i = 1, \dots, N_l$, which has a total size of at most 2^{-l} by (b). The second term corresponds to the sizes of the "adjustment layers".

By the uniform boundedness, we may pass to a weakly* converging subsequence of $(u_l^{(\delta)})_l$ with limit $u^{(\delta)} \in \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^2)$. Then, for a δ-independent $C > 0$,

$$u^{(\delta)}|_{\partial\Omega} = B_4 x, \qquad \|u^{(\delta)}\|_{\mathrm{W}^{1,\infty}} \leq C$$

and

$$\left|\left\{ x \in (0,1)^2 \,:\, \nabla u^{(\delta)}(x) \notin K_{\mathrm{T4}} \right\}\right| \leq \delta.$$

Now set $u_j := u^{(1/j)} \in \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^2)$, which is a uniformly bounded sequence with $u_j|_{\partial\Omega} = B_4 x$ and

$$\operatorname{dist}(\nabla u_j, K_{\mathrm{T4}}) \to 0 \quad \text{in measure}$$

as $j \to \infty$.

We can furthermore assume that $(\nabla u_j)_j$ converges weakly* in $\mathrm{W}^{1,\infty}$ and generates a homogeneous gradient Young measure $\mu \in \mathcal{M}^{\mathrm{qc}}(K_{\mathrm{T4}})$ with $[\mu] = B_4 \notin K_{\mathrm{T4}}$ (via Lemma 4.12). This in fact holds for the above construction, but one may also invoke Lemma 4.15 to ensure the homogeneity. Then, μ cannot be a Dirac mass, $K_{\mathrm{T4}}^{\mathrm{qc}} \supsetneq K_{\mathrm{T4}}$, and Proposition 10.9 (i) immediately yields that the inclusion $\nabla u \in K_{\mathrm{T4}}$ is not rigid for approximate solutions.

For the last step one can also argue by observing that if, up to selecting a subsequence, we had $\nabla u_j \to \nabla u$ in measure as $j \to \infty$ for some $u \in \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^2)$, then it would hold that $\nabla u \in K_{\mathrm{T4}}$ almost everywhere and $u|_{\partial\Omega} = B_4 x$, which is impossible since $\nabla u \in K_{\mathrm{T4}}$ is rigid for exact solutions by Problem 11.2 (or the Chlebík–Kirchheim Theorem 10.16). □

Theorem 11.1 provides a *negative* answer to the following question of historical importance.

Conjecture 11.2 (Tartar 1982 [324]) If a non-empty compact set $K \subset \mathbb{R}^{2\times 2}$ does not contain rank-one directions, then $\nabla u \in K$ is rigid for approximate solutions.

However, Tartar's conjecture turns out to be true for connected sets, see Problem 10.7. It is a remarkable result that rank-one connections and (generalized) T_4-configurations are in fact the only obstructions to rigidity for approximate solutions, as was shown by Faraco and Székelyhidi Jr. [136].

11.2 Hulls of Sets

In this and the next section we will consider several ways to compute, or at least bound, quasiconvex hulls.

For a non-empty compact set $K \subset \mathbb{R}^{m\times d}$ we define the set $\mathcal{M}^{\mathrm{lc}}(K)$ of **laminates of finite order** supported on K as

$$\mathcal{M}^{\mathrm{lc}}(K) := \bigcup_{i=0}^{\infty} \mathcal{M}^{\mathrm{lc},i}(K),$$

where

$$\begin{aligned}\mathcal{M}^{\mathrm{lc},0}(K) &:= \big\{\, \delta_A \;:\; A \in K \,\big\},\\ \mathcal{M}^{\mathrm{lc},i+1}(K) &:= \big\{\, \theta\mu_1 + (1-\theta)\mu_2 \;:\; \mu_1, \mu_2 \in \mathcal{M}^{\mathrm{lc},i}(K),\ \theta \in [0,1],\\ &\qquad\qquad \operatorname{rank}([\mu_1] - [\mu_2]) \le 1 \,\big\}.\end{aligned}$$

We call an element of $\mathcal{M}^{\mathrm{lc},i}(K) \setminus \bigcup_{j=0}^{i-1} \mathcal{M}^{\mathrm{lc},j}(K)$ a **laminate of order** i. Lemma 11.5 below will show that all elements of $\mathcal{M}^{\mathrm{lc}}(K)$ are homogeneous gradient Young measures. Problem 11.4 contains an alternative characterization of $\mathcal{M}^{\mathrm{lc}}(K)$.

The **lamination-convex hull** of K is defined to be

$$K^{\mathrm{lc}} := \big\{\, [\mu] \;:\; \mu \in \mathcal{M}^{\mathrm{lc}}(K) \,\big\}.$$

Thus, K^{lc} is obtained from K by inductively adding rank-one lines. Indeed, we could also write

$$K^{\mathrm{lc}} = \bigcup_{i=0}^{\infty} K^{\mathrm{lc},i},$$

where

$$\begin{aligned}K^{\mathrm{lc},0} &:= K,\\ K^{\mathrm{lc},i+1} &:= \big\{\, \theta A + (1-\theta)B \;:\; A, B \in K^{\mathrm{lc},i},\ \theta \in [0,1],\ \operatorname{rank}(A-B) \le 1 \,\big\}.\end{aligned}$$

We illustrate laminates of different orders in Figure 11.2.

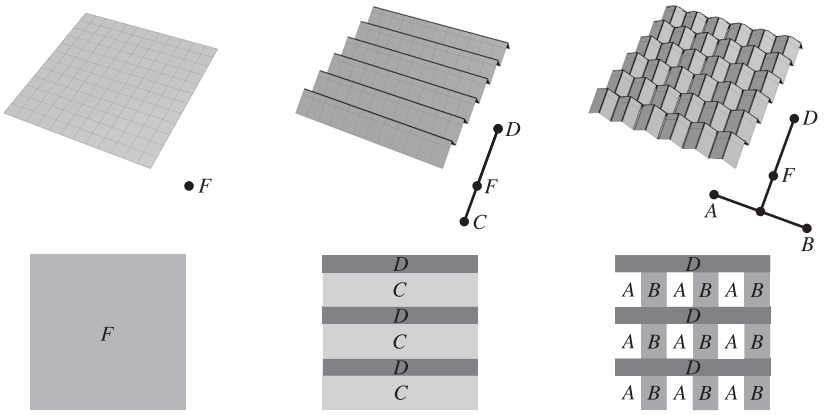


Fig. 11.2 Laminates of order 0, 1, and 2 (only one element of the generating sequence and its rank-one diagram as well as the gradient schematic are shown)

Example 11.3 Set for some $a \in \mathbb{R}^m$, $b \in \mathbb{R}^d$,

$$K := \{0, a \otimes b\} \subset \mathbb{R}^{m\times d}.$$

Then,

$$K^{\mathrm{lc}} = K^{\mathrm{qc}} = K^{**} = \left\{ \theta a \otimes b \ : \ \theta \in [0,1] \right\},$$

as can be easily verified.

We will see in Lemma 11.5 below that the lamination-convex hull K^{lc} is a lower bound on the quasiconvex hull K^{qc} in the sense that $K^{\mathrm{lc}} \subset K^{\mathrm{qc}}$. Unfortunately, K^{lc} is often strictly smaller than K^{qc}, limiting its usefulness. For instance, since there are no rank-one connections in the T_4-configuration K_{T4}, we get that $K^{\mathrm{lc}}_{\mathrm{T4}} = K_{\mathrm{T4}}$, so we have not come nearer to bounding $K^{\mathrm{qc}}_{\mathrm{T4}}$ from below. The reason for this is that the procedure employed in the proof of Theorem 11.1 to construct non-trivial elements of $\mathcal{M}^{\mathrm{qc}}(K_{\mathrm{T4}})$ and thus of points in $K^{\mathrm{qc}}_{\mathrm{T4}} \setminus K_{\mathrm{T4}}$, needs *infinitely* many lamination steps.

Consequently, we define the set of **laminates of infinite order** (also called **rank-one convex measures**) supported on a non-empty compact set $K \subset \mathbb{R}^{m\times d}$ as

$$\mathcal{M}^{\mathrm{rc}}(K) := \left\{ \mu \in \mathcal{M}^1(K) \ : \ \mu_j \overset{*}{\rightharpoonup} \mu \text{ for some } (\mu_j)_j \subset \mathcal{M}^{\mathrm{lc}}(K) \ \right\}.$$

The corresponding **rank-one-convex hull** of K is

$$K^{\mathrm{rc}} := \left\{ [\mu] \ : \ \mu \in \mathcal{M}^{\mathrm{rc}}(K) \right\}.$$

We have the following characterization of laminates of infinite order, which is similar in spirit to the Kinderlehrer–Pedregal Theorem 8.11:

Theorem 11.4 (Pedregal 1993 [271]) *Let $\mu \in \mathcal{M}^1(K)$. Then, $\mu \in \mathcal{M}^{\mathrm{rc}}(K)$ if and only if*

$$h([\mu]) \le \int h \, \mathrm{d}\mu$$

for all rank-one convex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$.

Thus,

$$\mathcal{M}^{\mathrm{rc}}(K) = \left\{ \mu \in \mathcal{M}^1(K) \ : \ h([\mu]) \le \int h \, \mathrm{d}\mu \text{ for all rank-one convex functions } h \in \mathrm{C}(\mathbb{R}^{m\times d}) \right\}.$$

We also define the set of **polyconvex measures** supported on K as

$$\mathcal{M}^{\mathrm{pc}}(K) := \left\{ \mu \in \mathcal{M}^1(K) \ : \ M([\mu]) = \langle M, \mu \rangle \text{ for all minors } M \ \right\},$$

where the minors are defined as in Section 5.2. Recall that for $m = d = 3$, we can equivalently choose $M \in \{\mathrm{id}, \mathrm{cof}, \det\}$. Since the minors are precisely the quasiaffine functions, see Corollary 5.4, we get immediately that

$$\mathcal{M}^{\mathrm{pc}}(K) = \Big\{ \mu \in \mathcal{M}^1(K) \ : \ h([\mu]) \le \int h \, \mathrm{d}\mu \text{ for all polyconvex functions } h \in \mathrm{C}(\mathbb{R}^{m\times d}) \Big\}.$$

Then, the **polyconvex hull** of a non-empty compact set $K \subset \mathbb{R}^{m\times d}$ is defined as

$$K^{\mathrm{pc}} := \big\{ [\mu] \ : \ \mu \in \mathcal{M}^{\mathrm{pc}}(K) \big\}.$$

There are further equivalent ways to define the various hulls, see Problem 11.5 and Problem 11.6.

Lemma 11.5 *For a compact set $K \subset \mathbb{R}^{m\times d}$ it holds that*

$$\mathcal{M}^{\mathrm{lc}}(K) \subset \mathcal{M}^{\mathrm{rc}}(K) \subset \mathcal{M}^{\mathrm{qc}}(K) \subset \mathcal{M}^{\mathrm{pc}}(K)$$

and thus

$$K \subset K^{\mathrm{lc}} \subset K^{\mathrm{rc}} \subset K^{\mathrm{qc}} \subset K^{\mathrm{pc}} \subset K^{**}.$$

Proof. The inclusion $\mathcal{M}^{\mathrm{lc}}(K) \subset \mathcal{M}^{\mathrm{rc}}(K)$ is trivial. Moreover, it follows from Corollary 5.12 in conjunction with Corollary 5.4 that for any homogeneous gradient Young measure $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ and any minor M we have $M([\mu]) = \langle M, \mu \rangle$, whereby $\mu \in \mathcal{M}^{\mathrm{pc}}(K)$.

It remains to show $\mathcal{M}^{\mathrm{lc}}(K) \subset \mathcal{M}^{\mathrm{qc}}(K)$, from which it follows that also $\mathcal{M}^{\mathrm{rc}}(K) \subset \mathcal{M}^{\mathrm{qc}}(K)$ by the weak*-closedness of $\mathcal{M}^{\mathrm{qc}}(K)$, which can be seen by a diagonal argument similar to the one in the proof of Lemma 8.13.

We proceed inductively. For $\mu \in \mathcal{M}^{\mathrm{lc},0}(K)$ there is nothing to show. For the inductive step assume that $\mu \in \mathcal{M}^{\mathrm{lc},i+1}(K)$ is of the form

$$\mu = \theta\mu_1 + (1-\theta)\mu_2, \qquad \theta \in [0,1],$$

where $\mu_1, \mu_2 \in \mathcal{M}^{\mathrm{lc},i}(K) \subset \mathcal{M}^{\mathrm{qc}}(K)$. Set $A_1 := [\mu_1]$, $A_2 := [\mu_2]$ and $F := \theta A_1 + (1-\theta)A_2$, such that $A_1 - A_2$ has rank at most one.

Apply the construction of the proof of Proposition 5.6 to get a norm-bounded sequence $(v_j)_j \subset \mathrm{W}^{1,\infty}_{Fx}(Q;\mathbb{R}^m)$, where $Q \subset \mathbb{R}^d$ is a rotated unit-volume cube, with

$$\nabla v_j \xrightarrow{\mathbf{Y}} \theta\delta_{A_1} + (1-\theta)\delta_{A_2}.$$

Now, on each solid piece of Q where ∇v_j is equal to A_1, say $R_1^{(j)}, \dots, R_M^{(j)} \subset Q$, we re-define ∇v_j to be equal to the n'th element of a generating sequence for μ_1 (by Lemma 4.15 we may choose the domain of definition freely and by Zhang's Lemma 8.14 we can ensure that the sequence has boundary values $A_1 x$). Similarly, if ∇v_j is equal to A_2 on the solid pieces $S_1^{(j)}, \dots, S_N^{(j)} \subset Q$, then re-define ∇v_j to be equal to the n'th element of a generating sequence for μ_2. These modifications yield a map $w_{j,n} \in \mathrm{W}^{1,\infty}_{Fx}(Q;\mathbb{R}^m)$ (n is the index of the "inner" sequences), which agrees with v_j outside the sets $R^{(j)} := \bigcup_m R_m^{(j)}$ and $S^{(j)} := \bigcup_m S_m^{(j)}$.

Let $(\varphi_k \otimes h_k)_k \subset \mathrm{C}_0(Q) \otimes \mathrm{C}_0(\mathbb{R}^{m\times d})$ be the countable set of integrands from Lemma 4.8. Then we may choose $n = n(j)$ sufficiently large so that

$$\left| \int_{R^{(j)}\cup S^{(j)}} h_k(\nabla w_{j,n(j)})\,\mathrm{d}x - |R^{(j)}| \cdot \int h_k\,\mathrm{d}\mu_1 - |S^{(j)}| \cdot \int h_k\,\mathrm{d}\mu_2\,\mathrm{d}x \right| \leq \frac{1}{j}$$

for all $k \leq j$. Since we also have that $|R^{(j)}| \to \theta$ and $|S^{(j)}| \to 1-\theta$ as well as

$$\left| \int_{Q\setminus(R^{(j)}\cup S^{(j)})} \varphi_k h_k(\nabla v_j)\,\mathrm{d}x \right| \to 0 \qquad \text{as } j \to \infty,$$

we infer via Lemma 4.8 that $\nabla w_{j,n(j)} \overset{\mathbf{Y}}{\to} \nu \in \mathbf{GY}^\infty(Q;\mathbb{R}^{m\times d})$ with

$$\int_Q \int h_k\,\mathrm{d}\nu_x\,\mathrm{d}x = \theta \int h_k\,\mathrm{d}\mu_1 + (1-\theta)\int h_k\,\mathrm{d}\mu_2 \qquad \text{for all } k \in \mathbb{N}.$$

Finally, using the averaging principle of Lemma 4.15 (this additional averaging is in fact not really necessary, as one can check through a closer inspection of the proof of Proposition 5.6), we get a sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}_{Fx}(Q;\mathbb{R}^m)$ such that

$$\nabla u_j \overset{\mathbf{Y}}{\to} \theta\mu_1 + (1-\theta)\mu_2 = \mu.$$

Thus, $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ and the proof is finished. □

As a first application of the preceding lemma, we may now calculate the various hulls of the T_4-configuration.

Proposition 11.6 *For $K_{\mathrm{T4}} = \{A_1, A_2, A_3, A_4\}$ from Theorem 11.1 we have (see Figure 11.3)*

$$K_{\mathrm{T4}} = K^{\mathrm{lc}}_{\mathrm{T4}} \subsetneq K^{\mathrm{rc}}_{\mathrm{T4}} = K^{\mathrm{qc}}_{\mathrm{T4}} = K^{\mathrm{pc}}_{\mathrm{T4}} = \{B_1, B_2, B_3, B_4\}^{**} \cup \bigcup_{i=1}^{4} \{A_i, B_i\}^{**},$$

where

$$B_1 := \begin{pmatrix} -1 & \\ & 1 \end{pmatrix}, \quad B_2 := \begin{pmatrix} -1 & \\ & -1 \end{pmatrix}, \quad B_3 := -B_1, \quad B_4 := -B_2.$$

Here, $\{B_1, B_2, B_3, B_4\}^{**}$ and $\{A_i, B_i\}^{**}$ denote the closed square with vertices B_1, B_2, B_3, B_4 and the closed line segment from A_i to B_i, respectively.

Proof. The fact that $K_{\mathrm{T4}} = K^{\mathrm{lc}}_{\mathrm{T4}}$ is clear since there are no rank-one connections in K_{T4}.

Step 1. We first compute $K^{\mathrm{rc}}_{\mathrm{T4}}$ as a lower bound on $K^{\mathrm{qc}}_{\mathrm{T4}}$ and $K^{\mathrm{pc}}_{\mathrm{T4}}$; this technique is sometimes called the *lamination method*. By (the proof of) Theorem 11.1 we have that the four intermediate matrices B_1, B_2, B_3, B_4 lie in $K^{\mathrm{rc}}_{\mathrm{T4}}$. Hence, also the rank-one lines between B_i and B_{i+1} lie in $K^{\mathrm{rc}}_{\mathrm{T4}}$, where $i \in \{1,2,3,4\}$ is a cyclic index. Since its edges are rank-one lines, the closed square $\{B_1, B_2, B_3, B_4\}^{**}$ then lies in $K^{\mathrm{rc}}_{\mathrm{T4}}$ as well. Clearly,

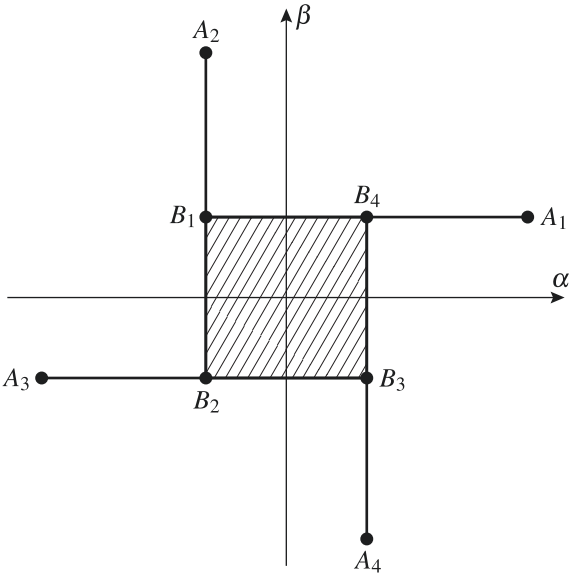


Fig. 11.3 The quasiconvex hull $K^{\mathrm{qc}}_{\mathrm{T4}}$ of the T_4-configuration (hatched area and solid lines) in the space of diagonal matrices with entries α, β

the additional rank-one lines $\{A_i, B_i\}^{**}$ also lie in that set. So, we have shown that

$$\widehat{K} := \{B_1, B_2, B_3, B_4\}^{**} \cup \bigcup_{i=1}^{4} \{A_i, B_i\}^{**} \subset K^{\mathrm{rc}}_{\mathrm{T4}} \subset K^{\mathrm{qc}}_{\mathrm{T4}}.$$

Step 2. Next, we show that $K^{\mathrm{qc}}_{\mathrm{T4}} \subset \widehat{K}$, for which we employ the *separation method*, which we have already seen in the proof of Theorem 10.11. We first observe that since $K^{\mathrm{qc}} \subset K^{**}$ and all matrices in K are diagonal, the same must hold for K^{qc}.

Let $F \in K^{\mathrm{qc}}$ and $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ with $F = [\mu]$. In Lemma 10.14 we saw that the function

$$\det{}^{++}(A) := \begin{cases} \det A & \text{if } A \text{ is positive semidefinite,} \\ 0 & \text{otherwise,} \end{cases} \qquad A \in \mathbb{R}^{2\times 2}_{\mathrm{sym}},$$

is quasiconvex on symmetric matrices, i.e.,

$$\det{}^{++}(A) \le \fint_{B_1} \det{}^{++}(A + \nabla^2 \psi(z))\, \mathrm{d}z$$

for all $A \in \mathbb{R}^{2\times 2}_{\mathrm{sym}}$ and all $\psi \in \mathrm{W}^{2,2}_c(B_1)$. Then, $A \mapsto \det{}^{++}(A - B_1)$ is also quasiconvex on symmetric matrices. This corresponds to shifting the whole T_4-configuration.

Via the Jensen-type inequality from Lemma 10.15, we thus get

$$\det{}^{++}(F - B_1) \le \int \det{}^{++}(A - B_1)\, \mathrm{d}\mu(A) = 0.$$

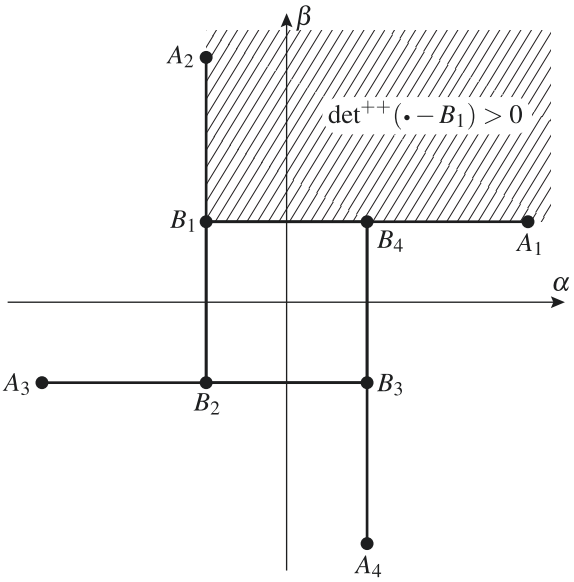


Fig. 11.4 One step in the determination of an upper bound for $K^{\mathrm{qc}}_{\mathrm{T4}}$

The equality on the right here follows since

$$\det^{++}(A_i - B_1) = 0 \qquad \text{for } i = 1, 2, 3, 4,$$

as can be checked directly. Therefore, we also must have $\det^{++}(F - B_1) = 0$, that is, $F - B_1$ is not positive definite. Thus, F cannot lie in the hatched region in Figure 11.4.

Repeating this argument after rotating the T_4-configuration, and further excluding the lines in direction $A_i - B_i$ starting from A_i ($i = 1, 2, 3, 4$), which do not even lie in the convex hull of K, we see that $F \in \widehat{K}$, whereby

$$K^{\mathrm{qc}}_{\mathrm{T4}} \subset \widehat{K}.$$

Together with Step 1 we have thus shown that $K^{\mathrm{rc}}_{\mathrm{T4}} = K^{\mathrm{qc}}_{\mathrm{T4}} = \widehat{K}$.

Step 3. We finally prove that $K^{\mathrm{pc}} \subset \widehat{K}$. In fact, the proof of this is the same as the previous step since

$$\det^{++}\left(\begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}\right) = \alpha^+ \beta^+,$$

where $s^+ := \max\{s, 0\}$ for $s \in \mathbb{R}$. So, $\det^{++}$ equals a polyconvex function on the subspace of diagonal matrices. □

11.3 Multi-Well Inclusions

Returning to the two-well problem in two dimensions, which we first studied in Section 10.7, we now investigate the non-rigid case. Thus, consider (after normalization)

$$K := \mathrm{SO}(2) \cup \mathrm{SO}(2)U, \quad U := \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}, \quad 0 < \alpha \le 1 \le \beta, \quad \alpha\beta \ge 1. \tag{11.2}$$

By Lemma 10.23, for $\alpha = 1$ the two wells are simply rank-one connected and for $\alpha < 1$ they are doubly rank-one connected.

In order to describe the hulls of K efficiently, it is convenient to introduce the so-called **conformal coordinates**: For any $A \in K^{**}$ we may write

$$A = \begin{pmatrix} y_1 & -y_2 \\ y_2 & y_1 \end{pmatrix} + \begin{pmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{pmatrix} U =: [\![y, z]\!],$$

where $y, z \in \mathbb{R}^2$ with

$$|y|^2 = y_1^2 + y_2^2 \le 1 \qquad \text{and} \qquad |z|^2 = z_1^2 + z_2^2 \le 1,$$

see Problem 10.9.

The following result then completely describes all the hulls of K:

Theorem 11.7 (Šverák 1993 [311]) *Let K be as in (11.2).*

(i) If $\alpha = 1$ (simple rank-one connection), then

$$\begin{aligned} K^{\mathrm{lc}} &= K^{\mathrm{rc}} = K^{\mathrm{qc}} = K^{\mathrm{pc}} \\ &= \left\{ QU_s \in \mathbb{R}^{2\times 2} \ : \ Q \in \mathrm{SO}(2),\ U_s = \begin{pmatrix} 1 & \\ & s \end{pmatrix},\ s \in [1, \beta] \right\}. \end{aligned}$$

(ii) If $\alpha < 1$ (double rank-one connection) and $\det U = \alpha\beta = 1$, then

$$\begin{aligned} K^{\mathrm{lc}} &= K^{\mathrm{rc}} = K^{\mathrm{qc}} = K^{\mathrm{pc}} \\ &= \left\{ A = [\![y, z]\!] \in K^{**} \ : \ |y| + |z| \le 1 \text{ and } \det A = 1 \right\}. \end{aligned}$$

(iii) If $\alpha < 1$ and $\det U = \alpha\beta > 1$, then

$$\begin{aligned} K^{\mathrm{lc}} &= K^{\mathrm{rc}} = K^{\mathrm{qc}} = K^{\mathrm{pc}} \\ &= \left\{ A = [\![y, z]\!] \in K^{**} \ : \ |y| \le \tfrac{\det U - \det A}{\det U - 1},\ |z| \le \tfrac{\det A - 1}{\det U - 1} \right\}. \end{aligned}$$

Proof. Ad (i). Let $F \in K^{\mathrm{pc}}$ and take $\mu \in \mathcal{M}^{\mathrm{pc}}(K)$ with $[\mu] = F$. We will show that $\mu \in \mathcal{M}^{\mathrm{lc},1}(K)$, which will allow us to conclude the assertion via Lemma 11.5. By the structure of K, we may write

$$\mu = \theta\gamma + (1-\theta) R(U)_{\#}\eta, \qquad \text{where} \qquad \gamma, \eta \in \mathcal{M}^1(\mathrm{SO}(2)),\ \theta \in [0,1].$$

Here, $R(U)_\# \eta(B) := \eta(B \cdot U^{-1})$ for any Borel set $B \subset \mathbb{R}^{2\times 2}$. Then, with $G := [\gamma]$, $H := [\eta]$, we have $F = \theta G + (1-\theta)HU$ and, since $G, H \in \mathrm{SO}(2)^{**}$, it holds that (see Appendix A.1)

$$G = \begin{pmatrix} g_1 & -g_2 \\ g_2 & g_1 \end{pmatrix}, \qquad H = \begin{pmatrix} h_1 & -h_2 \\ h_2 & h_1 \end{pmatrix}$$

with $g, h \in \mathbb{R}^2$ such that

$$\det G = |g|^2 = g_1^2 + g_2^2 \le 1 \qquad \text{and} \qquad \det H = |h|^2 = h_1^2 + h_2^2 \le 1.$$

Thus, we can compute, using Young's inequality,

$$\begin{aligned} G : \mathrm{cof}(HU) &= \begin{pmatrix} g_1 & -g_2 \\ g_2 & g_1 \end{pmatrix} : \begin{pmatrix} h_1 & -h_2 \\ \beta h_2 & \beta h_1 \end{pmatrix} \\ &= (1+\beta)(g_1 h_1 + g_2 h_2) \\ &\le \frac{1+\beta}{2}(|g|^2 + |h|^2) \\ &\le 1 + \beta. \end{aligned}$$

Observe also that the equality $G : \mathrm{cof}(HU) = 1 + \beta$ can only hold if $G = H$ and $\det G = \det H = 1$.

From the definition of polyconvex measures we further have

$$\det F = \theta + (1-\theta)\beta.$$

Then, using (10.16) and the above considerations, we compute

$$\begin{aligned} \det F &= \det(\theta G + (1-\theta)HU) \\ &= \theta^2 \det G + \theta(1-\theta) G : \mathrm{cof}(HU) + (1-\theta)^2 \det H \det U \\ &\le \theta^2 + \theta(1-\theta)(1+\beta) + (1-\theta)^2 \beta \\ &= \theta + (1-\theta)\beta \\ &= \det F. \end{aligned}$$

Hence, the above inequality is in fact an equality, from which we infer $G : \mathrm{cof}(HU) = 1 + \beta$, whereby $G = H$ and $\det G = \det H = 1$. This implies $G = H \in \mathrm{SO}(2)$ and

$$\mu = \theta \delta_G + (1-\theta)\delta_{GU} \in \mathcal{M}^{\mathrm{lc},1}(K) \tag{11.3}$$

since any probability measure $\sigma \in \mathcal{M}^1(\mathrm{SO}(2)^{**})$ with $[\sigma] \in \mathrm{SO}(2)$ is a Dirac mass. Indeed, for

$$A = \begin{pmatrix} a_1 & -a_2 \\ a_2 & a_1 \end{pmatrix} \in \mathrm{SO}(2)^{**}$$

we have $\det A = a_1^2 + a_2^2 \le 1$ and so the determinant is a strictly convex function on

$\mathrm{SO}(2)^{**}$. Thus, by Jensen's inequality,

$$1 = \det[\sigma] \le \int \det A \, \mathrm{d}\sigma(A) \le 1.$$

Consequently, σ is carried by $\mathrm{SO}(2)$, but it is elementary to see that any non-trivial convex combination of matrices in $\mathrm{SO}(2)$ has determinant strictly less than 1. Hence, $\operatorname{supp}\sigma$ must be a singleton.

Finally, from (11.3) it follows that

$$[\mu] = G(\theta \mathrm{Id} + (1-\theta)U),$$

which immediately implies the claimed formula for $K^{\mathrm{lc}} = K^{\mathrm{rc}} = K^{\mathrm{qc}} = K^{\mathrm{pc}}$.

Ad (ii). From the definition of the polyconvex hull we have that $\mu \in \mathcal{M}^{\mathrm{pc}}(K)$ if and only if $\mu \in \mathcal{M}^1(\mathbb{R}^{2\times 2})$ and

$$H([\mu], \det[\mu]) \le \int H(A, \det A) \, \mathrm{d}\mu(A)$$

for all convex $H \in \mathrm{C}(\mathbb{R}^{2\times 2} \times \mathbb{R})$. By Jensen's inequality, we thus get

$$K^{\mathrm{pc}} = \left\{ A \in \mathbb{R}^{2\times 2} \,:\, (A, \det A) \in (\mathbf{M}K)^{**} \right\},$$

where we have set $\mathbf{M}K := \{ (A, \det A) : A \in K \}$; also see Problem 11.6. Employing conformal coordinates, we can thus represent $A \in K$ as $[\![y, z]\!]$ for a pair $(y, z) \in \mathbb{R}^2 \times \mathbb{R}^2$, where either $|y| = 1$ and $z = 0$, or $y = 0$ and $|z| = 1$. To determine K^{pc} we then need to compute the convex hull G_κ^{**} of the set

$$\begin{aligned} G_\kappa := &\left\{ (y, 0, 1) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \,:\, |y| = 1 \right\} \\ &\cup \left\{ (0, z, \kappa) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \,:\, |z| = 1 \right\} \end{aligned}$$

in the case $\kappa = \det U = 1$. Since this set is invariant under rotations of the matrices represented by y, z, it suffices to compute this convex hull with $\mathbb{R}^1$ in place of $\mathbb{R}^2$, which is elementary. We get

$$G_1^{**} = \left\{ (y, z, 1) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \,:\, |y| + |z| \le 1 \right\}, \tag{11.4}$$

whereby

$$K^{\mathrm{pc}} = \left\{ A = [\![y, z]\!] \in K^{**} \,:\, |y| + |z| \le 1 \text{ and } \det A = 1 \right\}.$$

By Lemma 11.5 it remains to show that $K^{\mathrm{pc}} \subset K^{\mathrm{lc}}$.

Let $F \in K^{\mathrm{pc}} = K^{**} \cap \{\det = 1\}$, where here and in the following we simply write "$\{\det = 1\}$" for the set $\{ A \in \mathbb{R}^{2\times 2} : \det A = 1 \}$. The equality follows by considering G_1 again. We distinguish two cases:

(a) If $F = [\![\bar{y}, \bar{z}]\!] \in \partial K^{**} \cap \{\det = 1\}$, then by (11.4) it holds that $|\bar{y}| + |\bar{z}| = 1$. For $\bar{y} = 0$ or $\bar{z} = 0$ we immediately get $A \in K \subset K^{\mathrm{lc}}$. If $\bar{y} \neq 0$ and $\bar{z} \neq 0$, we define

$$q(t) := \det\left(t\left[\!\!\left[\frac{\bar{y}}{|\bar{y}|}, 0 \right]\!\!\right] + (1-t)\left[\!\!\left[0, \frac{\bar{z}}{|\bar{z}|} \right]\!\!\right]\right), \qquad t \in [0,1].$$

Clearly, q is a quadratic function in t and $q(|\bar{y}|)$ is the determinant of the matrix with representation $[\![\bar{y}, \bar{z}]\!]$, that is, our F. Thus, $q(0) = q(1) = q(|\bar{y}|) = 1$ and hence $q \equiv 1$, whereby $[\![\bar{y}/|\bar{y}|, 0]\!]$ and $[\![0, \bar{z}/|\bar{z}|]\!]$ are rank-one connected. We conclude that F lies on the rank-one line between the matrices $[\![\bar{y}/|\bar{y}|, 0]\!] \in K$ and $[\![0, \bar{z}/|\bar{z}|]\!] \in K$, that is, $F \in K^{\mathrm{lc}}$.

(b) Let now $F \in (K^{**} \setminus \partial K^{**}) \cap \{\det = 1\}$. It is always possible to find $a \in \mathbb{R}^2 \setminus \{0\}$, $n \in \mathbb{S}^{d-1}$ such that $\operatorname{cof} F : (a \otimes n) = a^{\mathsf{T}}(\operatorname{cof} F)n = 0$. Then, by Jacobi's formula (see Appendix A.1),

$$t \mapsto \det[F + t(a \otimes n)] \equiv \text{const},$$

whereby the determinant constraint is preserved along the rank-one line $F + \mathbb{R}(a \otimes n)$. This line intersects the set $\partial(K^{**}) \cap \{\det = 1\}$ for precisely two values of t. Since we have already shown that $\partial(K^{**}) \cap \{\det = 1\} \subset K^{\mathrm{lc}}$, we also get that $F \in K^{\mathrm{lc}}$.

Ad (iii). By a similar reasoning as at the beginning of Step (ii), we get

$$G_\kappa^{**} = \left\{ (y,z,s) \in \mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \,:\, |y| \le \tfrac{\kappa - s}{\kappa - 1}, |z| \le \tfrac{s-1}{\kappa - 1} \right\},$$

where this time $\kappa = \det U > 1$. Then, denoting by $\det[\![y, z]\!]$ the determinant of the matrix $[\![y, z]\!]$ with conformal coordinates (y, z), we have

$$K^{\mathrm{pc}} = \left\{ [\![y,z]\!] \in K^{**} \,:\, |y| \le \tfrac{\det U - \det[\![y,z]\!]}{\det U - 1}, |z| \le \tfrac{\det[\![y,z]\!] - 1}{\det U - 1} \right\}. \tag{11.5}$$

In the following we will show that $\partial K^{\mathrm{pc}} \subset K^{\mathrm{lc}}$. This suffices to prove the claim since every rank-one line through an interior point of K^{pc} intersects ∂K^{pc} in precisely two points.

So, let $F = [\![\bar{y}, \bar{z}]\!] \in \partial K^{\mathrm{pc}}$. Define

$$f(y,z) := (\det U - 1)|y| - \det U + \det[\![y,z]\!],$$
$$g(y,z) := (\det U - 1)|z| + 1 - \det[\![y,z]\!].$$

Then, (11.5) can be rewritten as

$$K^{\mathrm{pc}} = \left\{ [\![y,z]\!] \in \mathbb{R}^{2\times 2} \,:\, f(y,z) \le 0 \text{ and } g(y,z) \le 0 \right\}.$$

As $F = [\![\bar{y}, \bar{z}]\!] \in \partial K^{\mathrm{pc}}$, either $f(\bar{y}, \bar{z}) = 0$ or $g(\bar{y}, \bar{z}) = 0$. We distinguish three cases:

(a) If $f(\bar{y}, \bar{z}) = 0$ and $g(\bar{y}, \bar{z}) = 0$, then a simple calculation shows that $|\bar{y}| + |\bar{z}| = 1$. Along the line

$$M(t) := t\left[\!\!\left[\frac{\bar{y}}{|\bar{y}|}, 0 \right]\!\!\right] + (1-t)\left[\!\!\left[0, \frac{\bar{z}}{|\bar{z}|} \right]\!\!\right], \qquad t \in [0,1],$$

the functions f, g are quadratic in t and have three zeros, at $t = 0, 1, |\bar{y}|$. Thus, f, g are identically zero along the line $t \mapsto M(t)$. Since $\det U > 1$, from the definition of f (or g) we conclude that $\det[\![y, z]\!]$ grows linearly in $|t|$ along the line $t \mapsto M(t)$. Thus, $t \mapsto M(t)$ must be a rank-one line and so $F = M(|\bar{y}|) \in K^{\mathrm{lc}}$.

(b) If $f(\bar{y}, \bar{z}) < 0$ and $g(\bar{y}, \bar{z}) = 0$, we first assume without loss of generality that $\bar{z}_2 = 0$ by the $\mathrm{SO}(2)$-symmetry of K, K^{pc}. We also assume $\bar{z}_1 > 0$. The case $\bar{z}_1 < 0$ is similar. In the three-dimensional space $Y := \{(y_1, y_2, z_1, 0) \in \mathbb{R}^4\}$ the equation

$$\begin{aligned} 0 &= g(y_1, y_2, z_1, 0) \\ &= (\det U - 1) z_1 + 1 - \det[\![(y_1, y_2), (z_1, 0)]\!] \\ &= (\alpha\beta - 1) z_1 + 1 - (y_1, y_2, z_1) \begin{pmatrix} 1 & & \frac{\alpha+\beta}{2} \\ & 1 & \\ \frac{\alpha+\beta}{2} & & \alpha\beta \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ z_1 \end{pmatrix} \end{aligned}$$

defines a one-sheeted hyperboloid H since the matrix has two positive eigenvalues and one negative eigenvalue as well as $\alpha\beta > 1$. Since every one-sheeted hyperboloid has the property that through every point on it there pass two lines lying entirely inside it, we can thus find a line $t \mapsto M(t)$ in H with $M(0) = F$. On $t \mapsto M(t)$, however, the determinant is affine by construction of H and thus $t \mapsto M(t)$ must be a rank-one line. Write $M(t) = [\![y(t), z(t)]\!]$. Then, choosing $t_0 \in \mathbb{R}$ such that $z_1(t_0) = 0$, we see that $M(t_0) = [\![y(t_0), 0]\!] \in K$. We also observe

$$f(M(t)) = f(M(t)) + g(M(t)) = (\det B - 1)(|y(t)| + |z(t)| - 1) \to \infty$$

as $|t| \to \infty$. Combining this with the fact that $f(M(0)) = f(F) < 0$, there must be a $t_1 \in \mathbb{R}$ with the property that $f(M(t_1)) = g(M(t_1)) = 0$ and thus $M(t_1) \in K^{\mathrm{lc}}$ by (a). Consequently, since F lies on a rank-one line between $M(t_0) \in K$ and $M(t_1) \in K^{\mathrm{lc}}$, it holds that $F = M(0) \in K^{\mathrm{lc}}$.

(c) The case $f(y_0, z_0) = 0$ and $g(y_0, z_0) < 0$ is analogous to the previous one.

In all cases we have shown that $F \in K^{\mathrm{lc}}$ and thus $K^{\mathrm{pc}} \subset K^{\mathrm{lc}}$, which implies the claim. □

For more general inclusions in two dimensions, including the N-well problem, we quote the following result concerning the case of wells with the same determinant:

Theorem 11.8 (Bhattacharya–Dolzmann 2001 [53, 123]) *Let $\delta \geq 0$ and assume that the set*

$$K \subset \left\{ A \in \mathbb{R}^{2\times 2} \ : \ \det A = \delta \right\}$$

is compact and left $\mathrm{SO}(2)$*-invariant, i.e.,*

$$K = \mathrm{SO}(2)K = \left\{ QA \ : \ Q \in \mathrm{SO}(2), \ A \in K \right\}.$$

Then,

$$K^{\mathrm{lc}} = K^{\mathrm{rc}} = K^{\mathrm{qc}} = K^{\mathrm{pc}}$$
$$= \left\{ A \in \mathbb{R}^{2\times 2} \ : \ \det A = \delta \text{ and } |An|^2 \le \max_{B\in K} |Bn|^2 \text{ for all } n \in \mathbb{S}^1 \right\}.$$

For three dimensions, we only mention the following theorem.

Theorem 11.9 (Dolzmann–Kirchheim–Müller–Šverák 2000 [124]) *Let*

$$K := \mathrm{SO}(3) \cup \mathrm{SO}(3)U$$

with

$$U = \begin{pmatrix} \alpha & & \\ & \beta & \\ & & 1 \end{pmatrix}, \qquad 0 < \alpha \le 1 \le \beta, \qquad \alpha\beta \ge 1.$$

Then,

$$K^{\mathrm{lc}} = K^{\mathrm{rc}} = K^{\mathrm{qc}} = K^{\mathrm{pc}} = \left\{ Q \begin{pmatrix} \widehat{A} & \\ & 1 \end{pmatrix} \ : \ Q \in \mathrm{SO}(3), \ \widehat{A} \in \widehat{K}^{\mathrm{qc}} \right\},$$

where

$$\widehat{K} = \mathrm{SO}(2) \cup \mathrm{SO}(2) \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix},$$

and $\widehat{K}^{\mathrm{qc}}$ *can be explicitly calculated via Theorem 11.7.*

11.4 Non-Laminate Microstructures

All microstructure Young measures that we have explicitly constructed so far, like the T_4-configuration, have been laminates (of finite or infinite order). So, a natural question is whether there are microstructures that cannot be realized as laminates. By the Kinderlehrer–Pedregal Theorem 8.11 and Pedregal's characterization of laminates in Theorem 11.4, this question is intimately tied to Morrey's Conjecture 8.1.

Thanks to Švérak's counterexample of a rank-one convex, but not quasiconvex, function, see Theorem 8.4, we can also construct examples of gradient Young measures that are not laminates:

Example 11.10 Let $\varphi \in \mathrm{W}^{1,\infty}_{\mathrm{per}}((0,1)^2;\mathbb{R}^3)$ be the function constructed in the proof of Švérak's Theorem 8.4. Define the gradient Young measure

$$\overline{\delta[\nabla\varphi]} \in \mathbf{GY}^{\infty}((0,1)^2;\mathbb{R}^{3\times 2})$$

via the Riemann–Lebesgue Lemma 4.16 (φ has periodic boundary values), i.e.,

$$\left\langle h, \overline{\delta[\nabla\varphi]} \right\rangle = \fint_{(0,1)^2} h(\nabla\varphi(x)) \, \mathrm{d}x, \qquad h \in \mathrm{C}_0(\mathbb{R}^{3\times 2}).$$

Then, $\overline{\delta[\nabla\varphi]}$ is not a laminate: Indeed, in the proof of Theorem 8.4 a function $h = h_{\alpha,\beta}\colon \mathbb{R}^{3\times 2} \to \mathbb{R}$ was constructed that is rank-one convex, but not quasiconvex at $[\overline{\delta[\nabla\varphi]}] = 0$. More precisely, we showed in (8.2) that

$$\int_{(0,1)^2} h(\nabla\varphi)\,\mathrm{d}x < 0 = h(0) = h\big([\overline{\delta[\nabla\varphi]}]\big).$$

Hence, Theorem 11.4 implies that $\overline{\delta[\nabla\varphi]}$ cannot be a laminate.

Example 11.11 (James) The following is a more concrete example of a gradient Young measure that is not a laminate; it is still based on a variant of Šverák's construction, which is due to James. Let $\varphi\colon \mathbb{R} \to \mathbb{R}$ be the periodic sawtooth function

$$\varphi(t) := \begin{cases} t & \text{if } t - \lfloor t \rfloor \in [0, 1/4), \\ \frac{1}{2} - t & \text{if } t - \lfloor t \rfloor \in [1/4, 3/4), \\ t - 1 & \text{if } t - \lfloor t \rfloor \in [3/4, 1), \end{cases}$$

and define $u \in \mathrm{W}^{1,\infty}(\mathbb{R}^2;\mathbb{R}^3)$ via

$$u(x_1, x_2) := \begin{pmatrix} \varphi(x_1) \\ \varphi(x_2) \\ \varphi(x_1 + x_2) \end{pmatrix}.$$

Then, for $u_j(x) := u(jx)/j$, where $x \in (0,1)^2$, the gradients ∇u_j take values in the set

$$K := \left\{ \begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} \in \mathbb{R}^{3\times 2} \;:\; x, y, z \in \{-1, +1\} \right\}. \tag{11.6}$$

We refer to the eight elements of K by the signs of x, y, z; for example, the matrix corresponding to $(x, y, z) = (+1, -1, +1)$ is simply denoted by "$+-+$". See Figure 11.5 for an illustration of ∇u.

We have that $(\nabla u_j)_j$ generates a homogeneous gradient Young measure $\mu \in \mathbf{GY}^\infty((0,1)^2;\mathbb{R}^{3\times 2})$ and a simple counting argument gives

$$\begin{aligned} \mu = \frac{3}{16}\big[\delta_{+++} + \delta_{+--} + \delta_{-+-} + \delta_{--+}\big] \\ + \frac{1}{16}\big[\delta_{++-} + \delta_{+-+} + \delta_{-++} + \delta_{---}\big]. \end{aligned}$$

It is also straightforward to compute that $[\mu] = 0$. Notice that the matrices in the first block all have positive *parity*, which for the matrix corresponding to the signs x, y, z is defined to be the product xyz (i.e., the parity is positive if and only if the number of minuses is even). All matrices in the second block have negative parity.

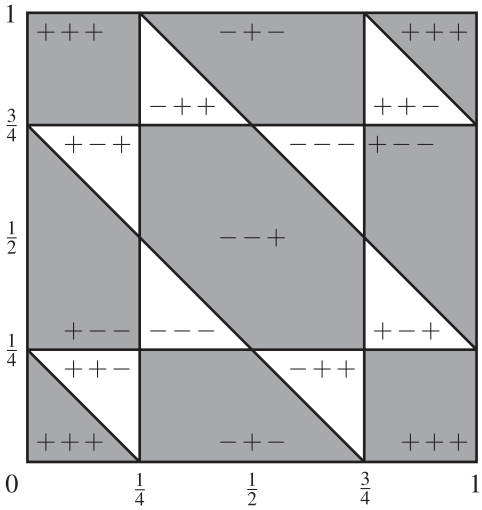


Fig. 11.5 James' example of a gradient Young measure that is not a laminate

For all $\alpha > 0$ there exists a $\beta > 0$ such that the function from the proof of Švérak's Theorem 8.4, namely

$$h_{\alpha,\beta}(A) := g(\mathbf{P}(A)) + \alpha\big(|A|^2 + |A|^4\big) + \beta|A - \mathbf{P}(A)|^2,$$

where

$$g\begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} := -xyz$$

and $\mathbf{P}\colon \mathbb{R}^{3\times 2} \to L$ is a linear projection onto

$$L := \left\{ \begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} \;:\; x, y, z \in \mathbb{R} \right\},$$

is rank-one convex. Thus, if μ was a laminate, Pedregal's Theorem 11.4 would imply

$$0 = h_{\alpha,\beta}([\mu]) \le \int h_{\alpha,\beta}(A)\, \mathrm{d}\mu(A).$$

As all matrices in L have the same (Frobenius) norm 2, we then get

$$\begin{aligned} 0 &\le \left(\sum_{\substack{A \in L \text{ with} \\ \text{parity } +1}} -\mu(A) \quad + \sum_{\substack{A \in L \text{ with} \\ \text{parity } -1}} \mu(A) \right) + 20\alpha \\ &= \left(-\frac{12}{16} + \frac{4}{16} \right) + 20\alpha = -\frac{1}{2} + 20\alpha. \end{aligned}$$

Since for α sufficiently small the right-hand side is negative, this is a contradiction. Hence, μ cannot be a laminate.

Unfortunately, while the preceding example shows that for the set K defined in (11.6) there exists a measure $\mu \in \mathcal{M}^{\mathrm{qc}}(K) \setminus \mathcal{M}^{\mathrm{rc}}(K)$ it turns out that nevertheless $K^{\mathrm{qc}} = K^{\mathrm{rc}}$, see Problem 11.8. Thus, it is reasonable to formulate a strong version of Morrey's conjecture:

Conjecture 11.12 There exists a compact set $K \subset \mathbb{R}^{m\times d}$ such that $K^{\mathrm{rc}} \neq K^{\mathrm{qc}}$.

Only some special cases are known. Milton showed that there is a compact set $K \subset \mathbb{R}^{3\times 2}$ (consisting of 7 matrices) with $K^{\mathrm{rc}} \neq K^{\mathrm{qc}}$ based on a modification of James' result, see [231]. There is also an argument of Šverák that shows that there is a compact set $K \subset \mathbb{R}^{6\times 2}$ with $K^{\mathrm{rc}} \neq K^{\mathrm{qc}}$, see Section 4.7 in [248]. Some recent developments are in [163].

11.5 Unbounded Microstructure

In this final section we will show how some lamination methods can be extended to unbounded sets. In particular, we will see that the Friesecke–James–Müller Theorem 10.20 and Korn's inequality (10.22) do not hold in L^1.

Theorem 11.13 (Conti–Faraco–Maggi 2005 [85]) *For all $N \in \mathbb{N}$ there exists a map $u \in \mathrm{W}^{1,\infty}(B_1;\mathbb{R}^d)$ with $u|_{\partial B_1} = x$ such that*

$$\inf_{Q\in \mathrm{SO}(d)} \int_{B_1} |\nabla u - Q| \,\mathrm{d}x \geq N \int_{B_1} \operatorname{dist}(\nabla u, \mathrm{SO}(d)) \,\mathrm{d}x.$$

The proof is based on the following explicit construction, where by $\mathcal{M}^{\mathrm{lc}}(\mathbb{R}^{m\times d})$ we denote the set of **unbounded (finite-order) laminates**, defined in complete analogy to $\mathcal{M}^{\mathrm{lc}}(K)$ for K compact, but without the restriction on the support.

Lemma 11.14 *There exists a sequence of finite-order laminates $(\mu_j)_j \subset \mathcal{M}^{\mathrm{lc}}(\mathbb{R}^{2\times 2})$ with*

$$[\mu_j] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \int \left| \frac{A + A^{\mathsf{T}}}{2} \right| \mathrm{d}\mu_j(A) = \sqrt{2},$$

and

$$\lim_{j\to\infty} \int |A| \,\mathrm{d}\mu_j(A) = \infty.$$

Proof. Step 1. We first construct for every $k \in \mathbb{N}$ a laminate $\gamma_k \in \mathcal{M}^{\mathrm{lc}}(\mathbb{R}^{2\times 2})$ such that

$$[\gamma_k] = \begin{pmatrix} 0 & k \\ k & 0 \end{pmatrix}, \quad \int \left| \frac{A + A^{\mathsf{T}}}{2} \right| \mathrm{d}\gamma_k(A) = \sqrt{2}k, \quad \int |A| \,\mathrm{d}\gamma_k(A) = \frac{5\sqrt{2}}{3}k.$$

Let $\delta_{\alpha,\beta}$ denote the Dirac mass at

$$M_{\alpha,\beta} := \begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix} \tag{11.7}$$

and for $k \in \mathbb{N}$ define

$$\gamma_k' := \frac{1}{3}\delta_{k,-k} + \frac{2}{3}\delta_{k,2k}.$$

Then, $[\gamma_k'] = M_{k,k}$ and γ_k' is a laminate since $M_{k,-k}$ and $M_{k,2k}$ are rank-one connected. Next, we set

$$\gamma_k'' := \frac{1}{4}\delta_{-2k,2k} + \frac{3}{4}\delta_{2k,2k},$$

for which $[\gamma_k''] = M_{k,2k}$. Again, γ_k'' is a laminate. Replacing the $\delta_{k,2k}$-term in the definition of γ_k' by γ_k'', we arrive at the second-order laminate

$$\gamma_k := \frac{1}{3}\delta_{k,-k} + \frac{1}{6}\delta_{-2k,2k} + \frac{1}{2}\delta_{2k,2k}.$$

It is easy to compute that γ_k has all the required properties.

Step 2. Let us now define $\mu_j \in \mathcal{M}^{\mathrm{lc}}(\mathbb{R}^{2\times 2})$ for $j = 1, 2, \ldots$. We set $\mu_1 := \gamma_1$. For $j = 2, 3, \ldots$ the μ_j are defined iteratively. By construction, μ_1 contains the term $\frac{1}{2}\delta_{2,2}$. We apply the first step to this term and replace it by a laminate γ_2, yielding μ_2. Generally, μ_j contains a term of the form $2^{-j}\delta_{2^j,2^j}$, and we may replace the mass $\delta_{2^j,2^j}$ via the first step of the proof with a laminate γ_{2^j}, giving μ_{j+1}. Clearly, this iteration does not change the barycenter of the μ_j, so

$$[\mu_j] = M_{1,1}.$$

Moreover, for the absolute symmetric first moment we get

$$\int \left| \frac{A + A^{\mathsf{T}}}{2} \right| \mathrm{d}\mu_j(A) = |M_{1,1}| = \sqrt{2}$$

since replacing $\delta_{2^j,2^j}$ with γ_{2^j} does not change the absolute symmetric first moment.

If we unwind the recursion, we have

$$\mu_j = \sum_{n=1}^{j} \left(\frac{2^{1-n}}{3}\delta_{2^{n-1},-2^{n-1}} + \frac{2^{1-n}}{6}\delta_{-2^n,2^n} \right) + 2^{-n}\delta_{2^n,2^n}.$$

Thus,

$$\int |A| \, \mathrm{d}\mu_j(A) \geq \sum_{n=1}^{j} \frac{2^{1-n}}{3} |M_{2^{n-1},-2^{n-1}}| = j\frac{\sqrt{2}}{3}$$

and consequently

$$\lim_{j\to\infty} \int |A| \, \mathrm{d}\mu_j(A) = \infty.$$

This finishes the proof. □

We next prove the following linearized version of Theorem 11.13, which is called *Ornstein's non-inequality* (or, more precisely, a special case of this quite general statement, see [195, 268]). It shows that an analogue of Korn's inequality (10.22) does not hold in $\mathrm{W}^{1,1}$.

Theorem 11.15 (Ornstein 1962 [268]) *For every $N \in \mathbb{N}$ there exists a map $u \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^d)$ such that*

$$\inf_{F\in\mathbb{R}^{d\times d}} \int_{B_1} |\nabla u - F| \,\mathrm{d}x \geq N \int_{B_1} \left| \frac{\nabla u + \nabla u^{\mathsf{T}}}{2} \right| \mathrm{d}x.$$

Proof. We only show the assertion for $d = 2$, the higher-dimensional cases follow from a natural embedding, see Problem 11.9.

Step 1. We first prove that for all $N \in \mathbb{N}$ there exists a map $v \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^2)$ with

$$\int_{B_1} |\nabla v| \,\mathrm{d}x \geq N \int_{B_1} \left| \frac{\nabla v + \nabla v^{\mathsf{T}}}{2} \right| \mathrm{d}x. \tag{11.8}$$

By the preceding lemma, we may find a finite-order laminate $\mu \in \mathcal{M}^{\mathrm{lc}}(\mathbb{R}^{2\times 2})$ such that

$$[\mu] = M_{1,1}, \quad \int |A| \,\mathrm{d}\mu(A) \geq 8\sqrt{2}N + \sqrt{2} \quad \int \left| \frac{A + A^{\mathsf{T}}}{2} \right| \mathrm{d}\mu(A) = \sqrt{2}.$$

with $M_{1,1}$ defined in (11.7). Then, by Lemma 11.5, μ is a homogeneous gradient Young measure on the domain B_1 (the fact that we may choose the domain B_1 follows from a covering argument as in Lemma 4.15). Thus, there exists a sequence $(v_j)_j \subset \mathrm{W}^{1,\infty}_{M_{1,1}x}(B_1;\mathbb{R}^2)$ with $\nabla v_j \xrightarrow{\mathbf{Y}} \mu$; in fact, the proof of Lemma 11.5 constructs this sequence explicitly. Note, however, that this sequence is not uniformly norm-bounded in $\mathrm{W}^{1,\infty}(B_1;\mathbb{R}^2)$.

Setting $v(x) := v_j(x) - M_{1,1}x$ for j large enough, we may assume

$$\int_{B_1} |\nabla v| \,\mathrm{d}x \geq \frac{1}{2} \int |A - M_{1,1}| \,\mathrm{d}\mu(A)$$

and

$$\int \left| \frac{A + A^{\mathsf{T}}}{2} - M_{1,1} \right| \mathrm{d}\mu(A) \geq \frac{1}{2} \int_{B_1} \left| \frac{\nabla v + \nabla v^{\mathsf{T}}}{2} \right| \mathrm{d}x.$$

Then,

$$\begin{aligned} \int_{B_1} |\nabla v| \,\mathrm{d}x &\geq \frac{1}{2} \int |A - M_{1,1}| \,\mathrm{d}\mu(A) \\ &\geq \frac{1}{2} \int |A| \,\mathrm{d}\mu(A) - \frac{\sqrt{2}}{2} \\ &\geq 4\sqrt{2}N \end{aligned}$$

$$
\begin{aligned}
&= 2N \int \left| \frac{A + A^T}{2} \right| \mathrm{d}\mu(A) + 2\sqrt{2}N \\
&\geq 2N \int \left| \frac{A + A^T}{2} - M_{1,1} \right| \mathrm{d}\mu(A) \\
&\geq N \int_{B_1} \left| \frac{\nabla v + \nabla v^T}{2} \right| \mathrm{d}x.
\end{aligned}
$$

Thus, (11.8) holds.

Step 2. Let $v \in W_0^{1,\infty}(B_1; \mathbb{R}^2)$ satisfy (11.8). We now show Ornstein's claim by scaling. Set for $0 < r < 1$,

$$
v_r(x) := r v\left(\frac{x}{r}\right), \qquad x \in B_1,
$$

where we consider v to be extended by zero to all of $\mathbb{R}^2$. We calculate for $r \leq 2^{-1/d}$ that

$$
\begin{aligned}
&\inf_{F \in \mathbb{R}^{2\times 2}} \int_{B_1} |\nabla v_r - F| \, \mathrm{d}x \\
&\quad \geq \inf_{F \in \mathbb{R}^{2\times 2}} \left(\int_{B_r} |\nabla v_r| \, \mathrm{d}x - \int_{B_r} |F| \, \mathrm{d}x + \int_{B_1 \setminus B_r} |F| \, \mathrm{d}x \right) \\
&\quad \geq \int_{B_r} |\nabla v_r| \, \mathrm{d}x + \inf_{F \in \mathbb{R}^{2\times 2}} \left(\omega_d (1 - 2r^d) |F| \right) \\
&\quad \geq \int_{B_1} |\nabla v_r| \, \mathrm{d}x.
\end{aligned}
$$

Since (11.8) is invariant under the scaling, the assertion of the theorem follows with $u := v_r$. □

Proof of Theorem 11.13. Again, we only consider the case $d = 2$. Let v be the function from Ornstein's Theorem 11.15 for the constant $2N$. Set

$$
u(x) := x + \varepsilon v(x), \qquad x \in B_1,
$$

for an $\varepsilon > 0$ to be determined later. We also observe, see (A.3), that

$$
\operatorname{dist}(\mathrm{Id} + A, \mathrm{SO}(2)) \leq \left| \frac{A + A^T}{2} \right| + C|A|^2.
$$

Thus, since $\nabla u = \mathrm{Id} + \varepsilon \nabla v$,

$$
\int_{B_1} \operatorname{dist}(\nabla u, \mathrm{SO}(2)) \, \mathrm{d}x \leq \varepsilon \int_{B_1} \left| \frac{\nabla v + \nabla v^T}{2} \right| \mathrm{d}x + \varepsilon^2 C \int_{B_1} |\nabla v|^2 \, \mathrm{d}x.
$$

Now choose $\varepsilon > 0$ so small that the second term is less than or equal to the first. Then,

$$\begin{aligned} N \int_{B_1} \operatorname{dist}(\nabla u, \mathrm{SO}(2))\, \mathrm{d}x &\le 2N\varepsilon \int_{B_1} \left| \frac{\nabla v + \nabla v^{\mathsf{T}}}{2} \right| \mathrm{d}x \\ &\le \varepsilon \inf_{F \in \mathbb{R}^{2\times 2}} \int_{B_1} |\nabla v - F|\, \mathrm{d}x \\ &= \inf_{G \in \mathbb{R}^{2\times 2}} \int_{B_1} |\nabla u - G|\, \mathrm{d}x \\ &\le \inf_{Q \in \mathrm{SO}(2)} \int_{B_1} |\nabla u - Q|\, \mathrm{d}x. \end{aligned}$$

This proves the claim. □

Notes and Historical Remarks

The task to compute quasiconvex hulls first arose in the theory of mathematical material science, in particular in the work of Ball and James [39, 40]. However, the explicit definitions and study of the rank-one convex hull and polyconvex hull as lower and upper bounds, respectively, only seem to have appeared later (even though the methods to compute them are older). Dacorogna contributed many results investigating the properties of the various hulls, see [91] for an overview and pointers to the literature. One can also define notions of quasiconvex, polyconvex, and rank-one convex sets, not just envelopes as we have done, and develop a corresponding theory, see Chapter 7 in [91].

In fact, constructions similar to the fundamental T_4-configuration were already employed by Scheffer in his 1974 PhD thesis [295] (which seems to be the first such construction in the literature), but its importance in the present context was only realized after Tartar's article [327], which refers to his work of 1983. Similar examples to Tartar's were found in [28, 54, 260] (in particular, [54] adapted Tartar's original example to the present version with diagonal matrices). Generalizations of the T_4-configuration to T_N-configurations with $N > 4$ were investigated in [193, 196, 320]; also see [108, 215, 292, 301] for some recent developments and further references.

For most of the material on the application to the two-well problems we follow [248, 250]. The proof of Proposition 11.6 is based on ideas from [308]. Theorem 11.7 is from [311].

Ornstein's non-inequality is in fact a much more general result that shows that many singular integral operators are not $(\mathrm{L}^1 \mapsto \mathrm{L}^1)$-bounded. An abstract approach to these questions is in [195]. Some recent rigidity results related to Korn's inequality can be found in [160–162].

Another aspect of microstructure which was not touched upon in this chapter is that of *uniqueness* and *stability*. Here, a microstructure $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ is called **unique** (in K) if $\widehat{\mu} \in \mathcal{M}^{\mathrm{qc}}(K)$ with $[\widehat{\mu}] = [\mu]$ implies $\widehat{\mu} = \mu$. Many ideas in this area go back to Luskin; a modern exposition, which also incorporates some new ideas with a view toward numerics is in Chapter 4 of [123].

Hulls of sets are also interesting for the divergence-free constraint, for instance in the theory of *shape-optimization*. Classical works in this area are [9–11, 198–200], most of which are surveyed in [8]; recent applications to the theory of shape-optimization can be found in [30, 264, 265].

Problems

11.1 Prove that the sequence $(u_j)_j \subset \mathrm{W}^{1,\infty}((0,1)^2;\mathbb{R}^m)$ that was constructed in Theorem 11.1 generates the homogeneous Young measure

$$\mu = \frac{8}{15}\delta_{A_1} + \frac{4}{15}\delta_{A_2} + \frac{2}{15}\delta_{A_3} + \frac{1}{15}\delta_{A_4}.$$

11.2 Even before the Chlebík–Kirchheim Theorem 10.16 was established, it was already known that the inclusion $\nabla u \in K_{\mathrm{T4}}$ with K_{T4} from Theorem 11.1 is rigid for exact inclusions. Prove this in an elementary way. *Hint:* Inspect the proof of rigidity for exact solutions in Theorem 10.11 for inspiration.

11.3 Let $K \subset \mathbb{R}^{m\times d}$ be a non-empty compact set. Show that K^{lc} is the smallest set $M \subset \mathbb{R}^{m\times d}$ that is closed under laminations, i.e., if $A, B \in M$ and $\operatorname{rank}(A-B) = 1$, then $\theta A + (1-\theta)B \in M$ for all $\theta \in (0,1)$.

11.4 Let $(t_k, A_k)_{k=1,\dots,n} \subset [0,1] \times \mathbb{R}^{m\times d}$ with $\sum_k t_k = 1$. We define:

(i) If $n = 2$ we say that $(t_k, A_k)_k$ satisfies the (H_2)**-condition** if $\operatorname{rank}(A_1 - A_2) \le 1$.

(ii) If $n > 2$ we say that $(t_k, A_k)_k$ satisfies the (H_n)**-condition** if possibly after a permutation of indices,

$$\operatorname{rank}(A_1 - A_2) \le 1$$

and with

$$s_1 = t_1 + t_2, \qquad B_1 = \frac{t_1}{s_1}A_1 + \frac{t_2}{s_1}A_2,$$
$$s_k = t_{k+1}, \qquad B_k = A_{k+1} \qquad \text{for} \quad k = 2, 3, \dots, n,$$

the collection $(s_k, B_k)_{k=1,\dots,n-1}$ satisfies the (H_{n-1})-condition.

Prove that

$$\mathcal{M}^{\mathrm{lc}}(K) = \Big\{ \mu \in \mathcal{M}^1(K) \ : \ \mu = \sum_{k=1}^{n} t_k \delta_{A_k}, \ (t_k, A_k)_k \text{ satisfies the } (H_n)\text{-condition for some } n \in \mathbb{N} \Big\}.$$

11.5 Show that for all non-empty compact sets $K \subset \mathbb{R}^{m\times d}$ it holds that

$$K^{\square} = \big\{ A \in \mathbb{R}^{m\times d} \ : \ h(A) \le \inf_K h \text{ for all } h\colon \mathbb{R}^{m\times d} \to \mathbb{R} \text{ with } h = h^{\square} \big\},$$

where $\square \in \{\mathrm{rc}, \mathrm{qc}, \mathrm{pc}, **\}$, and $h^{\square}$ is the respective envelope of h (i.e., $h = h^{\square}$ means that h has the respective convexity property).

11.6 Show that for a compact set $K \subset \mathbb{R}^{3\times 3}$ it holds that

$$K^{\mathrm{pc}} = \left\{ A \in \mathbb{R}^{3\times 3} \ : \ (A, \operatorname{cof} A, \det A) \in (\mathbf{M}K)^{**} \right\},$$

where

$$\mathbf{M}K := \left\{ (A, \operatorname{cof} A, \det A) \ : \ A \in K \right\}.$$

What is the corresponding formula for general dimensions?

11.7 Show that for all sufficiently small open sets $U \supset K_{\mathrm{T4}}$ there are no rank-one connections in U.

11.8 Show that for the set

$$K := \left\{ \begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} \ : \ x, y, z \in \{-1, 1\} \right\}$$

from Example 11.11 it holds that

$$K^{\mathrm{lc}} = K^{\mathrm{rc}} = K^{\mathrm{qc}} = K^{\mathrm{pc}} = K^{**} = \left\{ \begin{pmatrix} x & 0 \\ 0 & y \\ z & z \end{pmatrix} \ : \ x, y, z \in [-1, 1] \right\}.$$

11.9 Show how the proof of Theorem 11.15 for $d = 2$ also gives the result in all dimensions.

11.10 Show that for $\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain the space

$$\mathrm{LD}(\Omega) := \left\{ u \in \mathrm{L}^1(\Omega; \mathbb{R}^d) \ : \ \mathcal{E}u := \frac{1}{2}\left(\nabla u + \nabla u^{\mathsf{T}}\right) \in \mathrm{L}^1(\Omega; \mathbb{R}^{d\times d}_{\mathrm{sym}}) \right\}$$

is strictly larger than $\mathrm{W}^{1,1}(\Omega; \mathbb{R}^d)$.



Chapter 12
Convex Integration

Continuing on from the study of microstructure in the previous chapter, we now look at *exact* solutions to our prototypical differential inclusion, augmented with a boundary condition. So, consider

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \quad u|_{\partial\Omega} = v|_{\partial\Omega}, \\ \nabla u \in K \quad \text{a.e.}, \end{cases} \tag{12.1}$$

where $K \subset \mathbb{R}^{m\times d}$ is a non-empty compact set and $\in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$.

To illustrate the problem of exact solvability, let us for the moment assume that the prescribed boundary values are linear, that is, $v(x) = Fx$ for an $F \in \mathbb{R}^{m\times d}$. If $F \in K$, there obviously is an affine (exact) solution to (12.1). If $F \in \mathbb{R}^{m\times d} \setminus K^{\mathrm{qc}}$ there is not even an approximate solution: Indeed, if there was, we could pass to the Young measure limit and average the resulting Young measure via Lemma 4.15 to obtain $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ with $[\mu] = F$ (we use the linear boundary condition for this averaging step), contradicting $F \notin K^{\mathrm{qc}}$.

This chapter thus investigates the intermediate case

$$F \in K^{\mathrm{qc}} \setminus K,$$

which we assume to be non-empty. In numerous interesting applications it turns out that our differential inclusion (12.1) can in fact be solved *exactly* for many such F. This means that there exists a map $u \in \mathrm{W}^{1,\infty}_{Fx}(\Omega;\mathbb{R}^m)$ with

$$\nabla u(x) \in K \qquad \text{for a.e. } x \in \Omega.$$

Note that the gradient ∇u itself never takes the value $F \in K^{\mathrm{qc}} \setminus K$ (almost everywhere), even though u has the boundary values Fx in the sense of trace. Consequently, an exact solution u to (12.1) cannot be affine and, as we will see momentarily, it necessarily displays fine oscillations.

The method to construct such exact solutions is called *convex integration*. Starting from ideas first employed by John F. Nash and Mikhail Gromov, which were then transported to the PDE and Calculus of Variations context by Stefan Müller and Vladimir

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F. Rindler, *Calculus of Variations*, Graduate Texts in Mathematics 310,
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Šverák, this theory has led to many astonishing results. Here, we only discuss the applications to microstructure, but similar techniques also have profound implications in geometry and fluid dynamics.

There are several approaches to convex integration and we present three of them: the classical convex integration scheme based on in-approximations, the abstract and elegant Baire category method, and an approach based on Young measures.

Before we begin, however, let us observe that any theory of such "irregular" solutions to differential inclusions will require a fine analysis of K (and additional assumptions): For instance, let $K := K_{\mathrm{T}4}$ be the T_4-configuration from Theorem 11.1. We saw in Proposition 11.6 that $K_{\mathrm{T}4}^{\mathrm{qc}} \supsetneq K_{\mathrm{T}4}$. On the other hand, we know from the Chlebík–Kirchheim Theorem 10.16 (or the more elementary Problem 11.2) that the differential inclusion (12.1) is rigid for exact solutions in this case. Hence, only for $F \in K_{\mathrm{T}4}$ can we solve (12.1) exactly. The set $K_{\mathrm{T}4}^{\mathrm{qc}} \setminus K_{\mathrm{T}4}$ remains unreachable.

12.1 Exact Solutions to Differential Inclusions

As an introductory example, let us consider the following non-rigid two-well inclusion in two dimensions:

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^2), \quad u|_{\partial\Omega} = Fx, \quad 0 < \alpha < 1 < \beta, \quad \alpha\beta \geq 1, \\ \nabla u \in K := \mathrm{SO}(2) \cup \mathrm{SO}(2) \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}. \end{cases} \tag{12.2}$$

By Theorem 11.7 in conjunction with Lemma 10.23 (iii) we have $K^{\mathrm{qc}} \supsetneq K$. In fact, for every $Q \in \mathrm{SO}(2)$ there are precisely two solutions $R_1, R_2 \in \mathrm{SO}(2)$, $i = 1, 2$, of

$$\operatorname{rank}\left(Q - R \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}\right) \leq 1.$$

The definition of K^{qc} entails the approximate solvability of (12.2), with the solution sequences consisting of high-frequency laminations. One may hope to use such oscillatory sequences to also construct exact solutions. However, any finite-order laminate with affine boundary values is trivial (see Problem 12.1) and cut-off procedures, like the one in Step 2 of the proof of Proposition 5.6, destroy value constraints.

The fundamental idea of convex integration is to start with a map satisfying the given boundary conditions and then to iteratively add oscillations with high frequency and low amplitude in order to move the gradient towards K. The property of K that will allow us to implement this idea is the following: We say that an open and bounded set $G \subset \mathbb{R}^{m\times d}$ can be **piecewise-affinely reduced** to a compact and non-empty set $K \subset \mathbb{R}^{m\times d}$ if for all $F \in G$ there exists a sequence $(\psi_j)_j \subset \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$ of countably piecewise affine maps such that

(i) $F + \nabla\psi_j(x) \in G$ for almost every $x \in B_1$ and all $j \in \mathbb{N}$;

(ii) $\displaystyle\int_{B_1} \operatorname{dist}(F + \nabla\psi_j(x), K)\, \mathrm{d}x \to 0$ as $j \to \infty$.

Recall here that a map $\varphi\colon D \to \mathbb{R}^m$ is called countably piecewise affine if there exists a disjoint partition of D into countably many open sets D_k up to a negligible set, i.e., $D = Z \cup \bigcup_{k=1}^{\infty} D_k$, where $|Z| = 0$, such that $\varphi|_{D_k}$ is affine.

In the above definition the unit ball can be replaced by any bounded Lipschitz domain, see Problem 12.2.

Lemma 12.1 *If $G \subset \mathbb{R}^{m\times d}$ can be piecewise-affinely reduced to K, then $G \subset K^{\mathrm{qc}}$.*

Proof. Let $F \in G$ and let $(\psi_j)_j \subset \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$ be the corresponding sequence that witnesses the piecewise-affine reduction of G to K at the matrix F; in particular, $(\psi_j)_j$ satisfies (i), (ii) above. Up to taking a subsequence, we have that $F + \nabla\psi_j \xrightarrow{\mathbf{Y}} \nu$ for a gradient Young measure $\nu \in \mathbf{GY}^{\infty}(B_1;\mathbb{R}^{m\times d})$ with $[\nu] = F$.

Now observe that the averaging principle, Lemma 4.15, does not change the value distribution of the gradients being averaged. Consequently, we find a sequence $(\varphi_j)_j \subset \mathrm{W}_0^{1,\infty}(B_1;\mathbb{R}^m)$ satisfying (i), (ii) above and generating a homogeneous Young measure μ with $[\mu] = F$ as well as $\operatorname{supp}\mu \subset K$ (see Lemma 4.13). Thus, $\mu \in \mathcal{M}^{\mathrm{qc}}(K)$ (see the definition in Section 10.3), whereby $F \in K^{\mathrm{qc}}$. □

Example 12.2 For the T_4-configuration $K_{\mathrm{T4}} \subset \mathbb{R}^{2\times 2}$ from Section 11.1 we have that $K_{\mathrm{T4}}^{\mathrm{qc}}$ has empty interior (being contained in the set of diagonal matrices). Hence, no piecewise-affine reduction of any open set to K_{T4} can exist.

We are now ready to state the following abstract convex integration result, which in this form is due to Sychev [314], but a less general version was essentially contained in work by Gromov [170] and Müller–Šverák [250].

Theorem 12.3 *Let $K \subset \mathbb{R}^{m\times d}$ be a non-empty compact set and assume that the open and bounded set $G \subset \mathbb{R}^{m\times d}$ can be piecewise-affinely reduced to K. Suppose that either $v \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is countably piecewise affine or $v \in \mathrm{C}^1(\Omega;\mathbb{R}^m)$, and that*

$$\nabla v \in G \quad \text{a.e.}$$

Then, for every $\varepsilon > 0$ there exists a solution to the exact differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \quad u|_{\partial\Omega} = v|_{\partial\Omega}, \\ \nabla u \in K \quad \text{a.e.} \end{cases} \tag{12.3}$$

with $\|u - v\|_{\mathrm{L}^\infty} \le \varepsilon$.

In particular, this theorem applies to $v(x) = Fx$ for a matrix $F \in G \subset K^{\mathrm{qc}}$.

Proof. Let $\varepsilon > 0$. If v is already countably piecewise affine, then set $v_1 := v$. If v instead has C^1-regularity, then find a countably piecewise affine $v_1 \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with

$$\nabla v_1 \in G \quad \text{a.e.}, \qquad v_1|_{\partial\Omega} = v|_{\partial\Omega}, \qquad \|v - v_1\|_{\mathrm{L}^\infty} \le \frac{\varepsilon}{2}.$$

This construction requires the mesh on which v_1 is (countably) piecewise affine to become finer and finer the closer one approaches the boundary. We here furthermore used the assumption that G is open.

The idea of the proof is to use the piecewise affine reduction of G to K in conjunction with a "controlled" mollification to construct a sequence $(v_j)_j$ that converges *strongly* in $\mathrm{W}^{1,1}$ to the sought solution u of (12.3). So, let $(\eta_\delta)_{\delta>0}$ be a family of mollifiers and choose $\delta_1 > 0$ such that

$$\delta_1 \le \varepsilon \qquad \text{and} \qquad \|\nabla v_1 - \eta_{\delta_1} \star \nabla v_1\|_{\mathrm{L}^1} \le \frac{1}{2}.$$

Now, assuming that we have already defined v_j and δ_j for some $j \in \mathbb{N}$, we use the assumption that G can be piecewise-affinely reduced to K to find a countably piecewise affine map $v_{j+1} \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with

$$\nabla v_{j+1} \in G \quad \text{a.e.}, \qquad v_{j+1}|_{\partial\Omega} = v|_{\partial\Omega}, \qquad \|v_{j+1} - v_j\|_{\mathrm{L}^\infty} \le \frac{\delta_j}{2^{j+1}},$$

and

$$\int_\Omega \operatorname{dist}(\nabla v_{j+1}(x), K)\, \mathrm{d}x \le \frac{1}{2^{j+1}}. \tag{12.4}$$

Indeed, we can construct this map separately in each subdomain D_k of Ω from the partition with respect to which v_j is (countably) piecewise affine, say $v_j(x) = z_k + A_k x$ for $x \in D_k$, where $z_k \in \mathbb{R}^m$, $A_k \in \mathbb{R}^{m\times d}$ $(k \in \mathbb{N})$. From the reduction property choose $\psi_{j+1}^{(k)} \in \mathrm{W}_0^{1,\infty}(D_k;\mathbb{R}^m)$ countably piecewise affine with

$$A_k + \nabla\psi_{j+1}^{(k)} \in G \quad \text{a.e.}$$

and

$$\int_{D_k} \operatorname{dist}(A_k + \nabla\psi_{j+1}^{(k)}(x), K)\, \mathrm{d}x \le \frac{1}{2^{j+k+1}}.$$

Moreover, by a procedure similar to the proof of the averaging principle for Young measures, Lemma 4.15, we may assume that $\|\psi_{j+1}^{(k)}\|_{\mathrm{L}^\infty} \le 2^{-(j+1)}\delta_j$. Then, one may check that $v_{j+1}\colon \Omega \to \mathbb{R}^m$ defined via

$$v_{j+1}(x) := z_k + A_k x + \psi_{j+1}^{(k)}(x) \qquad \text{if } x \in D_k \ (k \in \mathbb{N})$$

has all the desired properties.

Furthermore, choose $0 < \delta_{j+1} \le \delta_j$ such that

$$\|\nabla v_{j+1} - \eta_{\delta_{j+1}} \star \nabla v_{j+1}\|_{\mathrm{L}^1} \le \frac{1}{2^{j+1}}, \tag{12.5}$$

where we consider the map v_{j+1} to be extended to all of $\mathbb{R}^d$ via a fixed extension of v (see Theorem A.26).

Having constructed the whole sequence $(v_j)_j$ in this way, for all $l \ge j$ we get

$$\|v_l - v_j\|_{\mathrm{L}^\infty} \le \sum_{m=j}^{l-1} \|v_{m+1} - v_m\|_{\mathrm{L}^\infty} \le \sum_{m=j}^{\infty} \frac{\delta_m}{2^{m+1}} \le \frac{\delta_j}{2^j} \le \frac{\varepsilon}{2^j}.$$

In particular, $(v_j)_j$ is an L^∞-Cauchy sequence and hence $v_j \to u$ in L^∞ for some $u \in \mathrm{L}^\infty(\Omega;\mathbb{R}^m)$ with

$$\|u - v_1\|_{\mathrm{L}^\infty} \le \frac{\varepsilon}{2} \qquad \text{and} \qquad \|u - v_j\|_{\mathrm{L}^\infty} \le \frac{\delta_j}{2^j}.$$

Crucially, also the gradients of the v_j's converge *strongly* in L^1: To this end we observe that since $\|\nabla\eta_{\delta_j}\|_{\mathrm{L}^1} \le C/\delta_j$ for some constant $C > 0$, on any $\Omega' \Subset \Omega$ it holds that

$$\|\eta_{\delta_j} \star (\nabla u - \nabla v_j)\|_{\mathrm{L}^1(\Omega')} = \|\nabla\eta_{\delta_j} \star (u - v_j)\|_{\mathrm{L}^1(\Omega')} \le \frac{C}{\delta_j} \cdot \frac{\delta_j}{2^j} = \frac{C}{2^j}$$

as soon as δ_j is sufficiently small. Then also

$$\begin{aligned}\|\nabla u - \nabla v_j\|_{\mathrm{L}^1(\Omega')} &\le \|\nabla u - \eta_{\delta_j} \star \nabla u\|_{\mathrm{L}^1(\Omega')} + \|\eta_{\delta_j} \star (\nabla u - \nabla v_j)\|_{\mathrm{L}^1(\Omega')} \\ &\quad + \|\eta_{\delta_j} \star \nabla v_j - \nabla v_j\|_{\mathrm{L}^1(\Omega')} \\ &\to 0,\end{aligned}$$

where we have also used (12.5). Thus, $v_j \to u$ in $\mathrm{W}^{1,1}_{\mathrm{loc}}$ and, selecting a subsequence (not relabeled), also $\nabla v_j \to \nabla u$ almost everywhere. Via (12.4), we thus arrive at the inclusion

$$\nabla u \in K \quad \text{a.e.}$$

Moreover, $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ since the above estimates are uniform in $\Omega' \Subset \Omega$, as well as $\|u - v\|_{\mathrm{L}^\infty} \le \varepsilon$ and $u|_{\partial\Omega} = v|_{\partial\Omega}$. So, u is indeed the sought exact solution to the differential inclusion (12.3). □

12.2 In-Approximations

Now we come to Gromov's original convex integration principle, which gives a convenient sufficient condition for the existence of piecewise-affine reductions, namely the following: For an open and bounded set $G \subset \mathbb{R}^{m\times d}$ define the **lamination-convex hull** just like for a compact set (cf. Section 11.2), that is,

$$G^{\mathrm{lc}} := \bigcup_{i=0}^{\infty} G^{\mathrm{lc},i},$$

where

$$\begin{aligned}G^{\mathrm{lc},0} &:= G,\\ G^{\mathrm{lc},i+1} &:= \big\{\, \theta A + (1-\theta)B \;:\; A, B \in G^{\mathrm{lc},i},\ \theta \in [0,1],\ \operatorname{rank}(A - B) \le 1 \,\big\}.\end{aligned}$$

Let $K \subset \mathbb{R}^{m\times d}$ be compact and non-empty. Then, a sequence $(G_k)_k$ of open, uniformly bounded sets $G_k \subset \mathbb{R}^{m\times d}$ ($k \in \mathbb{N}$) is called an **in-approximation** for K if the following two conditions hold:

(i) $G_k \subset G_{k+1}^{\mathrm{lc}}$ for all $k \in \mathbb{N}$;
(ii) if $A_k \in G_k$ and $A_k \to A$ as $k \to \infty$, then $A \in K$.

We then have the following fundamental result:

Theorem 12.4 (Gromov 1986 [170]) *Let $K \subset \mathbb{R}^{m\times d}$ be a non-empty compact set and assume that there exists an in-approximation $(G_k)_k$ for K. Suppose that either $v \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is countably piecewise affine or $v \in \mathrm{C}^1(\Omega;\mathbb{R}^m)$, and that*

$$\nabla v \in G_\infty := \bigcup_{l=1}^{\infty} G_l \quad a.e.$$

Then, for every $\varepsilon > 0$ there exist a solution to the exact differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \quad u|_{\partial\Omega} = v|_{\partial\Omega}, \\ \nabla u \in K \quad a.e. \end{cases}$$

with $\|u - v\|_{\mathrm{L}^\infty} \le \varepsilon$.

In order to apply the abstract Theorem 12.3, we need to show that, under the assumptions of Gromov's theorem, G_∞ can be piecewise-affinely reduced to K. This will be accomplished by successive laminations.

Lemma 12.5 *Let $A, B \in \mathbb{R}^{m\times d}$ with $\operatorname{rank}(A - B) = 1$, $\theta \in (0,1)$ and let $D \subset \mathbb{R}^d$ be a bounded Lipschitz domain. Then, for every $\varepsilon > 0$ there exists a countably piecewise affine map $u\colon D \to \mathbb{R}^m$ with $u|_{\partial D} = Fx$, where $F := \theta A + (1-\theta)B$, and*

$$\sup_{x\in D} \left[|u(x) - Fx| + \operatorname{dist}(\nabla u(x), \{A, B\})\right] \le \varepsilon. \tag{12.6}$$

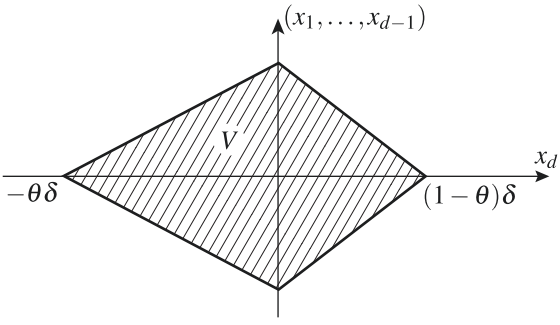


Fig. 12.1 The construction of the domain V

Proof. Let $\varepsilon > 0$.

Step 1. By a change of variables (which only results in an affine transformation of the resulting map u), we may assume that

$$A = -(1-\theta)a \otimes \mathrm{e}_d, \qquad B = \theta a \otimes \mathrm{e}_d, \qquad F = 0$$

for some $a \in \mathbb{R}^m \setminus \{0\}$.

We will first construct a special open and convex domain $V \subset \mathbb{R}^d$ such that (12.6) holds for $D = V$. In the auxiliary domain

$$U := (-1,1)^{d-1} \times (-\theta\delta, (1-\theta)\delta),$$

where $\delta > 0$ will be specified below, we define the functions $g, h \colon U \to \mathbb{R}$ by

$$g(x) := -\theta(1-\theta)\delta + \begin{cases} -(1-\theta)x_d & \text{if } x_d \le 0, \\ \theta x_d & \text{if } x_d > 0, \end{cases}$$
$$h(x) := \theta(1-\theta)\delta \sum_{i=1}^{d-1} |x_i|.$$

Then we let

$$V := \big\{ x \in U \,:\, g(x) + h(x) < 0 \big\},$$

which has the shape of a kite, see Figure 12.1. The map

$$v(x) := (g(x) + h(x))a, \qquad x \in V,$$

has the desired properties (in V). Indeed, by construction it is clear that v is countably piecewise affine and vanishes on ∂V. Moreover, as

$$\nabla g \in \big\{ -(1-\theta)\mathrm{e}_d, \theta \mathrm{e}_d \big\} \quad \text{a.e. in } U,$$

we can estimate

$$\sup_{x \in V} \operatorname{dist}(\nabla v(x), \{A, B\}) \le C\delta,$$

where $C > 0$. Thus, choosing $\delta > 0$ suitably, we can achieve (12.6) in $D = V$.

Step 2. From Vitali's Covering Theorem A.16 we find a covering

$$D = Z \cup \bigcup_{k=1}^{\infty} V(a_k, r_k),$$

where $V(a_k, r_k) := a_k + r_k V \subset D$ for $a_k \in D$, $0 < r_k \le \varepsilon/(1 + \|v\|_{\mathrm{L}^\infty})$, and $|Z| = 0$. Then define $u \colon D \to \mathbb{R}^m$ via

$$u(x) := r_k v\left(\frac{x - a_k}{r_k}\right) \qquad \text{if } x \in V(a_k, r_k) \ (k \in \mathbb{N}),$$

and observe

$$\nabla u(x) = \nabla v\left(\frac{x - a_k}{r_k}\right) \qquad \text{if } x \in V(a_k, r_k)\ (k \in \mathbb{N}).$$

Thus, u is countably piecewise affine, satisfies $u|_{\partial D} = 0$, and (12.6) holds. □

Lemma 12.6 *Let $U \subset \mathbb{R}^{m\times d}$ be open and bounded. Suppose $v \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is countably piecewise affine with*

$$\nabla v \in U^{\mathrm{lc}} \quad \text{a.e.}$$

Then, for every $\varepsilon > 0$ there exists a countably piecewise affine map $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ with

$$\nabla u \in U \quad \text{a.e.}, \qquad u|_{\partial\Omega} = v|_{\partial\Omega},$$

and $\|u - v\|_{\mathrm{L}^\infty} \le \varepsilon$.

Proof. We assume that v is affine; otherwise we apply the following argument separately in every set of the decomposition of Ω with respect to which v is countably piecewise affine. Moreover, we may add a constant to v to normalize to the situation where $v(x) = Fx$ for some $F \in \mathbb{R}^{m\times d}$.

By assumption, $F \in U^{\mathrm{lc}}$ and so, $F \in U^{\mathrm{lc},i}$ for some $i \in \mathbb{N} \cup \{0\}$. We proceed by induction over i. For $i = 0$ there is nothing to show. If $F \in U^{\mathrm{lc},i+1} \setminus U^{\mathrm{lc},i}$, then there are $A, B \in U^{\mathrm{lc},i}$ with $\operatorname{rank}(A - B) = 1$ and $\theta \in (0,1)$ such that $F = \theta A + (1-\theta)B$. By Lemma 12.5 there exists a countably piecewise affine map $w\colon \Omega \to \mathbb{R}^m$ with $w|_{\partial\Omega} = Fx = v|_{\partial\Omega}$ and

$$\|w - v\|_{\mathrm{L}^\infty} < \delta, \qquad \sup_{x\in\Omega} \operatorname{dist}(\nabla w(x), \{A, B\}) < \delta$$

for

$$\delta := \min\left\{\frac{\varepsilon}{2},\ \operatorname{dist}(A, \mathbb{R}^{m\times d} \setminus U^{\mathrm{lc},i}),\ \operatorname{dist}(B, \mathbb{R}^{m\times d} \setminus U^{\mathrm{lc},i})\right\}.$$

This δ is strictly positive because $U^{\mathrm{lc},i}$ is open (see Problem 12.3). Hence,

$$\nabla w \in U^{\mathrm{lc},i} \quad \text{a.e.}$$

Then, if $\Omega = Z \cup \bigcup_{k=1}^\infty D_k$, where $|Z| = 0$, is the decomposition of Ω with respect to which w is countably piecewise affine, we apply the induction hypothesis in every D_k to get countably piecewise affine maps $u_k \in \mathrm{W}^{1,\infty}(D_k;\mathbb{R}^m)$ with $u_k|_{\partial D_k} = w|_{\partial D_k}$,

$$\|u_k - w\|_{\mathrm{L}^\infty} < \frac{\varepsilon}{2} \qquad \text{and} \qquad \nabla u_k \in U \quad \text{a.e.}$$

Defining $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ via

$$u(x) := u_k(x) \qquad \text{if } x \in D_k\ (k \in \mathbb{N}),$$

we see that $u|_{\partial\Omega} = v|_{\partial\Omega}$, $\|u - v\|_{\mathrm{L}^\infty} \le \varepsilon$, and $\nabla u \in U$ almost everywhere. We thus conclude the proof of the inductive step and hence the proof of the lemma. □

Proof of Theorem 12.4. By Theorem 12.3 we only need to show that $G_\infty = \bigcup_{l=1}^\infty G_l$ can be piecewise-affinely reduced to K. Without loss of generality we assume $F \in G_1 \subset G_2^{\mathrm{lc}}$, where the inclusion holds by property (i) from the definition of an in-approximation. The proof for $F \in G_l$, $l = 2, 3, \ldots$, is analogous.

Let $n \in \mathbb{N}$. Lemma 12.6 (with $U := G_2$) allows us to construct a countably piecewise affine map $\psi_2^{(n)} \in \mathrm{W}_0^{1,\infty}(B_1; \mathbb{R}^m)$ with

$$F + \nabla\psi_2^{(n)} \in G_2 \subset G_3^{\mathrm{lc}} \quad \text{a.e.} \qquad \text{and} \qquad \|\psi_2^{(n)}\|_{\mathrm{L}^\infty} \leq \frac{1}{2^2 n}.$$

Since $\psi_2^{(n)}$ is countably piecewise affine, arguing separately in every piece, we may then find $\psi_3^{(n)} \in \mathrm{W}_0^{1,\infty}(B_1; \mathbb{R}^m)$ with

$$F + \nabla\psi_2^{(n)} + \nabla\psi_3^{(n)} \in G_3 \subset G_4^{\mathrm{lc}} \quad \text{a.e.} \qquad \text{and} \qquad \|\psi_3^{(n)}\|_{\mathrm{L}^\infty} \leq \frac{1}{2^3 n}.$$

Continuing this procedure, we construct a sequence $(\psi_k^{(n)})_k \subset \mathrm{W}^{1,\infty}(B_1; \mathbb{R}^m)$ of countably piecewise affine maps such that

$$F + \sum_{l=2}^{k} \nabla\psi_l^{(n)} \in G_k \subset G_{k+1}^{\mathrm{lc}} \quad \text{a.e.} \qquad \text{and} \qquad \|\psi_k^{(n)}\|_{\mathrm{L}^\infty} \leq \frac{1}{2^k n}.$$

For the countably piecewise affine map

$$\psi_j := \sum_{l=2}^{j} \psi_l^{(j)} \in \mathrm{W}^{1,\infty}(B_1; \mathbb{R}^m),$$

we thus have

$$F + \nabla\psi_j \in G_j \subset G_\infty \quad \text{a.e.} \qquad \text{and} \qquad \|\psi_j\|_{\mathrm{L}^\infty} \leq \sum_{l=2}^{j} \frac{1}{2^l j} \leq \frac{1}{j}.$$

It only remains to observe that

$$\int_{B_1} \operatorname{dist}(F + \nabla\psi_j(x), K)\, \mathrm{d}x \to 0 \qquad \text{as } j \to \infty.$$

This follows immediately since the integrand is uniformly bounded (as the G_k are uniformly bounded) and $F + \nabla\psi_j(x)$ converges pointwise almost everywhere to an element of K by property (ii) from the definition of an in-approximation. Thus, we have shown that G_1 can be affinely reduced to K and Theorem 12.3 yields the conclusion. □

We now apply Gromov's convex integration principle to the two-well problem in two dimensions, see (12.2). We only consider the case $\det U > 1$ (see [136, 251] for the case $\det U = 1$). That is, we will find exact solutions to the differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^2), \quad u|_{\partial\Omega} = Fx, \quad 0 < \alpha < 1 < \beta, \quad \alpha\beta > 1, \\ \nabla u \in K := \mathrm{SO}(2) \cup \mathrm{SO}(2)U, \quad U = \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix}. \end{cases} \tag{12.7}$$

Theorem 12.7 (Müller–Šverák 1993/1996 [250, 311]) *Let*

$$F \in \operatorname{int} K^{\mathrm{lc}} = \left\{ A = [\![y, z]\!] \in K^{**} \ : \ |y| < \tfrac{\det U - \det A}{\det U - 1}, \ |z| < \tfrac{\det A - 1}{\det U - 1} \right\}. \tag{12.8}$$

Then, there exists an exact solution to (12.7).

We remark that similar results have also been established by Dacorogna and Marcellini [94], see the notes at the end of this chapter.

Proof. Step 1. We first justify the formula for $\operatorname{int} K^{\mathrm{lc}}$. From Theorem 11.7 (iii) we know that

$$K^{\mathrm{lc}} = \left\{ A = [\![y, z]\!] \in K^{**} \ : \ |y| \le \tfrac{\det U - \det A}{\det U - 1}, \ |z| \le \tfrac{\det A - 1}{\det U - 1} \right\}.$$

By Problem 10.9 the conformal coordinate map $(y, z) \mapsto [\![y, z]\!]$ is a diffeomorphism and hence maps boundaries to boundaries. This immediately implies the formula (12.8).

Step 2. Define for any $J, V \in \mathbb{R}^{2\times 2}$ with $0 < \det J < \det V$ the set

$$L(J, V) := \left\{ A = [\![y, z]\!] \in K^{**} \ : \ |y| \le \tfrac{\det V - \det A}{\det V - \det J}, \ |z| \le \tfrac{\det A - \det J}{\det V - \det J} \right\}.$$

Note that $L(\mathrm{Id}, U) = K^{\mathrm{lc}}$. Moreover, the following continuity property holds: Let $(J_j, V_j)_j \subset \mathbb{R}^{2\times 2} \times \mathbb{R}^{2\times 2}$ be a converging sequence such that $(J_j, V_j) \to (J, V)$ as well as

$$0 < \det J_j < \det V_j \qquad \text{and} \qquad \lambda_*(V_j J_j^{-1}) < 1 < \lambda^*(V_j J_j^{-1}), \tag{12.9}$$

where $\lambda_*(V_j J_j^{-1}), \lambda^*(V_j J_j^{-1})$ are the small and the large eigenvalue of $V_j J_j^{-1}$, respectively. Let $P \subset \operatorname{int} L(J, V)$ be a compact set. Then, $P \subset L(J_j, V_j)$ for all sufficiently large j. Indeed, this follows by Step 1 and the fact that the conformal coordinates map is a diffeomorphism.

Step 3. We now prove the assertion of the theorem, for which we construct an in-approximation, starting with a neighborhood $G_1 \Subset K^{\mathrm{lc}}$ of $F \in \operatorname{int} K^{\mathrm{lc}}$. Assume that we have already constructed $G_k \Subset K^{\mathrm{lc}}$. We will show that we can find $G_{k+1} \Subset K^{\mathrm{lc}}$ with $G_k \subset G_{k+1}^{\mathrm{lc}}$ and such that $\sup_{H\in G_{k+1}} \operatorname{dist}(H, K) \le 1/k$.

Indeed, since $G_k \Subset K^{\mathrm{lc}} = L(\mathrm{Id}, U)$, by the continuity property from the previous step we may find $J_{k+1}, V_{k+1} \in \operatorname{int} K^{\mathrm{lc}}$ with

$$|J_{k+1} - \mathrm{Id}|, \ |V_{k+1} - U| \le \frac{1}{2k}$$

such that the above conditions (12.9) hold for J_{k+1}, V_{k+1} and

$$\overline{G_k} \subset L(J_{k+1}, V_{k+1}) = \big(\mathrm{SO}(2)J_{k+1} \cup \mathrm{SO}(2)V_{k+1}\big)^{\mathrm{lc}}.$$

Here, the equality follows analogously to the argument in the proof of Theorem 11.7 (iii). We also note that $\mathrm{Id}, U \in \partial K^{\mathrm{lc}}$.

We have that

$$\mathrm{SO}(2)J_{k+1} \cup \mathrm{SO}(2)V_{k+1} \subset \operatorname{int} K^{\mathrm{lc}}$$

since $J_{k+1}, V_{k+1} \in \operatorname{int} K^{\mathrm{lc}} = L(\mathrm{Id}, U)$ and the expression (12.8) for K^{lc} shows that this set is $\mathrm{SO}(2)$-invariant. Now take G_{k+1} to be a neighborhood of $\mathrm{SO}(2)J_{k+1} \cup \mathrm{SO}(2)V_{k+1}$ that is compactly contained in $\operatorname{int} K^{\mathrm{lc}}$, i.e., $G_{k+1} \Subset K^{\mathrm{lc}}$ and such that

$$\operatorname{dist}\big(H, \mathrm{SO}(2)J_{k+1} \cup \mathrm{SO}(2)V_{k+1}\big) \leq \frac{1}{2k} \qquad \text{for all } H \in G_{k+1}.$$

Then,

$$G_k \subset G_{k+1}^{\mathrm{lc}}$$

and for all $H \in G_{k+1}$ it holds that

$$\operatorname{dist}(H, K) \leq \frac{1}{2k} + \operatorname{dist}\big(\mathrm{SO}(2)J_{k+1} \cup \mathrm{SO}(2)V_{k+1}, K\big) \leq \frac{1}{k}.$$

This completes the induction.

The claim of the theorem then follows from Gromov's Convex Integration Theorem 12.4. □

We close this section by quoting the following result, which shows that exact solutions to (12.7) are *necessarily* very complicated. For the statement of the theorem we need a notion of boundary regularity: A set $E \subset \Omega$ is called a **set of finite perimeter in** Ω if the distributional derivative of the indicator function $\mathbb{1}_E$ in Ω is a finite Borel measure, i.e.,

$$\mathrm{Per}_\Omega(E) := \sup\left\{ \int_E \operatorname{div} \varphi \, \mathrm{d}x \;:\; \varphi \in \mathrm{C}_c^1(\Omega; \mathbb{R}^d),\ \|\varphi\|_{\mathrm{L}^\infty} \leq 1 \right\} < \infty. \tag{12.10}$$

For sets E with smooth boundary we have $\mathrm{Per}_\Omega(E) = \mathcal{H}^{d-1}(E \cap \Omega)$ (see Problem 12.9).

Theorem 12.8 (Dolzmann–Müller 1995 [126]) *Let $U \in \mathbb{R}^{d\times d}$ satisfy $\det U > 0$ and be such that every matrix in*

$$K := \mathrm{SO}(d) \cup \mathrm{SO}(d)U$$

is rank-one connected to precisely two other matrices in K. Then, every map $u \in \mathrm{W}^{1,\infty}(\Omega; \mathbb{R}^d)$ satisfying

$$\nabla u \in K \quad \text{a.e.}$$

such that

$$E := \big\{ x \in \Omega \;:\; \nabla u(x) \in \mathrm{SO}(d) \big\}$$

has finite perimeter in Ω, is locally an order-one ("simple") laminate. Furthermore, if u is affine on the boundary $\partial\Omega$, then u is itself an affine map.

12.3 Infinite-Order Laminates

In this section we quote an extension of the Gromov Convex Integration Theorem 12.4 that also works in situations where infinite-order laminates are necessary. For this, let again $K \subset \mathbb{R}^{m\times d}$ be compact and non-empty (merely closed sets are also possible if property (ii) below holds in a stronger form). Then, a sequence $(G_k)_k$ of open, uniformly bounded sets $G_k \subset \mathbb{R}^{m\times d}$ is called an **RC-in-approximation** for K if the following two conditions hold:

(i) $G_k \subset G_{k+1}^{\mathrm{rc}} := \bigcup\{\, S^{\mathrm{rc}} : S \subset G_{k+1} \text{ compact}\,\}$ for all $k \in N$;
(ii) if $A_k \in G_k$ and $A_k \to A$ as $k \to \infty$, then $A \in K$.

Theorem 12.9 (Müller–Šverák 2003 [252]) *Let $K \subset \mathbb{R}^{m\times d}$ be a non-empty compact set and assume that there exists an RC-in-approximation $(G_k)_k$ for K. Suppose that either $v \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m)$ is countably piecewise affine or $v \in \mathrm{C}^1(\Omega;\mathbb{R}^m)$, and that*

$$\nabla v \in G_\infty := \bigcup_{l=1}^{\infty} G_l \quad a.e.$$

Then, for every $\varepsilon > 0$ there exists a solution to the exact differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \quad u|_{\partial\Omega} = v|_{\partial\Omega}, \\ \nabla u \in K \quad a.e. \end{cases}$$

with $\|u - v\|_{\mathrm{L}^\infty} \le \varepsilon$.

The proof follows essentially the same strategy as Theorem 12.4, but there is an additional step to approximate infinite-order laminates with finite-order laminates, without changing the barycenter, see [252] for the details.

Recall that the Evans Partial Regularity Theorem 5.22 established the partial $\mathrm{C}^{1,\alpha}$-regularity ($\alpha \in (0,1)$) for minimizers of some quasiconvex integral functionals. It turns out that for weak solutions to the Euler–Lagrange equation that are *not* minimizers, the corresponding statement is false. In fact, there are many such pathological weak solutions, which can be proved using the theorem above:

Theorem 12.10 (Müller–Šverák 2003 [252]) *Let $\Omega \subset \mathbb{R}^2$ be a bounded Lipschitz domain. There exists a smooth and strongly 2-quasiconvex integrand $f_{\mathrm{MS}} \colon \mathbb{R}^{2\times 2} \to \mathbb{R}$ satisfying $\mathrm{D}^2 f(A)[B,B] \le M|B|^2$ for all $A, B \in \mathbb{R}^{m\times d}$ and a constant $M > 0$, such that the following property holds: For every $v \in \mathrm{C}^1(\Omega;\mathbb{R}^2)$ and every $\varepsilon > 0$ there exists a map $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^2)$ solving*

$$-\operatorname{div}[\mathrm{D}f_{\mathrm{MS}}(\nabla u)] = 0 \quad \text{in } \Omega \text{ (weakly)}$$

and $\|u - v\|_{\mathrm{L}^\infty} \le \varepsilon$, but $u \notin \mathrm{C}^1(U;\mathbb{R}^2)$ on any open subset $U \subset \Omega$.

For the proof, which makes use of generalized T_4-configurations, see [252]. There is also a (strongly) polyconvex integrand with the same property, as is shown in [320].

12.4 Crystalline Microstructure in 3D

The construction of exact solutions to multi-well differential inclusions in three space dimensions is an active but unfinished area of research. We only quote the following prototypical result.

Theorem 12.11 (Conti–Dolzmann–Kirchheim 2007 [83]) *Let*

$$K := \bigcup_{i=1}^{6} \mathrm{SO}(3)U_i,$$

where

$$U_1 = \begin{pmatrix} \alpha_1 & & \\ & \alpha_2 & \\ & & \alpha_3 \end{pmatrix}, \qquad 0 < \alpha_1 < \alpha_2 \leq \alpha_3, \qquad \alpha_1\alpha_2\alpha_3 = 1,$$

and the other matrices $U_2, \ldots, U_6$ are given by permuting the three values on the diagonal (if $\alpha_2 = \alpha_3$, this six-well problem reduces to a three-well problem). Then, there exists an $R > 0$ such that for all $v \in \mathrm{C}^{1,\gamma}(\Omega;\mathbb{R}^3)$, where $\gamma \in (0,1)$, that satisfy

$$|\nabla v(x) - \mathrm{Id}| < R, \quad \det \nabla v(x) = 1 \qquad \textit{for all } x \in \Omega,$$

the differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^3), \quad u|_{\partial\Omega} = v|_{\partial\Omega}, \\ \nabla u \in K \end{cases}$$

has at least one exact solution. Moreover, for any given $\varepsilon > 0$ this solution can be chosen such that $\|u - v\|_{\mathrm{L}^\infty} \leq \varepsilon$.

Example 12.12 With regard to the shape-memory effect observed in the NiAl alloy, which is described in Section 1.9, we have discussed in Example 10.10 that the material will try to satisfy the differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^3), \ \Omega \subset \mathbb{R}^3, \\ \nabla u \in K \quad \text{in } \Omega \end{cases}$$

as closely as possible, where K is the pointwise minimizer set of the integrand in the governing energy functional. We have

$$K = \begin{cases} \mathrm{SO}(3) & \text{above the critical temperature,} \\ \mathrm{SO}(3)U_1 \cup \mathrm{SO}(3)U_2 \cup \mathrm{SO}(3)U_3 & \text{below the critical temperature,} \end{cases}$$

and the matrices $U_1, \ldots, U_3$ are given explicitly in Section 1.9. This *symmetry breaking* from the cubic to the tetragonal phase explains the shape-memory effect: Above the critical temperature, the microstructure is rigid (see Reshetnyak's Rigidity Theorem 10.18)

and so, plastic deformations are reflected in changes to the crystal lattice bonds. This is the "memorized" original shape.

Once the material is cooled below the critical temperature, however, K turns out to have many rank-one connections, K^{qc} is "large" (but it is often unknown *how* large, see Problem 16 in [37]), and the material can accommodate the prescribed boundary conditions by arranging the different solid phase variants (corresponding to the wells) in a complex way. This change of shape is not a "plastic" deformation because the inter-atomic bonds are not broken in the process. When the specimen is heated above the critical temperature again, the different phases collapse into one and the specimen snaps back to its "memorized" shape.

By a convex integration result like Theorem 12.11 above (we remarked in Section 1.9 that $\det U_1 = \det U_2 = \det U_3 \approx 1$, so the above theorem is *almost* applicable) one sees that for given boundary values many exact solutions u to the above differential inclusion exist. Moreover, these solutions u can be chosen arbitrarily close to *any* prescribed (admissible) macroscopic deformation v. This is the mathematical manifestation of the aforementioned "deformability" without breaking any inter-atomic bonds.

For convex integration results applicable to the cubic-to-orthorhombic phase transition of CuAlNi, see [290] (in the linearized case).

12.5 Stability of Gradient Distributions

The theory of convex integration as presented so far is based on fairly concrete constructions. There is also a more abstract approach to solving differential inclusions, which is based on Baire category theory. This idea was first explored in this context by Dacorogna and Marcellini [94, 95] and then put into its final form by Kirchheim, who introduced the notion of *stability* for gradient distributions.

Let $K \subset \mathbb{R}^{m\times d}$ be compact and non-empty. As before, we aim to find exact solutions to

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^m), \quad u|_{\partial\Omega} = g, \\ \nabla u \in K, \end{cases} \tag{12.11}$$

where $g \in \mathrm{W}^{1,1/2}(\partial\Omega;\mathbb{R}^m)$.

We say that K is **(piecewise-affinely) stable** with respect to a bounded open set $G \subset \mathbb{R}^{m\times d}$ if for all $\eta > 0$ there exists a $\delta_\eta > 0$ such that the following holds: For any $F \in G$ with $\operatorname{dist}(F, K) > \eta$ there is a countably piecewise affine map $\psi \in \mathrm{W}^{1,\infty}_0(B_1;\mathbb{R}^m)$ such that

(i) $F + \nabla\psi(x) \in G$ for almost every $x \in B_1$;

(ii) $\fint_{B_1} |\nabla\psi(x)| \,\mathrm{d}x > \delta_\eta$.

As usual, from a standard covering argument we realize that in the above definition the unit ball can be replaced by any bounded Lipschitz domain.

Define the set

$$X_0 := \left\{ u \in \mathrm{W}^{1,2}_g(\Omega;\mathbb{R}^m) \ : \ u \text{ countably piecewise affine, } \nabla u \in G \text{ a.e.} \right\}$$

and assume that it is non-empty (which is a condition on G). Let furthermore

$$X := \text{w-clos}_{\mathrm{W}^{1,2}} X_0, \tag{12.12}$$

that is, X is the closure of X_0 with respect to the weak topology in $\mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$. The set X so defined together with the weak $\mathrm{W}^{1,2}$-topology is a complete metric space since the weak topology in $\mathrm{W}^{1,2}$ is metrizable on norm-bounded sets (G is bounded).

Then, a very general "convex integration" principle reads as follows:

Theorem 12.13 (Kirchheim 2003 [193]) *Assume that K is (piecewise-affinely) stable with respect to G. Then, the set of exact solutions of (12.11) is dense in X.*

In order to establish this theorem, we first recall the fundamental Baire category theorem from functional analysis:

Theorem 12.14 *Let X be a complete metric space.*

(i) If the sets $U_k \subset X$, $k \in \mathbb{N}$, are open and dense in X, then $\bigcap_{k=1}^{\infty} U_k$ is also dense in X.

*(ii) If $f : X \to \mathbb{R}$ is a **Baire-one function**, i.e., f is the pointwise limit of continuous functions, then the set of continuity points of f is dense in X.*

Proof. Ad (i). Let $V \subset X$ be open. We need to show that there is a $u \in V$ with $u \in \bigcap_{k=1}^{\infty} U_k$. Since U_1 is dense in X, it must intersect V. So, there is a closed ball

$$\overline{B_{r_1}(u_1)} \subset V \cap U_1 \qquad \text{for some } u_1 \in U_1 \text{ and } r_1 > 0.$$

We iterate this procedure to get $u_j \in X$, $j = 2, 3, \ldots$, and $r_j \in (0, 1/j)$ such that

$$\overline{B_{r_j}(u_j)} \subset B_{r_{j-1}}(u_{j-1}) \cap U_j.$$

Since $u_j \in B_{r_i}(u_i)$ if $j > i$, the sequence $(u_j)_j$ is Cauchy and thus converges to some $u \in X$. We have that $u \in B_{r_i}(u_i) \cap U_i$ for all i and thus

$$u \in V \cap \bigcap_{k=1}^{\infty} U_k.$$

Ad (ii). Define for the function f and $u \in X$ the quantity

$$\omega(u) := \lim_{\delta \downarrow 0} \left[\sup_{v \in B_\delta(u)} f(v) - \inf_{v \in B_\delta(u)} f(v) \right].$$

Let $\varepsilon > 0$ and assume that there is an open set $V \subset X$ such that $\omega(u) > 4\varepsilon$ for all $u \in \overline{V}$. In the remainder of the proof we will lead this to a contradiction. Thus, ω vanishes on a dense set in X, which is the set of points where f is continuous.

Suppose that $f_j \to f$ pointwise in X and define

$$E_n := \bigcap_{i,j \geq n} \big\{ u \in X \,:\, |f_i(u) - f_j(u)| \leq \varepsilon \big\}, \qquad n \in \mathbb{N},$$

which are closed, increasing sets with $\bigcup_n E_n = X$ since $f_j \to f$ pointwise. Thus,

$$\bigcap_{n=1}^{\infty} (\overline{V} \setminus E_n) = \emptyset.$$

By (i) not all of the relatively open sets $\overline{V} \setminus E_n$ can be dense in the complete metric space $\overline{V}$. Hence, there exists an $n \in \mathbb{N}$ such that $V \cap E_n$ contains a non-empty open set W. In particular, for all $w \in W$ (letting $i \to \infty$ and $j := n$),

$$|f(w) - f_n(w)| \leq \varepsilon.$$

Moreover, for any $u_0 \in W$ choose a ball $B_r(u_0) \subset W$ with the property that

$$|f_n(w) - f_n(u_0)| \leq \varepsilon \qquad \text{for all } w \in B_r(u_0).$$

This is possible by the continuity of f_n. Then, for all $u, v \in B_r(u_0)$,

$$\begin{aligned} |f(u) - f(v)| &\leq |f(u) - f_n(u)| + |f_n(u) - f_n(u_0)| + |f_n(u_0) - f_n(v)| \\ &\quad + |f_n(v) - f(v)| \\ &\leq 4\varepsilon. \end{aligned}$$

Thus, $\omega(u_0) \leq 4\varepsilon$ for all $u_0 \in W \subset V$. This is the sought contradiction. □

Returning to X as defined in (12.12), we denote by $S \subset X$ the set of **(weak-to-strong) stability points**, that is, those $u \in X$ such that for any sequence $(u_j)_j \subset X$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,2}$ it automatically holds that also $u_j \to u$ strongly in $\mathrm{W}^{1,2}$. On the face of it, this appears to be a very strong property. However, we have the following result:

Lemma 12.15 *The set S is dense in X.*

Proof. Let $(\eta_\delta)_{\delta>0} \subset \mathrm{C}_c^\infty(\mathbb{R}^d)$ be a family of mollifiers. We set

$$\mathcal{G}[u] := \int_\Omega |\nabla u|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}_g^{1,2}(\Omega;\mathbb{R}^m),$$

and

$$\mathcal{G}_n[u] := \int_\Omega |\eta_{1/n} \star \nabla u|^2 \,\mathrm{d}x, \qquad u \in \mathrm{W}_g^{1,2}(\Omega;\mathbb{R}^m),$$

where we consider u to be extended by some fixed extension of g outside of Ω. Then, all the $\mathcal{G}_n$ are continuous on X with respect to the weak convergence in $\mathrm{W}^{1,2}$. Indeed, let $(u_j)_j \subset X$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,2}$. So, for all $x \in \Omega$,

$$(\eta_{1/n} \star \nabla u_j)(x) = \int \eta_{1/n}(x-y)\nabla u_j(y)\,\mathrm{d}y$$
$$\to \int \eta_{1/n}(x-y)\nabla u(y)\,\mathrm{d}y = (\eta_{1/n} \star \nabla u)(x)$$

and, by Young's inequality for convolutions, see Lemma A.33,

$$\|\eta_{1/n} \star \nabla u_j\|_{\mathrm{L}^4} \le \|\eta_{1/n}\|_{\mathrm{L}^{4/3}} \cdot \|\nabla u_j\|_{\mathrm{L}^2}.$$

Thus, for fixed n, the family $(\eta_{1/n} \star \nabla u_j)_j$ is 2-equiintegrable and via Vitali's Convergence Theorem A.12 we may conclude that

$$\eta_{1/n} \star \nabla u_j \to \eta_{1/n} \star \nabla u \quad \text{in } \mathrm{L}^2$$

as $j \to \infty$ (with n held fixed). In particular, $\mathcal{G}_n[u_j] \to \mathcal{G}_n[u]$.

Next, observe that for all $u \in \mathrm{W}^{1,2}(\Omega;\mathbb{R}^m)$ we have

$$\mathcal{G}_n[u] \to \mathcal{G}[u] \qquad \text{as } n \to \infty$$

by standard properties of the mollification. This means that $\mathcal{G}$ is the pointwise limit of X-continuous functionals, i.e., a Baire-one functional. From the Baire Category Theorem 12.14 (ii) it follows that the set of continuity points of $\mathcal{G}$ is dense in X. For such a continuity point u we have that $u_j \rightharpoonup u$ in $\mathrm{W}^{1,2}$ implies $\|\nabla u_j\|_{\mathrm{L}^2} \to \|\nabla u\|_{\mathrm{L}^2}$, whereby $u_j \to u$ strongly in $\mathrm{W}^{1,2}$ (see the Radon–Riesz Theorem A.15). Thus, the set of continuity points of $\mathcal{G}$ is contained in S (in fact, these two sets are the same) and the claim follows. □

Proof of Theorem 12.13. Step 1. Let u be a (weak-to-strong) stability point. We will show that $\nabla u \in K$ almost everywhere, which will imply the claim of the theorem by Lemma 12.15. Assume to the contrary that

$$\mathcal{K}[u] := \int_\Omega \operatorname{dist}(\nabla u(x), K)\,\mathrm{d}x > 0$$

and take a sequence $(v_j)_j \subset X_0$ with $v_j \rightharpoonup u$ in $\mathrm{W}^{1,2}$. Then, since u is a stability point, it follows that $v_j \to u$ in $\mathrm{W}^{1,2}$. In particular, we may assume

$$\inf_{j\in\mathbb{N}} \mathcal{K}[v_j] > 0.$$

We will show below that for every v_j we can find a sequence $(v_{j,k})_k \subset X_0$ with $v_{j,k} \rightharpoonup v_j$ in $\mathrm{W}^{1,2}$ as $k \to \infty$ and

$$\|\nabla v_j - \nabla v_{j,k}\|_{\mathrm{L}^1} \ge \beta > 0,$$

where β is independent of j and k. Then, since all gradients $\nabla v_{j,k}$ are uniformly L^2-norm-bounded (because G is bounded), we may use the metrizability of the weak topology on norm-bounded sets to select a diagonal subsequence $(u_j)_j \subset X_0$ with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,2}$ and

$$\|\nabla u_j - \nabla u\|_{\mathrm{L}^1} \geq \|\nabla u_j - \nabla v_j\|_{\mathrm{L}^1} - \|\nabla v_j - \nabla u\|_{\mathrm{L}^1} \geq \frac{\beta}{2} > 0$$

for j sufficiently large. However, using again that u is a stability point, we must also have $u_j \to u$ in $\mathrm{W}^{1,2}$, a contradiction.

Step 2. By virtue of a standard weak compactness argument in $\mathrm{W}^{1,2}_g(\Omega;\mathbb{R}^m)$, it only remains to show that for any $v \in X_0$ with $\mathcal{K}[v] > 0$ and for any given $\varepsilon > 0$ we can find $w \in X_0$ with $\|v - w\|_{\mathrm{L}^2} \leq \varepsilon$ and

$$\|\nabla v - \nabla w\|_{\mathrm{L}^1} \geq \beta,$$

where $\beta > 0$ is a constant that does not depend on $\varepsilon > 0$ and that depends on v only through $\mathcal{K}[v]$.

Since $v \in X_0$ is countably piecewise affine, we can write

$$v(x) = \sum_{l=1}^{\infty} (z_l + A_l x)\mathbb{1}_{D_l}(x), \qquad x \in \Omega,$$

for some $z_l \in \mathbb{R}^m$, $A_l \in G$, and Lipschitz subdomains $D_l \subset \Omega$ that form a disjoint partition of Ω up to a negligible set. For $\eta > 0$ set

$$E_\eta := \left\{ x \in \Omega \,:\, \operatorname{dist}(\nabla v(x), K) > \eta \right\}$$

and estimate

$$\begin{aligned}\mathcal{K}[v] &= \int_{E_\eta} \operatorname{dist}(\nabla v(x), K)\, \mathrm{d}x + \int_{\Omega\setminus E_\eta} \operatorname{dist}(\nabla v(x), K)\, \mathrm{d}x \\ &\leq (|G|_\infty + |K|_\infty) \cdot |E_\eta| + \eta|\Omega|.\end{aligned}$$

Here, $|G|_\infty := \sup\{\,|A| : A \in G\,\}$ and likewise for $|K|_\infty$. Thus, choosing $\eta := \mathcal{K}[v]/(2|\Omega|) > 0$, we get

$$|E_\eta| \geq \frac{\mathcal{K}[v]}{2(|G|_\infty + |K|_\infty)} =: 2\alpha > 0.$$

Refining the (countably) piecewise affine partition for v if necessary, we may assume that

$$\sum_{D_l \subset E_\eta} |D_l| \geq \alpha.$$

Then use the (piecewise affine) stability of K with respect to G to find in each $D_l \subset E_\eta$ a countably piecewise affine map $\psi_l \in \mathrm{W}^{1,\infty}_0(D_l;\mathbb{R}^m)$ with

(a) $A_l + \nabla\psi_l(x) \in G$ for almost every $x \in D_l$;

(b) $\displaystyle\int_{D_l} |\nabla\psi_l(x)|\, \mathrm{d}x \geq \delta_\eta |D_l|$, where $\delta_\eta > 0$ is independent of l and ε;

(c) $\|\psi_l\|_{\mathrm{L}^2(D_l)} \leq \varepsilon \dfrac{|D_l|}{|\Omega|}$.

The last condition can be established via a "homogenization" argument similar to the one in Lemma 4.15. If $D_l \not\subset E_\eta$ set $\psi_l := 0$.

For

$$w(x) := \sum_{l=1}^{\infty} (z_l + A_l x + \psi_l(x)) \mathbb{1}_{D_l}(x), \qquad x \in \Omega,$$

it holds that $\|v - w\|_{\mathrm{L}^2} \leq \varepsilon$ and

$$\|\nabla v - \nabla w\|_{\mathrm{L}^1} \geq \sum_{D_l \subset E_\eta} \int_{D_l} |\psi_l(x)| \, \mathrm{d}x \geq \sum_{D_l \subset E_\eta} \delta_\eta |D_l| \geq \delta_\eta \alpha =: \beta.$$

Since β is independent of ε and depends on v only through $\mathcal{K}[v]$, the claim follows. □

Convex integration via Baire category theory has several advantages. First, checking stability can be easier than showing the existence of a countably piecewise affine reduction or an in-approximation as in the previous sections. Indeed, one only needs to show that away from K one can perturb affine maps in a suitable way. Second, the analysis of the differential inclusion $\nabla u \in K$ can be refined by introducing a notion of extremal points for K, see Chapter 3 of [193] for the details. This in fact explains the underlying workings of the whole Baire convex integration strategy: The functional $\mathcal{G}$ from the proof of Lemma 12.15 can only be continuous at $u \in X$ with respect to the weak convergence in $\mathrm{W}^{1,2}$ if ∇u is already "trapped" in the extreme points of K. Otherwise, the (piecewise affine) stability property would imply that a perturbation is possible, violating the continuity.

Applications of the stability approach, in particular to *inhomogeneous* differential inclusions, can be found in Chapters 3 and 4 of [193].

12.6 Convex Integration via Young Measures

In this section we will combine the convex integration method with the theory of gradient Young measures, at the same time also considering *unbounded* constraint sets like $\{\det = 1\}$.

Suppose $R\colon \mathbb{R}^{m\times d} \to [0,\infty)$ has linear growth, i.e., $R(A) \leq M(1 + |A|)$ for all $A \in \mathbb{R}^{m\times d}$ and a constant $M > 0$. Define the corresponding constraint set as

$$S_R := \big\{ A \in \mathbb{R}^{m\times d} \,:\, R(A) = 0 \big\}.$$

One may think of R as a type of distance function to the (in general unbounded) set S_R.

To illustrate the kind of result we have in mind, assume first that we are given a $\mathrm{W}^{1,p}$-gradient Young measure $\nu = (\nu_x)_x \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, where $p \in (1,\infty)$ and $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, with the property that there exists a sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ such that

$$\nabla u_j \xrightarrow{\mathbf{Y}} \nu \qquad \text{and} \qquad \nabla u_j \in S_R \quad \text{a.e.}$$

for all $j \in \mathbb{N}$. Then it is a direct consequence of Lemma 4.13 that

$$\operatorname{supp} \nu_x \subset S_R \qquad \text{for a.e. } x \in \Omega.$$

We now show that also the converse holds if R satisfies additional conditions, which are similar, in spirit at least, to the piecewise affine reducibility and stability conditions of the preceding sections.

To state the result, we call a function $R\colon \mathbb{R}^{m\times d} \to [0,\infty)$ with linear growth a **$\mathbf{GY}^p$-reducible constraint function**, where $p \in (1,\infty)$, if for every $F \in \mathbb{R}^{m\times d}$ there is a homogeneous $\mathrm{W}^{1,p}$-gradient Young measure $\mu^F = (\mu^F)_x \in \mathbf{GY}^p(B_1;\mathbb{R}^{m\times d})$ satisfying

$$[\mu^F] = F, \quad \operatorname{supp} \mu^F \subset S_R, \quad \int |A-F|^p \,\mathrm{d}\mu^F(A) \le C\cdot R(F)^p \tag{12.13}$$

for a constant $C > 0$, and if furthermore $x \mapsto \mu^{F(x)}$ is weakly*-measurable whenever $F\colon \mathbb{R}^n \to \mathbb{R}^{m\times d}$ is a Borel-measurable map ($n \in \mathbb{N}$).

Theorem 12.16 (Koumatos–Rindler–Wiedemann 2016 [201, 202]) *Let $R\colon \mathbb{R}^{m\times d} \to [0,\infty)$ be a $\mathbf{GY}^p$-reducible constraint function, where $p \in (1,\infty)$. Then, for every $\mathrm{W}^{1,p}$-gradient Young measure $\nu = (\nu_x)_x \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$ satisfying*

$$\operatorname{supp} \nu_x \subset S_R \qquad \textit{for a.e. } x \in \Omega,$$

there exists a sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with

$$\nabla u_j \xrightarrow{\mathbf{Y}} \nu \qquad \textit{and} \qquad \nabla u_j \in S_R \quad \textit{a.e.}$$

Furthermore, the sequence $(u_j)_j$ can be chosen such that its gradients $(\nabla u_j)_j$ are p-equiintegrable and $u_j - u \in \mathrm{W}^{1,p}_0(\Omega,\mathbb{R}^d)$, where $u \in \mathrm{W}^{1,p}(\Omega,\mathbb{R}^d)$ is any underlying deformation for ν (i.e., $[\nu_x] = \nabla u(x)$ for almost every $x \in \Omega$).

Remark 12.17 Also x-dependent constraints can be treated with only a bit of additional effort; see [201] for the details.

The proof of this theorem proceeds by a convergence argument in the style of convex integration together with an application of the Kinderlehrer–Pedregal Theorem 8.11. The core of the strategy is contained in the following lemma.

Lemma 12.18 *Let $R\colon \mathbb{R}^{m\times d} \to [0,\infty)$ be a $\mathbf{GY}^p$-reducible constraint function, where $p \in (1,\infty)$. Then, for every $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ there exists a map $v \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ such that*

(i) $\nabla v(x) \in S_R$ *for a.e.* $x \in \Omega$;
(ii) $v|_{\partial\Omega} = u|_{\partial\Omega}$;
(iii) $\|\nabla v - \nabla u\|_{\mathrm{L}^p}^p \le C \displaystyle\int_\Omega R(\nabla u(x))^p \,\mathrm{d}x,$

where $C > 0$ is a constant.

Proof. Assume without loss of generality that

$$\int_\Omega R(\nabla u(x))^p \,\mathrm{d}x > 0$$

because otherwise there is nothing to prove. We will inductively construct a uniformly norm-bounded sequence of maps $(w_l)_l \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with the following properties:

$$\operatorname{supp}(w_l - u) \Subset \Omega, \tag{12.14}$$

$$\int_\Omega R(\nabla w_l(x))^p \,\mathrm{d}x \le 2^{-lp} \int_\Omega R(\nabla u(x))^p \,\mathrm{d}x, \tag{12.15}$$

$$\int_\Omega |\nabla w_{l+1}(x) - \nabla w_l(x)|^p \,\mathrm{d}x \le 2^{-(l-1)p} C \int_\Omega R(\nabla u(x))^p \,\mathrm{d}x, \tag{12.16}$$

where $C > 0$ is a constant.

For $l = 0$, set $w_0 = u$ so that (12.14) and (12.15) are trivially satisfied.

For the inductive step, if $w_l \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ has been constructed to satisfy (12.14) and (12.15), we find w_{l+1} as follows: By the assumption that $R\colon \mathbb{R}^{m\times d} \to [0,\infty)$ is a $\mathbf{GY}^p$-reducible constraint function, for almost every $x \in \Omega$ these exists a homogeneous $\mathrm{W}^{1,p}$-gradient Young measure $\nu_x^l \in \mathcal{M}^1(\mathbb{R}^{m\times d})$ with

$$[\nu_x^l] = \nabla w_l(x), \qquad \operatorname{supp} \nu_x^l \subset S_R,$$

and

$$\int |A - \nabla w_l(x)|^p \,\mathrm{d}\nu_x^l(A) \le C \cdot R(\nabla w_l(x))^p. \tag{12.17}$$

In particular, for $x \in \Omega$ it holds that $R(\nabla w_l(x)) = 0$ if and only if $\nu_x^l = \delta_{\nabla w_l(x)}$.

If it happens that $R(\nabla w_l) \equiv 0$ almost everywhere, that is, $\nu_x^l = \delta_{\nabla w_l(x)}$ for almost every $x \in \Omega$, then set $w_{l+m} := w_l$ for all $m \in \mathbb{N}$, by which the whole sequence is already constructed.

Otherwise, since our assumptions entail the weak* measurability of $(\nu_x^l)_x$ with respect to x, we may consider $\nu^l = (\nu_x^l)_x$ as an (inhomogeneous) $\mathrm{W}^{1,p}$-gradient Young measure on Ω. Then invoke Lemma 5.11, individually for each homogeneous Young measure ν_x^l at almost every $x \in \Omega$, to obtain that for all quasiconvex functions $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with p-growth the Jensen-type inequality

$$h([\nu_x^l]) \le \int h \,\mathrm{d}\nu_x^l$$

holds. By the Kinderlehrer–Pedregal Theorem 8.11, applied to the *compound* Young measure $(\nu_x^l)_x$, in conjunction with Lemma 4.14 there exists $(w_{l,n})_n \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^{m\times d})$ such that $(\nabla w_{l,n})_n$ is a p-equiintegrable sequence generating ν^l, and $\operatorname{supp}(w_{l,n} - w_l) \Subset \Omega$. Hence, also $\operatorname{supp}(w_{l,n} - u) \Subset \Omega$ for all $n \in \mathbb{N}$.

By the Young measure limit representation for the integrand $A \mapsto R(A)^p$, as well as the fact that ν_x^l is supported in S_R, we may choose n large enough, say $n = N(l)$, so that for $\nabla w_{l+1} := \nabla w_{l,N(l)}$ it holds that

$$\int_\Omega R(\nabla w_{l+1}(x))^p \,\mathrm{d}x \le 2^{-(l+1)p} \int_\Omega R(\nabla u(x))^p \,\mathrm{d}x,$$

that is, (12.14) and (12.15) hold for w_{l+1}.

Taking $N(l)$ even larger if necessary, we can moreover ensure that

$$\int_\Omega |\nabla w_{l+1}(x) - \nabla w_l(x)|^p \,\mathrm{d}x \le 2^p \int_\Omega \int |A - \nabla w_l(x)|^p \,\mathrm{d}\nu_x^l(A) \,\mathrm{d}x. \tag{12.18}$$

Indeed, this follows again from the Young measure limit representation for the integrand $(x, A) \mapsto |A - \nabla w_l(x)|^p$ and the fact that we are in the case where the right-hand side is not zero.

Next, for any $l \in \mathbb{N}$, by (12.17) and (12.15) we infer that

$$\begin{aligned}\int_\Omega \int |A - \nabla w_l(x)|^p \,\mathrm{d}\nu_x^l(A) \,\mathrm{d}x &\le C \int_\Omega R(\nabla w_l(x))^p \,\mathrm{d}x \\ &\le 2^{-lp} C \int_\Omega R(\nabla u(x))^p \,\mathrm{d}x\end{aligned}$$

for a constant $C > 0$. Combining this with (12.18), we obtain the estimate

$$\int_\Omega |\nabla w_{l+1}(x) - \nabla w_l(x)|^p \,\mathrm{d}x \le 2^{-(l-1)p} C \int_\Omega R(\nabla u(x))^p \,\mathrm{d}x,$$

which is (12.16).

Having constructed the sequence $(w_l)_l$, we now observe via (12.16) together with the fixed boundary values $w_l|_{\partial\Omega} = u|_{\partial\Omega}$ (which are a consequence of (12.14)) and the Poincaré inequality (see Theorem A.27) that the sequence $(w_l)_l$ is Cauchy in $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ and therefore has a strong limit, which we denote by $v \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$. In particular, it holds that $v|_{\partial\Omega} = u|_{\partial\Omega}$, i.e. (ii). Using the triangle inequality and (12.16), we also deduce that

$$\begin{aligned}\|\nabla v - \nabla u\|_{\mathrm{L}^p} &\le \sum_{l=0}^\infty \|\nabla w_{l+1} - \nabla w_l\|_{\mathrm{L}^p} \\ &\le C^{1/p} \left(\int_\Omega R(\nabla u(x))^p \,\mathrm{d}x \right)^{1/p} \sum_{l=0}^\infty 2^{-(l-1)} \\ &\le 4C^{1/p} \left(\int_\Omega R(\nabla u(x))^p \,\mathrm{d}x \right)^{1/p},\end{aligned}$$

proving (iii).

Lastly, $(\nabla w_l)_l$ is p-equiintegrable (being Cauchy in L^p), and since

$$R(\nabla w_l(x))^p \le C(1 + |\nabla w_l(x)|)^p,$$

also $(R(\nabla w_l)^p)_l$ is seen to be a (1-)equiintegrable family. Therefore, by Vitali's Convergence Theorem A.12,

$$\int_\Omega R(\nabla v(x))^p \,\mathrm{d}x = \lim_{l\to\infty} \int_\Omega R(\nabla w_l(x))^p \,\mathrm{d}x = 0,$$

the last equality following from (12.15). This immediately implies $R(\nabla v(x)) = 0$ for almost every $x \in \Omega$, that is, (i). □

Proof of Theorem 12.16. Let ν be a $\mathrm{W}^{1,p}$-gradient Young measure with

$$\operatorname{supp} \nu_x \subset S_R \qquad \text{for a.e. } x \in \Omega.$$

Suppose that $u \in \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ is a underlying deformation for ν, that is, $\nabla u(x) = [\nu_x]$ for almost every $x \in \Omega$. Lemma 4.14 yields that there exists a sequence $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ with $u_j - u \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$ such that $(\nabla u_j)_j$ is p-equiintegrable and generates ν.

By the Young measure representation applied to the integrand $A \mapsto R(A)^p$ and the assumption on the support of ν, we may assume that

$$\int_\Omega R(\nabla u_j(x))^p \,\mathrm{d}x < \frac{1}{j}.$$

Applying the preceding Lemma 12.18 to each u_j, we obtain a new sequence $(v_j)_j$, such that $\nabla v_j \in S_R$ almost everywhere, $v_j - u \in \mathrm{W}^{1,p}_0(\Omega;\mathbb{R}^m)$, and

$$\|\nabla u_j - \nabla v_j\|^p_{\mathrm{L}^p} < \frac{C}{j}.$$

Hence, also $(\nabla v_j)_j$ is p-equiintegrable and generates ν. □

The most interesting application of this result is to the determinant constraint:

Proposition 12.19 *The function $R(A) := |r - \det A|^{1/d}$, $A \in \mathbb{R}^{d\times d}$ ($d \geq 2$), where $r \in \mathbb{R}$ is fixed, is a* $\mathbf{GY}^p$*-reducible constraint function for all $p \in (1, d)$.*

Proof. We prove this result for $d = 3$ only; the other cases are similar and the task of Problem 12.7.

Step 1. First, we transform an arbitrary matrix $F \in \mathbb{R}^{m\times d}$ to diagonal form using the real singular value decomposition (see Appendix A.1) and write $F = \widetilde{P}\widetilde{D}\widetilde{Q}^{\mathsf{T}}$ where $\widetilde{D} = \operatorname{diag}(\sigma_1, \sigma_2, \sigma_3)$ with $0 \leq \sigma_1 \leq \sigma_2 \leq \sigma_3$, and $\widetilde{P}, \widetilde{Q} \in \mathbb{R}^{3\times 3}$ are orthogonal matrices. If $\det F < 0$, either $\widetilde{P}$ or $\widetilde{Q}$ has negative determinant, say $\det \widetilde{P} < 0$. Then, $F = PDQ^{\mathsf{T}}$ where

$$D := \operatorname{diag}(\sigma_1, \sigma_2, -\sigma_3), \qquad P := \widetilde{P}\operatorname{diag}(1, 1, -1), \qquad Q := \widetilde{Q},$$

with $P, Q \in \mathrm{SO}(3)$ and $\det D < 0$. Similarly, if $\det F \geq 0$, we may write $F = PDQ^{\mathsf{T}}$, where $P, Q \in \mathrm{SO}(3)$ and $\det D \geq 0$ for

$$D := \operatorname{diag}(\sigma_1, \sigma_2, \sigma_3).$$

Note that if D can be written as a laminate then the same holds for F since $P(a\otimes b)Q^{\mathsf{T}} = (Pa)\otimes(Qb)$ for any $a, b \in \mathbb{R}^3$. We further remark that the matrices D and F have the same determinant and (Frobenius) matrix norm. Consequently, we may henceforth assume without loss of generality that

$$F = \operatorname{diag}(\sigma_1, \sigma_2, \pm\sigma_3).$$

with $0 \le \sigma_1 \le \sigma_2 \le \sigma_3$. In the following we distinguish the cases $\sigma_3 \ge 2^{-1/3}|r|^{1/3}$ and $\sigma_3 < 2^{-1/3}|r|^{1/3}$.

Step 2: The case $\sigma_3 \ge 2^{-1/3}|r|^{1/3}$. Set

$$\gamma := \frac{|r - \det F|^{1/2}}{\sigma_3^{1/2}}$$

and decompose F twice along rank-one lines as

$$\begin{aligned} F = {} & \frac{1}{4}\big[F + \gamma(\mathrm{e}_1\otimes\mathrm{e}_2) + \gamma(\mathrm{e}_2\otimes\mathrm{e}_1)\big] + \frac{1}{4}\big[F + \gamma(\mathrm{e}_1\otimes\mathrm{e}_2) - \gamma(\mathrm{e}_2\otimes\mathrm{e}_1)\big] \\ & + \frac{1}{4}\big[F - \gamma(\mathrm{e}_1\otimes\mathrm{e}_2) + \gamma(\mathrm{e}_2\otimes\mathrm{e}_1)\big] + \frac{1}{4}\big[F - \gamma(\mathrm{e}_1\otimes\mathrm{e}_2) - \gamma(\mathrm{e}_2\otimes\mathrm{e}_1)\big]. \end{aligned}$$

A direct computation yields that two of these four matrices (either those where both γ's have the same sign, or those where the γ's have different signs, depending on the sign of σ_3) have determinant r, and the other two have determinant $2\det F - r$. We call the former ones $F_{1,G1}$ and $F_{1,G2}$ and the latter ones $F_{1,B1}$ and $F_{1,B2}$, so that

$$F = \frac{1}{4}F_{1,B1} + \frac{1}{4}F_{1,G1} + \frac{1}{4}F_{1,G2} + \frac{1}{4}F_{1,B2}$$

with

$$\begin{aligned} \det F_{1,G1} &= \det F_{1,G2} = r, \\ \det F_{1,B1} &= \det F_{1,B2} = 2\det F - r. \end{aligned}$$

If $|\det F| \le |r|$, it holds that $2|r| \ge |r - \det F|$ and, taking into account that $\sigma_3 \ge 2^{-1/3}|r|^{1/3}$,

$$\sigma_3 \ge 4^{-1/3}|r - \det F|^{1/3}.$$

On the other hand, if $|\det F| > |r|$, we can use the fact that $\sigma_3 \ge |\det F|^{1/3}$ to infer that

$$\sigma_3 \ge |\det F|^{1/3} \ge 2^{-1/3}(|r| + |\det F|)^{1/3} \ge 2^{-1/3}|r - \det F|^{1/3}.$$

This implies that there is a constant $C > 0$, which is independent of F and r, such that in either case

$$\sigma_3 \ge C|r - \det F|^{1/3}.$$

For $\square = G1, G2, B1, B2$ it then follows that

$$|F_{1,\square} - F| = \left(\frac{2|r - \det F|}{\sigma_3}\right)^{1/2} \leq C|r - \det F|^{1/3} \tag{12.19}$$

with a constant $C > 0$ that is independent of F and r, and also

$$|r - \det F_{1,\square}| \leq 2|r - \det F|. \tag{12.20}$$

Moreover, the singular value σ_3 is not altered by this construction, and so there is still a singular value of $F_{1,B1}$ and $F_{1,B2}$ with modulus at least $2^{-1/3}|r|^{1/3}$. Therefore, we may recursively apply the above splitting procedure to decompose $F_{1,B1}$ and $F_{1,B2}$ in turn taking the role of F. This yields matrices $F_{2,G1}, \ldots, F_{2,G4}$, $F_{2,B1}, \ldots, F_{2,B4}$ such that

$$\begin{aligned} F_{1,B1} &= \frac{1}{4}F_{2,G1} + \frac{1}{4}F_{2,G2} + \frac{1}{4}F_{2,B1} + \frac{1}{4}F_{2,B2}, \\ F_{1,B2} &= \frac{1}{4}F_{2,G3} + \frac{1}{4}F_{2,G4} + \frac{1}{4}F_{2,B3} + \frac{1}{4}F_{2,B4} \end{aligned}$$

with $\det F_{2,Gj} = r$ for $j = 1, 2, 3, 4$.

Going on like this for $k = 1, 2, \ldots$, we obtain a sequence of laminates

$$\mu_k := \sum_{i=1}^{k} \sum_{j=1}^{2^i} \frac{1}{4^i} \delta_{F_{i,Gj}} + \sum_{j=1}^{2^k} \frac{1}{4^k} \delta_{F_{k,Bj}},$$

where $\det F_{i,Gj} = r$ for all i, j. Clearly, $[\mu_k] = F$ and one may compute

$$\int |A - F|^p \, \mathrm{d}\mu_k(A) = \sum_{i=1}^{k} \sum_{j=1}^{2^i} \frac{1}{4^i} |F_{i,Gj} - F|^p + \sum_{j=1}^{2^k} \frac{1}{4^k} |F_{k,Bj} - F|^p.$$

To estimate $|F_{i,Gj} - F|^p$ for fixed i, j, set $X_i := F_{i,Gj}$ and for $l = i, i-1, \ldots, 1$ let X_{l-1} be the matrix $F_{l-1,Bm}$ with $m \in \{1, \ldots, 2^{l-1}\}$ such that X_l is constructed from X_{l-1} by laminating as above (with the convention $F_{0,B1} := F$, so that $X_0 = F$). Likewise, set $Y_k := F_{k,Bj}$ and let Y_{l-1} be defined analogously to X_{l-1}. Then,

$$F_{i,Gj} - F = \sum_{l=1}^{i} X_l - X_{l-1}, \qquad F_{k,Bj} - F = \sum_{l=1}^{k} Y_l - Y_{l-1},$$

and

$$\int |A-F|^p \,\mathrm{d}\mu_k(A) \le \sum_{i=1}^{k} \sum_{j=1}^{2^i} \frac{1}{4^i} \left(\sum_{l=1}^{i} |X_l - X_{l-1}| \right)^p + \sum_{j=1}^{2^k} \frac{1}{4^k} \left(\sum_{l=1}^{k} |Y_l - Y_{l-1}| \right)^p .$$

In order to get bounds on $|X_l - X_{l-1}|$, we now use (12.19) and then (12.20) recursively to obtain

$$\begin{aligned} \sum_{l=1}^{i} |X_l - X_{l-1}| &\le \sum_{l=1}^{i} C\,|r - \det X_{l-1}|^{1/3} \\ &\le \sum_{l=1}^{i} 2^{(l-1)/3}\, C|r - \det F|^{1/3} + \frac{C\,|r-\det F|^{1/3}}{2^{1/3}-1} 2^{i/3} \end{aligned}$$

as well as a similar estimate for the sum involving the Y_l's. Hence,

$$\begin{aligned} \int |A-F|^p \,\mathrm{d}\mu_k(A) &\le C|r-\det F|^{p/3} \left[\sum_{i=1}^{k} (2^{p/3-1})^i + (2^{p/3-1})^k \right] \\ &\le C|r-\det F|^{p/3} \left[\frac{1}{1-\varrho} + \varrho^k \right], \end{aligned}$$

where $\varrho := 2^{p/3-1} < 1$ (since $p < 3$). So, there is a constant $C_p > 0$, which depends on p (but not on F or r) and which blows up as $p \uparrow 3$, such that

$$\int |A-F|^p \,\mathrm{d}\mu_k(A) \le C_p |r-\det F|^{p/3}, \tag{12.21}$$

Since every μ_k is a probability measure, by (12.21) we deduce in particular

$$\begin{aligned} \int |A|^p \,\mathrm{d}\mu_k(A) &\le 2^p \left[\int |A-F|^p \,\mathrm{d}\mu_k(A) + |F|^p \right] \\ &\le 2^p\, C_p |r-\det F|^{p/3} + 2^p |F|^p. \end{aligned} \tag{12.22}$$

Finally, the mass of μ_k carried by the matrices outside S_R is vanishes, that is,

$$\mu_k\big(\{\, A \in \mathbb{R}^{3\times 3} \;:\; \det A \neq r \,\}\big) = \frac{2^k}{4^k} \to 0 \tag{12.23}$$

as $k \to \infty$.

Step 3: The case $\sigma_3 < 2^{-1/3}|r|^{1/3}$. Again, we assume that F is given by

$$F = \operatorname{diag}(\sigma_1, \sigma_2, \pm\sigma_3)$$

with $0 < \sigma_1 \le \sigma_2 \le \sigma_3$, but now $\sigma_3 < 2^{-1/3}|r|^{1/3}$. We decompose F along a rank-one line as

$$F = \frac{1}{2}\left[F + \delta(e_3 \otimes e_3)\right] + \frac{1}{2}\left[F - \delta(e_3 \otimes e_3)\right] =: \frac{1}{2}F^+ + \frac{1}{2}F^-$$

and choose $\delta := 2 \cdot 2^{-1/3}|r|^{1/3}$. Then, the singular values $\sigma_3 + \delta$ and $\sigma_3 - \delta$ of F^+ and F^-, respectively, have absolute value at least $2^{-1/3}|r|^{1/3}$. Moreover,

$$|F^\pm - F| = 2\left(\frac{|r|}{2}\right)^{1/3} \leq 2|r - \det F|^{1/3}, \tag{12.24}$$

$$|r - \det F^\pm| \leq 3|r - \det F|. \tag{12.25}$$

Indeed, the first inequality follows from the observation that $|\det F| \leq \sigma_3^3 < |r|/2$, whereby $|r - \det F| > |r|/2$. For the second inequality, observe

$$|\det F^\pm - \det F| = \delta\sigma_1\sigma_2 \leq |r| \leq 2|r - \det F|$$

and use the triangle inequality.

We can treat $F^\pm$ exactly as in Step 2, which is now applicable. This gives us two sequences of laminates μ_k^+ and μ_k^- with $[\mu_k^\pm] = F^\pm$,

$$\nu_k^\pm\left(\left\{ A \in \mathbb{R}^{3\times 3} \ : \ \det A \neq r \right\}\right) \to 0 \tag{12.26}$$

as $k \to \infty$, and the estimate

$$\int |A - F^\pm|^p \, \mathrm{d}\mu_k^\pm(A) \leq C_p |r - \det F^\pm|^{p/3} \tag{12.27}$$

for $1 \leq p < 3$, where C_p does not depend on k, F or r. It follows that the probability measure μ_k defined by

$$\mu_k := \frac{1}{2}\mu_k^+ + \frac{1}{2}\mu_k^-$$

satisfies $[\mu_k] = F$ and

$$\mu_k\left(\left\{ A \in \mathbb{R}^{3\times 3} \ : \ \det A \neq r \right\}\right) \to 0$$

as $k \to \infty$. Moreover, combining (12.27) with (12.24) and (12.25), we have

$$\begin{aligned}
&\int |A - F|^p \, \mathrm{d}\mu_k(A) \\
&\quad = \frac{1}{2}\int |A - F|^p \, \mathrm{d}\mu_k^+(A) + \frac{1}{2}\int |A - F|^p \, \mathrm{d}\mu_k^-(A) \\
&\quad \leq C_p \Bigg[|F^+ - F|^p + |F^- - F|^p \\
&\qquad\qquad + \int |A - F^+|^p \, \mathrm{d}\mu_k^+(A) + \int |A - F^-|^p \, \mathrm{d}\mu_k^-(A) \Bigg] \\
&\quad \leq C_p \left[|r - \det F|^{p/3} + |r - \det F^\pm|^{p/3} \right] \\
&\quad \leq C_p |r - \det F|^{p/3},
\end{aligned} \tag{12.28}$$

where we have adjusted the constant C_p from line to line.

Step 4. By the bounds (12.22) and (12.28) in conjunction with Proposition 4.3 we obtain that there exists a subsequence (not relabeled) and a homogeneous gradient Young measure $\mu^F = (\mu^F)_x \in \mathbf{GY}(B_1; \mathbb{R}^{3\times 3})$ with $\mu_k \overset{*}{\rightharpoonup} \mu^F$. To see that μ^F is indeed generated by a sequence of gradients, one may employ a diagonal argument similar to Step 2 in the proof of Lemma 8.13.

That $[\mu^F] = F$ follows directly from the preceding construction. The second assertion in (12.13), namely $\operatorname{supp} \mu^F \subset S_R$, is a consequence of (12.23), (12.26) as well as testing the Young measure convergence with $A \mapsto \mathbb{1}_{\{\det A \neq r\}}(A)$. This characteristic function is lower semicontinuous, so we may obtain a lower Young measure limit representation by approximating h from below by continuous functions and then using the monotone convergence theorem (in a similar way to the proof of Proposition 4.7). The third assertion in (12.13), that is, $\int |A - F|^p \, \mathrm{d}\mu^F(A) \leq C \cdot R(F)^p$, is a consequence of (12.22) and (12.28) in conjunction with the lower Young measure limit representation for the integrand $A \mapsto |A - F|^p$. The weak* measurability assertion is the content of Problem 12.8. □

Recall the importance of the positivity of the Jacobian determinant in Chapter 6, in particular in Section 6.3. There, we worked in the Sobolev space $\mathrm{W}^{1,p}(\Omega; \mathbb{R}^3)$ with $p > 3$. In contrast, the following example shows that the Jacobian determinant constraint is "soft" in Sobolev spaces with integrability exponent below the dimension of the domain.

Example 12.20 Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain, $p \in (1, 3)$, and let $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega; \mathbb{R}^3)$ be arbitrary. Then, there exists a solution $u \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^3)$ to the following special case of the **prescribed Jacobian equation**:

$$\begin{cases} \det \nabla u \equiv 1 & \text{a.e. in } \Omega, \\ u|_{\partial\Omega} = g & \text{on } \partial\Omega. \end{cases}$$

Indeed, since the trace operator $u \mapsto u|_{\partial\Omega}$ is surjective from $\mathrm{W}^{1,p}(\Omega; \mathbb{R}^3)$ to $\mathrm{W}^{1-1/p,p}(\partial\Omega; \mathbb{R}^3)$ (see Theorem A.25), there exists a $w \in \mathrm{W}^{1,p}(\Omega; \mathbb{R}^3)$ such that $w|_{\partial\Omega} = g$. The statement then follows immediately from Theorem 12.16 applied to the elementary Young measure $\delta[\nabla w]$ (or just Lemma 12.18) together with Proposition 12.19. In particular, we could choose $\Omega = B_1$ and $g(x) := 2x$, showing that the *pointwise* incompressibility constraint $\det \nabla u \equiv 1$ (almost everywhere) is physically meaningless in this low-regularity situation. Intuitively, one might consider this example to be a Sobolev space version of the *Banach–Tarski paradox*.

Historically, observations like in the preceding example led to the introduction of the distributional determinant, which we already encountered in the proof of Lemma 6.7. More on these questions, which also relate to *cavitation* in elastic materials can be found in [177, 178, 201, 202, 249].

Notes and Historical Remarks

The basic idea of convex integration dates back to the famous Nash–Kuiper theorem in differential geometry [214,257]. These arguments were developed further until they culminated in Gromov's *h-principle*, which is detailed in his treatise [170] (also see [171]). There, the Lipschitz case, which is the one relevant to us, is also briefly mentioned; in particular, the term "in-approximation" is due to Gromov.

These ideas were transferred by Müller and Šverák [250, 252] to the multi-well inclusions that are of interest in material science. Simultaneously, Dacorogna and Marcellini [94, 95] adapted Cellina's Baire method [27, 71, 72] for ordinary differential inclusions to partial differential inclusions. Sychev [314, 317, 318] developed the convex integration method further to prove an existence theorem that only assumed the existence of what we call a "piecewise affine reduction". We remark that his original result, Theorem 1.1 in [317], is in fact a bit more general than our version.

The technical Lemma 12.5 is from [252], which itself is a variant of Theorem 2.4 in [140], but the result is already mentioned in Gromov's book [170]. While Theorem 12.7 as stated is due to Müller and Šverák [250], at the same time the Baire strategy by Dacorogna and Marcellini [94] also resulted in similar results.

Finally, Kirchheim's approach based on Baire-one functionals unified the different strands of development, see [193], and further related convex integration to the Banach–Mazur game. Our presentation mostly follows [193], but we also took some inspiration from [319]. We also remark that the term "convex integration" sometimes only designates the approach via an (RI-)in-approximation; Kirchheim speaks of "affine synthesis" in [193].

For convex integration in the inhomogeneous case, that is, where $K = K(x)$, we refer to [253]. Also the Young measure approach of [201, 202] can be extended to deal with inhomogeneous constraints. In this context we also mention [251], where additionally a uniform determinant constraint is respected in the convex integration procedure.

Further results on the *prescribed Jacobian equation* $\det \nabla u = g$ were obtained in [89, 96, 197], motivated by earlier work of Moser [243]. A strand of recent research activity centers on the question if one can also solve such equations with u a *homeomorphism*, see, e.g., [47, 111, 179, 185] for some recent developments.

One important and recurring theme in convex integration theory is that the regularity we require of solutions determines whether we can find solutions at all. The task is then to precisely identify the threshold between rigidity and non-rigidity. For instance, convex integration solutions with higher integrability for the two-well problem are in [115].

Convex integration has also been applied to the Euler equation to show the existence of very irregular solutions [109, 184, 295, 299].

Problems

12.1 Show that every finite-order laminate solution to the differential inclusion

$$\begin{cases} u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^2), \quad u|_{\partial\Omega} = Fx, \quad 0 < \alpha < 1 < \beta, \quad \alpha\beta \geq 1, \\ \nabla u \in K := \mathrm{SO}(2) \cup \mathrm{SO}(2) \begin{pmatrix} \alpha & \\ & \beta \end{pmatrix} \end{cases}$$

for $F \in K^{\mathrm{qc}} \setminus K$ is affine.

12.2 Prove that the open and bounded set $G \subset \mathbb{R}^{m\times d}$ can be piecewise-affinely reduced to a compact and non-empty set $K \subset \mathbb{R}^{m\times d}$ if for any bounded Lipschitz domain $\omega \subset \mathbb{R}^d$ for all $A \in G$ there exists a sequence $(\psi_j)_j \subset \mathrm{W}^{1,\infty}_0(\Omega;\mathbb{R}^m)$ of countably piecewise affine maps such that

(i) $A + \nabla\psi_j(x) \in G$ for almost every $x \in \Omega$ and all $j \in \mathbb{N}$;

(ii) $\displaystyle\int_\Omega \operatorname{dist}(A + \nabla\psi_j(x), K)\,\mathrm{d}x \to 0$ as $j \to \infty$.

12.3 Prove that the set $G^{\mathrm{lc},i}$ in the proof of Lemma 12.6 is open.

12.4 For an open set $G \subset \mathbb{R}^{m\times d}$ we define

$$G^{\mathrm{qc}} := \bigcup \left\{ K^{\mathrm{qc}} \,:\, K \subset G \text{ compact} \right\}.$$

Assume that $K \subset G^{\mathrm{qc}}$ is compact. Show that there exists a subset $G_0 \Subset G$ with $K \subset G_0^{\mathrm{qc}}$.

12.5 Use a convex integration procedure to show that there are continuous, bounded functions on the interval $[0,1]$ that are nowhere differentiable.

12.6 Show that if a compact set $K \subset \mathbb{R}^{m\times d}$ has an in-approximation consisting of open, uniformly bounded sets $G_k \subset \mathbb{R}^{m\times d}$ then K is piecewise-affinely stable with respect to $G_\infty := \bigcup_{l=1}^\infty G_l$.

12.7 Prove Proposition 12.19 for $d = 2$ and $d > 3$.

12.8 Show the measurability statement in Theorem 12.16 for the R from Proposition 12.19.

12.9 Show that for sets E with smooth boundary we have $\mathrm{Per}_\Omega(E) = \mathcal{H}^{d-1}(E \cap \Omega)$.

12.10 Show the following weaker version of the Dolzmann–Müller Theorem 12.8: Let $U \in \mathbb{R}^{d\times d}$ with $\det U > 0$ and such that with

$$K := \mathrm{SO}(d) \cup \mathrm{SO}(d)U$$

every matrix in K is rank-one connected to precisely two other matrices in K. Then, every $u \in \mathrm{W}^{1,\infty}(\Omega;\mathbb{R}^d)$ satisfying

$$\nabla u \in K \quad \text{a.e.}$$

with the property that

$$E := \left\{ x \in \Omega \,:\, \nabla u(x) \in \mathrm{SO}(d) \right\}$$

is a smooth manifold in Ω, is locally an order-one laminate.

Part III
Linear-Growth Theory

Chapter 13
Concentrations

All of the existence theorems for minimizers of integral functionals defined on a Sobolev space $\mathrm{W}^{1,p}(\Omega;\mathbb{R}^m)$ that we have seen so far required that $p > 1$. Extending the existence theory to the *linear-growth* case $p = 1$ turns out to be quite intricate and necessitates the development of new tools.

In order to illustrate this, consider the following minimization problem:

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega f(x, \nabla u(x))\,\mathrm{d}x \\ \text{over all} & u \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g. \end{cases} \tag{13.1}$$

Here, $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0,\infty)$ is a Carathéodory integrand (and $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, as usual). Moreover, $g \in \mathrm{L}^1(\partial\Omega;\mathbb{R}^m)$, which is the trace space for $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ (see Appendix A.5). Our hypothesis of *linear growth* on f means that there exists a constant $M > 0$ such that

$$f(x,A) \le M(1+|A|), \qquad (x,A) \in \Omega \times \mathbb{R}^{m\times d}.$$

If, for the moment, we also assume coercivity in the form

$$\mu|A| - \mu^{-1} \le f(x,A), \qquad (x,A) \in \Omega \times \mathbb{R}^{m\times d},$$

where $\mu > 0$, then a minimizing sequence $(u_j)_j \subset \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ satisfies

$$\limsup_{j\to\infty} \|u_j\|_{\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)} < \infty.$$

Here we also used the Poincaré inequality, see Theorem A.27 (iii), in conjunction with the fixed boundary values.

However, it is a well-known observation that in $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ a uniform norm-bound is not enough to deduce weak compactness, which is due to the non-reflexivity of $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$. For instance, take

$$u_j(x) := jx\mathbb{1}_{(0,1/j)}(x) + \mathbb{1}_{(1/j,1)}(x) \in \mathrm{W}^{1,1}(-1,1)$$

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F. Rindler, *Calculus of Variations*, Graduate Texts in Mathematics 310,
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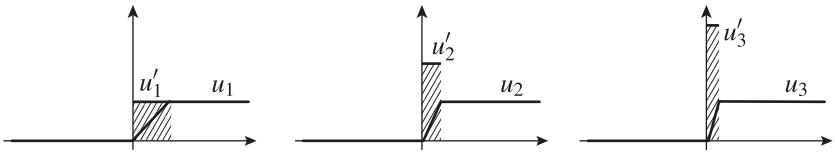


Fig. 13.1 A concentrating sequence

and observe that the u_j converge to $u = \mathbb{1}_{(0,1)} \notin \mathrm{W}^{1,1}(-1,1)$ pointwise and in L^1, see Figure 13.1. The decisive observation here is that the (weak) derivatives u_j' *concentrate*, meaning that the family $(u_j')_j$ is not equiintegrable. In fact, the measures $u_j' \mathcal{L}^d \llcorner (-1,1)$ converge weakly* to a measure that is not absolutely continuous with respect to Lebesgue measure, namely the Dirac mass δ_0.

We conclude that the space $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ is too small to serve as the set of candidate functions for our minimization problem (13.1). We need a space containing $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ such that a uniform norm-bound implies precompactness for a suitable (weak-type) convergence. Moreover, in order to retain the connection to our original problem, it is desirable that $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ is dense with respect to another notion of convergence that is strong enough to make the above functional $\mathcal{F}$ continuous. All of these requirements are fulfilled by the space $\mathrm{BV}(\Omega;\mathbb{R}^m)$ of *functions of bounded variation*.

However, before we can delve into the theory of integral functionals defined on BV-maps in the next chapter, and in particular consider how to extend (13.1) to this space, we first need to study *concentrations*, that is, the singularities that can occur in BV-derivatives and, more generally, the singular parts of PDE-constrained (vector) measures. With this aim, after introducing the so-called strict convergence of measures, we define tangent measures, which allow us to study the local structure of a given (vector) measure. Then we turn to the core of this chapter, namely to prove "rigidity" restrictions on the singular part of measures that satisfy a PDE-constraint. Finally, we present a fundamental result about convexity at singularities, which will become useful in Chapter 15, as well as a rigidity theorem in the linear-growth context.

13.1 Strict Convergence of Measures

In all of the following we will make constant use of the theory of vector measures, which is recalled in Appendix A.4.

Let $\Omega \subset \mathbb{R}^d$ be open (not necessarily bounded in this section). We say that a sequence of (vector) measures $(\mu_j)_j \subset \mathcal{M}(\Omega;\mathbb{R}^N)$ converges **strictly** to $\mu \in \mathcal{M}(\Omega;\mathbb{R}^N)$, written as "$\mu_j \to \mu$ strictly", if

$$\mu_j \overset{*}{\rightharpoonup} \mu \quad \text{in } \mathcal{M}(\Omega;\mathbb{R}^N) \qquad \text{and} \qquad |\mu_j|(\Omega) \to |\mu|(\Omega).$$

Lemma 13.1 *Let $\Omega \subset \mathbb{R}^d$ be open and let $\lambda_j \to \lambda$ strictly in $\mathcal{M}^+(\Omega)$. Then,*

$$\int g \, \mathrm{d}\lambda_j \to \int g \, \mathrm{d}\lambda \tag{13.2}$$

for all bounded functions $g \in \mathrm{C}(\Omega;\mathbb{R})$. The same conclusion holds if instead of strict convergence we only assume $\liminf_{j\to\infty} \lambda_j(U) \geq \lambda(U)$ for all open sets $U \subset \Omega$, and $\lambda_j(\Omega) \to \lambda(\Omega)$.

Proof. By considering $\alpha + \beta g$ for $\alpha, \beta \in \mathbb{R}$, we may without loss of generality assume that $g(x) \in [0,1]$ for all $x \in \Omega$. Let $E_t := \{ x \in \Omega : g(x) > t \}$ for $t \in [0,1]$ and observe via Fubini's theorem that

$$\int g \, \mathrm{d}\lambda = \int \int_0^1 \mathbb{1}_{E_t}(x) \, \mathrm{d}t \, \mathrm{d}\lambda(x) = \int_0^1 \lambda(E_t) \, \mathrm{d}t.$$

Since E_t is open, $\lambda_j \overset{*}{\rightharpoonup} \lambda$ implies (via Lemma A.20)

$$\liminf_{j\to\infty} \lambda_j(E_t) \geq \lambda(E_t) \qquad \text{for all } t \in [0,1].$$

Thus, by Fatou's lemma,

$$\liminf_{j\to\infty} \int g \, \mathrm{d}\lambda_j = \liminf_{j\to\infty} \int_0^1 \lambda_j(E_t) \, \mathrm{d}t \geq \int_0^1 \lambda(E_t) \, \mathrm{d}t = \int g \, \mathrm{d}\lambda.$$

Defining

$$a_j := \int g \, \mathrm{d}\lambda_j, \quad a := \int g \, \mathrm{d}\lambda, \quad b_j := \int 1 - g \, \mathrm{d}\lambda_j, \quad b := \int 1 - g \, \mathrm{d}\lambda,$$

the above arguments yield

$$\liminf_{j\to\infty} a_j \geq a, \qquad \liminf_{j\to\infty} b_j \geq b.$$

Moreover, since $\lambda_j(\Omega) \to \lambda(\Omega)$,

$$\lim_{j\to\infty} (a_j + b_j) = a + b.$$

Thus, it follows by elementary means that $a_j \to a$, that is, (13.2).

The additional statement is clear since we only used $\liminf_{j\to\infty} \lambda_j(U) \geq \lambda(U)$ for open sets $U \subset \Omega$ and $\lambda_j(\Omega) \to \lambda(\Omega)$. □

Corollary 13.2 *Let $\Omega \subset \mathbb{R}^d$ be open. If $\mu_j \to \mu$ strictly in $\mathcal{M}(\Omega;\mathbb{R}^N)$, then also $|\mu_j| \to |\mu|$ strictly in $\mathcal{M}^+(\Omega)$.*

Proof. It holds that $\liminf_{j\to\infty} |\mu_j|(U) \geq |\mu|(U)$ for all open sets $U \subset \Omega$ by the weak* convergence $\mu_j \overset{*}{\rightharpoonup} \mu$. This follows from the fact that the total variation for open sets can be written as the supremum of weakly*-continuous functions, see (A.4), whereby

it must be lower semicontinuous with respect to the weak* convergence. Since also $|\mu_j|(\Omega) \to |\mu|(\Omega)$ by assumption, we may apply the preceding lemma for $\lambda_j := |\mu_j|$, $\lambda := |\mu|$. Then, (13.2) for $g \in \mathrm{C}_0(\Omega)$ implies that indeed $|\mu_j| \overset{*}{\rightharpoonup} |\mu|$ in $\mathcal{M}^+(\Omega)$. □

The most important result about the strict convergence of measures is the following Reshetnyak continuity theorem. For a vector measure $\mu \in \mathcal{M}(\Omega;\mathbb{R}^N)$ denote by

$$\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|} \in \mathrm{L}^1(\Omega, |\mu|; \mathbb{R}^N)$$

the **polar** of μ, i.e., the Radon–Nikodým density of μ with respect to its own total variation measure $|\mu|$. This density is well-defined by the Besicovitch Differentiation Theorem A.24 and it holds that

$$\mu = \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}\,|\mu|, \qquad \left|\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}\right| = 1 \quad |\mu|\text{-a.e.},$$

the latter of which is the statement to be proved in Problem 13.1.

Theorem 13.3 (Reshetnyak 1967 [276]) *Let $\Omega \subset \mathbb{R}^d$ be open and let $g \in \mathrm{C}(\Omega \times \mathbb{S}^{N-1})$ be bounded. For any sequence $(\mu_j)_j \subset \mathcal{M}(\Omega;\mathbb{R}^N)$ with $\mu_j \to \mu$ strictly it holds that*

$$\int g\left(x, \frac{\mathrm{d}\mu_j}{\mathrm{d}|\mu_j|}(x)\right) \mathrm{d}|\mu_j|(x) \to \int g\left(x, \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)\right) \mathrm{d}|\mu|(x).$$

Proof. Define (in the spirit of Young measure theory, cf. the proof of Proposition 4.3)

$$\lambda_j := |\mu_j|(\mathrm{d}x) \otimes \delta_{\frac{\mathrm{d}\mu_j}{\mathrm{d}|\mu_j|}(x)} \in \mathcal{M}^+(\Omega \times \mathbb{S}^{N-1}),$$

that is,

$$\int_{\Omega\times\mathbb{S}^{N-1}} f(x,A)\,\mathrm{d}\lambda_j(x,A) = \int_\Omega f\left(x, \frac{\mathrm{d}\mu_j}{\mathrm{d}|\mu_j|}(x)\right) \mathrm{d}|\mu_j|(x)$$

for $f \in \mathrm{C}_0(\Omega \times \mathbb{S}^{N-1})$. We have

$$\sup_{j\in\mathbb{N}} |\lambda_j|(\Omega \times \mathbb{S}^{N-1}) = \sup_{j\in\mathbb{N}} |\mu_j|(\Omega) < \infty.$$

Thus, we may select a subsequence (not relabeled) with the property that

$$\lambda_j \overset{*}{\rightharpoonup} \lambda \quad \text{in } \mathcal{M}^+(\Omega \times \mathbb{S}^{N-1}).$$

Let $\pi\colon \Omega \times \mathbb{R}^N \to \mathbb{R}^d$ be the projection onto the first argument, $\pi(x,A) := x$. On the one hand, we get

$$\pi_\# \lambda_j \overset{*}{\rightharpoonup} \pi_\# \lambda,$$

where $\pi_\#\lambda := \lambda \circ \pi^{-1}$ is the push-forward of λ under π, see Appendix A.4. On the other

hand, Corollary 13.2 implies

$$\pi_\# \lambda_j = |\mu_j| \overset{*}{\rightharpoonup} |\mu|.$$

Hence, $\pi_\# \lambda = |\mu|$ and by the Disintegration Theorem 4.4 there exists a weakly* measurable family $(\nu_x)_{x\in\Omega} \subset \mathcal{M}^1(\mathbb{S}^{N-1})$ such that

$$\lambda = |\mu|(\mathrm{d}x) \otimes \nu_x.$$

Now we test the convergence $\lambda_j \overset{*}{\rightharpoonup} \lambda$ with functions of the form $(x, A) \mapsto \psi(x)A$, where $\psi \in \mathrm{C}_0(\Omega)$ (note that these functions lie in $\mathrm{C}_0(\Omega \times \mathbb{S}^{N-1})$) to see

$$\begin{aligned}
\int_\Omega \psi(x) \int A \,\mathrm{d}\nu_x(A)\,\mathrm{d}|\mu|(x) &= \int_{\Omega\times\mathbb{S}^{N-1}} \psi(x)A \,\mathrm{d}\lambda(x,A) \\
&= \lim_{j\to\infty} \int_{\Omega\times\mathbb{S}^{N-1}} \psi(x)A \,\mathrm{d}\lambda_j(x,A) \\
&= \lim_{j\to\infty} \int_\Omega \psi(x) \frac{\mathrm{d}\mu_j}{\mathrm{d}|\mu_j|}(x)\,\mathrm{d}|\mu_j|(x) \\
&= \lim_{j\to\infty} \int_\Omega \psi(x)\,\mathrm{d}\mu_j(x) \\
&= \int_\Omega \psi(x)\,\mathrm{d}\mu(x) \\
&= \int_\Omega \psi(x) \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)\,\mathrm{d}|\mu|(x).
\end{aligned}$$

Consequently, varying ψ, for the barycenter $[\nu_x]$ of ν_x we get

$$[\nu_x] = \int A \,\mathrm{d}\nu_x(A) = \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x) \qquad \text{for } |\mu|\text{-a.e. } x \in \Omega.$$

Then, at $|\mu|$-almost every $x \in \Omega$, it holds that

$$\begin{aligned}
&\frac{1}{2}\int_{\mathbb{S}^{N-1}} \left| A - \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)\right|^2 \mathrm{d}\nu_x(A) \\
&\quad = \frac{1}{2}\int_{\mathbb{S}^{N-1}} |A|^2 - 2\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)\cdot A + \left|\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)\right|^2 \mathrm{d}\nu_x(A) \\
&\quad = 1 - \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)\cdot[\nu_x] \\
&\quad = 0.
\end{aligned}$$

Hence,

$$\nu_x = \delta_{\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)} \qquad \text{for } |\mu|\text{-a.e. } x \in \Omega.$$

Finally, $\lambda_j \to \lambda$ strictly since

$$|\lambda_j|(\Omega \times \mathbb{S}^{N-1}) = |\mu_j|(\Omega) \to |\mu|(\Omega) = |\lambda|(\Omega \times \mathbb{S}^{N-1}).$$

Let g be as in the statement of the theorem. By Lemma 13.1 (technically applied in $\Omega \times U$ with U an open neighborhood of $\mathbb{S}^{N-1}$ and with g suitably extended) we thus get

$$\begin{aligned}\int g\left(x, \frac{\mathrm{d}\mu_j}{\mathrm{d}|\mu_j|}(x)\right) \mathrm{d}|\mu_j|(x) &= \int g(x, A)\ \mathrm{d}\lambda_j(x, A) \\ &\to \int g(x, A)\ \mathrm{d}\lambda(x, A) \\ &= \int g\left(x, \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x)\right) \mathrm{d}|\mu|(x)\end{aligned}$$

as $j \to \infty$. This yields the sought assertion since it identifies the limit for every subsequence. □

13.2 Tangent Measures

We now introduce a useful tool to study the local structure of measures. For $x_0 \in \mathbb{R}^d$ and $r > 0$ define the rescaling map

$$T^{(x_0,r)}(x) := \frac{x - x_0}{r}, \qquad x \in \mathbb{R}^d.$$

Let $\mu \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^N)$ be a vector-valued (local) Radon measure and let $x_0 \in \mathbb{R}^d$. A **tangent measure** to μ at x_0 is any (local) weak* limit in the space $\mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^N)$ of the rescaled measures

$$c_n T^{(x_0,r_n)}_{\#}\mu$$

for some sequence $r_n \downarrow 0$ of radii and rescaling constants $c_n > 0$. The definition of the push-forward $T^{(x_0,r)}_{\#}\mu$ (see Appendix A.4) here expands to

$$(T^{(x_0,r_n)}_{\#}\mu)(B) = \mu(x_0 + r_n B) \qquad \text{for any Borel set } B \subset \mathbb{R}^d.$$

The sequence $(c_n T^{(x_0,r_n)}_{\#}\mu)_n$ is called a **blow-up sequence** since it "magnifies" μ locally around x_0.

We collect all tangent measures to μ at a point x_0 in the set

$$\mathrm{Tan}(\mu, x_0) \subset \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^N).$$

Clearly, $0 \in \mathrm{Tan}(\mu, x_0)$ for all $x_0 \in \mathbb{R}^d$ and $\mathrm{Tan}(\mu, x_0) = \{0\}$ for all $x_0 \notin \operatorname{supp}\mu$. Moreover, for any $\alpha \geq 0$ and $\tau \in \mathrm{Tan}(\mu, x_0)$, also $\alpha\tau \in \mathrm{Tan}(\mu, x_0)$, so $\mathrm{Tan}(\mu, x_0)$ is a (one-sided) cone. We will see in Proposition 13.5 below that $\mathrm{Tan}(\mu, x_0)$ contains a non-zero measure for $|\mu|$-almost every $x_0 \in \mathbb{R}^d$.

For any non-zero $\tau \in \operatorname{Tan}(\mu, x_0)$ it turns out that, up to taking subsequences, we may always choose the rescaling constants c_n in the blow-up sequence $c_n T_\#^{(x_0,r_n)}\mu \overset{*}{\rightharpoonup} \tau$ as

$$\widetilde{c}_n := c\big[|\mu|(x_0 + r_n\overline{U})\big]^{-1} \tag{13.3}$$

for any (fixed) bounded open set $U \subset \mathbb{R}^d$ containing the origin such that $|\tau|(U) > 0$, and some constant $c > 0$ (which, of course, depends on τ and U). Indeed,

$$0 < |\tau|(U) \le \liminf_{n\to\infty} c_n|\mu|(x_0 + r_n U) \le \limsup_{n\to\infty} c_n|\mu|(x_0 + r_n\overline{U}) < \infty$$

by the (local) weak* convergence of $c_n T_\#^{(x_0,r_n)}|\mu|$. After the selection of a subsequence (not relabeled) we may assume $c_n|\mu|(x_0 + r_n\overline{U}) \to c$, and hence that $\widetilde{c}_n T_\#^{(x_0,r_n)}\mu \overset{*}{\rightharpoonup} \tau$.

The first fact we prove about tangent measures is a "polar decomposition" result:

Lemma 13.4 *Let $\mu \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$. At $|\mu|$-almost every point $x_0 \in \mathbb{R}^d$ and for all blow-up sequences $(c_n T_\#^{(x_0,r_n)}\mu)_n$, where $r_n \downarrow 0$, $c_n > 0$, it holds that*

$$\tau = \operatorname*{w*-lim}_{n\to\infty} c_n T_\#^{(x_0,r_n)}\mu \qquad \Longleftrightarrow \qquad |\tau| = \operatorname*{w*-lim}_{n\to\infty} c_n T_\#^{(x_0,r_n)}|\mu|.$$

In this case,

$$\tau = P_0\,|\tau| \qquad \textit{with} \qquad P_0 = \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x_0).$$

Thus,

$$\operatorname{Tan}(\mu, x_0) = \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x_0) \cdot \operatorname{Tan}(|\mu|, x_0)$$

for $|\mu|$-almost every $x_0 \in \mathbb{R}^d$.

Proof. Let $x_0 \in \mathbb{R}^d$ be a Lebesgue point of the polar $P := \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}$ of μ with respect to $|\mu|$, that is,

$$\fint_{B_{r_n R}(x_0)} |P(x) - P_0|\,\mathrm{d}|\mu|(x) \to 0 \qquad \text{as } n \to \infty.$$

By general measure theory results, $|\mu|$-almost every $x_0 \in \mathbb{R}^d$ has this property, see Theorem A.21.

Let $\varphi \in \mathrm{C}_c(\mathbb{R}^d)$. By (13.3) we may assume that $c_n = c[|\mu|(\overline{B_{r_n R}(x_0)})]^{-1}$ for some $c > 0$ and a large ball $B_R \ni \operatorname{supp}\varphi$. We have, using the transformation rule (A.5) for push-forwards of measures,

$$\begin{aligned}
&\int \varphi\,\mathrm{d}T_\#^{(x_0,r_n)}|\mu| - \int \varphi\, P_0 \cdot \mathrm{d}T_\#^{(x_0,r_n)}\mu \\
&\quad = \int \varphi(y)\big[1 - P_0 \cdot P(x_0 + r_n y)\big]\,\mathrm{d}T_\#^{(x_0,r_n)}|\mu|(y) \\
&\quad = \int \varphi\Big(\frac{x - x_0}{r_n}\Big)\big[1 - P_0 \cdot P(x)\big]\,\mathrm{d}|\mu|(x).
\end{aligned}$$

Since

$$|1 - P_0 \cdot P(x)| \le |P(x) - P_0|,$$

the Lebesgue point property of x_0 yields

$$\begin{aligned} c_n \left| \int \varphi \, \mathrm{d}T_{\#}^{(x_0,r_n)} |\mu| - \int \varphi \, P_0 \cdot \mathrm{d}T_{\#}^{(x_0,r_n)} \mu \right| \\ \le c \|\varphi\|_\infty \cdot \fint_{B_{r_n R}(x_0)} |P(x) - P_0| \, \mathrm{d}|\mu|(x) \\ \to 0 \end{aligned}$$

as $n \to \infty$. Therefore, if $c_n T_{\#}^{(x_0,r_n)} \mu$ converges weakly* to $\tau \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^N)$, then $c_n T_{\#}^{(x_0,r_n)} |\mu|$ converges weakly* to $P_0 \cdot \tau =: \sigma$. If we write $\tau = G|\tau|$ with the polar Radon–Nikodým density $G \in \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^d, |\tau|; \mathbb{S}^{N-1})$, then

$$|\tau| \le \operatorname*{w^*\text{-}lim}_{n\to\infty} c_n T_{\#}^{(x_0,r_n)} |\mu| = \sigma = P_0 \cdot \tau = (P_0 \cdot G)|\tau| \le |\tau|,$$

whereby $G = P_0$ almost everywhere with respect to $|\tau|$. Hence, $\sigma = |\tau|$.

On the other hand, if $c_n T_{\#}^{(x_0,r_n)} |\mu|$ converges weakly* to a measure $\sigma \in \mathcal{M}^+_{\mathrm{loc}}(\mathbb{R}^d)$, then we also get (selecting a subsequence if necessary) that

$$c_n T_{\#}^{(x_0,r_n)} \mu \overset{*}{\rightharpoonup} \tau \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^N).$$

Immediately, the previous argument applies and again yields $\sigma = |\tau|$. □

The following result shows that $|\mu|$-almost everywhere there exists a non-zero tangent measure.

Proposition 13.5 *Let $\mu \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^N)$. At $|\mu|$-almost every $x_0 \in \mathbb{R}^d$ there exists a non-zero tangent measure $\tau \in \mathrm{Tan}(\mu, x_0)$. Moreover, τ can be chosen to have polynomial growth of order $d + 2$, i.e., there exists a constant $C > 0$ (depending on τ) such that $|\tau|(B_r) \le C(1 + r^{d+2})$ for all $r > 0$.*

Proof. Restricting to a ball centered at the origin (tangent measures are local), we may assume without loss of generality that μ is finite. Furthermore, by the previous lemma we may suppose that μ is a positive measure.

Fix $\varepsilon > 0$ and set

$$\beta_k := \frac{2^d (k+1)^d k^2}{\varepsilon} \mu(\mathbb{R}^d) \qquad \text{for } k = 2, 3, \ldots$$

Define for $k = 2, 3, \ldots$ and $r > 0$ the set

$$A_{k,r} := \left\{ x \in \mathbb{R}^d \, : \, \mu(B_{kr}(x)) \ge \beta_k \, \mu(B_r(x)) \right\}.$$

Assume that for some ball $B_{r/2}(x) \subset \mathbb{R}^d$ it holds that $B_{r/2}(x) \cap A_{k,r} \neq \emptyset$. Then, take $z \in B_{r/2}(x) \cap A_{k,r}$ and estimate

$$\beta_k \mu(B_{r/2}(x)) \leq \beta_k \mu(B_r(z)) \leq \mu(B_{kr}(z)) \leq \mu(B_{(k+1)r}(x)).$$

By Fubini's theorem,

$$\begin{aligned}
\mu(B) &= \frac{1}{\omega_d r^d} \int \mathbb{1}_B(y) \int \mathbb{1}_{B_r(y)}(x) \, \mathrm{d}x \, \mathrm{d}\mu(y) \\
&= \frac{1}{\omega_d r^d} \int \mathbb{1}_B(y) \int \mathbb{1}_{B_r(x)}(y) \, \mathrm{d}x \, \mathrm{d}\mu(y) \\
&= \frac{1}{\omega_d r^d} \int \mu(B \cap B_r(x)) \, \mathrm{d}x
\end{aligned}$$

for any Borel set $B \subset \mathbb{R}^d$ (recall that $\omega_d := |B_1|$). Thus,

$$\begin{aligned}
\mu(A_{k,r}) &= \frac{1}{\omega_d (r/2)^d} \int \mu(A_{k,r} \cap B_{r/2}(x)) \, \mathrm{d}x \\
&\leq \frac{2^d (k+1)^d}{\beta_k} \cdot \frac{1}{\omega_d (k+1)^d r^d} \int \mu(B_{(k+1)r}(x)) \, \mathrm{d}x \\
&= \frac{2^d (k+1)^d}{\beta_k} \mu(\mathbb{R}^d) \\
&= \frac{\varepsilon}{k^2}.
\end{aligned}$$

Next, for

$$E_r := \left\{ x \in \mathbb{R}^d \ : \ \text{there exists a } k \in \{2, 3, \ldots\} \text{ such that } x \in A_{k,r} \right\}$$

we have

$$\mu(E_r) \leq \sum_{k=2}^{\infty} \frac{\varepsilon}{k^2} \leq \frac{\pi^2}{6} \varepsilon.$$

Defining

$$E := \bigcup_{n=1}^{\infty} \bigcap_{l=n}^{\infty} E_{1/l},$$

we see from the continuity of μ from below that

$$\mu(E) \leq \frac{\pi^2}{6} \varepsilon.$$

Let $x_0 \in \operatorname{supp} \mu \setminus E$. For all $n \in \mathbb{N}$ by definition there exists an $l_n \geq n$ such that $x_0 \in \mathbb{R}^d \setminus E_{1/l_n}$, that is,

$$\mu(B_{k/l_n}(x_0)) < \beta_k \mu(B_{1/l_n}(x_0)) \qquad \text{for all } k = 2, 3, \ldots.$$

Set r_n to be $1/l_n$ for the l_n so chosen at n. Then,

$$\limsup_{n\to\infty} \frac{\mu(B_{kr_n}(x_0))}{\mu(B_{r_n}(x_0))} \le \beta_k = \frac{2^d(k+1)^d k^2}{\varepsilon}\,\mu(\mathbb{R}^d)$$

for $k = 2, 3, \ldots$ Consequently, with $c_n := \mu(B_{r_n}(x_0))^{-1}$ we have

$$\limsup_{n\to\infty} (c_n T_\#^{(x_0,r_n)}\mu)(B_k) \le C_\varepsilon (k+1)^{d+2}$$

for some ε-dependent constant $C_\varepsilon > 0$. First, this shows that we may select a subsequence of the r_n's (not relabeled) such that it holds that

$$c_n T_\#^{(x_0,r_n)}\mu \overset{*}{\rightharpoonup} \tau \in \mathrm{Tan}(\mu, x_0) \qquad \text{in } \mathcal{M}^+_{\mathrm{loc}}(\mathbb{R}^d).$$

Second, τ is non-zero since

$$\tau(\overline{B_1}) \ge \limsup_{n\to\infty} (c_n T_\#^{(x_0,r_n)}\mu)(\overline{B_1}) \ge 1.$$

Finally,

$$\tau(B_k) \le \liminf_{n\to\infty} (c_n T_\#^{(x_0,r_n)}\mu)(B_k) \le C_\varepsilon (k+1)^{d+2}.$$

In conclusion, we have shown that τ is a non-zero tangent measure to μ at x_0 with polynomial growth.

The existence of a τ as above holds at μ-almost every $x_0 \in \mathbb{R}^d$ since we can make $\mu(E) \le C\varepsilon$ arbitrarily small, whereby the exceptional set must be μ-negligible. □

The following observation shows that we may even find blow-up sequences that converge *strictly*.

Lemma 13.6 *Let $\mu \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$. At $|\mu|$-almost every $x_0 \in \mathbb{R}^d$ and for every open, bounded, and convex set $C \subset \mathbb{R}^d$ with $0 \in C$ the following assertions hold:*

(i) There exists a tangent measure $\tau \in \mathrm{Tan}(\mu, x_0)$ with $|\tau|(C) = 1$ and $|\tau|(\partial C) = 0$.
(ii) There exists a blow-up sequence $\gamma_n := c_n T_\#^{(x_0,r_n)}\mu$ such that $\gamma_n \to \tau$ strictly in $\mathcal{M}(\overline{C};\mathbb{R}^N)$.

Proof. Choose $\eta > 0$, $K \in \mathbb{N}$ such that $B_1 \Subset \eta C \Subset B_K$. At $|\mu|$-almost every point $x_0 \in \operatorname{supp}\mu$ there exists a sequence $r_n \downarrow 0$ such that

$$\limsup_{n\to\infty} \frac{|\mu|(B_{Kr_n}(x_0))}{|\mu|(B_{r_n}(x_0))} \le \beta_K \tag{13.4}$$

for a constant $\beta_K > 0$; this is proved in Proposition 13.5.

At such an x_0, let $b_n := |\mu|(B_{r_n}(x_0))^{-1}$. Then we have from (13.4) that, up to selecting a subsequence,

$$b_n T_\#^{(x_0,r_n)}\mu \overset{*}{\rightharpoonup} \sigma \quad \text{in } \mathcal{M}(B_K;\mathbb{R}^N).$$

Hence,

$$|\sigma|(\overline{\eta C}) \geq \limsup_{n\to\infty} (b_n T_{\#}^{(x_0,r_n)}|\mu|)(\overline{\eta C}) \geq 1.$$

From Lemma 13.4 (we now additionally require that x_0 is chosen to make the conclusion of that result valid) we infer furthermore that

$$b_n T_{\#}^{(x_0,r_n)}|\mu| \overset{*}{\rightharpoonup} |\sigma|.$$

We can also assume $|\sigma|(\partial(\eta C)) = 0$ by increasing η slightly and without changing K since $|\sigma|$ is a finite measure (see Problem 13.2).

For the rescaled measure

$$\tau := \frac{1}{|\sigma|(\eta C)} T_{\#}^{(0,\eta)}\sigma = \frac{\sigma(\eta\,\bullet)}{|\sigma|(\eta C)}$$

it holds with $c_n := |\sigma|(\eta C)^{-1} b_n$ that

$$\begin{aligned} \gamma_n := c_n T_{\#}^{(x_0,\eta r_n)}\mu &\overset{*}{\rightharpoonup} \tau \qquad \text{in } \mathcal{M}(\overline{C};\mathbb{R}^N), \\ c_n T_{\#}^{(x_0,\eta r_n)}|\mu| &\overset{*}{\rightharpoonup} |\tau| \qquad \text{in } \mathcal{M}^+(\overline{C}). \end{aligned}$$

Since $|\tau|(\partial C) = |\sigma|(\eta C)^{-1}|\sigma|(\partial(\eta C)) = 0$, we get from standard results in measure theory (see Lemma A.23) that

$$|\gamma_n|(C) \to |\tau|(C) = 1.$$

Thus, $\tau \in \mathrm{Tan}(\mu, x_0)$ satisfies assertion (i) in the statement of the lemma and the blow-up sequence $\gamma_n = c_n T_{\#}^{(x_0,\eta r_n)}\mu$ satisfies assertion (ii). □

13.3 Functions of Bounded Variation

In this section we give an overview over some aspects of the theory of BV-maps. We refer to [18, 134, 219, 344] for a more thorough exposition and for proofs.

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain. The space $\mathrm{BV}(\Omega;\mathbb{R}^m)$ of functions of **bounded variation** is defined to contain all maps $u \in \mathrm{L}^1(\Omega;\mathbb{R}^m)$ such that there exists a matrix-valued Radon measure

$$Du = [Du]_j^i = [\partial_j u^i]_j^i \in \mathcal{M}(\Omega;\mathbb{R}^{m\times d}),$$

henceforth called the **(weak) derivative** of u, with the property that for all $i = 1,\dots,m$ and $j = 1,\dots,d$ the integration-by-parts formula

$$\int_\Omega u^i(x)\frac{\partial\psi}{\partial x_j}(x)\,\mathrm{d}x = -\int_\Omega \psi(x)\,\mathrm{d}[Du]_j^i(x), \qquad \psi \in \mathrm{C}_c^1(\Omega),$$

holds. One observes that $\mathrm{BV}(\Omega;\mathbb{R}^m)$ is a Banach space under the norm

$$\|u\|_{\mathrm{BV}(\Omega;\mathbb{R}^m)} := \|u\|_{\mathrm{L}^1(\Omega;\mathbb{R}^m)} + |Du|(\Omega),$$

but the norm topology turns out to be too strong for most purposes; for instance, smooth maps are not norm-dense in $\mathrm{BV}(\Omega;\mathbb{R}^m)$.

Therefore, one is led to consider the following notions of convergence: We say that a sequence $(u_j)_j \subset \mathrm{BV}(\Omega;\mathbb{R}^m)$ **converges weakly*** to $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$, denoted as "$u_j \overset{*}{\rightharpoonup} u$", if

(i) $u_j \to u$ in L^1 and

(ii) $Du_j \overset{*}{\rightharpoonup} Du$ (as measures).

If we add the mass-conservation hypothesis

(iii) $|Du_j|(\Omega) \to |Du|(\Omega)$,

then the u_j are said to converge to u **strictly**, written as "$u_j \to u$ strictly"; in fact, (ii) follows from (i) and (iii) by a compactness argument in conjunction with Corollary 13.2.

An even stronger notion of convergence is the **area-strict convergence**, where (iii) is replaced by

(iii') $\langle Du_j\rangle(\Omega) \to \langle Du\rangle(\Omega)$.

Here, for $\mu \in \mathcal{M}(\Omega,\mathbb{R}^N)$ with Lebesgue–Radon–Nikodým decomposition

$$\mu = \frac{\mathrm{d}\mu}{\mathrm{d}\mathcal{L}^d}\,\mathcal{L}^d \llcorner \Omega + \mu^s, \qquad \mu^s \text{ singular with respect to } \mathcal{L}^d,$$

the quantity $\langle\mu\rangle(\Omega)$ is the **(reduced) area-functional**, which is defined via

$$\langle\mu\rangle(B) := \int_B \sqrt{1 + \left|\tfrac{\mathrm{d}\mu}{\mathrm{d}\mathcal{L}^d}\right|^2}\,\mathrm{d}x + |\mu^s|(B) \tag{13.5}$$

for any Borel set $B \subset \Omega$. It can be shown that area-strict convergence implies strict convergence, see Problem 13.5. Smooth functions are area-strictly dense (hence also strictly dense) in $\mathrm{BV}(\Omega;\mathbb{R}^m)$, as will be proved in Lemma 13.7 below.

The **compactness theorem** for weak* convergence in $\mathrm{BV}(\Omega;\mathbb{R}^m)$ says that a sequence $(u_j)_j \subset \mathrm{BV}(\Omega;\mathbb{R}^m)$ with $\sup_j \|u_j\|_{\mathrm{BV}} < \infty$ has a weakly* converging subsequence.

It holds that if $\mathcal{H}^{d-1}(S) = 0$, then also $Du(S) = 0$. Furthermore, the derivative Du of $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ has the Lebesgue–Radon–Nikodým decomposition

$$Du = D^a u + D^s u, \qquad D^a u = \nabla u\,\mathcal{L}^d \llcorner \Omega,$$

where the Lebesgue density $\nabla u \in \mathrm{L}^1(\Omega;\mathbb{R}^{m\times d})$ is called the **approximate gradient** of u and the measure $D^s u \in \mathcal{M}(\Omega;\mathbb{R}^{m\times d})$ is the **singular part** of Du. The latter further splits as

$$D^s u = D^j u + D^c u,$$

where $D^j u$ is the **jump part** and $D^c u$ is the **Cantor part**.

The jump part is of the form

$$D^j u = (u^+ - u^-) \otimes n_{J_u} \, \mathcal{H}^{d-1} \llcorner J_u,$$

where $J_u \subset \Omega$ is the $\mathcal{H}^{d-1}$-rectifiable **jump set**, n_{J_u} is a normal on J_u, and $u^\pm$ are the one-sided traces of u on J_u in positive and negative n_{J_u}-direction, respectively; we refer to [18] for the definition of rectifiability (whose precise form is not absolutely important for our purposes here). In fact, J_u is the $\mathcal{H}^{d-1}$-rectifiable (Borel) set of points $x_0 \in \Omega$, where both of the two one-sided traces $u^\pm(x_0)$, defined via

$$\lim_{r \downarrow 0} \frac{1}{r^d} \int_{\{\, x \in B_r(x_0) \,:\, x \cdot n_{J_u} \gtrless 0 \,\}} |u(y) - u^\pm(x_0)| \, \mathrm{d}y = 0,$$

exist, but differ.

For every $u \in \mathrm{BV}(\Omega; \mathbb{R}^m)$ there is an $\mathcal{L}^d$-negligible Borel set $S_u \subset \Omega$ such that every $x_0 \in \Omega \setminus S_u$ is an **approximate continuity point** of u, that is, there exists a $\widetilde{u}(x_0) \in \mathbb{R}^m$ such that

$$\lim_{r \downarrow 0} \frac{1}{r^d} \int_{B_r(x_0)} |u(y) - \widetilde{u}(x_0)| \, \mathrm{d}y = 0.$$

Consequently, S_u is called the **approximate discontinuity set** of u, and the Borel-measurable map $\widetilde{u} \colon \Omega \setminus S_u \to \mathbb{R}^m$ is called the **precise representative** of u. It can be shown that $\mathcal{H}^{d-1}(S_u \setminus J_u) = 0$.

The Lebesgue density ∇u of the derivative Du can also be interpreted pointwise: The Calderón–Zygmund theorem entails that at $\mathcal{L}^d$-almost every approximate continuity point $x_0 \in \Omega \setminus S_u$ the map u is **approximately differentiable**, that is,

$$\lim_{r \downarrow 0} \fint_{B_r(x_0)} \frac{|u(x) - \widetilde{u}(x_0) - \nabla u(x_0)(x - x_0)|}{r} \, \mathrm{d}x = 0.$$

From the Besicovitch Differentiation Theorem A.24 it further follows that

$$\nabla u(x_0) = \lim_{r \downarrow 0} \frac{Du(B_r(x_0))}{\omega_d r^d} = \lim_{r \downarrow 0} \frac{D^a u(B_r(x_0))}{\omega_d r^d}$$

at $\mathcal{L}^d$-almost every $x_0 \in \Omega$. Denote by D_u the **approximate differentiability set**, i.e., the collection of all approximate differentiability points $x_0 \in \Omega$ such that $\nabla u(x_0)$ satisfies the two conditions above. At all $x_0 \in D_u$,

$$\lim_{r \downarrow 0} \frac{|D^s u|(B_r(x_0))}{r^d} = 0. \tag{13.6}$$

On the other hand, $|D^s u|$-almost every $x_0 \in \Omega$ satisfies

$$\lim_{r \downarrow 0} \frac{|Du|(B_r(x_0))}{r^d} = \infty, \tag{13.7}$$

again by the Besicovitch Differentiation Theorem A.24.

The **trace** $u|_{\partial\Omega}\colon \partial\Omega \to \mathbb{R}^m$ of $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ on the boundary of Ω is defined for $\mathcal{H}^{d-1}$-almost every $x_0 \in \partial\Omega$ via

$$\lim_{r\downarrow 0} \frac{1}{r^d} \int_{B_r(x_0)\cap\Omega} \left|u(y) - u|_{\partial\Omega}(x_0)\right| \mathrm{d}x = 0.$$

It can be shown that $u|_{\partial\Omega} \in \mathrm{L}^1(\partial\Omega;\mathbb{R}^m)$.

We define the one-sided traces on Lipschitz subdomains $D \Subset \Omega$ in the same manner as the boundary traces. The inner and outer one-sided trace disagree on a non-$\mathcal{H}^{d-1}$-negligible subset of ∂D precisely if $|Du|(\partial D) > 0$. It is easy to see that the trace operator $u \mapsto u|_{\partial\Omega}$ is not continuous with respect to weak* convergence in $\mathrm{BV}(\Omega;\mathbb{R}^m)$, but it turns out to be continuous with respect to the strict convergence and hence also with respect to the area-strict convergence.

We also remark that any bounded Lipschitz domain $\Omega \subset \mathbb{R}^d$ is a BV-extension domain in the sense that for every $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ there exists a $\bar{u} \in \mathrm{BV}(\mathbb{R}^d;\mathbb{R}^m)$ with compact support such that $u = \bar{u}$ almost everywhere in Ω and $|D\bar{u}|(\partial\Omega) = 0$.

In BV, variants of the usual **Poincaré inequalities** hold: For all $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$,

$$\|u\|_{\mathrm{BV}} \le C\left(|Du|(\Omega) + \int_{\partial\Omega} |u| \,\mathrm{d}\mathcal{H}^{d-1}\right),$$
$$\|u - [u]_\Omega\|_{\mathrm{BV}} \le C|Du|(\Omega) \qquad \text{with} \qquad [u]_\Omega := \fint_\Omega u \,\mathrm{d}x,$$

where u in the boundary integral is to be understood in the sense of trace. Here, $C = C(\Omega, p) > 0$ is a constant.

We will employ the following "gluing" procedure many times in the sequel: Given a Lipschitz subdomain $D \Subset \Omega$ as well as $u \in \mathrm{BV}(D;\mathbb{R}^m)$, $v \in \mathrm{BV}(\Omega \setminus \overline{D};\mathbb{R}^m)$, the "glued" map $w := u\mathbb{1}_D + v\mathbb{1}_{\Omega\setminus\overline{D}}$ lies in $\mathrm{BV}(\Omega;\mathbb{R}^m)$ and

$$Dw = Du\llcorner D + Dv\llcorner(\Omega \setminus \overline{D}) + (u|_{\partial D} - v|_{\partial(\mathbb{R}^d\setminus\overline{D})}) \otimes n_D \,\mathcal{H}^{d-1}\llcorner\partial D,$$

where n_D is the measure-theoretic unit inner normal to ∂D, that is,

$$n_D := \frac{\mathrm{d}D\mathbb{1}_D}{\mathrm{d}|D\mathbb{1}_D|}.$$

For $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$, extended to all of $\mathbb{R}^d$, in the following we will often consider the push-forward maps $T^{(x_0,r)}_{\#}u$ under the rescaling $T^{(x_0,r)}(x) := (x - x_0)/r$ (where $x_0 \in \Omega$ and $r > 0$), that is,

$$(T^{(x_0,r)}_{\#}u)(y) := u(x_0 + ry), \qquad y \in \mathbb{R}^d.$$

Then,

$$D(T^{(x_0,r)}_{\#}u)(B) = \frac{(T^{(x_0,r)}_{\#}Du)(B)}{r^{d-1}} = \frac{Du(x_0 + rB)}{r^{d-1}} \tag{13.8}$$

for any Borel set $B \subset \Omega$, see Problem 13.3.

13.4 Approximation of BV-Maps

In preparation for our investigation of linear-growth integral functionals, we first need to consider how to approximate BV-maps with smooth ones in a suitable sense.

Lemma 13.7 *Let $\Omega \subset \mathbb{R}^d$ be a bounded open set. For each $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ there exists a sequence $(v_j)_j \subset (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^m)$ with $v_j|_{\partial\Omega} = u|_{\partial\Omega}$ and $v_j \to u$ area-strictly. If $u \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$, we may additionally require that $v_j \to u$ in $\mathrm{W}^{1,1}$.*

Note that for this lemma we do not assume any boundary regularity on Ω. See Problem 13.4 for a simpler proof in the case when Ω is a Lipschitz domain.

Proof. Let $\varepsilon > 0$ and take a collection of open sets $\Omega_i \subset \Omega$, where $i \in \mathbb{N} \cup \{0\}$, with

$$\Omega_i \Subset \Omega_{i+1} \qquad \text{and} \qquad \bigcup_{i=0}^{\infty} \Omega_i = \Omega.$$

Furthermore, we assume

$$|Du|(\Omega \setminus \Omega_0) < \varepsilon.$$

Let $U_0 := \Omega_1$ and set

$$U_i := \Omega_{i+1} \setminus \overline{\Omega_{i-1}} \qquad \text{for } i \in \mathbb{N}.$$

Then, the U_i, $i \in \mathbb{N} \cup \{0\}$, form an open cover of Ω such that every point of Ω lies in at most two of the sets. Let $(\varrho_i)_{i\in\mathbb{N}\cup\{0\}} \subset \mathrm{C}^\infty(\Omega;[0,1])$ be a smooth partition of unity subordinate to the cover $(U_i)_i$, so that $\varrho_i \in \mathrm{C}_c^\infty(U_i;[0,1])$ and $\sum_{i=0}^\infty \varrho_i = \mathbb{1}_\Omega$. The construction of such partitions of unity is detailed, for instance, in Lemma 2.3.1 of [344].

Now take a family of mollifiers $(\eta_\delta)_{\delta>0} \in \mathrm{C}_c^\infty(B_1)$. Then, for each $i \in \mathbb{N} \cup \{0\}$ choose $\delta_i > 0$ such that

$$\operatorname{supp}\big(\eta_{\delta_i} \star (\varrho_i u)\big) \subset U_i, \tag{13.9}$$

$$\int_\Omega |\eta_{\delta_i} \star (\varrho_i u) - \varrho_i u| \,\mathrm{d}x < \frac{\varepsilon}{2^{i+1}}, \tag{13.10}$$

$$\int_\Omega |\eta_{\delta_i} \star (u \otimes \nabla\varrho_i) - u \otimes \nabla\varrho_i| \,\mathrm{d}x < \frac{\varepsilon}{2^{i+1}}, \tag{13.11}$$

and

$$\int_\Omega \varrho_i \big[\eta_{\delta_i} \star \langle \varrho_i Du\rangle\big] \,\mathrm{d}x \le \int_\Omega \varrho_i \,\mathrm{d}\langle \varrho_i Du\rangle + \varepsilon \fint_\Omega \varrho_i \,\mathrm{d}x, \tag{13.12}$$

where $\varrho_i u$ and $u \otimes \nabla\varrho_i$ are understood to be extended by zero to all of $\mathbb{R}^d$. Let

$$v_\varepsilon := \sum_{i=0}^{\infty} \eta_{\delta_i} \star (\varrho_i u),$$

which lies in $C^\infty(\Omega)$ and, since $u = \sum_{i=0}^\infty \varrho_i u$, we get from (13.10) that

$$\int_\Omega |v_\varepsilon - u| \,\mathrm{d}x \le \sum_{i=0}^\infty \int_\Omega |\varrho_i u - \eta_{\delta_i} \star (\varrho_i u)| \,\mathrm{d}x < \varepsilon. \tag{13.13}$$

By the product rule,

$$\nabla v_\varepsilon = \sum_{i=0}^\infty \eta_{\delta_i} \star (\varrho_i Du) + \sum_{i=0}^\infty \big[\eta_{\delta_i} \star (u \otimes \nabla \varrho_i) - u \otimes \nabla \varrho_i\big]$$

since $\sum_{i=0}^\infty \nabla \varrho_i = 0$. For $x \in U_i$ at most two terms in each sum are non-zero, so that

$$\nabla v_\varepsilon = \eta_{\delta_i} \star (\varrho_i Du) + E_i \qquad \text{on } U_i,$$

where the $E_i \colon U_i \to \mathbb{R}^{m\times d}$ are error terms, whose sum we may estimate using (13.11) as follows:

$$\sum_{i=0}^\infty \int_{U_i} |E_i| \,\mathrm{d}x \le 2|Du|(\Omega \setminus \overline{\Omega_0}) + \varepsilon < 3\varepsilon. \tag{13.14}$$

Next, the elementary estimate $\langle A + B\rangle \le \langle A\rangle + |B|$ for all $A, B \in \mathbb{R}^{m\times d}$ implies

$$\int_\Omega \varrho_i \langle \nabla v_\varepsilon\rangle \,\mathrm{d}x \le \int_\Omega \varrho_i \langle \eta_{\delta_i} \star (\varrho_i Du)\rangle \,\mathrm{d}x + \int_{U_i} |E_i| \,\mathrm{d}x.$$

Also, by Jensen's inequality and (13.12),

$$\begin{aligned}\int_\Omega \varrho_i \langle \eta_{\delta_i} \star (\varrho_i Du)\rangle \,\mathrm{d}x &\le \int_\Omega \varrho_i \big[\eta_{\delta_i} \star \langle \varrho_i Du\rangle\big] \,\mathrm{d}x \\ &\le \int_\Omega \varrho_i \,\mathrm{d}\langle \varrho_i Du\rangle + \varepsilon \fint_\Omega \varrho_i \,\mathrm{d}x \\ &\le \int_\Omega \varrho_i \,\mathrm{d}\langle Du\rangle + \varepsilon \fint_\Omega \varrho_i \,\mathrm{d}x.\end{aligned}$$

Combining the last two estimates, summing over $i \in \mathbb{N} \cup \{0\}$, and employing (13.14), we arrive at

$$\langle Dv_\varepsilon\rangle(\Omega) < \langle Du\rangle(\Omega) + 4\varepsilon.$$

Since the (reduced) area functional $\langle \bullet\rangle(\Omega)$ is lower semicontinuous with respect to the weak* convergence of measures (see Problem 13.5 (ii)),

$$\langle Du\rangle(\Omega) \le \liminf_{\varepsilon\downarrow 0} \langle Dv_\varepsilon\rangle(\Omega) \le \limsup_{\varepsilon\downarrow 0} \langle Dv_\varepsilon\rangle(\Omega) \le \langle Du\rangle(\Omega).$$

Hence, $\langle Dv_\varepsilon\rangle(\Omega) \to \langle Du\rangle(\Omega)$ as $\varepsilon \downarrow 0$. As we have already established that $\|u - v_\varepsilon\|_{L^1} < \varepsilon$, see (13.13), this shows the area-strict convergence of v_ε to u.

If additionally $u \in W^{1,1}(\Omega;\mathbb{R}^m)$ is assumed, then in the preceding argument we choose δ_i so that in addition to (13.9)–(13.12) also

$$\int_{U_i} \left|\nabla(\varrho_i u) - \nabla(\eta_{\delta_i} \star (\varrho_i u))\right| \,\mathrm{d}x < \frac{\varepsilon}{2^{i+1}}$$

holds for each $i \in \mathbb{N} \cup \{0\}$. Hence,

$$\int_\Omega |\nabla u - \nabla v_\varepsilon| \,\mathrm{d}x < \varepsilon$$

and the strong convergence $v_\varepsilon \to u$ in $\mathrm{W}^{1,1}$ follows.

It remains to show that $v_\varepsilon \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ with $v_\varepsilon|_{\partial\Omega} = u|_{\partial\Omega}$, which will complete the proof. By construction, $v_\varepsilon \in \mathrm{W}^{1,1}(D;\mathbb{R}^m)$ for all Lipschitz subdomains $D \Subset \Omega$. We will prove that

$$w := \begin{cases} v_\varepsilon - u & \text{on } \Omega, \\ 0 & \text{on } \mathbb{R}^d \setminus \overline{\Omega}, \end{cases}$$

lies in $\mathrm{BV}(\mathbb{R}^d;\mathbb{R}^m)$ and that $|Dw|(\partial\Omega) = 0$, which implies the claim.

Take any $\psi \in \mathrm{C}_c^\infty(\mathbb{R}^d)$. From (13.9) we derive for $j = 1,\dots,d$ and $k = 1,\dots,m$ that

$$\begin{aligned}
\int_{\mathbb{R}^d} w^k \frac{\partial\psi}{\partial x_j} \,\mathrm{d}x &= \int_\Omega [v_\varepsilon - u]^k \frac{\partial\psi}{\partial x_j} \,\mathrm{d}x \\
&= \sum_{i=0}^\infty \int_\Omega \left[\eta_{\delta_i} \star (\varrho_i u) - \varrho_i u\right]^k \frac{\partial\psi}{\partial x_j} \,\mathrm{d}x \\
&= \sum_{i=0}^\infty \left[\int_{U_i} \psi\varrho_i \,\mathrm{d}[Du]_j^k - \int_{U_i} \psi \left[\eta_{\delta_i} \star (\varrho_i [Du]_j^k)\right] \mathrm{d}x \right] \\
&\qquad + \sum_{i=0}^\infty \int_{U_i} \psi \left[u^k \frac{\partial\varrho_i}{\partial x_j} - \eta_{\delta_i} \star \left(u^k \frac{\partial\varrho_i}{\partial x_j} \right) \right] \mathrm{d}x. \qquad (13.15)
\end{aligned}$$

Combining this with (13.11), we arrive at

$$\left| \int_{\mathbb{R}^d} w^k \frac{\partial\psi}{\partial x_j} \,\mathrm{d}x \right| \le C\big(|Du|(\Omega) + 1\big)\|\psi\|_\infty.$$

Thus, $w \in \mathrm{BV}(\mathbb{R}^d;\mathbb{R}^m)$.

Next, assume that ψ is supported near $\partial\Omega$, say $\psi \equiv 0$ on U_i for all $i < i_0$. Then, by (13.15) we get

$$\begin{aligned}
\int_{\mathbb{R}^d} w^k \frac{\partial\psi}{\partial x_j} \,\mathrm{d}x &= \sum_{i=i_0}^\infty \left[\int_{U_i} \psi\varrho_i \,\mathrm{d}[Du]_j^k - \int_{U_i} \psi \left[\eta_{\delta_i} \star (\varrho_i [Du]_j^k)\right] \mathrm{d}x \right] \\
&\qquad + \sum_{i=i_0}^\infty \int_{U_i} \psi \left[u^k \frac{\partial\varrho_i}{\partial x_j} - \eta_{\delta_i} \star \left(u^k \frac{\partial\varrho_i}{\partial x_j} \right) \right] \mathrm{d}x,
\end{aligned}$$

and so, letting $i_0 \to \infty$, we conclude that $|Dw|(\partial\Omega) = 0$. □

13.5 Structure of Singularities

Recall the decomposition of the derivative Du of $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$, namely

$$Du = \nabla u\, \mathcal{L}^d \llcorner \Omega + D^j u + D^c u.$$

Here,

$$D^j u = (u^+ - u^-) \otimes n\, \mathcal{H}^{d-1} \llcorner J_u$$

is the jump part of Du, as specified in the previous section.

For the jump part the local structure is clear. In particular, the above expression for $D^j u$ together with the elementary fact that a matrix $A \in \mathbb{R}^{m\times d}$ has rank at most one if and only if $A = a \otimes n$ for some $a \in \mathbb{R}^m$, $n \in \mathbb{R}^d$, immediately implies that

$$\operatorname{rank} \frac{\mathrm{d}D^j u}{\mathrm{d}|D^j u|}(x) = 1 \qquad \text{for } |D^j u|\text{-a.e. } x \in \Omega.$$

Intuitively, this means that locally around $x_0 \in J_u$ the map u is almost *one-directional* (locally, the set J_u is "almost straight").

A natural but deep question is whether the same property also holds for the Cantor part. This was conjectured by Ambrosio–De Giorgi [17] and proved first by Giovanni Alberti. The result is now usually called Alberti's rank-one theorem.

Theorem 13.8 (Alberti 1993 [4]) *Let $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$. Then,*

$$\operatorname{rank} \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x) = 1 \qquad \textit{for } |D^s u|\textit{-a.e. } x \in \Omega.$$

As a consequence of this fundamental result, we get that even at points $x_0 \in \Omega$ around which $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ has a Cantor-type (e.g., fractal) structure, the "slope" of u has a well-defined *direction*, given by the vector $n(x_0) \in \mathbb{S}^{d-1}$ in

$$\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x_0) = a(x_0) \otimes n(x_0),$$

where $a(x_0) \in \mathbb{R}^{m\times d}\setminus\{0\}$. This is made precise in the following important consequence of Alberti's theorem.

Corollary 13.9 *Let $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$. Then, at $|D^s u|$-almost every $x_0 \in \Omega$ every tangent measure $\tau \in \mathrm{Tan}(D^s u, x_0)$ is one-directional in the sense that there exists a direction $n \in \mathbb{S}^{d-1}$ such that*

$$\tau(B + v) = \tau(B)$$

for all bounded Borel sets $B \subset \mathbb{R}^d$ and all $v \in \mathbb{R}^d$ orthogonal to n.

This corollary is illustrated in Figure 13.2.

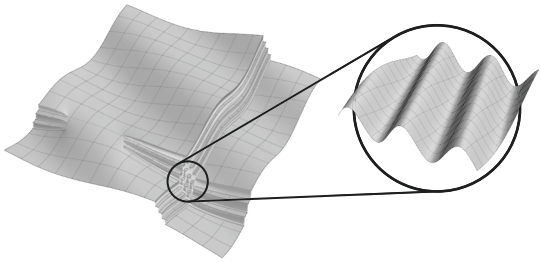


Fig. 13.2 The blow-up of a BV-map at a singular point

Proof. By Lemma 13.4 and Alberti's Rank-One Theorem 13.8 we know that at $|D^s u|$-almost every $x_0 \in \Omega$,

$$\tau = P_0 \, |\tau| \qquad \text{with} \qquad \operatorname{rank} P_0 = 1.$$

We write $P_0 = a \otimes n$ for some $a \in \mathbb{R}^m \setminus \{0\}$ and $n \in \mathbb{S}^{d-1}$. If $\gamma_n := c_n T_{\#}^{(x_0, r_n)} D^s u$ is a blow-up sequence for τ, then set

$$v_n(y) := r_n^{d-1} c_n u(x_0 + r_n y) + d_n, \qquad y \in \mathbb{R}^d,$$

where the $d_n \in \mathbb{R}^m$ are chosen such that $[v_n] = \fint_{B_1} v_n \, \mathrm{d}x = 0$. Then,

$$Dv_n = \gamma_n \overset{*}{\rightharpoonup} \tau \quad \text{in } \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m \times d})$$

and thus, by the Poincaré inequality in BV (see Section 13.3), $v_n \overset{*}{\rightharpoonup} w$ in $\mathrm{BV}_{\mathrm{loc}}$ for some $w \in \mathrm{BV}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^m)$, for which $Dw = \tau = P_0 \, |\tau|$.

A mollification argument shows that we may assume that $Dw = \nabla w \mathcal{L}^d$ with a smooth map $w \in \mathrm{C}^\infty(\mathbb{R}^d; \mathbb{R}^m)$ such that

$$\nabla w(x) = P_0 \, |\nabla w(x)| \qquad \text{for a.e. } x \in \mathbb{R}^d.$$

For any $v \in \mathbb{R}^d$ that is orthogonal to n we have

$$\frac{\mathrm{d}}{\mathrm{d}t} w(x + tv)\Big|_{t=0} = \nabla w(x) v = a n^{\mathsf{T}} v \, |\nabla w(x)| = 0.$$

This implies that w is constant in direction v. As v was an arbitrary orthogonal vector to n, $w(x)$ can only depend on $x \cdot n$. This directly yields the assertion of the lemma. □

One important application of Alberti's theorem is that it allows one to exclude certain types of singularities in BV-maps, as is illustrated in the following example.

Example 13.10 Let $S \subset \mathbb{R}^2$ be the Sierpiński triangle in the plane, which is constructed by repeatedly removing (inverted) maximal equilateral (solid) triangles from a given equilateral (solid) triangle, see Figure 13.3. Denote the sets in this construction by $S_0, S_1, \ldots$, whereby $S = \bigcap_{n=0}^{\infty} S_n$. It is shown, for instance, in Example 9.4 of [135] that S has Hausdorff-dimension $\log 3 / \log 2 \approx 1.58$ and that $0 < \mathcal{H}^{\log 3/\log 2}(S) < \infty$. It

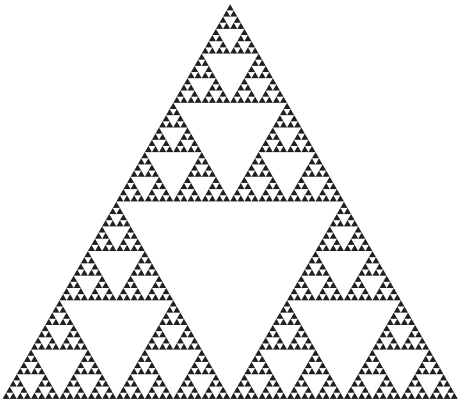


Fig. 13.3 The Sierpiński triangle

is also easy to see that S is self-similar,

$$S = \bigcup_{n=1}^{3} \psi_n(S),$$

where the $\psi_1, \psi_2, \psi_3 \colon \mathbb{R}^2 \to \mathbb{R}^2$ are the homotheties of rate $1/2$ that keep one of the three vertices of S_0 fixed. Call S_∞ the result of covering all of $\mathbb{R}^2$ with copies of S. Define

$$\mu_S := \mathcal{H}^{\log 3/\log 2} \llcorner S_\infty$$

and observe that μ_S is not one-directional in the sense of Corollary 13.9: For the blow-up sequence $\gamma_n := c_n T_\#^{(x_0, r_n)} \mu_S$ with $x_0 \in S_\infty$, $r_n = 2^{-n}$, and $c_n > 0$ chosen suitably, we have that the γ_n are just translates of μ_S with uniformly bounded translation distances (because of the self-similarity). In particular, they are locally uniformly bounded in the total variation norm and so a subsequence converges weakly* to a tangent measure $\tau \in \operatorname{Tan}(\mu_S, x_0)$. However, this τ can only be a translation of μ_S and so τ is not one-directional. Thus, from Corollary 13.9 we infer that there is no map $u \in \mathrm{BV}_{\mathrm{loc}}(\mathbb{R}^2; \mathbb{R}^m)$ for any $m \in \mathbb{N}$ with $|D^s u| = \mu_S$. A straightforward extension of this argument shows that also locally no Sierpiński triangle can be found in $|D^s u|$.

Here, we consider Alberti's rank-one theorem to be the consequence of a more general result about singularities in measures satisfying a linear PDE-constraint: Let $\Omega \subset \mathbb{R}^d$ be open (not necessarily bounded) and consider $\mu \in \mathcal{M}_{\mathrm{loc}}(\Omega; \mathbb{R}^N)$ that satisfies the k'th-order linear constant-coefficient PDE

$$\mathcal{A}\mu := \sum_{|\alpha| \le k} A_\alpha \partial^\alpha \mu = \sigma \qquad \text{in } \mathrm{C}_c^\infty(\Omega; \mathbb{R}^M)^*, \tag{13.16}$$

where $A_\alpha \in \mathbb{R}^{M \times N}$, $\partial^\alpha = \partial_1^{\alpha_1} \cdots \partial_d^{\alpha_d}$ for each multi-index $\alpha = (\alpha_1, \ldots, \alpha_d) \in (\mathbb{N} \cup \{0\})^d$ with length $|\alpha| := \alpha_1 + \cdots + \alpha_d \le k$ ($k \in \mathbb{N}$), and $\sigma \in \mathcal{M}_{\mathrm{loc}}(\Omega; \mathbb{R}^M)$.

This means that we require

$$\sum_{|\alpha|\le k} (-1)^{|\alpha|} \int_\Omega (A_\alpha^\top \partial^\alpha \psi) \cdot \mathrm{d}\mu = \int_\Omega \psi \cdot \mathrm{d}\sigma, \qquad \psi \in \mathrm{C}_c^\infty(\Omega; \mathbb{R}^M).$$

Note that in general there may be many solutions to (13.16). Prototypical instances of this setup are $\mathcal{A} = \mathrm{curl}$ (defined as in Example 8.6) and $\mathcal{A} = \mathrm{div}$.

For $\mathcal{A}$ we define the **wave cone** (cf. the definition in Section 8.2 for first-order operators)

$$\Lambda_{\mathcal{A}} := \bigcup_{\xi \in \mathbb{S}^{d-1}} \ker \mathbb{A}^k(\xi) \subset \mathbb{R}^N, \tag{13.17}$$

where, with $\xi^\alpha := \xi_1^{\alpha_1} \cdots \xi_d^{\alpha_d} \in \mathbb{R}$,

$$\mathbb{A}^k(\xi) := (2\pi \mathrm{i})^k \sum_{|\alpha|=k} A_\alpha \xi^\alpha \in \mathbb{R}^{M\times N},$$

is the **principal symbol** of $\mathcal{A}$.

We have already seen in Section 8.2 that the wave cone plays a fundamental role in the study of oscillations under the constraint (13.16). Shortly, we will see that the wave cone also determines the admissible *concentrations* in measures satisfying (13.16).

The underlying idea can best be understood via a heuristic argument: Assume for simplicity that $\sigma = 0$ and that $\mathcal{A}$ is a first-order homogeneous PDE operator, i.e., $\mathcal{A} = \sum_{l=1}^d A_l \partial_l$ (for instance $\mathcal{A} = \mathrm{curl}$ or $\mathcal{A} = \mathrm{div}$). If

$$\mathcal{A}\mu = 0$$

then at $|\mu^s|$-almost every point $x_0 \in \Omega$ and for all singular tangent measures

$$\tau = P_0|\tau| \in \mathrm{Tan}(\mu^s, x_0), \qquad \text{where} \qquad P_0 = \frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}(x_0),$$

it holds that

$$\mathcal{A}\tau = 0$$

since τ is the weak* limit of a blow-up sequence and our PDE-constraint is scaling-invariant. Here, the fact that $\tau = P_0|\tau|$ follows from Lemma 13.4.

We formally Fourier-transform the above PDE for τ to get (here, $\mathbb{A}(\xi) = \mathbb{A}^1(\xi)$)

$$\mathbb{A}(\xi)\widehat{\tau}(\xi) = \mathbb{A}(\xi) P_0 \, \widehat{|\tau|}(\xi), \qquad \xi \in \mathbb{R}^d,$$

where $\widehat{\tau}$, $\widehat{|\tau|}$ denote the Fourier transforms of τ, $|\tau|$, respectively. Note, however, that these Fourier transforms are only defined (distributionally) if τ happens to be a *tempered distribution*, see [164]. The tangent measures τ with polynomial growth that were constructed in Proposition 13.5 in fact have this property, but there could be others (e.g., with exponential growth). Ignoring this issue for the moment, the above equation

yields that

$$P_0 \in \ker \mathbb{A}(\xi) \subset \Lambda_{\mathcal{A}} \qquad \text{for all } \xi \in \mathbb{R}^d \setminus \{0\} \text{ where } \widehat{|\tau|}(\xi) \neq 0.$$

In particular, if $P_0 \notin \Lambda_{\mathcal{A}}$, then

$$\widehat{|\tau|}(\xi) = 0 \qquad \text{for all } \xi \in \mathbb{R}^d \setminus \{0\},$$

whereby $\operatorname{supp} \widehat{|\tau|} \subset \{0\}$ and thus τ is absolutely continuous with respect to $\mathcal{L}^d$.

Based on the intuition that tangent measures describe the "local shape" of a measure, this suggests that x_0 cannot be a singular point of μ. Hence, one may conjecture that $\frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}(x_0)$ lies in the wave cone $\Lambda_{\mathcal{A}}$ for $|\mu^s|$-a.e. $x_0 \in \Omega$.

However, the absolute continuity of a singular tangent measure $\tau \neq 0$ with respect to $\mathcal{L}^d$ is not by itself a contradiction. Indeed, Preiss provided an example of a purely singular measure that has only multiples of Lebesgue measure as tangent measures, see Example 5.9 (1) in [274].

Yet, the singular polar indeed only attains values in the wave cone:

Theorem 13.11 (De Philippis–Rindler 2016 [112]) *Let $\mu \in \mathcal{M}_{\mathrm{loc}}(\Omega;\mathbb{R}^N)$ be a solution of (13.16) with Lebesgue–Radon–Nikodým decomposition*

$$\mu = \mu^a + \frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}\,|\mu^s|, \qquad \mu^s \text{ singular with respect to } \mathcal{L}^d.$$

Then,

$$\frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}(x) \in \Lambda_{\mathcal{A}} \qquad \text{for } |\mu^s|\text{-a.e. } x \in \Omega.$$

For the proof we will need the following two auxiliary lemmas (the first one being an extension of Lemma A.37):

Lemma 13.12 *The Bessel potential*

$$(\mathrm{id} - \Delta)^{-s/2} u := \mathcal{F}^{-1}\big[(1 + 4\pi^2|\xi|^2)^{-s/2}\widehat{u}(\xi)\big], \qquad u \in \mathrm{C}_c^\infty(\mathbb{R}^d),$$

of order $s > 0$ extends to a compact operator from $\mathrm{L}^1(B_1)$ to $\mathrm{L}^p(\mathbb{R}^d)$ whenever $1 \leq p < p(d,s)$, where

$$p(d,s) := \begin{cases} \frac{d}{d-s} & \text{if } s < d, \\ \infty & \text{if } s \geq d. \end{cases}$$

Here, as usual, the compactness of the operator $(\mathrm{id} - \Delta)^{-s/2}$ means that it maps norm-bounded sets into (strongly) precompact sets.

Proof. For $u \in \mathrm{C}_c^\infty(\mathbb{R}^d)$ we can write the application of the Bessel potential via a convolution,

$$(\mathrm{id} - \Delta)^{-s/2} u = u \star b_{s,d}, \qquad \text{where} \qquad b_{s,d} := \mathcal{F}^{-1}\big[(1 + 4\pi^2|\xi|^2)^{-s/2}\big].$$

It can be computed, see, for instance, Section 6.1.2 of [165], that $b_{s,d}$ is smooth in $\mathbb{R}^d \setminus \{0\}$ and that

$$b_{s,d}(x) \sim \begin{cases} |x|^{-d+s} & \text{if } |x| \le 2, \\ \mathrm{e}^{-|x|/2} & \text{if } |x| \ge 2. \end{cases}$$

Thus, $b_{s,d} \in \mathrm{L}^p(\mathbb{R}^d)$ for $1 \le p < p(d,s)$. Young's inequality for convolutions, see Lemma A.33, then gives for $u \in \mathrm{L}^1(B_1)$ that

$$(\mathrm{id} - \Delta)^{-s/2} u \in \mathrm{L}^p(\mathbb{R}^d) \qquad \text{for } 1 \le p < p(d,s).$$

For every $\varepsilon > 0$ we can furthermore write

$$b_{s,d} = b_{1,\varepsilon} + b_{2,\varepsilon} \qquad \text{with} \qquad b_{1,\varepsilon} \in \mathrm{C}^1_c(\mathbb{R}^d) \quad \text{and} \quad \|b_{2,\varepsilon}\|_{\mathrm{L}^1} < \varepsilon.$$

Indeed, for a smooth radial cut-off function $\varrho \in \mathrm{C}^\infty_c(B_1; [0,1])$ with $\varrho \equiv 1$ in a neighborhood of the origin we may set

$$b_{1,\varepsilon} := \big(\varrho(\delta x) - \varrho(\delta^{-1} x)\big) b_{s,d}, \qquad b_{2,\varepsilon} := \big(1 - \varrho(\delta x) + \varrho(\delta^{-1} x)\big) b_{s,d},$$

where $\delta \in (0,1)$ is chosen sufficiently small. Then,

$$(\mathrm{id} - \Delta)^{-s/2} u = u \star b_{1,\varepsilon} + u \star b_{2,\varepsilon} =: T^{(1)}_\varepsilon u + T^{(2)}_\varepsilon u.$$

Because $b_{1,\varepsilon} \in \mathrm{C}^1_c(\mathbb{R}^d)$, the operator $T^{(1)}_\varepsilon$ is compact from $\mathrm{L}^1(B_1)$ to $\mathrm{L}^1(\mathbb{R}^d)$.

Moreover, again by Young's inequality for convolutions,

$$\|(\mathrm{id} - \Delta)^{-s/2} - T^{(1)}_\varepsilon\|_{\mathrm{L}^1 \to \mathrm{L}^1} = \|T^{(2)}_\varepsilon\|_{\mathrm{L}^1 \to \mathrm{L}^1} \le \varepsilon,$$

so that $(\mathrm{id} - \Delta)^{-s/2}$ is the limit in the uniform topology of compact operators and thus compact as well.

Indeed, the last statement can be seen as follows: Let $(u_j)_j \subset \mathrm{L}^1(B_1)$ be a norm-bounded sequence. The compactness of $T^{(1)}_\varepsilon$ means that for all $\varepsilon > 0$ the sequence $(T^{(1)}_\varepsilon u_j)_j$ is precompact in $\mathrm{L}^1(\mathbb{R}^d)$. Fix a sequence $\varepsilon_k \downarrow 0$ and let $(u_{1,j})_j$ be a subsequence of $(u_{0,j})_j := (u_j)_j$ such that $(T^{(1)}_{\varepsilon_1} u_{1,j})_j$ is Cauchy in $\mathrm{L}^1(B_1)$. In particular, we may require that $\|T^{(1)}_{\varepsilon_1} u_{1,1} - T^{(1)}_{\varepsilon_1} u_{1,l}\|_{\mathrm{L}^1} \le \varepsilon_1$ for all $l \in \mathbb{N}$. In this manner we construct for every $k \in \mathbb{N}$ a subsequence $(u_{k,j})_j$ of $(u_{k-1,j})_{j \ge k}$ such that $(T^{(1)}_{\varepsilon_k} u_{k,j})_j$ is Cauchy and $\|T^{(1)}_{\varepsilon_k} u_{k,k} - T^{(1)}_{\varepsilon_k} u_{k,l}\|_{\mathrm{L}^1} \le \varepsilon_k$ for all $l \ge k$. Setting $v_j := u_{j,j}$, we then have for $l \ge k$ that

$$\begin{aligned} &\|(\mathrm{id} - \Delta)^{-s/2} v_k - (\mathrm{id} - \Delta)^{-s/2} v_l\|_{\mathrm{L}^1} \\ &\quad \le \|(\mathrm{id} - \Delta)^{-s/2} v_k - T^{(1)}_{\varepsilon_k} v_k\|_{\mathrm{L}^1} + \|T^{(1)}_{\varepsilon_k} v_k - T^{(1)}_{\varepsilon_k} v_l\|_{\mathrm{L}^1} \\ &\qquad + \|T^{(1)}_{\varepsilon_k} v_l - (\mathrm{id} - \Delta)^{-s/2} v_l\|_{\mathrm{L}^1} \\ &\quad \le 2\varepsilon_k \sup_{j \in \mathbb{N}} \|v_j\|_{\mathrm{L}^1} + \varepsilon_k. \end{aligned}$$

Thus, the sequence $((\mathrm{id}-\Delta)^{-s/2}v_j)_j$ is Cauchy in $\mathrm{L}^1(B_1)$, whereby it has a (unique) limit.

The conclusion of the lemma now follows since strong L^1-convergence together with L^q-boundedness also implies strong L^p-convergence for all $1 \le p < q$ (this follows from Hölder's inequality). □

Lemma 13.13 *Let $(f_j)_j \subset \mathrm{L}^1(B_1)$ be such that*

(i) $f_j \overset{*}{\rightharpoonup} 0$ in $\mathrm{C}_c^\infty(B_1)^*$;
(ii) the negative parts $f_j^- := \max\{-f_j, 0\}$ converge to zero in measure, i.e.,

$$\lim_{j\to\infty} \left|\left\{ x \in B_1 \,:\, f_j^-(x) > \delta \right\}\right| = 0 \qquad \text{for every } \delta > 0;$$

(iii) the family of negative parts $(f_j^-)_j$ is equiintegrable.

Then, $f_j \to 0$ in $\mathrm{L}^1_{\mathrm{loc}}(B_1)$.

Proof. Let $\varphi \in \mathrm{C}_c^\infty(B_1; [0,1])$. Then,

$$\begin{aligned}\int_{B_1} \varphi |f_j| \,\mathrm{d}x &= \int_{B_1} \varphi f_j \,\mathrm{d}x + 2\int_{B_1} \varphi f_j^- \,\mathrm{d}x \\ &\le \int_{B_1} \varphi f_j \,\mathrm{d}x + 2\int_{B_1} f_j^- \,\mathrm{d}x.\end{aligned}$$

The first term on the right-hand side vanishes as $j \to \infty$ by assumption (i). Vitali's Convergence Theorem A.12 in conjunction with assumptions (ii) and (iii) further gives that the second term also tends to zero in the limit. □

Proof of Theorem 13.11. We will only show the theorem for the first-order homogeneous equation with zero on the right-hand side, i.e.,

$$\mathcal{A}\mu = \sum_{l=1}^{d} A_l \partial_l \mu = 0 \qquad \text{in } \mathrm{C}_c^\infty(\Omega; \mathbb{R}^M)^*$$

since this is the only case that is needed in the sequel; see [112] for the full proof. In this situation,

$$\Lambda_{\mathcal{A}} = \bigcup_{\xi \in \mathbb{S}^{d-1}} \ker \mathbb{A}(\xi), \qquad \mathbb{A}(\xi) = (2\pi \mathrm{i}) \sum_{l=1}^{d} A_l \xi_l.$$

Step 1. Assume to the contrary that there is a point $x_0 \in \Omega$ and a sequence $r_n \downarrow 0$ such that

(a) $\displaystyle\lim_{n\to\infty} \frac{|\mu^a|(B_{r_n}(x_0))}{|\mu^s|(B_{r_n}(x_0))} = 0;$

(b) $\displaystyle\lim_{n\to\infty} \fint_{B_{r_n}(x_0)} \left| \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x_0) - \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x) \right| \mathrm{d}|\mu^s|(x) = 0;$

(c) there exists a $\tau \in \mathrm{Tan}(|\mu^s|, x_0)$ with $\tau(B_{1/2}) > 0$ and

$$\gamma_n := c_n T_{\#}^{(x_0, r_n)} |\mu^s| \overset{*}{\rightharpoonup} \tau \quad \text{in } \mathcal{M}(B_1), \qquad c_n := |\mu^s|(B_{r_n}(x_0))^{-1};$$

(d) for the polar at x_0 it holds that

$$P_0 := \frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}(x_0) \notin \Lambda_{\mathcal{A}}$$

and hence there is $c > 0$ such that $|\mathbb{A}(\xi) P_0| \geq c|\xi|$ for all $\xi \in \mathbb{R}^d$.

Indeed, (a), (b) hold at $|\mu^s|$-almost every point by the Besicovitch Differentiation Theorem A.24 and the fact that $|\mu^s|$-almost every $x_0 \in \Omega$ is a Lebesgue point of $\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}$. Assertion (c) follows since for $|\mu^s|$-almost every $x \in \Omega$ the space of tangent measures $\mathrm{Tan}(|\mu^s|, x_0)$ is non-trivial by Proposition 13.5. Note that via (13.3) we may define the c_n as above (choose the radii r_n such that $|\mu^s|(\partial B_{r_n}(x_0)) = 0$) and via a rescaling argument we can ensure that $\tau(B_{1/2}) > 0$. Finally, (d) is the assumption that will lead to a contradiction.

We will show below that (a)–(d) imply the following two assertions:

(I) $\tau \llcorner B_{1/2}$ is absolutely continuous with respect to $\mathcal{L}^d$;
(II) $\lim_{n\to\infty} |\gamma_n - \tau|(B_{1/2}) = 0$.

Then, we may conclude as follows: The γ_n are singular with respect to $\mathcal{L}^d$ by construction. So, by (I) we may find Borel sets $E_n \subset B_{1/2}$ with $\mathcal{L}^d(E_n) = \tau(E_n) = 0$ and $\gamma_n(E_n) = \gamma_n(B_{1/2})$. Then, by (II),

$$\gamma_n(B_{1/2}) = \gamma_n(E_n) \leq |\gamma_n - \tau|(B_{1/2}) + \tau(E_n) = |\gamma_n - \tau|(B_{1/2}) \to 0.$$

Hence, $\tau(B_{1/2}) = 0$, contradicting (c) above. We are thus left to prove (I) and (II).

Step 2. Since $\mathcal{A}$ is rescaling-invariant, we have for all $r > 0$ that

$$\mathcal{A}(T_{\#}^{(x_0, r)} \mu) = 0.$$

Therefore,

$$\mathcal{A}(P_0 \gamma_n) = \mathcal{A}\big(P_0 \gamma_n - c_n T_{\#}^{(x_0, r_n)} \mu\big). \tag{13.18}$$

Let $(\eta_\delta)_{\delta>0} \subset \mathrm{C}_c^\infty(\mathbb{R}^d)$ be a family of positive mollifiers. The total variation on open sets is lower semicontinuous, whereby

$$|\gamma_n - \tau|(B_{1/2}) \leq \liminf_{\delta\to 0} |\gamma_n \star \eta_\delta - \tau|(B_{1/2}).$$

Thus, for every $n \in \mathbb{N}$ there exists a $\delta_n \in (0, 1/n)$ with

$$|\gamma_n - \tau|(B_{1/2}) \leq |\gamma_n \star \eta_{\delta_n} - \tau|(B_{1/2}) + \frac{1}{n}. \tag{13.19}$$

We now convolve (13.18) with η_{δ_n}. In this way, with

$$u_n := \gamma_n \star \eta_{\delta_n} \overset{*}{\rightharpoonup} \tau \quad \text{in } \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d), \qquad V_n := \left[P_0\gamma_n - c_n T_{\#}^{(x_0,r_n)}\mu\right] \star \eta_{\delta_n},$$

we get

$$\mathcal{A}(P_0 u_n) = \mathcal{A}(V_n). \tag{13.20}$$

Note that u_n, V_n are smooth and $u_n \geq 0$ since our family of mollifiers was chosen to be positive.

Let $\varrho \in \mathrm{C}_c^\infty(B_{3/4}; [0,1])$ be a cut-off function with $\varrho \equiv 1$ on $B_{1/2}$. Then,

$$\mathcal{A}(P_0 \varrho u_n) = \varrho \mathcal{A}(P_0 u_n) + [\mathcal{A}, \varrho](P_0 u_n)$$

where $[\mathcal{A}, \varrho] = \mathcal{A} \circ \varrho - \varrho \mathcal{A}$ is the *commutator* between $\mathcal{A}$ and ϱ. We compute

$$[\mathcal{A}, \varrho](P_0 u_n) = \sum_{l=1}^{d} A_l(\partial_l \varrho) P_0 u_n = \mathcal{A}(P_0 \varrho) u_n$$

since the terms with derivatives on u_n cancel. In particular, $[\mathcal{A}, \varrho]$ is a differential operator of order *zero*. So, combining the above with (13.20),

$$\begin{aligned} \mathcal{A}(P_0 \varrho u_n) &= \varrho \mathcal{A}(V_n) + \mathcal{A}(P_0 \varrho) u_n \\ &= \mathcal{A}(\varrho V_n) - [\mathcal{A}, \varrho](V_n) + \mathcal{A}(P_0 \varrho) u_n \\ &= \mathcal{A}(\varrho V_n) + R_n \end{aligned} \tag{13.21}$$

with

$$R_n := \mathcal{A}(P_0 \varrho) u_n - \sum_{l=1}^{d} A_l(\partial_l \varrho) V_n \in \mathrm{C}_c^\infty(B_1; \mathbb{R}^M).$$

Now, for $\delta_n \leq 1/4$ it holds that

$$\begin{aligned} \int_{B_{3/4}} |V_n| \, \mathrm{d}x &\leq c_n \left| P_0 \, T_{\#}^{(x_0,r_n)} |\mu^s| - T_{\#}^{(x_0,r_n)} \mu \right| (B_1) \\ &\leq \frac{|P_0 \, |\mu^s| - \mu^s|(B_{r_n}(x_0))}{|\mu^s|(B_{r_n}(x_0))} + \frac{|\mu^a|(B_{r_n}(x_0))}{|\mu^s|(B_{r_n}(x_0))} \\ &= \fint_{B_{r_n}(x_0)} \left| \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x_0) - \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x) \right| \mathrm{d}|\mu^s|(x) + \frac{|\mu^a|(B_{r_n}(x_0))}{|\mu^s|(B_{r_n}(x_0))}. \end{aligned}$$

From (a), (b) we thus deduce

$$\lim_{n \to \infty} \int_{B_{3/4}} |V_n| \, \mathrm{d}x = 0. \tag{13.22}$$

Hence, the remainder terms R_n are uniformly L^1-bounded.

Step 3. We now Fourier transform (13.21) to obtain

$$\mathbb{A}(\xi)P_0\,\widehat{\varrho u_n}(\xi) = \mathbb{A}(\xi)\widehat{\varrho V_n}(\xi) + \widehat{R}_n(\xi), \qquad \xi \in \mathbb{R}^d.$$

Multiplying on the left by $[\mathbb{A}(\xi)P_0]^* = \overline{[\mathbb{A}(\xi)P_0]^{\mathsf{T}}}$ and adding $\widehat{\varrho u_n}(\xi)$ to both sides of the above equation yields

$$(1 + |\mathbb{A}(\xi)P_0|^2)\,\widehat{\varrho u_n}(\xi) = [\mathbb{A}(\xi)P_0]^*\mathbb{A}(\xi)\,\widehat{\varrho V_n}(\xi) + \widehat{\varrho u_n}(\xi) + [\mathbb{A}(\xi)P_0]^*\widehat{R}_n(\xi)$$

or, equivalently,

$$\begin{aligned}\widehat{\varrho u_n}(\xi) = {} & \frac{[\mathbb{A}(\xi)P_0]^*\mathbb{A}(\xi)}{1 + |\mathbb{A}(\xi)P_0|^2}\,\widehat{\varrho V_n}(\xi) \\ & + \frac{1 + 4\pi^2|\xi|^2}{1 + |\mathbb{A}(\xi)P_0|^2} \cdot \frac{1}{1 + 4\pi^2|\xi|^2}\,\widehat{\varrho u_n}(\xi) \\ & + \frac{(1 + 4\pi^2|\xi|^2)^{1/2}[\mathbb{A}(\xi)P_0]^*}{1 + |\mathbb{A}(\xi)P_0|^2} \cdot \frac{1}{(1 + 4\pi^2|\xi|^2)^{1/2}}\,\widehat{R}_n(\xi).\end{aligned}$$

Thus,

$$\begin{aligned}\varrho u_n = {} & T_0[\varrho V_n] + \big(T_1 \circ (\mathrm{id} - \Delta)^{-1}\big)[\varrho u_n] + \big(T_2 \circ (\mathrm{id} - \Delta)^{-1/2}\big)[R_n] \\ =: {} & f_n + g_n + h_n\end{aligned} \tag{13.23}$$

with the Fourier multiplier operators

$$\begin{aligned} T_0[V] &:= \mathcal{F}^{-1}\big[m_0(\xi)\widehat{V}(\xi)\big], & m_0(\xi) &= \frac{[\mathbb{A}(\xi)P_0]^*\mathbb{A}(\xi)}{1 + |\mathbb{A}(\xi)P_0|^2}, \\ T_1[u] &:= \mathcal{F}^{-1}\big[m_1(\xi)\widehat{u}(\xi)\big], & m_1(\xi) &= \frac{1 + 4\pi^2|\xi|^2}{1 + |\mathbb{A}(\xi)P_0|^2}, \\ T_2[R] &:= \mathcal{F}^{-1}\big[m_2(\xi)\widehat{R}(\xi)\big], & m_2(\xi) &= \frac{(1 + 4\pi^2|\xi|^2)^{1/2}[\mathbb{A}(\xi)P_0]^*}{1 + |\mathbb{A}(\xi)P_0|^2}, \end{aligned}$$

and with $(\mathrm{id} - \Delta)^{-s/2}$ the Bessel potential of order s, as defined in Lemma 13.12.

By the *ellipticity estimate*

$$|\mathbb{A}(\xi)P_0| \geq c|\xi| \qquad \text{for } \xi \in \mathbb{R}^d$$

from (d) above, T_0 is an operator associated with a Mihlin multiplier, see Theorem A.36. The weak-type estimate from this theorem in conjunction with (13.22) gives

$$\sup_{t \geq 0} t\Big|\big\{\, x \in \mathbb{R}^d \;:\; |f_n(x)| > t \,\big\}\Big| \leq C\|\varrho V_n\|_{\mathrm{L}^1} \to 0. \tag{13.24}$$

By Lemma 13.12, for every $s > 0$ the operator $(\mathrm{id} - \Delta)^{-s/2}$ is compact from $\mathrm{L}^1(B_1)$ to $\mathrm{L}^p(\mathbb{R}^d)$ for $1 \leq p < p(d,s)$ and the operators T_1, T_2 are bounded from L^p to L^p by the Mihlin Multiplier Theorem A.36 and (d) again. Thus, the family

$$g_n + h_n = \big(T_1 \circ (\mathrm{id} - \Delta)^{-1}\big)[\varrho u_n] + \big(T_2 \circ (\mathrm{id} - \Delta)^{-1/2}\big)[R_n]$$

is precompact in $\mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^d)$. Since $\varrho u_n \geq 0$, we also obtain from (13.23) that

$$f_n^- := \max\{-f_n, 0\} \leq |g_n + h_n|.$$

Thus, the family $(f_n^-)_n$ is precompact in $\mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^d)$, hence equiintegrable by the Dunford–Pettis Theorem A.13.

Via (13.22),

$$\big\langle f_n, \varphi \big\rangle = \big\langle T_0[\varrho V_n], \varphi \big\rangle = \big\langle \varrho V_n, T_0^*[\varphi] \big\rangle \to 0$$

for every $\varphi \in \mathrm{C}_c^\infty(\mathbb{R}^d)$, where T_0^* is the adjoint of T_0. Now apply Lemma 13.13, using also (13.24), to see that

$$f_n \to 0 \quad \text{in } \mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^d).$$

Combining, we conclude from (13.23) that the family $(\varrho u_n)_n$ is precompact in $\mathrm{L}^1_{\mathrm{loc}}(\mathbb{R}^d)$ and hence the convergence $\varrho u_n \overset{*}{\rightharpoonup} \varrho\tau$ improves to

$$\varrho u_n \to \varrho\tau \quad \text{in } \mathrm{L}^1(\mathbb{R}^d).$$

In particular, $\varrho\tau \in \mathrm{L}^1(\mathbb{R}^d)$. This immediately yields (I) and then, combining this with (13.19), also (II). □

The proof of Alberti's rank-one theorem now follows by simple linear algebra:

Proof of Theorem 13.8. Let $u \in \mathrm{BV}(\Omega; \mathbb{R}^m)$. One sees directly from the definition of weak derivatives that

$$\partial_i [Du]^k_j = \partial_i \partial_j u^k = \partial_j \partial_i u^k = \partial_j [Du]^k_i$$

for all $i, j = 1, \dots, d$ and all $k = 1, \dots, m$. We already computed in Example 8.6 that for

$$\mathcal{A}\mu := \operatorname{curl} \mu := \big(\partial_j \mu_i^k - \partial_i \mu_j^k\big)_{i,j=1,\dots,d}^{k=1,\dots,m}$$

we have

$$\Lambda_{\mathrm{curl}} = \bigcup_{\xi \in \mathbb{S}^{d-1}} \ker \mathbb{A}^1(\xi) = \big\{ a \otimes \xi \in \mathbb{R}^{m \times d} \ : \ a \in \mathbb{R}^m, \xi \in \mathbb{S}^{d-1} \big\}.$$

Consequently, the assertion of Theorem 13.8 follows immediately from Theorem 13.11. □

13.6 Rigidity for the Two-Point Inclusion

For general two-state inclusions, which parallel the two-gradient inclusions of Section 10.1, we have the following rigidity result. This extends the Ball–James Rigidity

Theorem 7.3 (also see Theorem 10.1) to the case of merely L^1-bounded sequences and also to general PDE-constraints.

Theorem 13.14 (De Philippis–Palmieri–Rindler 2018 [110]) *Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $\mathcal{A}, \Lambda_{\mathcal{A}}$ be as in (13.16), (13.17), respectively. For $A, B \in \mathbb{R}^N$ with $A - B \notin \Lambda_{\mathcal{A}}$ the following assertions hold:*

*(i) Suppose that $V \in \mathrm{L}^\infty(\Omega; \mathbb{R}^N)$ satisfies $\mathcal{A}V = 0$ (in the sense of distributions) and the **exact two-point inclusion***

$$V(x) \in \{A, B\} \quad \text{for a.e. in } \Omega$$

holds. Then, either $V \equiv A$ or $V \equiv B$ almost everywhere.

*(ii) Suppose that $(V_j)_j \subset \mathrm{L}^1(\Omega; \mathbb{R}^N)$ is a uniformly norm-bounded sequence of maps such that $\mathcal{A}V_j = 0$ for all $j \in \mathbb{N}$, and that the **approximate two-point inclusion***

$$\lim_{j \to \infty} \int_\Omega \operatorname{dist}(V_j(x), \{A, B\}) \, \mathrm{d}x = 0 \tag{13.25}$$

holds. Then, up to extracting a subsequence, either

$$\int_\Omega |V_j(x) - A| \, \mathrm{d}x \to 0 \quad \text{or} \quad \int_\Omega |V_j(x) - B| \, \mathrm{d}x \to 0$$

as $j \to \infty$.

See [110] for the proof.

Remark 13.15 Note that in comparison to Theorem 7.3 (ii), the condition (13.25) has been strengthened from requiring mere convergence in measure to strong convergence. Indeed, (13.25) entails the equiintegrability of the sequence $(V_j)_j$, so the second assertion of the theorem is precisely about the situation *without* concentrations. However, this strengthening of the assumption is necessary, see Problem 13.10. Moreover, we have excluded the non-rigid case $A - B \in \Lambda_{\mathcal{A}}$ in the theorem since no universal structure of such solutions is known for general operators $\mathcal{A}$ except in special cases, see [151, 269, 270] for $\mathcal{A} = \operatorname{div}$ and [114] for symmetric gradients.

13.7 Λ-Convexity for 1-Homogeneous Functions

The final result in this chapter shows that positively 1-homogeneous functions that are convex in some directions have a surprising global convexity property. To set the stage, in all of the following let $\Lambda \subset \mathbb{R}^N$ be a **balanced cone**, that is, for $Z \in \Lambda$ and $\gamma \in \mathbb{R}$ also $\gamma Z \in \Lambda$. Furthermore assume that $\operatorname{span} \Lambda = \mathbb{R}^N$. We call $h \colon \mathbb{R}^N \to \mathbb{R}$ **Λ-convex** if it is convex along any direction in Λ, that is,

$$h(\theta A + (1-\theta)B) \le \theta h(A) + (1-\theta)h(B)$$

for all $A, B \in \mathbb{R}^N$ with $A - B \in \Lambda$ and all $\theta \in (0,1)$.

Example 13.16 The most important example is the rank-one cone

$$\Lambda_{\mathrm{curl}} = \left\{ a \otimes \xi \in \mathbb{R}^{m\times d} \ : \ a \in \mathbb{R}^m, \xi \in \mathbb{S}^{d-1} \right\},$$

which was computed in the proof of Theorem 13.8, since Λ_{curl}-convexity coincides with the notion of rank-one convexity.

Theorem 13.17 (Kirchheim–Kristensen 2016 [195]) *Let $h^\infty : \mathbb{R}^N \to \mathbb{R}$ be positively 1-homogeneous and Λ-convex. Then, h^∞ is convex at every $F \in \Lambda$, i.e., there exists a linear function $L_F : \mathbb{R}^N \to \mathbb{R}$ with*

$$h^\infty(F) = L_F(F) \qquad \text{and} \qquad h^\infty \ge L_F.$$

We first show two auxiliary results.

Lemma 13.18 *Let $g : \mathbb{R}^N \to \mathbb{R}$ be positively 1-homogeneous and Λ-convex. If $X \in \Lambda$, then*

$$g(X+Y) \le g(X) + g(Y), \qquad Y \in \mathbb{R}^N.$$

Proof. Observe by the Λ-convexity and positive 1-homogeneity of g that for all $t \ge 1$ we have

$$g(X+Y) - g(Y) \le \frac{g(tX+Y) - g(Y)}{t} = g\left(X + \frac{Y}{t}\right) - \frac{g(Y)}{t}.$$

Since g is (globally) Lipschitz by a straightforward analogue of Lemma 5.9 (the proof is a straightforward generalization), the right-hand side converges to $g(X)$ as $t \to \infty$, which proves the claim. □

Lemma 13.19 *Let $h^\infty : \mathbb{R}^N \to \mathbb{R}$ be positively 1-homogeneous and Λ-convex. Then, for every matrix $A \in \mathbb{R}^N$ with $A \in \Lambda$ there exists a positively 1-homogeneous and Λ-convex **subcone** $g : \mathbb{R}^N \to \mathbb{R}$ to h^∞ at A, that is, $g(A) = h^\infty(A)$ and*

$$h^\infty(A) + g(B-A) \le h^\infty(B) \qquad \text{for all } B \in \mathbb{R}^N. \tag{13.26}$$

Proof. Throughout the proof of the lemma we assume without loss of generality that $h^\infty(A) = 0$. Indeed, we may add a linear function to h^∞ to achieve this, which we then subtract from the resulting subcone again.

We first show that for all $B \in \mathbb{R}^N$ with $B \ne A$ and all $\theta \in (0,1)$ it holds that

$$\frac{h^\infty(A + \theta(B-A))}{\theta} \le h^\infty(B). \tag{13.27}$$

Indeed, applying Lemma 13.18 for h^∞ at $X := (\theta^{-1} - 1)A$, $Y := B$, we get

$$\frac{h^\infty(A+\theta(B-A))}{\theta} \le h^\infty\big((\theta^{-1}-1)A\big) + h^\infty(B) = h^\infty(B),$$

which is (13.27).

Define for $s \ge 1$ the function

$$g_s(M) := sh^\infty\left(A + \frac{M}{s}\right), \qquad M \in \mathbb{R}^N.$$

For any $M_1, M_2 \in \mathbb{R}^N$ we have

$$|g_s(M_1) - g_s(M_2)| \le sL\left|\frac{M_1}{s} - \frac{M_2}{s}\right| = L|M_1 - M_2|,$$

where $L > 0$ is the Lipschitz constant of h^∞, which is finite by the aforementioned analogue of Lemma 5.9. So, the functions g_s are uniformly Lipschitz continuous.

Now let $t \ge s$. From (13.27) with $B := A + s^{-1}M$ and $\theta := s/t$ we get

$$g_t(M) = th^\infty\left(A + \frac{M}{t}\right) \le sh^\infty\left(A + \frac{M}{s}\right) = g_s(M).$$

Thus, $g_s(M)$ is monotonically decreasing in s for every fixed $M \in \mathbb{R}^N$. Define

$$g(M) := \lim_{s\to\infty} g_s(M) = \inf_{s>0} g_s(M), \qquad M \in \mathbb{R}^N.$$

As the pointwise limit of uniformly Lipschitz continuous and Λ-convex functions, g is also Lipschitz continuous and Λ-convex. Moreover, g is positively 1-homogeneous since for all $\alpha \ge 0$ and $M \in \mathbb{R}^N$ it holds that

$$\begin{aligned} g(\alpha M) &= \lim_{s\to\infty}\left[sh^\infty\left(A + \frac{\alpha M}{s}\right)\right] \\ &= \alpha \lim_{s\to\infty}\left[\frac{s}{\alpha}h^\infty\left(A + \frac{M}{s/\alpha}\right)\right] \\ &= \alpha g(M). \end{aligned}$$

Finally, for $M := B - A$, $B \in \mathbb{R}^N$, we get

$$g(B-A) \le g_1(B-A) = h^\infty(B).$$

Thus, (13.26) holds and we have proved the lemma. □

Proof of Theorem 13.17. Set $A_{md} := F$ and extend this to an orthonormal basis $\{A_1, \dots, A_{md}\} \subset \Lambda$ of $\mathbb{R}^N$. This is possible since span $\Lambda = \mathbb{R}^N$.

First, we construct a sequence

$$h^\infty =: g_{md} \ge g_{md-1} \ge \cdots \ge g_0$$

of positively 1-homogeneous Λ-convex functions $g_k\colon \mathbb{R}^N \to \mathbb{R}$ as follows: Set $g_{md} := h^\infty$, Then, given g_k (where $k = md, md-1, \dots, 2$), we inductively define $g_{k-1}\colon \mathbb{R}^N \to \mathbb{R}$ to be a subcone to g_k at the k'th basis matrix A_k via Lemma 13.19. From (13.26) for g_k and with $A := A_k$, $B := A_k + M$, where $M \in \mathbb{R}^N$ is arbitrary,

$$g_k(A_k) + g_{k-1}(M) \le g_k(A_k + M).$$

On the other hand, Lemma 13.18 for g_k at $X := A_k$, $Y := M$ yields

$$g_k(A_k + M) \le g_k(A_k) + g_k(M).$$

Combining these two estimates, we conclude $g_{k-1} \le g_k$.

With all the g_k constructed via the above scheme, define linear maps

$$L_k\colon \operatorname{span}\{A_1, \dots, A_k\} \to \mathbb{R}$$

iteratively for $k = 0, 1, \dots, md$ according to the following procedure: Let $L_0\colon \{0\} \to \mathbb{R}$ be the trivial linear map. In step k we assume inductively that

$$L_k \le g_k \quad \text{on } \operatorname{span}\{A_1, \dots, A_k\} \tag{13.28}$$

and, if $k \ge 1$,

$$L_k(A_k) = g_k(A_k). \tag{13.29}$$

Then set

$$L_{k+1}(\alpha A_{k+1} + B) := \alpha g_{k+1}(A_{k+1}) + L_k(B),$$

where $B \in \operatorname{span}\{A_1, \dots, A_k\}$, $\alpha \in \mathbb{R}$.

By definition, (13.29) holds for step $k+1$. Moreover, from (13.28) and the fact that g_k is a subcone to g_{k+1} at A_{k+1} we obtain the estimate

$$\begin{aligned} L_{k+1}(A_{k+1} + B) &= g_{k+1}(A_{k+1}) + L_k(B) \\ &\le g_{k+1}(A_{k+1}) + g_k(B) \\ &\le g_{k+1}(A_{k+1} + B). \end{aligned}$$

Since g_{k+1} is positively 1-homogeneous, this further implies for $\alpha \ge 0$ that

$$\begin{aligned} L_{k+1}(\alpha A_{k+1} + B) &= \alpha L_{k+1}(A_{k+1} + \alpha^{-1} B) \\ &\le \alpha g_{k+1}(A_{k+1} + \alpha^{-1} B) \\ &= g_{k+1}(\alpha A_{k+1} + B). \end{aligned}$$

For $\alpha < 0$ we have by Lemma 13.18 for g_{k+1} at $X := (1-\alpha)A_{k+1}$, $Y := \alpha A_{k+1} + B$ that

$$\begin{aligned} g_{k+1}(A_{k+1}) + L_k(B) &= L_{k+1}(A_{k+1} + B) \\ &\le g_{k+1}(A_{k+1} + B) \\ &\le g_{k+1}((1-\alpha)A_{k+1}) + g_{k+1}(\alpha A_{k+1} + B) \\ &= (1-\alpha) g_{k+1}(A_{k+1}) + g_{k+1}(\alpha A_{k+1} + B). \end{aligned}$$

Thus, by the definition of L_{k+1}, also in the case $\alpha < 0$ it holds that

$$L_{k+1}(\alpha A_{k+1} + B) = \alpha g_{k+1}(A_{k+1}) + L_k(B) \le g_{k+1}(\alpha A_{k+1} + B)$$

and we have established (13.28) at $k + 1$.

Now set $L_F := L_{md}$, for which $L_F \le g_{md} = h^\infty$ by (13.28), and also

$$L_F(F) = g_{md}(F) = h^\infty(F)$$

by (13.29) together with the fact that $A_{md} = F$, as required. □

If we apply the preceding theorem to the wave cone Λ_{curl} of the curl-operator, see Example 13.16, we obtain the following result:

Corollary 13.20 *Let $h^\infty : \mathbb{R}^{m\times d} \to \mathbb{R}$ be positively 1-homogeneous and rank-one convex. Then, h^∞ is convex at every $F \in \mathbb{R}^{m\times d}$ with* $\operatorname{rank} F \le 1$*, i.e., there exists a linear function $L_F : \mathbb{R}^{m\times d} \to \mathbb{R}$ with*

$$h^\infty(F) = L_F(F) \qquad \text{and} \qquad h^\infty \ge L_F.$$

As an immediate consequence, for all positively 1-homogeneous, rank-one convex $h^\infty : \mathbb{R}^{m\times d} \to \mathbb{R}$ and all probability measures $\mu \in \mathcal{M}^1(\mathbb{R}^{m\times d})$ with the property that the barycenter $[\mu] = \langle \mathrm{id}, \mu\rangle$ is a matrix of rank at most one, the *Jensen-type inequality*

$$h^\infty([\mu]) \le \int h^\infty \, \mathrm{d}\mu \tag{13.30}$$

holds.

Notes and Historical Remarks

The Reshetnyak Continuity Theorem 13.3 was first proved in [277], whereas our argument is from [18]. The paper [116] contains more on different notions of convergence for measures.

Often, tangent measures are only considered as defined on the unit ball instead of the whole of the space (see, for example, Section 2.7 in [18]). This is, however, sometimes restrictive. Here, we essentially use Preiss's original theory as developed in [274], also see Chapter 14 of [227]. The original definition, however, explicitly excluded the zero-measure from $\mathrm{Tan}(\mu, x_0)$, which here we include. Proposition 13.5 is a slight improvement of Preiss' existence theorem for non-zero tangent measures, which was first proved as Theorem 2.5 in [274] (the polynomial growth seems to have been first observed in the appendix of [278]). The proof of Lemma 13.4 is adapted from Theorem 2.44 in [18]. Lemma 13.6 is originally due to Larsen, see Lemma 5.1 in [217].

As was already mentioned, the study of properties of a measure via its tangent measures has its limits. Most strikingly, Preiss constructed a purely singular positive measure on a bounded interval (in particular a BV-derivative) such that each of its tan-

gent measures is a multiple of Lebesgue measure, see Example 5.9 (1) in [274]. Also see [267] for a measure that has *every* local measure as a tangent measure at almost every point.

The proof of Lemma 13.7 is from the appendix of [209] (the case for Lipschitz domains can be found in Lemma B.1 of [55]). The notion of area-strict convergence in BV seems to be somewhat less well known than it deserves. However, as shown in the next chapter, it is the right one when considering integral functionals.

Alberti's original proof [4] of what is now called Alberti's rank-one theorem is via a "decomposition technique" together with the BV-coarea formula; a streamlined version of his proof can be found in [107]. Another proof in two dimensions was announced in [5]. A very nice and short geometric argument is in [225]. For more on the wave cone we refer to [121, 254, 255, 280, 323, 324]. A recasting of Theorem 13.11 into the language of microlocal analysis is in [45].

For $\mathcal{A} = \operatorname{div}$ and some other first-order operators, Theorem 13.14 was already shown by Garroni and Nesi [151] in the case $p > 1$ (and with a different proof). More general version of this theorem with non-homogeneous operators $\mathcal{A}$ and non-zero (but asymptotically vanishing) right-hand sides in the constraint PDE can be found in the original reference [110]. A quantitative rigidity result was established in [22]. A refinement of the techniques of [112] also allows one to prove various results about the rectifiability of BV-derivatives (and various other PDE-constrained measures), see [21].

The Kirchheim–Kristensen Theorem 13.17 was announced in [194] with a simpler proof in a special case. The full proof appeared in [195].

Problems

13.1 Let $\mu \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^N)$. Show that

$$\left| \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x) \right| = 1 \quad \text{for } |\mu|\text{-a.e. } x \in \mathbb{R}^d.$$

13.2 Let $\Lambda \in \mathcal{M}^+(B_1)$ be a finite and positive Borel measure. Then, we have $\Lambda(\partial B_r) = 0$ for all but countably many $r \in (0, 1)$.

13.3 Prove (13.8) using the integration-by-parts definition of BV-maps.

13.4 Find a simpler proof of Lemma 13.7 in the case when Ω is a bounded Lipschitz domain as follows: For $u \in \mathrm{BV}(\Omega; \mathbb{R}^m)$ define

$$u_\delta(x) := \int \eta(y) u(x - \delta \varrho(x) y) \, \mathrm{d}y, \qquad \delta > 0,$$

where $\eta \in \mathrm{C}_c^\infty(B_1)$ is a standard mollifying kernel and ϱ is a regularized distance to the boundary of Ω, i.e., $\varrho \in \mathrm{C}^\infty(\Omega)$, $C^{-1} \operatorname{dist}(x, \partial\Omega) \le \varrho(x) \le C \operatorname{dist}(x, \partial\Omega)$ ($C > 0$), and $|\nabla \varrho(x)| \le C$ for all $x \in \Omega$. See, for instance, [303], p. 171, for the construction of such a ϱ.

13.5 Let $\Omega \subset \mathbb{R}^d$ be open.

(i) Let $(\mu_j)_j \subset \mathcal{M}(\Omega;\mathbb{R}^N)$ with $\mu_j \to \mu$ area-strictly, i.e., $\mu_j \overset{*}{\rightharpoonup} \mu$ and $\langle \mu_j\rangle(\Omega) \to \langle\mu\rangle(\Omega)$. Show that then also $\mu_j \to \mu$ strictly. *Hint:* If $\mu = a\mathcal{L}^d + \mu^s$ is the Lebesgue–Radon–Nikodým decomposition of μ, then define $\widetilde{\mu} \in \mathcal{M}(\Omega;\mathbb{R}^N \times \mathbb{R})$ as $\widetilde{\mu}(B) := (a,0)\mathcal{L}^d + (0,1)\mu^s$ and use the Reshetnyak continuity theorem from the preceding problem.

(ii) Show that the mapping $\mu \in \mathcal{M}(\Omega;\mathbb{R}^N) \mapsto \langle\mu\rangle(\Omega)$ is weakly* lower semicontinuous.

13.6 Let $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$. Show that for $|Du|$-almost every $x_0 \in \Omega$, every $\tau \in \mathrm{Tan}(Du,x_0)$ is a BV-derivative, $\tau = Dv$ for some $v \in \mathrm{BV}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^m)$, and with $P_0 := \frac{\mathrm{d}Du}{\mathrm{d}|Du|}(x_0)$ it holds that:

(i) If $\operatorname{rank} P_0 \geq 2$, then $v(x) = v_0 + \alpha P_0 x$, where $\alpha \in \mathbb{R}$, $v_0 \in \mathbb{R}^m$.

(ii) If $P_0 = a \otimes n$ ($a \in \mathbb{R}^m$, $n \in \mathbb{S}^{d-1}$), then there exist $h \in \mathrm{BV}(\mathbb{R})$, $v_0 \in \mathbb{R}^m$ such that $v(x) = v_0 + h(x\cdot n)a$.

What does Alberti's Rank-One Theorem imply about (i)?

13.7 Let $\mu \in \mathcal{M}(\Omega;\mathbb{R}^{d\times d})$ be a matrix-valued measure such that (in the distributional, i.e., $\mathrm{C}_c^\infty(\Omega)^*$-weak sense)

$$\operatorname{div}\mu \in \mathcal{M}(\Omega;\mathbb{R}^d).$$

Prove that then

$$\operatorname{rank}\frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x) \leq d-1 \qquad \text{for } |\mu^s|\text{-a.e. } x \in \Omega.$$

13.8 A map $u \in \mathrm{L}^1(\Omega;\mathbb{R}^d)$ ($\Omega \subset \mathbb{R}^d$ a bounded Lipschitz domain) is called a function of **bounded deformation** if the symmetric part of its distributional derivative is a measure,

$$Eu := \frac{Du + (Du)^{\mathsf{T}}}{2} \in \mathcal{M}(\Omega;\mathbb{R}^{d\times d}_{\mathrm{sym}}).$$

We collect all these functions into the set $\mathrm{BD}(\Omega)$, see [15,330,331] for the theory of this space. Let $\mu = (\mu_j^k) \in \mathcal{M}(\Omega,\mathbb{R}^{d\times d}_{\mathrm{sym}})$ be the symmetrized derivative of some $u \in \mathrm{BD}(\Omega)$, $\mu = Eu$. Verify that then

$$0 = \mathcal{A}\mu := \left[\sum_{i=1}^d \partial_i\partial_k\mu_i^j + \partial_i\partial_j\mu_i^k - \partial_j\partial_k\mu_i^i - \partial_i\partial_i\mu_j^k\right]_{j,k=1,\dots,d}$$

in the sense of $\mathrm{C}_c^\infty(\Omega)^*$. These equations are often called the *Saint-Venant compatibility conditions* in applications.

13.9 In the situation of the previous problem and denoting the Lebesgue–Radon–Nikodým decomposition of the symmetrized derivative Eu of $u \in \mathrm{BD}(\Omega)$ by

$$Eu = \mathcal{E}u\,\mathcal{L}^d \llcorner \Omega + E^s u,$$

prove that for $|E^s u|$-almost every $x \in \Omega$, there exist $a(x), b(x) \in \mathbb{R}^d \setminus \{0\}$ such that

$$\frac{\mathrm{d}E^s u}{\mathrm{d}|E^s u|}(x) = a(x) \odot b(x),$$

where we define the symmetrized tensor product as $a \odot b := (a \otimes b + b \otimes a)/2$ for $a, b \in \mathbb{R}^d$. *Hint:* Use Theorem 13.11 and the previous problem. This result was one of the main motivations for Theorem 13.11.

13.10 Show that the assumption (13.25) cannot be omitted by finding a uniformly norm-bounded sequence $(u_j)_j \subset \mathrm{L}^1(\Omega; \mathbb{R}^2)$ on any bounded Lipschitz domain $\Omega \subset \mathbb{R}^2$ with $\operatorname{div} u_j = 0$ for all $j \in \mathbb{N}$, $\operatorname{dist}(u_j, \{0, \mathrm{Id}\}) \to 0$ in measure, but

$$\lim_{j\to\infty} \int_{B_1} |u_j(x) - v| \, \mathrm{d}x > 0$$

for all $v \in \mathbb{R}^2$. *Hint:* Start from a singular loop (like $\tau \mathcal{H}^1 \llcorner \partial B_1$ with τ a choice of tangent vectors).

Chapter 14
Linear-Growth Functionals

After the preparations in the previous chapter, we now return to the task at hand, namely to analyze the following minimization problem for an integral functional with linear growth:

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_\Omega f(x, \nabla u(x))\, \mathrm{d}x \\ \text{over all} & u \in \mathrm{W}^{1,1}(\Omega; \mathbb{R}^m) \text{ with } u|_{\partial\Omega} = g. \end{cases}$$

As usual, we assume $\Omega \subset \mathbb{R}^d$ to be a bounded Lipschitz domain and $f\colon \Omega \times \mathbb{R}^{m\times d} \to [0, \infty)$ to be a Carathéodory integrand. We also suppose that there exists a constant $M > 0$ such that

$$f(x, A) \le M(1 + |A|), \qquad (x, A) \in \Omega \times \mathbb{R}^{m\times d}.$$

Furthermore, we let $g \in \mathrm{L}^1(\partial\Omega; \mathbb{R}^m)$.

As we have seen in the introduction to the previous chapter, $\mathrm{W}^{1,1}(\Omega; \mathbb{R}^m)$ is too small to describe concentration effects. Our first task is thus to extend $\mathcal{F}$ in a natural way to a functional $\widetilde{\mathcal{F}}$ defined on $\mathrm{BV}(\Omega; \mathbb{R}^m)$ such that this extension is continuous with respect to a notion of convergence in which $\mathrm{W}^{1,1}(\Omega; \mathbb{R}^m)$ is dense in $\mathrm{BV}(\Omega; \mathbb{R}^m)$. As we will see shortly, this extension turns out to be

$$\widetilde{\mathcal{F}}[u] := \int_\Omega f(x, \nabla u(x))\, \mathrm{d}x + \int_\Omega f^\infty\left(x, \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right) \mathrm{d}|D^s u|(x),$$

where f^∞ is the *(strong) recession function*, defined in (14.1) below. The notion of convergence in which $\widetilde{\mathcal{F}}$ is continuous and in which $\mathrm{W}^{1,1}(\Omega; \mathbb{R}^m)$ is dense in $\mathrm{BV}(\Omega; \mathbb{R}^m)$ can be found in the area-strict convergence. This intermediate notion of convergence is weaker than convergence in norm, but stronger than weak* convergence. In particular, we will conclude that the infimum of $\mathcal{F}$ over $\mathrm{W}^{1,1}(\Omega; \mathbb{R}^m)$ and the infimum of $\widetilde{\mathcal{F}}$ over $\mathrm{BV}(\Omega; \mathbb{R}^m)$ agree.

If we furthermore assume (linear-growth) coercivity on f, then we obtain a uniform $\mathrm{W}^{1,1}$-norm bound on any minimizing sequence $(u_j)_j \subset \mathrm{W}^{1,1}(\Omega; \mathbb{R}^m)$ for $\mathcal{F}$. Invoking the compactness theorem for the weak* convergence in $\mathrm{BV}(\Omega; \mathbb{R}^m)$, we se-

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lect a (not relabeled) subsequence with $u_j \overset{*}{\rightharpoonup} u_* \in \mathrm{BV}(\Omega;\mathbb{R}^m)$. One observes that $(u_j)_j \subset \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ is still a minimizing sequence for $\widetilde{\mathcal{F}}$. In order to conclude that u_* is a minimizer of $\widetilde{\mathcal{F}}$, and hence may be called a "generalized minimizer" of the original functional $\mathcal{F}$, we thus need to investigate the BV-weak* lower semicontinuity of our extended functional $\widetilde{\mathcal{F}}$. This will be our second task. Here, we focus on the "classical" approach to these questions, whereas the next chapter presents a more abstract perspective.

Finally, we will also consider a sharp-interface problem, continuing our study of Γ-convergence from Chapter 9.

14.1 Extension of Functionals

Recall that for a sequence $(u_j)_j \subset \mathrm{BV}(\Omega;\mathbb{R}^m)$ we say that $u_j \to u$ area-strictly if $u_j \to u$ in L^1, $Du_j \overset{*}{\rightharpoonup} Du$ in $\mathcal{M}(\Omega;\mathbb{R}^{m\times d})$, and $\langle Du_j\rangle(\Omega) \to \langle Du\rangle(\Omega)$, where $\langle\bullet\rangle$ is the (reduced) area functional, which was defined in (13.5).

If one wants to determine the extension of the functional $\mathcal{F}$ from the space $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ to the space $\mathrm{BV}(\Omega;\mathbb{R}^m)$, one needs to deal with concentrations. In order to understand the behavior of the (linear-growth) integrand f "at infinity", we define the following notions: Let $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$ be a Carathéodory integrand with linear growth. The **(strong) recession function** $f^\infty\colon \overline{\Omega}\times\mathbb{R}^N \to \mathbb{R}$ is

$$f^\infty(x,A) := \lim_{\substack{x'\to x\\ A'\to A\\ t\to\infty}} \frac{f(x',tA')}{t}, \qquad (x,A)\in\overline{\Omega}\times\mathbb{R}^N, \tag{14.1}$$

if this limit exists (in $\mathbb{R}$) for all such x, A. The function f^∞ is easily seen to be positively 1-homogeneous in A, that is,

$$f^\infty(x,\alpha A) = \alpha f^\infty(x,A) \qquad \text{for all } \alpha\ge 0,\ (x,A)\in\overline{\Omega}\times\mathbb{R}^N.$$

If f is Lipschitz continuous in its second argument and the Lipschitz constant is uniform with respect to x, then the definition of f^∞ simplifies to

$$f^\infty(x,A) = \lim_{\substack{x'\to x\\ t\to\infty}} \frac{f(x',tA)}{t}.$$

Clearly, the recession function f^∞ does not necessarily exist (not even for quasiconvex f, see Theorem 2 of [245]). However, for f with linear growth, we can always define the **upper weak recession function** $f^\#\colon \overline{\Omega}\times\mathbb{R}^N \to \mathbb{R}$ and the **lower weak recession function** $f_\#\colon \overline{\Omega}\times\mathbb{R}^N \to \mathbb{R}$, respectively, by

$$f^\#(x,A) := \limsup_{\substack{x'\to x\\ A'\to A\\ t\to\infty}} \frac{f(x',tA')}{t}, \qquad f_\#(x,A) := \liminf_{\substack{x'\to x\\ A'\to A\\ t\to\infty}} \frac{f(x',tA')}{t},$$

where $x \in \overline{\Omega}$, $A \in \mathbb{R}^{m\times d}$. We remark that here we employed the usual definition of lim sup, lim inf for functions on metric spaces, i.e.,

$$f^{\#}(x,A) := \lim_{\varepsilon\downarrow 0} \sup R_\varepsilon, \qquad f_{\#}(x,A) := \lim_{\varepsilon\downarrow 0} \inf R_\varepsilon,$$

where

$$R_\varepsilon := \left\{ \frac{f(x',tA')}{t} \ : \ 0<|x'-x|<\varepsilon,\ 0<|A'-A|<\varepsilon,\ t>\frac{1}{\varepsilon} \right\}.$$

It is immediate that $f^{\#}$, $f_{\#}$ are finite and positively 1-homogeneous in their second argument. If f is Lipschitz continuous in its second argument and the Lipschitz constant is uniform with respect to x, then

$$f^{\#}(x,A) = \limsup_{\substack{x'\to x\\ t\to\infty}} \frac{f(x',tA)}{t}, \qquad f_{\#}(x,A) = \liminf_{\substack{x'\to x\\ t\to\infty}} \frac{f(x',tA)}{t},$$

where $x \in \overline{\Omega}$, $A \in \mathbb{R}^N$.

The extension question for $\mathcal{F}$ is then settled by the following result.

Theorem 14.1 *Let $f\colon \Omega\times\mathbb{R}^{m\times d} \to \mathbb{R}$ be a continuous integrand with linear growth and such that the strong recession function f^∞ exists. Then, the area-strictly continuous extension of the functional*

$$\mathcal{F}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m),$$

to $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ is given by

$$\widetilde{\mathcal{F}}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x + \int_\Omega f^\infty\left(x, \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right) \mathrm{d}|D^s u|(x).$$

In particular, $\widetilde{\mathcal{F}}$ is area-strictly continuous.

In this context recall the Lebesgue–Radon–Nikodým decomposition of Du, given as

$$Du = \nabla u\,\mathcal{L}^d + \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}\,|D^s u|.$$

Remark 14.2 It turns out that the preceding result remains valid with the upper weak recession function $f^{\#}$ in place of f^∞ if f is (jointly) continuous, rank-one convex (or rank-one concave), and moreover the strong recession function exists in all rank-one directions, that is

$$f^{\#}(x,A) = f_{\#}(x,A) = (f(x,\cdot))^{\#}(A) = \limsup_{\substack{A'\to A\\ t\to\infty}} \frac{f(x,tA')}{t}$$

for all $(x, A) \in \Omega \times \mathbb{R}^{m\times d}$ such that rank $A \le 1$. The proof of this extension is the task of Problem 14.4; also see Problem 14.5 for a condition on how to verify the above equality between recession functions.

Proof of Theorem 14.1. We will prove the general statement that for any sequence $(\mu_j)_j \subset \mathcal{M}(\Omega;\mathbb{R}^N)$ with $\mu_j \to \mu$ **area-strictly**, i.e., $\mu_j \overset{*}{\rightharpoonup} \mu$ and $\langle \mu_j\rangle(\Omega) \to \langle\mu\rangle(\Omega)$, it holds that

$$\int_\Omega f\left(x, \frac{\mathrm{d}\mu_j}{\mathrm{d}\mathcal{L}^d}(x)\right)\mathrm{d}x + \int_\Omega f^\infty\left(x, \frac{\mathrm{d}\mu_j^s}{\mathrm{d}|\mu_j^s|}(x)\right)\mathrm{d}|\mu_j^s|(x)$$
$$\to \int_\Omega f\left(x, \frac{\mathrm{d}\mu}{\mathrm{d}\mathcal{L}^d}(x)\right)\mathrm{d}x + \int_\Omega f^\infty\left(x, \frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}(x)\right)\mathrm{d}|\mu^s|(x).$$

This can be seen as an extension of Reshetnyak's Continuity Theorem 13.3. Together with Lemma 13.7, this assertion then immediately implies the theorem.

To see the claim, for an integrand f as in the statement of the theorem define the **perspective integrand** $Pf\colon \Omega \times \mathbb{R}^N \times \mathbb{R} \to \mathbb{R}$ by

$$Pf(x, A, s) := \begin{cases} sf(x, s^{-1}A) & \text{if } s \neq 0,\\ f^\infty(x, A) & \text{if } s = 0,\end{cases} \qquad (x, A, s) \in \Omega \times \mathbb{R}^N \times \mathbb{R}.$$

Clearly, Pf is continuous and $(A, s) \mapsto Pf(x, A, s)$ is positively 1-homogeneous (jointly in (A, s)) for fixed $x \in \Omega$. Likewise, for $\mu \in \mathcal{M}(\Omega;\mathbb{R}^N)$ we define $P^*\mu \in \mathcal{M}(\Omega;\mathbb{R}^N \times \mathbb{R})$ via

$$P^*\mu(B) := \big(\mu(B), \mathcal{L}^d(B)\big) \qquad \text{for any Borel set } B \subset \Omega.$$

With the Lebesgue–Radon–Nikodým decomposition $\mu = a\mathcal{L}^d + \mu^s$ of μ we have

$$P^*\mu = (a, 1)\,\mathcal{L}^d + (1, 0)\,\mu^s.$$

Then,

$$|P^*\mu|(\Omega) = \int_\Omega \langle a(x)\rangle\,\mathrm{d}x + |\mu^s|(\Omega) = \langle\mu\rangle(\Omega)$$

and hence the area-strict convergence of a sequence $(\mu_j)_j \subset \mathcal{M}(\Omega;\mathbb{R}^N)$ is equivalent to the strict convergence of $(P^*\mu_j)_j \subset \mathcal{M}(\Omega;\mathbb{R}^N \times \mathbb{R})$. Thus, by Reshetnyak's Continuity Theorem 13.3, the mapping

$$\mu \mapsto \mathcal{G}[\mu] := \int_\Omega Pf\left(x, \frac{\mathrm{d}P^*\mu}{\mathrm{d}|P^*\mu|}(x)\right)\mathrm{d}|P^*\mu|(x)$$

is area-strictly continuous. Since

$$\begin{aligned}\mathcal{G}[\mu] &= \int_\Omega Pf\left(x, \frac{(a(x),1)}{\langle a(x)\rangle}\right)\langle a(x)\rangle\,\mathrm{d}x + \int_\Omega Pf\left(x, \frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}(x), 0\right)\mathrm{d}|\mu^s|(x)\\ &= \int_\Omega f\left(x, \frac{\mathrm{d}\mu}{\mathrm{d}\mathcal{L}^d}(x)\right)\mathrm{d}x + \int_\Omega f^\infty\left(x, \frac{\mathrm{d}\mu^s}{\mathrm{d}|\mu^s|}(x)\right)\mathrm{d}|\mu^s|(x),\end{aligned}$$

the claim follows. □

Example 14.3 Let $w \in \mathrm{BV}_{\mathrm{loc}}(\mathbb{R})$ be the (shifted) staircase function

$$w(x) := \left\lfloor x - \frac{1}{2}\right\rfloor, \qquad x \in \mathbb{R}.$$

For the sequence $(u_j)_j \subset \mathrm{BV}(0,1)$ defined as $u_j(x) := w(jx)/j$ for $x \in (0,1)$, we have $u_j \overset{*}{\rightharpoonup} u$ with $u(x) = x$. Moreover, one computes

$$|Du_j|((0,1)) = 1, \qquad \langle Du_j\rangle((0,1)) = 2,$$

where $\langle\bullet\rangle$ is the reduced area-functional defined in (13.5) (note that the derivatives Du_j are purely singular), but

$$|Du|((0,1)) = 1, \qquad \langle Du\rangle((0,1)) = \sqrt{2},$$

whereby $u_j \to u$ strictly but the u_j do not converge area-strictly to u. Thus, strict continuity fails for $\widetilde{\mathcal{F}} := \langle\bullet\rangle(\Omega)$ (the extended functional for the integrand $f(A) := \sqrt{1+|A|^2}$, which has the strong recession function $f^\infty(A) = |A|$). We conclude that the area-strict convergence cannot be replaced with mere strict convergence in the above theorem.

Example 14.4 In general, the existence of the *strong* recession function cannot be dispensed with: On $\Omega := (-1,1)$ and $\mathbb{R}^{m\times d} = \mathbb{R}$ define $f(a) := |a|\sin a$, $a \in \mathbb{R}$. Then,

$$f^\#(\pm 1) = \limsup_{t\to\infty}\frac{f(\pm t)}{t} = 1, \qquad f_\#(\pm 1) = \liminf_{t\to\infty}\frac{f(\pm t)}{t} = -1.$$

Now set

$$u_j(x) := \beta_j x \mathbb{1}_{(0,\beta_j^{-1})}(x) + \mathbb{1}_{(\beta_j^{-1},1)}(x), \qquad \text{where} \qquad \beta_j := 2\pi j - \pi/2.$$

Denote by $\widetilde{\mathcal{F}}^\#$ the functional $\widetilde{\mathcal{F}}$ with f^∞ replaced by $f^\#$. We compute

$$u_j' = \beta_j \mathbb{1}_{(0,\beta_j^{-1})} \to \delta_0 \quad \text{area-strictly,}$$

whereby $u_j \to \mathbb{1}_{(0,1)}$ area-strictly in BV, but

$$\mathcal{F}[u_j] = \widetilde{\mathcal{F}}^\#[u_j] = \int_0^{\beta_j^{-1}} \beta_j \sin\beta_j\,\mathrm{d}x = -1 \neq 1 = \widetilde{\mathcal{F}}^\#[\mathbb{1}_{(0,1)}].$$

Further, one can convince oneself with a similar example that also for $f_\#$ we do not have the extension property. Note that this example is not in contradiction to Remark 14.2.

14.2 Lower Semicontinuity

For a Carathéodory integrand $f\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with linear growth we now (re)define

$$\mathcal{F}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x + \int_\Omega f^\#\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x),$$

where $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$. Note that here we use the upper weak recession function, which always exists. If even the strong recession function f^∞ exists, then by Theorem 14.1 the $\mathcal{F}$ so defined is the area-strictly continuous extension of our original $\mathcal{F}$ (defined on $\mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$). However, we will show in the next section that, even without the existence of the strong recession function, the above $\mathcal{F}$ is always the *relaxation* of the extended-valued functional

$$\mathcal{F}^\infty[u] := \begin{cases} \displaystyle\int_\Omega f(\nabla u(x))\,\mathrm{d}x & \text{if } u \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m),\\ +\infty & \text{if } u \in (\mathrm{BV}\setminus \mathrm{W}^{1,1})(\Omega;\mathbb{R}^m).\end{cases}$$

Thus, our choice of extension for $\mathcal{F}$ is justified even if the strong recession function f^∞ does not exist.

The following is the fundamental lower semicontinuity theorem in BV:

Theorem 14.5 (Ambrosio–Dal Maso, Fonseca–Müller 1992 [16, 144]) *Assume that $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ is a continuous and quasiconvex integrand with linear growth. Then, $\mathcal{F}$ is weakly* lower semicontinuous on the space $\mathrm{BV}(\Omega;\mathbb{R}^m)$.*

In the following we only prove Theorem 14.5 under the additional assumption that the strong recession function f^∞ exists; see Remark 14.15 below for the general case. We start with the following simple gluing lemma.

Lemma 14.6 *Let $U, V \subset \mathbb{R}^d$ be bounded Lipschitz domains with $U \Subset V$ and let $u \in \mathrm{BV}(U;\mathbb{R}^m)$, $v \in \mathrm{BV}(V\setminus\overline{U};\mathbb{R}^m)$. For*

$$w := u\mathbb{1}_U + v\mathbb{1}_{V\setminus\overline{U}} \in \mathrm{BV}(V;\mathbb{R}^m)$$

there exists a sequence $(w_j)_j \subset (\mathrm{W}^{1,1}\cap \mathrm{C}^\infty)(V;\mathbb{R}^m)$ that converges area-strictly to w on V. Moreover, for all continuous linear-growth integrands $f\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ such that the strong recession function f^∞ exists, it holds that

$$\lim_{j\to\infty} \mathcal{F}[w_j;V] = \mathcal{F}[u;U] + \mathcal{F}[v;V\setminus\overline{U}] + \int_{\partial U} f^\infty\left(\frac{u-v}{|u-v|}\otimes n_U\right)|u-v|\,\mathrm{d}\mathcal{H}^{d-1},$$

where the values of u and v on ∂U are to be understood as one-sided traces (that is, $u|_{\partial U}$ and $v|_{\partial(V\setminus\overline{U})}$) and n_U denotes the unit inner normal on ∂U.

Proof. By the BV-theory (see Section 13.3), we have that $w \in \mathrm{BV}(V;\mathbb{R}^m)$ and satisfies

$$Dw = Du\llcorner U + Dv\llcorner (V \setminus \overline{U}) + (u-v)\otimes n_U\, \mathcal{H}^{d-1}\llcorner \partial U.$$

Hence, the conclusion follows immediately from Theorem 14.1 in conjunction with Lemma 13.7. □

Next, we present a characterization of quasiconvexity in BV.

Lemma 14.7 *Let $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ be a Borel function with linear growth and such that the strong recession function h^∞ exists. Then, h is quasiconvex if and only if for one (hence all) bounded Lipschitz domains $D \subset \mathbb{R}^d$, all $u \in \mathrm{BV}(D;\mathbb{R}^m)$, and all affine maps $a\colon \mathbb{R}^d \to \mathbb{R}^m$ it holds that*

$$\begin{aligned} |D|h(\nabla a) \le \int_D h(\nabla u)\,\mathrm{d}x + \int_D h^\infty\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}\right)\mathrm{d}|D^s u| \\ + \int_{\partial D} h^\infty\left(\frac{u-a}{|u-a|}\otimes n_D\right)|u-a|\,\mathrm{d}\mathcal{H}^{d-1}, \end{aligned}$$

where n_D denotes the unit inner normal on ∂D and $u = u|_{\partial D}$ is the inner trace in the last integral.

Proof. First, we notice that for $u \in \mathrm{W}^{1,\infty}_a(D;\mathbb{R}^m) \subset \mathrm{BV}(D;\mathbb{R}^m)$ the inequality in the lemma reads as

$$|D|h(\nabla a) \le \int_D h(\nabla u)\,\mathrm{d}x,$$

which is equivalent to the quasiconvexity of h.

Turning to the other implication, we assume that h is quasiconvex. Let $r \in (0,1)$ and define the bounded Lipschitz domain

$$D_r = \left\{ x \in \mathbb{R}^d \;:\; \operatorname{dist}(x;D) < r \right\}.$$

Take a sequence $r_n \downarrow 0$, so that

$$D \Subset D_{r_n} \qquad \text{and} \qquad \bigcap_{n\in\mathbb{N}} D_{r_n} = \overline{D}.$$

Via Lemma 14.6 with $U := D$, $V := D_{r_n}$, $u \in \mathrm{BV}(U,\mathbb{R}^m)$, and $v = a$ we find maps $w_n \in W^{1,\infty}_a(D_{r_n};\mathbb{R}^m)$ $(n \in \mathbb{N})$ with

$$\begin{aligned} \int_{D_{r_n}} h(\nabla w_n)\,\mathrm{d}x \le \int_D h(\nabla u)\,\mathrm{d}x + \int_D h^\infty\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}\right)\mathrm{d}|D^s u| \\ + \int_{\partial D} h^\infty\left(\frac{u-a}{|u-a|}\otimes n_D\right)|u-a|\,\mathrm{d}\mathcal{H}^{d-1} \\ + \int_{D_{r_n}\setminus\overline{D}} h(\nabla a)\,\mathrm{d}x + \frac{1}{n}. \end{aligned}$$

The quasiconvexity of h implies

$$|D_{r_n}|h(\nabla a) \le \int_{D_{r_n}} h(\nabla w_n)\,\mathrm{d}x.$$

On the other hand,

$$\int_{D_{r_n}\setminus\overline{D}} h(\nabla a)\,\mathrm{d}x \le M(1+|\nabla a|)|D_{r_n}\setminus\overline{D}| \to 0 \qquad \text{as } n\to\infty.$$

Thus, letting $n\to\infty$,

$$\begin{aligned}|D|h(\nabla a) \le \int_D h(\nabla u)\,\mathrm{d}x + \int_D h^\infty\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}\right)\mathrm{d}|D^s u| \\ + \int_{\partial D} h^\infty\left(\frac{u-a}{|u-a|}\otimes n_D\right)|u-a|\,\mathrm{d}\mathcal{H}^{d-1},\end{aligned}$$

which is the claim. □

We will analyze $\mathcal{F}$ through the "blow-up" behavior of the auxiliary functional

$$\mathcal{J}[u;D] := \inf\left\{\mathcal{F}[w;D] \;:\; w\in \mathrm{BV}(D;\mathbb{R}^m) \text{ with } w|_{\partial D} = u|_{\partial D}\right\}, \tag{14.2}$$

where $D\subset\Omega$ is a Lipschitz subdomain of Ω and

$$\mathcal{F}[u;D] := \int_D f(\nabla u(x))\,\mathrm{d}x + \int_D f^\infty\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x)$$

for $u\in\mathrm{BV}(\Omega;\mathbb{R}^m)$. We start with the following "maximum principle" for $\mathcal{J}$:

Lemma 14.8 *For all $u,v\in\mathrm{BV}(\Omega;\mathbb{R}^m)$ and all convex Lipschitz subdomains $D\subset\Omega$ it holds that*

$$\left|\mathcal{J}[u;D]-\mathcal{J}[v;D]\right| \le M\int_{\partial D}|u-v|\,\mathrm{d}\mathcal{H}^{d-1},$$

where $M>0$ is the linear-growth constant of f.

Proof. For reasons of notational simplicity we assume that $D=B$ is a ball. The general proof proceeds along the same lines.

Since the trace spaces for $\mathrm{BV}(B;\mathbb{R}^m)$ and $(\mathrm{W}^{1,1}\cap\mathrm{C}^\infty)(B;\mathbb{R}^m)$ are both equal to $\mathrm{L}^1(\partial B;\mathbb{R}^m)$ and since $\mathcal{J}$ only depends on its argument's boundary values, we may without loss of generality assume that $u,v\in(\mathrm{W}^{1,1}\cap\mathrm{C}^\infty)(B;\mathbb{R}^m)$. Below we will establish that

$$\mathcal{J}[u;B] \le \mathcal{F}[w;B] + M\int_{\partial B}|u-v|\,\mathrm{d}\mathcal{H}^{d-1} \tag{14.3}$$

for all $w\in\mathrm{BV}(B;\mathbb{R}^m)$ with $w|_{\partial B}=v|_{\partial B}$. The conclusion of the lemma then follows by taking the infimum over all such w and by switching the roles of u and v.

To show (14.3) we may suppose that $w \in (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(B;\mathbb{R}^m)$ with $w|_{\partial B} = v|_{\partial B}$ by virtue of Theorem 14.1 in conjunction with Lemma 13.7. Let $B = B_R(x_0)$ for some $x_0 \in \mathbb{R}^d$ and $R > 0$. For $\delta \in (0,1)$ denote the concentric subball with radius δR by $B_\delta := B_{\delta R}(x_0)$. Let $w_\delta \in \mathrm{BV}(B;\mathbb{R}^m)$ be defined as

$$w_\delta(x) := \begin{cases} w(x) & \text{if } x \in B_\delta, \\ u(x) & \text{if } x \in B \setminus \overline{B_\delta}, \end{cases}$$

for which $w_\delta|_{\partial B} = u|_{\partial B}$ and

$$Dw_\delta = Dw \llcorner B_\delta + Du \llcorner (B \setminus \overline{B_\delta}) + (w-u) \otimes n_{B_\delta}\, \mathcal{H}^{d-1} \llcorner \partial B_\delta,$$

where n_{B_δ} is the unit inner normal on ∂B_δ. Using the linear growth of f (with growth constant $M > 0$), we estimate

$$\begin{aligned} \mathcal{J}[u;B] &\le \mathcal{F}[w_\delta;B] \\ &= \mathcal{F}[w;B_\delta] + \mathcal{F}[u;B \setminus \overline{B_\delta}] \\ &\quad + \int_{\partial B_\delta} f^\infty\left(\frac{w-u}{|w-u|} \otimes n_{B_\delta}\right)|w-u|\,\mathrm{d}\mathcal{H}^{d-1} \\ &\le \mathcal{F}[w;B] + M\big(2\mathcal{L}^d + |Dw| + |Du|\big)(B \setminus \overline{B_\delta}) \\ &\quad + M \int_{\partial B_\delta} |w-u|\,\mathrm{d}\mathcal{H}^{d-1}. \end{aligned}$$

As $\delta \uparrow 1$, the second term vanishes and

$$\begin{aligned} \int_{\partial B_\delta} |w(x)-u(x)|\,\mathrm{d}\mathcal{H}^{d-1}(x) &= \delta^{d-1} \int_{\partial B} |w(\delta y)-u(\delta y)|\,\mathrm{d}\mathcal{H}^{d-1}(y) \\ &\to \int_{\partial B} |w(y)-u(y)|\,\mathrm{d}\mathcal{H}^{d-1}(y) \end{aligned}$$

by the continuity of the trace operator. Thus, we conclude (14.3). □

Next, we have the following local lower semicontinuity property of $\mathcal{J}$:

Lemma 14.9 *Let $u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{BV}(\Omega;\mathbb{R}^m)$ and $|Du_j| \overset{*}{\rightharpoonup} \Lambda$ in $\mathcal{M}^+(\overline{\Omega})$. Then, for every Lipschitz subdomain $D \subset \Omega$ with $\Lambda(\partial D) = 0$ it holds that*

$$\mathcal{J}[u;D] \le \liminf_{j\to\infty} \mathcal{F}[u_j;D].$$

Proof. As before, we only consider the case of a ball, $D = B_R(x_0) \subset \Omega$ for some $x_0 \in \Omega$, $R > 0$. In view of Theorem 14.1 in conjunction with Lemma 13.7 we may without loss of generality suppose that $u_j \in (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^m)$.

Take a sequence of radii $r_k \uparrow R$ such that $\Lambda(\partial B_{r_k}(x_0)) = 0$ and, possibly after selecting a subsequence (not relabeled),

$$\int_{\partial B_{r_k}(x_0)} |u_j - u| \, \mathrm{d}\mathcal{H}^{d-1} \to 0 \qquad \text{as } j \to \infty$$

for all $k \in \mathbb{N}$. The existence of such radii can be seen as follows: By a curvilinear version of Fubini's theorem (i.e., the coarea formula),

$$\int_0^R \int_{\partial B_r(x_0)} |u_j - u| \, \mathrm{d}\mathcal{H}^{d-1} \, \mathrm{d}r = \int_{B_R(x_0)} |u_j - u| \, \mathrm{d}x \to 0.$$

Thus, we may select a subsequence of the u_j's such that

$$\int_{\partial B_r(x_0)} |u_j - u| \, \mathrm{d}\mathcal{H}^{d-1} \to 0 \qquad \text{for a.e. } r \in (0, R).$$

Also observing that the set of all radii $r \in (0, R)$ with the property $\Lambda(\partial B_r(x_0)) \neq 0$ is at most countable because Λ is a finite measure (see Problem 13.2), we then choose appropriate r_k satisfying the two properties above.

Owing to the inequality $|Du| \leq \mathrm{w}^*\text{-}\lim_{j\to\infty} |Du_j| = \Lambda$, it also holds that $|Du|(\partial B_{r_k}(x_0)) = 0$ and therefore the inner and outer one-sided traces on $\partial B_{r_k}(x_0)$ coincide, that is,

$$u|_{\partial B_{r_k}(x_0)} = u|_{\partial(\Omega \setminus \overline{B_{r_k}(x_0)})} \quad \mathcal{H}^{d-1}\text{-a.e.}$$

We now fix k, set $B := B_R(x_0)$, $B_k := B_{r_k}(x_0)$, and define

$$w_j(x) := \begin{cases} u_j(x) & \text{if } x \in B_k, \\ u(x) & \text{if } x \in B \setminus \overline{B_k}, \end{cases}$$

which lies in $\mathrm{BV}(B; \mathbb{R}^m)$ and satisfies $w_j|_{\partial B} = u|_{\partial B}$ as well as

$$Dw_j = Du_j \llcorner B_k + Du \llcorner (B \setminus \overline{B_k}) + (u_j - u) \otimes n_{B_k} \, \mathcal{H}^{d-1} \llcorner \partial B_k.$$

From the additivity of $\mathcal{F}[u; \bullet]$ for disjoint sets and the linear growth of f with growth constant $M > 0$ we see that

$$\begin{aligned} \mathcal{J}[u; B] &\leq \mathcal{F}[w_j; B] \\ &= \mathcal{F}[u_j; B_k] + \mathcal{F}[u; B \setminus \overline{B}_k] \\ &\qquad + \int_{\partial B_k} f^\infty\left(\frac{u_j - u}{|u_j - u|} \otimes n_{B_k}\right) |u_j - u| \, \mathrm{d}\mathcal{H}^{d-1} \\ &\leq \mathcal{F}[u_j; B] + M\big(2\mathcal{L}^d + |Du| + |Du_j|\big)(B \setminus \overline{B_k}) \\ &\qquad + M \int_{\partial B_k} |u_j - u| \, \mathrm{d}\mathcal{H}^{d-1}. \end{aligned}$$

Taking the lower limit as $j \to \infty$ (and keeping k fixed), we deduce from $\Lambda(\partial B_k) = 0$ and $|Du| \le \Lambda$ that

$$\mathcal{J}[u; B] \le \liminf_{j\to\infty} \mathcal{F}[u_j; B] + 2M(\mathcal{L}^d + \Lambda)(\overline{B} \setminus B_k).$$

Now let $k \to \infty$. The claim of the lemma follows for our subsequence of u_j's since $\Lambda(\overline{B} \setminus B_k) \to \Lambda(\partial B) = 0$.

It remains to prove the result for our original sequence $(u_j)_j$. For this, select a subsequence $j(l)$ such that

$$\lim_{l\to\infty} \mathcal{F}[u_{j(l)}; B] = \liminf_{j\to\infty} \mathcal{F}[u_j; B].$$

Then, by the proof above, we get for a further subsequence (whose indices are still denoted by $j(l)$) that

$$\mathcal{J}[u; B] \le \liminf_{l\to\infty} \mathcal{F}[u_{j(l)}; B] = \liminf_{j\to\infty} \mathcal{F}[u_j; B].$$

This finishes the proof of the lemma. □

Next, we investigate the fine structure of $\mathcal{F}$ around regular and singular points. We will show below that for $(\mathcal{L}^d + |Du|)$-almost every $x_0 \in \Omega$ and for every $\varepsilon > 0$ there exists an $r_0(x_0) > 0$ such that

$$\mathcal{F}[u; U(x_0, r)] \le \mathcal{J}[u; U(x_0, r)] + \varepsilon(\mathcal{L}^d + |Du|)(U(x_0, r)) \tag{14.4}$$

for almost all $r \in (0, r_0(x_0))$, where $U(x_0, r) \subset \Omega$ is an open convex set that contains x_0 and satisfies

$$(\mathcal{L}^d + |Du| + \Lambda)(\partial U(x_0, r)) = 0, \qquad B_r(x_0) \subset U(x_0, r) \subset B_{\kappa r}(x_0)$$

for a fixed constant $\kappa \ge 1$ and Λ as defined in Lemma 14.9. In fact, $U(x_0, r)$ will either be the open ball $B_r(x_0)$ or a cube with side length $2r$; hence we may choose $\kappa = \sqrt{d}$.

Lemma 14.10 *The estimate* (14.4) *holds for* $\mathcal{L}^d$*-almost every point* $x_0 \in \Omega$ *with* $U(x_0, r) = B_r(x_0)$.

Proof. Let $x_0 \in \Omega$ be such that

(a) u is approximately differentiable at x_0 (see Section 13.3);
(b) $\lim_{r\downarrow 0} \dfrac{Du(B_r(x_0))}{\omega_d r^d} = \dfrac{\mathrm{d}Du}{\mathrm{d}\mathcal{L}^d}(x_0) = \nabla u(x_0)$;
(c) $\lim_{r\downarrow 0} \dfrac{|Du|(B_r(x_0))}{\omega_d r^d} = \dfrac{\mathrm{d}|Du|}{\mathrm{d}\mathcal{L}^d}(x_0) = |\nabla u(x_0)|$;
(d) x_0 is an $\mathcal{L}^d$-Lebesgue point of ∇u.

By the results of Section 13.3, $\mathcal{L}^d$-almost every $x_0 \in \Omega$ has these properties.

The maps

$$u_r(y) := \frac{u(x_0 + ry) - \widetilde{u}(x_0)}{r}, \qquad y \in B_1,$$

where we denote the precise representative of u by $\widetilde{u}$ (see Section 13.3), satisfy

$$u_r \to u_0 \quad \text{in } \mathrm{L}^1, \qquad Du_r \to Du_0 \quad \text{strictly} \qquad \text{as } r \downarrow 0$$

with

$$u_0(y) := \nabla u(x_0)y, \qquad y \in B_1.$$

Indeed, the L^1-strong convergence follows from the approximate differentiability after a change of variables. The strict convergence of Du_r to Du_0 can be seen by calculating, using (13.8),

$$\lim_{r\downarrow 0} |Du_r|(B_1) = \omega_d \cdot \lim_{r\downarrow 0} \frac{|Du|(B_r(x_0))}{|B_r(x_0)|} = \omega_d |\nabla u(x_0)| = |Du_0|(B_1).$$

From Lemma 5.9 we know that the integrand f is (globally) Lipschitz continuous, say with Lipschitz constant $L > 0$. Since x_0 is a Lebesgue point of ∇u and since $r^{-d}|D^s u|(B_r(x_0)) \to 0$ by the approximate differentiability of u in x_0 (see (13.6)), we get

$$\begin{aligned}
&\big|\mathcal{F}[u; B_r(x_0)] - \mathcal{F}[u_0; B_r(x_0)]\big| \\
&\quad \le L \int_{B_r(x_0)} |\nabla u(y) - \nabla u(x_0)| \,\mathrm{d}y + M|D^s u|(B_r(x_0)) \\
&\quad \le \frac{\varepsilon}{2}|B_r(x_0)|
\end{aligned}$$

for $r > 0$ sufficiently small. Lemma 14.7 (i.e., quasiconvexity) thus implies for all $w \in \mathrm{BV}(B_r(x_0); \mathbb{R}^m)$ with $w = u_0$ on $\partial B_r(x_0)$ that

$$\mathcal{F}[u; B_r(x_0)] \le \int_{B_r(x_0)} f(\nabla u_0) \,\mathrm{d}x + \frac{\varepsilon}{2}|B_r(x_0)| \le \mathcal{F}[w; B_r(x_0)] + \frac{\varepsilon}{2}|B_r(x_0)|.$$

Taking the infimum over all such w, we arrive at

$$\mathcal{F}[u; B_r(x_0)] \le \mathcal{J}[u_0; B_r(x_0)] + \frac{\varepsilon}{2}|B_r(x_0)|. \tag{14.5}$$

On the other hand, one may prove the transformation rule

$$\mathcal{J}\left[y \mapsto \frac{u(x_0 + ry)}{r}; D\right] = \frac{1}{r^d}\mathcal{J}[u; x_0 + rD] \tag{14.6}$$

for all $x_0 \in \Omega$, all $r > 0$, and all Lipschitz subdomains $D \subset \Omega$. An easy way to see this is by approximating u with smooth maps via Lemma 13.7 and employing a change of variables. Consequently,

$$\mathcal{J}[u_0; B_r(x_0)] = r^d \mathcal{J}[u_0; B_1], \qquad \mathcal{J}[u; B_r(x_0)] = r^d \mathcal{J}[u_r; B_1].$$

We thus infer from Lemma 14.8 together with the strict convergence of the Du_r to Du_0 and the strict continuity of the trace operator (recalled in Section 13.3) that

$$\left|\mathcal{J}[u_0; B_r(x_0)] - \mathcal{J}[u; B_r(x_0)]\right| \le \frac{\varepsilon}{2}|B_r(x_0)|$$

for r sufficiently small. Together with (14.5), the estimate (14.4) follows with $U(x_0, r) := B_r(x_0)$ after selecting $r > 0$ such that $(\mathcal{L}^d + |Du| + \Lambda)(\partial B_r(x_0)) = 0$. □

Lemma 14.11 *The estimate* (14.4) *holds for* $|D^s u|$*-almost every point* $x_0 \in \Omega$ *with* $U(x_0, r)$ *a (rotated) cube.*

Proof. We claim that (14.4) holds for all $x_0 \in \Omega$ such that

(a) $\dfrac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x_0) = a \otimes n$ for some $a \in \mathbb{S}^{m-1}$, $n \in \mathbb{S}^{d-1}$;

(b) $\alpha_r := \dfrac{|Du|(Q_n(x_0, r))}{r^d} \to \infty$ as $r \downarrow 0$, where $Q_n(x_0, r)$ is a open cube with midpoint $x_0 \in \Omega$, side-length $2r$, and two faces orthogonal to n (and with fixed orientation).

By virtue of Alberti's Rank-One Theorem 13.8, the Besicovitch Differentiation Theorem A.24, and (13.7), these properties hold for $|D^s u|$-almost every $x_0 \in \Omega$. During the course of the proof we will require further properties of x_0, but every time the exceptional set will be $|D^s u|$-negligible. Also, we need to work with concrete sequences $r_k \downarrow 0$ (as $k \to \infty$) and take (several) subsequences thereof in the course of the proof, but we still always just write "$r \downarrow 0$" to keep the notation manageable.

Step 1. Let

$$u_r(y) := \frac{1}{\alpha_r} \cdot \frac{u(x_0 + ry) - [u]_{Q_n(x_0,r)}}{r}, \qquad y \in Q_n(0,1),\ r > 0,$$

whereby

$$[u_r]_{Q_1} = \fint_{Q_1} u_r \,\mathrm{d}x = 0.$$

One calculates, using (13.8),

$$Du_r(B) = \frac{Du(x_0 + rB)}{r^d \alpha_r} = \frac{Du(x_0 + rB)}{|Du|(Q_n(x_0, r))}$$

for any Borel set $B \subset Q_n(0,1)$. Hence, $|Du_r|(Q_n(0,1)) = 1$. By Corollary 13.9 (a consequence of Alberti's Rank-One Theorem 13.8) and the Poincaré inequality in BV (see Section 13.3), up to taking a subsequence, we may assume that $u_r \overset{*}{\rightharpoonup} u_0$ in $\mathrm{BV}(Q_n(0,1); \mathbb{R}^m)$ for

$$u_0(y) = a\psi(y \cdot n), \qquad y \in Q_n(0,1), \tag{14.7}$$

with a bounded and increasing function $\psi\colon (-1,1)\to\mathbb{R}$; also see Problem 14.8. Now apply Lemma 13.6 on blow-ups without loss of mass to the measure $|Du|$ to see that we may furthermore assume that

(c) Du_r converges strictly to Du_0 on $Q_n(0,1)$ and $|Du_0|(Q_n(0,1))=1$.

So,

$$Du_0(Q_n(0,1)) = a\otimes n.$$

Define the functions $f_r\colon \mathbb{R}^{m\times d}\to\mathbb{R}$, $r>0$, as

$$f_r(A) := \frac{f(\alpha_r A)}{\alpha_r}, \qquad A\in\mathbb{R}^{m\times d},$$

which satisfy $|f_r(A)|\le M(1+|A|)$ for r small enough (with the same M as in the linear growth assumption on f). The f_r are quasiconvex (by Problem 14.6) and

$$f_r(A)\to f^\infty(A) \qquad \text{as } r\downarrow 0$$

by the rank-one convexity of f.

Next, define the auxiliary functionals $\mathcal{F}_r, \mathcal{J}_r$ for Lipschitz subdomains $D\subset\Omega$ and $v\in \mathrm{BV}(D;\mathbb{R}^m)$ via

$$\begin{aligned}\mathcal{F}_r[v;D] &:= \int_D f_r(\nabla v)\,\mathrm{d}x + \int_D f_r^\infty\left(\frac{\mathrm{d}D^s v}{\mathrm{d}|D^s v|}\right)\mathrm{d}|D^s v|,\\ \mathcal{J}_r[v;D] &:= \inf\left\{\mathcal{F}_r[w;D]\ :\ w\in \mathrm{BV}(D;\mathbb{R}^m) \text{ with } w|_{\partial D} = v|_{\partial D}\right\}.\end{aligned}$$

We observe that Lemma 14.8 (for $\mathcal{J}_r$) implies

$$\left|\mathcal{J}_r[u_r;Q_n(0,1)] - \mathcal{J}_r[u_0;Q_n(0,1)]\right| \le M\int_{\partial Q_n(0,1)} |u_r-u_0|\,\mathrm{d}\mathcal{H}^{d-1}\to 0, \tag{14.8}$$

where the convergence follows by the strict convergence of Du_r to Du_0 and the strict continuity of the trace operator in BV.

Step 2. We next claim that

$$\mathcal{J}_r[u_0;Q_n(0,1)] \ge f_r\left(\frac{a\otimes n}{|Q_n(0,1)|}\right)|Q_n(0,1)|. \tag{14.9}$$

To see this, observe first that with ψ from (14.7) we have

$$Du_0 = (a\otimes n)|Du_0| = (a\otimes n)D\psi,$$

whereby, using (c) above, $1=|Du_0|(Q_n(0,1)) = 2^{d-1}|D\psi|(-1,1)$. Thus,

$$|D\psi|(-1,1) = \lim_{t\uparrow 1}\psi(t) - \lim_{t\downarrow -1}\psi(t) = 2^{1-d}.$$

Define the staircase function

$$v(y) := a\psi\left(y \cdot n - 2\left\lfloor \frac{y \cdot n + 1}{2} \right\rfloor\right) + 2^{1-d} a \left\lfloor \frac{y \cdot n + 1}{2} \right\rfloor, \qquad y \in \mathbb{R}^d,$$

and set

$$w_k(y) := \frac{v(ky)}{k}, \qquad y \in \overline{Q_n(0,1)},$$

for $k \in \mathbb{N}$. We have that $w_k \to w$ uniformly in $\overline{Q_n(0,1)}$ as $k \to \infty$, where

$$w(y) := 2^{-d}(a \otimes n)y, \qquad y \in \mathbb{R}^d,$$

because ψ is bounded. Moreover, the trace of w_k on $\partial Q_n(0,1)$ converges to the trace of w and hence Lemma 14.8 implies

$$\left|\mathcal{J}_r[w_k; Q_n(0,1)] - \mathcal{J}_r[w; Q_n(0,1)]\right| \le M \int_{\partial Q_n(0,1)} |w_k - w| \, \mathrm{d}\mathcal{H}^{d-1} \to 0. \quad (14.10)$$

Now disjointly split

$$Q_n(0,1) = Z \cup \bigcup_{l=1}^{k^d} Q_l^{(k)}, \qquad |Z| = 0,$$

in the canonical way into a grid of k^d open cubes with two faces orthogonal to n and with side length $2/k$. From (14.6) we infer that

$$\begin{aligned} \mathcal{J}_r[u_0; Q_n(0,1)] &= \mathcal{J}_r[v; Q_n(0,1)] \\ &= k^d \mathcal{J}_r[w_k; (-1/k, 1/k)^d] \\ &= \sum_{l=1}^{k^d} \mathcal{J}_r[w_k; Q_l^{(k)}] \\ &\ge \mathcal{J}_r[w_k; Q_n(0,1)] \end{aligned}$$

for all $k \in \mathbb{N}$. The last inequality follows since we may combine admissible functions in the definition of $\mathcal{J}_r[w_k; Q_l^{(k)}]$ into an admissible function in the definition of $\mathcal{J}_r[w_k; Q_n(0,1)]$. Now let $k \to \infty$ and employ (14.10) together with Lemma 14.7 to see that

$$\mathcal{J}_r[u_0; Q_n(0,1)] \ge \mathcal{J}_r[w; Q_n(0,1)] \ge |Q_n(0,1)| f_r(\nabla w).$$

The claim (14.9) now follows from $\nabla w \equiv 2^{-d}(a \otimes n)$.

Step 3. From (14.9) and (c) we infer that

$$\mathcal{J}_r[u_0; Q_n(0,1)] \ge f_r\left(\frac{a \otimes n}{|Q_n(0,1)|}\right) |Q_n(0,1)| \to f^\infty(a \otimes n)$$

as $r \downarrow 0$. Combining this with (14.8), we arrive at

$$\liminf_{r\downarrow 0} \mathcal{J}_r[u_r; Q_n(0,1)] \geq f^\infty(a \otimes n).$$

We will show below that

$$\frac{\mathcal{F}[u; Q_n(x_0,r)]}{|Du|(Q_n(x_0,r))} \to f^\infty(a \otimes n) \qquad \text{as } r \downarrow 0 \tag{14.11}$$

for $|D^s u|$-a.e. $x_0 \in \Omega$. Then, for r sufficiently small,

$$\begin{aligned}
\frac{\mathcal{J}[u; Q_n(x_0,r)]}{|Du|(Q_n(x_0,r))} &= r^{-d}\, \mathcal{J}_r[\alpha_r^{-1} u; Q_n(x_0,r)] \\
&= \mathcal{J}_r[u_r; Q_n(0,1)] \\
&\geq f^\infty(a \otimes n) - \frac{\varepsilon}{2} \\
&\geq \frac{\mathcal{F}[u; Q_n(x_0,r)]}{|Du|(Q_n(x_0,r))} - \varepsilon,
\end{aligned}$$

where we have used (14.6) once more. Thus, (14.4) follows with $U(x_0,r) := Q_n(x_0,r)$.

Step 4. To finish the proof of the lemma, it remains to show (14.11) at $|D^s u|$-almost every point $x_0 \in \Omega$. We now additionally assume that x_0 satisfies

(d) $\lim_{r\downarrow 0} \dfrac{|D^a u|(Q_n(x_0,r))}{|Du|(Q_n(x_0,r))} = \dfrac{\mathrm{d}|D^a u|}{\mathrm{d}|Du|}(x_0) = 0$;
(e) x_0 is a $|D^s u|$-Lebesgue point of $\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}$.

This is no restriction because

$$\frac{\mathrm{d}|D^a u|}{\mathrm{d}|Du|}(x_0) \leq \frac{\mathrm{d}|D^a u|}{\mathrm{d}|D^s u|}(x_0) = 0$$

for $|D^s u|$-almost every $x_0 \in \Omega$ by the Besicovitch Differentiation Theorem A.24. So,

$$\frac{\mathrm{d}|D^s u|}{\mathrm{d}|Du|} = \frac{\mathrm{d}|D^a u|}{\mathrm{d}|Du|} + \frac{\mathrm{d}|D^s u|}{\mathrm{d}|D^s u|} = \frac{\mathrm{d}|Du|}{\mathrm{d}|Du|} = 1 \qquad |D^s u|\text{-a.e.}$$

By the Lebesgue point property (e) of x_0,

$$\begin{aligned}
&\frac{1}{|D^s u|(Q_n(x_0,r)} \int_{Q_n(x_0,r))} \left| f^\infty\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}\right) - f^\infty(a \otimes n) \right| \mathrm{d}|D^s u| \\
&\quad \leq \frac{L}{|D^s u|(Q_n(x_0,r)} \int_{Q_n(x_0,r))} \left| \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(y) - \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x_0) \right| \mathrm{d}|D^s u|(y) \\
&\quad \to 0
\end{aligned}$$

as $r \downarrow 0$, where $L > 0$ is the Lipschitz constant of f^∞ (which is the same as the Lipschitz constant of f). As seen above, $\frac{\mathrm{d}|D^s u|}{\mathrm{d}|Du|}(x_0) = 1$ and hence we can replace the denominator in the leading fraction by $|Du|(Q_n(x_0,r))$. Finally,

$$\begin{aligned}&\frac{1}{|Du|(Q_n(x_0,r))}\int_{Q_n(x_0,r)}|f(\nabla u)|\,\mathrm{d}x\\&\quad\le\frac{M}{|Du|(Q_n(x_0,r))}\int_{Q_n(x_0,r)}1+|\nabla u|\,\mathrm{d}x\\&\quad\to 0\end{aligned}$$

as $r\downarrow 0$, since $\alpha_r\to\infty$ and $\frac{\mathrm{d}|D^a u|}{\mathrm{d}|Du|}(x_0)=0$. Together, these assertions yield the claim (14.11). □

Combining the last two lemmas, we get:

Lemma 14.12 *Let $\Lambda\in\mathcal{M}^+(\Omega)$ and $\varepsilon>0$. Then, there exist countably many disjoint convex open sets $(U_k)_k$ (balls or cubes) with $\Lambda(\partial U_k)=0$ that cover Ω up to an $(\mathcal{L}^d+|Du|)$-negligible set such that*

$$\mathcal{F}[u;\Omega]\le\sum_{k=1}^{\infty}\mathcal{J}[u;U_k]+\varepsilon.$$

Proof. We have shown in the last two lemmas that $(\mathcal{L}^d+|D^s u|)$-almost every $x_0\in\Omega$ satisfies (14.4) for sufficiently small radii $r>0$ and some convex open set $U(x_0,r)$. More precisely, there is an $\mathcal{L}^d$-negligible Borel set $N_1\subset\Omega$ and a $|D^s u|$-negligible Borel set N_2 such that (14.4) holds at all $x_0\in\Omega\setminus(N_1\cap N_2)$. Thus, at such x_0 the assumptions of the following Morse covering theorem hold:

Theorem 14.13 *Let $B\subset\mathbb{R}^d$ be a bounded Borel set, $\mu\in\mathcal{M}^+(\mathbb{R}^d)$, $\kappa\ge 1$, and let*

$$\mathcal{C}\subset\big\{x+K\,:\,x\in B,\,K\subset\mathbb{R}^d\text{ convex and compact}\big\}$$

be a family of sets such that for all $x\in B\setminus N$, where $N\subset B$ is a Borel set with $\mu(N)=0$, and for all $r\in(0,r_0(x_0))$ ($r_0(x_0)>0$ given for every $x_0\in B$), there exists an $x+K\in\mathcal{C}$ with

$$B_r(x)\subset x+K\subset B_{\kappa r}(x).$$

Then, there exists a disjoint countable family $\mathcal{C}'\subset\mathcal{C}$ with

$$\mu\Big(B\setminus\bigcup\mathcal{C}'\Big)=0.$$

Via this theorem, which is proved in Theorem 1.147 of [142], we can cover Ω up to a $(\mathcal{L}^d+|Du|)$-negligible set by countably many mutually disjoint sets U_k ($k\in\mathbb{N}$) satisfying (14.4). Note that the original result only holds for the *closures* $\overline{U_k}$, but as $(\mathcal{L}^d+|Du|)(\partial U_k)=0$, this is equivalent.

Clearly, as an integral functional, $\mathcal{F}[u;\bullet]$ is countably additive and vanishes on $(\mathcal{L}^d+|Du|)$-negligible sets. Thus,

$$\mathcal{F}[u;\Omega]\le\sum_{k=1}^{\infty}\mathcal{J}[u;U_k]+\varepsilon(\mathcal{L}^d+|Du|)(\Omega).$$

This immediately yields the claim of the lemma after adjusting ε. □

We can now complete the proof of the main result of this section.

Proof of Theorem 14.5. Let $(u_j)_j \subset \mathrm{BV}(\Omega;\mathbb{R}^m)$ with $u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{BV}(\Omega;\mathbb{R}^m)$. By Lemma 13.7 in conjunction with Theorem 14.1 we may assume that in fact $u_j \in (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^m)$ and that (after selecting a not relabeled subsequence)

$$f(\nabla u_j)\,\mathcal{L}^d \llcorner \Omega \overset{*}{\rightharpoonup} \lambda, \quad |Du_j| \overset{*}{\rightharpoonup} \Lambda \qquad \text{in } \mathcal{M}^+(\overline{\Omega}).$$

By the linear growth assumption (15.2), we have

$$0 \le \lambda \le M(\mathcal{L}^d \llcorner \Omega + \Lambda).$$

Furthermore, f is Lipschitz continuous by Lemma 5.9 with Lipschitz constant $L > 0$, say. We also assume that $|f(0)| \le L$.

Via Lemma 14.12 we construct a countable family of open disjoint sets U_k ($k \in \mathbb{N}$) with $\Lambda(\partial U_k) = 0$ for all $k \in \mathbb{N}$,

$$(\mathcal{L}^d + |Du|)\Big(\Omega \setminus \bigcup_{k\in\mathbb{N}} U_k\Big) = 0,$$

and such that

$$\mathcal{F}[u;\Omega] \le \sum_{k=1}^{\infty} \mathcal{J}[u;U_k] + \varepsilon \le \sum_{k=1}^{\infty} \liminf_{j\to\infty} \mathcal{F}[u_j;U_k] + \varepsilon,$$

where we used Lemma 14.9 for each of the U_k's. Since $\Lambda(\partial U_k) = 0$, it holds that

$$\lim_{j\to\infty} \mathcal{F}[u_j;U_k] = \lambda(U_k).$$

Thus,

$$\mathcal{F}[u;\Omega] \le \lambda\Big(\bigcup_{k\in\mathbb{N}} U_k\Big) + \varepsilon = \lambda(\Omega) + \varepsilon.$$

By standard results in measure theory (see Lemma A.20), we infer that (note $f \ge 0$)

$$\lambda(\Omega) \le \liminf_{j\to\infty} \mathcal{F}[u_j;\Omega],$$

whereby

$$\mathcal{F}[u;\Omega] \le \liminf_{j\to\infty} \mathcal{F}[u_j;\Omega] + \varepsilon.$$

Thus, as $\varepsilon > 0$ was arbitrary, we have proved that $\mathcal{F} = \mathcal{F}[\bullet;\Omega]$ is lower semicontinuous. □

Extending all functions $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ to $\widetilde{u} \in \mathrm{BV}(\widetilde{\Omega};\mathbb{R}^m)$ on a larger domain $\widetilde{\Omega} \supseteq \Omega$ by setting $\widetilde{u}|_{\widetilde{\Omega}\setminus\overline{\Omega}} := w$ for some $w \in \mathrm{BV}(\widetilde{\Omega}\setminus\overline{\Omega};\mathbb{R}^m)$ with $w|_{\partial(\widetilde{\Omega}\setminus\overline{\Omega})} = g$ for $g \in \mathrm{L}^1(\partial\Omega;\mathbb{R}^m)$, and applying Theorem 14.5 in $\widetilde{\Omega}$, we also immediately get the following weak* lower semicontinuity result:

Corollary 14.14 *Assume that $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ is a quasiconvex integrand with linear growth and let $g \in \mathrm{L}^1(\partial\Omega;\mathbb{R}^m)$. Then, the functional*

$$\mathcal{F}_{\mathrm{ext}}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x + \int_\Omega f^{\#}\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x) + \int_{\partial\Omega} f^{\#}\big((u(x)-g(x))\otimes n_\Omega(x)\big)\,\mathrm{d}\mathcal{H}^{d-1}(x),$$

where $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$, is weakly lower semicontinuous.*

Remark 14.15 We stated Theorem 14.5 and Corollary 14.14 also in the case when the strong recession function f^∞ does not exist, but only proved them under this additional existence assumption. In the general case (with $f^{\#}$ in the definition of $\mathcal{F}$), the proof of Theorem 14.5 and Corollary 14.14 is the same except that the use of Theorem 14.1 has to be replaced by Remark 14.2 (i.e., the solution to Problem 14.4).

As an existence theorem for minimizers we then get:

Theorem 14.16 *Assume that $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ is a quasiconvex integrand that satisfies the coercivity and linear growth estimate*

$$\mu|A| - \mu^{-1} \le f(A) \le M(1+|A|) \qquad A \in \mathbb{R}^{m\times d},$$

for some $\mu, M > 0$ and let $g \in \mathrm{L}^1(\partial\Omega;\mathbb{R}^m)$. Then, the functional

$$\mathcal{F}_{\mathrm{ext}}[u] := \int_\Omega f(\nabla u(x))\,\mathrm{d}x + \int_\Omega f^{\#}\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x) + \int_{\partial\Omega} f^{\#}\big((u(x)-g(x))\otimes n_\Omega(x)\big)\,\mathrm{d}\mathcal{H}^{d-1}(x),$$

where $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$, has a minimizer over the space $\mathrm{BV}(\Omega;\mathbb{R}^m)$.

Note that since the trace operator on $\mathrm{BV}(\Omega;\mathbb{R}^m)$ is not weakly* continuous, we cannot expect that boundary values are preserved along a minimizing sequence.

Proof. The theorem follows directly from the preceding lower semicontinuity result via the Direct Method and the usual coercivity arguments, in particular the Poincaré inequality in BV (see Section 13.3). □

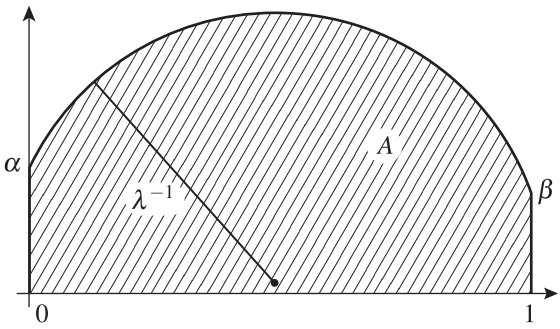


Fig. 14.1 The solution to the isoperimetric problem

Example 14.17 The isoperimetric problem from Section 1.2 leads to the following minimization problem:

$$\begin{cases} \text{Minimize} & \mathcal{F}[u] := \displaystyle\int_0^1 \sqrt{1 + (u(s)')^2}\, \mathrm{d}s + |u(0) - \alpha| + |u(1) - \beta| \\ \text{over all} & u \in \mathrm{BV}(0,1) \quad \text{with} \quad \displaystyle\int_0^1 u(s)\, \mathrm{d}s = A, \end{cases}$$

where $\alpha, \beta, A > 0$ are given. Here, note that the strong recession function of the integrand $f(a) := \sqrt{1 + a^2}$ is $f^\infty(a) = |a|$ and that we have already transformed the strict boundary conditions into the *penalty terms* $|u(0) - \alpha|$ and $|u(1) - \beta|$. By the preceding theorem, with the side constraint incorporated like in Section 2.5, there exists a solution to this problem.

Let us also explicitly identify this solution under the assumption that it is of class $\mathrm{W}^{2,1}$ inside the domain $(0,1)$. In this case we can use Theorem 3.22 on Lagrange multipliers to see that u must solve the differential equation

$$\left(\frac{u'}{\sqrt{1 + (u')^2}} \right)' = \lambda \qquad \text{in } (0,1)$$

for some $\lambda \in \mathbb{R}$. The term on the left is the inverse curvature radius of the curve $\gamma(s) := (s, u(s))^\mathsf{T}$, which is therefore constant. From geometric reasoning we obtain that u is part of a circle that is open from below, see Figure 14.1.

In the special case when $\alpha = \beta$ we get that the radius of the circle is $1/2$ and thus u is a semicircle. Then, the area under the graph is

$$A_{\max} = \alpha + \frac{\pi}{8}.$$

Consequently, if the prescribed area A is larger than $\pi/8$, we must have a jump in u in the left and right endpoints.

14.3 Relaxation

In analogy to Chapter 7 we now consider the situation where the integrand f of the functional $\mathcal{F}$ is not quasiconvex. Then, $\mathcal{F}$ cannot be weakly* lower semicontinuous by Proposition 5.18 (which is applicable here despite $p = 1$, since weak* lower semicontinuity in BV implies weak lower semicontinuity in $\mathrm{W}^{1,p}$ for any $p > 1$) and so we need to compute its relaxation. For $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ a continuous integrand and any Lipschitz subdomain $D \subset \Omega$ let

$$\mathcal{F}[u;D] := \int_D f(\nabla u(x))\,\mathrm{d}x, \qquad u \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m).$$

Then, for $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ we define the **relaxation** $\mathcal{F}_*[\bullet;D]$ of $\mathcal{F}[\bullet;D]$ as

$$\mathcal{F}_*[u;D] := \inf\left\{ \liminf_{j\to\infty} \mathcal{F}[u_j;D] \;:\; (u_j)_j \subset \mathrm{W}^{1,1}(D;\mathbb{R}^m),\, u_j \overset{*}{\rightharpoonup} u \text{ in BV} \right\}$$

and we also set $\mathcal{F}_*[u] := \mathcal{F}_*[u;\Omega]$. Note that this definition differs from the abstract definition of the relaxation in Chapter 7. Indeed, there we identified the relaxation with the (weakly*) lower semicontinuous envelope, which here is

$$\widetilde{\mathcal{F}}_*[u] := \sup\left\{ \mathcal{H}[u] \;:\; \mathcal{H} \le \mathcal{F} \text{ and } \mathcal{H} \text{ is weakly* lower semicontinuous} \right\}.$$

However, we will show in Theorem 14.18 below that $\mathcal{F}_*$ as defined above is weakly* lower semicontinuous. Then, for all $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ it holds that

$$\begin{aligned}
\widetilde{\mathcal{F}}_*[u] &\le \inf\left\{ \liminf_{j\to\infty} \widetilde{\mathcal{F}}_*[u_j] \;:\; (u_j)_j \subset \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m),\, u_j \overset{*}{\rightharpoonup} u \text{ in BV} \right\} \\
&\le \inf\left\{ \liminf_{j\to\infty} \mathcal{F}[u_j] \;:\; (u_j)_j \subset \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m),\, u_j \overset{*}{\rightharpoonup} u \text{ in BV} \right\} \\
&= \mathcal{F}_*[u] \\
&\le \widetilde{\mathcal{F}}_*[u]
\end{aligned}$$

and we conclude that in fact $\widetilde{\mathcal{F}}_* = \mathcal{F}_*$. Thus, we may work with the more convenient definition of the relaxation given above. In this context also see Problem 7.6.

The main theorem of this section is the following.

Theorem 14.18 *Assume that $f\colon \mathbb{R}^{m\times d} \to [0,\infty)$ is a continuous integrand with*

$$\mu|A| - \mu^{-1} \le f(A) \le M(1+|A|), \qquad A \in \mathbb{R}^{m\times d},$$

for some $\mu, M > 0$. Denote the quasiconvex envelope of f by Qf. Then, the integral representation

$$\mathcal{F}_*[u] = \int_\Omega Qf(\nabla u(x))\,\mathrm{d}x + \int_\Omega (Qf)^{\#}\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x)$$

holds for all $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$. *In particular, the functional* $\mathcal{F}_*$ *is weakly* lower semicontinuous on* $\mathrm{BV}(\Omega;\mathbb{R}^m)$.

Remark 14.19 The result continues to hold if instead of the lower bound $\mu|A| - \mu^{-1} \le f(A)$ we only assume that $Q(f - \delta|\cdot|) > -\infty$ for some $\delta > 0$, see Problem 14.9.

Proof. We know from Proposition 7.1 that Qf is finite, quasiconvex, and has linear growth with the same growth constant $M > 0$, so that

$$Qf(\nabla u) \le M(1 + |\nabla u|) \qquad \text{and} \qquad (Qf)^{\#}\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}\right) \le M.$$

Denote by $\mathcal{F}^{\mathrm{qc}}[u]$ the functional on the right-hand side in the statement of the theorem. The inequality

$$\mathcal{F}^{\mathrm{qc}} \le \mathcal{F}_* \tag{14.12}$$

follows immediately from $Qf \le f$ in conjunction with Theorem 14.5, which entails that $\mathcal{F}^{\mathrm{qc}}$ is weakly* lower semicontinuous. It remains to show the reverse inequality.

Step 1. We claim that we may relax the convergence requirement in the definition of $\mathcal{F}_*[u]$ to

$$\mathcal{F}_*[u] = \inf\left\{ \liminf_{j\to\infty} \mathcal{F}[u_j] \;:\; (u_j)_j \subset \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m),\, u_j \to u \text{ in } \mathrm{L}^1 \right\}.$$

Indeed, if this were false, we could find $(u_j)_j \subset \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ with $u_j \to u$ in L^1 and

$$\mathcal{F}_*[u] > \lim_{j\to\infty} \int_\Omega f(\nabla u_j(x))\,\mathrm{d}x \ge \mu \cdot \limsup_{j\to\infty} \|\nabla u_j\|_{\mathrm{L}^1} - \mu^{-1}|\Omega|.$$

So, the ∇u_j are in fact uniformly L^1-bounded, whereby $u_j \overset{*}{\rightharpoonup} u$ in BV, which directly leads to a contradiction.

Step 2. Using Lemma 13.7 choose a sequence $(u_j)_j \subset \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ with $u_j|_{\partial\Omega} = u|_{\partial\Omega}$ and $u_j \to u$ area-strictly. Furthermore, let $\Omega_0 \Subset \Omega$ be a Lipschitz subdomain with $(\mathcal{L}^d + |Du|)(\partial\Omega_0) = 0$. Since countably piecewise affine functions are dense in $\mathrm{W}^{1,1}(\Omega_0;\mathbb{R}^m)$ under given boundary values (see Theorem A.30), via Theorem 14.1 we may assume that u_j is countably piecewise affine in Ω_0, say $\nabla u_j = A_i^{(j)}$ almost everywhere in $D_i^{(j)} \subset \Omega_0$ ($A_i^{(j)} \in \mathbb{R}^{m\times d}$, $i \in \mathbb{N}$). Here, the $D_i^{(j)}$ are open and disjoint and for every $j \in \mathbb{N}$ it holds that $\Omega_0 = Z^{(j)} \cup \bigcup_i D_i^{(j)}$ for some $\mathcal{L}^d$-negligible set $Z^{(j)}$.

Employing the formula (7.1) for the quasiconvex envelope, we can pick maps $\psi_i^{(j)} \in W_0^{1,\infty}(D_i^{(j)};\mathbb{R}^m)$ with

$$\int_{D_i^{(j)}} |\psi_i^{(j)}(x)|\,\mathrm{d}x \le \frac{|D_i^{(j)}|}{j}$$

and

$$\int_{D_i^{(j)}} f(A_i^{(j)} + \nabla\psi_i^{(j)}(x))\,\mathrm{d}x < \left(Qf(A_i^{(j)}) + \frac{1}{j}\right)|D_i^{(j)}|.$$

Let $v_j : \Omega \to \mathbb{R}^m$ be defined as

$$v_j := \begin{cases} u_j + \psi_i^{(j)} & \text{in } D_i^{(j)} \ (i \in \mathbb{N}), \\ u_j & \text{in } \Omega \setminus \overline{\Omega_0}. \end{cases}$$

We have $v_j \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$ by the coercivity of f, $v_j|_{\partial\Omega} = u|_{\partial\Omega}$ and $v_j \to u$ in L^1. Thus,

$$\mathcal{F}_*[u] \le \liminf_{j\to\infty} \int_\Omega f(\nabla v_j(x))\,\mathrm{d}x.$$

Using also $v_j = u_j$ on $\Omega \setminus \overline{\Omega_0}$, we may then estimate

$$\int_\Omega f(\nabla v_j(x))\,\mathrm{d}x \le \int_{\Omega_0} Qf(\nabla u_j(x))\,\mathrm{d}x + \frac{|\Omega|}{j} + M\int_{\Omega\setminus\overline{\Omega_0}} 1 + |\nabla u_j(x)|\,\mathrm{d}x.$$

Now, by Alberti's Rank-One Theorem 13.8,

$$\operatorname{rank} \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|} = 1 \qquad |D^s u|\text{-a.e.}$$

Moreover, the strong recession function $(Qf)^\infty = (Qf)^\#$ exists at rank-one matrices due to the rank-one convexity of Qf, see Proposition 5.6 in conjunction with Problem 14.6. Using Theorem 14.1 and Remark 14.2 we can then pass to the limit as $j \to \infty$ to get

$$\begin{aligned}\mathcal{F}_*[u] \le \int_{\Omega_0} Qf(\nabla u(x))\,\mathrm{d}x + \int_{\Omega_0} (Qf)^\#\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x) \\ + M(\mathcal{L}^d + |Du|)(\Omega \setminus \Omega_0).\end{aligned}$$

Now let $\Omega_0 \uparrow \Omega$ (in the sense that $\sup_{x\in\Omega_0} \operatorname{dist}(x,\partial\Omega) \to 0$) to see that

$$\mathcal{F}_*[u] \le \int_\Omega Qf(\nabla u(x))\,\mathrm{d}x + \int_\Omega (Qf)^\#\left(\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x) = \mathcal{F}^{\mathrm{qc}}[u].$$

Together with (14.12) this concludes the proof of the theorem. □

14.4 Sharp-Interface Limits

As the final topic of this chapter, we return to minimization problems for functionals that depend on a parameter and their variational convergence. More precisely, we now we consider *singularly-perturbed problems*, where concentration phenomena may appear, so that the limit functional will only be defined on the space of functions of bounded variation. This investigation is set within the framework of Γ-convergence and as such continues the theory developed in Chapter 9.

As a prototypical instance of a singularly-perturbed functional we consider the following phase transition example from Section 1.11: For $u \in \mathrm{L}^1(\Omega)$ we define the **Allen–Cahn (or Modica–Mortola) functional**

$$\mathcal{F}_\varepsilon[u] := \begin{cases} \displaystyle\int_\Omega \frac{1}{\varepsilon} f(u(x)) + \varepsilon |\nabla u(x)|^2 \,\mathrm{d}x & \text{if } u \in \mathrm{W}^{1,2}(\Omega) \text{ and } \int_\Omega u \,\mathrm{d}x = \gamma, \\ +\infty & \text{otherwise.} \end{cases}$$

Here, $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain, $f : \mathbb{R} \to [0, \infty)$ is a continuous double-well potential with zeros (only) at $\alpha, \beta \in \mathbb{R}$ ($\alpha < \beta$), $\gamma \in (\alpha|\Omega|, \beta|\Omega|)$, and $\varepsilon > 0$. Note that in Section 1.11 we had $\alpha = -1$, $\beta = 1$ and also additionally required that $u(x) \in [-1, 1]$ for almost every $x \in \Omega$. Let us also remark that due to the positivity of the integrand, $\mathcal{F}_\varepsilon[u]$ is always well-defined, even if $f \circ u \notin \mathrm{L}^1(\Omega)$.

As the parameter ε tends to zero, the regularizing gradient term becomes weaker and (approximate) minimizers may develop sharp interfaces, i.e., jumps, in the limit. It is thus realistic to expect that the limit functional $\mathcal{F}_0$ is of a fundamentally different nature. To express this idea rigorously, recall from (12.10) the definition of the perimeter of a Borel set $E \subset \Omega$, namely,

$$\mathrm{Per}_\Omega(E) = |D\mathbb{1}_E|(\Omega) = \sup \left\{ \int_E \operatorname{div} \varphi \,\mathrm{d}x \;:\; \varphi \in \mathrm{C}^1_c(\Omega; \mathbb{R}^d),\ \|\varphi\|_\infty \le 1 \right\}.$$

If E has a smooth boundary, one can show that $\mathrm{Per}_\Omega(E) = \mathcal{H}^{d-1}(\partial E \cap \Omega)$. Then set for $u \in \mathrm{L}^1(\Omega)$,

$$\mathcal{F}_0[u] := \begin{cases} \sigma_0 \,\mathrm{Per}_\Omega\big(\{u = \alpha\}\big) & \text{if } u \in \mathrm{BV}(\Omega; \{\alpha, \beta\}),\ \int_\Omega u \,\mathrm{d}x = \gamma, \\ +\infty & \text{otherwise,} \end{cases}$$

where

$$\sigma_0 := 2 \int_\alpha^\beta \sqrt{f(s)} \,\mathrm{d}s.$$

The main result of this section is the following:

Theorem 14.20 (Modica–Mortola 1977 [233, 234]) *The functionals $\mathcal{F}_\varepsilon$ Γ-converge to $\mathcal{F}_0$ as $\varepsilon \downarrow 0$ with respect to the strong L^1-topology.*

For the proof without loss of generality we assume that $\alpha = -1$ and $\beta = 1$, which can be accomplished along the lines of the transformation explained in Section 1.11. To establish the sought Γ-convergence, we need to show the following:

(I) The *lim inf-inequality*

$$\mathcal{F}_0[u] \le \liminf_{\varepsilon \downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon] \tag{14.13}$$

holds for all sequences $u_\varepsilon \to u$ in $\mathrm{L}^1(\Omega)$.

(II) For all $u \in \mathrm{L}^1(\Omega)$ there exists a *recovery sequence* $(u_\varepsilon)_\varepsilon \subset \mathrm{L}^1(\Omega)$ such that $u_\varepsilon \to u$ in L^1 as $\varepsilon \downarrow 0$ and

$$\mathcal{F}_0[u] = \lim_{\varepsilon \downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon].$$

The lim inf-inequality turns out to be fairly straightforward to prove:

Proof of the lim inf-inequality in Theorem 14.20. Let $u_\varepsilon \to u$ in $\mathrm{L}^1(\Omega)$ and assume that $\liminf_{\varepsilon \downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon] < \infty$ (otherwise there is nothing to show), whereby we may in particular require that $u_\varepsilon \in \mathrm{W}^{1,2}(\Omega)$ for all $\varepsilon > 0$. Then, there exists a sequence $\varepsilon_n \downarrow 0$ (as $n \to \infty$) such that $u_{\varepsilon_n} \to u$ almost everywhere. Fatou's lemma implies

$$\int_\Omega f(u(x))\,\mathrm{d}x \le \liminf_{n\to\infty} \int_\Omega f(u_{\varepsilon_n}(x))\,\mathrm{d}x \le \liminf_{n\to\infty} \varepsilon_n \mathcal{F}_{\varepsilon_n}[u_{\varepsilon_n}] = 0.$$

Thus, $u(x) \in \{-1, 1\}$ almost everywhere.

Next, we observe from Young's inequality that

$$\begin{aligned}\liminf_{n\to\infty} \mathcal{F}_{\varepsilon_n}[u_{\varepsilon_n}] &\ge \liminf_{n\to\infty}\ 2 \int_\Omega \sqrt{f(u_{\varepsilon_n}(x))} \cdot |\nabla u_{\varepsilon_n}(x)|\,\mathrm{d}x \\ &= \liminf_{n\to\infty}\ 2 \int_\Omega |\nabla h(u_{\varepsilon_n}(x))|\,\mathrm{d}x,\end{aligned}$$

where we have set

$$h(t) := \int_0^t \sqrt{f(s)}\,\mathrm{d}s, \qquad t \in \mathbb{R},$$

As usual, we here interpret "$\int_0^s$" as "$-\int_s^0$" for $s < 0$.

From the L^1-strong lower semicontinuity of the total variation norm (see Problem 14.10) we then get, with $D(h \circ u)$ denoting the derivative of the map $h \circ u$ in the BV-sense, that

$$\begin{aligned}\liminf_{n\to\infty} \mathcal{F}_{\varepsilon_n}[u_{\varepsilon_n}] &\ge 2|D(h \circ u)|(\Omega) \\ &= 2(h(1) - h(-1))\,\mathrm{Per}_\Omega\big(\{\, x \in \Omega : u(x) = -1 \,\}\big) \\ &= \mathcal{F}_0[u]\end{aligned}$$

since $\sigma_0 = 2(h(1) - h(-1))$ by definition. In particular, $u \in \mathrm{BV}(\Omega; \{-1, 1\})$. This establishes the lim inf-inequality (14.13). □

For the existence of a recovery sequence, we first need a few technical preparations.

Lemma 14.21 *Let the Borel set $E \subset \Omega$ be of finite perimeter, i.e., $\mathrm{Per}_\Omega(E) < \infty$, and assume that*

$$E \text{ and } \Omega \setminus E \text{ both contain a non-empty open ball.} \tag{14.14}$$

Then, there exists a sequence of open and bounded sets $E_n \subset \mathbb{R}^d$ with smooth boundaries such that

$$|E_n \cap \Omega| = |E|, \tag{14.15}$$

the measure-theoretic transversality condition

$$\mathcal{H}^{d-1}(\partial E_n \cap \partial\Omega) = 0 \tag{14.16}$$

holds, and

$$|(E \,\Delta\, E_n) \cap \Omega| \to 0, \qquad \mathrm{Per}_\Omega(E_n) \to \mathrm{Per}_\Omega(E) \tag{14.17}$$

as $n \to \infty$.

Here, $A \,\Delta\, B := (A \setminus B) \cup (B \setminus A)$ denotes the **symmetric difference** between the sets A and B.

Proof. The idea is to mollify $\mathbb{1}_E$ and then to select for the E_n suitable superlevel sets. The technical details, however, are somewhat involved.

Step 1. Let $u \in (\mathrm{BV} \cap \mathrm{L}^\infty)(\mathbb{R}^d)$ be an extension of $\mathbb{1}_E \in (\mathrm{BV} \cap \mathrm{L}^\infty)(\Omega)$ with the property that $|Du|(\partial\Omega) = 0$. Since Ω was assumed to be a bounded Lipschitz domain, such an extension of $\mathbb{1}_E$ always exists; this result was recalled in Section 13.3.

Let $(\eta_\delta)_{\delta>0} \subset \mathrm{C}_c^\infty(\mathbb{R}^d)$ be a family of mollifiers as in Appendix A.5; in particular, $\operatorname{supp}\eta_\delta \subset B_\delta$. Set

$$u_\delta := \eta_\delta \star u,$$

for which it holds that $u_\delta \to u$ in L^1 and in measure, as well as $|Du_\delta|(\mathbb{R}^d) \to |Du|(\mathbb{R}^d)$ as $\delta \downarrow 0$. The latter fact can be seen by similar arguments as in the proof of Lemma 13.7.

By assumption, there exists an $\varepsilon > 0$ and $x_1, x_2 \in \mathbb{R}^d$ such that

$$B_{2\varepsilon}(x_1) \subset E, \qquad B_{2\varepsilon}(x_2) \subset \Omega \setminus E.$$

Then,

$$u_\delta = u \quad \text{on } \overline{B_\varepsilon(x_1)} \cup \overline{B_\varepsilon(x_2)} \qquad \text{if } \delta < \varepsilon.$$

Now, since $u_\delta \to u$ in measure, for every $n \in \mathbb{N}$ we may choose a positive $\delta_n < \min\{n^{-1}, \varepsilon\}$ such that

$$\left|\left\{ x \in \Omega \,:\, \left|u_{\delta_n}(x) - u(x)\right| \geq \frac{1}{n} \right\}\right| \leq \frac{1}{n}. \tag{14.18}$$

Let

$$p_n := \operatorname*{ess\,inf}_{t \in (n^{-1}, 1-n^{-1})} \mathrm{Per}_\Omega\big(\{ x \in \mathbb{R}^d : u_{\delta_n}(x) > t \}\big),$$

which is defined to be the largest almost-everywhere lower bound of the function $t \mapsto \mathrm{Per}_\Omega(\{ x \in \mathbb{R}^d : u_{\delta_n}(x) > t \})$ in the interval $(n^{-1}, 1 - n^{-1})$. Here we note in passing that this function is Lebesgue-measurable in t since it can be written as the supremum over countably many continuous functions.

For each $n \in \mathbb{N}$ choose $t_n \in (n^{-1}, 1 - n^{-1})$ such that the following three conditions hold:

(a) $\mathrm{Per}_\Omega\big(\{ x \in \mathbb{R}^d : u_{\delta_n}(x) > t_n \}\big) \leq p_n + \dfrac{1}{n}$;
(b) $\nabla u_{\delta_n}(x) \neq 0$ for all $x \in \mathbb{R}^d$ such that $u_{\delta_n}(x) = t_n$;
(c) $\mathcal{H}^{d-1}\big(\{ x \in \partial\Omega : u_{\delta_n}(x) = t_n \}\big) = 0$.

By the definition of p_n, (a) holds for a non-negligible set of t's and (b) is satisfied for a Lebesgue-full set of t's by the Sard Theorem A.18. The fact that $\mathcal{H}^{d-1}(\partial\Omega) < \infty$ implies (c) for all but countably many t's. Then, with t_n so defined, set

$$D_n := \left\{ x \in \mathbb{R}^d \,:\, u_{\delta_n}(x) > t_n \right\}, \qquad \lambda_n := |D_n \cap \Omega| - |E|,$$

and, letting $r_n > 0$ such that $|B_{r_n}(x_1)| = |B_{r_n}(x_2)| = |\lambda_n|$,

$$E_n := \begin{cases} D_n \setminus \overline{B_{r_n}(x_1)} & \text{if } \lambda_n > 0, \\ D_n & \text{if } \lambda_n = 0, \\ D_n \cup B_{r_n}(x_2) & \text{if } \lambda_n < 0. \end{cases}$$

Clearly, the sets E_n are open and bounded. It remains to show the assertions (14.15), (14.16), (14.17), as well as the smoothness of ∂E_n.

Step 2. If $x \in (D_n \cap \Omega) \setminus E$, then $u_{\delta_n}(x) > t_n > n^{-1}$ and $u(x) = 0$, whereas if $x \in E \setminus (D_n \cap \Omega)$, then $u_{\delta_n}(x) \le t_n < 1 - n^{-1}$ and $u(x) = 1$. Hence, also using (14.18),

$$|\lambda_n| \le |(D_n \cap \Omega) \,\Delta\, E| \le \left| \left\{ x \in \Omega \,:\, \left| u_{\delta_n}(x) - u(x) \right| \ge \frac{1}{n} \right\} \right| \le \frac{1}{n}. \tag{14.19}$$

Consequently,

$$r_n \to 0 \qquad \text{as } n \to \infty. \tag{14.20}$$

For n large enough, $r_n < \varepsilon/2$, so that

$$\overline{B_{r_n}(x_1)} \subset B_\varepsilon(x_1) \Subset D_n \cap \Omega \qquad \text{and} \qquad \overline{B_{r_n}(x_2)} \subset B_\varepsilon(x_2) \Subset \Omega \setminus D_n.$$

Since $\delta_n < \varepsilon$, we have $\overline{B_\varepsilon(x_1)} \subset D_n \cap \Omega$ and $\overline{B_\varepsilon(x_2)} \subset \Omega \setminus D_n$. Then, from (b) we infer that ∂E_n is smooth. Moreover,

$$\begin{aligned} |E_n \cap \Omega| &= \begin{cases} |D_n \cap \Omega| - |\overline{B_{r_n}(x_1)}| & \text{if } \lambda_n > 0, \\ |D_n \cap \Omega| & \text{if } \lambda_n = 0, \\ |D_n \cap \Omega| + |B_{r_n}(x_2)| & \text{if } \lambda_n < 0 \end{cases} \\ &= |E|, \end{aligned}$$

which shows (14.15) after discarding some elements at the beginning of the sequence $(E_n)_n$.

Step 3. In a similar fashion to the previous step we deduce that if $\lambda_n \neq 0$, then

$$\partial E_n \cap \partial\Omega = (\partial D_n \cup \partial B_{r_n}(x_i)) \cap \partial\Omega \qquad \text{for either } i = 1 \text{ or } i = 2,$$

depending on whether $\lambda_n > 0$ or $\lambda_n < 0$. On the other hand, $\partial B_{r_n}(x_i) \cap \partial\Omega = \emptyset$, so that by (c) we see that

$$\mathcal{H}^{d-1}(\partial E_n \cap \partial\Omega) = \mathcal{H}^{d-1}(\partial D_n \cap \partial\Omega) = 0.$$

This shows the transversality condition (14.16).

Step 4. To prove (14.17) we first recall that $\overline{B_\varepsilon(x_1)} \subset D_n \cap \Omega$ and $\overline{B_\varepsilon(x_2)} \subset \Omega \setminus D_n$. Then, by (14.19),

$$|(E_n \cap \Omega)\,\Delta(D_n \cap \Omega)| = |\lambda_n| \to 0 \qquad \text{as } n \to \infty.$$

Thus, again by (14.19),

$$\lim_{n\to\infty} |(E\,\Delta\,E_n) \cap \Omega| = \lim_{n\to\infty} |(E_n \cap \Omega)\,\Delta\,E| = \lim_{n\to\infty} |(D_n \cap \Omega)\,\Delta\,E| = 0.$$

This is the first part of (14.17).

Because $\overline{B_\varepsilon(x_1)} \subset D_n \cap \Omega$ and $\overline{B_\varepsilon(x_2)} \subset \Omega \setminus D_n$, we also have

$$\mathrm{Per}_\Omega(E_n) = \mathrm{Per}_\Omega(D_n) + \mathcal{H}^{d-1}(\partial B_{r_n}(x_i)) \tag{14.21}$$

for $i = 1$ or $i = 2$ (depending on the sign of λ_n). Moreover,

$$\mathbb{1}_{E_n} \to \mathbb{1}_E \quad \text{in } \mathrm{L}^1(\Omega)$$

by the first part of (14.17), which was shown above. Observe that the perimeter is lower semicontinuous with respect to this convergence, see Problem 14.10. Consequently,

$$\begin{aligned}
\mathrm{Per}_\Omega(E) &\le \liminf_{n\to\infty} \mathrm{Per}_\Omega(E_n) \\
&= \liminf_{n\to\infty} \left[\mathrm{Per}_\Omega(D_n) + \mathcal{H}^{d-1}(\partial B_{r_n}(x_i))\right] \\
&= \liminf_{n\to\infty} \mathrm{Per}_\Omega(D_n),
\end{aligned} \tag{14.22}$$

where we also employed (14.20). On the other hand, by (a) above,

$$\mathrm{Per}_\Omega(D_n) \le p_n + \frac{1}{n} \le \mathrm{Per}_\Omega\big(\{\, x \in \mathbb{R}^d : u_{\delta_n}(x) > t \,\}\big) + \frac{1}{n} \tag{14.23}$$

for all $n \in \mathbb{N}$ and almost every $t \in (n^{-1}, 1 - n^{-1})$.

The **Fleming–Rishel coarea formula** states that for every $u \in \mathrm{BV}(\Omega)$ the set $\{\, x \in \Omega : u(x) > t \,\}$ has finite perimeter for $\mathcal{L}^1$-almost every $t \in \mathbb{R}$ and

$$|Du|(\Omega) = \int_{-\infty}^{+\infty} \mathrm{Per}_\Omega(\{\, x \in \Omega : u(x) > t \,\})\, \mathrm{d}t.$$

A proof of this fundamental fact can be found in Theorem 3.40 of [18].

We can now integrate (14.23) from n^{-1} to $1 - n^{-1}$ and employ said coarea formula to get

$$\left(1 - \frac{2}{n}\right) \mathrm{Per}_\Omega(D_n) \le \int_\Omega |\nabla u_{\delta_n}|\, \mathrm{d}x + \frac{1}{n}\left(1 - \frac{2}{n}\right). \tag{14.24}$$

On the other hand, from $|Du_{\delta_n}|(\mathbb{R}^d) \to |Du|(\mathbb{R}^d)$, which we justified at the beginning of the proof, we deduce that

$$\lim_{n\to\infty} \int_\Omega |\nabla u_{\delta_n}| \, \mathrm{d}x = |Du|(\Omega) = \mathrm{Per}_\Omega(E).$$

We then see from (14.24) that

$$\limsup_{n\to\infty} \mathrm{Per}_\Omega(D_n) \le \mathrm{Per}_\Omega(E).$$

Combining this with (14.22), we arrive at

$$\mathrm{Per}_\Omega(E) \le \liminf_{n\to\infty} \mathrm{Per}_\Omega(D_n) \le \limsup_{n\to\infty} \mathrm{Per}_\Omega(D_n) \le \mathrm{Per}_\Omega(E),$$

and so, also using (14.21),

$$\lim_{n\to\infty} \mathrm{Per}_\Omega(E_n) = \lim_{n\to\infty} \mathrm{Per}_\Omega(D_n) = \mathrm{Per}_\Omega(E).$$

This is the second part of (14.17). □

If we dispense with the assumption (14.14), we can still prove the following result.

Lemma 14.22 *Let the Borel set $E \subset \Omega$ be of finite perimeter, i.e., $\mathrm{Per}_\Omega(E) < \infty$. Then, there exists a sequence of sets $D_n \subset \mathbb{R}^d$ with smooth boundaries such that*

$$|(E \,\Delta\, D_n) \cap \Omega| \to 0, \qquad \mathrm{Per}_\Omega(D_n) \to \mathrm{Per}_\Omega(E)$$

as $n \to \infty$.

Proof. The proof is contained in the parts of the proof for the previous lemma relating to the sets D_n, for which the assumption (14.14) is not used. □

Lemma 14.23 *Let $E \subset \mathbb{R}^d$ be open with a smooth, compact, non-empty boundary and suppose that the measure-theoretic transversality condition*

$$\mathcal{H}^{d-1}(\partial E \cap \partial\Omega) = 0 \tag{14.25}$$

holds. Define the signed distance function $\delta_E \colon \mathbb{R}^d \to \mathbb{R}$ via

$$\delta_E(x) := \begin{cases} -\operatorname{dist}(x, \partial E) & \text{if } x \in E, \\ \operatorname{dist}(x, \partial E) & \text{if } x \notin E. \end{cases} \tag{14.26}$$

Then, δ_E is Lipschitz continuous,

$$|\nabla \delta_E| \equiv 1 \quad \text{a.e.},$$

and for all $t \in \mathbb{R}$ it holds that

$$\lim_{t\to 0} \mathcal{H}^{d-1}(S_t \cap \Omega) = \mathcal{H}^{d-1}(\partial E \cap \Omega), \tag{14.27}$$

where $S_t := \{ x \in \mathbb{R}^d : \delta_E(x) = t \}$.

Proof. Step 1. We first show for the sets

$$F_r := \big\{ x \in E \ : \ \operatorname{dist}(x, \partial E) = r \big\}, \qquad r > 0,$$

that

$$\lim_{r \downarrow 0} \mathcal{H}^{d-1}(F_r) = \mathcal{H}^{d-1}(\partial E). \tag{14.28}$$

For this we recall the geometric fact, which is proved in detail in Appendix A of [157], that for small $r > 0$ there exists a diffeomorphism φ from

$$V_r := \big\{ x \in E \ : \ 0 < \operatorname{dist}(x, \partial E) < r \big\}$$

to the d-dimensional manifold $\partial E \times (0, r)$ with

$$\det \nabla \varphi(x) = \prod_{i=1}^{d-1} \big(1 - \kappa_i(\widehat{\varphi}(x)) \operatorname{dist}(x, \partial E)\big) \geq \mu > 0, \tag{14.29}$$

where $\widehat{\varphi}(x)$ is the component of $\varphi(x)$ on ∂E (more precisely, one needs to choose suitable parametrizations and express $\nabla \varphi$ in those) and $\kappa_1(\widehat{x}), \dots, \kappa_{d-1}(\widehat{x})$ denote the principal curvatures of ∂E at $\widehat{x} \in \partial E$. Also, $x \mapsto \operatorname{dist}(x, \partial E)$ is smooth on $\overline{V}_r$ and, with the unit outward normal vector n on ∂E,

$$\nabla[\operatorname{dist}(x, \partial E)] = -n(\widehat{\varphi}(x)) \qquad \text{for all } x \in \overline{V}_r.$$

Finally, if m_r is the normal vector to F_r, oriented outwards with respect to V_r, then

$$\nabla[\operatorname{dist}(x, \partial E)] = m_r(x) \qquad \text{for all } x \in F_r.$$

By the Gauss–Green theorem, noticing that $\partial V_r = F_r \cup \partial E$ (disjointly),

$$\begin{aligned} \int_{V_r} \Delta[\operatorname{dist}(\bullet, \partial E)] \, \mathrm{d}x &= \int_{F_r} \nabla[\operatorname{dist}(\bullet, \partial E)] \cdot m_r \, \mathrm{d}\mathcal{H}^{d-1} \\ &\quad + \int_{\partial E} \nabla[\operatorname{dist}(\bullet, \partial E)] \cdot n \, \mathrm{d}\mathcal{H}^{d-1} \\ &= \mathcal{H}^{d-1}(F_r) - \mathcal{H}^{d-1}(\partial E). \end{aligned}$$

Using the lower bound in (14.29), we have $\det \nabla \varphi^{-1} \leq \mu^{-1}$ and thus

$$|V_r| = \int_{\partial E} \int_0^r \det \nabla \varphi^{-1}(y, s) \, \mathrm{d}s \, \mathrm{d}\mathcal{H}^{d-1}(y) \leq r \cdot \frac{\mathcal{H}^{d-1}(\partial E)}{\mu} \to 0 \tag{14.30}$$

as $r \downarrow 0$. This directly implies (14.28).

Step 2. Next, we show that if the transversality condition (14.25) holds, then

$$\lim_{r \downarrow 0} \mathcal{H}^{d-1}(F_r \cap \Omega) = \mathcal{H}^{d-1}(\partial E \cap \Omega). \tag{14.31}$$

By virtue of (14.30) we obtain that $\mathbb{1}_{E\setminus V_r} \to \mathbb{1}_E$ in L^1. From the lower semicontinuity of the perimeter (see Problem 14.10) we thus infer that

$$\begin{aligned}\mathcal{H}^{d-1}(\partial E \cap \Omega) &= \mathrm{Per}_\Omega(E)\\ &\le \liminf_{r\downarrow 0} \mathrm{Per}_\Omega(E \setminus V_r)\\ &= \liminf_{r\downarrow 0} \mathcal{H}^{d-1}(F_r \cap \Omega), \end{aligned} \tag{14.32}$$

where the last equality follows since $\partial(E \setminus V_r) = F_r$, which is a smooth manifold.

On the other hand, we have

$$\mathcal{H}^{d-1}(F_r \cap \Omega) \le \mathcal{H}^{d-1}(F_r) - \mathcal{H}^{d-1}(F_r \cap (\mathbb{R}^d \setminus \Omega)).$$

By a similar argument as before,

$$\mathcal{H}^{d-1}(\partial E \cap (\mathbb{R}^d \setminus \overline{\Omega})) \le \liminf_{r\downarrow 0} \mathcal{H}^{d-1}(F_r \cap (\mathbb{R}^d \setminus \overline{\Omega})).$$

Thus, using (14.28) and the transversality condition (14.25),

$$\begin{aligned}\limsup_{r\downarrow 0} \mathcal{H}^{d-1}(F_r \cap \Omega) &\le \limsup_{r\downarrow 0} \mathcal{H}^{d-1}(F_r) - \liminf_{r\downarrow 0} \mathcal{H}^{d-1}(F_r \cap (\mathbb{R}^d \setminus \overline{\Omega}))\\ &\le \mathcal{H}^{d-1}(\partial E) - \mathcal{H}^{d-1}(\partial E \cap (\mathbb{R}^d \setminus \overline{\Omega}))\\ &= \mathcal{H}^{d-1}(\partial E \cap \Omega).\end{aligned}$$

Combining this with (14.32), we arrive at (14.31).

Step 3. Finally, we prove the statements about δ_E. Clearly, we have for all $x, y \in \mathbb{R}^d$ that $\delta_E(y) \le \delta_E(x) + |x - y|$, so δ_E is Lipschitz continuous. Hence δ_E is differentiable almost everywhere by Rademacher's Theorem A.31 and $|\nabla \delta_E| \le 1$ almost everywhere. On the other hand, for every $x \in \mathbb{R}^d$ there exists an $\bar{x} \in \partial E$ such that $|\delta_E(x)| = |x - \bar{x}|$. For every y on the connecting line between x and $\bar{x}$, it holds that $|\delta_E(y)| = |y - \bar{x}|$ and so $|\nabla \delta_E| \equiv 1$ almost everywhere. Finally, (14.27) follows from Step 2 (i.e., from (14.31)) applied to E and $\mathbb{R}^d \setminus \overline{E}$. □

We can now complete the proof of the Modica–Mortola theorem.

Proof of the existence of a recovery sequence in Theorem 14.20. Since $\mathcal{F}_0[u] = +\infty$ for $u \in (\mathrm{L}^1 \setminus \mathrm{BV})(\Omega)$ or if u takes values other than ± 1, we only need to explicitly construct a recovery sequence for maps $u \in \mathrm{BV}(\Omega; \{-1, 1\})$, which we henceforth assume.

Step 1. Set

$$E := E_{-1} = \big\{ x \in \Omega \,:\, u(x) = -1 \big\} \subset \Omega,$$

so that

$$u = -\mathbb{1}_{E\cap\Omega} + \mathbb{1}_{\Omega\setminus E}, \qquad |E \cap \Omega| = |\Omega \setminus E| - \gamma,$$

where we recall that $\gamma \in (-|\Omega|, |\Omega|)$ is the parameter such that $\int_\Omega u \, \mathrm{d}x = \gamma$. Since u has bounded variation, we know that E is of finite perimeter in Ω. Via Lemma 14.22

we may approximate u by

$$v = -\mathbb{1}_{F\cap\Omega} + \mathbb{1}_{\Omega\setminus F},$$

where $F \subset \Omega$ is of finite perimeter and additionally has a smooth boundary. By the properties of this approximation, for any fixed $\eta > 0$ we may choose v to also satisfy

$$\big|\mathcal{F}_0[u] - \mathcal{F}_0[v]\big| < \eta.$$

Since $\gamma \in (-|\Omega|, |\Omega|)$, both $F \cap \Omega$ and $F \setminus \Omega$ contain a non-empty open ball (this also uses the smoothness of the boundary of ∂F). Thus, Lemma 14.21 becomes applicable and via another approximation step we find

$$w = -\mathbb{1}_{G\cap\Omega} + \mathbb{1}_{\Omega\setminus G},$$

where $G \subset \Omega$ has finite perimeter and a smooth boundary. Moreover, the measure-theoretic transversality condition

$$\mathcal{H}^{d-1}(\partial G \cap \partial\Omega) = 0$$

holds and

$$\big|\mathcal{F}_0[v] - \mathcal{F}_0[w]\big| < \eta.$$

The preceding approximation arguments show that we only need to find a recovery sequence for $u \in \mathrm{BV}(\Omega; \{-1, 1\})$ with the property that $E = E_{-1} \subset \Omega$ is of finite perimeter, has a smooth boundary inside Ω, and the measure-theoretic transversality condition

$$\mathcal{H}^{d-1}(\partial E \cap \partial\Omega) = 0$$

holds. Finally, we may further assume that E is in fact open since the boundary ∂E is a Lebesgue negligible set, on which we may modify the representative of our u freely.

Step 2. Define for $\varepsilon > 0$ and $s \in [-1, 1]$, $t \in \mathbb{R}$ the functions

$$\varphi_\varepsilon(s) := \int_{-1}^{s} \frac{\varepsilon}{\sqrt{\varepsilon + f(r)}}\,\mathrm{d}r, \qquad \psi_\varepsilon(t) := \begin{cases} -1 & \text{if } t \le 0, \\ \varphi_\varepsilon^{-1}(t) & \text{if } 0 < t < \varphi_\varepsilon(1), \\ 1 & \text{if } \varphi_\varepsilon(1) \le t. \end{cases}$$

This is possible since φ_ε is strictly increasing, hence invertible. Then, with $\delta_E \colon \mathbb{R}^d \to \mathbb{R}$ defined as in (14.26) for our set E as above, we set

$$u_\varepsilon(x) := \psi_\varepsilon(\delta_E(x) + \eta_\varepsilon), \qquad x \in \Omega,$$

where $\eta_\varepsilon \in [0, \varphi_\varepsilon(1)]$ is chosen such that

$$\int_\Omega u_\varepsilon(x)\,\mathrm{d}x = \int_\Omega u(x)\,\mathrm{d}x.$$

Indeed, since

$$\psi_\varepsilon(\delta_E(x)) \le u(x) \le \psi_\varepsilon(\delta_E(x) + \varphi_\varepsilon(1)), \qquad x \in \Omega,$$

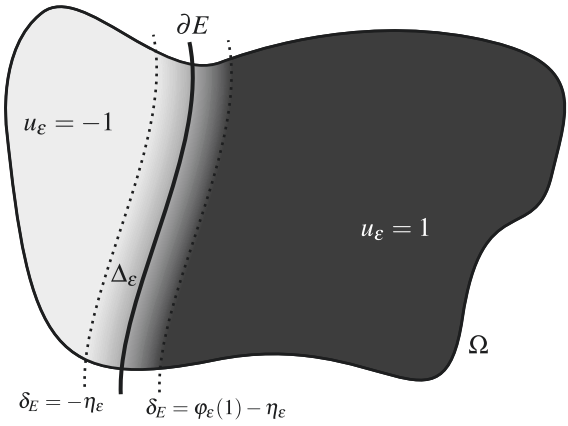


Fig. 14.2 The transition layer construction

and as ψ_ε is continuous, such an η_ε always exists. Figure 14.2 illustrates this *transition layer* construction. We will show in the following that u_ε is a recovery sequence for u.

Step 3. We first prove that $u_\varepsilon \to u$ in L^1. For this we recall **Federer's coarea formula**, which generalizes Fubini's theorem: Let $w \in \mathrm{L}^1(\Omega)$ and $h\colon \Omega \to \mathbb{R}$ be Lipschitz. Then,

$$\int_\Omega w(x)\,|\nabla h(x)|\,\mathrm{d}x = \int_{-\infty}^{+\infty} \int_{h^{-1}(t)} w(x)\,\mathrm{d}\mathcal{H}^{d-1}(x)\,\mathrm{d}t.$$

A proof of this result (in a more general form) can be found in Section 2.12 of [18].

In our situation, we first note that if $\delta(x) \le -\eta_\varepsilon$ or $\delta_E(x) \ge \varphi_\varepsilon(1) - \eta_\varepsilon$ then $u_\varepsilon(x) = u(x)$. Define the *transition region* $\Delta_\varepsilon \subset \Omega$ via

$$\Delta_\varepsilon := \big\{ x \in \Omega \,:\, -\eta_\varepsilon \le \delta_E(x) \le \varphi_\varepsilon(1) - \eta_\varepsilon \big\}.$$

Using that $|\nabla \delta_E| \equiv 1$ almost everywhere by Lemma 14.23, and Federer's coarea formula, we get

$$\begin{aligned}
\int_\Omega |u_\varepsilon(x) - u(x)|\,\mathrm{d}x &= \int_{\Delta_\varepsilon} \big|\psi_\varepsilon(\delta_E(x) + \eta_\varepsilon) - u(x)\big|\,\mathrm{d}x \\
&\le \int_{\Delta_\varepsilon} \big(|\psi_\varepsilon(\delta_E(x) + \eta_\varepsilon)| + 1\big) \cdot |\nabla \delta_E(x)|\,\mathrm{d}x \\
&= \int_{-\eta_\varepsilon}^{\varphi_\varepsilon(1)-\eta_\varepsilon} \big(|\psi_\varepsilon(t + \eta_\varepsilon)| + 1\big) \cdot \mathcal{H}^{d-1}(S_t \cap \Omega)\,\mathrm{d}t,
\end{aligned}$$

where as in Lemma 14.23 we have set $S_t := \{ x \in \mathbb{R}^d : \delta_E(x) = t \}$. Then, with

$$g(s) := \sup_{t \in [-s,s]} \mathcal{H}^{d-1}(S_t \cap \Omega),$$

we can further estimate (also note $0 \le \eta_\varepsilon \le \varphi_\varepsilon(1)$ and $\varphi_\varepsilon(1) \le 2\sqrt{\varepsilon}$)

$$\int_\Omega |u_\varepsilon(x) - u(x)| \, \mathrm{d}x \le 2\varphi_\varepsilon(1) g(\varphi_\varepsilon(1)) \le 4\sqrt{\varepsilon} g(2\sqrt{\varepsilon}).$$

On the other hand, by Lemma 14.23, $g(s) \to \mathcal{H}^{d-1}(\partial E \cap \Omega) < \infty$ as $s \to 0$. Thus, we conclude that $u_\varepsilon \to u$ as $\varepsilon \downarrow 0$.

Step 4. It remains to show that $\mathcal{F}_\varepsilon[u_\varepsilon] \to \mathcal{F}_0[u]$ as $\varepsilon \downarrow 0$. In fact, in light of the already established lim inf-inequality, it suffices to prove

$$\limsup_{\varepsilon \downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon] \le \mathcal{F}_0[u]. \tag{14.33}$$

Again using Federer's coarea formula and $|\nabla \delta_E| \equiv 1$ almost everywhere, we estimate

$$\begin{aligned}
\mathcal{F}_\varepsilon[u_\varepsilon] &= \int_{\Delta_\varepsilon} \frac{1}{\varepsilon} f(\psi_\varepsilon(\delta_E(x) + \eta_\varepsilon)) + \varepsilon |\psi_\varepsilon'(\delta_E(x) + \eta_\varepsilon)|^2 \, \mathrm{d}x \\
&= \int_{-\eta_\varepsilon}^{\varphi_\varepsilon(1) - \eta_\varepsilon} \left[\frac{1}{\varepsilon} f(\psi_\varepsilon(t + \eta_\varepsilon)) + \varepsilon |\psi_\varepsilon'(t + \eta_\varepsilon)|^2 \right] \cdot \mathcal{H}^{d-1}(S_t \cap \Omega) \, \mathrm{d}t \\
&\le g(\varphi_\varepsilon(1)) \int_0^{\varphi_\varepsilon(1)} \frac{1}{\varepsilon} f(\psi_\varepsilon(t)) + \varepsilon |\psi_\varepsilon'(t)|^2 \, \mathrm{d}t.
\end{aligned}$$

We also compute

$$\psi_\varepsilon'(t) = \frac{1}{\varphi_\varepsilon'(\varphi_\varepsilon^{-1}(t))} = \frac{\sqrt{\varepsilon + f(\psi_\varepsilon(t))}}{\varepsilon} \qquad \text{for } 0 < t < \varphi_\varepsilon(1).$$

Then, continuing the above estimate,

$$\begin{aligned}
\mathcal{F}_\varepsilon[u_\varepsilon] &\le g(\varphi_\varepsilon(1)) \int_0^{\varphi_\varepsilon(1)} \frac{1}{\varepsilon} f(\psi_\varepsilon(t)) + \frac{\varepsilon + f(\psi_\varepsilon(t))}{\varepsilon} \, \mathrm{d}t \\
&\le 2g(\varphi_\varepsilon(1)) \int_0^{\varphi_\varepsilon(1)} \frac{\varepsilon + f(\psi_\varepsilon(t))}{\varepsilon} \, \mathrm{d}t \\
&= 2g(\varphi_\varepsilon(1)) \int_0^{\varphi_\varepsilon(1)} \sqrt{\varepsilon + f(\psi_\varepsilon(t))} \cdot \psi_\varepsilon'(t) \, \mathrm{d}t \\
&= 2g(\varphi_\varepsilon(1)) \int_{-1}^{1} \sqrt{\varepsilon + f(s)} \, \mathrm{d}s.
\end{aligned}$$

Taking the upper limit of the last expression as $\varepsilon \downarrow 0$ using Lemma 14.23, we arrive at

$$\limsup_{\varepsilon \downarrow 0} \mathcal{F}_\varepsilon[u_\varepsilon] \le 2\,\mathcal{H}^{d-1}(\partial E \cap \Omega) \int_{-1}^{1} \sqrt{f(s)} \, \mathrm{d}s = \sigma_0 \operatorname{Per}_\Omega(E) = \mathcal{F}_0[u].$$

Here we also employed the fact that $\mathcal{H}^{d-1}(\partial E \cap \Omega) = \operatorname{Per}_\Omega(E)$ since $\partial E \cap \Omega$ is smooth. Thus, (14.33) holds. □

Example 14.24 In our phase transition example from Section 1.11, we can now apply the Modica–Mortola Theorem 14.20 to see that the approximate functionals $\mathcal{F}_\varepsilon$ Γ-converge to $\mathcal{F}_0$ as $\varepsilon \downarrow 0$ with respect to L^1-convergence. We note that the additional constraint that $u(x) \in [\alpha, \beta]$ causes no problems since the recovery sequence constructed in the proof above satisfies this additional requirement anyway. We have thus shown that the regularized energy functionals $\mathcal{F}_\varepsilon$ converge to their sharp-interface limit $\mathcal{F}_0$ as the strength of the regularization tends to zero.

Notes and Historical Remarks

Theorem 14.1 in the extended form of Remark 14.2 is from [210]. Results in this direction already appeared in [55]. A version of this theorem also holds true for a class of u-dependent integrands, see [281].

The blow-up method in the proof of Theorem 14.5 was first introduced by Fonseca–Müller in [143], also see [59] for a systematic approach to the idea of proving lower semicontinuity and relaxation theorems for integral functionals via the auxiliary functional $\mathcal{J}[u; U]$ from (14.2). In fact, the idea of Lemma 14.12 dates back to [102] where it was used in the context of Sobolev spaces. In the BV-context it seems to have been used for the first time in [58] and [59]. A proof of Theorem 14.5 that does not use Alberti's rank-one theorem was discovered in [279].

We note that the work by Fonseca–Müller [144] additionally considered u-dependent integrands (under suboptimal assumptions). A more general approach to this problem using *liftings* can be found in [282, 283].

The reader is pointed to Problem 14.2 and also to [343] for the construction of a non-convex quasiconvex function with linear growth and to [245] for an example of a non-convex quasiconvex function that is even positively 1-homogeneous. Theorem 8.1 of [204] shows, in a non-constructive fashion, that "many" quasiconvex functions with linear growth must exist.

The notation for recession functions is not consistent in the literature. In many works, the upper weak recession function $f^\#$ is written as f^∞ and simply called the "recession function". We refer to [29], in particular Section 2.5, for a more systematic approach to recession functions and their associated cones.

For the proof of the Modica–Mortola Theorem 14.20 we closely follow the original works [233, 234], with some minor modifications.

The theory of sharp-interface limits for higher-order functionals and for vector-valued problems is far from complete. The works [86, 87, 103, 215, 216] contain some recent progress in this direction as well as many historical references.

Problems

14.1 Show that for a continuous convex function $f\colon \mathbb{R}^N \to \mathbb{R}$ with linear growth it holds that $f^\# = f^\infty$.

14.2 Prove that there exists a non-convex quasiconvex integrand $f\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with linear growth. Also prove that the strong recession function f^∞ exists. *Hint:* Extend Lemma 7.5 to the case $p = 1$.

14.3 Let $f \in \mathrm{C}(\overline{\Omega}\times\mathbb{R}^{m\times d})$ with linear growth. Show that there exists a sequence $(f_k)_k$ of continuous integrands for which the strong recession functions f_k^∞ exists and such that

$$\sup_{k\in\mathbb{N}} f_k(x,A) = f(x,A) \qquad \text{and} \qquad \sup_{k\in\mathbb{N}} f_k^\infty(x,A) = f_\#(x,A).$$

14.4 Prove the following strengthened version of Theorem 14.1: Let $f\colon \Omega\times\mathbb{R}^{m\times d} \to [0,\infty)$ be continuous with linear growth such that $A \mapsto f(x,A)$ is rank-one convex (or rank-one concave) for almost every $x \in \Omega$ and moreover

$$f^\#(x,A) = f_\#(x,A) = (f(x,\bullet))^\#(A) = \limsup_{\substack{A'\to A\\ t\to\infty}} \frac{f(x,tA')}{t}$$

for all $(x,A) \in \Omega\times\mathbb{R}^{m\times d}$ such that rank $A \le 1$. Then, the functional

$$\mathcal{F}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x + \int_\Omega f^\#\left(x, \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x),$$

where $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$, is continuous with respect to the area-strict convergence on $\mathrm{BV}(\Omega,\mathbb{R}^m)$. *Hint:* Use Problem 14.3 and also Alberti's Rank-One Theorem 13.8.

14.5 Show that if $f\colon \Omega\times\mathbb{R}^{m\times d} \to \mathbb{R}$ is rank-one convex and the function

$$(f(x,\bullet))^\#(A) = \limsup_{\substack{A'\to A\\ t\to\infty}} \frac{f(x,tA')}{t}$$

is continuous in $x \in \Omega$ for fixed A, then

$$f^\#(x,A) = f_\#(x,A) = (f(x,\bullet))^\#(A)$$

for all $(x,A) \in \Omega\times\mathbb{R}^{m\times d}$ such that rank $A \le 1$. *Hint:* Use Dini's Theorem, which asserts that if a monotone sequence of continuous functions converges pointwise on a compact space and if the limit function is also continuous, then the convergence is uniform.

14.6 Show that for a quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with linear growth the rescaled functions

$$h_r(A) := \frac{h(rA)}{r}, \qquad A \in \mathbb{R}^{m\times d},$$

are quasiconvex for all $r > 0$. Conclude that the upper weak recession function

$$h^\#(A) := \limsup_{t\to\infty} \frac{h(tA)}{t}$$

is quasiconvex. Show also that if rank $A \le 1$, then this upper limit is in fact a proper limit and $h^\infty(A) = h^\#(A)$ (with h^∞ now defined only where its defining limit exists in $\mathbb{R}$).

14.7 Show that Theorem 14.5 cannot be extended to integrands f taking negative values without restricting the class of admissible BV-sequences.

14.8 Let $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ and assume that $x_0 \in \Omega$ is such that

$$\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x_0) = a \otimes n \qquad \text{for some } a \in \mathbb{S}^{m-1},\, n \in \mathbb{S}^{d-1}.$$

Set

$$u_r(y) := \frac{r^d}{|Du|(Q_n(x_0,r))} \cdot \frac{u(x_0+ry) - [u]_{Q_n(x_0,r)}}{r}, \qquad y \in Q_n(0,1),\ r>0,$$

where $Q_n(x_0,r)$ is an open cube with midpoint $x_0 \in \Omega$, side-length $2r$, and two faces orthogonal to n. Then, show that for $|D^s u|$-almost every such x_0, up to taking a concrete sequence of r's, it holds that $u_r \overset{*}{\rightharpoonup} u_0$ in $\mathrm{BV}(Q_n(0,1);\mathbb{R}^m)$ with $u_0(y) = a\psi(y\cdot n)$ for a bounded function $\psi\colon (-1,1) \to \mathbb{R}$. Further show that ψ can be chosen to be *increasing*. *Hint:* Use Corollary 13.9.

14.9 Show that Theorem 14.18 continues to hold if instead of the lower bound $\mu|A| - \mu^{-1} \le f(A)$ we only assume that $Q(f - \delta|\bullet|) > -\infty$ for some $\delta > 0$. *Hint:* Use the Kirchheim–Kristensen Theorem 13.17.

14.10 Show that for a sequence $(u_j)_j \subset \mathrm{BV}(\Omega)$ and $u \in \mathrm{L}^1(\Omega)$ with $u_j \to u$ in L^1 it holds that

$$|Du|(\Omega) \le \liminf_{j\to\infty} |Du|(\Omega).$$

Also show that the perimeter $\mathrm{Per}_\Omega(E)$ is lower semicontinuous with respect to the convergence of sets $E_j \to E$ defined as $\mathbb{1}_{E_j} \to \mathbb{1}_E$ in $\mathrm{L}^1(\Omega)$. *Hint:* Write the total variation norm and the perimeter as a supremum over L^1-continuous functionals.



Chapter 15
Generalized Young Measures

Recall that if $p \in (1,\infty]$ then for a standard $\mathrm{W}^{1,p}$-gradient Young measure we can always find a p-equiintegrable (bounded if $p = \infty$) generating sequence; see Lemma 4.14 for $p < \infty$ and Zhang's Lemma 8.14 for $p = \infty$. Thus, for $p > 1$, we do not need to worry about (L^p-)concentrations in generating sequences of gradients. However, as has already become apparent in Chapters 13 and 14, the linear-growth case $p = 1$ is special. Here, concentrations cannot be truncated off and have to be taken into account.

In this final chapter we study the theory of *generalized Young measures*, which extends the standard theory of Young measures developed in Chapter 4. Besides quantifying oscillations (like classical Young measures), this theory allows one to quantify *concentrations* as well, thus providing a rich toolbox for investigating linear-growth functionals.

The main new idea allowing one to also capture concentration effects is to employ a *compactification* of the target space. For this, we transform maps $u\colon \Omega \to \mathbb{R}^N$ into maps $\widetilde{u}$ taking values in $\mathbb{B}^N$, the (open) unit ball in $\mathbb{R}^N$. This can be achieved by choosing a homeomorphism $\varphi\colon \mathbb{R}^N \to \mathbb{B}^N$ and setting $\widetilde{u} := \varphi \circ u$. Then, the behavior of u "at infinity" can be understood by studying $\widetilde{u}$ near $\partial\mathbb{B}^N$.

After developing the basic framework of generalized Young measures, including the introduction of a suitable set of "test integrands", we turn to investigating the class of generalized Young measures that are generated by BV-derivatives. In particular, localization ("blow-up") principles will play a prominent role, just like they did for classical Young measures. Finally, we will consider how these tools can be used to establish Jensen-type inequalities, which will yield lower semicontinuity results for integral functionals.

15.1 Functional Analysis Setup

We first define the space $\mathbf{E}(\Omega;\mathbb{R}^N)$, whose elements are the "test integrands" for generalized Young measures. In order to do so, we introduce for $f \in \mathrm{C}(\overline{\Omega}\times\mathbb{R}^N)$ and $g \in \mathrm{C}(\overline{\Omega}\times\mathbb{B}^N)$, where by $\mathbb{B}^N$ we denote the open unit ball in $\mathbb{R}^N$, the following linear

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F. Rindler, *Calculus of Variations*, Graduate Texts in Mathematics 310,
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transformations:

$$(Sf)(x,\widehat{A}) := (1-|\widehat{A}|)\, f\Big(x, \frac{\widehat{A}}{1-|\widehat{A}|}\Big), \qquad x \in \overline{\Omega},\ \widehat{A} \in \mathbb{B}^N, \tag{15.1}$$

$$(S^{-1}g)(x,A) := (1+|A|)\, g\Big(x, \frac{A}{1+|A|}\Big), \qquad (x,A) \in \overline{\Omega} \times \mathbb{R}^N.$$

Clearly, $S^{-1} \circ S$ and $S \circ S^{-1}$ are the identities on $\mathrm{C}(\overline{\Omega}\times\mathbb{R}^N)$ and $\mathrm{C}(\overline{\Omega}\times\mathbb{B}^N)$, respectively. Then we set

$$\mathbf{E}(\Omega;\mathbb{R}^N) := \big\{\, f \in \mathrm{C}(\overline{\Omega}\times\mathbb{R}^N) \ :\ Sf \in \mathrm{C}(\overline{\Omega\times\mathbb{B}^N}) \,\big\}.$$

Here, the condition "$Sf \in \mathrm{C}(\overline{\Omega\times\mathbb{B}^N})$" is to be understood in the sense that $Sf \in \mathrm{C}(\overline{\Omega}\times\mathbb{B}^N)$ has a (necessarily unique) continuous extension to $\overline{\Omega\times\mathbb{B}^N}$, which is also denoted by Sf. As the norm on $\mathbf{E}(\Omega;\mathbb{R}^N)$ we use the natural choice

$$\|f\|_{\mathbf{E}(\Omega;\mathbb{R}^N)} := \|Sf\|_{\mathrm{C}(\overline{\Omega\times\mathbb{B}^N})} = \sup_{(x,\widehat{A})\in\overline{\Omega\times\mathbb{B}^N}} |Sf(x,\widehat{A})|,$$

under which $\mathbf{E}(\Omega;\mathbb{R}^N)$ becomes a Banach space and the operator

$$S\colon \mathbf{E}(\Omega;\mathbb{R}^N) \to \mathrm{C}(\overline{\Omega\times\mathbb{B}^N})$$

is an isometric isomorphism. In particular, $\mathbf{E}(\Omega;\mathbb{R}^N)$ inherits the separability of $\mathrm{C}(\overline{\Omega\times\mathbb{B}^N})$.

Since

$$|f(x,A)| = (1+|A|)\left|Sf\Big(x,\frac{A}{1+|A|}\Big)\right|, \qquad (x,A) \in \overline{\Omega}\times\mathbb{R}^N,$$

all $f \in \mathbf{E}(\Omega;\mathbb{R}^N)$ have linear growth, i.e., there exists a constant $M > 0$ with

$$|f(x,A)| \le M(1+|A|), \qquad (x,A) \in \overline{\Omega}\times\mathbb{R}^N. \tag{15.2}$$

Moreover, our definition of $\mathbf{E}(\Omega;\mathbb{R}^N)$ is designed so that the (strong) recession function $f^\infty\colon \overline{\Omega}\times\mathbb{R}^N \to \mathbb{R}$ exists, which was defined in (14.1) as

$$f^\infty(x,A) := \lim_{\substack{x'\to x\\ A'\to A\\ t\to\infty}} \frac{f(x',tA')}{t}, \qquad (x,A) \in \overline{\Omega}\times\mathbb{R}^N.$$

Note that f^∞ agrees with Sf on $\overline{\Omega}\times\partial\mathbb{B}^N$, as can be seen by substituting $t = s/(1-s)$, $s \in (0,1)$, and letting $s \to 1$. We also recall that f^∞ is positively 1-homogeneous.

A **generalized Young measure** on the bounded open set $\Omega \subset \mathbb{R}^d$ with values in $\mathbb{R}^N$ is a triple $\nu = (\nu_x, \lambda_\nu, \nu_x^\infty)$ consisting of

(i) a parametrized family of probability measures $(\nu_x)_{x\in\Omega} \subset \mathcal{M}^1(\mathbb{R}^N)$, called the **oscillation measure**;
(ii) a positive finite measure $\lambda_\nu \in \mathcal{M}^+(\overline{\Omega})$, called the **concentration measure**;
(iii) a parametrized family of probability measures $(\nu_x^\infty)_{x\in\overline{\Omega}} \subset \mathcal{M}^1(\partial\mathbb{B}^N)$, called the **concentration-direction measure**;

and satisfying the following conditions:

(iv) the map $x \mapsto \nu_x$ is weakly* measurable with respect to $\mathcal{L}^d$, i.e., the function $x \mapsto \langle f(x,\bullet), \nu_x\rangle$ is $\mathcal{L}^d$-measurable for all bounded Borel functions $f\colon \Omega\times\mathbb{R}^N \to \mathbb{R}$;
(v) the map $x \mapsto \nu_x^\infty$ is weakly* measurable with respect to λ_ν, i.e., the function $x \mapsto \langle f^\infty(x,\bullet), \nu_x^\infty\rangle$ is λ_ν-measurable for all bounded Borel functions $f\colon \Omega \times \partial\mathbb{B}^N \to \mathbb{R}$;
(vi) $x \mapsto \langle |\bullet|, \nu_x\rangle \in \mathrm{L}^1(\Omega)$.

We identify generalized Young measures μ, ν if $\mu_x = \nu_x$ for $\mathcal{L}^d$-almost every $x \in \Omega$, $\lambda_\mu = \lambda_\nu$, and $\mu_x^\infty = \nu_x^\infty$ for λ_μ-almost every $x \in \overline{\Omega}$. All these (equivalence classes of) generalized Young measures are collected in the set

$$\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N).$$

If for the target dimension we have $N = 1$, then we simply write $\mathbf{Y}^{\mathcal{M}}(\Omega)$ instead of $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R})$. In all of the following we usually refer to generalized Young measures simply as "Young measures".

The **duality pairing** between $f \in \mathbf{E}(\Omega;\mathbb{R}^N)$ and $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ is defined as

$$\begin{aligned}\langle\!\langle f, \nu\rangle\!\rangle &:= \int_\Omega \big\langle f(x,\bullet), \nu_x\big\rangle \,\mathrm{d}x + \int_{\overline{\Omega}} \big\langle f^\infty(x,\bullet), \nu_x^\infty\big\rangle \,\mathrm{d}\lambda_\nu(x)\\ &= \int_\Omega \int_{\mathbb{R}^N} f(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x + \int_{\overline{\Omega}} \int_{\partial\mathbb{B}^N} f^\infty(x,A)\,\mathrm{d}\nu_x^\infty(A)\,\mathrm{d}\lambda_\nu(x).\end{aligned}$$

In this way, the space $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ can be considered a part of the dual space $\mathbf{E}(\Omega;\mathbb{R}^N)^*$ of $\mathbf{E}(\Omega;\mathbb{R}^N)$. A sequence of Young measures $(\nu_j)_j \subset \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ **converges weakly*** to $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$, written as

$$\nu_j \overset{*}{\rightharpoonup} \nu,$$

if for every $f \in \mathbf{E}(\Omega;\mathbb{R}^N)$ it holds that $\langle\!\langle f, \nu_j\rangle\!\rangle \to \langle\!\langle f, \nu\rangle\!\rangle$.

The **barycenter** of a Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ is the measure $[\nu] \in \mathcal{M}(\overline{\Omega};\mathbb{R}^N)$ given by

$$[\nu] := [\nu_x]\,\mathcal{L}^d_x \llcorner \Omega + [\nu_x^\infty]\,\lambda_\nu(\mathrm{d}x), \tag{15.3}$$

where $[\mu] := \int A \,\mathrm{d}\mu(A)$ and we wrote "$\mathcal{L}^d_x$", "$\lambda_\nu(\mathrm{d}x)$" to emphasize that the measures $\mathcal{L}^d, \lambda_\nu$ act with respect to the x-variable. It is not hard to see that if $\nu_j \overset{*}{\rightharpoonup} \nu$ in $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$, then $[\nu_j] \overset{*}{\rightharpoonup} [\nu]$ in $\mathcal{M}(\overline{\Omega};\mathbb{R}^N)$.

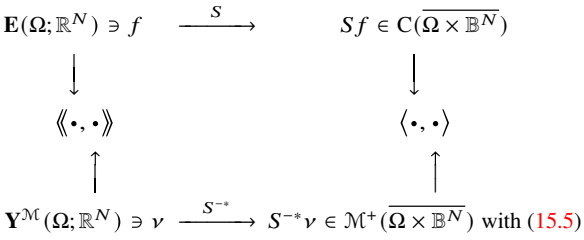


Fig. 15.1 Duality relationships.

In the following we will establish the fundamental compactness result for the weak* convergence in $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$. Notice first that since $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N) \subset \mathbf{E}(\Omega;\mathbb{R}^N)^*$ via the duality pairing $\langle\!\langle \bullet, \bullet \rangle\!\rangle$, a sequence of Young measures that is suitably bounded (to be detailed below) has a weakly*-converging subsequence in $\mathbf{E}(\Omega;\mathbb{R}^N)^*$. However, it is not a priori clear that the limit is also a Young measure. Here and in the following we always identify $\mathrm{C}(\overline{\Omega \times \mathbb{B}^N})^*$ with $\mathcal{M}(\overline{\Omega \times \mathbb{B}^N})$ via the Riesz Representation Theorem A.22.

We first observe that the linear transformation $S\colon \mathbf{E}(\Omega;\mathbb{R}^N) \to \mathrm{C}(\overline{\Omega \times \mathbb{B}^N})$ defined in (15.1) is an isomorphism, hence the operator

$$S^{-*} := (S^{-1})^*\colon \mathbf{E}(\Omega;\mathbb{R}^N)^* \to \mathcal{M}(\overline{\Omega \times \mathbb{B}^N}),$$

that is, the dual operator of the inverse of S, acts on a Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ as follows:

$$\begin{aligned} \langle \Phi, S^{-*}\nu \rangle &= \langle\!\langle S^{-1}\Phi, \nu \rangle\!\rangle \\ &= \int_\Omega \langle S^{-1}\Phi(x, \bullet), \nu_x \rangle \, \mathrm{d}x + \int_{\overline{\Omega}} \langle \Phi(x, \bullet), \nu_x^\infty \rangle \, \mathrm{d}\lambda_\nu(x) \end{aligned} \tag{15.4}$$

for any $\Phi \in \mathrm{C}(\overline{\Omega \times \mathbb{B}^N})$ (noticing that $(S^{-1}\Phi)^\infty|_{\overline{\Omega}\times\partial\mathbb{B}^N} = \Phi|_{\overline{\Omega}\times\partial\mathbb{B}^N}$). In particular,

$$\langle Sf, S^{-*}\nu \rangle = \langle\!\langle f, \nu \rangle\!\rangle \qquad \text{for all } f \in \mathbf{E}(\Omega;\mathbb{R}^N).$$

Figure 15.1 presents a diagram of the duality relationships.

Lemma 15.1 *The set $S^{-*}(\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)) \subset \mathcal{M}(\overline{\Omega \times \mathbb{B}^N})$ consists precisely of all the positive measures $\mu \in \mathcal{M}^+(\overline{\Omega \times \mathbb{B}^N})$ that satisfy*

$$\int_{\overline{\Omega\times\mathbb{B}^N}} \varphi(x)(1 - |A|) \, \mathrm{d}\mu(x, A) = \int_\Omega \varphi(x) \, \mathrm{d}x \qquad \textit{for all } \varphi \in \mathrm{C}(\overline{\Omega}). \tag{15.5}$$

Proof. Step 1. For $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ and $\Phi = \mathbb{1}$, (15.4) gives

$$(S^{-*}\nu)(\overline{\Omega \times \mathbb{B}^N}) = \langle \mathbb{1}, S^{-*}\nu \rangle = \int_\Omega \langle 1 + |\bullet|, \nu_x \rangle \, \mathrm{d}x + \lambda_\nu(\overline{\Omega}) < \infty$$

by the assumptions on ν. Thus, $S^{-*}\nu$ is a finite measure on $\overline{\Omega \times \mathbb{B}^N}$. Moreover, for $\varphi \in \mathrm{C}(\overline{\Omega})$ set $\Phi(x, A) := \varphi(x)(1 - |A|)$, so that

$$\int_{\overline{\Omega\times\mathbb{B}^N}} \varphi(x)(1-|A|)\,\mathrm{d}(S^{-*}\nu)(x,A) = \big\langle \Phi, S^{-*}\nu \big\rangle = \big\langle\!\big\langle \varphi \otimes \mathbb{1}, \nu \big\rangle\!\big\rangle = \int_\Omega \varphi(x)\,\mathrm{d}x.$$

This implies (15.5).

The positivity of $S^{-*}\nu$ follows from the fact that for $\Phi \geq 0$ we have

$$\big\langle \Phi, S^{-*}\nu \big\rangle = \big\langle\!\big\langle S^{-1}\Phi, \nu \big\rangle\!\big\rangle \geq 0.$$

Step 2. To prove the converse, let $\mu \in \mathcal{M}^+(\overline{\Omega \times \mathbb{B}^N})$ be such that (15.5) holds. We need to construct a Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega; \mathbb{R}^N)$ with $\mu = S^{-*}\nu$, that is,

$$\big\langle Sf, \mu \big\rangle = \big\langle Sf, S^{-*}\nu \big\rangle = \big\langle\!\big\langle f, \nu \big\rangle\!\big\rangle \qquad \text{for all } f \in \mathbf{E}(\Omega; \mathbb{R}^N).$$

By the Disintegration Theorem 4.4 we infer the existence of a measure $\kappa \in \mathcal{M}^+(\overline{\Omega})$ and a weakly* κ-measurable family $(\eta_x)_{x\in\overline{\Omega}} \subset \mathcal{M}^1(\overline{\mathbb{B}^N})$ such that

$$\mu = \kappa(\mathrm{d}x) \otimes \eta_x.$$

Let

$$\kappa = g\mathcal{L}^d \llcorner \Omega + \kappa^s, \qquad g \in \mathrm{L}^1(\Omega),\ \kappa^s \text{ singular to } \mathcal{L}^d,$$

be the Lebesgue–Radon–Nikodým decomposition of κ with respect to $\mathcal{L}^d$. From (15.5) we get that

$$\big\langle 1 - |\bullet|, \eta_x \big\rangle \big[g(x)\,\mathcal{L}^d_x \llcorner \Omega + \kappa^s(\mathrm{d}x) \big] = \mathcal{L}^d \llcorner \Omega.$$

Thus, $\langle 1 - |\bullet|, \eta_x \rangle\, \kappa^s(\mathrm{d}x) = 0$, whereby η_x is concentrated in $\partial\mathbb{B}^N$ for κ^s-almost every $x \in \overline{\Omega}$. Consequently, for $f \in \mathbf{E}(\Omega; \mathbb{R}^N)$ (recall $Sf = f^\infty$ on $\overline{\Omega} \times \partial\mathbb{B}^N$),

$$\begin{aligned}\big\langle Sf(x,\bullet), \eta_x \big\rangle\, \kappa(\mathrm{d}x) = {} & \left(g(x) \int_{\mathbb{B}^N} Sf(x,\bullet)\,\mathrm{d}\eta_x \right) \mathcal{L}^d_x \llcorner \Omega \\ & + \left(\eta_x(\partial\mathbb{B}^N) g(x) \fint_{\partial\mathbb{B}^N} f^\infty(x,\bullet)\,\mathrm{d}\eta_x \right) \mathcal{L}^d_x \llcorner \Omega \\ & + \left(\eta_x(\partial\mathbb{B}^N) \fint_{\partial\mathbb{B}^N} f^\infty(x,\bullet)\,\mathrm{d}\eta_x \right) \kappa^s(\mathrm{d}x). \end{aligned} \tag{15.6}$$

Define the measures $\nu_x \in \mathcal{M}^+(\mathbb{R}^N)$ for $\mathcal{L}^d$-almost every $x \in \Omega$ by

$$\big\langle h, \nu_x \big\rangle := g(x) \int_{\mathbb{B}^N} Sh\,\mathrm{d}\eta_x, \qquad h \in \mathrm{C}_0(\mathbb{R}^N);$$

the measure $\lambda_\nu \in \mathcal{M}^+(\overline{\Omega})$ by

$$\lambda_\nu := \eta_x(\partial\mathbb{B}^N)\,\kappa(\mathrm{d}x);$$

and the measures $\nu_x^\infty \in \mathcal{M}^+(\partial\mathbb{B}^N)$ for λ_ν-almost every $x \in \overline{\Omega}$ by

$$\left\langle h^\infty, \nu_x^\infty \right\rangle := \fint_{\partial\mathbb{B}^N} h^\infty \, \mathrm{d}\eta_x, \qquad h^\infty \in \mathrm{C}(\partial\mathbb{B}^N).$$

Then, (15.6) becomes

$$\begin{aligned} \left\langle Sf, \mu \right\rangle &= \left\langle Sf(x,\bullet), \eta_x \right\rangle \kappa(\mathrm{d}x) \\ &= \left\langle f(x,\bullet), \nu_x \right\rangle \mathcal{L}_x^d \llcorner \Omega + \left\langle f^\infty(x,\bullet), \nu_x^\infty \right\rangle \lambda_\nu(\mathrm{d}x) \\ &= \left\langle\!\left\langle f, \nu \right\rangle\!\right\rangle. \end{aligned} \tag{15.7}$$

Step 3. It remains to show that ν_x and ν_x^∞ are indeed probability measures. For ν_x^∞ this is obvious. For ν_x, at $\mathcal{L}^d$-almost every $x \in \Omega$ observe that by (15.7) with $f(x,A) := \varphi(x)$ and (15.5) we infer that

$$\begin{aligned} \int_\Omega \varphi(x) \left\langle \mathbb{1}, \nu_x \right\rangle \mathrm{d}x &= \int_\Omega \varphi(x) \left\langle 1 - |\bullet|, \eta_x \right\rangle \mathrm{d}\kappa(x) \\ &= \int_{\overline{\Omega\times\mathbb{B}^N}} \varphi(x)(1-|A|) \, \mathrm{d}\mu(x,A) \\ &= \int_\Omega \varphi(x) \, \mathrm{d}x \end{aligned}$$

for every $\varphi \in \mathrm{C}(\overline{\Omega})$. Thus, $\langle \mathbb{1}, \nu_x \rangle = 1$ for $\mathcal{L}^d$-almost every $x \in \Omega$.

Finally, using $f(x,A) := 1 + |A|$ in (15.7),

$$\int_\Omega \left\langle 1 + |\bullet|, \nu_x \right\rangle \mathrm{d}x + \lambda_\nu(\overline{\Omega}) = \left\langle \mathbb{1}, \mu \right\rangle = \mu(\overline{\Omega\times\mathbb{B}^N}) < \infty$$

since $Sf(x,A) = \mathbb{1}$. Therefore, $x \mapsto \langle |\bullet|, \nu_x \rangle \in \mathrm{L}^1(\Omega)$ and $\lambda_\nu(\overline{\Omega}) < \infty$. □

Corollary 15.2 *The set $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ is weakly* closed (considered as a subset of $\mathbf{E}(\Omega;\mathbb{R}^N)^*$).*

Proof. It suffices to observe that condition (15.5) is weakly* continuous (in the topological sense). Indeed, since $\varphi(x)(1-|A|)$ for $\varphi \in \mathrm{C}(\overline{\Omega})$ is an admissible test function for the weak* topology on $\mathcal{M}(\overline{\Omega\times\mathbb{B}^N}) \cong \mathrm{C}(\overline{\Omega\times\mathbb{B}^N})^*$, the map

$$\mu \in \mathcal{M}(\overline{\Omega\times\mathbb{B}^N}) \quad \mapsto \quad \int_{\overline{\Omega\times\mathbb{B}^N}} \varphi(x)(1-|A|) \, \mathrm{d}\mu(x,A)$$

is continuous in the (locally convex) weak* topology and thus the set

$$S^{-*}(\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)) \subset \mathcal{M}^+(\overline{\Omega\times\mathbb{B}^N})$$

is weakly* closed in $\mathrm{C}(\overline{\Omega\times\mathbb{B}^N})^*$. Via the isometric isomorphism S^* the weak* closedness is then transported to the set $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$. □

The following is the basic *compactness principle* for generalized Young measures.

Theorem 15.3 *Let $(\nu_j)_j \subset \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ be a sequence of Young measures such that*

$$\sup_{j\in\mathbb{N}} \left\langle\!\!\left\langle \mathbb{1}\otimes|\bullet|, \nu_j \right\rangle\!\!\right\rangle < \infty$$

or, equivalently,

(i) the functions $x \mapsto \langle|\bullet|, (\nu_j)_x\rangle$ are uniformly bounded in $\mathrm{L}^1(\Omega)$ and
(ii) the sequence $(\lambda_{\nu_j}(\overline{\Omega}))_j$ is uniformly bounded.

Then, there exists a subsequence (not relabeled) such that

$$\nu_j \overset{*}{\rightharpoonup} \nu$$

for a Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$.

Proof. Let $\Phi \in \mathrm{C}(\overline{\Omega\times\mathbb{B}^N})$ with $\|\Phi\|_\infty \le 1$. Then,

$$\begin{aligned}
\left|\left\langle \Phi, S^{-*}\nu_j\right\rangle\right| &= \left|\left\langle\!\!\left\langle S^{-1}\Phi, \nu_j\right\rangle\!\!\right\rangle\right| \\
&\le \int_\Omega \int_{\mathbb{R}^N} (1+|A|)\left|\Phi\left(x, \frac{A}{1+|A|}\right)\right| \mathrm{d}(\nu_j)_x(A)\,\mathrm{d}x \\
&\qquad + \int_{\overline{\Omega}} \int_{\partial\mathbb{B}^N} |\Phi(x,A)|\,\mathrm{d}(\nu_j)_x^\infty(A)\,\mathrm{d}\lambda_\nu(x) \\
&\le \sup_{j\in\mathbb{N}} \left(\int_\Omega \left\langle 1+|\bullet|, (\nu_j)_x\right\rangle \mathrm{d}x + \lambda_{\nu_j}(\overline{\Omega})\right) \\
&< \infty.
\end{aligned}$$

Thus, the sequence $(S^{-*}\nu_j)_j$ is uniformly norm-bounded in $\mathcal{M}^+(\overline{\Omega\times\mathbb{B}^N})$ and hence there exists a weakly* converging subsequence, say $S^{-*}\nu_j \overset{*}{\rightharpoonup} \mu$ in $\mathcal{M}^+(\overline{\Omega\times\mathbb{B}^N})$. By Corollary 15.2, the limit μ is again the transformation under S^{-*} of a Young measure, that is, $\mu = S^{-*}\nu$ for a Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$. Finally, via the isomorphism S^* the weak* convergence is transported to the Young measure side. □

15.2 Generation and Examples

Let $\gamma \in \mathcal{M}(\overline{\Omega};\mathbb{R}^N)$ be a (finite) Radon measure with Lebesgue–Radon–Nikodým decomposition

$$\gamma = g\mathcal{L}^d \llcorner \Omega + \gamma^s,$$

where $g \in \mathrm{L}^1(\Omega;\mathbb{R}^N)$ and γ^s is singular with respect to $\mathcal{L}^d$. To γ we associate an **elementary Young measure** $\delta[\gamma] \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ via

$$\begin{aligned} \delta[\gamma]_x &:= \delta_{g(x)} && \text{for } \mathcal{L}^d\text{-a.e. } x \in \Omega, \\ \lambda_{\delta[\gamma]} &:= |\gamma^s|, \\ \delta[\gamma]^\infty_x &:= \delta_{P(x)} && \text{for } |\gamma^s|\text{-a.e. } x \in \overline{\Omega}, \end{aligned}$$

where

$$P := \frac{\mathrm{d}\gamma^s}{\mathrm{d}|\gamma^s|} \in \mathrm{L}^1(\overline{\Omega}, |\gamma^s|; \partial\mathbb{B}^N).$$

We say that a sequence $(\gamma_j)_j \subset \mathcal{M}(\overline{\Omega}; \mathbb{R}^N)$ with $\sup_j |\gamma_j|(\overline{\Omega}) < \infty$ **generates** the Young measure $\nu = (\nu_x, \lambda_\nu, \nu^\infty_x) \in \mathbf{Y}^{\mathcal{M}}(\Omega; \mathbb{R}^N)$, in symbols

$$\gamma_j \xrightarrow{\mathbf{Y}} \nu,$$

if $\delta[\gamma_j] \overset{*}{\rightharpoonup} \nu$ in $\mathbf{Y}^{\mathcal{M}}(\Omega; \mathbb{R}^N)$, that is,

$$\langle\!\langle f, \delta[\gamma_j] \rangle\!\rangle \to \langle\!\langle f, \nu \rangle\!\rangle \qquad \text{for all } f \in \mathbf{E}(\Omega; \mathbb{R}^N).$$

Unwinding the definitions, $\gamma_j \xrightarrow{\mathbf{Y}} \nu$ thus means that for all $f \in \mathbf{E}(\Omega; \mathbb{R}^N)$,

$$\begin{aligned} &f\left(x, \frac{\mathrm{d}\gamma_j}{\mathrm{d}\mathcal{L}^d}(x)\right) \mathcal{L}^d_x \llcorner \Omega \;+\; f^\infty\left(x, \frac{\mathrm{d}\gamma^s_j}{\mathrm{d}|\gamma^s_j|}(x)\right) |\gamma^s_j|(\mathrm{d}x) \\ &\quad \overset{*}{\rightharpoonup} \left\langle f(x, \bullet), \nu_x \right\rangle \mathcal{L}^d_x \llcorner \Omega \;+\; \left\langle f^\infty(x, \bullet), \nu^\infty_x \right\rangle \lambda_\nu(\mathrm{d}x) \quad \text{in } \mathcal{M}(\overline{\Omega}). \end{aligned}$$

If $V_j \, \mathcal{L}^d \llcorner \Omega \xrightarrow{\mathbf{Y}} \nu$ for a sequence of uniformly L^1-bounded maps $(V_j)_j \subset \mathrm{L}^1(\Omega; \mathbb{R}^N)$, we simply write $V_j \xrightarrow{\mathbf{Y}} \nu$.

Proposition 15.4 *Let $(\gamma_j)_j \subset \mathcal{M}(\overline{\Omega}; \mathbb{R}^N)$. Then, $\gamma_j \to \gamma$ area-strictly in $\overline{\Omega}$ if and only if $\gamma_j \xrightarrow{\mathbf{Y}} \delta[\gamma]$.*

Proof. If $\gamma_j \to \gamma$ area-strictly in $\mathcal{M}(\overline{\Omega}; \mathbb{R}^N)$, then $\gamma_j \xrightarrow{\mathbf{Y}} \delta[\gamma]$ follows by the same strategy as in the proof of Theorem 14.1, which is based on Reshetnyak's Continuity Theorem 13.3. For the converse, test the weak* convergence $\delta[\gamma_j] \overset{*}{\rightharpoonup} \delta[\gamma]$ with the integrand $f(x, A) := \sqrt{1 + |A|^2}$, which lies in $\mathbf{E}(\Omega; \mathbb{R}^N)$. □

The following is customarily called the *Fundamental Theorem* of the generalized Young measure theory.

Theorem 15.5 *Let $(\gamma_j)_j \subset \mathcal{M}(\overline{\Omega}; \mathbb{R}^N)$ be a sequence of Radon measures such that*

$$\sup_{j \in \mathbb{N}} |\gamma_j|(\overline{\Omega}) < \infty.$$

Then, there exists a subsequence (not relabeled) with $\gamma_j \xrightarrow{\mathbf{Y}} \nu$ for some $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega; \mathbb{R}^N)$.

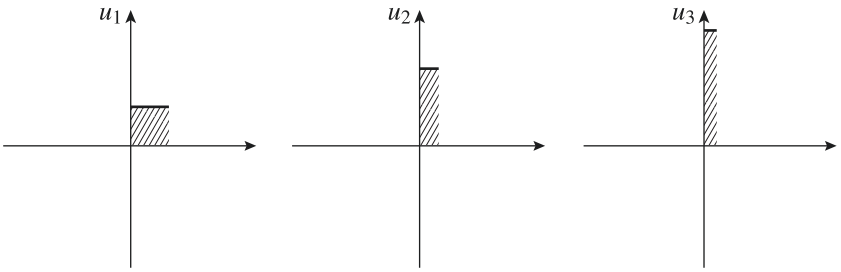


Fig. 15.2 A concentrating sequence

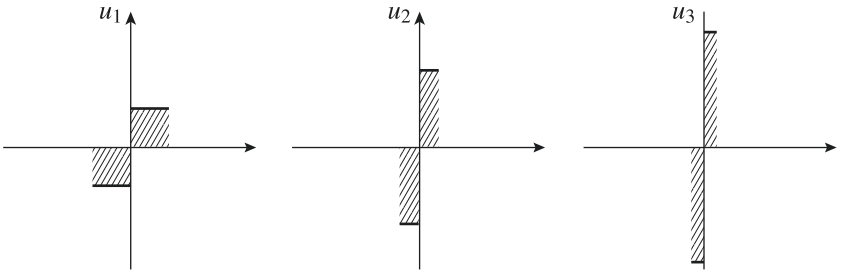


Fig. 15.3 Another concentrating sequence

Proof. Set $\nu_j := \delta[\gamma_j]$, the elementary Young measure associated with γ_j and apply the compactness result from Theorem 15.3, whose assumptions are satisfied since the quantities $\langle\langle \mathbb{1} \otimes |\bullet|, \delta[\gamma_j]\rangle\rangle = |\gamma_j|(\overline{\Omega})$ are uniformly bounded. □

We have already seen many examples of oscillation effects, most directly in the examples of classical Young measures from Section 4.2. All these examples carry over to the present theory of generalized Young measures (with zero concentration measure). The following examples will illustrate model cases of concentration effects.

Example 15.6 Take $\Omega := (0,1)$ and set $u_j := j\mathbb{1}_{(0,1/j)}$, see Figure 15.2. Then, $u_j \overset{\mathbf{Y}}{\to} \nu \in \mathbf{Y}^{\mathcal{M}}((0,1))$ with

$$\nu_x = \delta_0 \quad \text{a.e.}, \qquad \lambda_\nu = \delta_0, \qquad \nu_0^\infty = \delta_{+1}.$$

Example 15.7 On $\Omega := (-1,1)$ define $u_j := j(\mathbb{1}_{(0,1/j)} - \mathbb{1}_{(-1/j,0)})$, see Figure 15.3. Then, $u_j \overset{\mathbf{Y}}{\to} \nu \in \mathbf{Y}^{\mathcal{M}}((-1,1))$ with

$$\nu_x = \delta_0 \quad \text{a.e.}, \qquad \lambda_\nu = 2\delta_0, \qquad \nu_0^\infty = \frac{1}{2}\delta_{+1} + \frac{1}{2}\delta_{-1}.$$

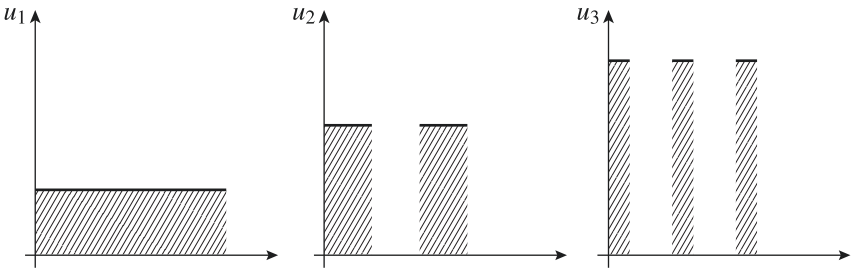


Fig. 15.4 A diffuse concentration

Example 15.8 (Diffuse concentration) On $\Omega := (0,1)$ define the uniformly L^1-bounded sequence

$$u_j := \sum_{k=0}^{j-1} j \mathbb{1}_{\left(\frac{k}{j}, \frac{k}{j}+\frac{1}{j^2}\right)}, \qquad j \in \mathbb{N},$$

see Figure 15.4. The u_j converge to zero almost everywhere and in the *biting sense*, i.e., there exists an increasing sequence of subsets $\Omega_k \subset \Omega$ with $|\Omega_k| \uparrow |\Omega|$ as $k \to \infty$ and $u_j \rightharpoonup 0$ in $\mathrm{L}^1(\Omega_k)$ for all $k \in \mathbb{N}$ (see Section 6.4 in [272] for more on this). On the other hand, $u_j \overset{*}{\rightharpoonup} \mathbb{1}$. Consequently, the $(u_j)_j$ cannot be equiintegrable; otherwise, by Vitali's Convergence Theorem A.12 the limits would agree. We can also verify this directly, namely

$$\lim_{h\to\infty} \sup_{j\in\mathbb{N}} \int_{\{|u_j|\geq h\}} |u_j| \, \mathrm{d}x = 1.$$

Furthermore, $(u_j)_j$ has no L^1-weakly converging subsequence: Since L^1-weak convergence implies weak* convergence in the sense of measures, the L^1-weak limit would have to be $\mathbb{1}$, contradicting the biting limit 0. It is the task of Problem 15.3 to show that $u_j \xrightarrow{\mathbf{Y}} \nu \in \mathbf{Y}^{\mathcal{M}}((0,1))$ with

$$\nu_x = \delta_0 \quad \text{a.e.}, \qquad \lambda_\nu = \mathcal{L}^1 \llcorner (0,1), \qquad \nu_x^\infty = \delta_{+1}.$$

Example 15.9 The last example can be modified, now with $\mathbb{R}^2$ as target space, to

$$v_j(x) := \sum_{k=0}^{j-1} j \mathbb{1}_{\left(\frac{k}{j}, \frac{k}{j}+\frac{1}{j^2}\right)}(x) \begin{bmatrix} \cos(2\pi j^2 x) \\ \sin(2\pi j^2 x) \end{bmatrix}, \qquad x \in (0,2).$$

One can compute that $v_j \xrightarrow{\mathbf{Y}} \nu$ for $\nu \in \mathbf{Y}^{\mathcal{M}}((0,1);\mathbb{R}^2)$ given as

$$\nu_x = \delta_0 \quad \mathcal{L}^d\text{-a.e.}, \qquad \lambda_\nu = \mathcal{L}^1 \llcorner (0,1), \qquad \nu_x^\infty = \frac{1}{2\pi} \mathcal{H}^1 \llcorner \mathbb{S}^1.$$

Indeed, the v_j concentrate in all directions uniformly, hence ν_x^∞ must be the uniform probability measure on $\mathbb{S}^1$.

We conclude this section with a useful density result.

Lemma 15.10 *There exists a countable set $(f_k)_k = (\varphi_k \otimes h_k)_k \subset \mathbf{E}(\Omega;\mathbb{R}^N)$ with $\varphi_k \in \mathrm{C}(\overline{\Omega})$ and $h_k \in \mathrm{C}(\mathbb{R}^N)$ such that for $\nu_j, \nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$, where $j \in \mathbb{N}$, the condition*

$$\langle\!\langle f_k, \nu_j \rangle\!\rangle \to \langle\!\langle f_k, \nu \rangle\!\rangle \qquad \text{for all } k \in \mathbb{N}$$

implies $\nu_j \overset{*}{\rightharpoonup} \nu$. Moreover, all the h_k can be chosen to be Lipschitz continuous.

Proof. First, assuming that $f := \mathbb{1} \otimes |\bullet|$ is in the collection, we may always suppose that the sequence $(\nu_j)_j$ converges weakly* (up to a subsequence, but then also entirely once the limit is uniquely identified).

Take countable sets $\mathcal{A} \subset \mathrm{C}(\overline{\Omega})$, $\mathcal{B} \subset \mathrm{C}^1_c(\mathbb{R}^N)$ that are dense in $\mathrm{C}(\overline{\Omega})$ and $\mathrm{C}_0(\mathbb{R}^N)$, respectively, with respect to the $\|\bullet\|_\infty$-norm. Furthermore, let $\mathcal{C} \subset \mathrm{C}^1(\partial\mathbb{B}^N)$ be countable and dense in $\mathrm{C}(\partial\mathbb{B}^N)$ with $\mathbb{1} \in \mathcal{C}$. Set for $h^\infty \in \mathcal{C}$,

$$(Gh^\infty)(A) := |A| h^\infty\left(\frac{A}{|A|}\right), \qquad A \in \mathbb{R}^N,$$

and define

$$(\varphi_k \otimes h_k)_k := (\mathcal{A} \otimes \mathcal{B}) \cup (\mathcal{A} \otimes G(\mathcal{C})) \subset \mathbf{E}(\Omega;\mathbb{R}^N),$$

where the tensor product is understood to act elementwise. By standard results of measure theory, the values of

$$\langle\!\langle \varphi \otimes h, \nu \rangle\!\rangle = \int_\Omega \varphi(x) \langle h, \nu_x \rangle \, \mathrm{d}x \qquad \text{for all } \varphi \in \mathcal{A},\ h \in \mathcal{B}$$

determine the $\mathrm{L}^1(\Omega)$-map $x \mapsto \langle h, \nu_x \rangle$ and then in turn the measures ν_x for $\mathcal{L}^d$-almost every $x \in \Omega$. Testing with $\mathcal{A} \otimes (G\mathbb{1})$ gives for all $\varphi \in \mathrm{C}(\overline{\Omega})$ that

$$\langle\!\langle \varphi \otimes G\mathbb{1}, \nu \rangle\!\rangle = \int_\Omega \varphi(x) \langle G\mathbb{1}, \nu_x \rangle \, \mathrm{d}x + \int_{\overline{\Omega}} \varphi(x) \langle \mathbb{1}, \nu_x^\infty \rangle \, \mathrm{d}\lambda_\nu(x).$$

Since the first integral is already identified and the second integral reduces to $\int_{\overline{\Omega}} \varphi \, \mathrm{d}\lambda_\nu$, the measure λ_ν is also uniquely determined. The identification of ν_x^∞ (up to a λ_ν-negligible set) is similar to that of ν_x.

By construction, every $h \in \mathcal{B}$ is Lipschitz continuous: Let $h = Gh^\infty$ with $h^\infty \in \mathcal{C}$. There is a constant $C > 0$ with

$$\left| h^\infty\left(\frac{A}{|A|}\right) - h^\infty\left(\frac{B}{|B|}\right) \right| \le C \left| \frac{A}{|A|} - \frac{B}{|B|} \right|$$

for all $A, B \in \mathbb{R}^N \setminus \{0\}$. Using the elementary estimate

$$\left| \frac{A}{|A|} - \frac{B}{|B|} \right| \cdot |B| = \left| \frac{A}{|A|}|B| - B \right| = \left| \frac{A}{|A|}|B| - \frac{A}{|A|}|A| \right| + |A - B| \le 2|A - B|,$$

we then get

$$\begin{aligned}\big|h(A)-h(B)\big| &= \left||A|h^\infty\left(\frac{A}{|A|}\right)-|B|h^\infty\left(\frac{B}{|B|}\right)\right| \\ &\le \left|h^\infty\left(\frac{A}{|A|}\right)-h^\infty\left(\frac{B}{|B|}\right)\right||B| + \left|h^\infty\left(\frac{A}{|A|}\right)\right||\,|A|-|B|\,| \\ &\le \Big(2C+\max_{\partial\mathbb{B}^N}|h^\infty|\Big)|A-B|,\end{aligned}$$

whereby h is seen to be Lipschitz continuous. □

15.3 Extended Representation

We now extend the limit representation via Young measures to a larger class of integrands than $\mathbf{E}(\Omega;\mathbb{R}^N)$, called the **representation integrands** and defined as follows:

$$\mathbf{R}(\Omega;\mathbb{R}^N) := \Big\{ f\colon \overline{\Omega}\times\mathbb{R}^N\to\mathbb{R} \;:\; f \text{ Carathéodory with linear growth,} \\ f^\infty \in \mathrm{C}(\overline{\Omega}\times\mathbb{R}^N) \text{ exists as in (14.1)} \Big\}.$$

Proposition 15.11 *Let $\nu_j \overset{*}{\rightharpoonup} \nu$ in $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ and assume either*

(i) $f\in\mathbf{R}(\Omega;\mathbb{R}^N)$ or
(ii) $f(x,A)=\mathbb{1}_B(x)g(x,A)$, where $g\in\mathbf{E}(\Omega;\mathbb{R}^N)$ and $B\subset\overline{\Omega}$ is a Borel set with $(\mathcal{L}^d+\lambda_\nu)(\partial B)=0$.

Then, $\langle\!\langle f,\nu_j\rangle\!\rangle \to \langle\!\langle f,\nu\rangle\!\rangle$.

Before we come to the proof, we note that for $f\in\mathbf{R}(\Omega;\mathbb{R}^N)$ the expression

$$\big\langle\!\big\langle f,\nu\big\rangle\!\big\rangle = \int_\Omega\int_{\mathbb{R}^N} f(x,A)\,\mathrm{d}\nu_x(A)\,\mathrm{d}x + \int_{\overline{\Omega}}\int_{\partial\mathbb{B}^N} f^\infty(x,A)\,\mathrm{d}\nu_x^\infty(A)\,\mathrm{d}\lambda_\nu(x)$$

is well-defined by the weak* measurability of $(\nu_x)_x$ with respect to $\mathcal{L}^d$ and the weak* measurability of $(\nu_x^\infty)_x$ with respect to λ_ν. In case (ii), where f^∞ is not continuous, the discontinuity set is Borel-measurable and $(\mathcal{L}^d+\lambda_\nu)$-negligible, so the above expression is still well-defined.

Proof. We will show the representation for a Carathéodory integrand $f\colon\overline{\Omega}\times\mathbb{R}^N\to\mathbb{R}$ that possesses a jointly continuous recession function $f^\infty\colon(\overline{\Omega}\setminus Z)\times\mathbb{R}^N\to\mathbb{R}$ in the sense of (14.1), where $Z\subset\overline{\Omega}$ is a Borel set with $(\mathcal{L}^d+\lambda_\nu)(Z)=0$. This implies both (i) and (ii).

Step 1. First, we assume that $f(x,\bullet)$ has uniformly bounded support, that is, $\operatorname{supp} f(x,\bullet)\Subset B_R$ for all $x\in\overline{\Omega}$ and a fixed $R>0$. Since $f^\infty\equiv 0$, we need to show

$$\int_\Omega \big\langle f(x,\bullet),(\nu_j)_x\big\rangle\,\mathrm{d}x \to \int_\Omega \big\langle f(x,\bullet),\nu_x\big\rangle\,\mathrm{d}x.$$

Let $(\psi_k)_k \subset \mathrm{C}_0(B_{R/(1+R)})$ be a countable and dense family and fix $\varepsilon > 0$. The set

$$E_k := \left\{ x \in \overline{\Omega} \,:\, \|Sf(x,\bullet) - \psi_k\|_\infty \le \varepsilon \right\}, \qquad k \in \mathbb{N},$$

is measurable since, by the continuity of $f(x,\bullet)$ for fixed $x \in \overline{\Omega}$,

$$E_k = \bigcap_{A \in \mathbb{Q}^N} \left\{ x \in \overline{\Omega} \,:\, |Sf(x,A) - \psi_k(A)| \le \varepsilon \right\}.$$

We have that $\bigcup_{k\in\mathbb{N}} E_k = \overline{\Omega}$ because for any fixed $x \in \overline{\Omega}$ it holds that $Sf(x,\bullet) \in \mathrm{C}_0(B_{R/(1+R)})$ and thus $x \in E_k$ for at least one $k \in \mathbb{N}$. Consequently, the inductively defined sets

$$F_k := E_k \setminus \bigcup_{i=1}^{k-1} F_i, \qquad k \in \mathbb{N},$$

form a measurable disjoint partition of $\overline{\Omega}$.

Define

$$g_\varepsilon(x,A) := \sum_{k=1}^{\infty} \mathbb{1}_{F_k}(x) S^{-1}\psi_k(A), \qquad (x,A) \in \overline{\Omega} \times \mathbb{R}^N.$$

Then, $\|Sf - Sg_\varepsilon\|_\infty \le \varepsilon$, whereby in particular

$$\|g_\varepsilon\|_\infty \le (1+R)(\|Sf\|_\infty + \varepsilon).$$

For each fixed $k \in \mathbb{N}$ we infer from $\nu_j \overset{*}{\rightharpoonup} \nu$ that for all $\varphi \in \mathrm{C}^\infty(\overline{\Omega})$ it holds that

$$\int \varphi(x) \left\langle S^{-1}\psi_k, (\nu_j)_x \right\rangle \mathrm{d}x \to \int \varphi(x) \left\langle S^{-1}\psi_k, \nu_x \right\rangle \mathrm{d}x.$$

Thus,

$$\int_{F_k} \left\langle S^{-1}\psi_k, (\nu_j)_x \right\rangle \mathrm{d}x \to \int_{F_k} \left\langle S^{-1}\psi_k, \nu_x \right\rangle \mathrm{d}x \tag{15.8}$$

because we can L^1-approximate $\mathbb{1}_{F_k}$ by smooth functions. Since

$$\left| \int_{F_k} \left\langle S^{-1}\psi_k, (\nu_j)_x \right\rangle \mathrm{d}x \right| \le (1+R)(\|Sf\|_\infty + \varepsilon)|F_k|,$$

we can then use the dominated convergence theorem for sums and (15.8) to compute

$$\begin{aligned} \lim_{j\to\infty} \int_\Omega \left\langle g_\varepsilon(x,\bullet), (\nu_j)_x \right\rangle \mathrm{d}x &= \lim_{j\to\infty} \sum_{k=1}^{\infty} \int_{F_k} \left\langle S^{-1}\psi_k, (\nu_j)_x \right\rangle \mathrm{d}x \\ &= \sum_{k=1}^{\infty} \lim_{j\to\infty} \int_{F_k} \left\langle S^{-1}\psi_k, (\nu_j)_x \right\rangle \mathrm{d}x \end{aligned}$$

$$= \sum_{k=1}^{\infty} \int_{F_k} \langle S^{-1}\psi_k, \nu_x \rangle \, \mathrm{d}x$$
$$= \int_{\Omega} \langle g_\varepsilon(x, \bullet), \nu_x \rangle \, \mathrm{d}x.$$

Moreover, $\|Sf - Sg_\varepsilon\|_\infty \le \varepsilon$, and so,

$$\left| \int_\Omega \langle g_\varepsilon(x, \bullet), (\nu_j)_x \rangle \, \mathrm{d}x - \int_\Omega \langle f(x, \bullet), (\nu_j)_x \rangle \, \mathrm{d}x \right| \le \varepsilon \int_\Omega \langle 1 + |\bullet|, (\nu_j)_x \rangle \, \mathrm{d}x$$
$$\le \varepsilon C$$

for some j-independent constant $C > 0$; the same holds with ν in place of ν_j. Combining these arguments,

$$\left| \lim_{j\to\infty} \langle\!\langle f, \nu_j \rangle\!\rangle - \langle\!\langle f, \nu \rangle\!\rangle \right| \le \left| \lim_{j\to\infty} \langle\!\langle g_\varepsilon, \nu_j \rangle\!\rangle - \langle\!\langle g_\varepsilon, \nu \rangle\!\rangle \right| + 2\varepsilon C = 2\varepsilon C.$$

As $\varepsilon > 0$ was arbitrary, we have proved the assertion in the case when $f(x, \bullet)$ has uniformly bounded support.

Step 2. Next, we extend the representation to Carathéodory integrands $f : \overline{\Omega} \times \mathbb{R}^N \to \mathbb{R}$ with $f^\infty \equiv 0$. Fix $\varepsilon > 0$ and let $r \in (0, 1)$ be so large that $|Sf(x, A)| \le \varepsilon$ for all $x \in \overline{\Omega}$ and $A \in \mathbb{B}^N$ with $|A| \ge r$. We can see that such an $r > 0$ must exist since otherwise we could find sequences $(x_n)_n \subset \overline{\Omega}$ and $(A_n)_n \subset \mathbb{B}^N$ with $|A_n| \ge 1 - n^{-1}$ and $Sf(x_n, A_n) \ge \varepsilon$. Without loss of generality we may assume $x_n \to x \in \overline{\Omega}$, $A_n \to A \in \partial\mathbb{B}^N$ and so f^∞ could not be zero everywhere.

Select a cut-off function $\varrho \in \mathrm{C}_c^\infty(\mathbb{R}^N; [0, 1])$ with $\varrho \equiv 1$ on $B_{r/(1-r)}$. Then, writing $f\varrho$ for the function $(x, A) \mapsto f(x, A)\varrho(A)$, where $x \in \overline{\Omega}$, $A \in \mathbb{R}^N$, we have

$$\left| \langle\!\langle f\varrho - f, \nu_j \rangle\!\rangle \right| \le \varepsilon \left[\int_\Omega \langle 1 + |\bullet|, (\nu_j)_x \rangle \, \mathrm{d}x + \lambda_{\nu_j}(\overline{\Omega}) \right] \le \varepsilon C$$

for some constant $C > 0$ and all $j \in \mathbb{N}$; the same holds with ν in place of ν_j. By the previous step,

$$\langle\!\langle f\varrho, \nu_j \rangle\!\rangle \to \langle\!\langle f\varrho, \nu \rangle\!\rangle$$

and so, combining with the previous estimate, the conclusion follows in the case when $f^\infty \equiv 0$.

Step 3. Finally, when f^∞ is not identically zero, we write

$$f = g + f^\infty \qquad \text{with} \qquad g^\infty \equiv 0.$$

The last step applies to g and we obtain

$$\langle\!\langle g, \nu_j \rangle\!\rangle \to \langle\!\langle g, \nu \rangle\!\rangle. \tag{15.9}$$

To investigate the convergence for the positively 1-homogeneous functions f^∞, we define the measures $\mu_j := S^{-*}\nu_j \in \mathcal{M}^+(\overline{\Omega \times \mathbb{B}^N})$. Then,

$$\mu_j \overset{*}{\rightharpoonup} \mu := S^{-*}\nu$$

and

$$\langle \Phi, \mu_j \rangle = \int_\Omega \langle S^{-1}\Phi(x, \bullet), (\nu_j)_x \rangle \, \mathrm{d}x + \int_{\overline{\Omega}} \langle \Phi(x, \bullet), (\nu_j)_x^\infty \rangle \, \mathrm{d}\lambda_{\nu_j}(x)$$

for all $\Phi \in \mathrm{C}(\overline{\Omega \times \mathbb{B}^N})$. Testing the convergence with all Φ such that $\|\Phi\|_\infty \le 1$, we get

$$|\mu_j| \le \langle 1 + |\bullet|, (\nu_j)_x \rangle \, \mathcal{L}^d \llcorner \Omega + \lambda_{\nu_j} \overset{*}{\rightharpoonup} \langle 1 + |\bullet|, \nu_x \rangle \, \mathcal{L}^d \llcorner \Omega + \lambda_\nu.$$

Denote by Λ the weak* limit of the $|\mu_j|$ in $\mathcal{M}^+(\overline{\Omega})$. Then, $(\mathcal{L}^d + \lambda_\nu)(Z) = 0$ implies $\Lambda(Z \times \overline{\mathbb{B}^N}) = 0$. By standard results in measure theory (see Lemma A.23), for any bounded Borel function $\Psi \colon \overline{\Omega \times \mathbb{B}^N} \to \mathbb{R}$ with a Λ-negligible set of discontinuity points we have

$$\langle \Psi, \mu_j \rangle \to \langle \Psi, \mu \rangle.$$

For $\Psi := Sf^\infty$, which by assumption is continuous outside $Z \times \overline{\mathbb{B}^N}$, this gives

$$\langle\!\langle f^\infty, \nu_j \rangle\!\rangle = \langle Sf^\infty, \mu_j \rangle \to \langle Sf^\infty, \mu \rangle = \langle\!\langle f^\infty, \nu \rangle\!\rangle.$$

Combining this with (15.9), we arrive at

$$\langle\!\langle f, \nu_j \rangle\!\rangle = \langle\!\langle g, \nu_j \rangle\!\rangle + \langle\!\langle f^\infty, \nu_j \rangle\!\rangle \to \langle\!\langle g, \nu \rangle\!\rangle + \langle\!\langle f^\infty, \nu \rangle\!\rangle = \langle\!\langle f, \nu \rangle\!\rangle.$$

This shows the assertion at the beginning of the proof and thus (i) and (ii). □

Example 15.12 Let $\Omega := (-1, 1)$ and $f(x, A) := \mathbb{1}_{(0,1)}(x)|A|$ for $x \in \Omega$, $A \in \mathbb{R}$, so that $f^\infty(x, A) = \mathbb{1}_{(0,1)}(x)|A|$. Then, for $\nu_j \in \mathbf{Y}^{\mathcal{M}}(\Omega)$ given as

$$(\nu_j)_x := \delta_0 \quad \text{a.e.}, \qquad \lambda_{\nu_j} := \delta_{1/j}, \qquad (\nu_j)_{1/j}^\infty := \delta_{+1},$$

we have $\nu_j \overset{*}{\rightharpoonup} \nu$ for

$$\nu_x = \delta_0 \quad \text{a.e.}, \qquad \lambda_\nu = \delta_0, \qquad \nu_0^\infty = \delta_{+1}.$$

On the other hand,

$$\lim_{j\to\infty} \langle\!\langle f, \nu_j \rangle\!\rangle = \lim_{j\to\infty} \int_{(0,1)} \mathbb{1}_{(0,1)}(x) \, \mathrm{d}\delta_{1/j}(x) = 1 \neq 0 = \langle\!\langle f, \nu \rangle\!\rangle.$$

Here, the discontinuity set $\{0, 1\}$ of f^∞ is not negligible with respect to $\mathcal{L}^d + \delta_0$. This example therefore shows that in Proposition 15.11 (ii) the assumption $(\mathcal{L}^d + \lambda_\nu)(\partial B) = 0$ is necessary.

15.4 Strong Precompactness of Sequences

By Vitali's Convergence Theorem A.12, the absence of oscillations and concentrations implies strong precompactness of an L^1-bounded sequence. In this section we study in more detail how compactness properties are reflected in the generated Young measure.

We start with oscillations:

Lemma 15.13 *Let $(V_j)_j \subset \mathrm{L}^1(\Omega;\mathbb{R}^N)$ with $V_j \overset{\mathbf{Y}}{\to} \nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$. Then, the sequence $(V_j)_j$ converges in measure to $V \in \mathrm{L}^1(\Omega;\mathbb{R}^N)$ if and only if $\nu_x = \delta_{V(x)}$ almost everywhere.*

Proof. The proof is very similar to that of Lemma 4.13 once we observe that the integrand

$$\chi(x,A) := \frac{|A - V(x)|}{1 + |A - V(x)|}, \qquad (x,A) \in \Omega \times \mathbb{R}^N,$$

lies in $\mathbf{R}(\Omega;\mathbb{R}^N)$ and $\chi^\infty \equiv 0$. □

Next, we show how the concentration parts $\lambda_\nu, (\nu_x^\infty)_x$ of a Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ reflect the equiintegrability properties of the generating sequence.

Lemma 15.14 *Let $(V_j)_j \subset \mathrm{L}^1(\Omega;\mathbb{R}^N)$ with $V_j \overset{\mathbf{Y}}{\to} \nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$.*

(i) The sequence $(V_j)_j$ is equiintegrable if and only if $\lambda_\nu = 0$.

(ii) For $f \in \mathbf{E}(\Omega;\mathbb{R}^N)$ let $F_j(x) := f(x, V_j(x))$ $(x \in \Omega)$. Then, the sequence $(F_j)_j$ is equiintegrable if and only if $\langle |f^\infty(x,\cdot)|, \nu_x^\infty \rangle = 0$ for λ_ν-almost every $x \in \overline{\Omega}$.

Proof. We only need to show (ii), for (i) take $f(x,A) := |A|$. So, let $f \in \mathbf{E}(\Omega;\mathbb{R}^N)$. Set, for $R > 0$,

$$\eta_R := \limsup_{j\to\infty} \int_{\{|F_j| \geq R\}} |F_j(x)|\,\mathrm{d}x, \qquad \eta_\infty := \lim_{R\uparrow\infty} \eta_R.$$

The family $(F_j)_j$ is equiintegrable if and only if $\eta_\infty = 0$ (see Appendix A.3). In the following we will prove the formula

$$\eta_\infty = \int_{\overline{\Omega}} \left\langle |f^\infty(x,\cdot)|, \nu_x^\infty \right\rangle \mathrm{d}\lambda_\nu(x), \tag{15.10}$$

which implies (ii).

Let, for $t \geq 0$,

$$h(t) := \begin{cases} 0 & \text{if } 0 < t < \frac{1}{2}, \\ 2t - 1 & \text{if } \frac{1}{2} \leq t \leq 1, \\ t & \text{if } t > 1, \end{cases} \qquad h_R(t) := Rh\left(\frac{t}{R}\right).$$

We have

$$h_{2R}(t) \leq t\mathbb{1}_{[R,\infty)}(t) \leq h_R(t) \quad \text{for all } t \geq 0, \qquad (h_R \circ |f|)^\infty = |f^\infty|.$$

This allows us to estimate

$$\int_\Omega h_{2R}(|F_j(x)|)\,\mathrm{d}x \le \int_{\{|F_j|\ge R\}} |F_j(x)|\,\mathrm{d}x \le \int_\Omega h_R(|F_j(x)|)\,\mathrm{d}x.$$

Letting $j \to \infty$, we arrive at

$$\begin{aligned}\int_\Omega \big\langle h_{2R}\circ|f(x,\bullet)|,\nu_x\big\rangle\,\mathrm{d}x + \int_{\overline{\Omega}}\big\langle|f^\infty(x,\bullet)|,\nu_x^\infty\big\rangle\,\mathrm{d}\lambda_\nu(x)\\ \le \eta_R\\ \le \int_\Omega \big\langle h_R\circ|f(x,\bullet)|,\nu_x\big\rangle\,\mathrm{d}x + \int_{\overline{\Omega}}\big\langle|f^\infty(x,\bullet)|,\nu_x^\infty\big\rangle\,\mathrm{d}\lambda_\nu(x).\end{aligned}$$

For $R \to \infty$ the first integral in both the first and the last expression vanishes and (15.10) follows. □

We can now combine the last two lemmas via Vitali's Convergence Theorem A.12 to decide whether an L^1-bounded sequences is strongly precompact:

Corollary 15.15 *Let $(V_j)_j \subset \mathrm{L}^1(\Omega;\mathbb{R}^N)$ with $V_j \xrightarrow{\mathbf{Y}} \nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$. Then, $V_j \to V$ in L^1 if and only if $\nu_x = \delta_{V(x)}$ for ($\mathcal{L}^d$-)almost every $x \in \Omega$ and $\lambda_\nu = 0$.*

15.5 BV-Young Measures

Like for classical Young measures, the most important subclass of generalized Young measures are those generated by gradients. Here, we consider gradients of $\mathrm{W}^{1,1}$-maps or, more generally, BV-derivatives. We thus specialize the framework defined in the preceding sections to $\mathbb{R}^N = \mathbb{R}^{m\times d}$ and we denote by $\mathbb{B}^{m\times d}$ and $\partial\mathbb{B}^{m\times d}$ the unit ball and the unit sphere in $\mathbb{R}^{m\times d}$, respectively.

For $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ we associate with $Du \in \mathcal{M}(\Omega;\mathbb{R}^{m\times d})$ the elementary Young measure $\delta[Du] \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^{m\times d})$ as before, that is,

$$\begin{aligned}\delta[Du]_x &:= \delta_{\nabla u(x)} && \text{for } \mathcal{L}^d\text{-a.e. } x\in\Omega\\ \lambda_{\delta[Du]} &:= |D^s u|,\\ \delta[Du]_x^\infty &:= \delta_{P(x)} && \text{for } |D^s u|\text{-a.e. } x\in\Omega,\end{aligned}$$

where

$$P := \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|} \in \mathrm{L}^1(\Omega,|D^s u|;\partial\mathbb{B}^{m\times d}).$$

For $(u_j)_j \subset \mathrm{BV}(\Omega;\mathbb{R}^m)$ the sequence $(Du_j)_j$ generates the Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^{m\times d})$, in symbols

$$Du_j \xrightarrow{\mathbf{Y}} \nu,$$

if

$$\delta[Du_j] \overset{*}{\rightharpoonup} \nu \quad \text{in } \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^{m\times d}).$$

We collect all these **BV-Young measures** ν in the set

$$\mathbf{BVY}(\Omega;\mathbb{R}^{m\times d}) \subset \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^{m\times d}),$$

where we again identify equivalent Young measures, just like we did for $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^{m\times d})$.

The Fundamental Theorem 15.5 adapts to our BV-context as follows:

Theorem 15.16 *Let $(u_j)_j \subset \mathrm{BV}(\Omega;\mathbb{R}^m)$ be a uniformly norm-bounded sequence. Then, there exists a subsequence (not relabeled) such that $Du_j \overset{\mathbf{Y}}{\to} \nu$ for some $\nu \in \mathbf{BVY}(\Omega;\mathbb{R}^{m\times d})$.*

All the examples from Section 15.2 are in fact BV-Young measures since the generating sequences are defined on a one-dimensional domain and thus are trivially derivatives.

The restriction of the barycenter $[\nu]$ to Ω is a BV-derivative since for a sequence $u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{BV}(\Omega;\mathbb{R}^m)$ and the integrand $f(x,A) := A$, it follows from (15.3) that

$$Du_j \overset{*}{\rightharpoonup} [\nu] = [\nu_x]\,\mathcal{L}^d_x \llcorner \Omega + [\nu^\infty_x]\,\lambda_\nu(\mathrm{d}x) \quad \text{in } \mathcal{M}(\overline{\Omega};\mathbb{R}^{m\times d}).$$

On the other hand, $Du_j \overset{*}{\rightharpoonup} Du$ in $\mathcal{M}(\Omega;\mathbb{R}^{m\times d})$, and so

$$Du = [\nu]\llcorner\Omega = [\nu_x]\,\mathcal{L}^d_x\llcorner\Omega + [\nu^\infty_x]\,(\lambda_\nu\llcorner\Omega)(\mathrm{d}x).$$

Any $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ with $Du = [\nu]\llcorner\Omega$ is called an **underlying deformation** of ν.

Let

$$\lambda_\nu = \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}\,\mathcal{L}^d\llcorner\Omega + \lambda^s_\nu, \qquad \lambda^s_\nu \text{ singular to } \mathcal{L}^d,$$

be the Lebesgue–Radon–Nikodým decomposition of λ_ν with respect to $\mathcal{L}^d\llcorner\Omega$ and let

$$\lambda^s_\nu\llcorner\Omega = g\,|D^s u| + \lambda^*_\nu, \qquad g \in \mathrm{L}^1(\Omega,|D^s u|),\ \lambda^*_\nu \text{ singular to } |D^s u|,$$

be the Lebesgue–Radon–Nikodým decomposition of $\lambda^s_\nu\llcorner\Omega$ with respect to $|D^s u|$. Then

$$\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x) = [\nu^\infty_x]\,g(x) \qquad \text{for } |D^s u|\text{-a.e. } x\in\Omega$$

and

$$[\nu^\infty_x] = 0 \qquad \text{for } \lambda^*_\nu\text{-a.e. } x\in\Omega.$$

By Alberti's Rank-One Theorem 13.8, the matrix $[\nu^\infty_x]$ has rank one for $|D^s u|$-almost every $x\in\Omega$ (note that $g(x)\neq 0$ for $|D^s u|$-almost every $x\in\Omega$). Thus,

$$\operatorname{rank}[\nu^\infty_x] \le 1 \qquad \text{for } \lambda^s_\nu\text{-a.e. } x\in\Omega. \tag{15.11}$$

Before we proceed with the abstract theory, we give another example of a BV-Young measure.

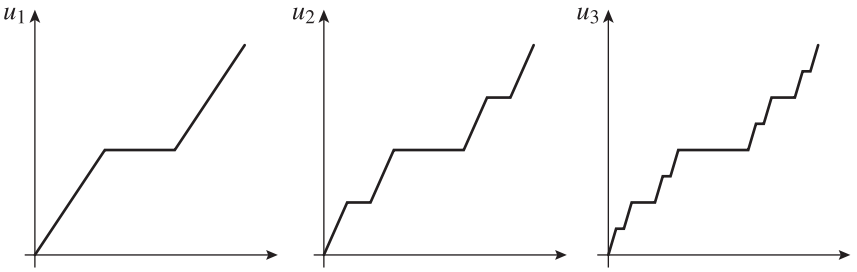


Fig. 15.5 The construction of the standard 1/3-Cantor function

Example 15.17 (Cantor functions) Fix $\delta \in (0, 1/2)$ and let C_j be the j'th set in the construction of the δ-Cantor set, i.e., $C_0 := (0, 1)$ and we obtain C_{j+1} from C_j by removing from each interval in C_j the centered open interval of length $\delta^j(1 - 2\delta)$. Then, the **δ-Cantor set** is

$$C := \bigcap_{j\in\mathbb{N}} C_j.$$

The usual "middle-third" Cantor set is obtained for $\delta = 1/3$. The maps

$$u_j(x) := \frac{1}{(2\delta)^j} \int_0^x \mathbb{1}_{C_j}\, \mathrm{d}y,$$

see Figure 15.5, converge area-strictly in $\mathrm{BV}(0, 1)$ to the **δ-Cantor function** $\chi_\delta \in \mathrm{C}((0, 1))$, where

$$\chi_\delta(x) = \alpha^{-1}\mathcal{H}^\gamma(C \cap [0, x]), \qquad D\chi_\delta = \alpha^{-1}\mathcal{H}^\gamma \llcorner C,$$

for

$$\gamma = \frac{\ln 2}{\ln(1/\delta)}, \qquad \alpha = \mathcal{H}^\gamma(C) = 2^{-\gamma}\frac{\pi^{\gamma/2}}{\Gamma(1 + \gamma/2)},$$

with Γ being Euler's Γ-function, see p. 60 ff. in [227] for the details. By Proposition 15.4 the Young measure $\nu \in \mathbf{BVY}((0, 1))$ generated by $(Du_j)_j$ is

$$\nu_x = \delta_0 \quad \mathcal{L}^d\text{-a.e.}, \qquad \lambda_\nu = \alpha^{-1}\,\mathcal{H}^\gamma \llcorner C, \qquad \nu_x^\infty = \delta_{+1} \quad \mathcal{H}^\gamma\text{-a.e.}$$

The following result on good generating sequences is often useful.

Proposition 15.18 *Let $\nu \in \mathbf{BVY}(\Omega; \mathbb{R}^{m\times d})$.*

(i) There exists a sequence $(u_j)_j \subset (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega; \mathbb{R}^m)$ with $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$.
(ii) If $\lambda_\nu(\partial\Omega) = 0$, then the u_j from (i) can be chosen to satisfy $u_j|_{\partial\Omega} = u|_{\partial\Omega}$ for any underlying deformation $u \in \mathrm{BV}(\Omega; \mathbb{R}^m)$ of ν.

See Problem 15.5 for the necessity of the assumption $\lambda_\nu(\partial\Omega) = 0$ in (ii).

Proof. Ad (i). Let $(v_j)_j \subset \mathrm{BV}(\Omega;\mathbb{R}^m)$ be a generating sequence for ν, that is, $Dv_j \overset{\mathbf{Y}}{\to} \nu$. We also take a countable collection $(f_k)_k \subset \mathbf{E}(\Omega;\mathbb{R}^{m\times d})$ that determines the Young measure convergence as in Lemma 15.10. Then, from the area-strict density of smooth functions in $\mathrm{BV}(\Omega;\mathbb{R}^m)$, see Lemma 13.7, in conjunction with Proposition 15.4 we construct for each $j \in \mathbb{N}$ a map $u_j \in (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^m)$ with

$$\left| \int_\Omega f_k(x, \nabla v_j(x))\,\mathrm{d}x + \int_\Omega f_k^\infty\left(x, \frac{\mathrm{d}D^s v_j}{\mathrm{d}|D^s v_j|}(x)\right) \mathrm{d}|D^s v_j|(x) - \int_\Omega f_k(x, \nabla u_j(x))\,\mathrm{d}x \right| \le \frac{1}{j}$$

whenever $k \le j$. Hence,

$$\int_\Omega f_k(x, \nabla u_j(x))\,\mathrm{d}x \to \langle\!\langle f_k, \nu \rangle\!\rangle \qquad \text{as } j \to \infty$$

for all $k \in \mathbb{N}$, and Lemma 15.10 thus implies $\nabla u_j \overset{\mathbf{Y}}{\to} \nu$.

Ad (ii). From (i) we know that there exist a sequence $(v_j)_j \subset (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^m)$ with $Dv_j \overset{\mathbf{Y}}{\to} \nu$. We may add constants to the v_j and take another subsequence so that in addition $v_j \to u$ in L^1. Then choose a sequence $(\varrho_n)_n \subset \mathrm{C}_c^\infty(\Omega;[0,1])$ of cut-off functions with $\varrho_n \uparrow \mathbb{1}_\Omega$ pointwise as $n \to \infty$ and such that for

$$K_n := \left\{ x \in \Omega \,:\, \varrho_n(x) = 1 \right\}$$

it holds that

$$(\mathcal{L}^d + \lambda_\nu)(\partial K_n) = 0.$$

Now employ Lemma 13.7 to get a sequence $(w_j)_j \subset (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^m)$ such that $w_j|_{\partial\Omega} = u|_{\partial\Omega}$ and $w_j \to u$ area-strictly. Set

$$u_{j,n} := \varrho_n v_j + (1 - \varrho_n) w_j \in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m).$$

Then, $u_{j,n}|_{\partial\Omega} = u|_{\partial\Omega}$,

$$\nabla u_{j,n} = \varrho_n \nabla v_j + (1 - \varrho_n)\nabla w_j + (v_j - w_j) \otimes \nabla \varrho_n,$$

and

$$\lim_{n\to\infty} \lim_{j\to\infty} \|u_{j,n} - u\|_{\mathrm{L}^1(\Omega;\mathbb{R}^m)} = 0.$$

Thus, for $f \in \mathbf{E}(\Omega;\mathbb{R}^{m\times d})$ with linear-growth constant $M > 0$ we can estimate

$$\begin{aligned} &\left| \langle\!\langle f, \delta[Du_{j,n}] \rangle\!\rangle - \langle\!\langle f, \delta[Dv_j] \rangle\!\rangle \right| \\ &\quad \le \int_{\Omega\setminus K_n} |f(\nabla u_{j,n})| + |f(\nabla v_j)|\,\mathrm{d}x \\ &\quad \le M \int_{\Omega\setminus K_n} 2 + 2|\nabla v_j| + |\nabla w_j| + |v_j - w_j||\nabla \varrho_n|\,\mathrm{d}x. \end{aligned}$$

Hence, since $(\mathcal{L}^d + \lambda_\nu)(\partial K_n) = 0$,

$$\begin{aligned}
&\limsup_{n\to\infty} \limsup_{j\to\infty} \left| \langle\!\langle f, \delta[Du_{j,n}] \rangle\!\rangle - \langle\!\langle f, \delta[Dv_j] \rangle\!\rangle \right| \\
&\quad \le \limsup_{n\to\infty} 2M \left(\int_{\overline{\Omega}\setminus K_n} \left\langle 1 + |\bullet|, \nu_x \right\rangle \mathrm{d}x + \lambda_\nu(\overline{\Omega} \setminus K_n) + |Du|(\Omega \setminus K_n) \right) \\
&\quad = 2M\lambda_\nu(\partial\Omega) \\
&\quad = 0.
\end{aligned}$$

We can then select a diagonal sequence $u_j := u_{j,n(j)}$ with the desired properties. □

Like for classical Young measures, the question arises whether one can characterize the subclass of BV-Young measures $\mathbf{BVY}(\Omega;\mathbb{R}^{m\times d})$ in the space $\mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^{m\times d})$. The analogue of the Kinderlehrer–Pedregal Theorem 8.11 is the following:

Theorem 15.19 (Kristensen–Rindler 2010 [195, 209]) *For any $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^{m\times d})$ with $\lambda_\nu(\partial\Omega) = 0$ it holds that $\nu \in \mathbf{BVY}(\Omega;\mathbb{R}^{m\times d})$ if and only if there exists a map $u \in \mathrm{BV}(\Omega;\mathbb{R}^m)$ with $[\nu] = Du$ and for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with linear growth the **(regular) Jensen-type inequality***

$$h\left([\nu_x] + [\nu_x^\infty] \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x) \right) \le \left\langle h, \nu_x \right\rangle + \left\langle h^\#, \nu_x^\infty \right\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)$$

holds at almost every $x \in \Omega$.

Remark 15.20 It turns out that in the situation of the preceding theorem for all quasiconvex $h\colon \mathbb{R}^{m\times d} \to \mathbb{R}$ with linear growth also the **singular Jensen-type inequality**

$$h^\#([\nu_x^\infty]) \le \left\langle h^\#, \nu_x^\infty \right\rangle$$

holds at λ_ν^s-almost every $x \in \Omega$, see Proposition 15.27 below.

A proof of Theorem 15.19 that is similar in spirit to our proof of the classical Kinderlehrer–Pedregal Theorem 8.11 can be found in [280]. This proof requires also the above singular Jensen-type inequality, which is, however, in fact always satisfied by the Kirchheim–Kristensen Theorem 13.17 in conjunction with Alberti's Rank-One Theorem 13.8. A refinement with regard to the boundary assumption $\lambda_\nu(\partial\Omega) = 0$ can be found in [31]. Recently, this characterization result has been extended to symmetric gradients in [113] and then to general PDE-constraints simultaneously in [20] and [208] (which also give new proofs of the above theorem).

15.6 Localization

In Proposition 5.13 we saw how we can "blow-up" or "localize" a classical gradient Young measure $\nu = (\nu_x)_x \in \mathbf{GY}^p(\Omega;\mathbb{R}^{m\times d})$, $p \in (1,\infty)$. As a consequence, for almost every $x_0 \in \Omega$ the probability measure ν_{x_0} is a homogeneous gradient Young measure in its own right. The discussion in this section parallels these developments for

generalized Young measures. As the barycenter is now a measure instead of a function, however, we need to employ the theory of tangent measures from Section 13.2 and we will also distinguish between *regular* and *singular* blow-ups.

Before we come to the localization principles, we need to introduce local versions of generalized Young measures. Define $\mathbf{Y}^{\mathcal{M}}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$ in the same way as $\mathbf{Y}^{\mathcal{M}}(\mathbb{R}^d;\mathbb{R}^N)$ but with λ_ν only in $\mathcal{M}^+_{\mathrm{loc}}(\mathbb{R}^d)$ and $x \mapsto \langle|\bullet|,\nu_x\rangle \in \mathrm{L}^1_{\mathrm{loc}}(\Omega)$. Now, $\mathbf{Y}^{\mathcal{M}}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$ can be seen as part of the dual space to $\mathbf{E}_c(\mathbb{R}^d;\mathbb{R}^N)$, which is defined like $\mathbf{E}(\mathbb{R}^d;\mathbb{R}^N)$, but requiring in addition that for $f \in \mathbf{E}_c(\mathbb{R}^d;\mathbb{R}^N) \subset \mathrm{C}(\mathbb{R}^d \times \mathbb{R}^N)$ there exists a compact set $K \subset \mathbb{R}^d$ with $\operatorname{supp} f(\bullet, A) \subset K$ for all $A \in \mathbb{R}^N$. Likewise, the **(local) weak* convergence** $\nu_j \overset{*}{\rightharpoonup} \nu$ in $\mathbf{Y}^{\mathcal{M}}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$ is defined with respect to $\mathbf{E}_c(\mathbb{R}^d;\mathbb{R}^N)$, that is, $\nu_j \overset{*}{\rightharpoonup} \nu$ if $\langle\!\langle f,\nu_j\rangle\!\rangle \to \langle\!\langle f,\nu\rangle\!\rangle$ for all $f \in \mathbf{E}_c(\mathbb{R}^d;\mathbb{R}^N)$. All of the results from the preceding sections also hold *mutatis mutandis* in $\mathbf{Y}^{\mathcal{M}}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$. In particular, we have the following compactness result.

Theorem 15.21 *Let $(\nu_j)_j \subset \mathbf{Y}^{\mathcal{M}}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$ satisfy*

$$\sup_{j\in\mathbb{N}} \big\langle\!\big\langle \varphi \otimes |\bullet|, \nu_j \big\rangle\!\big\rangle < \infty \qquad \text{for all } \varphi \in \mathrm{C}_c(\mathbb{R}^d).$$

Then, there exists a subsequence (not relabeled) such that

$$\nu_j \overset{*}{\rightharpoonup} \nu$$

for a Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^N)$.

Finally, define $\mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^{m\times d})$ as the space of all those local (generalized) Young measures that are generated by derivatives of sequences in $\mathrm{BV}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^m)$.

We are now in a position to state and prove the first localization principle.

Proposition 15.22 *Let $\nu \in \mathbf{BVY}(\Omega;\mathbb{R}^{m\times d})$. Then, for $\mathcal{L}^d$-almost every point $x_0 \in \Omega$ there exists a **regular tangent BV-Young measure** $\sigma \in \mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d;\mathbb{R}^{m\times d})$, that is,*

$$[\sigma] \in \operatorname{Tan}([\nu],x_0), \qquad \sigma_y = \nu_{x_0} \quad \text{for } \mathcal{L}^d\text{-a.e. } y, \tag{15.12}$$

$$\lambda_\sigma = \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0)\,\mathcal{L}^d \in \operatorname{Tan}(\lambda_\nu,x_0), \qquad \sigma^\infty_y = \nu^\infty_{x_0} \quad \text{for } \lambda_\sigma\text{-a.e. } y. \tag{15.13}$$

Proof. Step 1. Let the family $(\varphi_k \otimes h_k)_k \subset \mathbf{E}_c(\mathbb{R}^d;\mathbb{R}^{m\times d})$ determine the (local) weak* Young measure convergence; the construction of this family is analogous to the proof of Lemma 15.10.

Choose $x_0 \in \Omega$ with the following properties:

(a) There exists a sequence $r_n \downarrow 0$ such that with $P_0 := \frac{\mathrm{d}[\nu]}{\mathrm{d}\mathcal{L}^d}(x_0)$ it holds that

$$\gamma_n := r_n^{-d} T^{(x_0,r_n)}_{\#}[\nu] \overset{*}{\rightharpoonup} P_0\,\mathcal{L}^d \in \operatorname{Tan}([\nu],x_0),$$

where $T^{(x_0,r_n)}_{\#}[\nu]$ denotes the push-forward of the measure $[\nu]$ under the rescaling map $T^{(x_0,r_n)}(x) := (x-x_0)/r_n$ as in Section 13.2;

(b) $\lim_{r\downarrow 0} r^{-d}\lambda_\nu^s(B_r(x_0)) = 0$, where λ_ν^s is the Lebesgue-singular part of λ_ν, and

$$\frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0)\,\mathcal{L}^d \in \mathrm{Tan}(\lambda_\nu, x_0);$$

(c) the point x_0 is an $\mathcal{L}^d$-Lebesgue point for the functions

$$x \mapsto \langle h_k, \nu_x\rangle + \langle h_k^\infty, \nu_x^\infty\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x), \qquad k \in \mathbb{N}.$$

Indeed, at $\mathcal{L}^d$-almost every $x_0 \in \Omega$ condition (a) follows from Lemma 13.4; condition (b) is a consequence of the Besicovitch Differentiation Theorem A.24 and Proposition 13.5; finally, condition (c) is implied by Theorem A.21.

Step 2. By virtue of Proposition 15.18 take a sequence $(u_j)_j \subset (\mathrm{W}^{1,1}\cap \mathrm{C}^\infty)(\Omega;\mathbb{R}^m)$ with $\nabla u_j \xrightarrow{\mathbf{Y}} \nu$ and let $\widetilde{u}_j \in \mathrm{BV}(\mathbb{R}^d;\mathbb{R}^m)$ be the extension of u_j by zero to all of $\mathbb{R}^d$. Define

$$v_j^{(n)}(y) := \frac{\widetilde{u}_j(x_0 + r_n y)}{r_n}, \qquad y \in \mathbb{R}^d.$$

The following transformation rules hold for $T_\#^{(x_0,r_n)}\mu$, where $\mu \in \mathcal{M}(\Omega;\mathbb{R}^N)$:

$$\frac{\mathrm{d}T_\#^{(x_0,r_n)}\mu}{\mathrm{d}\mathcal{L}^d} = r_n^d \frac{\mathrm{d}\mu}{\mathrm{d}\mathcal{L}^d}(x_0 + r_n\bullet), \qquad \frac{\mathrm{d}T_\#^{(x_0,r_n)}\mu}{\mathrm{d}|T_\#^{(x_0,r_n)}\mu|} = \frac{\mathrm{d}\mu}{\mathrm{d}|\mu|}(x_0 + r_n\bullet).$$

We may then compute

$$\begin{aligned} Dv_j^{(n)} &= r_n^{-d} T_\#^{(x_0,r_n)} D\widetilde{u}_j \\ &= \nabla u_j(x_0 + r_n\bullet)\,\mathcal{L}^d + r_n^{-1}\big(u_j(x_0 + r_n\bullet)|_{\partial\Omega_n} \otimes n_{\Omega_n}\big)\,\mathcal{H}^{d-1}\llcorner\partial\Omega_n. \end{aligned}$$

Here, $\Omega_n := r_n^{-1}(\Omega - x_0)$, $n_{\Omega_n}\colon \partial\Omega_n \to \mathbb{S}^{d-1}$ is the unit inner normal to $\partial\Omega_n$, and $u_j(x_0 + r_n\bullet)|_{\partial\Omega_n}$ is the (inner) trace of $y \mapsto u_j(x_0 + r_n y)$ on $\partial\Omega_n$. Then, also using the BV-Poincaré inequality and the boundedness of the BV-trace operator, see Section 13.3,

$$\begin{aligned} \|v_j^{(n)}\|_{\mathrm{BV}(\mathbb{R}^d;\mathbb{R}^m)} &\le C(n)|Dv_j^{(n)}|(\mathbb{R}^d) \\ &= C(n)|D\widetilde{u}_j|(\mathbb{R}^d) \\ &\le C(n)\|u_j\|_{\mathrm{BV}(\Omega;\mathbb{R}^m)}, \end{aligned} \tag{15.14}$$

where we have absorbed all n-dependent constants (including r_n^{-d}) into $C(n) > 0$. If we hold n fixed, this expression is j-uniformly bounded. Consequently, we may select an n-dependent subsequence of the j's (not relabeled) such that $Dv_j^{(n)} \xrightarrow{\mathbf{Y}} \sigma^{(n)}$ for some $\sigma^{(n)} \in \mathbf{BVY}(\mathbb{R}^d;\mathbb{R}^{m\times d})$.

Step 3. Fix $\varphi_k \otimes h_k \in \mathbf{E}_c(\mathbb{R}^d;\mathbb{R}^{m\times d})$ from the Young measure-determining collection above and choose $n \in \mathbb{N}$ so large that $\operatorname{supp}\varphi_k \Subset \Omega_n$. Then, the boundary measure in $Dv_j^{(n)}$ (the part supported on $\partial\Omega_n$) can be neglected in the following calculation:

$$
\begin{aligned}
\Big\langle\!\Big\langle \varphi_k \otimes h_k, \sigma^{(n)} \Big\rangle\!\Big\rangle &= \lim_{j\to\infty} \int \varphi_k(y) h_k\big(\nabla v_j^{(n)}(y)\big)\,\mathrm{d}y \\
&= \lim_{j\to\infty} \int \varphi_k(y) h_k\big(\nabla u_j(x_0 + r_n y)\big)\,\mathrm{d}y \\
&= \lim_{j\to\infty} \frac{1}{r_n^d} \int \varphi_k\Big(\frac{x-x_0}{r_n}\Big) h_k(\nabla u_j(x))\,\mathrm{d}x \\
&= \frac{1}{r_n^d} \Big\langle\!\Big\langle \varphi_k\Big(\frac{\bullet - x_0}{r_n}\Big) \otimes h_k, \nu \Big\rangle\!\Big\rangle \\
&= \frac{1}{r_n^d} \int \varphi_k\Big(\frac{x-x_0}{r_n}\Big) \Big[\big\langle h_k, \nu_x \big\rangle + \big\langle h_k^\infty, \nu_x^\infty \big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x) \Big]\,\mathrm{d}x \\
&\quad + \frac{1}{r_n^d} \int \varphi_k\Big(\frac{x-x_0}{r_n}\Big) \big\langle h_k^\infty, \nu_x^\infty \big\rangle\,\mathrm{d}\lambda_\nu^s(x).
\end{aligned}
$$

We call the last two integrals the regular and the singular parts respectively.

For the regular part we have

$$
\begin{aligned}
&\frac{1}{r_n^d} \int \varphi_k\Big(\frac{x-x_0}{r_n}\Big) \Big[\big\langle h_k, \nu_x \big\rangle + \big\langle h_k^\infty, \nu_x^\infty \big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x) \Big]\,\mathrm{d}x \\
&\quad = \int \varphi_k(y) \Big[\big\langle h_k, \nu_{x_0 + r_n y} \big\rangle + \big\langle h^\infty, \nu_{x_0 + r_n y}^\infty \big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0 + r_n y) \Big]\,\mathrm{d}y \\
&\quad \to \int \varphi_k(y) \Big[\big\langle h_k, \nu_{x_0} \big\rangle + \big\langle h_k^\infty, \nu_{x_0}^\infty \big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0) \Big]\,\mathrm{d}y
\end{aligned}
$$

as $n \to \infty$ ($r_n \downarrow 0$) by the Lebesgue point property (c) above.

Turning to the singular part, let $N \in \mathbb{N}$ be so large that $\operatorname{supp} \varphi_k \subset B_N$. By assumption (b) on x_0,

$$
\Big| \frac{1}{r_n^d} \int \varphi_k\Big(\frac{x-x_0}{r_n}\Big) \big\langle h_k^\infty, \nu_x^\infty \big\rangle\,\mathrm{d}\lambda_\nu^s(x) \Big| \le M \|\varphi_k\|_\infty \cdot \frac{\lambda_\nu^s(B_{Nr_n}(x_0))}{r_n^d} \to 0
$$

as $n \to \infty$, where $M := \sup \{\, |h^\infty(A)| : A \in \partial\mathbb{B}^{m\times d} \,\}$.

Step 4. We may assume that the integrands $\varphi \otimes |\bullet|$ for a dense set of $\varphi \in \mathrm{C}_c(\mathbb{R}^d)$ (in the $\|\bullet\|_\infty$-norm) are contained in the collection $(\varphi_k \otimes h_k)_k$. Then, the above arguments imply

$$
\sup_{n\in\mathbb{N}} \Big| \Big\langle\!\Big\langle \varphi \otimes |\bullet|, \sigma^{(n)} \Big\rangle\!\Big\rangle \Big| < \infty \qquad \text{for all } \varphi \in \mathrm{C}_c(\mathbb{R}^d).
$$

Hence, the compactness result from Theorem 15.21 applies and we may select a subsequence (not relabeled) with

$$
\sigma^{(n)} \overset{*}{\rightharpoonup} \sigma \in \mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m\times d}).
$$

Here we also used that $\mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m\times d})$ is sequentially (locally) weakly* closed, see Problem 15.9. Since $[\sigma^{(n)}] = \gamma_n + \eta_n$, where η_n is a jump part measure that moves out to infinity in the limit, we furthermore get that $[\sigma] \in \mathrm{Tan}([\nu], x_0)$. This implies the first assertion in (15.12).

It also follows from the preceding arguments that

$$\langle\!\langle \varphi_k \otimes h, \sigma \rangle\!\rangle = \int \varphi_k(y) \left[\langle h_k, \nu_{x_0} \rangle + \langle h_k^\infty, \nu_{x_0}^\infty \rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0) \right] \mathrm{d}y$$

for all $\varphi_k \otimes h_k$ from the family exhibited at the beginning of the proof. Since the $\varphi_k \otimes h_k$ are dense in $\mathbf{E}_c(\mathbb{R}^d; \mathbb{R}^{m\times d})$ (in the sense that they determine Young measures), we have $\sigma_y = \nu_{x_0}$ and $\sigma_y^\infty = \nu_{x_0}^\infty$ for $\mathcal{L}^d$-almost every $y \in \mathbb{R}^d$, that is, the second assertion of (15.12) and the second assertion of (15.13) hold. Finally, the first assertion of (15.13) follows since

$$\lambda_\sigma = \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0)\, \mathcal{L}^d \in \mathrm{Tan}(\lambda_\nu, x_0)$$

by (b). □

Next, we investigate localization at singular points.

Proposition 15.23 *Let $\nu \in \mathbf{BVY}(\Omega; \mathbb{R}^{m\times d})$. Then, for λ_ν^s-almost every point $x_0 \in \Omega$ there exists a **singular tangent BV-Young measure** $\sigma \in \mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m\times d})$, that is,*

$$[\sigma] \in \mathrm{Tan}([\nu], x_0), \qquad \sigma_y = \delta_0 \quad \mathcal{L}^d\text{-a.e.}, \tag{15.15}$$

$$\lambda_\sigma \in \mathrm{Tan}(\lambda_\nu^s, x_0) \setminus \{0\}, \qquad \sigma_y^\infty = \nu_{x_0}^\infty \quad \lambda_\sigma\text{-a.e.} \tag{15.16}$$

Proof. Step 1. Let $(g_k)_k \subset \mathrm{C}(\mathbb{R}^{m\times d})$ be a countable set of positively 1-homogeneous functions whose restrictions to $\partial\mathbb{B}^{m\times d}$ are dense in $\mathrm{C}(\partial\mathbb{B}^{m\times d})$. Let $x_0 \in \Omega$ be such that the following properties hold:

(a) There exist sequences $r_n \downarrow 0$, $c_n > 0$ and $\lambda_0 \in \mathrm{Tan}(\lambda_\nu^s, x_0) \setminus \{0\}$ such that

$$c_n T_\#^{(x_0, r_n)} \lambda_\nu^s \overset{*}{\rightharpoonup} \lambda_0; \tag{15.17}$$

(b) it holds that

$$\lim_{r\downarrow 0} \frac{1}{\lambda_\nu^s(B_r(x_0))} \int_{B_r(x_0)} 1 + \left\langle |\bullet|, \nu_x \right\rangle + \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\, \mathrm{d}x = 0; \tag{15.18}$$

(c) the point x_0 is a λ_ν^s-Lebesgue point for the functions

$$x \mapsto [\nu_x^\infty] \qquad \text{and} \qquad x \mapsto \left\langle g_k, \nu_x^\infty \right\rangle, \quad k \in \mathbb{N}.$$

By Proposition 13.5 condition (a) holds at λ_ν^s-almost every $x_0 \in \Omega$, condition (b) follows from the Besicovitch Differentiation Theorem A.24, and condition (c) is a direct consequence of Theorem A.21. Further, by (13.3), the constants c_n in (15.17) can be chosen to be

$$c_n = c\left[\lambda_\nu^s(\overline{B_{Rr_n}(x_0)})\right]^{-1}$$

for any (henceforth fixed) $R > 0$ and corresponding $c > 0$ with the property that $\lambda_0(B_R) > 0$. We also choose R such that $\lambda_\nu^s(\partial B_{Rr_n}(x_0)) = 0$ for all n, which is always possible by the finiteness of the measure λ_ν^s (see Problem 13.2). Then, we get

from (15.17) that there exist constants $\beta_N > 0$, $N = 1, 2, \ldots$, such that

$$\limsup_{n\to\infty} c_n \lambda_\nu^s(B_{Nr_n}(x_0)) = \limsup_{n\to\infty} c \cdot \frac{\lambda_\nu^s(B_{Nr_n}(x_0))}{\lambda_\nu^s(B_{Rr_n}(x_0))} \le \beta_N. \tag{15.19}$$

Thus, with (15.18), we compute

$$\begin{aligned}
&\limsup_{n\to\infty} c_n \left\langle\!\!\left\langle \mathbb{1}_{B_{Nr_n}(x_0)} \otimes |\bullet|, \nu \right\rangle\!\!\right\rangle \\
&\quad = \limsup_{n\to\infty} \left[\frac{c}{\lambda_\nu^s(B_{Rr_n}(x_0))} \int_{B_{Nr_n}(x_0)} \left\langle |\bullet|, \nu_x \right\rangle + \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\, \mathrm{d}x \right. \\
&\qquad\qquad \left. + c \cdot \frac{\lambda_\nu^s(B_{Nr_n}(x_0))}{\lambda_\nu^s(B_{Rr_n}(x_0))} \right] \\
&\quad \le 0 + \beta_N.
\end{aligned}$$

Similarly,

$$\limsup_{n\to\infty} \left(c_n T_\#^{(x_0,r_n)} |[\nu]|\right)(B_N) \le \beta_N \qquad \text{for all } N \in \mathbb{N}.$$

After selecting a subsequence (not relabeled) of the r_n, we may thus suppose that

$$c_n T_\#^{(x_0,r_n)} [\nu] \overset{*}{\rightharpoonup} \tau \in \mathrm{Tan}([\nu], x_0). \tag{15.20}$$

We note that it is possible that $\tau = 0$ (clearly, $\tau \neq 0$ for $[\nu]$-almost every $x_0 \in \operatorname{supp}[\nu]$, but not necessarily for λ_ν^s-almost every $x_0 \in \Omega$).

Step 2. Let $(u_j)_j \subset (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(\Omega; \mathbb{R}^m)$ be such that $\nabla u_j \overset{\mathbf{Y}}{\to} \nu$, see Proposition 15.18. Denote by $\widetilde{u}_j \in \mathrm{BV}(\mathbb{R}^d; \mathbb{R}^m)$ the extension of u_j by zero and define

$$v_j^{(n)}(y) := r_n^{d-1} c_n \widetilde{u}_j(x_0 + r_n y), \qquad y \in \mathbb{R}^d.$$

We have

$$\begin{aligned}
Dv_j^{(n)} &= c_n T_\#^{(x_0,r_n)} D\widetilde{u}_j \\
&= c_n r_n^d \nabla u_j(x_0 + r_n \bullet)\mathcal{L}^d + c_n r_n^{d-1} u_j(x_0 + r_n \bullet)|_{\partial\Omega_n} \otimes n_{\Omega_n} \mathcal{H}^{d-1} \llcorner \Omega_n,
\end{aligned}$$

where $\Omega_n := r_n^{-1}(\Omega - x_0)$. Similarly to (15.14), we get

$$\|v_j^{(n)}\|_{\mathrm{BV}(\mathbb{R}^d;\mathbb{R}^m)} \le C(n) \|u_j\|_{\mathrm{BV}(\Omega;\mathbb{R}^m)}.$$

Hence, holding n fixed, we may assume, up to taking an n-dependent subsequence of the j's, that $Dv_j^{(n)} \overset{\mathbf{Y}}{\to} \sigma^{(n)}$ for some $\sigma^{(n)} \in \mathbf{BVY}(\mathbb{R}^d; \mathbb{R}^{m\times d})$ as $j \to \infty$.

Step 3. Fix a positively 1-homogeneous $g \in \mathrm{C}(\mathbb{R}^{m\times d})$ and $\varphi \in \mathrm{C}_c(\mathbb{R}^d)$. For all n so large that $\operatorname{supp}\varphi \Subset \Omega_n$,

$$\begin{aligned}
\left\langle\!\left\langle \varphi \otimes g, \sigma^{(n)} \right\rangle\!\right\rangle &= \lim_{j\to\infty} \int \varphi(y) g\big(\nabla v_j^{(n)}(y)\big)\,\mathrm{d}y \\
&= \lim_{j\to\infty} r_n^d c_n \int \varphi(y) g\big(\nabla u_j(x_0 + r_n y)\big)\,\mathrm{d}y \\
&= \lim_{j\to\infty} c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big) g(\nabla u_j(x))\,\mathrm{d}y \\
&= c_n \left\langle\!\left\langle \varphi\Big(\frac{\bullet - x_0}{r_n}\Big) \otimes g, \nu \right\rangle\!\right\rangle \\
&= c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big)\Big[\big\langle g, \nu_x\big\rangle + \big\langle g, \nu_x^\infty\big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\Big]\,\mathrm{d}x \\
&\quad + c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big)\big\langle g, \nu_x^\infty\big\rangle\,\mathrm{d}\lambda_\nu^s(x). \qquad (15.21)
\end{aligned}$$

As before, we call the last two integrals the regular and singular parts, respectively.

The regular part is estimated as follows: Set $M := \sup\{\,|g(A)| : A \in \partial\mathbb{B}^{m\times d}\,\}$ and pick $N \in \mathbb{N}$ large enough such that $\operatorname{supp}\varphi \subset B_N$. Via (15.19) we assume n to be so large that

$$c_n \lambda_\nu^s(B_{Nr_n}(x_0)) \le \beta_N + 1.$$

Then,

$$\begin{aligned}
&\left| c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big)\Big[\big\langle g, \nu_x\big\rangle + \big\langle g, \nu_x^\infty\big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\Big]\,\mathrm{d}x \right| \\
&\quad \le c_n M \|\varphi\|_\infty \int_{B_{Nr_n}(x_0)} \big\langle |\bullet|, \nu_x\big\rangle + \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\,\mathrm{d}x \\
&\quad \le \frac{M\|\varphi\|_\infty(\beta_N+1)}{\lambda_\nu^s(B_{Nr_n}(x_0))} \int_{B_{Nr_n}(x_0)} \big\langle |\bullet|, \nu_x\big\rangle + \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\,\mathrm{d}x \\
&\quad \to 0 \qquad (15.22)
\end{aligned}$$

as $n \to \infty$. Here, the convergence in the last line follows from (15.18). Plugging this back into (15.21), we get

$$\limsup_{n\to\infty} \left\langle\!\left\langle \varphi \otimes g, \sigma^{(n)} \right\rangle\!\right\rangle = \limsup_{n\to\infty}\, c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big)\big\langle g, \nu_x^\infty\big\rangle\,\mathrm{d}\lambda_\nu^s(x). \qquad (15.23)$$

For $g = |\bullet|$, also using (15.17), this gives

$$\begin{aligned}\limsup_{n\to\infty} \langle\!\langle \varphi \otimes |\bullet|, \sigma^{(n)} \rangle\!\rangle &= \limsup_{n\to\infty} c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big)\, \mathrm{d}\lambda_\nu^s(x) \\ &= \limsup_{n\to\infty} \int \varphi\, \mathrm{d}\big(c_n T_\#^{(x_0,r_n)} \lambda_\nu^s\big) \\ &= \int \varphi\, \mathrm{d}\lambda_0.\end{aligned}$$

In particular,

$$\limsup_{n\to\infty} \langle\!\langle \varphi \otimes |\bullet|, \sigma^{(n)} \rangle\!\rangle \le \|\varphi\|_\infty\, \lambda_0(\operatorname{supp}\varphi).$$

Thus, the Young measure compactness criterion, see Theorem 15.21, implies that there exists a subsequence of the r_n's (not relabeled) with

$$\sigma^{(n)} \overset{*}{\rightharpoonup} \sigma \in \mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m\times d}),$$

where we also used that $\mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m\times d})$ is sequentially weakly* closed, see Problem 15.9. Furthermore, (15.23) implies

$$\langle\!\langle \varphi \otimes g, \sigma \rangle\!\rangle = \lim_{n\to\infty} c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big) \big\langle g, \nu_x^\infty \big\rangle\, \mathrm{d}\lambda_\nu^s(x). \tag{15.24}$$

Step 4. We have

$$[\sigma^{(n)}] = c_n T_\#^{(x_0,r_n)}[\nu] + \eta_n,$$

where $\eta_n \in \mathcal{M}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m\times d})$ is a measure carried by the set $\partial\Omega_n$, whereby $\eta_n \overset{*}{\rightharpoonup} 0$. From (15.20), we thus infer that $[\sigma^{(n)}] \overset{*}{\rightharpoonup} \tau$ as $n \to \infty$ and

$$[\sigma] = \tau \in \mathrm{Tan}([\nu], x_0),$$

which is the first statement in (15.15). We now take functions $\varphi \in \mathrm{C}_c(\mathbb{R}^d)$, $\chi \in \mathrm{C}_c(\mathbb{R}^{m\times d})$ and in a similar fashion to (15.21) derive that

$$\begin{aligned}\langle\!\langle \varphi \otimes |\bullet|\chi(\bullet), \sigma^{(n)} \rangle\!\rangle &= \lim_{j\to\infty} c_n \int \varphi\Big(\frac{x-x_0}{r_n}\Big) |\nabla u_j(x)| \chi\big(r_n^d c_n \nabla u_j(x)\big)\, \mathrm{d}x \\ &= c_n \Big\langle\!\Big\langle \varphi\Big(\frac{\bullet - x_0}{r_n}\Big) \otimes |\bullet|\chi(r_n^d c_n \bullet), \nu \Big\rangle\!\Big\rangle \\ &\to 0\end{aligned}$$

as $n \to \infty$. Here, the last convergence follows analogously to (15.22) since χ has compact support in $\mathbb{R}^{m\times d}$. Thus,

$$\langle\!\langle \varphi \otimes |\bullet|\chi(\bullet), \sigma \rangle\!\rangle = 0$$

for all φ, χ as above. From this, varying φ and χ, we conclude that $\sigma_y = \delta_0$ for $\mathcal{L}^d$-almost every $y \in \mathbb{R}^d$, which is the second statement of (15.15).

Moving on to the first assertion of (15.16), use $g := |\bullet|$ in (15.24) and the previously shown fact $\sigma_y = \delta_0$ almost everywhere to derive

$$\begin{aligned}\int \varphi \, \mathrm{d}\lambda_\sigma &= \langle\!\langle \varphi \otimes |\bullet|, \sigma \rangle\!\rangle \\ &= \lim_{n\to\infty} c_n \int \varphi\Big(\frac{x - x_0}{r_n}\Big) \, \mathrm{d}\lambda_\nu^s(x) \\ &= \lim_{n\to\infty} \int \varphi \, \mathrm{d}\big(c_n T_\#^{(x_0, r_n)} \lambda_\nu^s\big) \\ &= \int \varphi \, \mathrm{d}\lambda_0\end{aligned}$$

for any $\varphi \in \mathrm{C}_c(\mathbb{R}^d)$. Here, the last equality is a consequence of (15.17). Thus, indeed $\lambda_\sigma = \lambda_0 \in \mathrm{Tan}(\lambda_\nu^s, x_0)$.

Finally, to establish the second statement of (15.16), we first prove the following assertion: Let $U \subset \mathbb{R}^d$ be open and bounded with $(\mathcal{L}^d + \lambda_\sigma)(\partial U) = 0$ and let $g \in \mathrm{C}(\mathbb{R}^{m\times d})$ be positively 1-homogeneous. Then,

$$\langle\!\langle \mathbb{1}_U \otimes g, \sigma \rangle\!\rangle = \big\langle g, \nu_{x_0}^\infty \big\rangle \, \lambda_\sigma(U). \tag{15.25}$$

Once this is established we may vary U and g to see that $\sigma_y^\infty = \nu_{x_0}^\infty$ for λ_σ-almost every $y \in \mathbb{R}^d$, which is the second claim of (15.16).

To prove (15.25), we assume $\lambda_\sigma(U) > 0$ since the case $\lambda_\sigma(U) = 0$ is trivial. Then use $\varphi = \mathbb{1}_U$ in (15.24) via Proposition 15.11 (ii) to get

$$\int_U \big\langle g, \sigma_y^\infty \big\rangle \, \mathrm{d}\lambda_\sigma(y) = \langle\!\langle \mathbb{1}_U \otimes g, \sigma \rangle\!\rangle = \lim_{n\to\infty} c_n \int_{x_0 + r_n U} \big\langle g, \nu_x^\infty \big\rangle \, \mathrm{d}\lambda_\nu^s(x).$$

Since $\lambda_\sigma \in \mathrm{Tan}(\lambda_\nu^s, x_0)$ and $\lambda_\sigma(U) > 0$, (13.3) implies

$$c_n = \frac{\widetilde{c}(U)}{\lambda_\nu^s(x_0 + r_n U)}$$

for some $\widetilde{c}(U) > 0$. Consequently,

$$\begin{aligned}\lim_{n\to\infty} c_n \int_{x_0 + r_n U} \big\langle g, \nu_x^\infty \big\rangle \, \mathrm{d}\lambda_\nu^s(x) &= \widetilde{c}(U) \cdot \lim_{n\to\infty} \fint_{x_0 + r_n U} \big\langle g, \nu_x^\infty \big\rangle \, \mathrm{d}\lambda_\nu^s(x) \\ &= \widetilde{c}(U) \, \big\langle g, \nu_{x_0}^\infty \big\rangle\end{aligned}$$

by the Lebesgue point property (c) of x_0 (first for $g = g_k$ and then by density for the general case). Thus, we have

$$\int_U \big\langle g, \sigma_y^\infty \big\rangle \, \mathrm{d}\lambda_\sigma(y) = \langle\!\langle \mathbb{1}_U \otimes g, \sigma \rangle\!\rangle = \widetilde{c}(U) \, \big\langle g, \nu_{x_0}^\infty \big\rangle.$$

With $g = |\bullet|$ we see that $\widetilde{c}(U) = \lambda_\sigma(U)$ and (15.25) follows. □

15.7 Lower Semicontinuity

We now show how the theory of generalized Young measures can be used to investigate the weak* lower semicontinuity properties of functionals with linear-growth integrands. First, we extend Theorem 14.1 as follows:

Proposition 15.24 *Let $f \in \mathbf{R}(\Omega;\mathbb{R}^{m\times d})$. Then, the functional*

$$\widetilde{\mathcal{F}}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x + \int_\Omega f^\infty\left(x,\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x) + \int_{\partial\Omega} f^\infty\big(x,u(x)\otimes n_\Omega(x)\big)\,\mathrm{d}\mathcal{H}^{d-1}(x), \qquad u\in \mathrm{BV}(\Omega;\mathbb{R}^m),$$

with $n_\Omega\colon \partial\Omega \to \mathbb{S}^{d-1}$ the unit inner normal to $\partial\Omega$, is the area-strictly continuous extension to the space $\mathrm{BV}(\Omega;\mathbb{R}^m)$ of the functional

$$\mathcal{F}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x + \int_{\partial\Omega} f^\infty\big(x,u(x)\otimes n_\Omega(x)\big)\,\mathrm{d}\mathcal{H}^{d-1}(x),$$

where $u\in \mathrm{W}^{1,1}(\Omega;\mathbb{R}^m)$.

Proof. Extend u by zero to a larger Lipschitz domain $\Omega' \ni \Omega$. The space $\mathrm{W}^{1,1}(\Omega';\mathbb{R}^m)$ is area-strictly dense in $\mathrm{BV}(\Omega';\mathbb{R}^m)$ by Lemma 13.7. The result then follows by Proposition 15.4, applied on the domain Ω', in conjunction with Proposition 15.11. □

The following is the main lower semicontinuity result of this chapter. In comparison to Theorem 14.5 we here allow the integrand to be x-dependent, but we need to require that the recession function exists in the strong sense (it should be remarked that already [144] treated x-dependent integrands, albeit under further technical assumptions).

Theorem 15.25 *Let $f \in \mathbf{R}(\Omega;\mathbb{R}^{m\times d})$ be positive and quasiconvex in the second argument, that is, we suppose that $f\colon \overline{\Omega}\times\mathbb{R}^{m\times d}\to[0,\infty)$ is such that the following assumptions hold:*

(i) f is a Carathéodory integrand;
(ii) $f(x,A)\le M(1+|A|)$ for some $M>0$ and all $x\in\overline{\Omega}$, $A\in\mathbb{R}^{m\times d}$;
(iii) the strong recession function f^∞ exists in the sense of (14.1) and is (jointly) continuous on $\overline{\Omega}\times\mathbb{R}^{m\times d}$;
(iv) $f(x,\bullet)$ is quasiconvex for all $x\in\overline{\Omega}$.

Then, the functional

$$\mathcal{F}[u] := \int_\Omega f(x,\nabla u(x))\,\mathrm{d}x + \int_\Omega f^\infty\left(x,\frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x) + \int_{\partial\Omega} f^\infty\big(x,u(x)\otimes n_\Omega(x)\big)\,\mathrm{d}\mathcal{H}^{d-1}(x), \qquad u\in \mathrm{BV}(\Omega;\mathbb{R}^m),$$

is lower semicontinuous with respect to the weak convergence in $\mathrm{BV}(\Omega;\mathbb{R}^m)$.*

Remark 15.26 The boundary term in $\mathcal{F}$ may be omitted, see Problem 15.10.

The proof proceeds by showing Jensen-type inequalities for BV-Young measures at regular and singular points.

Proposition 15.27 *Let $\nu \in \mathbf{BVY}(\Omega;\mathbb{R}^{m\times d})$. Then, for all quasiconvex functions $h\colon \mathbb{R}^{m\times d}\to[0,\infty)$ with linear growth, it holds that*

(i) $h\left([\nu_x]+[\nu_x^\infty]\frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\right)\le\left\langle h,\nu_x\right\rangle+\left\langle h^\#,\nu_x^\infty\right\rangle\frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)$ *for $\mathcal{L}^d$-a.e. $x\in\Omega$;*

(ii) $h^\#([\nu_x^\infty])\le\left\langle h^\#,\nu_x^\infty\right\rangle$ *for λ_ν^s-a.e. $x\in\Omega$, where λ_ν^s is the singular part of λ_ν with respect to $\mathcal{L}^d$.*

In the proof of the proposition we will need an approximation lemma, which we state in slightly more generality than is strictly necessary here (we only need the x-independent case) because of the independent interest of the result.

Lemma 15.28 *For every function $f\in \mathrm{C}(\overline{\Omega}\times\mathbb{R}^N)$ with linear growth there is a decreasing sequence $(f_n)_n\subset\mathbf{E}(\Omega;\mathbb{R}^N)$ with*

$$\inf_{n\in\mathbb{N}} f_n=\lim_{n\to\infty} f_n=f,\qquad \inf_{n\in\mathbb{N}} f_n^\infty=\lim_{n\to\infty} f_n^\infty=f^\#\qquad \textit{(pointwise).}$$

Furthermore, if $M\ge 0$ is such that $|f(x,A)|\le M(1+|A|)$ for all $(x,A)\in\overline{\Omega}\times\mathbb{R}^N$, then the f_n can also be chosen to satisfy $|f_n(x,A)|\le M(1+|A|)$.

Proof. We denote by $(Sf)^{\mathrm{usc}}\colon\overline{\Omega\times\mathbb{B}^N}\to\mathbb{R}$ the upper semicontinuous extension of $Sf\colon\overline{\Omega}\times\mathbb{B}^N\to\mathbb{R}$ to $\overline{\Omega\times\mathbb{B}^N}$, that is,

$$(Sf)^{\mathrm{usc}}(x,\widehat{A}):=\limsup_{\substack{x_n\to x\\ \widehat{A}_n\to\widehat{A}}} Sf(x_n,\widehat{A}_n),\qquad (x,\widehat{A})\in\overline{\Omega\times\mathbb{B}^N}.$$

Since Sf is continuous on $\overline{\Omega}\times\mathbb{B}^N$, we have $(Sf)^{\mathrm{usc}}|_{\overline{\Omega}\times\mathbb{B}^N}=Sf$. For sequences $x_n\to x$ in $\overline{\Omega}$, $(\widehat{A}_n)\to\widehat{A}\in\mathbb{B}^N$, $\widehat{A}\in\partial\mathbb{B}^N$, and $t_n\to\infty$, it holds that

$$\limsup_{n\to\infty}\frac{f(x_n,t_n\widehat{A}_n)}{t_n}=\limsup_{n\to\infty}(t_n^{-1}+|\widehat{A}_n|)Sf\left(x_n,\frac{\widehat{A}_n}{t_n^{-1}+|\widehat{A}_n|}\right).$$

Hence, $f^\#=(Sf)^{\mathrm{usc}}|_{\overline{\Omega}\times\partial\mathbb{B}^N}$. Indeed, "$\le$" holds since $(Sf)^{\mathrm{usc}}$ is upper semicontinuous and "$\ge$" follows by taking a sequence $(x_n,\widehat{A}_n)\to(x,\widehat{A})$ with $Sf(x_n,\widehat{A}_n)\to(Sf)^{\mathrm{usc}}(x,\widehat{A})$ and setting $t_n:=(1-|\widehat{A}_n|)^{-1}$.

Let now $(g_k)_k\subset\mathrm{C}(\overline{\Omega\times\mathbb{B}^N})$ be a decreasing sequence with $g_k\downarrow(Sf)^{\mathrm{usc}}$ pointwise in $\overline{\Omega\times\mathbb{B}^N}$ and $|g_k|\le M$. Then set $f_k:=S^{-1}g_k$ for which $f^k\downarrow f$, f_k^∞ exists in the sense of (14.1), and

$$f_k^\infty=g_k|_{\overline{\Omega}\times\partial\mathbb{B}^N}\downarrow(Sf)^{\mathrm{usc}}|_{\overline{\Omega}\times\partial\mathbb{B}^N}=f^\#.$$

Finally, assuming $|f(x,A)|\le M(1+|A|)$ for some $M\ge 0$, it holds that $|Sf|\le M$, whereby $|(Sf)^{\mathrm{usc}}|\le M$ as well. □

Proof of Proposition 15.27. Ad (i). Let $\sigma \in \mathbf{BVY}(B_1;\mathbb{R}^{m\times d})$ be a regular tangent Young measure to ν at a suitable $x_0 \in \Omega$ as in Proposition 15.22, which we consider to be restricted to B_1 (see Problem 15.8). Then,

$$[\sigma] = F\,\mathcal{L}^d \llcorner B_1, \qquad \text{where} \qquad F = [\nu_{x_0}] + [\nu^\infty_{x_0}]\frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0).$$

Via Proposition 15.18 (ii) we obtain a sequence $(v_n)_n \subset (\mathrm{W}^{1,1} \cap \mathrm{C}^\infty)(B_1;\mathbb{R}^m)$ with $Dv_n \overset{\mathbf{Y}}{\to} \sigma$ and $v_n = Fx$ on ∂B_1.

Quasiconvex functions $h\colon \mathbb{R}^{m\times d} \to [0,\infty)$ with linear growth are Lipschitz continuous by Proposition 5.6 and Lemma 5.9. Hence, the application of Lemma 15.28 yields a sequence $(\mathbb{1} \otimes h_k)_k \subset \mathbf{E}(B_1;\mathbb{R}^{m\times d})$ such that $h_k \downarrow h$, $h_k^\infty \downarrow h^\#$ pointwise, and all h_k have uniformly bounded linear growth constants. Then, using the quasiconvexity, we get

$$\begin{aligned} h(F) &\le \limsup_{n\to\infty} \fint_{B_1} h(\nabla v_n)\,\mathrm{d}x \\ &\le \lim_{n\to\infty} \fint_{B_1} h_k(\nabla v_n)\,\mathrm{d}x \\ &= \frac{1}{\omega_d}\Big\langle\!\Big\langle \mathbb{1}\otimes h_k, \sigma \Big\rangle\!\Big\rangle \\ &= \big\langle h_k, \nu_{x_0}\big\rangle + \big\langle h_k^\infty, \nu^\infty_{x_0}\big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x_0) \end{aligned}$$

for all $k \in \mathbb{N}$, where we used (15.12) and (15.13). We may then invoke the monotone convergence theorem and utilize $h_k \downarrow h$, $h_k^\infty \downarrow h^\#$ to conclude.

Ad (ii). From (15.11) we know

$$\operatorname{rank}[\nu^\infty_x] \le 1 \qquad \text{for } \lambda^s_\nu\text{-a.e. } x \in \Omega.$$

By the Kirchheim–Kristensen Theorem 13.17 in the version of Corollary 13.20, $h^\#$ is convex at matrices of rank at most one, hence (ii) follows immediately from the Jensen-type inequality stated in (13.30). □

Proof of Theorem 15.25. Given a sequence $u_j \overset{*}{\rightharpoonup} u$ in $\mathrm{BV}(\Omega;\mathbb{R}^m)$, we consider all u_j, u to be extended to the whole space by zero, whereby

$$Du_j = Du_j \llcorner \Omega + (u_j \otimes n_\Omega)\,\mathcal{H}^{d-1} \llcorner \partial\Omega.$$

Select a subsequence $(u_{j(l)})_l$ such that

$$\liminf_{j\to\infty} \mathcal{F}[u_j] = \lim_{l\to\infty} \mathcal{F}[u_{j(l)}]$$

and

$$Du_{j(l)} \overset{\mathbf{Y}}{\to} \nu \in \mathbf{BVY}(\mathbb{R}^d;\mathbb{R}^{m\times d}),$$

the latter being possible by Theorem 15.16. It suffices to show lower semicontinuity along the sequence $(u_{j(l)})_l$, which in the following we just denote by $(u_j)_j$.

We have

$$[\nu] = Du + (u \otimes n_\Omega)\, \mathcal{H}^{d-1} \llcorner \partial\Omega.$$

Let λ_ν^* be the singular part of λ_ν with respect to $|D^s u| + |u|\, \mathcal{H}^{d-1} \llcorner \partial\Omega$; in particular, λ_ν^* is carried by a $(|D^s u| + |u|\, \mathcal{H}^{d-1} \llcorner \partial\Omega)$-negligible set. We compute using (15.11) and the arguments preceding it that

$$\begin{aligned}
[\nu_x] + [\nu_x^\infty] \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x) &= \begin{cases} \nabla u(x) & \text{for } \mathcal{L}^d\text{-a.e. } x \in \Omega, \\ 0 & \text{for } \mathcal{L}^d\text{-a.e. } x \in \mathbb{R}^d \setminus \Omega; \end{cases} \\
\frac{[\nu_x^\infty]}{|[\nu_x^\infty]|} &= \begin{cases} \dfrac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x) & \text{for } |D^s u|\text{-a.e. } x \in \Omega, \\ \dfrac{u|_{\partial\Omega}(x)}{|u|_{\partial\Omega}(x)|} \otimes n_\Omega(x) & \text{for } |u|\, \mathcal{H}^{d-1}\text{-a.e. } x \in \partial\Omega; \end{cases} \\
[\nu_x^\infty] &= 0 \qquad \text{for } \lambda_\nu^*\text{-a.e. } x \in \mathbb{R}^d; \\
|[\nu_x^\infty]|\, \lambda_\nu^s &= |D^s u| + |u|\, \mathcal{H}^{d-1} \llcorner \partial\Omega; \\
[\nu_x] &= 0 \qquad \text{for all } x \in \mathbb{R}^d \setminus \overline{\Omega}; \\
\lambda_\nu \llcorner (\mathbb{R}^d \setminus \overline{\Omega}) &= 0.
\end{aligned}$$

Also extend f to $\mathbb{R}^d \times \mathbb{R}^{m\times d}$ as follows: First extend f^∞ restricted to $\overline{\Omega} \times \partial\mathbb{B}^{m\times d}$ continuously to $\mathbb{R}^d \times \partial\mathbb{B}^{m\times d}$ and then set $f(x, A) := f^\infty(x, A)$ for $x \in \mathbb{R}^d \setminus \overline{\Omega}$ and $A \in \mathbb{R}^{m\times d}$. This extended f is still a Carathéodory integrand, f^∞ is jointly continuous and $f(x, 0) = 0$ for all $x \in \mathbb{R}^d \setminus \overline{\Omega}$.

Then, Proposition 15.11 on representation of limits with integrands of class **R** and the Jensen-type inequalities in Proposition 15.27 (applied in a larger bounded Lipschitz domain $\Omega' \ni \Omega$) give

$$\begin{aligned}
\liminf_{j\to\infty} \mathcal{F}[u_j] &= \int_\Omega \big\langle f(x, \cdot), \nu_x \big\rangle + \big\langle f^\infty(x, \cdot), \nu_x^\infty \big\rangle \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\, \mathrm{d}x \\
&\quad + \int_{\overline{\Omega}} \big\langle f^\infty(x, \cdot), \nu_x^\infty \big\rangle\, \mathrm{d}\lambda_\nu^s(x) \\
&\geq \int_\Omega f\left(x, [\nu_x] + [\nu_x^\infty] \frac{\mathrm{d}\lambda_\nu}{\mathrm{d}\mathcal{L}^d}(x)\right) \mathrm{d}x + \int_{\overline{\Omega}} f^\infty\big(x, [\nu_x^\infty]\big)\, \mathrm{d}\lambda_\nu^s(x) \\
&= \mathcal{F}[u].
\end{aligned}$$

This finishes the proof. □

We thus have the following existence theorem.

Theorem 15.29 *Let $f\colon \overline{\Omega} \times \mathbb{R}^{m\times d} \to [0, \infty)$ be such that the following assumptions hold:*

(i) f is a Carathéodory integrand;
(ii) $f(x, A) \leq M(1 + |A|)$ for some $M > 0$ and all $x \in \overline{\Omega}$, $A \in \mathbb{R}^{m\times d}$;

(iii) *the strong recession function* f^∞ *exists in the sense of* (14.1) *and is (jointly) continuous on* $\overline{\Omega} \times \mathbb{R}^{m\times d}$;
(iv) $f(x, \bullet)$ *is quasiconvex for all* $x \in \overline{\Omega}$;
(v) $\mu|A| - \mu^{-1} \le f(x, A)$ *for some* $\mu > 0$ *and all* $(x, A) \in \overline{\Omega} \times \mathbb{R}^{m\times d}$.

Then, the functional

$$\mathcal{F}[u] := \int_\Omega f(x, \nabla u(x))\,\mathrm{d}x + \int_\Omega f^\infty\left(x, \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right)\mathrm{d}|D^s u|(x) + \int_{\partial\Omega} f^\infty\big(x, u(x)\otimes n_\Omega(x)\big)\,\mathrm{d}\mathcal{H}^{d-1}(x), \qquad u \in \mathrm{BV}(\Omega;\mathbb{R}^m),$$

has a minimizer over the space $\mathrm{BV}(\Omega;\mathbb{R}^m)$.

Proof. The proof is by the Direct Method and the usual coercivity arguments in conjunction with Theorem 15.25. □

Notes and Historical Remarks

In the context presented here, the theory of generalized Young measures started with the article [122] by DiPerna and Majda, who were interested in turbulence concentrations in fluid dynamics (their measures, however, described L^2-concentrations). Also Balder's work [33] developed related tools, but it is more concerned with optimal control theory (where Young measures first originated, see the notes to Chapter 4). The L^1-framework was first introduced by Alibert & Bouchitté [6] and then developed into the form presented here in [209], partly inspired by the approaches to classical Young measures in the papers [50, 315]. In particular, Lemma 15.14 and Lemma 15.28 are adapted from [6]. Many authors have developed the theory further, a selection of some relevant papers is [141, 188, 212, 213, 315, 321].

In our setting we only work with the sphere compactification of $\mathbb{R}^N$, as explained in the opening remarks of this chapter; also see Problem 15.2. The theory can be extended to much more general target spaces and compactifications. For instance, $\mathbb{R}^N$ may be replaced by a Banach space X with the analytic Radon–Nikodým property, that is, the validity of the Radon–Nikodým theorem for X-valued measures, see [33]. Further, we may employ another compactification of $\mathbb{R}^N$, for instance, the compactification generated by a separable, complete ring of continuous bounded functions, see Section 4.8 in [139] and also [73], or even the Stone–Čech compactification $\beta\mathbb{R}^N$. Finally, Ω may be replaced by a general finite measure space. Such generalizations are discussed in [6, 33, 70, 122, 213].

A generalized Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}(\Omega;\mathbb{R}^N)$ may be described alternatively in the spirit of Berliocchi–Lasry [50] as

$$\nu := (\mathcal{L}^d_x \llcorner \Omega) \otimes \nu_x \qquad \text{and} \qquad \nu^\infty := \lambda_\nu \otimes \nu^\infty_x.$$

Then,

$$\nu(D \times \mathbb{R}^N) = |\Omega \cap D| \qquad \text{and} \qquad \nu^\infty(D \times \partial\mathbb{B}^N) = \lambda_\nu(D)$$

for all Borel sets $D \subset \overline{\Omega}$. So, ν can be understood as a classical Young measure with respect to $\mathcal{L}^d \llcorner \Omega$ and target space $\mathbb{R}^N$, and ν^∞ can be understood as a classical Young measure with respect to λ_ν and target space $\partial\mathbb{B}^N$.

An alternative approach to quantify concentration effects is the theory of varifolds [12–14, 146, 333]. Time-dependent generalized Young measures have also been developed for quasistatic evolution in plasticity theory [99–101]; in this context also see [117].

A slight refinement of the Characterization Theorem 15.19 for BV-Young measures can be found in [195]. We also mention the work [31], where the Souček space is considered as a more natural space of underlying deformations for BV-Young measures with boundary concentrations. A predecessor of the characterization results for general PDE-constrained Young measures established in [20, 208] can be found in [32] (under restrictive assumptions, however).

The approach to BV-lower semicontinuity via generalized Young measures in Theorem 15.25, in particular using the regular and singular localization results of Propositions 15.22, 15.23, was first implemented in this form in [279]; the version we present here includes the shortening possible by the Kirchheim–Kristensen Theorem 13.17.

We refer to [19] for analogous weak* lower semicontinuity results for functionals defined on PDE-constrained measures. For a very general approach to weak* lower semicontinuity for u-dependent integrand that gives essentially optimal results using so-called *liftings* of BV-maps, and Young measures thereof, see [282, 283].

Problems

15.1 Show that $f \in \mathrm{C}(\overline{\Omega} \times \mathbb{R}^N)$ is an element of $\mathbf{E}(\Omega; \mathbb{R}^N)$ if and only if f^∞ exists in the sense of (14.1).

15.2 Show that every element of $\mathbf{Y}^{\mathcal{M}}(\Omega; \mathbb{R}^N)$ can be identified with a classical L^∞-Young measure with values in the *sphere compactification* $\sigma\mathbb{R}^N \cong \mathbb{B}^N$ of $\mathbb{R}^N$ (where $\mathbb{R}^N$ is embedded into $\sigma\mathbb{R}^N$ via the map $v \mapsto (1 + |v|)^{-1} v$).

15.3 In the situation of Example 15.8, show that $u_j' \overset{\mathbf{Y}}{\to} \nu$ with

$$\nu_x = \delta_0 \quad \text{a.e.}, \qquad \lambda_\nu = \mathcal{L}^1 \llcorner (0,1), \qquad \nu_x^\infty = \delta_{+1} \quad \text{a.e.}$$

15.4 The limit $x' \to x$ in the definition of f^∞ cannot be omitted without breaking important parts of the generalized Young measure theory:

(i) For $\Omega := (-1, 1)$ find a Carathéodory integrand $f : \overline{\Omega} \times \mathbb{R} \to \mathbb{R}$ such that $f^\infty(0, \bullet)$ is not well-defined in the sense of (14.1), but

$$\widetilde{f^\infty}(x, A) := \lim_{\substack{A' \to A \\ t \to \infty}} \frac{f(x, tA')}{t} = 0, \qquad (x, A) \in \Omega \times \mathbb{R}^N.$$

Note the omission of the limit $x' \to x$ in contrast to the definition (14.1) of f^∞.

(ii) Find a uniformly norm-bounded sequence $(v_j)_j \subset \mathrm{L}^1(-1,1)$ generating the Young measure $\nu \in \mathbf{Y}^{\mathcal{M}}((-1,1))$ with

$$\nu_x = \delta_0 \quad \text{a.e.}, \qquad \lambda_\nu = 2\delta_0, \qquad \nu_0^\infty = \delta_{+1}$$

and

$$\widetilde{\langle\!\langle f, \delta[v_j] \rangle\!\rangle} \overset{*}{\rightharpoonup} \widetilde{\langle\!\langle f, \nu \rangle\!\rangle},$$

where $\widetilde{\langle\!\langle \bullet, \bullet \rangle\!\rangle}$ is defined like the usual duality product $\langle\!\langle \bullet, \bullet \rangle\!\rangle$, but with f^∞ replaced by $\widetilde{f^\infty}$. Conclude that Proposition 15.11 cannot hold for $\widetilde{\langle\!\langle \bullet, \bullet \rangle\!\rangle}$ in place of $\langle\!\langle \bullet, \bullet \rangle\!\rangle$.

15.5 Let $u_j := \mathbb{1}_{(0,1/j)}$ on $\Omega := (0,1)$, for which $Du_j = -\delta_{1/j}$, $u_j \overset{*}{\rightharpoonup} 0$ in $\mathrm{BV}(0,1)$, and $Du_j \overset{\mathbf{Y}}{\to} \nu \in \mathbf{BVY}((0,1))$ with

$$\nu_x = \delta_0 \quad \text{a.e.}, \qquad \lambda_\nu = \delta_0, \qquad \nu_0^\infty = \delta_{-1} \quad \text{a.e.}$$

Prove that no sequence $(v_j)_j \subset \mathrm{BV}((0,1))$ with $v_j(0) = v_j(1) = 0$ can generate this ν. This shows that in Proposition 15.18 (ii) the assumption $\lambda_\nu(\partial\Omega) = 0$ cannot be omitted.

15.6 Prove that for any $\lambda \in \mathcal{M}^+(\overline{\Omega})$, there exists a BV-Young measure $\nu \in \mathbf{BVY}(\Omega; \mathbb{R}^{m\times d})$ with $[\nu] = 0$ and $\lambda_\nu = \lambda$.

15.7 Use the result of Problem 13.6 to derive the singular Jensen inequality in Proposition 15.27 without the use of the Kirchheim–Kristensen theorem or Alberti's Rank-One Theorem. Deduce from this the Lower Semicontinuity Theorem 15.25. This shows that neither Alberti's rank-one theorem nor the Kirchheim–Kristensen theorem are necessary to prove weak* lower semicontinuity in BV.

15.8 Let $\Omega_0, \Omega \subset \mathbb{R}^d$ be two bounded Lipschitz domains with $\Omega_0 \Subset \Omega$ and let $\nu \in \mathbf{BVY}(\Omega; \mathbb{R}^{m\times d})$. If $\lambda_\nu(\partial\Omega_0) = 0$, then the restriction $\nu \llcorner \Omega_0$, that is,

$$(\nu \llcorner \Omega_0)_x = \nu_x, \qquad (\nu \llcorner \Omega_0)_x^\infty = \nu_x^\infty \quad \text{for all } x \in \overline{\Omega_0}$$

and

$$\lambda_{\nu \llcorner \Omega_0} := \lambda_\nu \llcorner \Omega_0,$$

lies in $\mathbf{BVY}(\Omega; \mathbb{R}^{m\times d})$.

15.9 Show that $\mathbf{BVY}(\mathbb{R}^d; \mathbb{R}^{m\times d})$ and $\mathbf{BVY}_{\mathrm{loc}}(\mathbb{R}^d; \mathbb{R}^{m\times d})$ are sequentially weakly* closed.

15.10 Show that in the situation of Theorem 15.25 also the functional (without boundary term)

$$\mathcal{F}[u] := \int_\Omega f(x, \nabla u(x))\, \mathrm{d}x + \int_\Omega f^\#\left(x, \frac{\mathrm{d}D^s u}{\mathrm{d}|D^s u|}(x)\right) \mathrm{d}|D^s u|(x),$$

where $u \in \mathrm{BV}(\Omega; \mathbb{R}^m)$, is lower semicontinuous with respect to weak* convergence in the space $\mathrm{BV}(\Omega; \mathbb{R}^m)$.

Appendix A
Prerequisites

This appendix recalls some notation and results that are needed throughout the book.

A.1 Linear Algebra

We first review some facts from linear algebra and matrix analysis, see, e.g., [183] for an advanced course.

For an $(m \times d)$-matrix $A \in \mathbb{R}^{m\times d}$ we denote by A_k^j the element in the j'th row and k'th column ($j = 1, \dots, m$; $k = 1, \dots, d$). In this book the matrix space $\mathbb{R}^{m\times d}$ always comes equipped with the **Frobenius matrix (inner) product**

$$A : B := \operatorname{tr}(A^\mathsf{T} B) = \operatorname{tr}(AB^\mathsf{T}) = \sum_{j,k} A_k^j B_k^j, \qquad A, B \in \mathbb{R}^{m\times d},$$

which is just the Euclidean inner product if we identify matrices with vectors in $\mathbb{R}^{md}$. This inner product induces the **Frobenius matrix norm**

$$|A| := \sqrt{\sum_{j,k} (A_k^j)^2}, \qquad A \in \mathbb{R}^{m\times d}.$$

While of course all norms on the finite-dimensional space $\mathbb{R}^{m\times d}$ are equivalent, some finer arguments require us to specify a matrix norm; if nothing else is stated, we always use the Frobenius norm.

The Frobenius norm can also be expressed as

$$|A| = \sqrt{\sum_i \sigma_i(A)^2}, \qquad A \in \mathbb{R}^{m\times d}, \tag{A.1}$$

where $\sigma_i(A) \geq 0$ is the i'th singular value of A, $i = 1, \dots, \min\{d, m\}$. For this, recall that every matrix $A \in \mathbb{R}^{m\times d}$ has a **(real) singular value decomposition**

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F. Rindler, *Calculus of Variations*, Graduate Texts in Mathematics 310,
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$$A = P\Sigma Q^\mathsf{T}$$

for orthogonal matrices $P \in \mathbb{R}^{m\times m}$, $Q \in \mathbb{R}^{d\times d}$ ($P^{-1} = P^\mathsf{T}$, $Q^{-1} = Q^\mathsf{T}$), and a diagonal matrix

$$\Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_{\min\{d,m\}}) \in \mathbb{R}^{m\times d}.$$

This is to be understood as the matrix having the positive entries

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0 = \sigma_{r+1} = \sigma_{r+2} = \cdots = \sigma_{\min\{d,m\}},$$

on the main diagonal, called the **singular values** of A, and zeros everywhere else. Here, $r = \operatorname{rank} A$ is the rank of A.

As a consequence of (A.1), it follows immediately that the Frobenius norm is orthogonally invariant, that is, for all $A \in \mathbb{R}^{m\times d}$ and all orthogonal $P \in \mathbb{R}^{m\times m}$, $Q \in \mathbb{R}^{d\times d}$ it holds that

$$|PA| = |A| = |AQ|.$$

The **tensor product** of the vectors $a \in \mathbb{R}^m$, $b \in \mathbb{R}^d$ is defined as

$$a \otimes b := ab^\mathsf{T} \in \mathbb{R}^{m\times d}.$$

Occasionally, we will also use $a \otimes b$ for a a column vector and b a row vector to denote the matrix product ab. While technically incorrect, this notation emphasizes that the result is a *matrix*.

We recall the following elementary fact: Let $A \in \mathbb{R}^{m\times d}$. Then, $\operatorname{rank} A \leq 1$ if and only if there exist vectors $a \in \mathbb{R}^m$, $b \in \mathbb{R}^d$ such that $A = a \otimes b$. The tensor product also interacts well with the Frobenius norm:

$$|a \otimes b| = |a| \cdot |b|, \qquad a \in \mathbb{R}^m,\ b \in \mathbb{R}^d,$$

where in $\mathbb{R}^m$ and $\mathbb{R}^d$ we use the usual Euclidean norm.

Let $A \in \mathbb{R}^{d\times d}$ with columns $A_j \in \mathbb{R}^d$ ($j = 1, \dots, d$). Then, its determinant $\det A$ satisfies the **Hadamard inequality**

$$|\det A| \leq \prod_{j=1}^{d} |A_j| \leq |A|^d.$$

An analogous formula holds with the rows of A.

For $A \in \mathbb{R}^{d\times d}$ the **cofactor matrix** $\operatorname{cof} A \in \mathbb{R}^{d\times d}$ of A is the matrix whose (j,k)'th entry is $(-1)^{j+k} M_{\neg k}^{\neg j}(A)$ with $M_{\neg k}^{\neg j}(A)$ being the (j,k)**-minor** of A, i.e., the determinant of the matrix that originates from A by deleting the j'th row and the k'th column. For

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

we have

$$\operatorname{cof} A = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}.$$

Cramer's rule entails that

$$A(\operatorname{cof} A)^{\mathsf{T}} = (\operatorname{cof} A)^{\mathsf{T}} A = (\det A)\mathrm{Id}, \qquad A \in \mathbb{R}^{d\times d},$$

where Id denotes the identity matrix. In particular, if A is invertible,

$$A^{-1} = \frac{(\operatorname{cof} A)^{\mathsf{T}}}{\det A}. \tag{A.2}$$

From this we deduce in particular that cof A is invertible if A is. Sometimes, the matrix $(\operatorname{cof} A)^{\mathsf{T}}$ is called the **adjugate matrix** to A (we will not use this terminology here, however).

One important formula concerning the derivative of the determinant (and another way to define the cofactor matrix) is

$$\frac{\mathrm{d}}{\mathrm{d}A} \det A = \operatorname{cof} A.$$

Combining this with Cramer's rule, one obtains **Jacobi's formula**, which entails that for any continuously differentiable function $A(t)\colon \mathbb{R} \to \mathbb{R}^{d\times d}$ it holds that

$$\frac{\mathrm{d}}{\mathrm{d}t} \det A(t) = \operatorname{cof} A(t) : \frac{\mathrm{d}A(t)}{\mathrm{d}t} = \operatorname{tr}\left[(\operatorname{cof} A(t))^{\mathsf{T}} \frac{\mathrm{d}A(t)}{\mathrm{d}t}\right].$$

Recall that a matrix $Q \in \mathbb{R}^{d\times d}$ is called **orthogonal** if it is invertible and $Q^{-1} = Q^{\mathsf{T}}$. The **special orthogonal group** $\mathrm{SO}(d)$ is defined as

$$\mathrm{SO}(d) := \left\{ Q \in \mathbb{R}^{d\times d} \ : \ Q \text{ invertible}, Q^{-1} = Q^{\mathsf{T}}, \det Q = 1 \right\}.$$

It has the following useful property, which can be verified via (A.2):

$$\operatorname{cof} Q = Q \qquad \text{for all } Q \in \mathrm{SO}(d).$$

Any $Q \in \mathrm{SO}(2) \subset \mathbb{R}^{2\times 2}$ is a rotation,

$$Q = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

for some $\theta \in [0, 2\pi)$. For the convex hull $\mathrm{SO}(2)^{**}$ of $\mathrm{SO}(2)$ one may compute

$$\mathrm{SO}(2)^{**} = \left\{ A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \ : \ a^2 + b^2 \le 1 \right\}.$$

Moreover, a Taylor expansion and the fact that the Lie algebra of the Lie group $\mathrm{SO}(2)$ is the vector space of all skew-symmetric matrices yields

$$\operatorname{dist}(\mathrm{Id} + A, \mathrm{SO}(2)) \le \left| \frac{A + A^{\mathsf{T}}}{2} \right| + C|A|^2 \tag{A.3}$$

for some $C > 0$. See [176] for more on matrix Lie groups.

We also recall a special case of the theorem on the **(real) Jordan normal form** for real (2×2)-matrices: Let $A \in \mathbb{R}^{2\times 2}$. Then, there exists an invertible matrix $S \in \mathbb{R}^{2\times 2}$ such that

$$S^{-1}AS = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \qquad \text{or} \qquad S^{-1}AS = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

with $a, c \in \mathbb{R}$ and $b \in \{0, 1\}$

Finally, any square matrix $A \in \mathbb{R}^{d\times d}$ has a real **polar decomposition**

$$A = QS,$$

where Q is orthogonal and S is symmetric and positive definite.

In a few instances we also deal with fourth-order **tensors** $\mathbf{T} = \mathbf{T}^{ik}_{jl}$ $(i, k = 1, \dots, m;$ $j, l = 1, \dots, d)$. They define bilinear forms on $\mathbb{R}^{m\times d}$ via

$$A : \mathbf{T}B := \sum_{i,j,k,l} \mathbf{T}^{ik}_{jl} A^i_j B^k_l, \qquad A, B \in \mathbb{R}^{m\times d}.$$

We call $\mathbf{T}$ symmetric and positive definite if the corresponding bilinear form has these properties. More precisely, the symmetry means that $A : \mathbf{T}B = B : \mathbf{T}A$ for all $A, B \in \mathbb{R}^{m\times d}$ and a constant $C > 0$. The positive-definiteness entails that $A : \mathbf{T}A \geq c|A|^2$ for some $c > 0$ and all $A \in \mathbb{R}^{m\times d}$.

Finally, for a matrix $A \in \mathbb{R}^{d\times d}$ denote by

$$\operatorname{sym} A := \frac{1}{2}(A + A^{\mathsf{T}})$$

its **symmetric part**. We collect all symmetric $(d \times d)$-matrices in the space $\mathbb{R}^{d\times d}_{\text{sym}}$.

A.2 Functional Analysis

We assume that the reader has a solid foundation in the basic notions of functional analysis such as Banach spaces and their duals, weak/weak* convergence (and, rarely, topology), reflexivity, and weak/weak* compactness. Thorough references for most of this material are [88] and [289].

Let X be a Banach space. By $B_r(x_0) \subset X$ we denote the open ball with center x_0 and radius $r > 0$; if $x_0 = 0$ we just write $B_r := B_r(0)$. The application of $x^* \in X^*$ to $x \in X$ is often expressed via the **duality pairing** $\langle x, x^* \rangle := x^*(x)$.

We write $x_j \rightharpoonup x$ in X for **weak convergence**, that is, $\langle x_j, x^* \rangle \to \langle x, x^* \rangle$ for all $x^* \in X^*$, and $x^*_j \overset{*}{\rightharpoonup} x^*$ in X^* for **weak* convergence**, that is, $\langle x, x^*_j \rangle \to \langle x, x^* \rangle$ for all $x \in X$. The weak* topology is metrizable on norm-bounded sets in the dual to a separable Banach space. Likewise, in reflexive and separable Banach spaces the weak topology is metrizable on norm-bounded sets. In this context we note that in general Banach spaces topological weak compactness is equivalent to sequential weak

compactness by the Eberlein–Šmulian theorem. Also recall that the norm in a Banach space is lower semicontinuous with respect to weak convergence: If $x_j \rightharpoonup x$ in X, then $\|x\| \le \liminf_{j\to\infty} \|x_j\|$.

Theorem A.1 (Weak compactness) *Let X be a separable, reflexive Banach space. Then, norm-bounded sets in X are sequentially weakly precompact.*

Theorem A.2 (Banach–Alaoglu) *Let X be a separable Banach space. Then, norm-bounded sets in the dual space X^* are weakly* sequentially precompact.*

Weak convergence can be "improved" to strong convergence in the following way:

Lemma A.3 (Mazur) *Let $x_j \rightharpoonup x$ in a Banach space X. Then, there exists a sequence $(y_j)_j \subset X$ of convex combinations,*

$$y_j = \sum_{n=j}^{N(j)} \theta_n^{(j)} x_n, \qquad \text{where} \qquad \theta_n^{(j)} \in [0,1], \quad \sum_{n=j}^{N(j)} \theta_n^{(j)} = 1,$$

such that $y_j \to x$ in X.

At one point we will also need the following fact about locally convex topologies on vector spaces, whose proof can be found in Theorem 3.10 of [289] (together with definitions and properties of the involved topological notions):

Lemma A.4 *Let X be a vector space and let X' be a vector space of linear functionals on X that separates the points of X, that is, for $x_1, x_2 \in X$ with $x_1 \neq x_2$ there is a functional $x' \in X'$ such that $\langle x_1, x' \rangle \neq \langle x_2, x' \rangle$. Let τ be the weakest topology on X that makes all the functionals $x' \in X'$ continuous on X. Then, (X, τ) is a locally convex topological vector space with X' as its dual space, $(X, \tau)^* = X'$.*

Finally, we recall the following fundamental result:

Theorem A.5 (Hahn–Banach separation theorem) *Let X be a Banach space or a locally convex topological vector space. Let $K, F \subset X$ be disjoint, non-empty, and convex subsets of X such that K is compact and F is closed. Then, K and F can be separated by a hyperplane, that is, there exists an $x^* \in X^*$ such that*

$$\sup_{x\in K} \langle x, x^* \rangle < \inf_{x\in F} \langle x, x^* \rangle.$$

A.3 Measure Theory

We assume that the reader is familiar with the notion of Lebesgue- and Borel-measurability, negligible sets, and L^p-spaces; a good introduction is in [288]. For the d-dimensional Lebesgue measure we write $\mathcal{L}^d$, or $\mathcal{L}^d_x$ if we want to stress the integration variable. Often, however, the Lebesgue measure of a Borel- or Lebesgue-measurable set

$A \subset \mathbb{R}^d$ is simply denoted by $|A|$. We also write $\omega_d := \mathcal{L}^d(B_1)$ for the volume of the d-dimensional unit ball $B_1 \subset \mathbb{R}^d$.

The **indicator function** of a subset $A \in \mathbb{R}^d$ is

$$\mathbb{1}_A(x) := \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{otherwise,} \end{cases} \qquad x \in \mathbb{R}^d.$$

In the following we recall some basic results that are needed throughout the book.

Lemma A.6 *Let $B \subset \mathbb{R}^d$ be a Borel set. A function $f\colon B \to \mathbb{R}^N$ is Lebesgue-measurable if and only if there exists a sequence of simple functions*

$$f_j := \sum_{k=1}^{K(j)} v_k^{(j)} \mathbb{1}_{B_k^{(j)}},$$

where $K(j) \in \mathbb{N}$ ($j \in \mathbb{N}$), the $B_k^{(j)} \subset B$ are disjoint Lebesgue-measurable sets with $\bigcup_{k=1}^{K(j)} B_k^{(j)} = B$, and $v_k^{(j)} \in \mathbb{R}^N$ for all j, k, such that $f_j \to f$ pointwise as $j \to \infty$. Moreover, the pointwise limit of a sequence of Lebesgue-measurable functions (if it exists) is itself Lebesgue-measurable.

Lemma A.7 (Fatou) *Let $f_j\colon \mathbb{R}^d \to [0, +\infty]$, $j \in \mathbb{N}$, be Lebesgue-measurable functions. Then,*

$$\int \liminf_{j\to\infty} f_j(x)\, \mathrm{d}x \le \liminf_{j\to\infty} \int f_j(x)\, \mathrm{d}x.$$

Lemma A.8 (Monotone convergence) *Let $f_j\colon \mathbb{R}^d \to [0, +\infty]$, $j \in \mathbb{N}$, be Lebesgue-measurable functions with $f_j(x) \uparrow f(x)$ for almost every $x \in \Omega$. Then, $f\colon \mathbb{R}^d \to [0, +\infty]$ is measurable and*

$$\int f(x)\, \mathrm{d}x = \lim_{j\to\infty} \int f_j(x)\, \mathrm{d}x.$$

Lemma A.9 *If $f_j \to f$ (strongly) in $\mathrm{L}^p(\mathbb{R}^d)$, $p \in [1, \infty]$, that is,*

$$\|f_j - f\|_{\mathrm{L}^p} \to 0 \qquad \text{as } j \to \infty,$$

then $f_j \to f$ in measure, that is, for all $\delta > 0$,

$$\left|\left\{ x \in \mathbb{R}^d \,:\, |f_j(x) - f(x)| > \delta \right\}\right| \to 0 \qquad \text{as } j \to \infty.$$

Moreover, there exists a subsequence (not relabeled) such that $f_j \to f$ pointwise almost everywhere.

Theorem A.10 (Dominated convergence theorem) *Let $f_j\colon \mathbb{R}^d \to \mathbb{R}^N$, $j \in \mathbb{N}$, be Lebesgue-measurable functions such that there exists an L^p-integrable majorant $g \in \mathrm{L}^p(\mathbb{R}^d)$ for some $p \in [1, \infty)$, that is,*

$$|f_j| \le g \qquad \text{for all } j \in \mathbb{N}.$$

If $f_j \to f$ pointwise almost everywhere for some $f\colon \mathbb{R}^d \to \mathbb{R}$, then also $f_j \to f$ in L^p; in particular, $f \in \mathrm{L}^p(\mathbb{R}^d)$.

The following strengthening of Lebesgue's theorem is often useful:

Theorem A.11 (Pratt) *Let $f_j\colon \mathbb{R}^d \to \mathbb{R}^N$, $j \in \mathbb{N}$, be Lebesgue-measurable functions. If $f_j \to f$ pointwise almost everywhere (or in measure) and there exists a sequence $(g_j)_j \subset \mathrm{L}^1(\mathbb{R}^d)$ with $g_j \to g$ in L^1 such that $|f_j| \le g_j$, then $f_j \to f$ in L^1.*

The following convergence theorem is of fundamental significance:

Theorem A.12 (Vitali) *Let $\Omega \subset \mathbb{R}^d$ be bounded and $(f_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^m)$, $p \in [1,\infty)$. Assume furthermore that the following two conditions hold:*

(i) No oscillations: *$f_j \to f$ in measure or almost everywhere.*
(ii) No concentrations: *the family $(f_j)_j$ is p-equiintegrable.*

Then, $f_j \to f$ in L^p.

Here, $(f_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^m)$ is called **p-equiintegrable**, $p \in [1,\infty)$, if one of the following equivalent conditions is satisfied:

(i) $\lim\limits_{R\uparrow\infty} \sup\limits_{j\in\mathbb{N}} \int_{\{|f_j|\ge R\}} |f_j|^p \,\mathrm{d}x = 0$;

(ii) $\lim\limits_{R\uparrow\infty} \limsup\limits_{j\to\infty} \int_{\{|f_j|\ge R\}} |f_j|^p \,\mathrm{d}x = 0$;

(iii) for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all Borel sets $B \subset \Omega$ with $|B| < \delta$ we have
$$\sup_{j\in\mathbb{N}} \int_B |f_j|^p \,\mathrm{d}x < \varepsilon.$$

The notion of 1-equiintegrability is also just called "equiintegrability".

It is straightforward to see that a sequence that is uniformly norm-bounded in $\mathrm{L}^q(\mathbb{R}^d)$ is p-equiintegrability for all $p < q$. Indeed, if $f \in \mathrm{L}^q(\mathbb{R}^d)$ and $R > 0$,

$$\int_{\{|f|\ge R\}} |f|^p \,\mathrm{d}x \le \left|\left\{ x \in \mathbb{R}^d \,:\, |f(x)| \ge R \right\}\right|^{(q-p)/q} \cdot \|f\|_{\mathrm{L}^q}^p$$

by Hölder's inequality (with exponents $q/(q-p)$ and q/p). Moreover,

$$\left|\left\{ x \in \mathbb{R}^d \,:\, |f(x)| \ge R \right\}\right| \le \frac{1}{R^q} \|f\|_{\mathrm{L}^q}^q$$

by **Markov's inequality**, namely that for any $g\colon \mathbb{R}^d \to \mathbb{R}$ and any $\varphi\colon (0,\infty) \to (0,\infty)$ such that $\varphi(|g|) \in \mathrm{L}^1(\mathbb{R}^d)$ it holds that

$$\left|\left\{ x \in \mathbb{R}^d \,:\, |g(x)| \ge \lambda \right\}\right| \le \frac{1}{\varphi(\lambda)} \int \varphi(|g|) \,\mathrm{d}x, \qquad \lambda > 0.$$

Combining these inequalities, we see that

$$\int_{\{|f|\geq R\}} |f|^p \,\mathrm{d}x \leq \frac{1}{R^{q-p}} \|f\|_{\mathrm{L}^q}^q$$

and hence criterion (i) above is satisfied for a uniformly norm-bounded family $(f_j)_j \subset \mathrm{L}^q(\Omega;\mathbb{R}^m)$.

Theorem A.13 (Dunford–Pettis) *Let $\Omega \subset \mathbb{R}^d$ be bounded and open. A norm-bounded family $(f_j)_j \subset \mathrm{L}^1(\Omega)$ is (1-)equiintegrable if and only if it is weakly sequentially precompact in $\mathrm{L}^1(\Omega)$.*

We remark that the usual formulation of the Dunford–Pettis theorem only mentions *topological* precompactness. The statement above follows by also utilizing the Eberlein–Šmulian theorem (see Chapter V in [88]).

Theorem A.14 (Egorov) *Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set and let $f_j \colon \Omega \to \mathbb{R}^N$, $j \in \mathbb{N}$, be Lebesgue-measurable functions. If $f_j \to f$ pointwise almost everywhere, then for every $\varepsilon > 0$ there exists a compact set $K_\varepsilon \subset \Omega$ with $|\Omega \setminus K_\varepsilon| \leq \varepsilon$ and such that $f_j \to f$ uniformly in K_ε.*

Theorem A.15 (Radon–Riesz) *Let $p \in (1,\infty)$ and let $(f_j)_j \subset \mathrm{L}^p(\Omega;\mathbb{R}^m)$ with $f_j \rightharpoonup f$ (weak convergence) as well as $\|f_j\|_{\mathrm{L}^p} \to \|f\|_{\mathrm{L}^p}$. Then, $f_j \to f$ (strongly) in L^p.*

The following *covering theorem* is a handy tool for several constructions, see Theorem 2.19 in [18] for a proof:

Theorem A.16 (Vitali covering theorem) *Let $\Omega, D \subset \mathbb{R}^d$ be open and bounded. Then, there exist $a_k \in \Omega$, $r_k > 0$ ($k \in \mathbb{N}$) such that we may write Ω as the disjoint union*

$$\Omega = Z \cup \bigcup_{k=1}^{\infty} \overline{D(a_k, r_k)}, \qquad D(a_k, r_k) := a_k + r_k D,$$

with $Z \subset \Omega$ a Lebesgue-negligible set ($|Z| = 0$). Moreover, if for almost every $x \in \Omega$ we are given a real number $r(x) > 0$, then we may additionally require that $r_k < r(a_k)$ for all $k \in \mathbb{N}$.

Theorem A.17 (Lusin) *Let $\Omega \subset \mathbb{R}^d$ be a bounded Borel set and let $f \colon \Omega \to \mathbb{R}^N$ be Lebesgue-measurable. Then, for every $\varepsilon > 0$ there exists a compact set $K \subset \Omega$ such that $|\Omega \setminus K| \leq \varepsilon$ and $f|_K$ is uniformly continuous.*

Theorem A.18 (Sard) *Let $f \colon \mathbb{R}^d \to \mathbb{R}$ be d times continuously differentiable. Then,*

$$\mathcal{L}^1(f(S)) = 0, \qquad \text{where} \qquad S := \big\{ x \in \mathbb{R}^d \,:\, \nabla f(x) = 0 \big\}.$$

We also need other measures than Lebesgue measure on subsets of $\mathbb{R}^N$ (this will usually be either $\mathbb{R}^d$ or a matrix space $\mathbb{R}^{m\times d}$, which is identified with $\mathbb{R}^{md}$). All of these abstract measures will be **Borel measures**, that is, they are defined on the **Borel σ-algebra** $\mathfrak{B}(\mathbb{R}^N)$ of $\mathbb{R}^N$, which is the smallest σ-algebra that contains all the open sets.

All positive Borel measures defined on $\mathbb{R}^N$ that do not take the value $+\infty$ are collected in the set $\mathcal{M}^+(\mathbb{R}^N)$ of **(finite, positive) Radon measures**; its subclass of **probability measures** is $\mathcal{M}^1(\mathbb{R}^N)$.

We remark that all σ-finite measures on $\mathbb{R}^N$ are in fact **inner regular**, meaning that for every Borel set $B \subset \mathbb{R}^N$ it holds that

$$\mu(B) = \sup\big\{\, \mu(K) \;:\; K \subset B \text{ compact} \,\big\}.$$

A **local (positive) Radon measure** μ is a set function $\mu\colon \mathfrak{B}(\mathbb{R}^N) \to [0, +\infty]$ such that μ restricted to (subsets of) any compact set is a finite Radon measure. In this case we write $\mu \in \mathcal{M}^+_{\mathrm{loc}}(\mathbb{R}^N)$. We also use $\mathcal{M}^+(B), \mathcal{M}^+_{\mathrm{loc}}(B), \mathcal{M}^1(B)$ for the subclass of measures that only charge a Borel set $B \subset \mathbb{R}^N$, that is, the measure of the complement of B is zero.

For $\mu \in \mathcal{M}^+(\mathbb{R}^N)$ and $h\colon \mathbb{R}^N \to \mathbb{R}$ Borel-measurable (or μ-measurable) we define the **duality pairing**

$$\big\langle h, \mu \big\rangle := \int h(A)\, \mathrm{d}\mu(A)$$

whenever this integral makes sense, which is in particular the case if either $h \geq 0$ or if

$$h \in \mathrm{L}^1(\mathbb{R}^N, \mu) := \left\{ h\colon \mathbb{R}^N \to \mathbb{R} \ \mu\text{-measurable} \;:\; \int |h|\, \mathrm{d}\mu < \infty \right\}.$$

The following notation is convenient for the **barycenter** of a (finite, positive) Borel measure $\mu \in \mathcal{M}^+(\mathbb{R}^N)$:

$$[\mu] := \big\langle \mathrm{id}, \mu \big\rangle = \int A\, \mathrm{d}\mu(A).$$

We also define the **support** of $\mu \in \mathcal{M}^+(\mathbb{R}^N)$ by

$$\operatorname{supp} \mu := \big\{ x \in \mathbb{R}^N \;:\; \mu(B_r(x)) > 0 \text{ for all } r > 0 \big\}.$$

The **restriction** of a Borel measure $\mu \in \mathcal{M}^+(\mathbb{R}^N)$ to a Borel set $A \subset \mathbb{R}^N$ is

$$(\mu \llcorner A)(B) := \mu(A \cap B) \qquad \text{for any Borel set } B \subset \mathbb{R}^N.$$

Probability measures and convex functions interact well:

Lemma A.19 (Jensen inequality) *For all probability measures $\mu \in \mathcal{M}^1(\mathbb{R}^N)$ and all convex $h\colon \mathbb{R}^N \to \mathbb{R}$ it holds that*

$$h([\mu]) \leq \int h(A)\, \mathrm{d}\mu(A).$$

We say that a sequence $(\mu_j)_j \subset \mathcal{M}^+(\mathbb{R}^N)$ **converges weakly*** in $\mathcal{M}^+(\mathbb{R}^N)$ to $\mu \in \mathcal{M}^+(\mathbb{R}^N)$, in symbols

$$\mu_j \overset{*}{\rightharpoonup} \mu,$$

if $\langle \psi, \mu_j \rangle \to \langle \psi, \mu \rangle$ for all $\psi \in \mathrm{C}_0(\mathbb{R}^N)$. We speak of **local weak* convergence** if $\langle \psi, \mu_j \rangle \to \langle \psi, \mu \rangle$ for all $\psi \in \mathrm{C}_c(\mathbb{R}^N)$. A sequence $(\mu_j)_j \subset \mathcal{M}^+(\mathbb{R}^N)$ with $\sup_j \mu_j(\mathbb{R}^N) < \infty$ has a weakly* converging subsequence by Theorem A.1 and Theorem A.22 below.

We also recall a useful convergence lemma:

Lemma A.20 *Let $\mu_j \overset{*}{\rightharpoonup} \mu$ in $\mathcal{M}^+_{\mathrm{loc}}(\mathbb{R}^N)$. Then for every lower semicontinuous function $g\colon \mathbb{R}^N \to [0,\infty]$ it holds that*

$$\int g \,\mathrm{d}\mu \le \liminf_{j\to\infty} \int g \,\mathrm{d}\mu_j,$$

and for every upper semicontinuous function $h\colon \mathbb{R}^N \to [0,\infty)$ with compact support it holds that

$$\int h \,\mathrm{d}\mu \ge \limsup_{j\to\infty} \int h \,\mathrm{d}\mu_j.$$

In particular, for $U \subset \mathbb{R}^N$ open and $K \subset \mathbb{R}^N$ compact,

$$\mu(U) \le \liminf_{j\to\infty} \mu_j(U) \qquad \text{and} \qquad \mu(K) \ge \limsup_{j\to\infty} \mu_j(K).$$

Similar definitions and statements apply to $\mathcal{M}^+(U), \mathcal{M}^+_{\mathrm{loc}}(U), \mathcal{M}^1(U)$.

Finally, we recall a very useful "continuity" property of measurable functions:

Theorem A.21 *Let $f \in \mathrm{L}^1(\mathbb{R}^N, \mu; \mathbb{R}^m)$, that is, $f\colon \mathbb{R}^N \to \mathbb{R}^m$ is μ-integrable, where $\mu \in \mathcal{M}^+(\mathbb{R}^N)$. Then, μ-almost every $x_0 \in \mathbb{R}^N$ is a **Lebesgue point** of f with respect to μ, that is,*

$$\lim_{r\downarrow 0} \fint_{B_r(x_0)} |f(x) - f(x_0)| \,\mathrm{d}\mu(x) = 0,$$

where $\fint_{B_r(x_0)} := |B_r(x_0)|^{-1} \int_{B_r(x_0)}$.

We denote by $\mathcal{H}^s$ the s-dimensional Hausdorff measure, $0 \le s < \infty$. For the definition of this measure and the associated notion of **$\mathcal{H}^s$-rectifiable sets** (whose precise form is not important for most of this book), we refer to [18].

A.4 Vector Measures

In this section we exhibit a few aspects of the theory of **vector (Radon) measures** (often just called "measures" in this book), which are σ-additive set functions $\mu\colon \mathfrak{B}(\mathbb{R}^d) \to \mathbb{R}^N$ (in particular, $\mu(\emptyset) = 0$). All such μ are collected in the space $\mathcal{M}(\mathbb{R}^d; \mathbb{R}^N)$; likewise define $\mathcal{M}(\Omega; \mathbb{R}^N)$ for a Borel set $B \subset \mathbb{R}^d$. We will also use **local vector measures**, defined analogously to the case of positive measures, which are collected in the set $\mathcal{M}_{\mathrm{loc}}(B; \mathbb{R}^N)$. Standard references for vector measure theory are [119, 120].

If for the target dimension we have $N = 1$, then we simply write $\mathcal{M}(\Omega)$ instead of $\mathcal{M}(\Omega;\mathbb{R})$; the elements of this space are called **signed (Radon) measures**, but in this case we also usually just speak of "measures".

The **total variation measure** of $\mu \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^N)$ is the positive measure $|\mu| \in \mathcal{M}^+(\mathbb{R}^d)$ defined as

$$|\mu|(B) := \sup\left\{ \sum_{k=1}^{\infty} |\mu(B_k)| \;:\; B = \bigcup_{k=1}^{\infty} B_k,\, B_k \text{ Borel},\, B_k \cap B_l = \emptyset \text{ if } k \neq l \right\}.$$

It can be shown that for all open sets $U \subset \mathbb{R}^d$ it holds that

$$|\mu|(U) = \sup\left\{ \int \psi \cdot \mathrm{d}\mu \;:\; \psi \in \mathrm{C}_c(U;\mathbb{R}^N),\; \|\psi\|_\infty \leq 1 \right\}, \tag{A.4}$$

where the dot "$\cdot$" indicates that μ's values are to be scalar-multiplied with the values of ψ (e.g., $\int (v_0 \mathbb{1}_B) \cdot \mathrm{d}\mu = v_0 \cdot \mu(B)$ for $v_0 \in \mathbb{R}^N$ and B a Borel set). See Proposition 1.47 in [18] for a proof. We also set $\operatorname{supp}\mu := \operatorname{supp}|\mu|$.

There is an alternative, dual, view on vector measures, expressed in the following important theorem:

Theorem A.22 (Riesz representation theorem) *The space of vector Radon measures $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^N)$ is isometrically isomorphic to the dual space $\mathrm{C}_0(\mathbb{R}^d;\mathbb{R}^N)^*$ via the duality pairing*

$$\langle \varphi, \mu \rangle = \int \varphi \cdot \mathrm{d}\mu, \qquad \varphi \in \mathrm{C}_0(\mathbb{R}^d;\mathbb{R}^N),\; \mu \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^N).$$

As a convenient consequence, we can *define* measures through their action on $\mathrm{C}_0(\mathbb{R}^d;\mathbb{R}^N)$. Note, however, that we then need to check the boundedness $|\langle \varphi, \mu \rangle| \leq \|\varphi\|_\infty$ for all $\varphi \in \mathrm{C}_0(\mathbb{R}^d;\mathbb{R}^N)$.

If an element $\mu \in \mathrm{C}_0(\mathbb{R}^d)^*$ is additionally positive, that is, $\langle \varphi, \mu \rangle \geq 0$ whenever $\varphi \geq 0$, and normalized, that is, $\langle \mathbb{1}, \mu \rangle = 1$ (here, $\mathbb{1} \equiv 1$ on the whole space), then the μ from the Riesz representation theorem is a *probability measure*, $\mu \in \mathcal{M}^1(\mathbb{R}^N)$.

The weak* convergence of vector measures is defined exactly as for positive measures, namely by considering vector measures as elements of $\mathrm{C}_0(\mathbb{R}^d;\mathbb{R}^N)^*$. A sequence $(\mu_j)_j \subset \mathcal{M}(\Omega;\mathbb{R}^N)$ with $\sup_j |\mu_j|(\Omega) < \infty$ has a weakly* converging subsequence by Theorem A.1. Sometimes, for a norm-bounded sequence $(v_j)_j \subset \mathrm{L}^1(\Omega;\mathbb{R}^N)$, we will say that "$v_j \overset{*}{\rightharpoonup} \mu$ in $\mathcal{M}(\Omega;\mathbb{R}^N)$" when really we mean $v_j \mathcal{L}^d \llcorner \Omega \overset{*}{\rightharpoonup} \mu$ in $\mathcal{M}(\Omega;\mathbb{R}^N)$.

The following lemma is proved in Proposition 1.62 (b) of [18].

Lemma A.23 *Let $\mu_j \overset{*}{\rightharpoonup} \mu$ in $\mathcal{M}(\mathbb{R}^d;\mathbb{R}^N)$ and assume that $|\mu_j| \overset{*}{\rightharpoonup} \Lambda \in \mathcal{M}^+(\mathbb{R}^d)$. If $K \subset \mathbb{R}^d$ is compact and $\Lambda(\partial K) = 0$, then $\mu_j(K) \to \mu(K)$. Moreover, if $h\colon \mathbb{R}^d \to \mathbb{R}$ is a bounded Borel function with compact support and a Λ-negligible set of discontinuity points, then*

$$\int h \,\mathrm{d}\mu_j \to \int h \,\mathrm{d}\mu.$$

Of fundamental importance is the following theorem:

Theorem A.24 (Besicovitch differentiation theorem) *Given $\mu \in \mathcal{M}(\mathbb{R}^d;\mathbb{R}^N)$ and $\nu \in \mathcal{M}^+(\mathbb{R}^d)$, for ν-almost every $x_0 \in \mathbb{R}^d$ in the support of ν, the limit*

$$\frac{\mathrm{d}\mu}{\mathrm{d}\nu}(x_0) := \lim_{r\downarrow 0} \frac{\mu(B_r(x_0))}{\nu(B_r(x_0))}$$

*exists in $\mathbb{R}^N$; it is called the **Radon–Nikodým derivative** of μ with respect to ν. Moreover, the **Lebesgue–Radon–Nikodým decomposition** splits μ as*

$$\mu = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\nu + \mu^s$$

*with $\mu^s = \mu \llcorner E$ **singular** with respect to ν (that is, μ^s is concentrated on a ν-negligible set), where*

$$E := (\mathbb{R}^d \setminus \operatorname{supp}\nu) \cup \left\{ x \in \operatorname{supp}\nu \;:\; \lim_{r\downarrow 0} \frac{|\mu|(B_r(x))}{\nu(B_r(x))} = \infty \right\}.$$

Finally, for a Borel measure $\mu \in \mathcal{M}(B)$ and a surjective Borel map $\varphi\colon B \to B'$, with $B \subset \mathbb{R}^d$, $B' \subset \mathbb{R}^m$ Borel sets, we define the **push-forward measure** of μ under φ via

$$\varphi_\# \mu := \mu \circ \varphi^{-1} \in \mathcal{M}(B').$$

We then have the following transformation formula for any measurable map $g\colon B' \to \mathbb{R}$:

$$\int_{B'} g \,\mathrm{d}(\varphi_\#\mu) = \int_B g \circ \varphi \,\mathrm{d}\mu, \tag{A.5}$$

provided these integrals are defined.

A.5 Sobolev and Other Function Spaces

We give a brief overview of Sobolev spaces, see [219] or [133] for more detailed accounts and proofs.

In all of the following we assume that $\Omega \subset \mathbb{R}^d$ is a **bounded Lipschitz domain**, that is, Ω is open, bounded, connected, and such that locally in a neighborhood around every boundary point $x \in \partial\Omega$, Ω is the set of points above a Lipschitz graph on a hyperplane through x. As usual, we denote by $\mathrm{C}(\Omega) = \mathrm{C}^0(\Omega), \mathrm{C}^k(\Omega)$, $k = 1, 2, \ldots$, the spaces of continuous and k times continuously differentiable functions. The spaces $\mathrm{C}^k(\overline{\Omega})$ contain the $\mathrm{C}^k(\Omega)$-maps such that all l'th-order derivatives for $l \le k$ (including $l = 0$, that is, the map itself) can be continuously extended to $\overline{\Omega}$. As norms in these spaces we have

$$\|u\|_{\mathrm{C}^k} := \max_{|\alpha|\le k} \|\partial^\alpha u\|_\infty, \qquad u \in \mathrm{C}^k(\Omega), \qquad k = 0, 1, 2, \ldots,$$

where $\|\cdot\|_\infty$ is the supremum norm. Here, the sum is over all **multi-indices** $\alpha \in (\mathbb{N} \cup \{0\})^d$ with $|\alpha| := \alpha_1 + \cdots + \alpha_d \le k$, and

$$\partial^\alpha := \partial_1^{\alpha_1} \partial_2^{\alpha_2} \cdots \partial_d^{\alpha_d}$$

is the α**-derivative** operator.

Similarly, we define the linear space $\mathrm{C}^\infty(\Omega)$ of infinitely-often differentiable functions, but this cannot be equipped with a complete norm. A subscript "$_c$" indicates that all functions u in the respective function space (e.g., $\mathrm{C}_c^\infty(\Omega)$) must have their **support**

$$\operatorname{supp} u := \overline{\{\, x \in \Omega : u(x) \neq 0 \,\}}$$

compactly contained in Ω (so, $\operatorname{supp} u \subset \Omega$ and $\operatorname{supp} u$ compact). For the compact containment of a bounded set A in an open set B we write $A \Subset B$, which means that $\overline{A} \subset B$. In this way, the previous condition could be written as $\operatorname{supp} u \Subset \Omega$. We denote by $\mathrm{C}_0^k(\Omega)$ the closure of $\mathrm{C}_c^\infty(\Omega)$ in $\mathrm{C}^k(\Omega)$. All the spaces $\mathrm{C}_0^k(\Omega)$, $k = 0, 1, \ldots$, are separable.

For $k \in \mathbb{N}$ a positive integer and $p \in [1, \infty]$, the **Sobolev space** $\mathrm{W}^{k,p}(\Omega)$ is defined to contain all functions $u \in \mathrm{L}^p(\Omega)$ such that the **weak derivative** $\partial^\alpha u$ exists and lies in $\mathrm{L}^p(\Omega)$ for all multi-indices $\alpha \in (\mathbb{N} \cup \{0\})^d$ with $|\alpha| \le k$. This means that for every such α, there is a (unique) function $v_\alpha \in \mathrm{L}^p(\Omega)$ satisfying

$$\int v_\alpha \cdot \psi \, \mathrm{d}x = (-1)^{|\alpha|} \int u \cdot \partial^\alpha \psi \, \mathrm{d}x \qquad \text{for all } \psi \in \mathrm{C}_c^\infty(\Omega),$$

and we write "$\partial^\alpha u$" for this v_α. The uniqueness follows from the Fundamental Lemma 3.11 of the Calculus of Variations. Clearly, if $u \in \mathrm{C}^k(\Omega)$, then all k'th-order weak derivatives coincide with their classical counterparts. For $u \in \mathrm{W}^{1,p}(\Omega)$ we further define the **weak gradient** and **weak divergence**, via, respectively,

$$\nabla u := (\partial_1 u, \partial_2 u, \ldots, \partial_d u), \qquad \operatorname{div} u := \partial_1 u + \partial_2 u + \cdots + \partial_d u.$$

As norm in $\mathrm{W}^{k,p}(\Omega)$, $p \in [1, \infty)$, we use

$$\|u\|_{\mathrm{W}^{k,p}} := \left(\sum_{|\alpha| \le k} \|\partial^\alpha u\|_{\mathrm{L}^p}^p \right)^{1/p}, \qquad u \in \mathrm{W}^{k,p}(\Omega).$$

For $p = \infty$, we set

$$\|u\|_{\mathrm{W}^{k,\infty}} := \max_{|\alpha| \le k} \|\partial^\alpha u\|_{\mathrm{L}^\infty}, \qquad u \in \mathrm{W}^{k,\infty}(\Omega).$$

Under these norms, the sets $\mathrm{W}^{k,p}(\Omega)$ become Banach spaces.

Concerning the boundary values of Sobolev functions we have:

Theorem A.25 (Trace) *For $p \in [1,\infty]$ there exists a linear* ***trace operator***

$$\mathrm{tr}_\Omega \colon \mathrm{W}^{1,p}(\Omega) \to \mathrm{L}^p(\partial\Omega)$$

such that

$$\mathrm{tr}_\Omega(\varphi) = \varphi|_{\partial\Omega} \qquad \text{if } \varphi \in \mathrm{C}(\overline{\Omega}).$$

We write $\mathrm{tr}_\Omega(u)$ simply as $u|_{\partial\Omega}$. For $p \in (1,\infty)$ the operator tr_Ω is bounded and weakly (weakly if $p = \infty$) continuous between $\mathrm{W}^{1,p}(\Omega)$ and $\mathrm{L}^p(\partial\Omega)$.*

For $p \in (1,\infty)$ denote the image of $\mathrm{W}^{1,p}(\Omega)$ under tr_Ω by $\mathrm{W}^{1-1/p,p}(\partial\Omega)$, which is called the **trace space** of $\mathrm{W}^{1,p}(\Omega)$. The norms on $\mathrm{W}^{1-1/p,p}(\partial\Omega)$) involve fractional derivatives, see [219] for details. For $p = 1$, the trace space is $\mathrm{L}^1(\partial\Omega, \mathcal{H}^{d-1} \llcorner \partial\Omega)$, which we will denote by just $\mathrm{L}^1(\partial\Omega)$.

We write $\mathrm{W}^{1,p}_0(\Omega)$ for the linear subspace of $\mathrm{W}^{1,p}(\Omega)$ consisting of all $\mathrm{W}^{1,p}$-maps with zero boundary values (in the sense of trace). More generally, we use $\mathrm{W}^{1,p}_g(\Omega)$ with $g \in \mathrm{W}^{1-1/p,p}(\partial\Omega)$ (with the convention $\mathrm{W}^{0,1}(\partial\Omega) := \mathrm{L}^1(\partial\Omega)$ in the case $p = 1$) for the affine subspace of all $\mathrm{W}^{1,p}(\Omega)$-maps with boundary trace g.

The following are some properties of Sobolev spaces, stated for simplicity only for the first-order space $\mathrm{W}^{1,p}(\Omega)$.

Theorem A.26 (Extension) *Every $u \in \mathrm{W}^{1,p}(\Omega)$, where $p \in [1,\infty]$, can be extended to $\bar{u} \in \mathrm{W}^{1,p}(\mathbb{R}^d)$ with $\|\bar{u}\|_{\mathrm{W}^{1,p}(\mathbb{R}^d)} \le C\|u\|_{\mathrm{W}^{1,p}(\mathbb{R}^d)}$ and $\operatorname{supp}\bar{u}$ compact, where $C = C(\Omega,p) > 0$ is a constant.*

Theorem A.27 (Poincaré inequalities) *Let $u \in \mathrm{W}^{1,p}(\Omega)$, where $p \in [1,\infty]$.*

(i) If $u|_{\partial\Omega} = 0$ then

$$\|u\|_{\mathrm{L}^p} \le C\|\nabla u\|_{\mathrm{L}^p},$$

where $C = C(\Omega,p) > 0$ is a constant.

(ii) Setting $[u]_\Omega := \fint_\Omega u \,\mathrm{d}x$,

$$\|u - [u]_\Omega\|_{\mathrm{L}^p} \le C\|\nabla u\|_{\mathrm{L}^p},$$

where $C = C(\Omega,p) > 0$ is a constant.

(iii) If $p \in (1,\infty)$ then

$$\|u\|_{\mathrm{L}^p} \le C\big(\|\nabla u\|_{\mathrm{L}^p} + \|u|_{\partial\Omega}\|_{\mathrm{W}^{1-1/p,p}}\big).$$

For $p = 1$ it holds that

$$\|u\|_{\mathrm{L}^1} \le C\big(\|\nabla u\|_{\mathrm{L}^1} + \|u|_{\partial\Omega}\|_{\mathrm{L}^1}\big).$$

Theorem A.28 (Sobolev embedding) *Let $u \in \mathrm{W}^{1,p}(\Omega)$, where $p \in [1,\infty]$.*

(i) If $p < d$, then $u \in \mathrm{L}^{p^*}(\Omega)$, where

$$p^* := \frac{dp}{d-p},$$

and there is a constant $C = C(\Omega, p) > 0$ such that

$$\|u\|_{\mathrm{L}^{p^*}} \le C\|u\|_{\mathrm{W}^{1,p}}.$$

(ii) If $p = d$, then $u \in \mathrm{L}^q(\Omega)$ for all $1 \le q < \infty$ and

$$\|u\|_{\mathrm{L}^q} \le C\|u\|_{\mathrm{W}^{1,p}},$$

where $C = C(\Omega, p, q) > 0$ is a constant.

(iii) If $p > d$, then $u \in \mathrm{C}(\Omega)$ and

$$\|u\|_{\infty} \le C\|u\|_{\mathrm{W}^{1,p}},$$

where $C = C(\Omega, p) > 0$ is a constant.

The second part can in fact be made more precise by considering embeddings into Hölder spaces, see Section 5.6.3 in [133] for details.

Theorem A.29 (Rellich–Kondrachov) *Let $(u_j)_j \subset \mathrm{W}^{1,p}(\Omega)$, where $p \in [1,\infty]$, with $u_j \rightharpoonup u$ in $\mathrm{W}^{1,p}$.*

(i) If $p < d$, then $u_j \to u$ in $\mathrm{L}^q(\Omega)$ for any $q < p^* = dp/(d-p)$.
(ii) If $p = d$, then $u_j \to u$ in $\mathrm{L}^q(\Omega)$ for any $q < \infty$.
(iii) If $p > d$, then $u_j \to u$ uniformly (i.e., in the supremum norm).

We denote by $\mathrm{W}^{-k,q}(\Omega;\mathbb{R}^m)$ the dual space to $\mathrm{W}_0^{k,p}(\Omega;\mathbb{R}^m)$, where $1/p + 1/q = 1$. Analogous statements to the above embedding theorems hold for these negative Sobolev spaces as well, as can be seen by a dualization argument; see [219] for the details.

Theorem A.30 (Density) *For every $u \in \mathrm{W}^{1,p}(\Omega)$, where $p \in [1,\infty]$, and all $\varepsilon > 0$ there exists a map $v \in \mathrm{W}^{1,p}(\Omega) \cap \mathrm{C}^\infty(\overline{\Omega})$ with $v|_{\partial\Omega} = u|_{\partial\Omega}$ and $\|u - v\|_{\mathrm{W}^{1,p}} < \varepsilon$. Moreover, there also exists a countably piecewise affine $w \in (\mathrm{W}^{1,p} \cap \mathrm{C})(\Omega)$ with $w|_{\partial\Omega} = u|_{\partial\Omega}$ and $\|u - w\|_{\mathrm{W}^{1,p}} < \varepsilon$.*

Here, a map $w\colon \Omega \to \mathbb{R}$ is called **(countably) piecewise affine** if there exists a disjoint partition of Ω into countably many open sets D_k ($k \in \mathbb{N}$), up to a negligible set, i.e., $\Omega = Z \cup \bigcup_{k=1}^\infty D_k$, where $|Z| = 0$, such that $w|_{D_k}$ is affine.

For $0 < \gamma \le 1$ a function $u\colon \Omega \to \mathbb{R}$ is **γ-Hölder continuous function**, in symbols $u \in \mathrm{C}^{0,\gamma}(\Omega)$, if

$$\|u\|_{\mathrm{C}^{0,\gamma}} := \|u\|_\infty + \sup_{\substack{x,y\in\Omega \\ x\neq y}} \frac{|u(x) - u(y)|}{|x-y|^\gamma} < \infty.$$

Functions in $\mathrm{C}^{0,1}(\Omega)$ are called **Lipschitz continuous**. We construct the higher-order spaces $\mathrm{C}^{k,\gamma}(\Omega)$ analogously.

Theorem A.31 (Rademacher) *Let $\Omega \subset \mathbb{R}^d$ be an open, bounded, and* convex *set. Then, the space $\mathrm{W}^{1,\infty}(\Omega)$ consists precisely of all Lipschitz maps on Ω, the Lipschitz constant is equal to the $\mathrm{W}^{1,\infty}$-norm, and for $u \in \mathrm{W}^{1,\infty}(\Omega)$ the classical gradient ∇u exists almost everywhere in Ω and agrees with the weak gradient.*

Next, we define a **family of mollifiers** as follows: Let $\eta \in \mathrm{C}_c^\infty(\mathbb{R}^d)$ be radially symmetric and positive. Then, define the family $(\eta_\delta)_{\delta>0}$ by

$$\eta_\delta(x) := \frac{1}{\delta^d}\eta\left(\frac{x}{\delta}\right), \qquad x \in \mathbb{R}^d.$$

For $u \in \mathrm{W}^{k,p}(\mathbb{R}^d)$, where $k \in \mathbb{N} \cup \{0\}$, and $p \in [1,\infty]$, we define the **mollification** $u_\delta \in \mathrm{W}^{k,p}(\mathbb{R}^d)$ of u as the **convolution** between η_δ and u, i.e.,

$$u_\delta(x) := (\eta_\delta \star u)(x) := \int \eta_\delta(x-y)u(y)\,\mathrm{d}y, \qquad x \in \mathbb{R}^d.$$

Lemma A.32 *For every $u \in \mathrm{W}^{k,p}(\mathbb{R}^d)$, where $k \in \{0,1,2,\dots\}$ and $p \in [1,\infty)$, it holds that*

$$u_\delta \to u \quad \text{in } \mathrm{W}^{k,p}$$

as $\delta \downarrow 0$. The same holds in $\mathrm{W}^{k,p}_{\mathrm{loc}}(\Omega)$ if $\Omega \subset \mathbb{R}^d$ is a bounded Lipschitz domain.

Analogous results also hold for continuously differentiable functions.

Lemma A.33 (Young's inequality for convolutions) *Let $u \in \mathrm{L}^p(\mathbb{R}^d)$, $v \in \mathrm{L}^q(\mathbb{R}^d)$ and let $p,q,r \in [1,\infty]$ be such that*

$$1 + \frac{1}{r} = \frac{1}{p} + \frac{1}{q}.$$

Then,

$$\|u \star v\|_{\mathrm{L}^r} \le \|u\|_{\mathrm{L}^p} \cdot \|v\|_{\mathrm{L}^q}.$$

Finally, all the above notions and theorems continue to hold for vector-valued functions $u = (u^1,\dots,u^m)^\mathsf{T} : \Omega \to \mathbb{R}^m$ and in this case we set

$$\nabla u := \begin{pmatrix} \partial_1 u^1 & \partial_2 u^1 & \cdots & \partial_d u^1 \\ \partial_1 u^2 & \partial_2 u^2 & \cdots & \partial_d u^2 \\ \vdots & \vdots & & \vdots \\ \partial_1 u^m & \partial_2 u^m & \cdots & \partial_d u^m \end{pmatrix}.$$

We use the spaces $\mathrm{C}(\Omega;\mathbb{R}^m)$, $\mathrm{C}^k(\Omega;\mathbb{R}^m)$, $\mathrm{W}^{k,p}(\Omega;\mathbb{R}^m)$, $\mathrm{C}^{k,\gamma}(\Omega;\mathbb{R}^m)$ with analogous definitions as in the scalar-valued case; for matrices like ∇u we use the Frobenius matrix norm and similarly for higher-order tensors.

Occasionally, we employ *local* versions of the spaces defined above, namely $\mathrm{C}_{\mathrm{loc}}(\Omega)$, $\mathrm{C}^k_{\mathrm{loc}}(\Omega)$, $\mathrm{W}^{k,p}_{\mathrm{loc}}(\Omega)$, $\mathrm{C}^{k,\gamma}_{\mathrm{loc}}(\Omega)$, where the defining norm is only finite on every compact subset of Ω.

Finally, we quote the following two classical results about extensions of functions:

Theorem A.34 (Tietze) *Let X be a metric space, let $F \subset X$ be closed, and assume that $f\colon F \to \mathbb{R}^m$ is continuous. Then, f can be extended to a continuous $\bar{f}\colon X \to \mathbb{R}^m$. If f is uniformly bounded, then $\bar{f}$ can also be chosen uniformly bounded with the same supremum norm.*

Theorem A.35 (Kirszbraun) *Let $\Omega \subset \mathbb{R}^d$ and let $f\colon \Omega \to \mathbb{R}^m$ be a Lipschitz continuous map. Then, f can be extended to $\bar{f}\colon \mathbb{R}^d \to \mathbb{R}^m$ with the same Lipschitz constant as f.*

A.6 Harmonic Analysis

We only need a few basics of Fourier analysis and the Mihlin multiplier theorem. A thorough introduction can be found in [164, 165].

Define for $u \in \mathrm{L}^1(\mathbb{R}^d)$ (or vector-valued u) the **Fourier transform** $\widehat{u} = \mathcal{F}u \in \mathrm{L}^\infty(\mathbb{R}^d)$ as follows:

$$\widehat{u}(\xi) := \mathcal{F}u(\xi) := \int_{\mathbb{R}^d} u(x)\mathrm{e}^{-2\pi \mathrm{i} x\cdot\xi}\,\mathrm{d}x, \qquad \xi \in \mathbb{R}^d.$$

We also define the **inverse Fourier transform** $\check{v} = \mathcal{F}^{-1}v$ for $v \in \mathrm{L}^1(\mathbb{R}^d)$ to be

$$\check{u}(x) := \mathcal{F}^{-1}v(x) := \int_{\mathbb{R}^d} v(\xi)\mathrm{e}^{2\pi \mathrm{i} x\cdot\xi}\,\mathrm{d}\xi, \qquad x \in \mathbb{R}^d.$$

For $u, v \in (\mathrm{L}^1 \cap \mathrm{L}^2)(\mathbb{R}^d)$ we have the **Plancherel identity**,

$$\|\widehat{u}\|_{\mathrm{L}^2} = \|u\|_{\mathrm{L}^2} \tag{A.6}$$

and the **Parseval relation**

$$\int u \cdot \bar{v}\,\mathrm{d}x = \int \widehat{u} \cdot \bar{\widehat{v}}\,\mathrm{d}\xi. \tag{A.7}$$

The Plancherel identity then allows one to extend $\mathcal{F}, \mathcal{F}^{-1}$ to the space $\mathrm{L}^2(\mathbb{R}^d)$ by a density argument.

The following is a classical result concerning the $(\mathrm{L}^p \to \mathrm{L}^p)$-boundedness of Fourier multiplier operators, see, for instance, [164] and [49] (Theorem 6.1.6) for a proof.

Theorem A.36 (Mihlin multiplier theorem) *Let $m \in \mathrm{C}^{\lfloor d/2 \rfloor + 1}(\mathbb{R}^d \setminus \{0\}; \mathbb{C})$ satisfy*

$$|\partial^\alpha m(\xi)| \le K|\xi|^{-|\alpha|}, \qquad \xi \in \mathbb{R}^d \setminus \{0\},$$

for all multi-indices $\alpha \in \mathbb{N}_0^d$ with $|\alpha| := |\alpha_1| + \cdots + |\alpha_d| \le \lfloor d/2 \rfloor + 1$ (here, $\lfloor t \rfloor$ denotes the largest integer less than or equal to $t \in \mathbb{R}$) and some $K > 0$. Then,

$$Tu := \mathcal{F}^{-1}[m(\xi)\widehat{u}(\xi)],$$

which for $u \in \mathrm{L}^2(\mathbb{R}^d)$ is well-defined via the Plancherel identity (A.6), extends to a bounded operator $T\colon \mathrm{L}^p(\mathbb{R}^d) \to \mathrm{L}^p(\mathbb{R}^d)$ for all $p \in (1, \infty)$ satisfying

$$\|T\|_{\mathrm{L}^p \to \mathrm{L}^p} \le C \max\{p, (p-1)^{-1}\}K,$$

where $C = C(d) > 0$ is a constant. Furthermore, for $p = 1$ the "weak-type" estimate

$$\left|\left\{ x \in \mathbb{R}^d \ : \ |(Tu)(x)| \ge t \right\}\right| \le \frac{CK}{t}\|u\|_{\mathrm{L}^1}$$

holds for all $t > 0$ and a constant $C = C(d) > 0$.

As a special case, the conclusions of the preceding theorem hold for any positively 0-homogeneous smooth multiplier $m\colon \mathbb{R}^d \setminus \{0\} \to \mathbb{C}$.

Define the **Bessel potential** of order $s > 0$ via

$$(\mathrm{id} - \Delta)^{-s/2} u := \mathcal{F}^{-1}\left[(1 + 4\pi^2|\xi|^2)^{-s/2}\widehat{u}(\xi)\right], \qquad u \in \mathrm{C}_c^\infty(\mathbb{R}^d).$$

We then have (see [164] for a proof):

Proposition A.37 *The operators $(\mathrm{id} - \Delta)^{-s/2}$, $s > 0$, extend to bounded linear operators*

$$(\mathrm{id} - \Delta)^{-s/2}\colon \mathrm{W}^{k,p}(\mathbb{R}^d) \to \mathrm{W}^{k+s,p}(\mathbb{R}^d),$$

where $k \in \mathbb{N} \cup \{0\}$ (with $\mathrm{W}^{0,p} := \mathrm{L}^p$) and $p \in (1, \infty)$.

We will also on occasion use the **(centered) maximal function** $\mathrm{M}f\colon \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ of $f\colon \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$, which is defined as

$$(\mathrm{M}f)(x_0) := \sup_{r>0} \fint_{B_r(x_0)} |f(x)| \, \mathrm{d}x, \qquad x_0 \in \mathbb{R}^d.$$

We quote the following results about the maximal function, whose proofs can be found in [164, 220, 304] (in particular, the less well-known statement (iii) is contained in Lemma 1.68 of [220]):

Theorem A.38 *Let $f \in \mathrm{L}^p(\mathbb{R}^d)$, where $p \in [1, \infty]$.*

(i) If $p \in (1, \infty]$, then

$$\|\mathrm{M}f\|_{\mathrm{L}^p} \le C\|f\|_{\mathrm{L}^p},$$

where $C = C(d, p) > 0$ is a constant.

(ii) If $p \in [1, \infty)$, then the weak-type estimate

$$\left|\left\{ x \in \mathbb{R}^d \ : \ |\mathrm{M}f| \geq t \right\}\right| \leq \frac{C}{t^p} \int_{\{|f| \geq t/2\}} |f|^p \, \mathrm{d}x \leq \frac{C}{t^p} \|u\|_{\mathrm{L}^p}^p$$

holds for all $t > 0$ and a constant $C = C(d) > 0$.

(iii) Assume additionally that $f \in \mathrm{W}^{1,p}(\mathbb{R}^d)$, where $p \in (1, \infty]$. Then, for every $K > 0$, f is Lipschitz continuous on the set $\{\mathrm{M}(|f| + |\nabla f|) < K\}$ and its Lipschitz constant is bounded by CK, where $C = C(d, m, p) > 0$ is a constant.

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